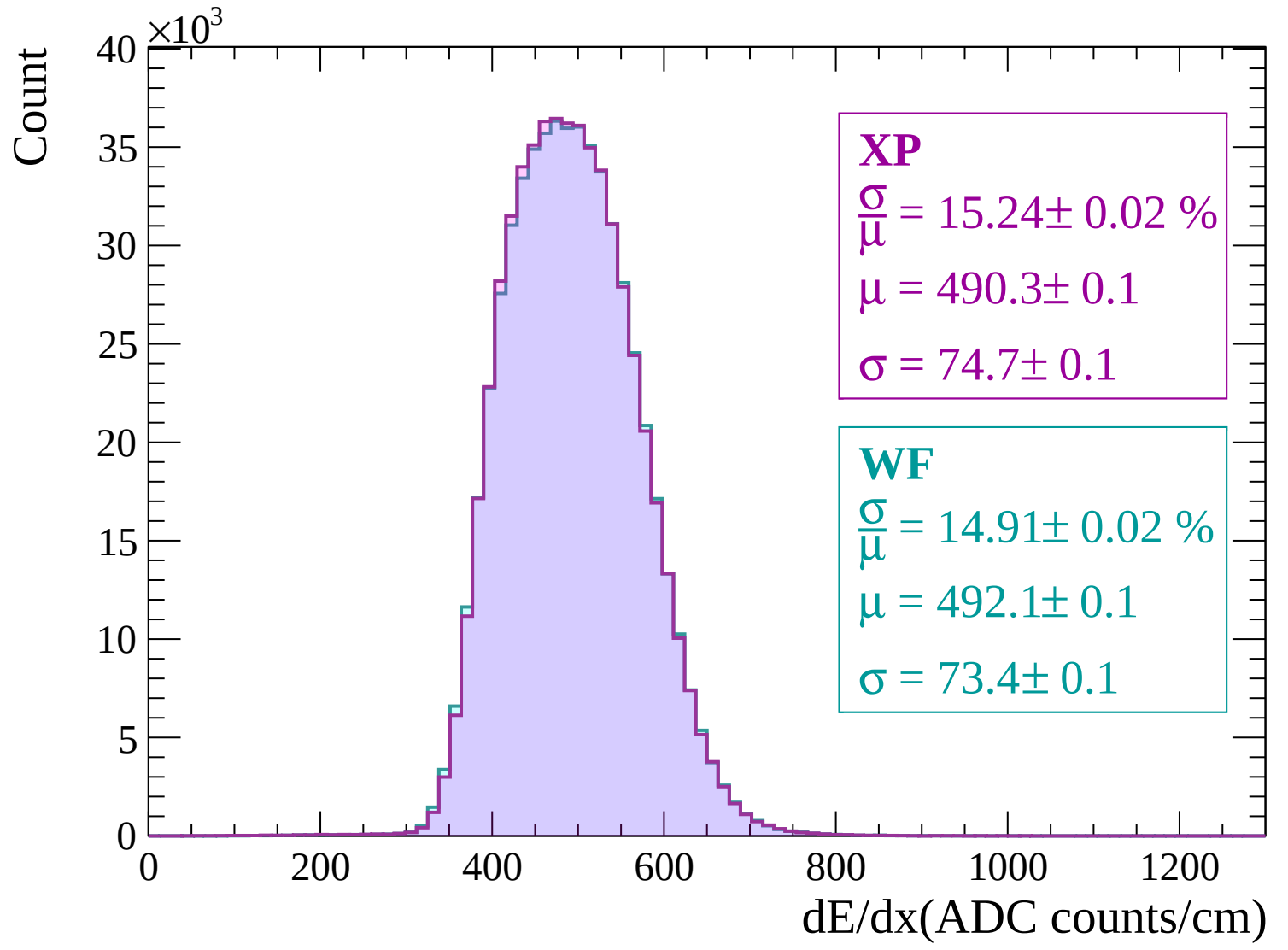
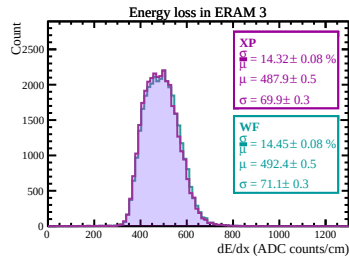
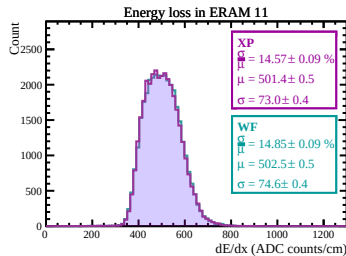
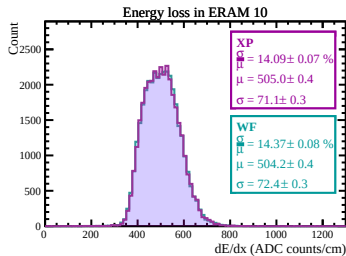
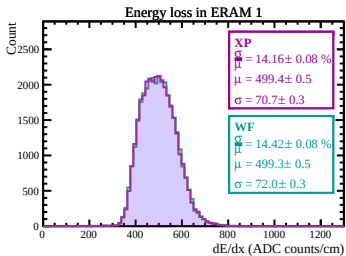
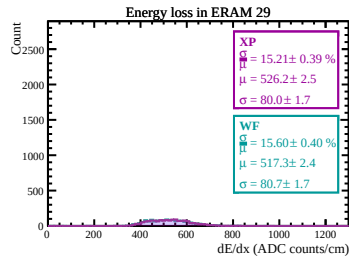
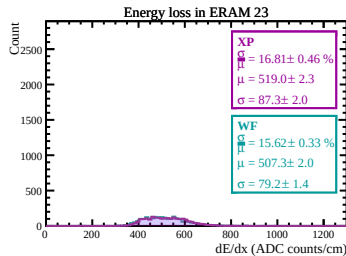
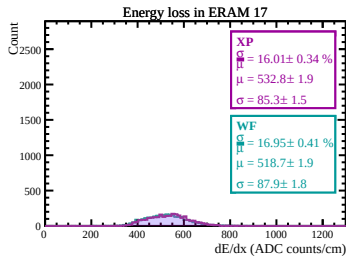
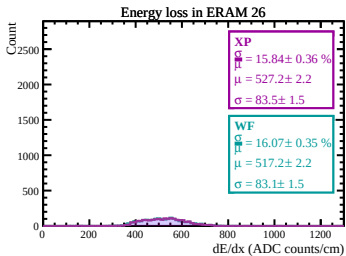
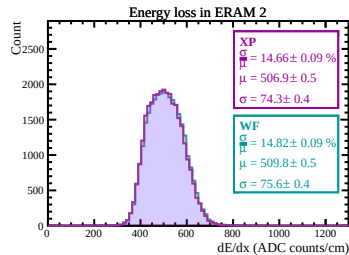
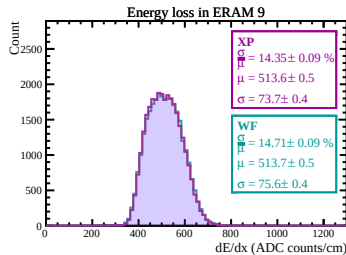
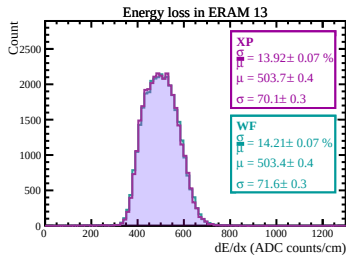
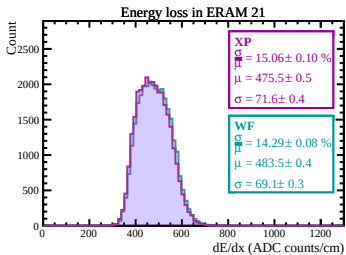
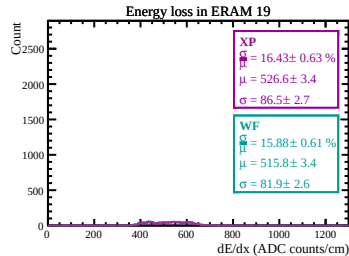
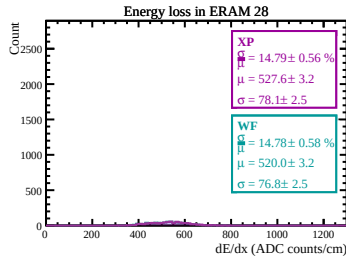
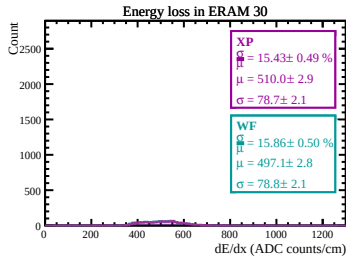
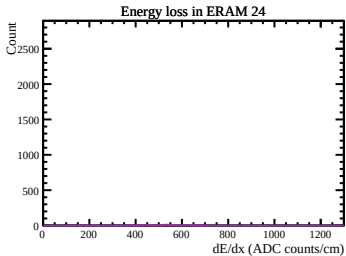
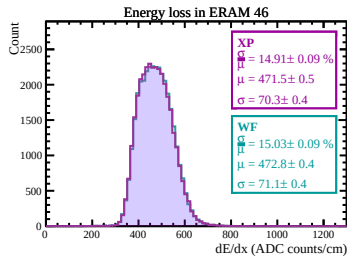
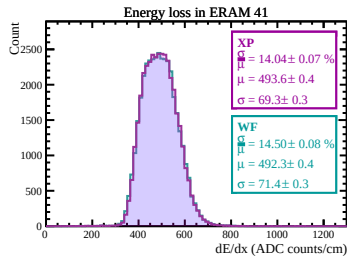
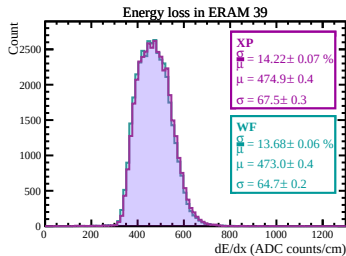
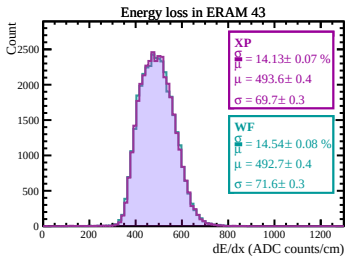
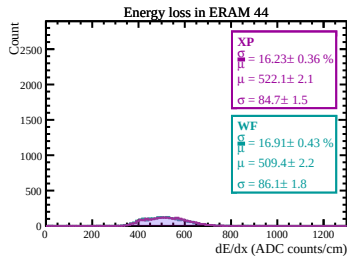
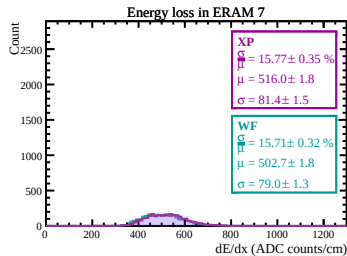
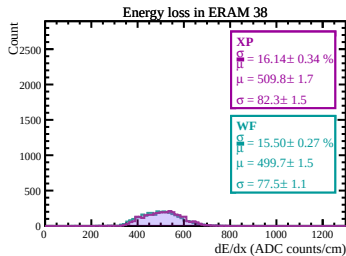
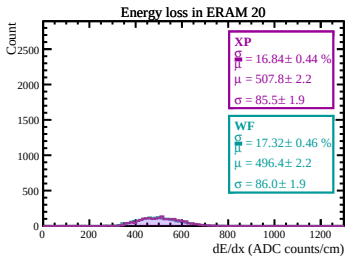
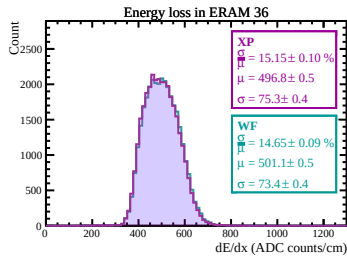
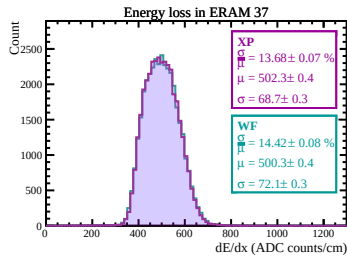
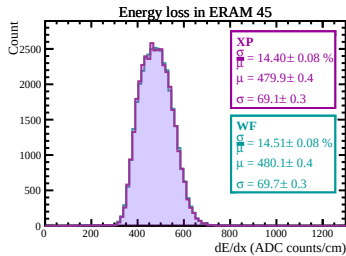
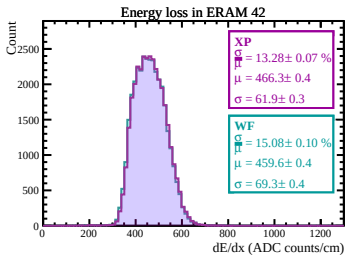
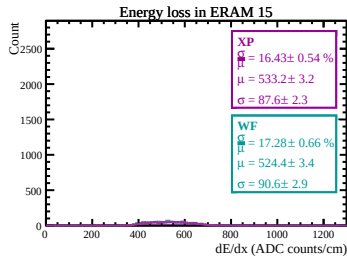
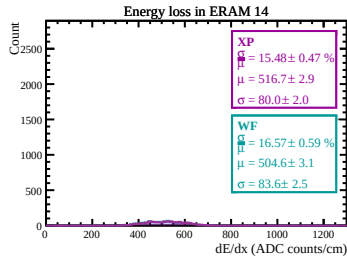
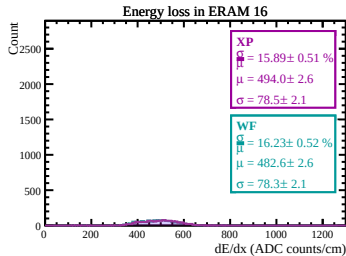
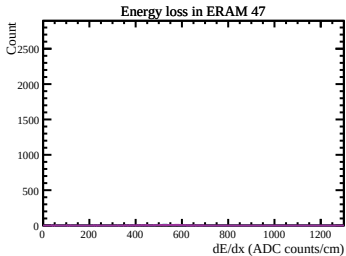
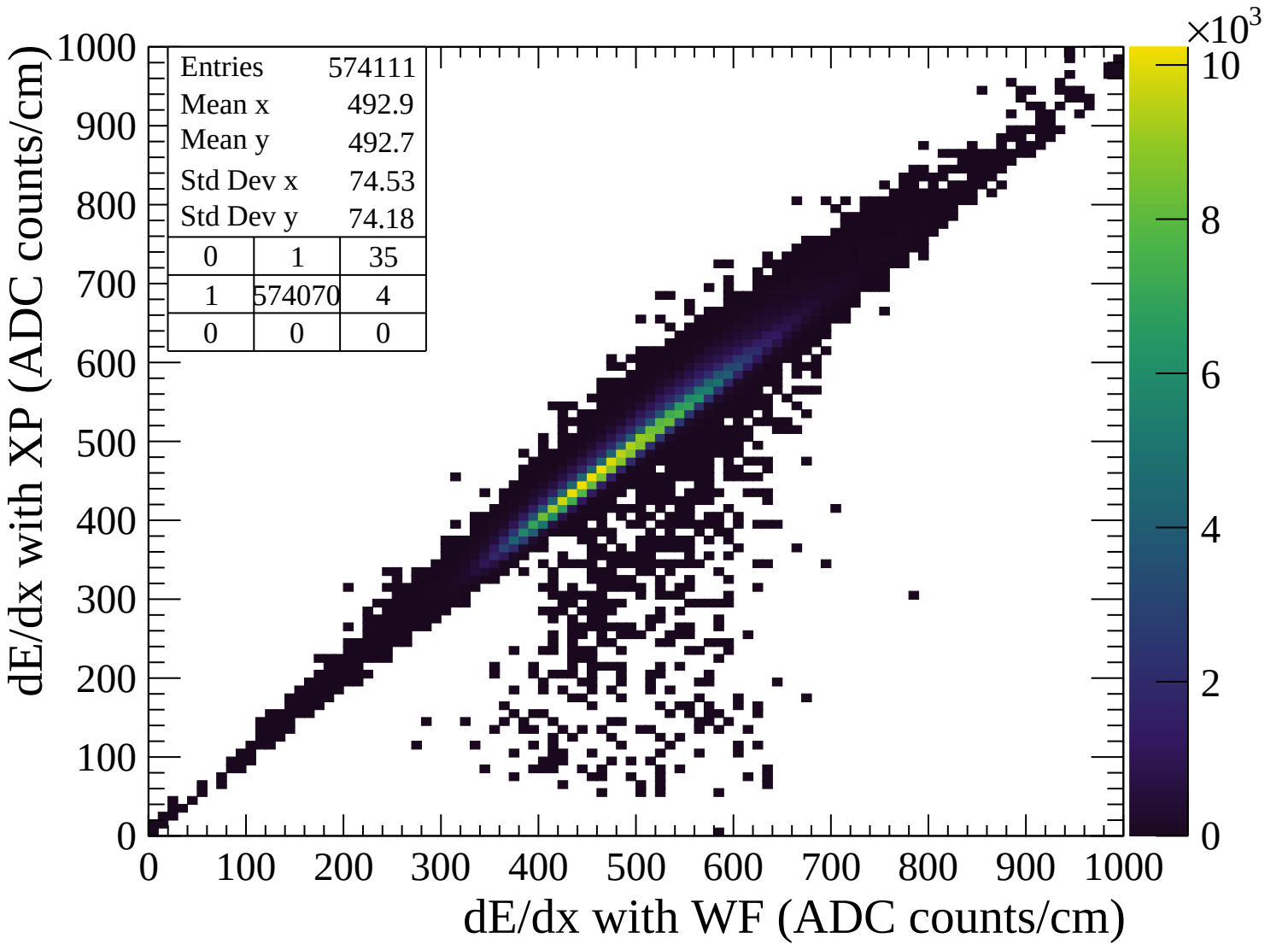
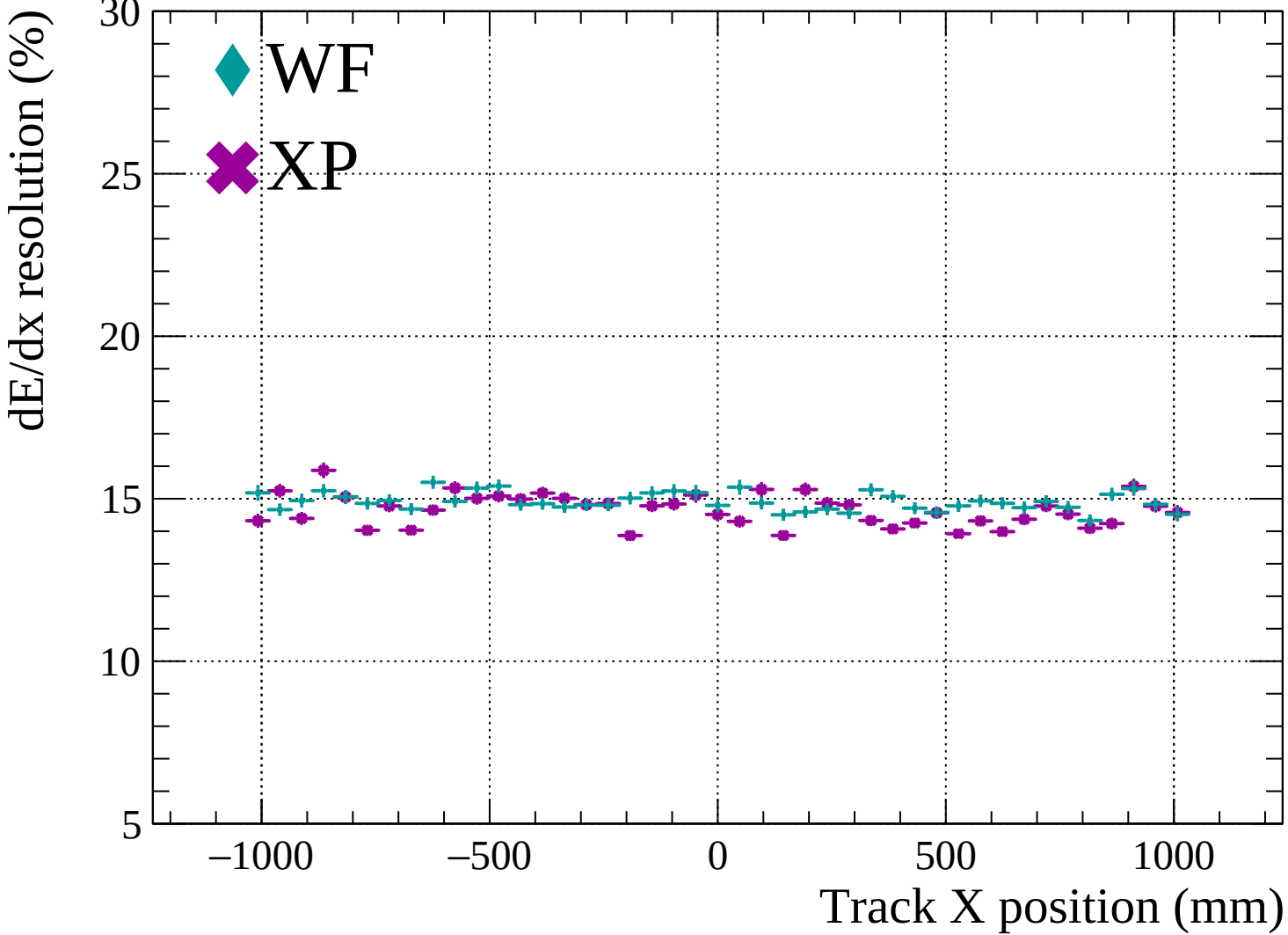


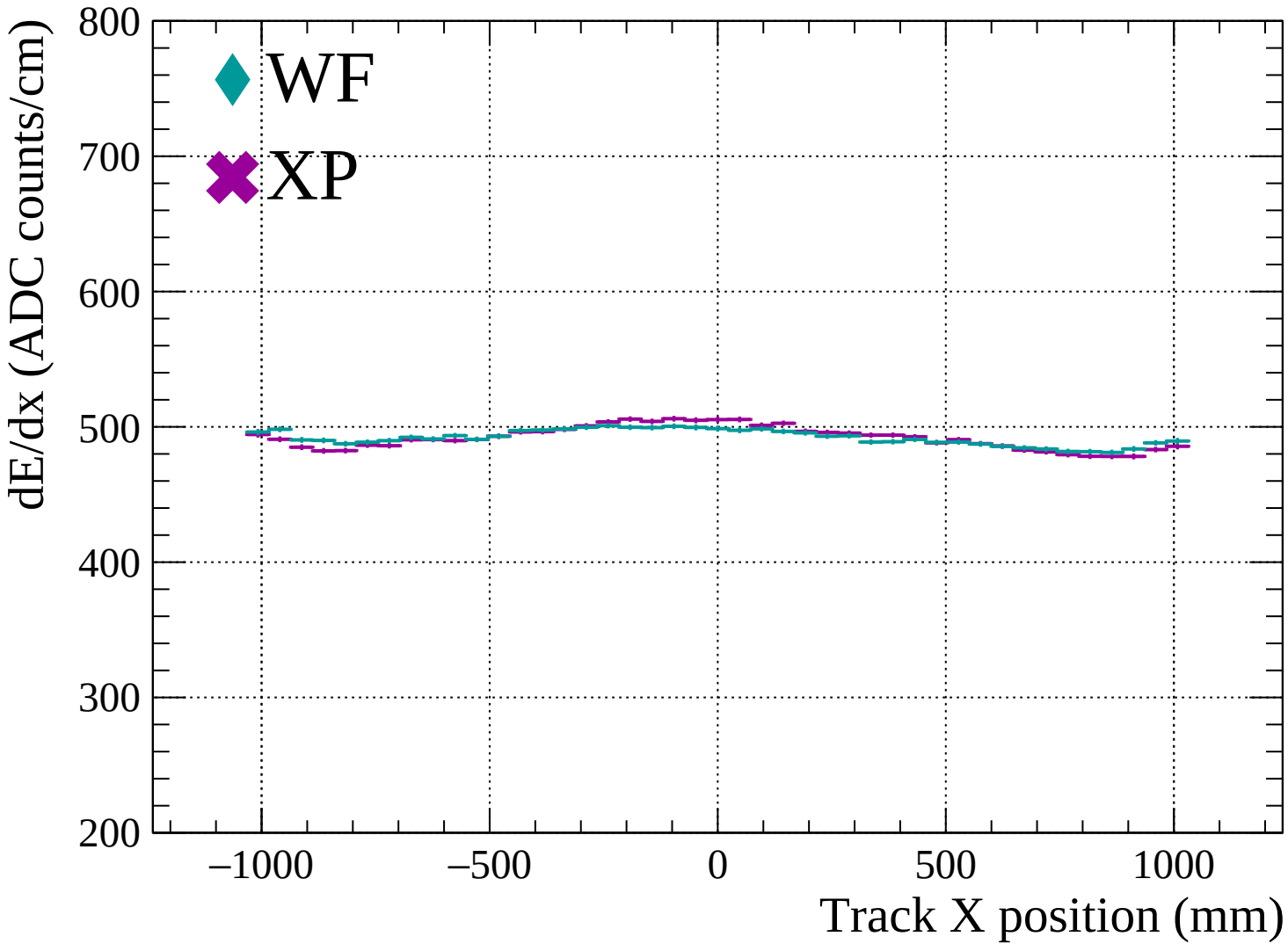


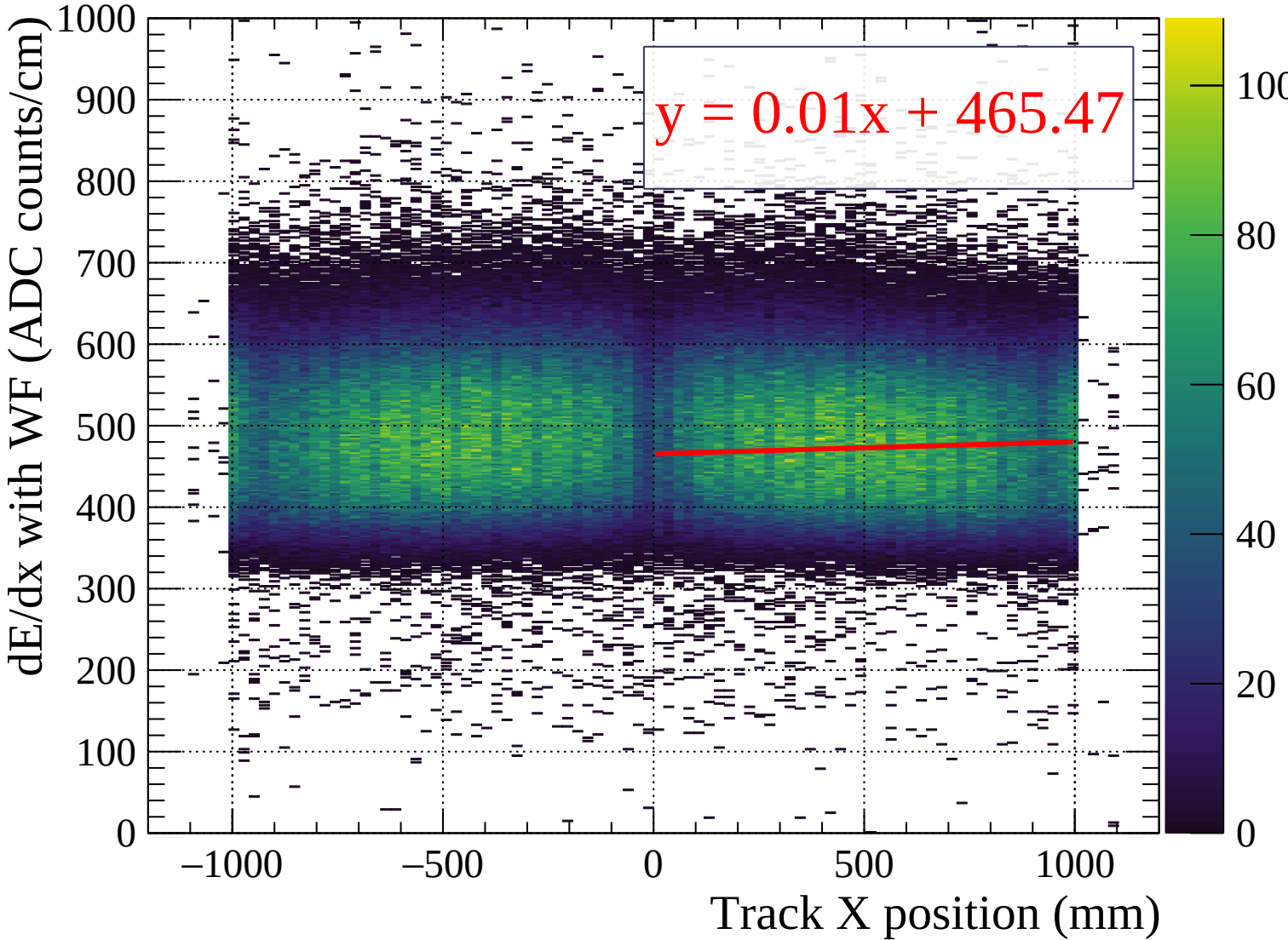


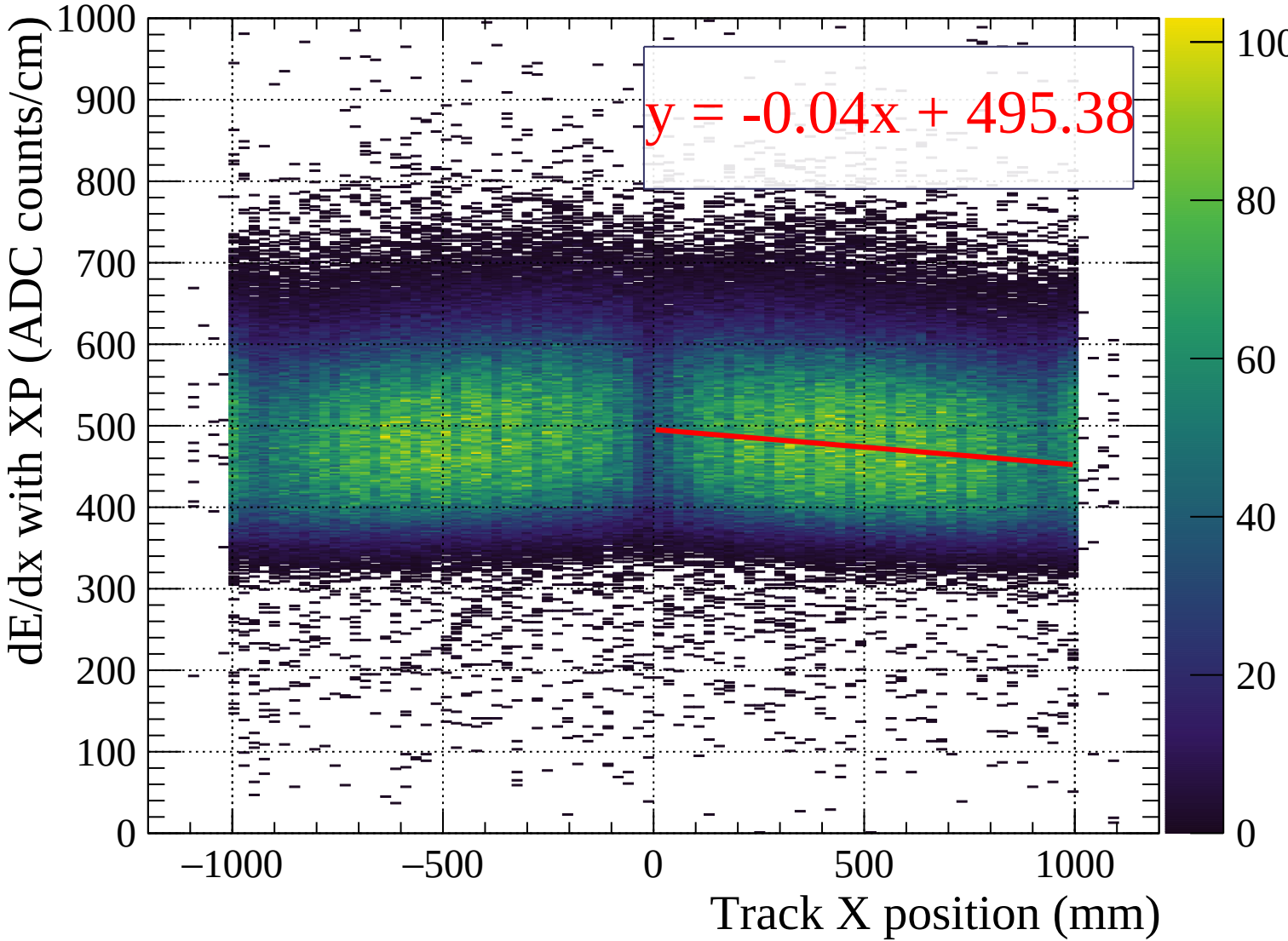


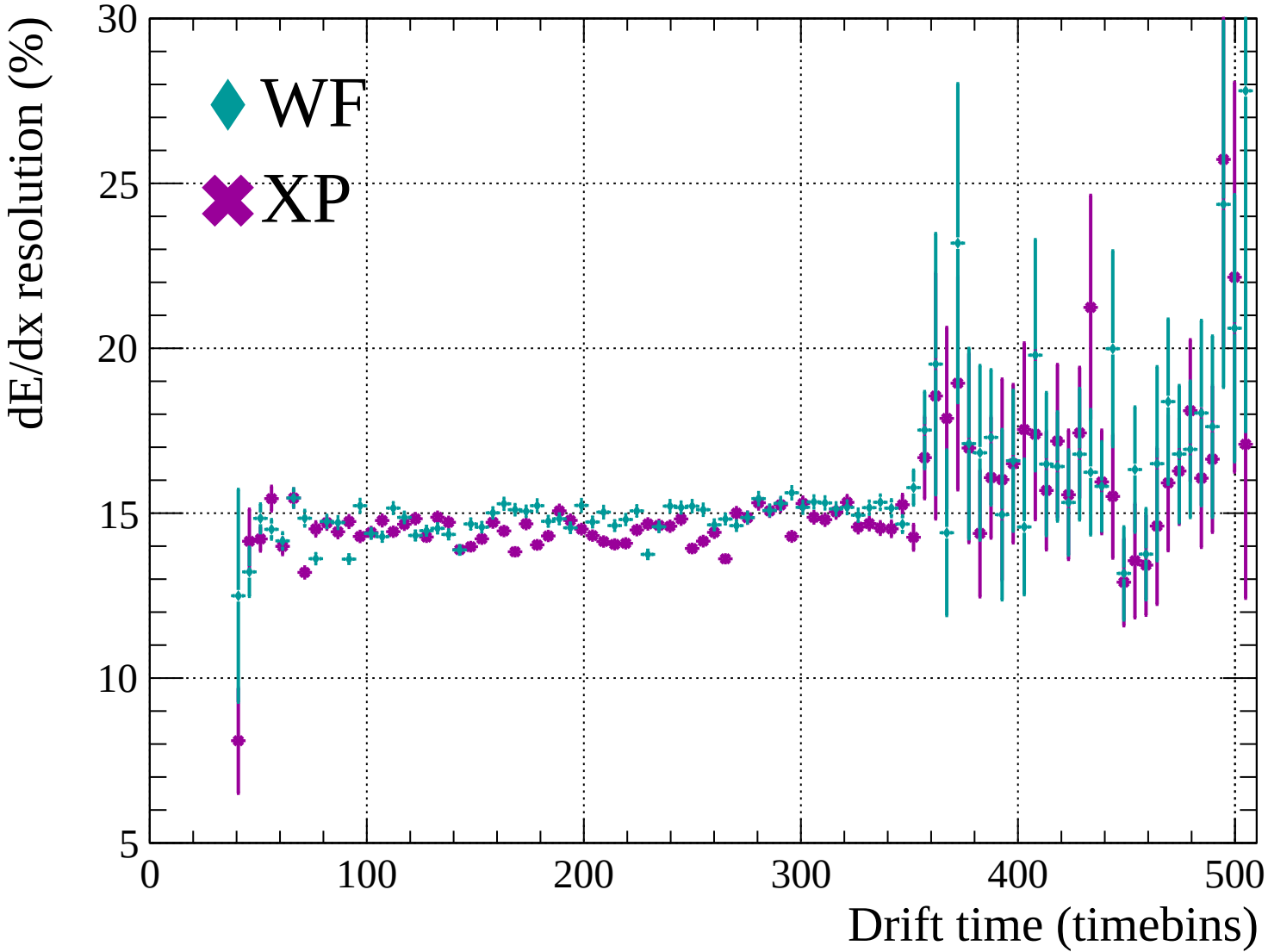


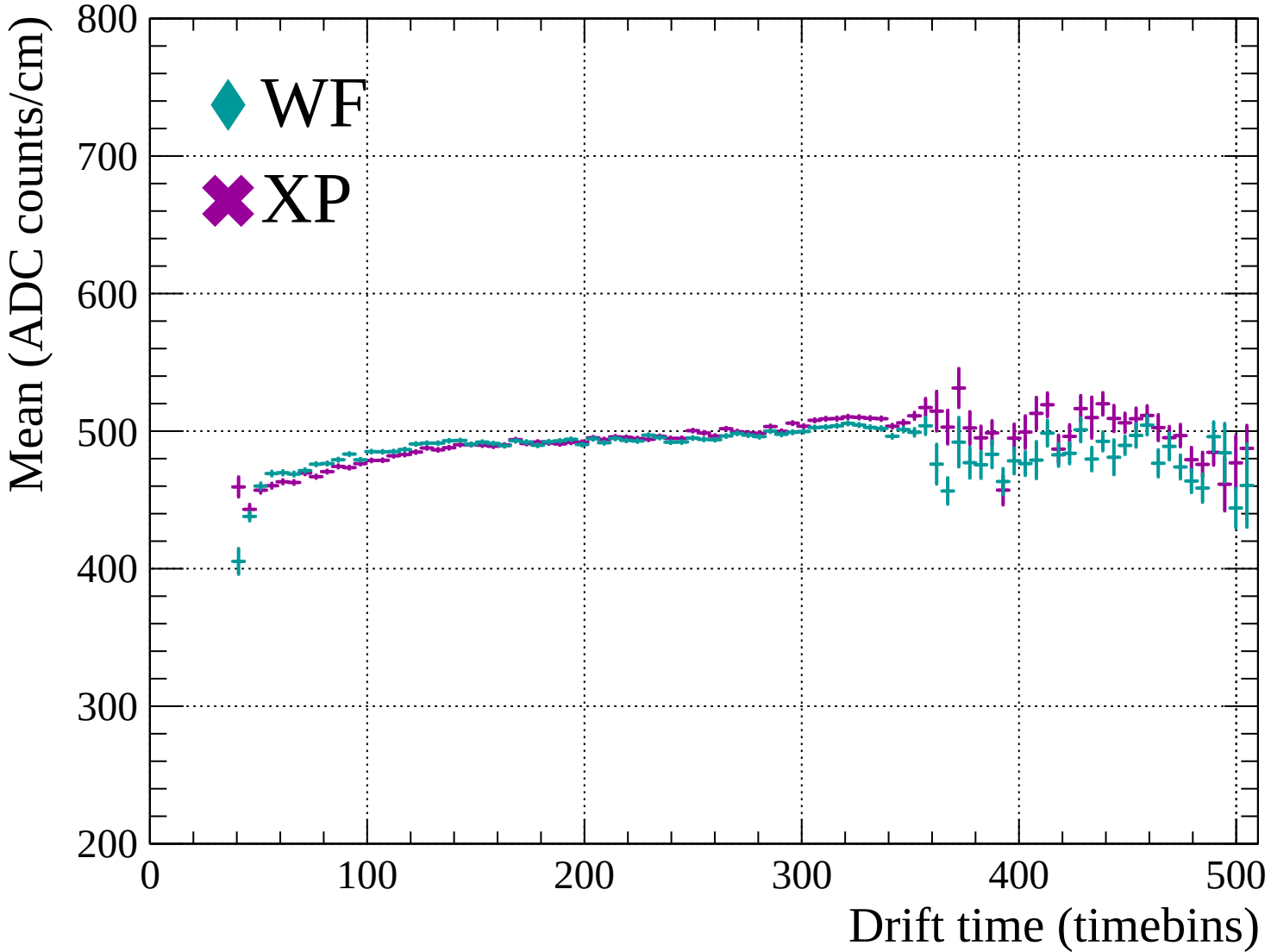


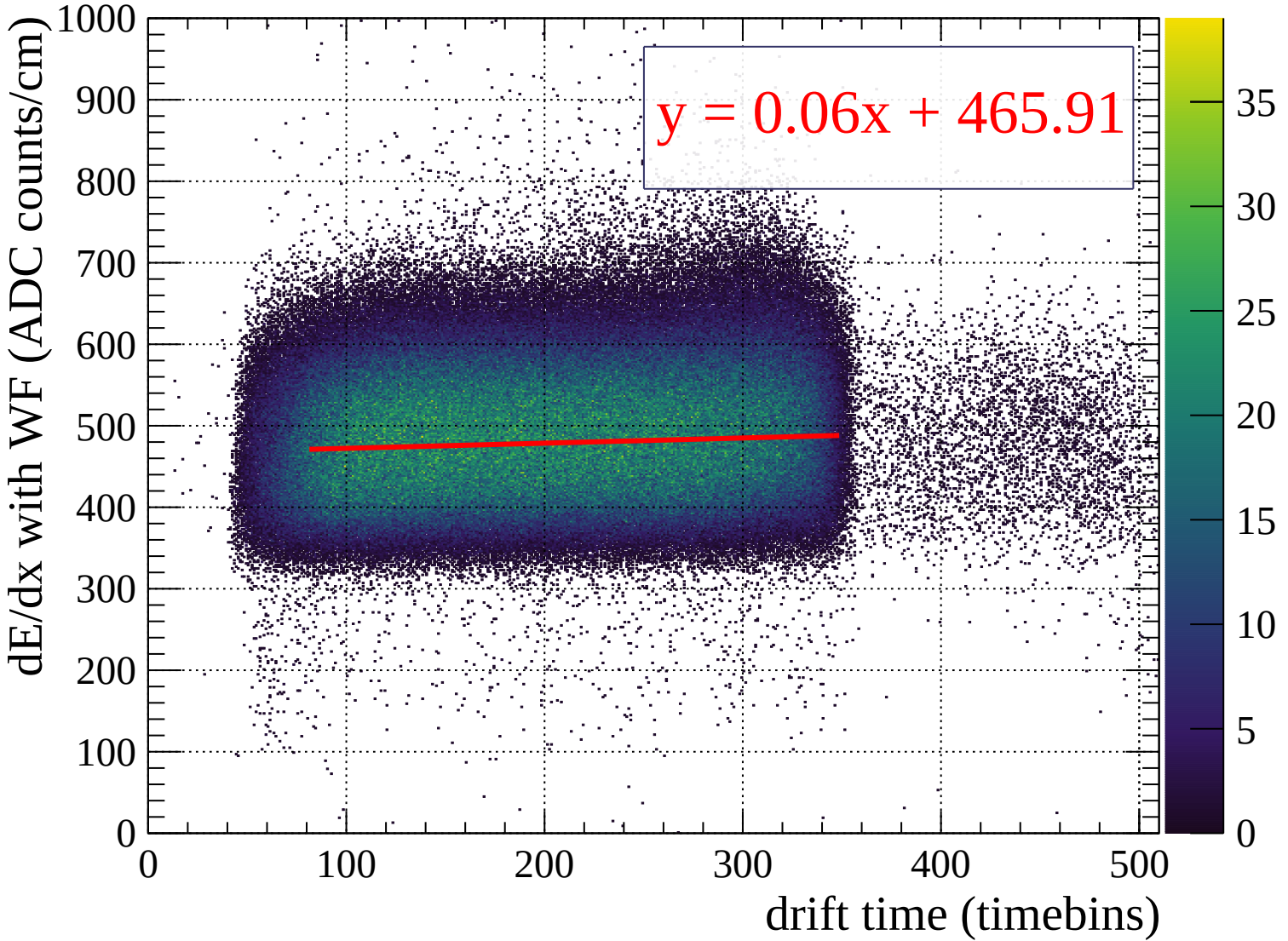


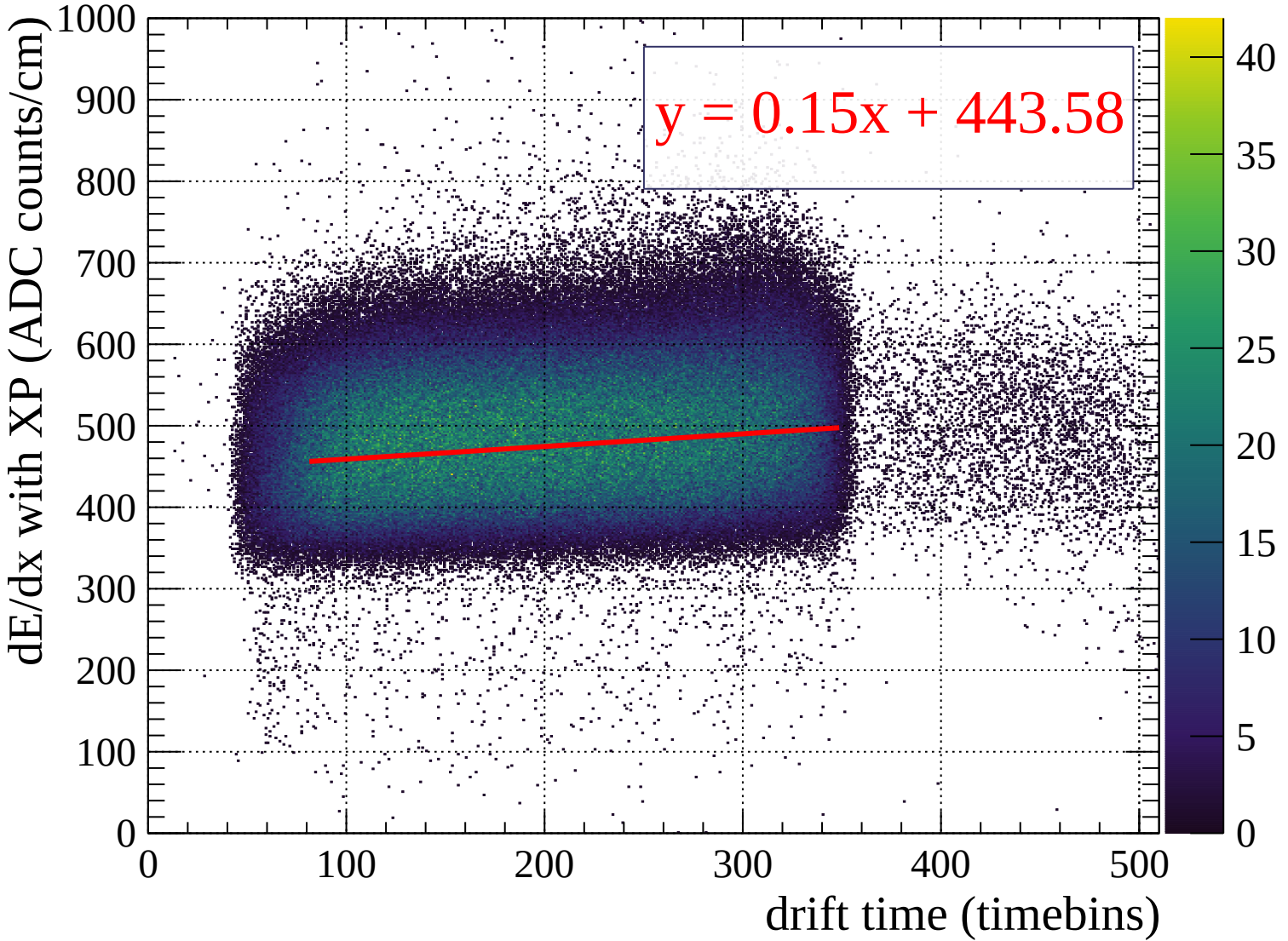


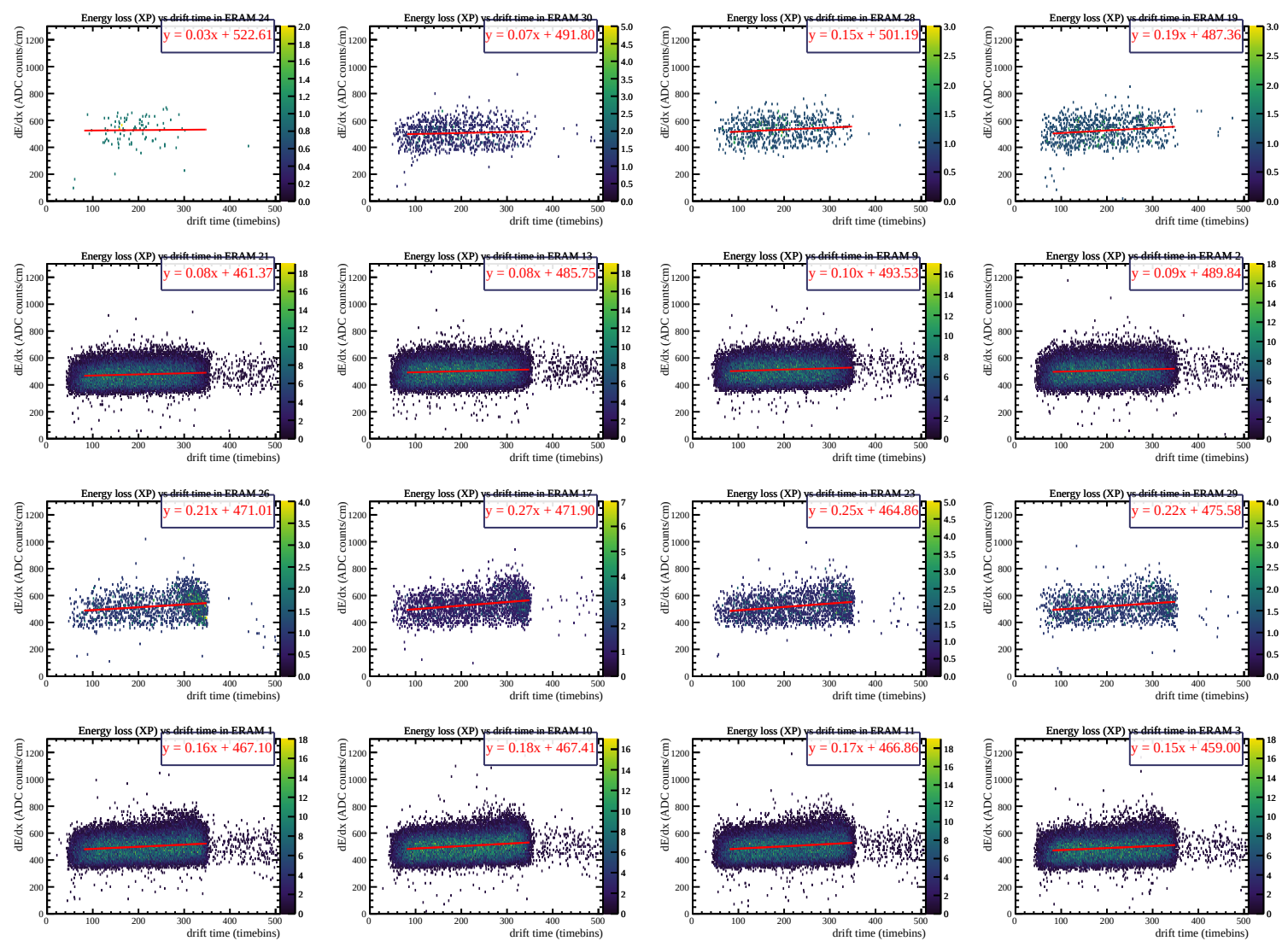


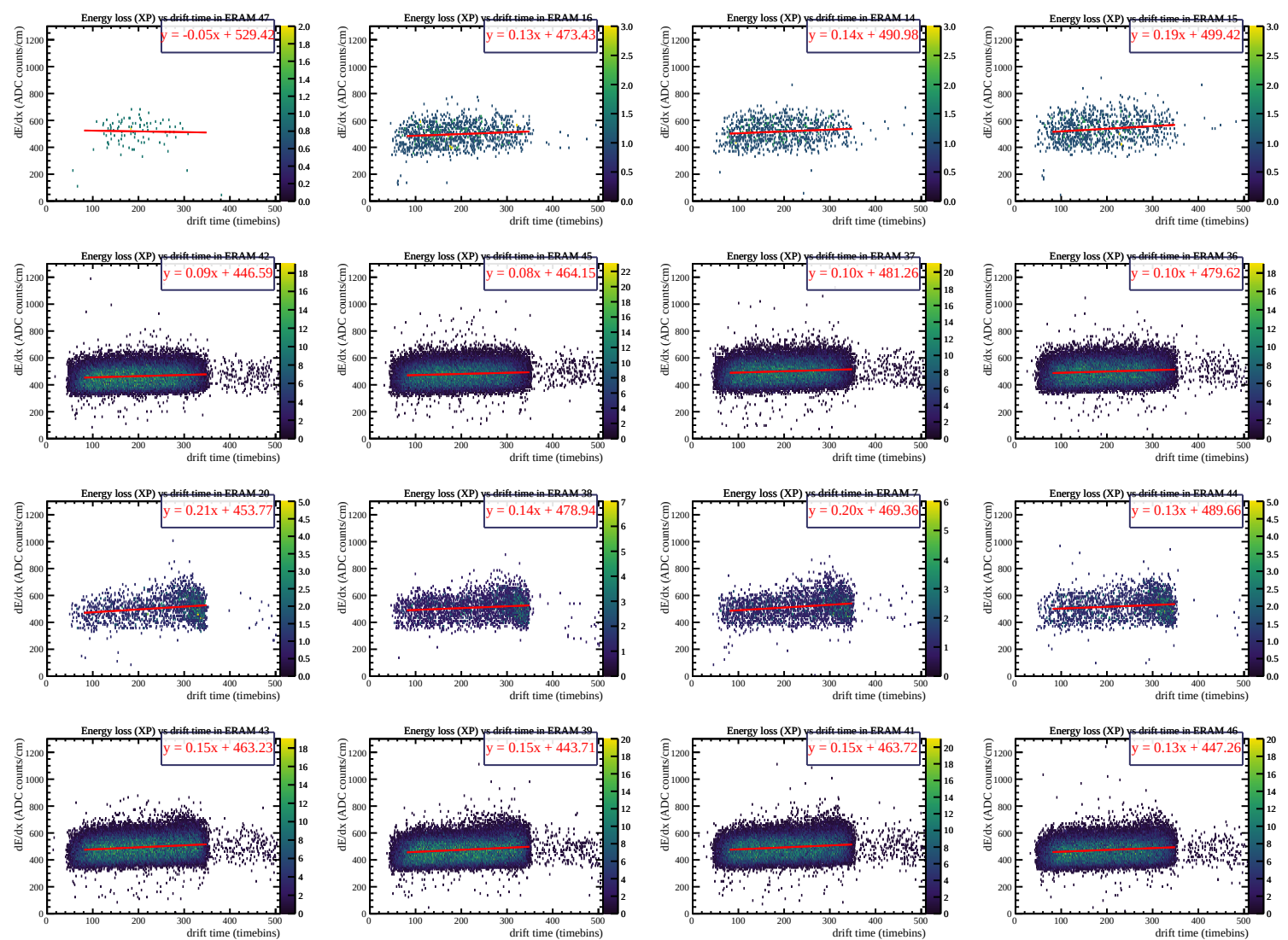


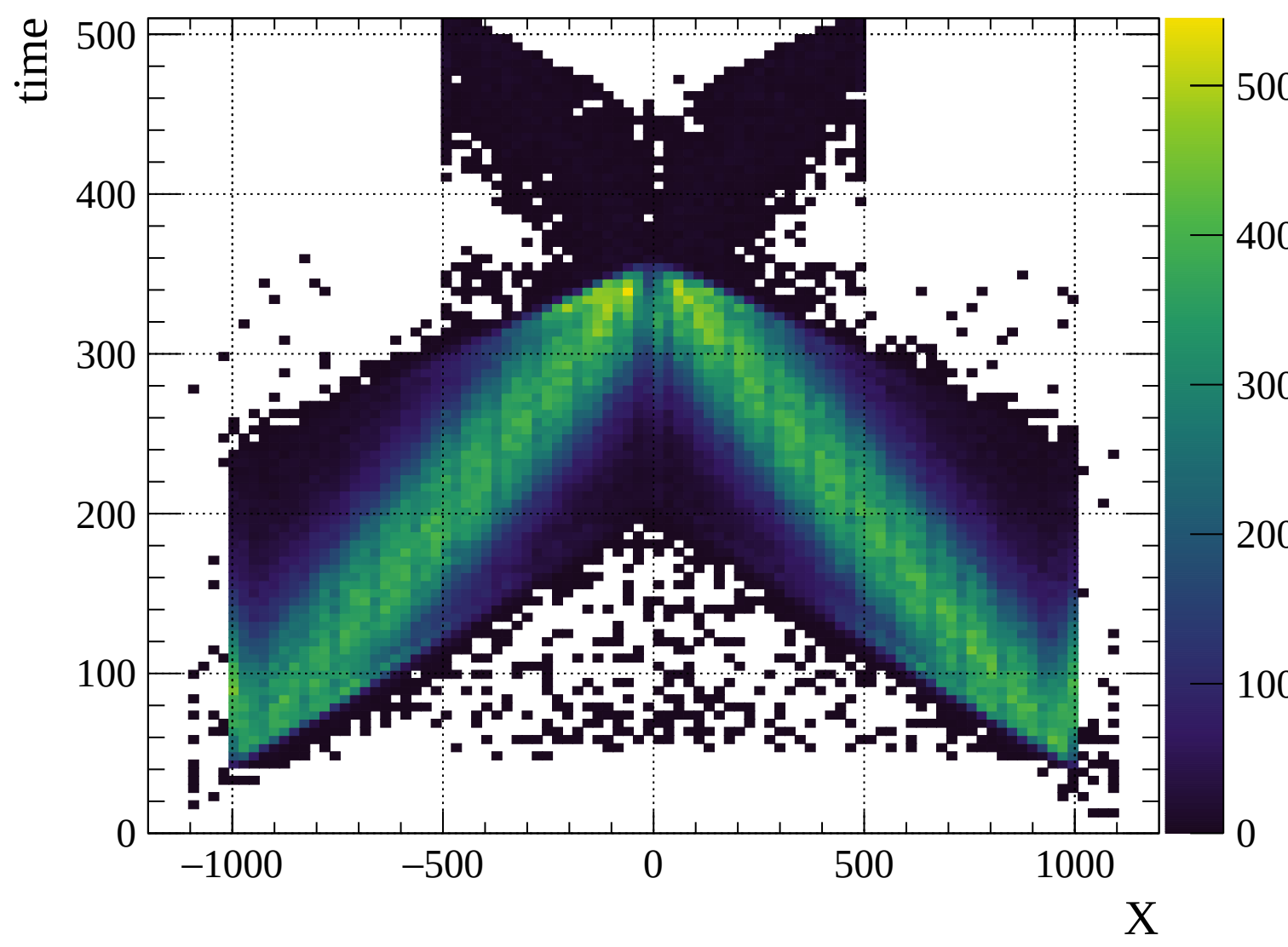


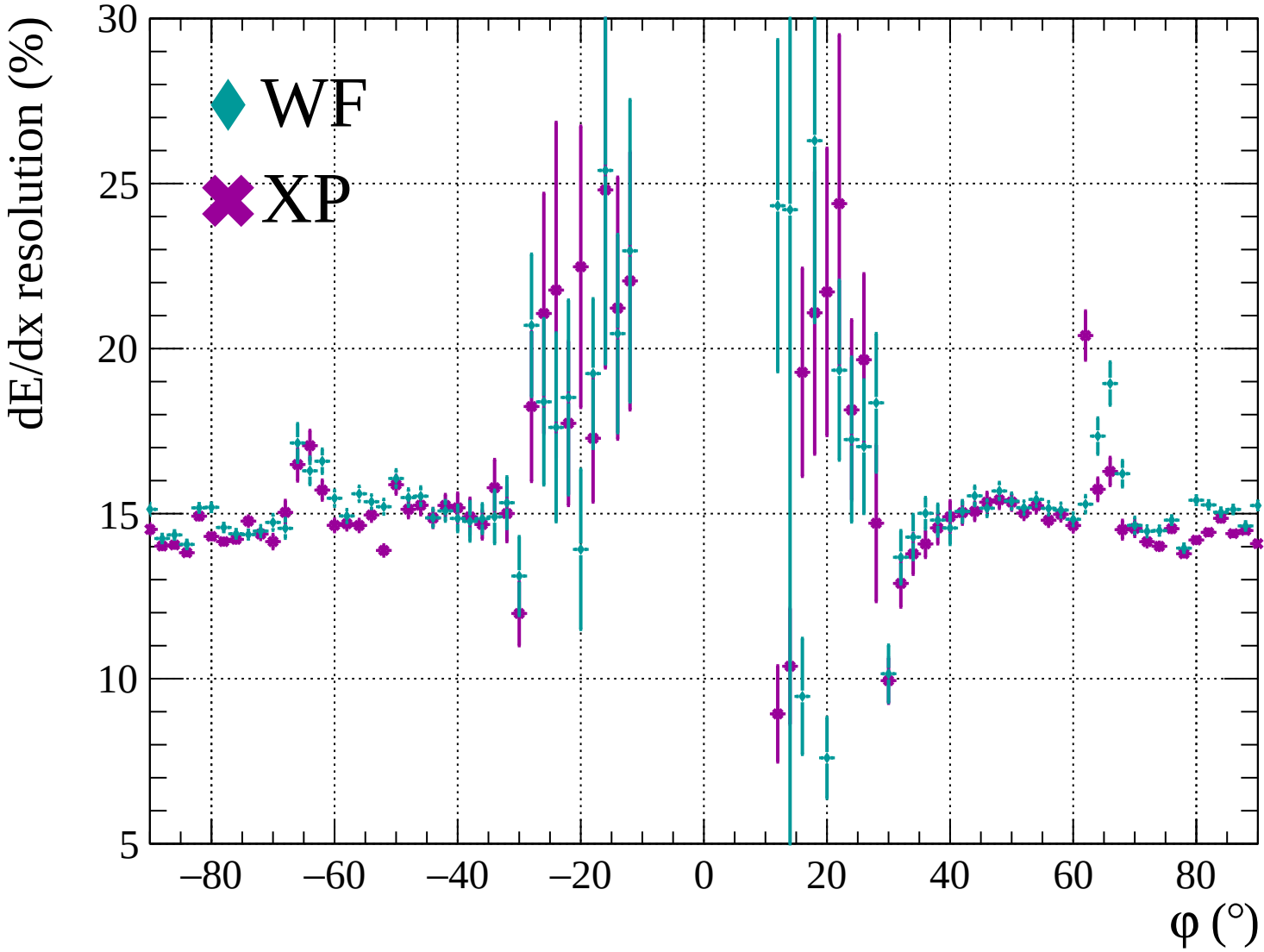


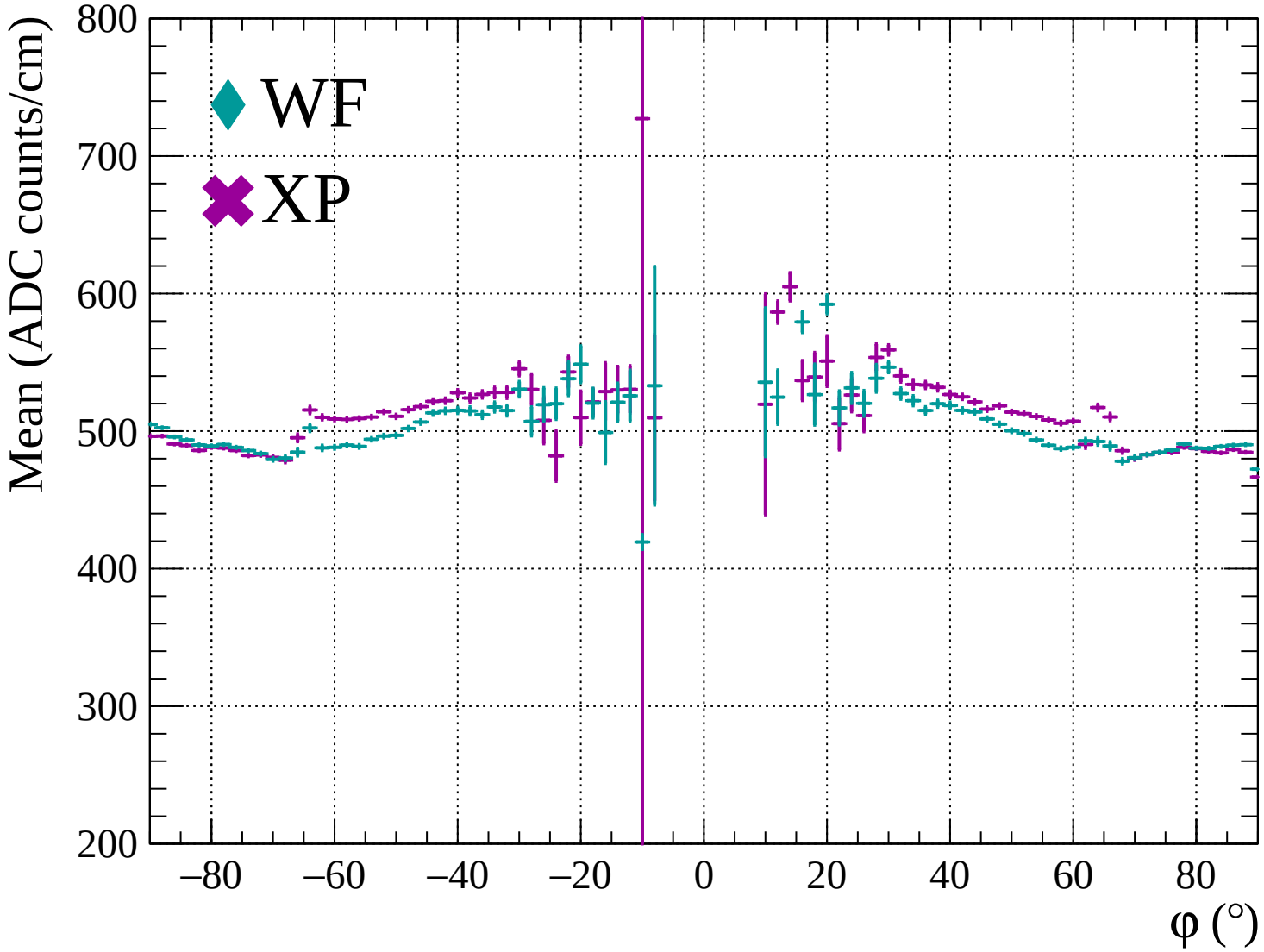


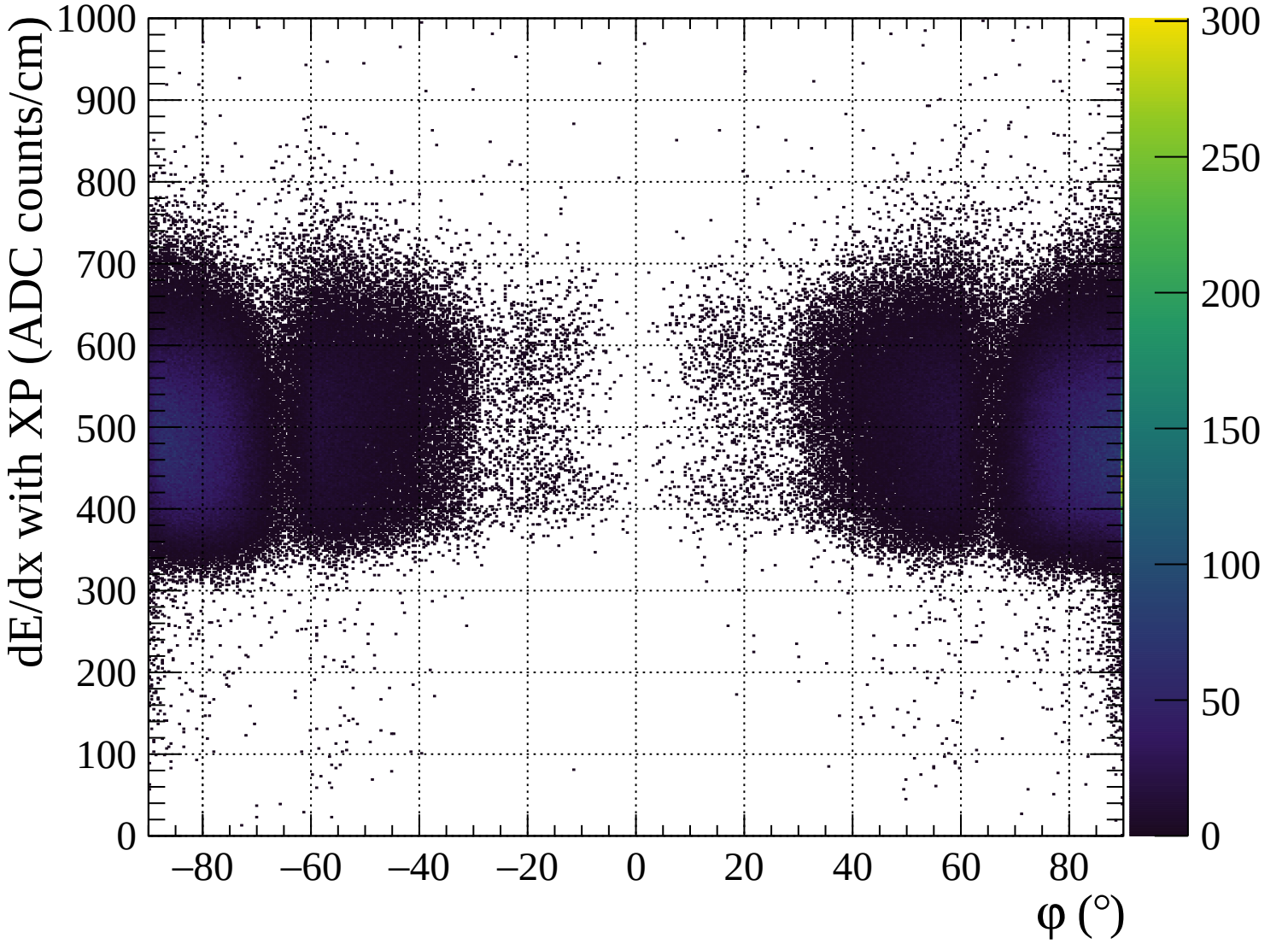


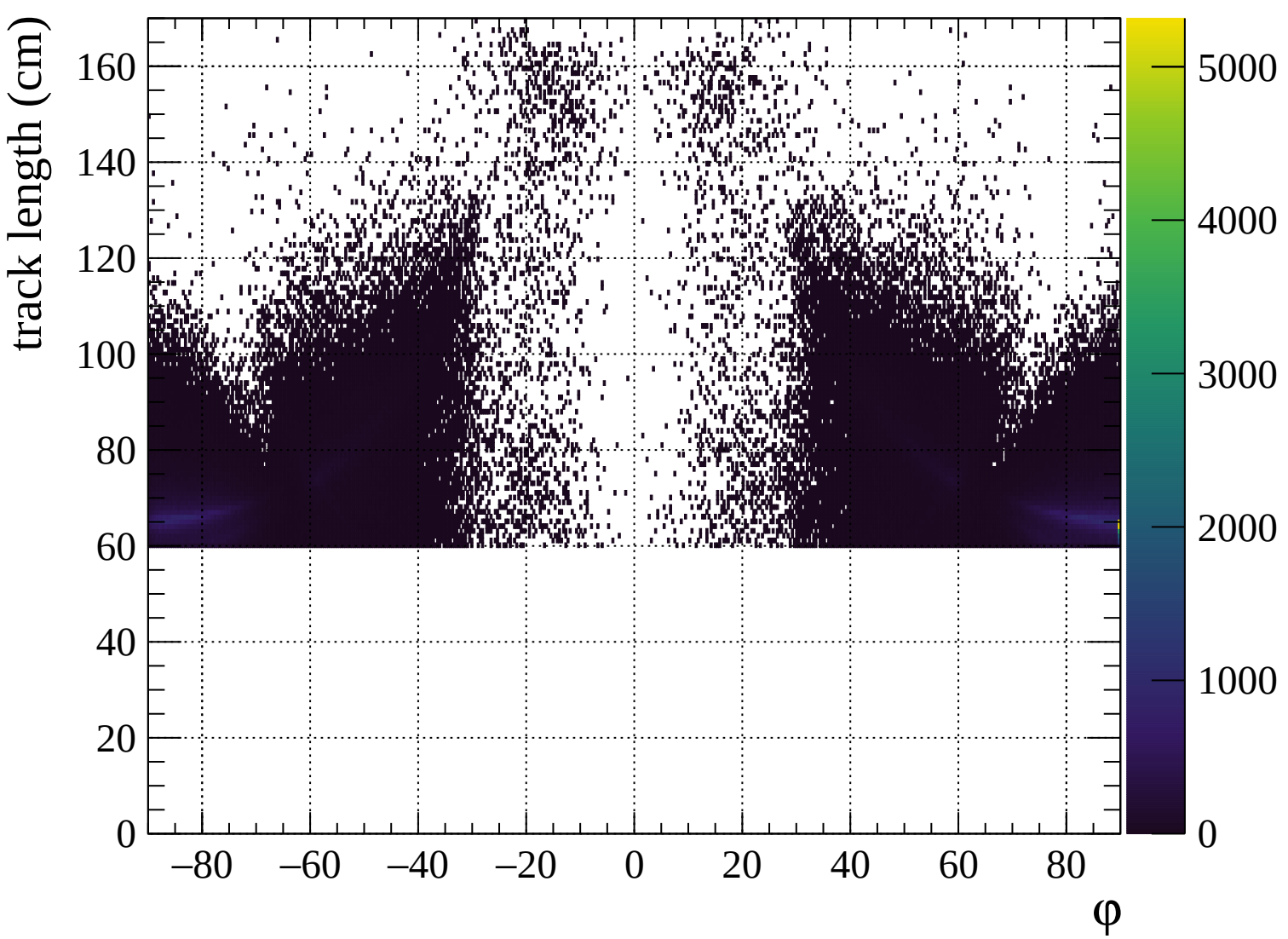


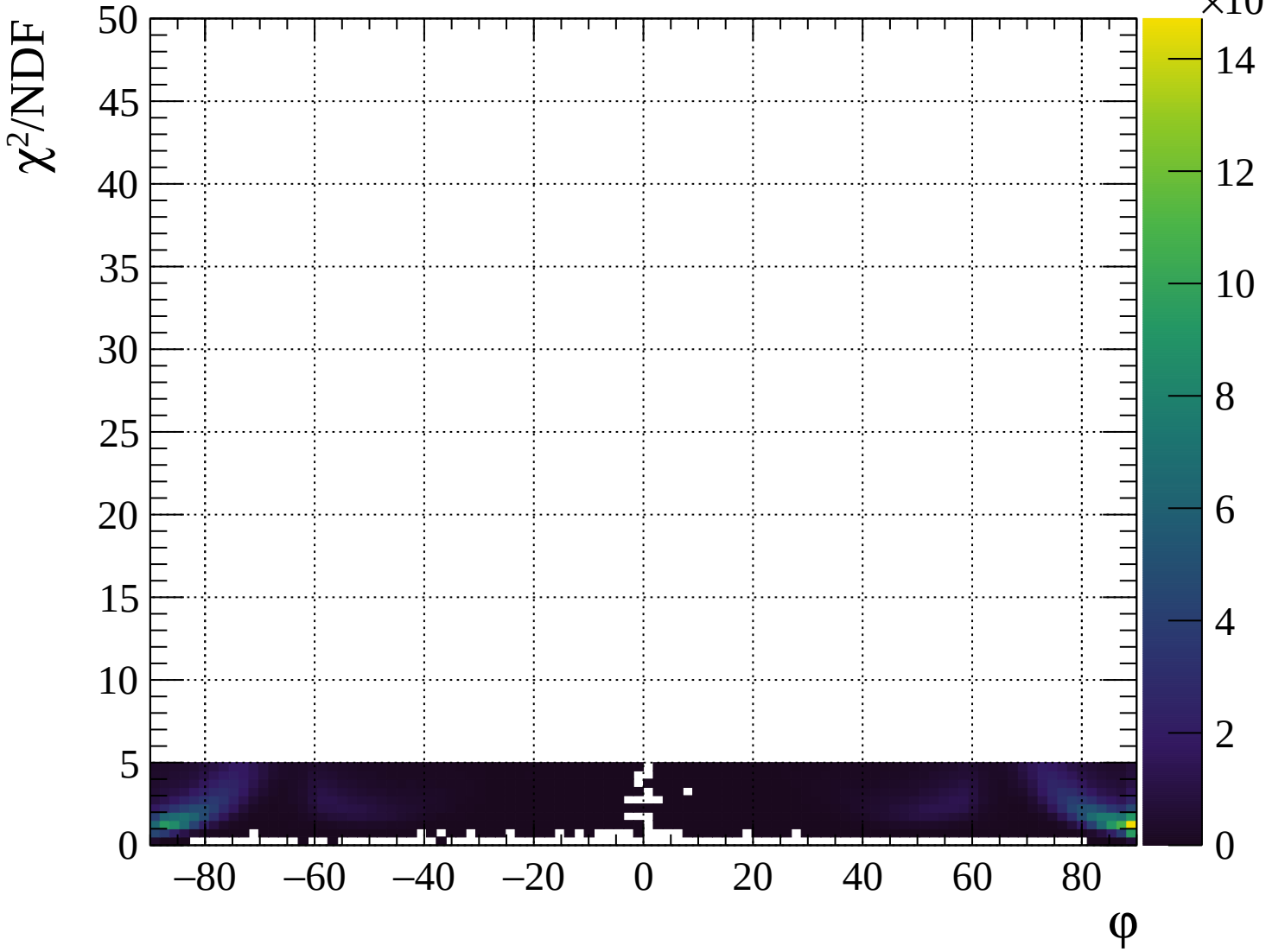


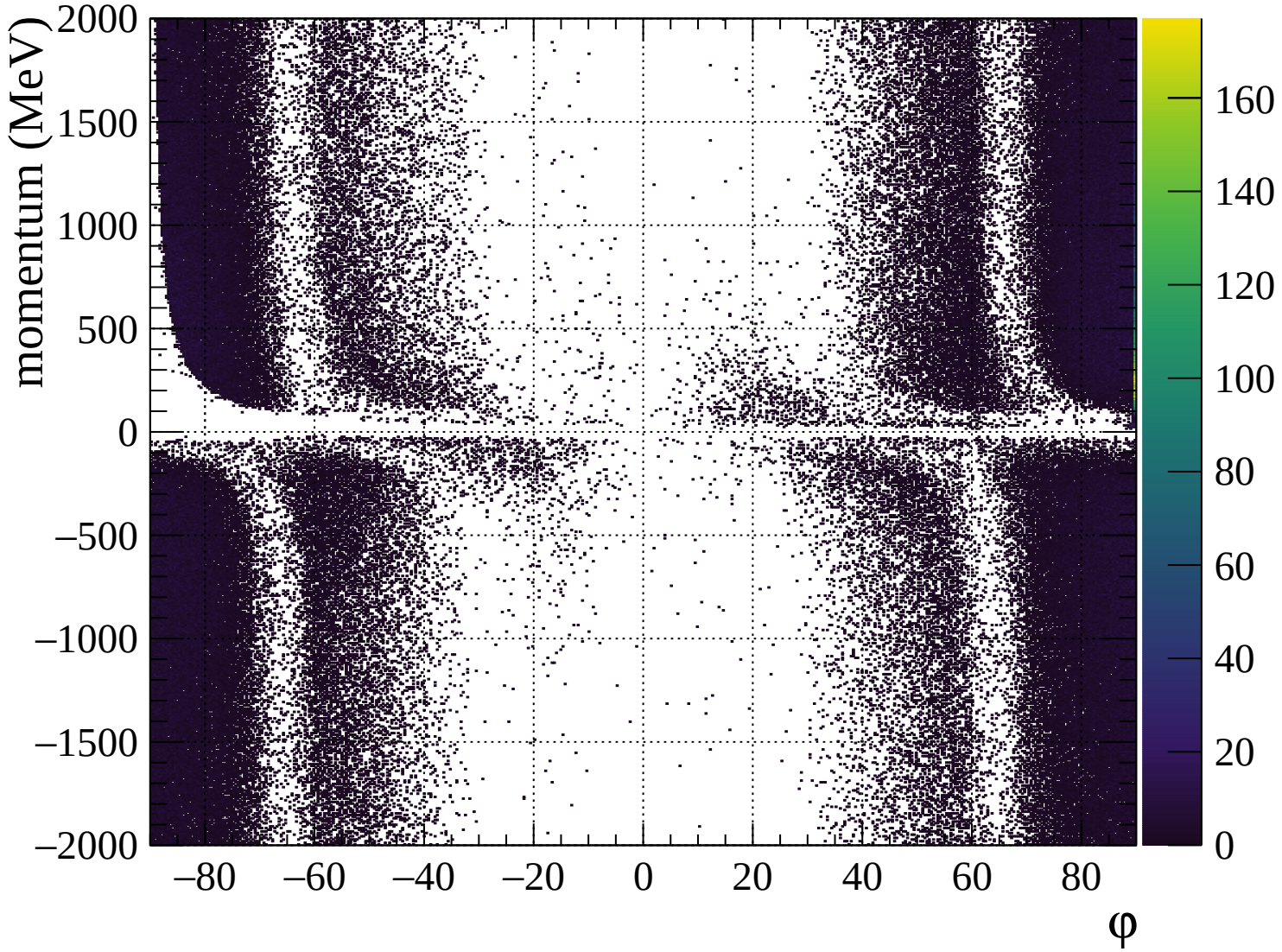


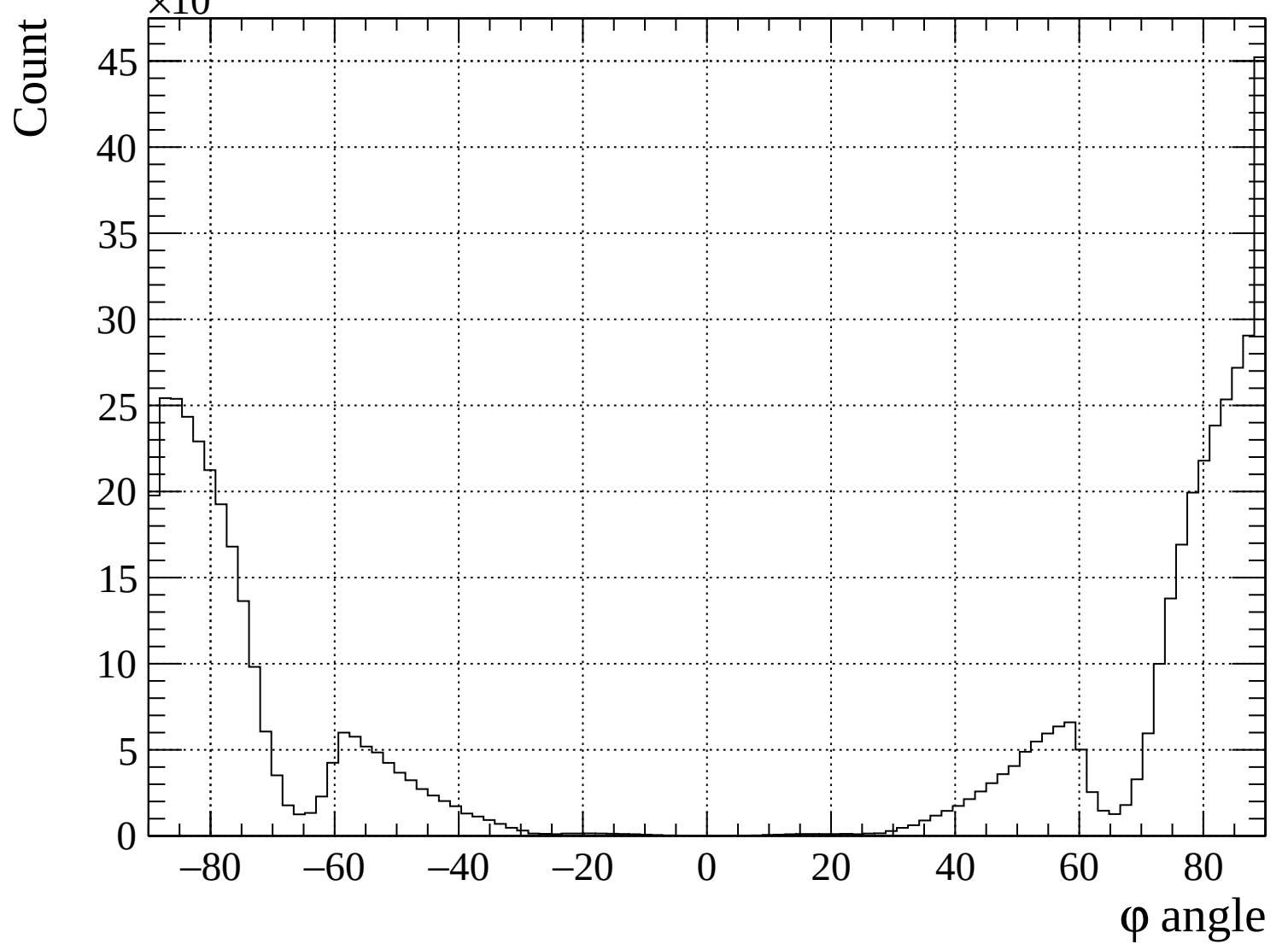


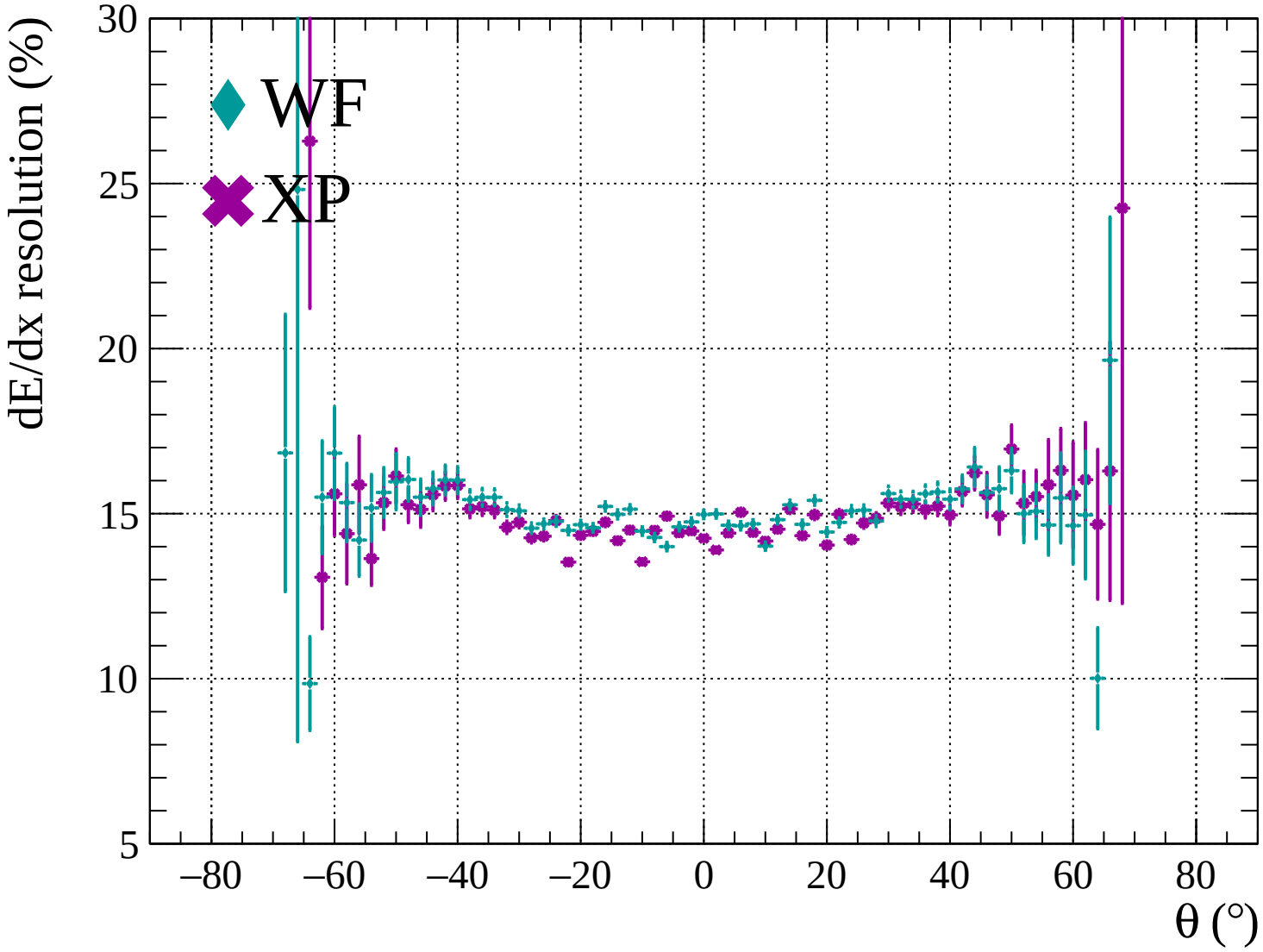


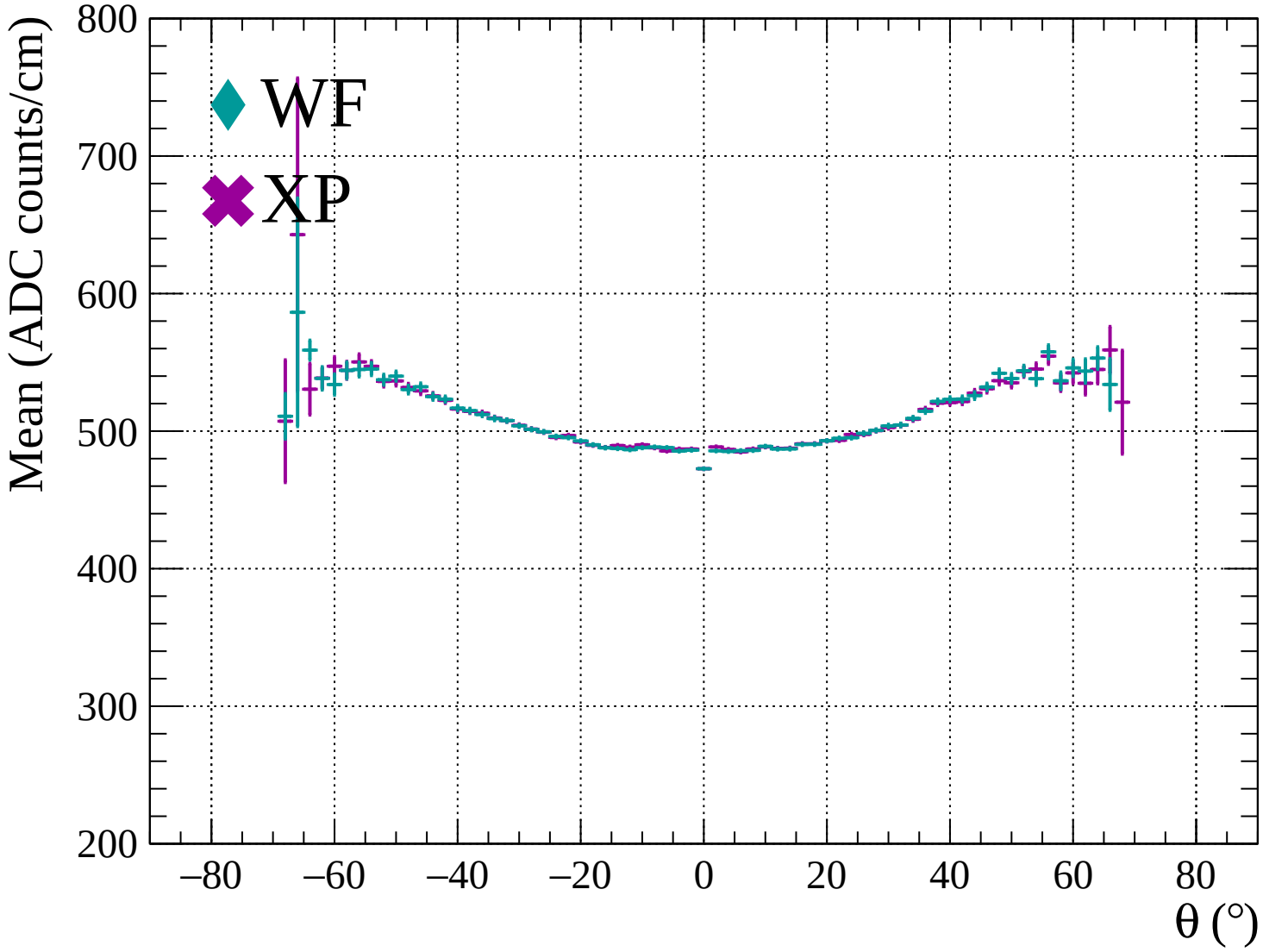


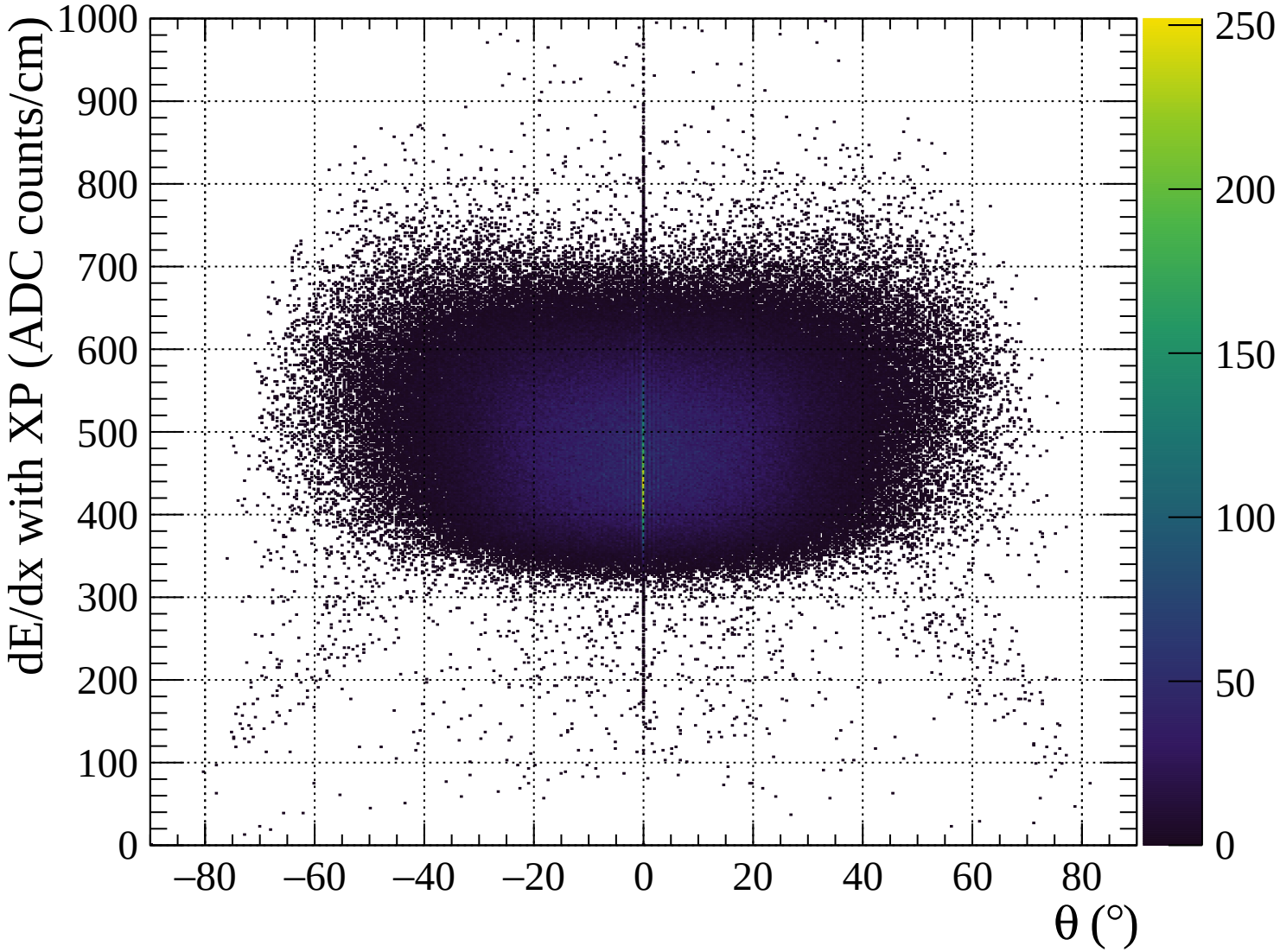


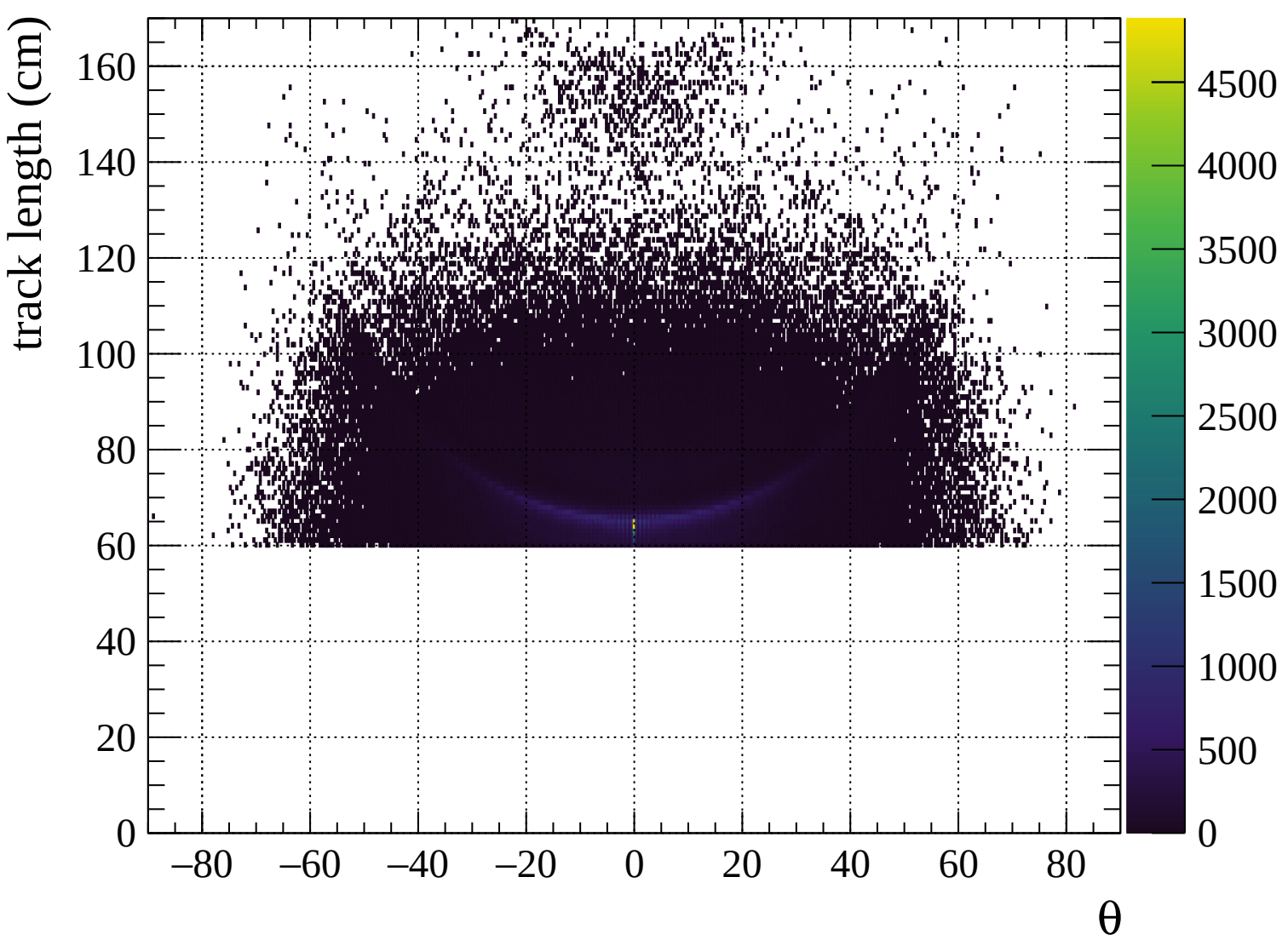


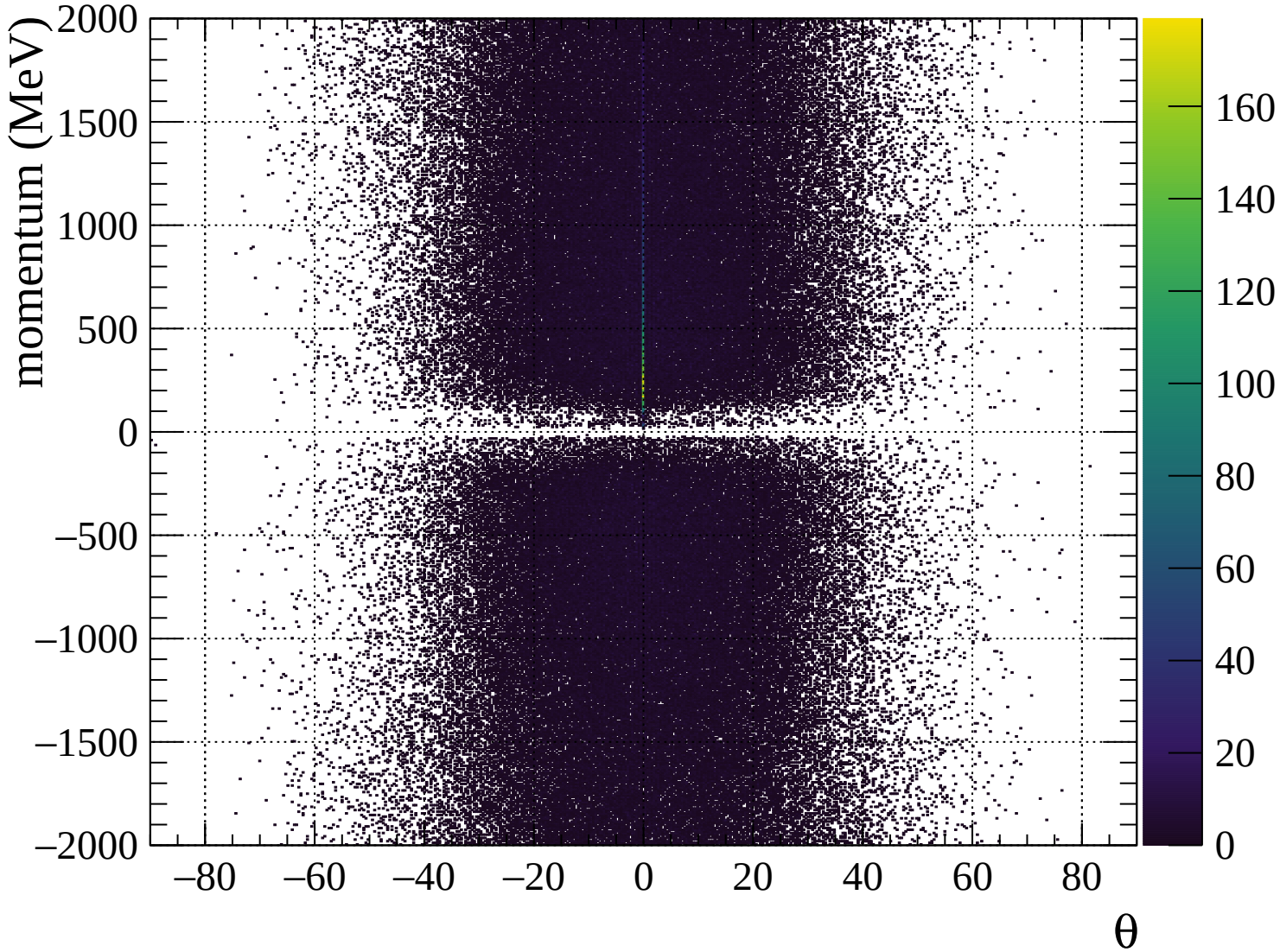


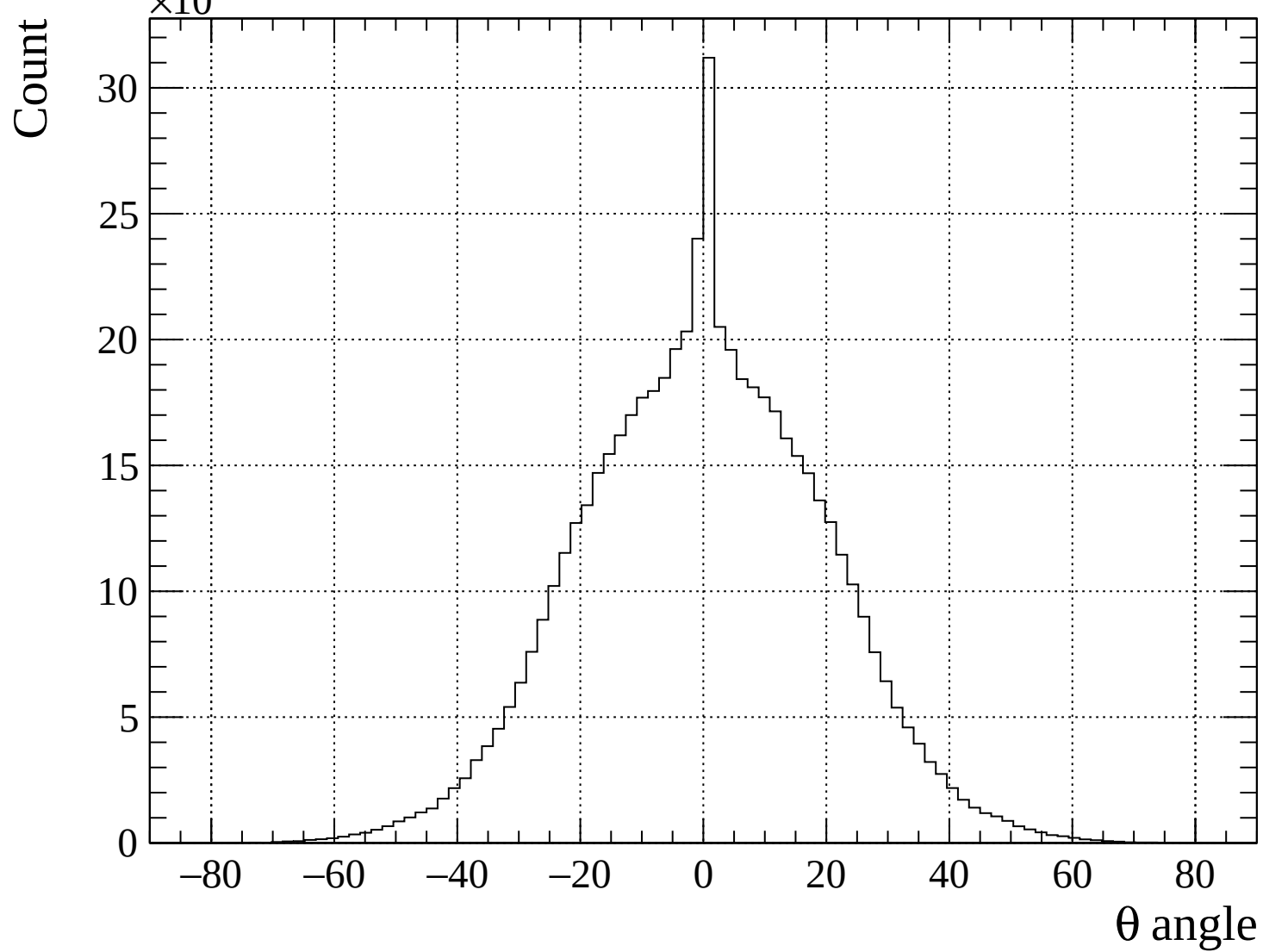


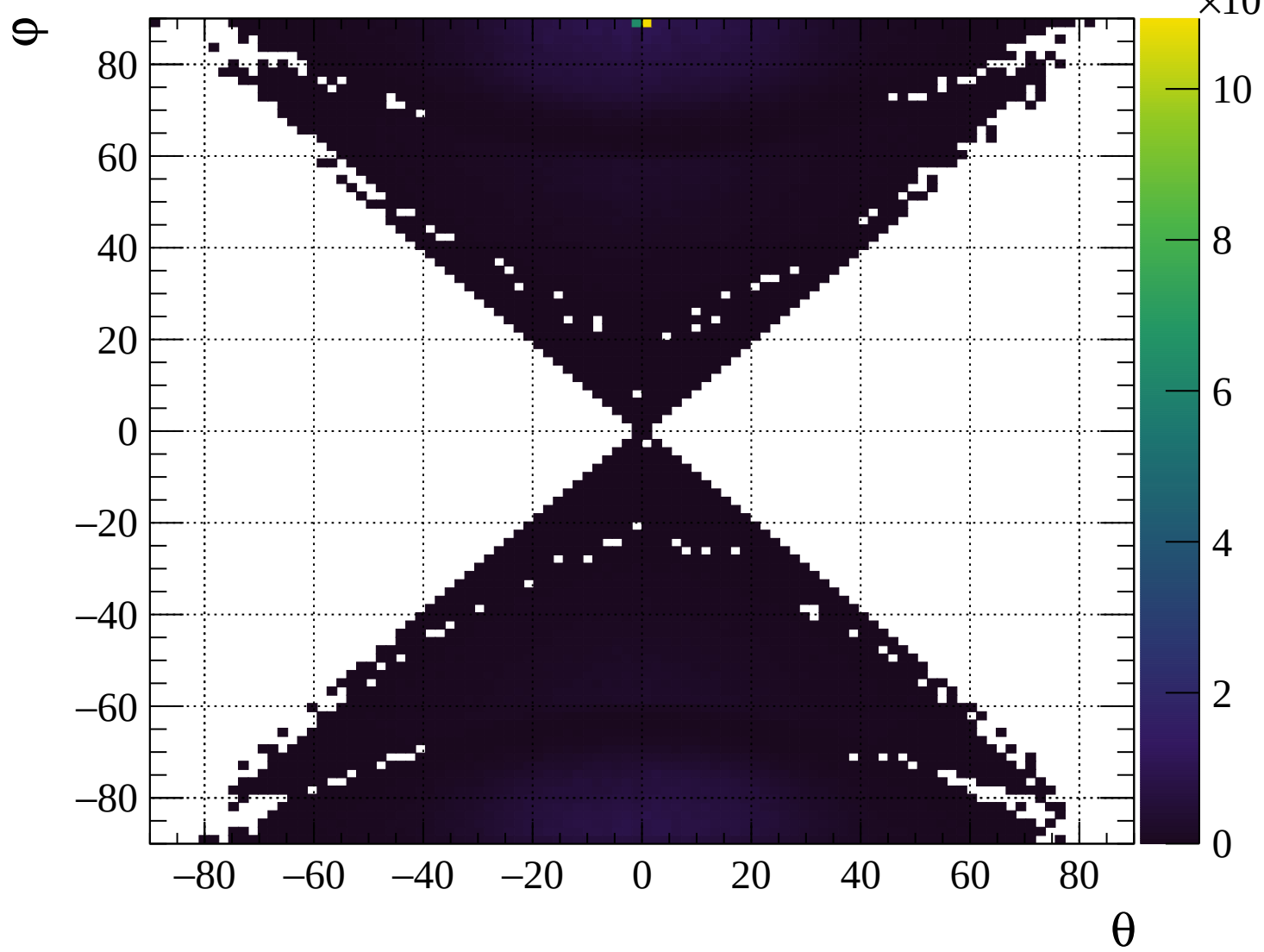


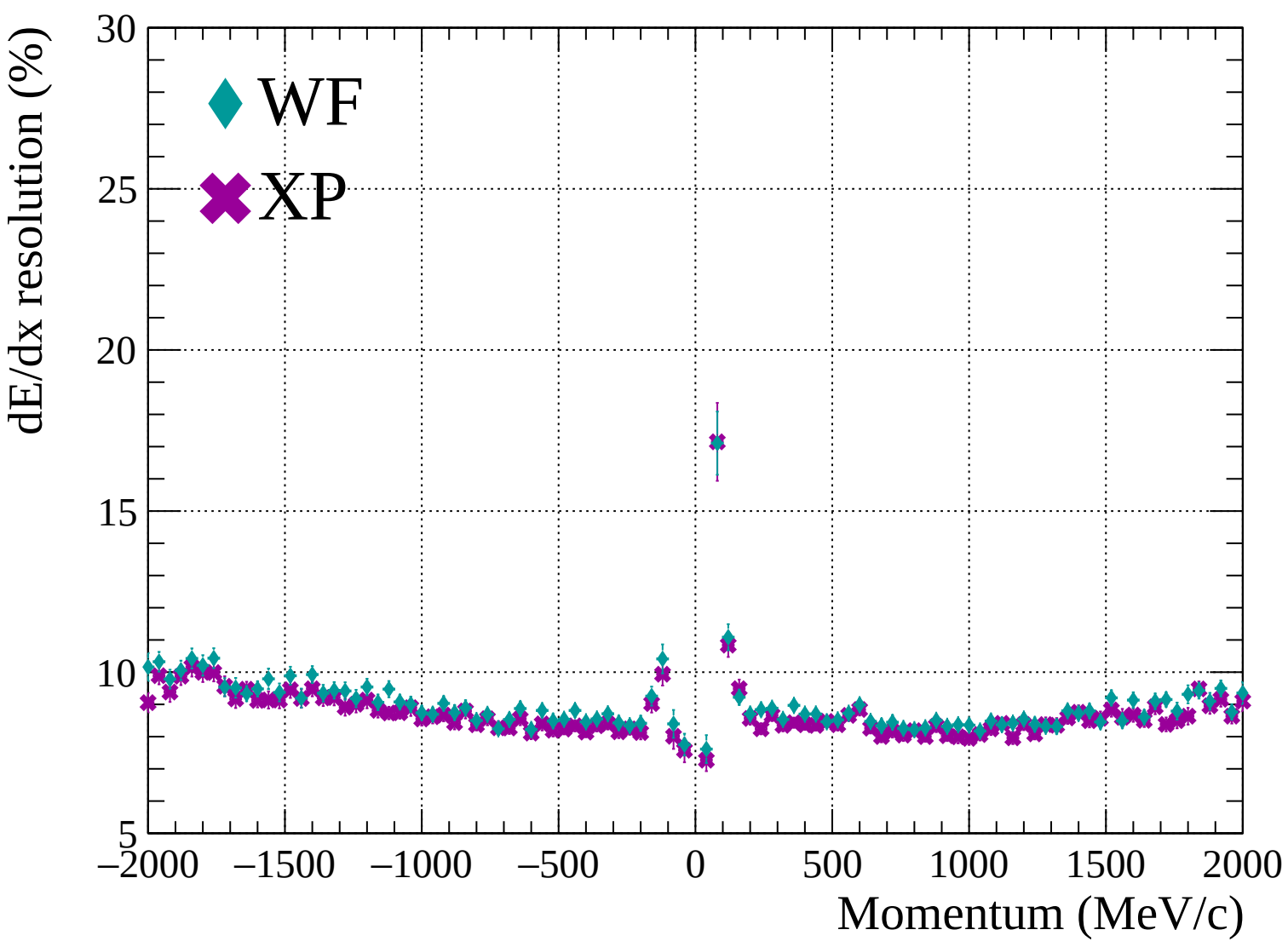


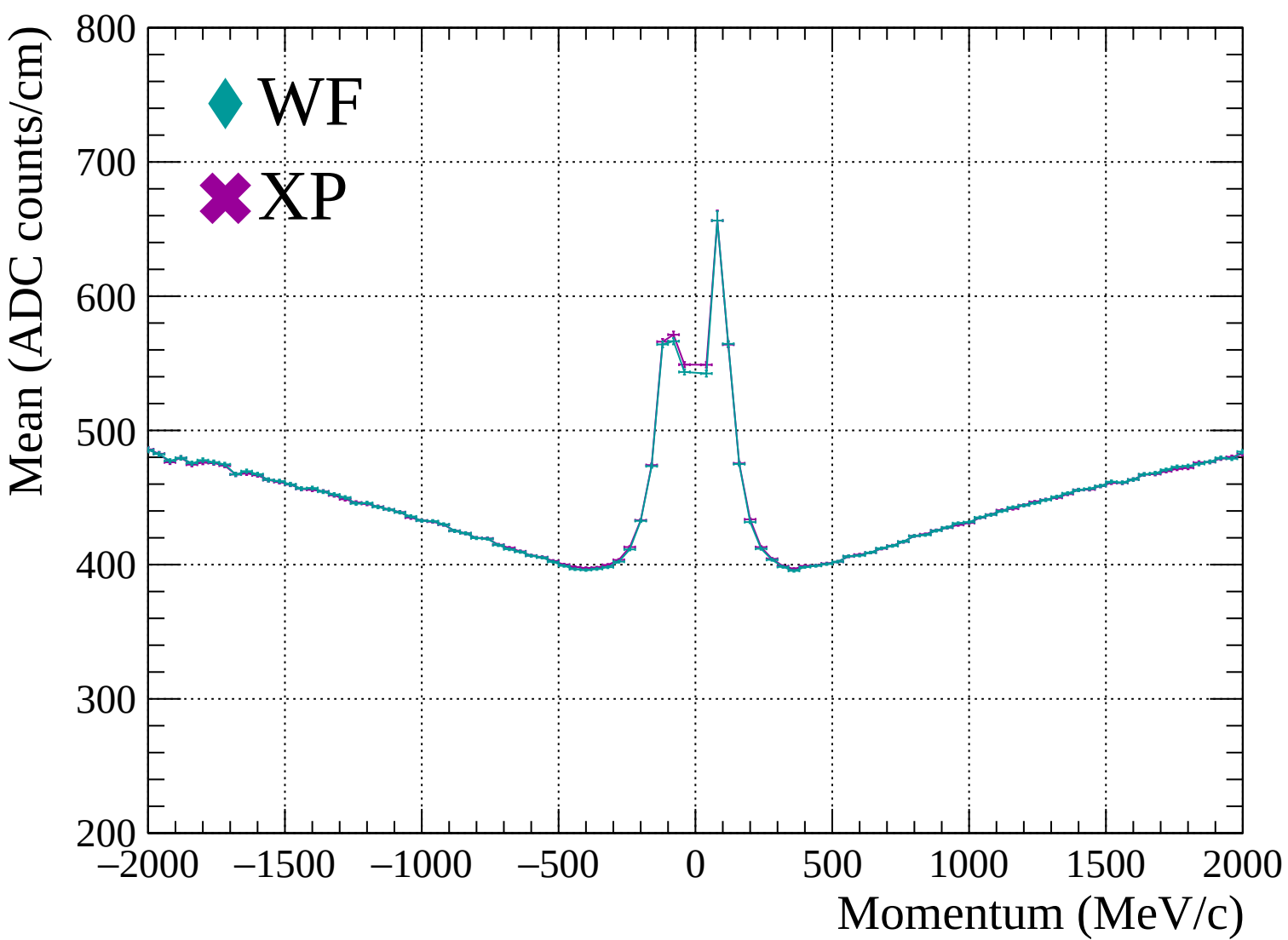


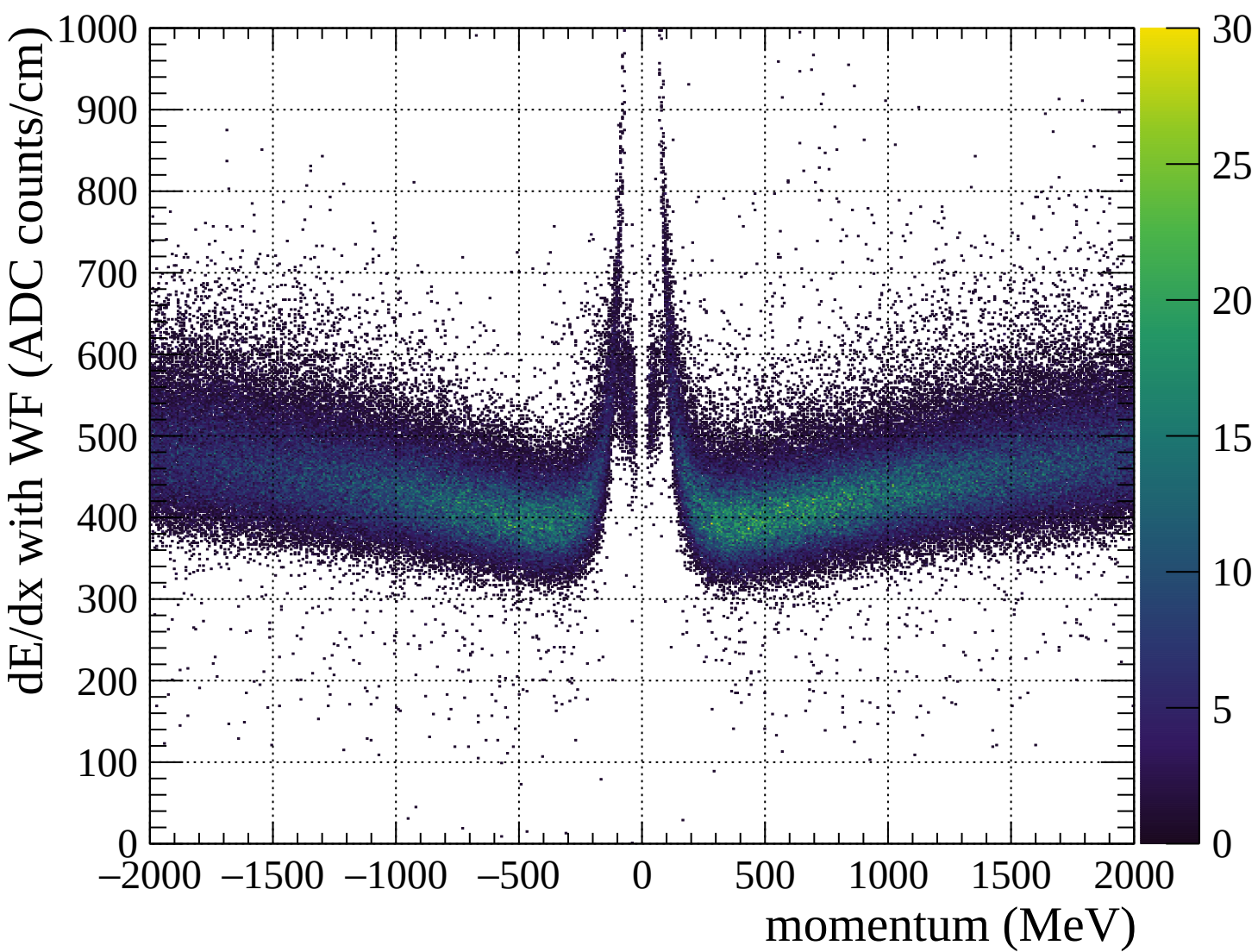


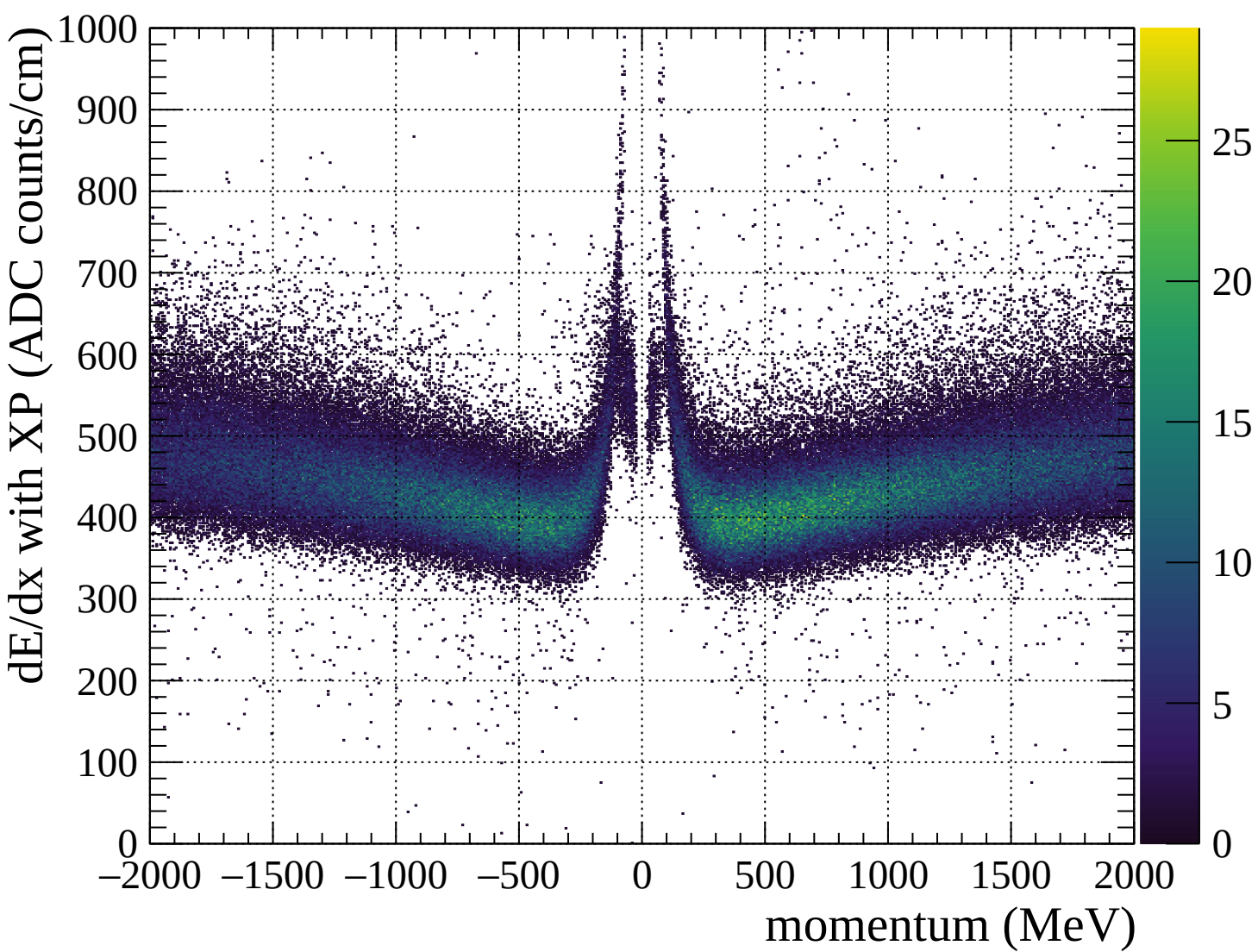


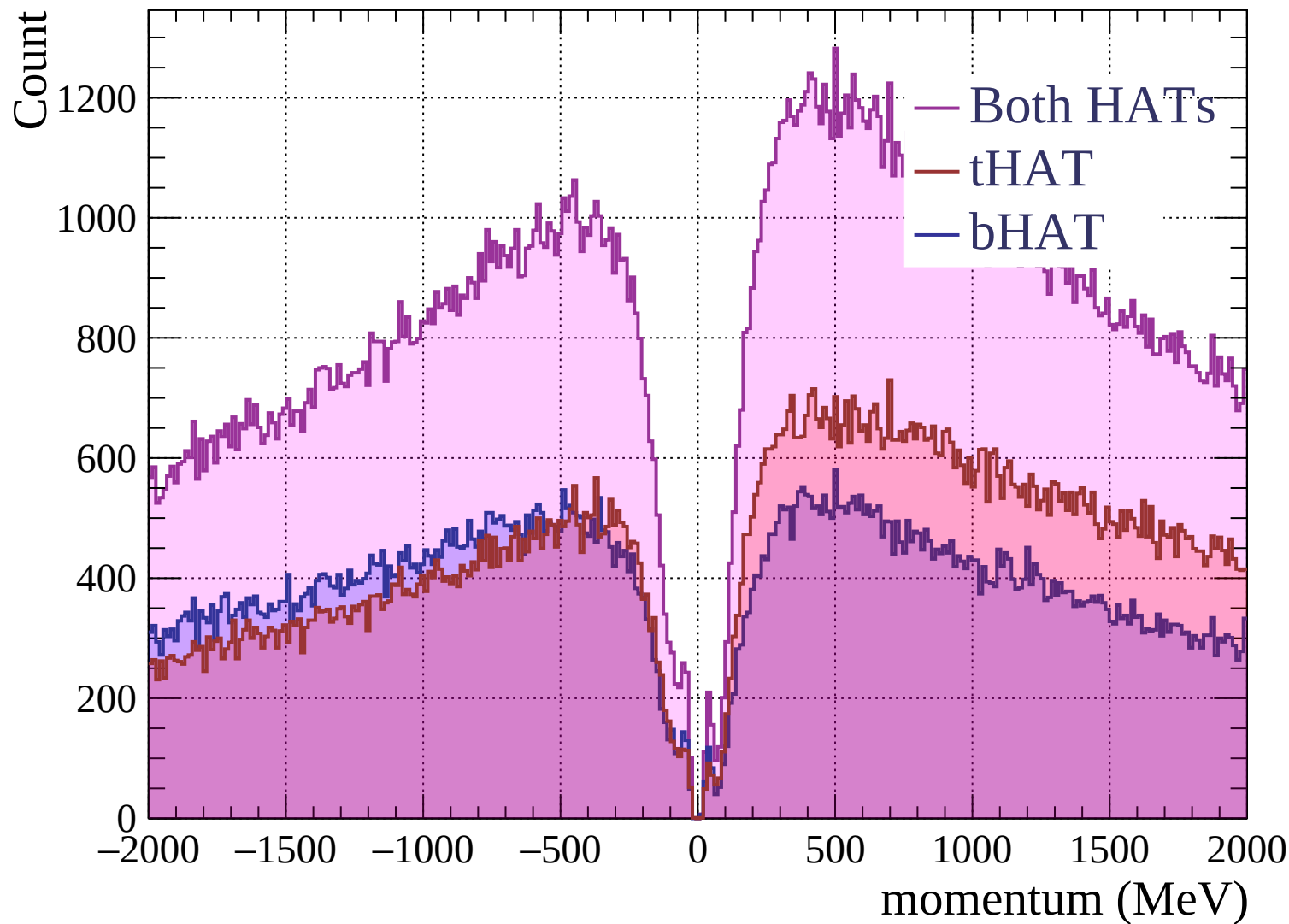


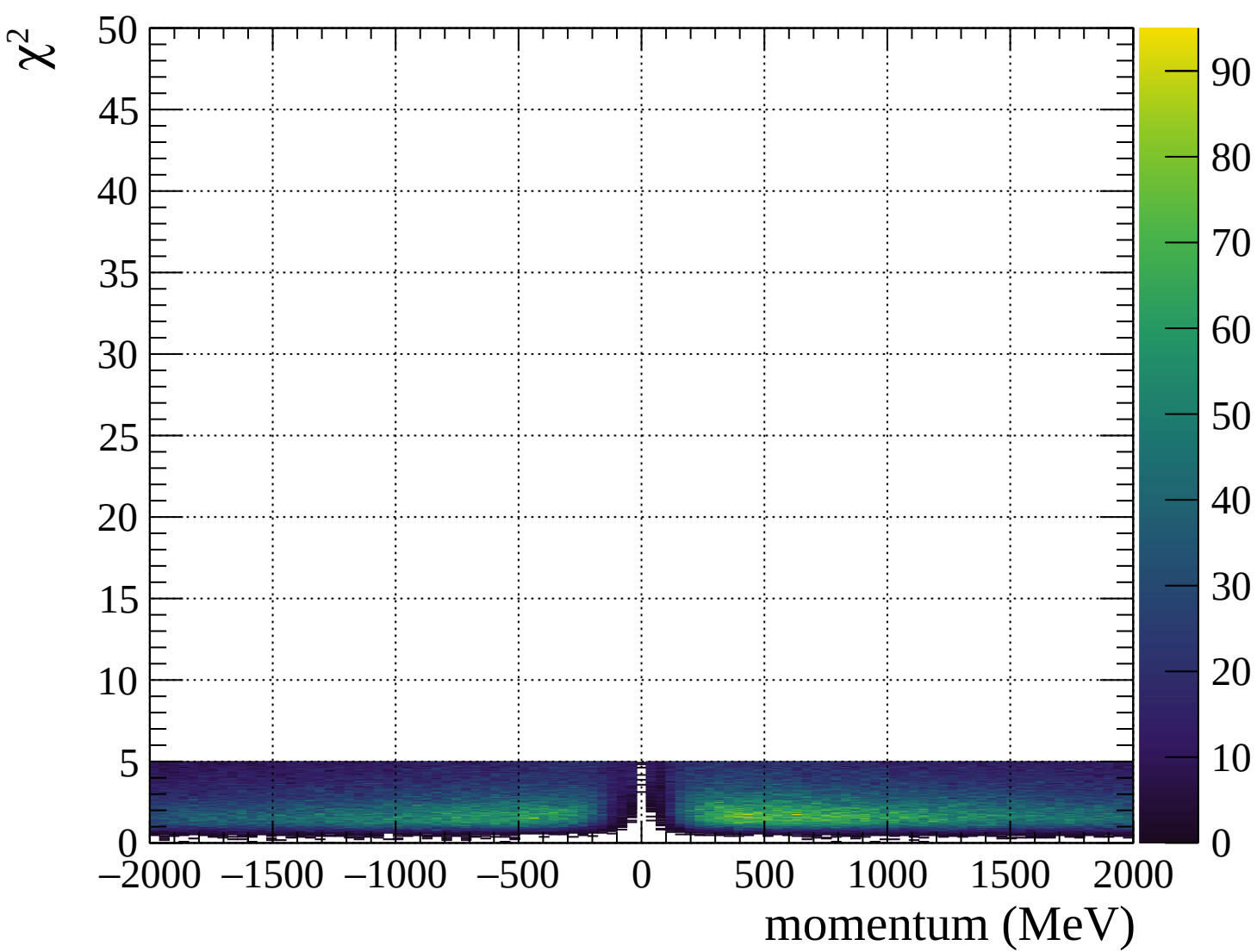




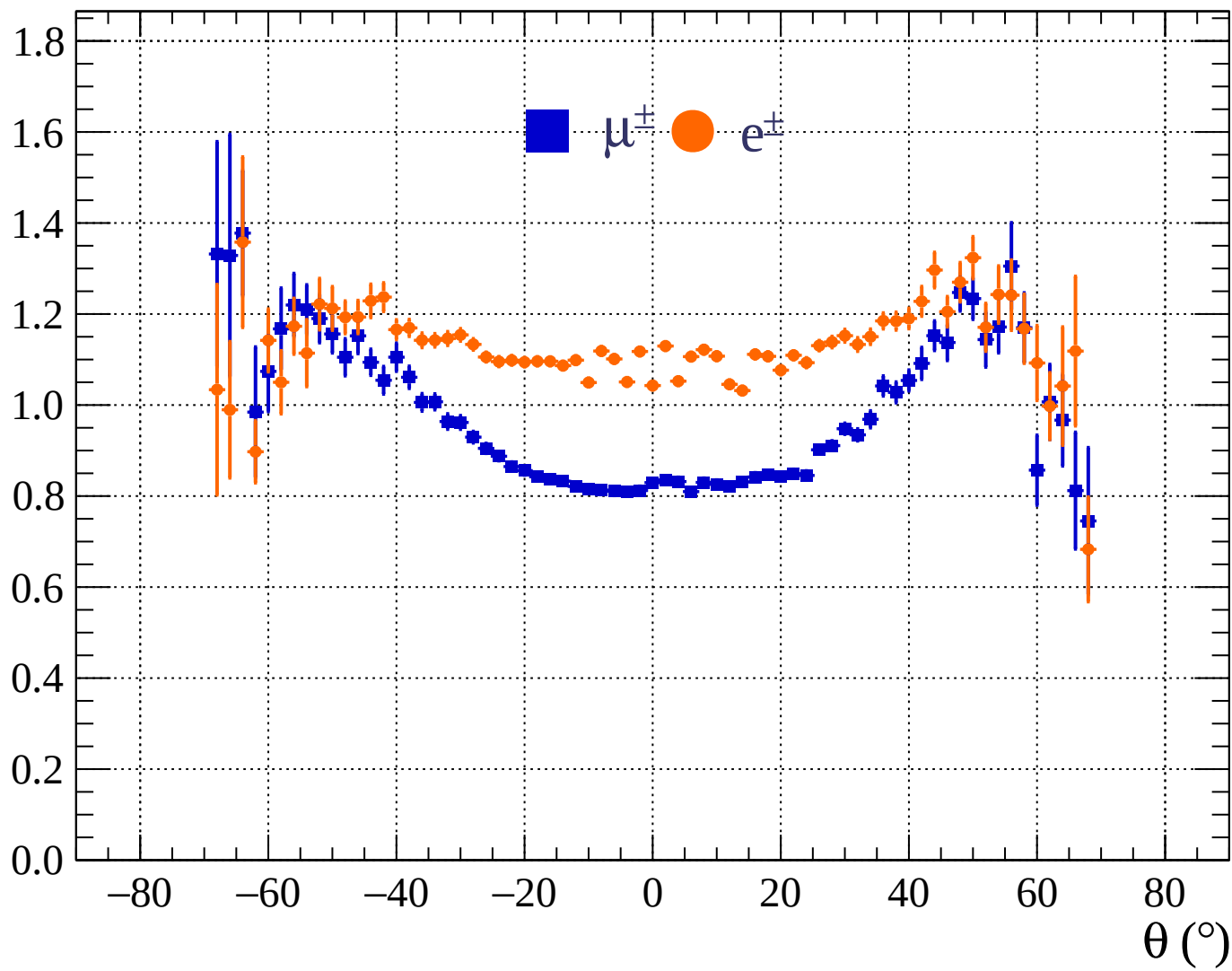




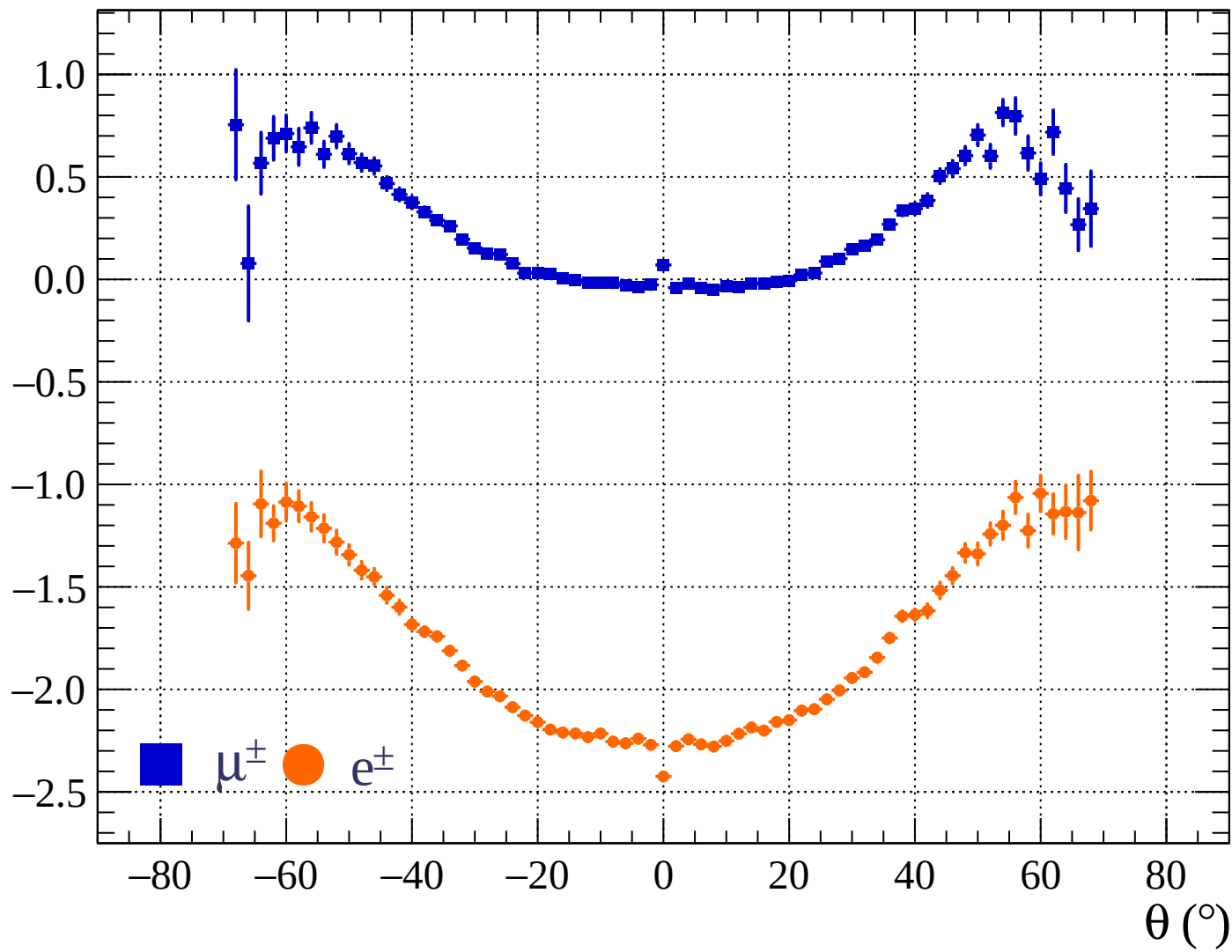




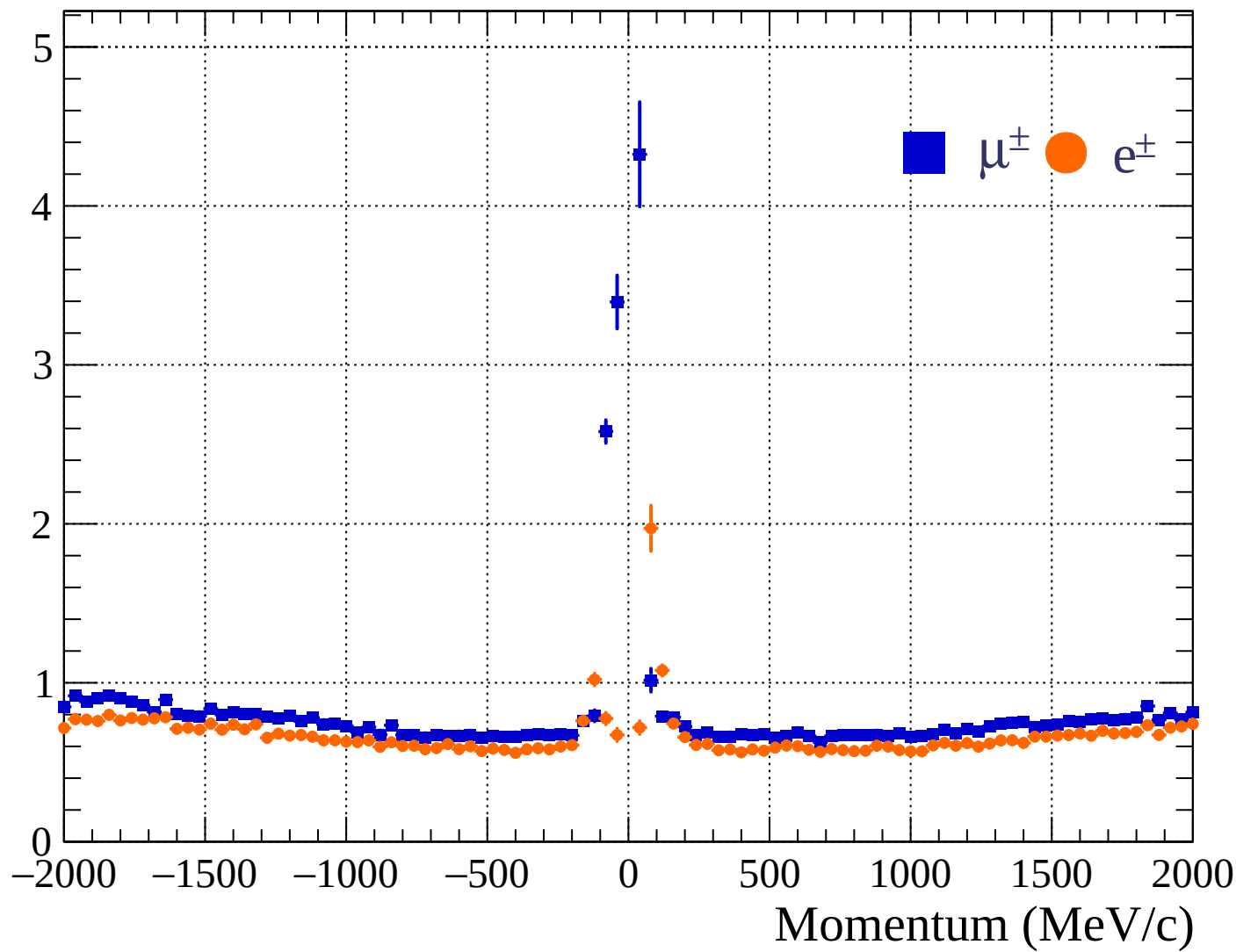
Pulls standard deviation



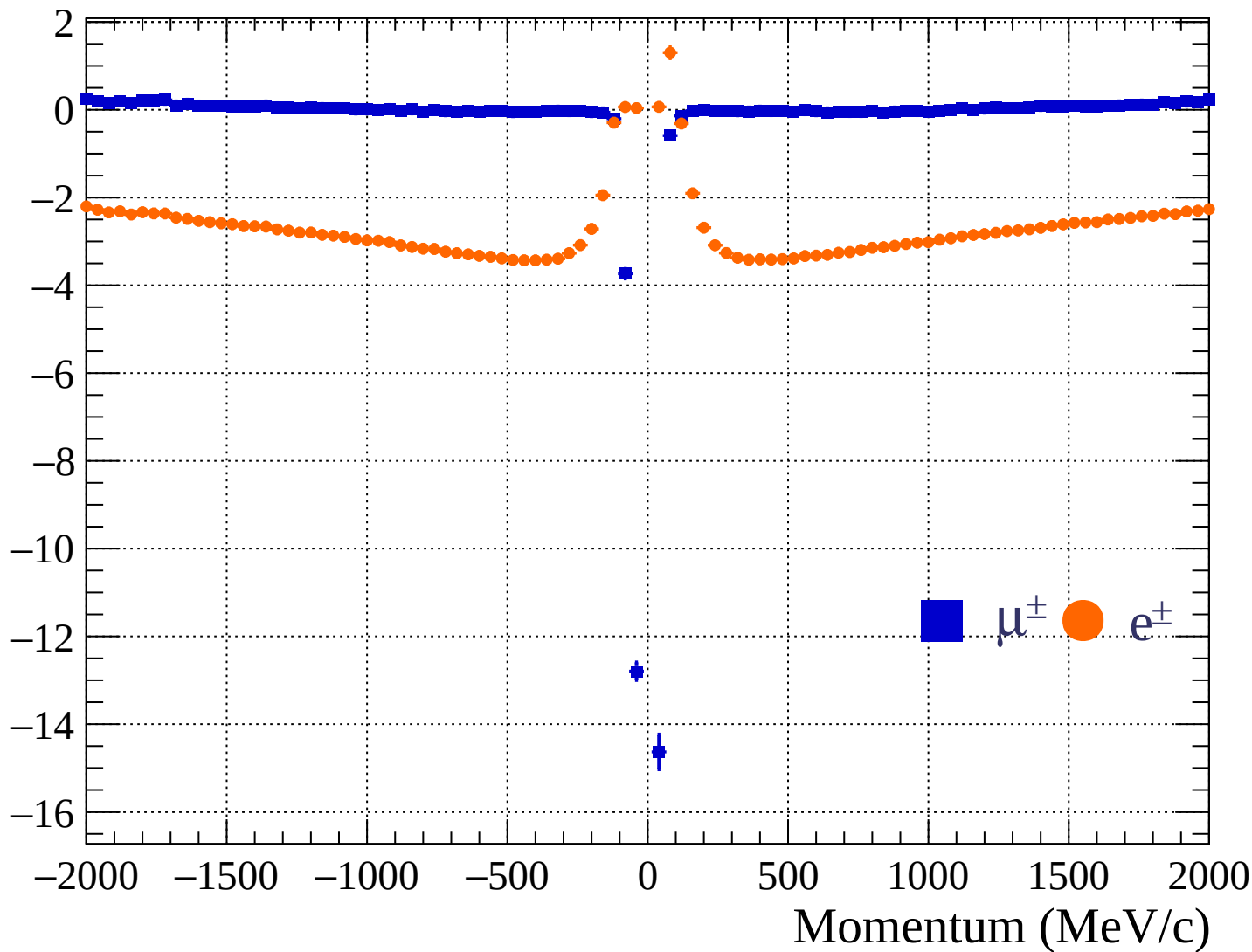
Pulls mean

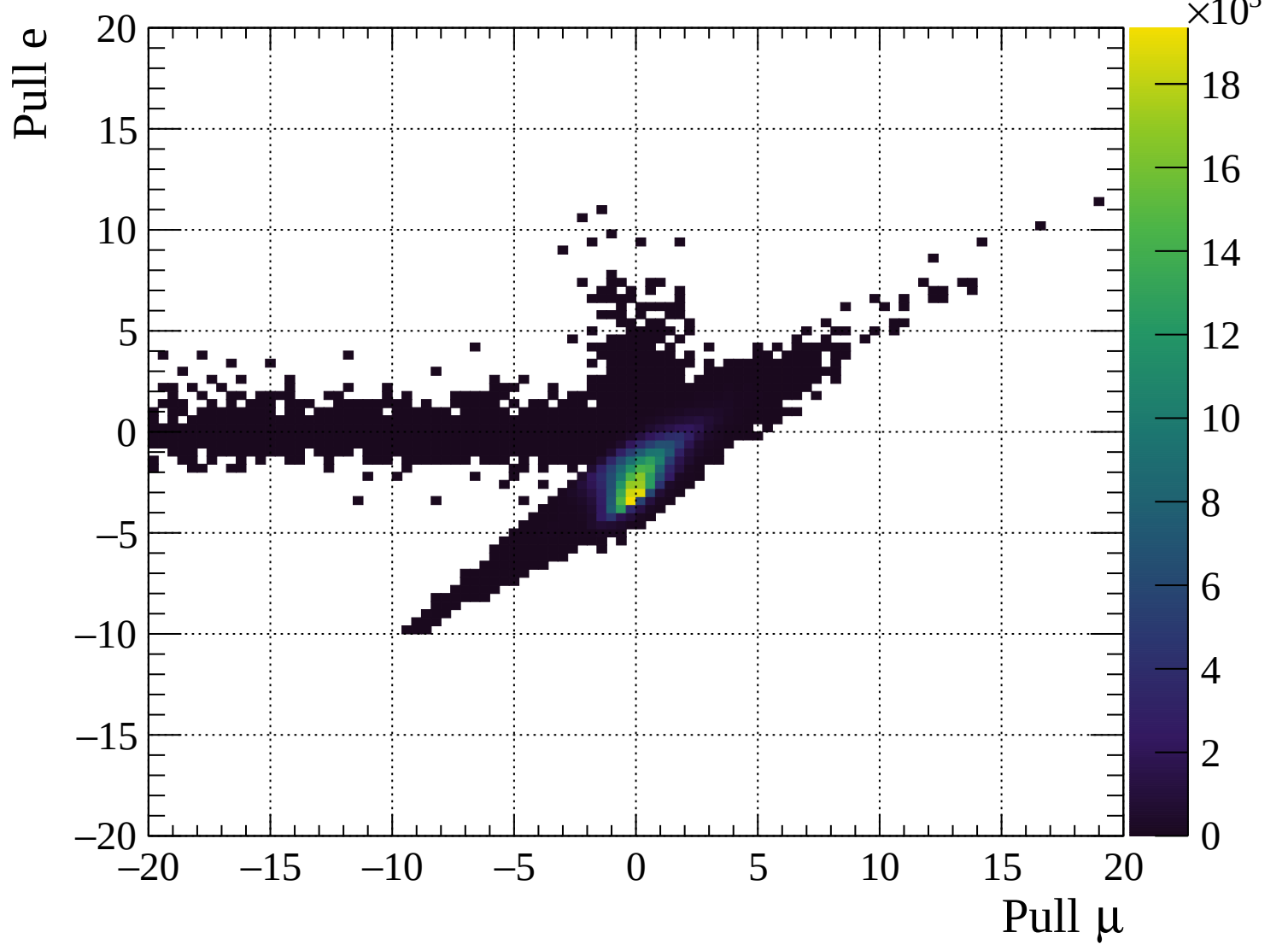


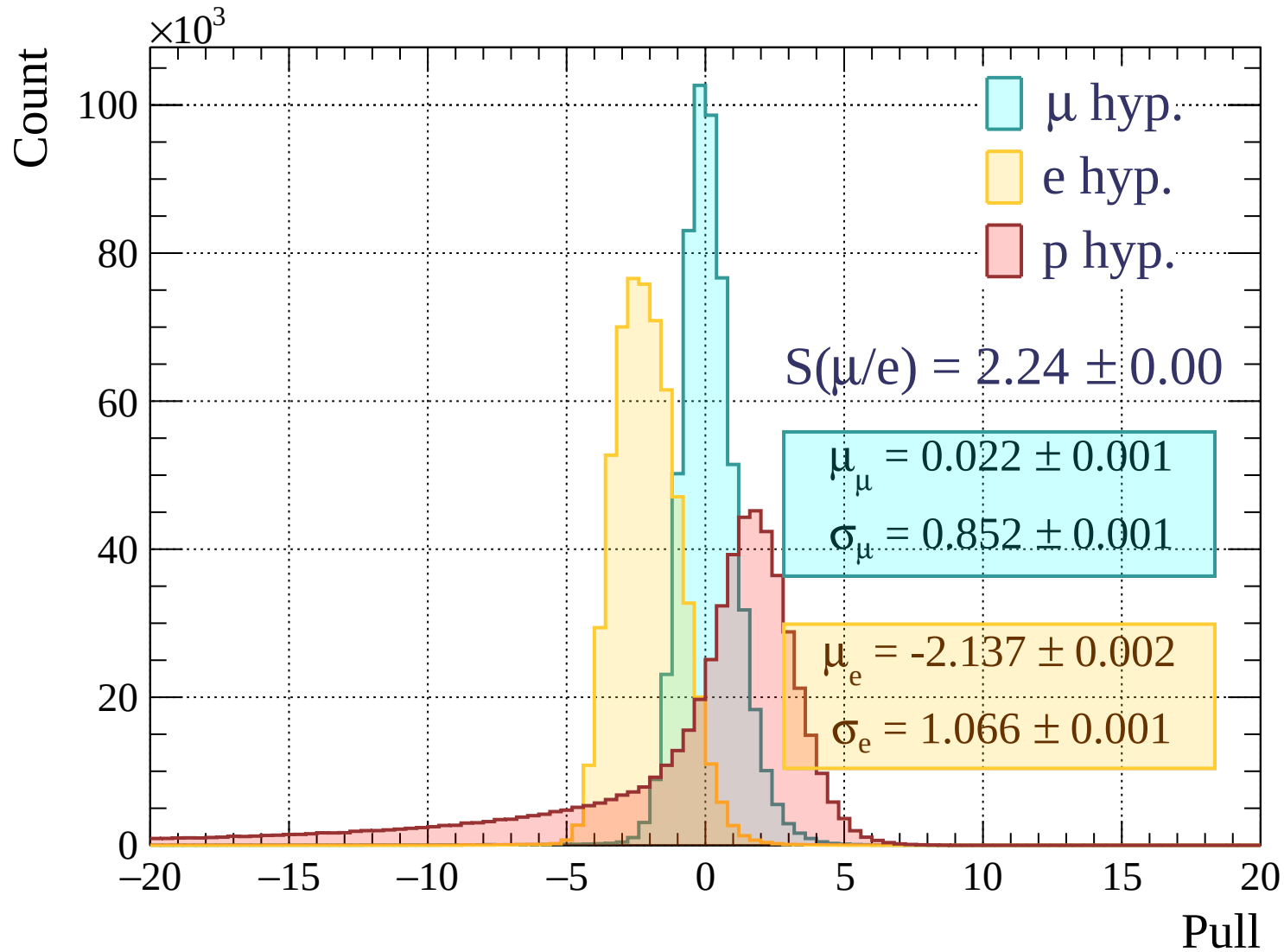
Pulls standard deviation

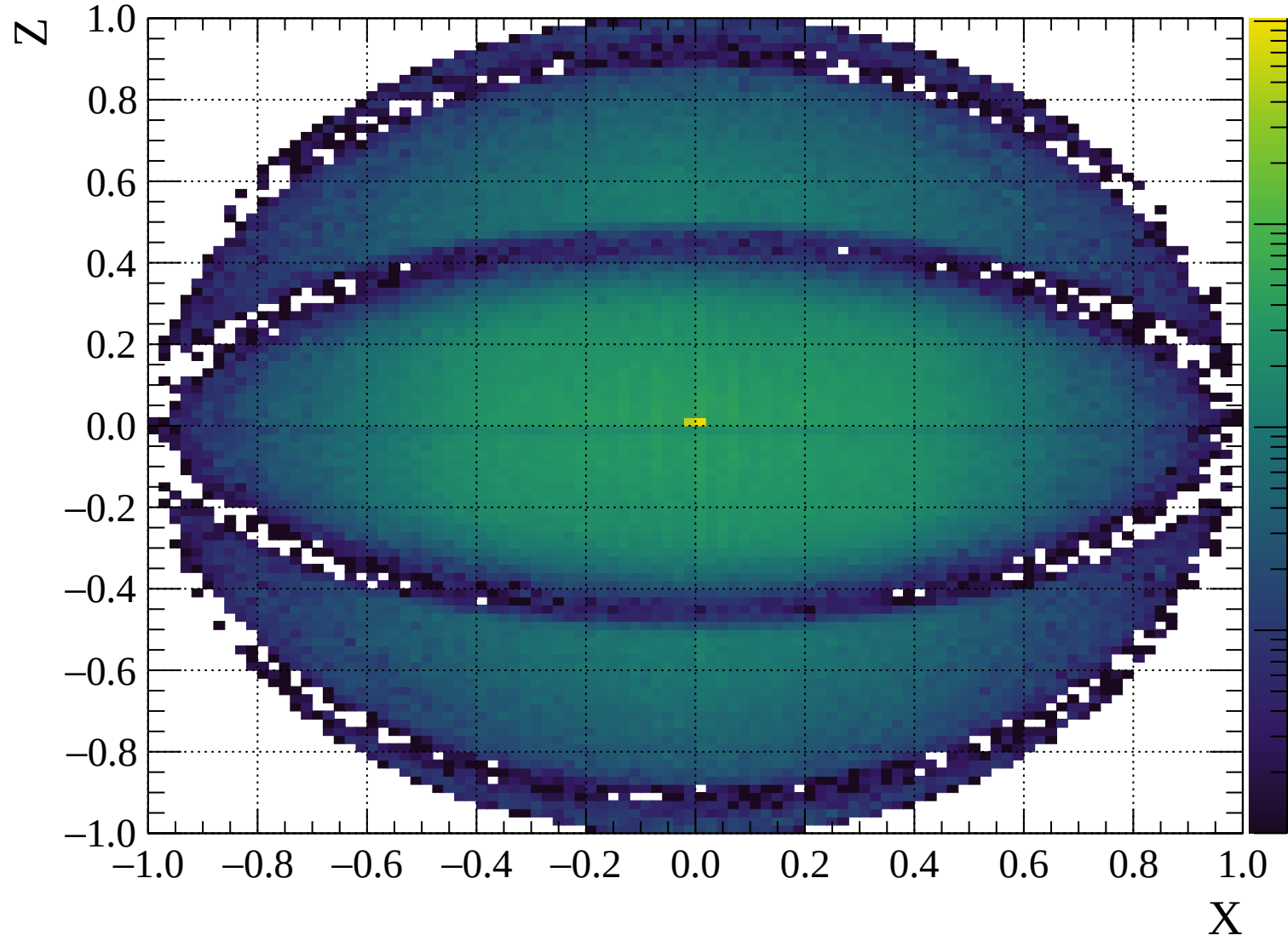


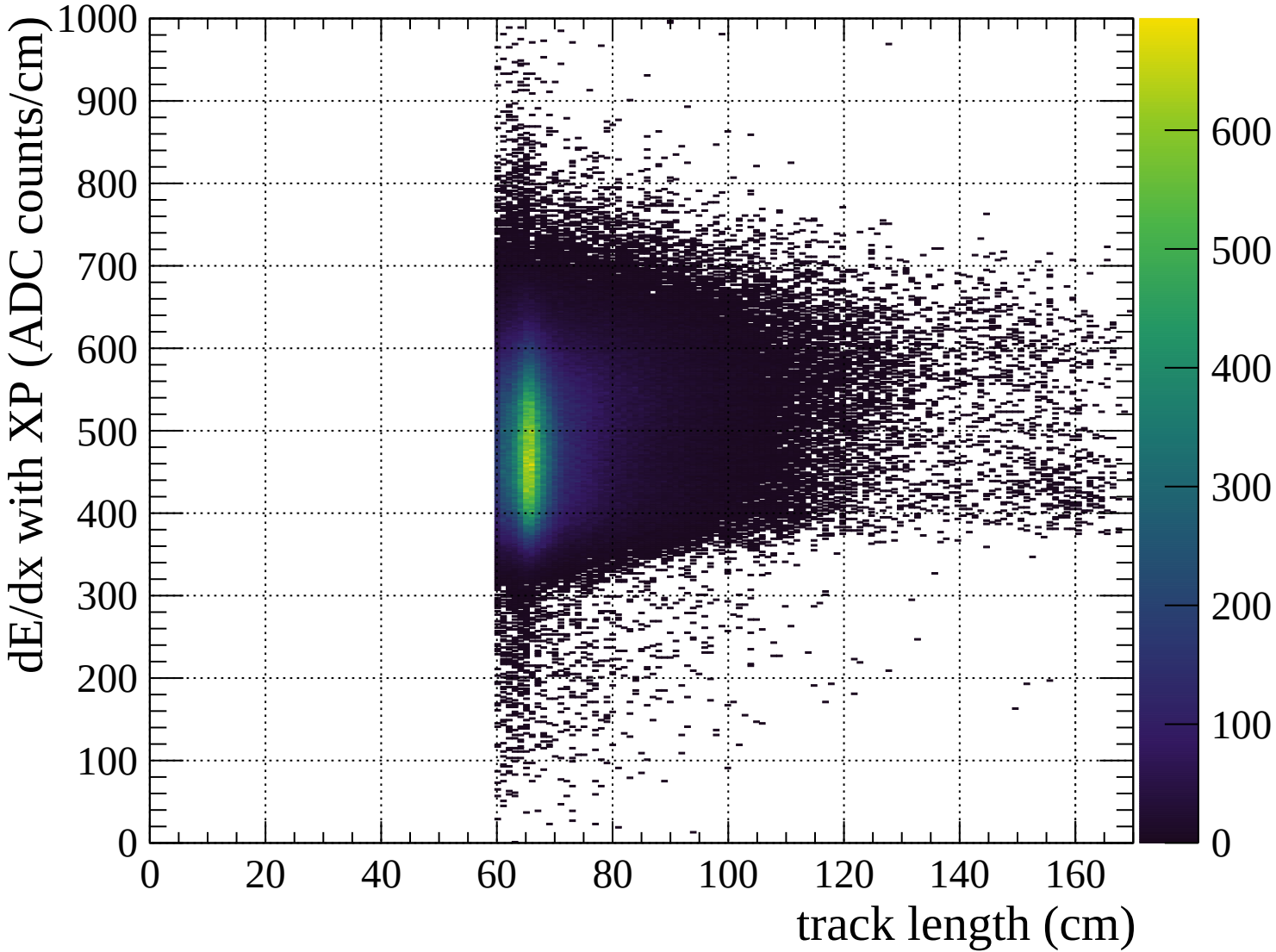
Pulls mean

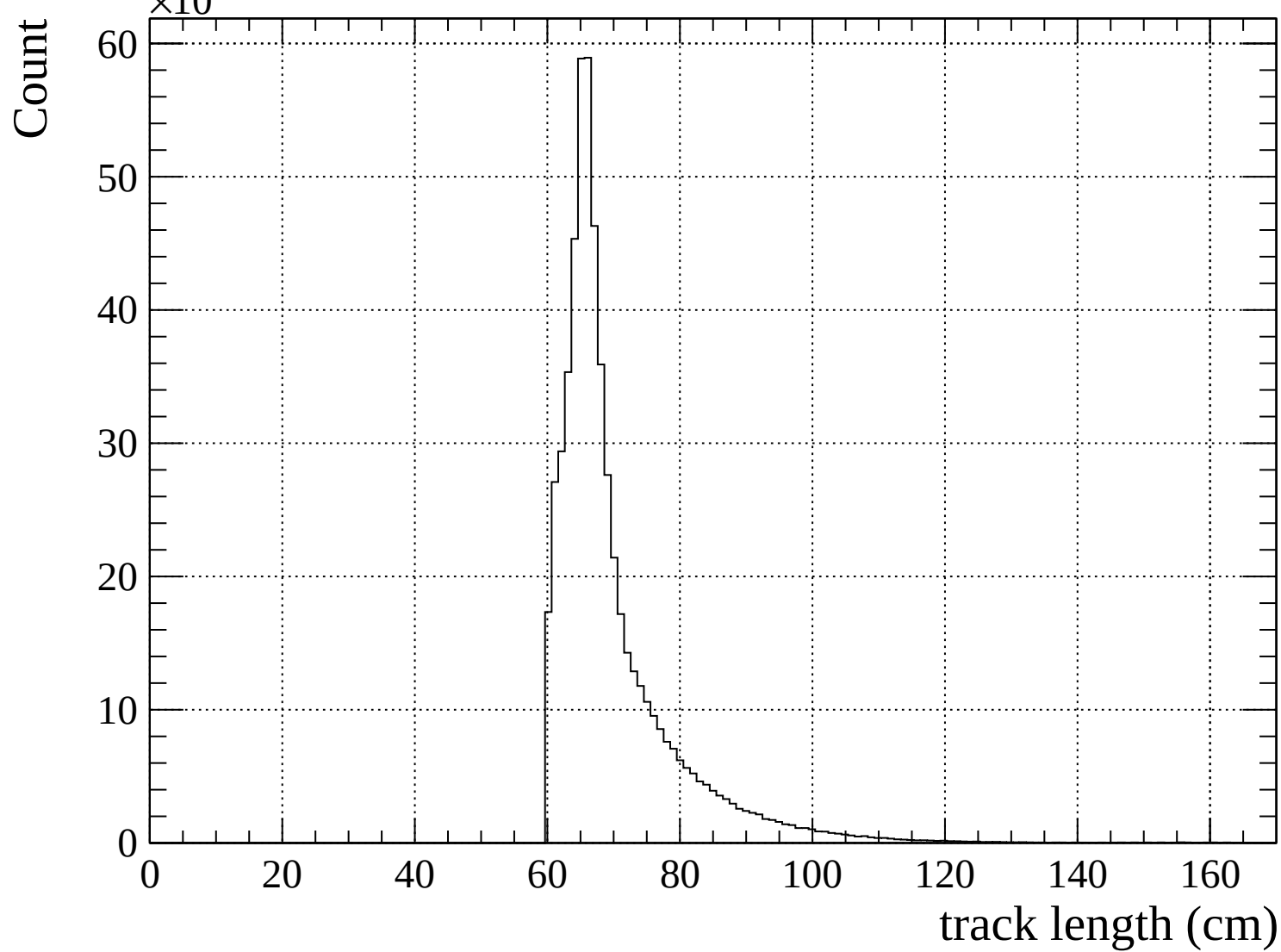


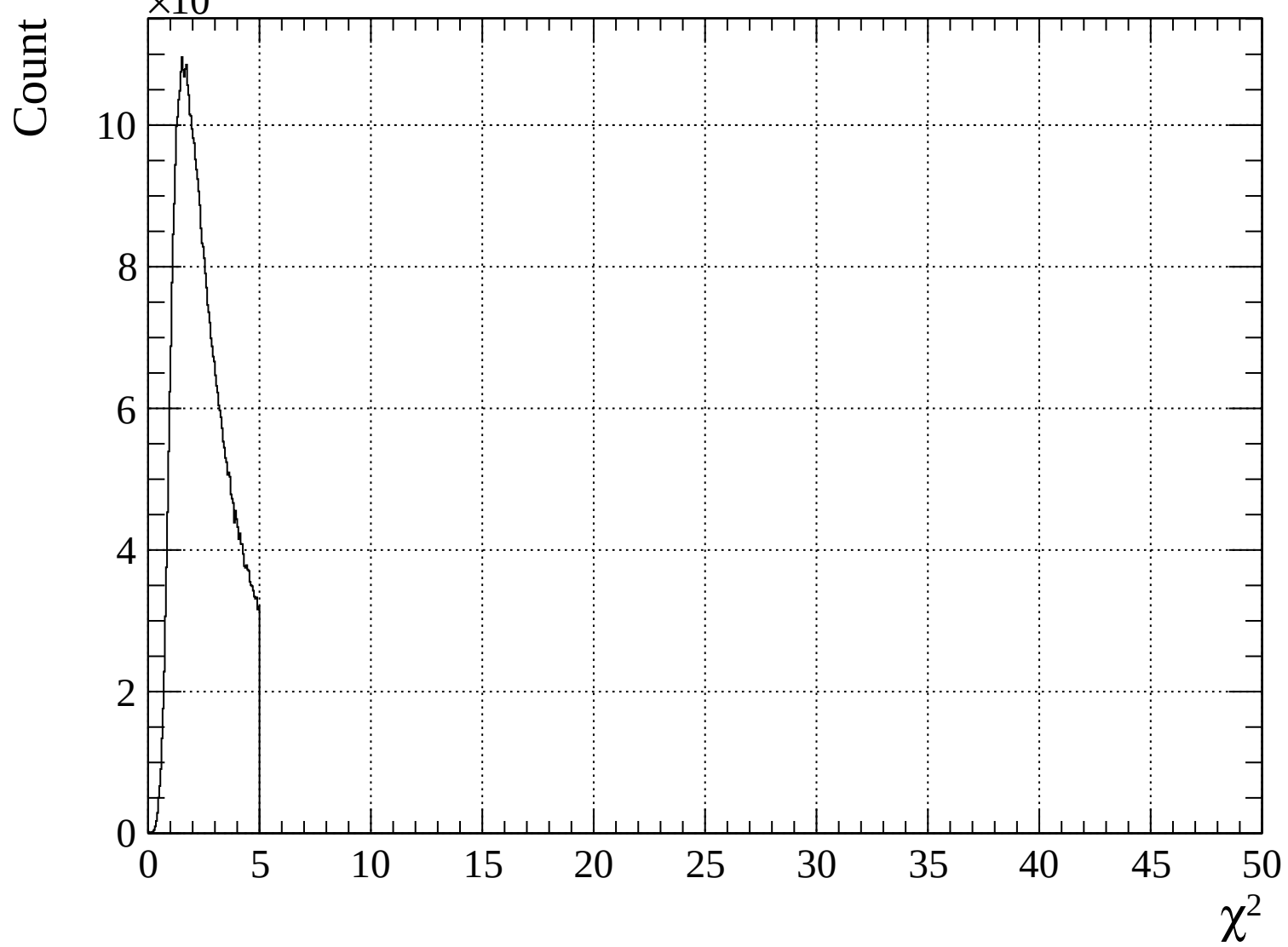


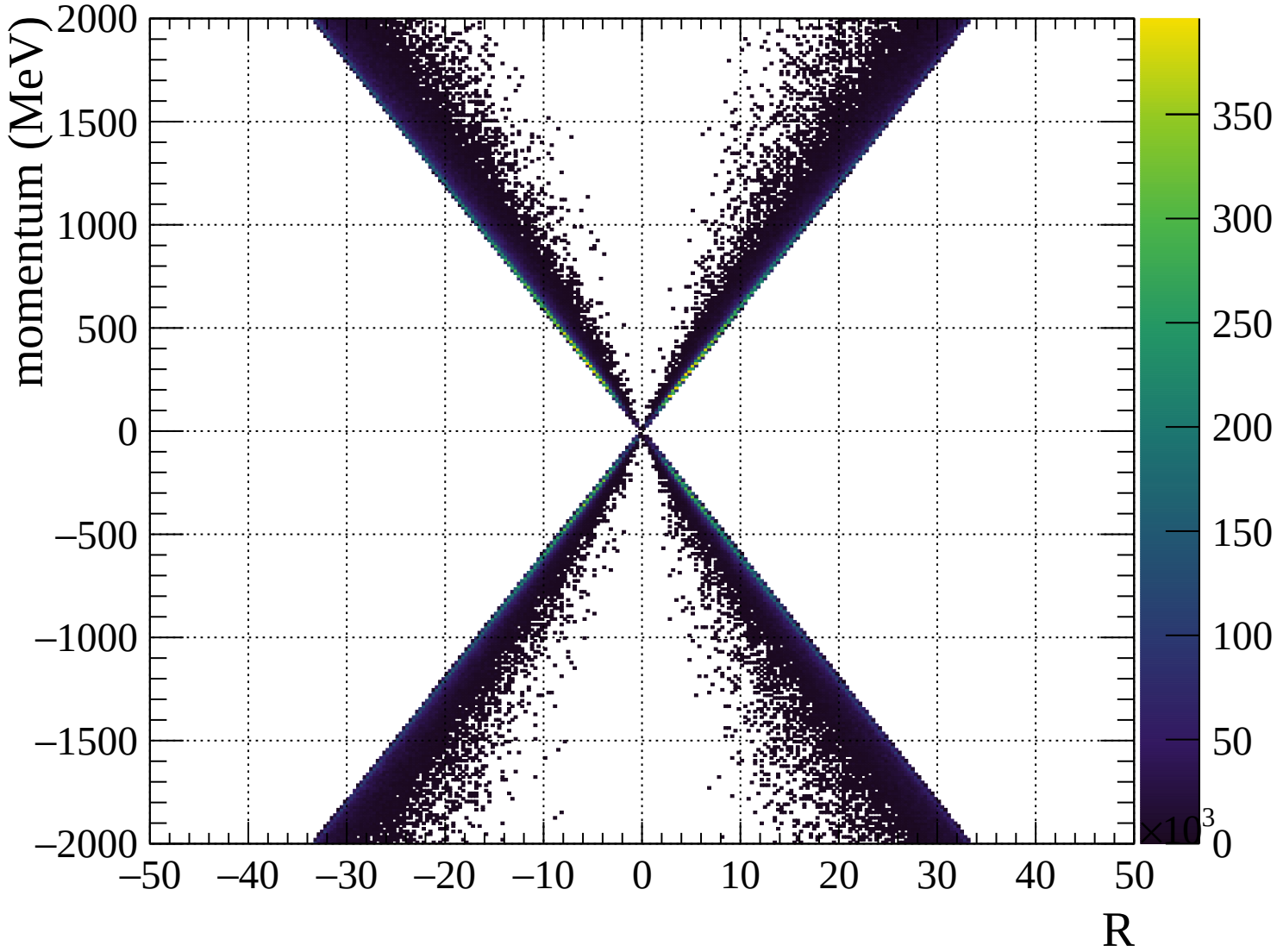


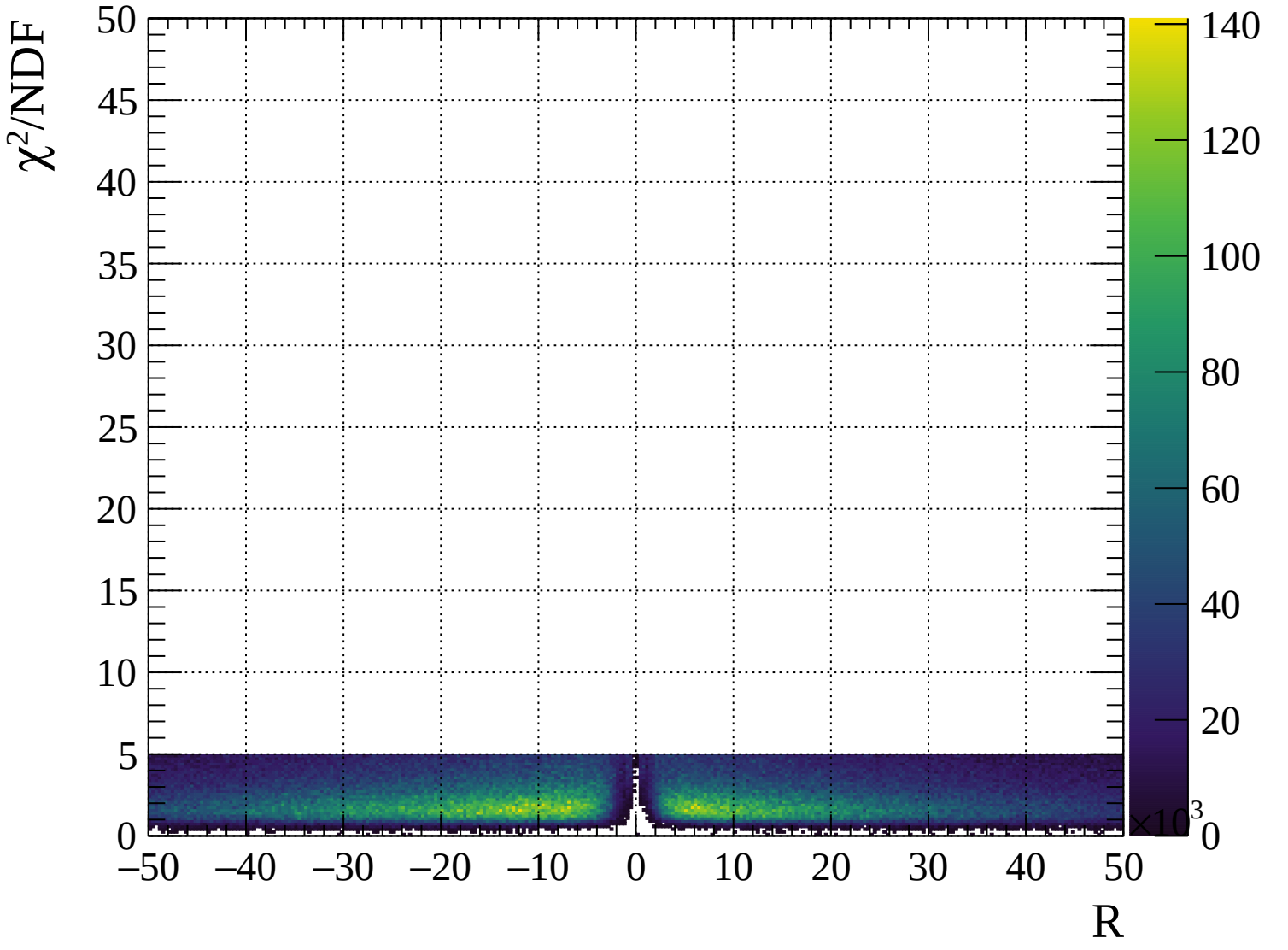






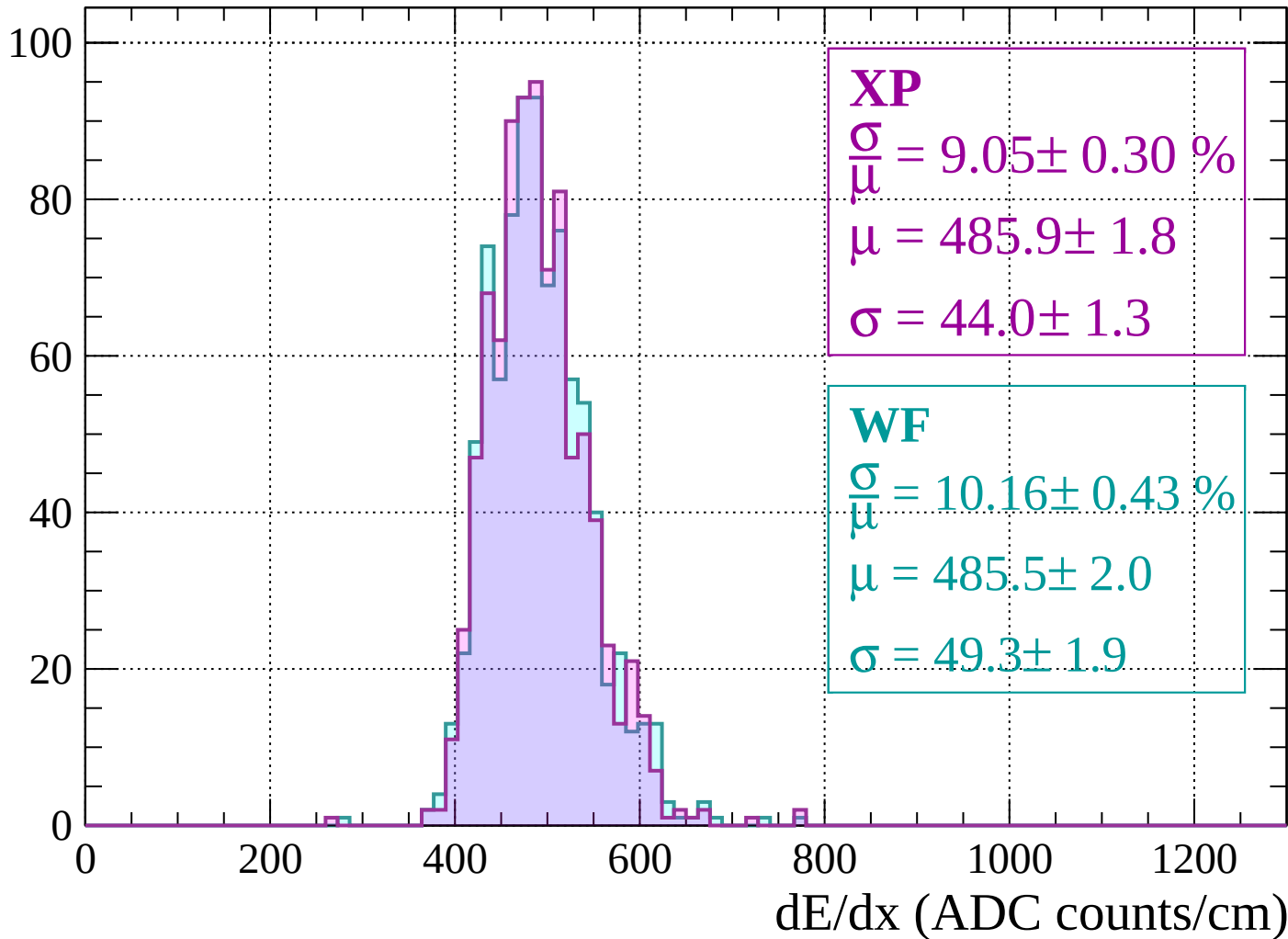




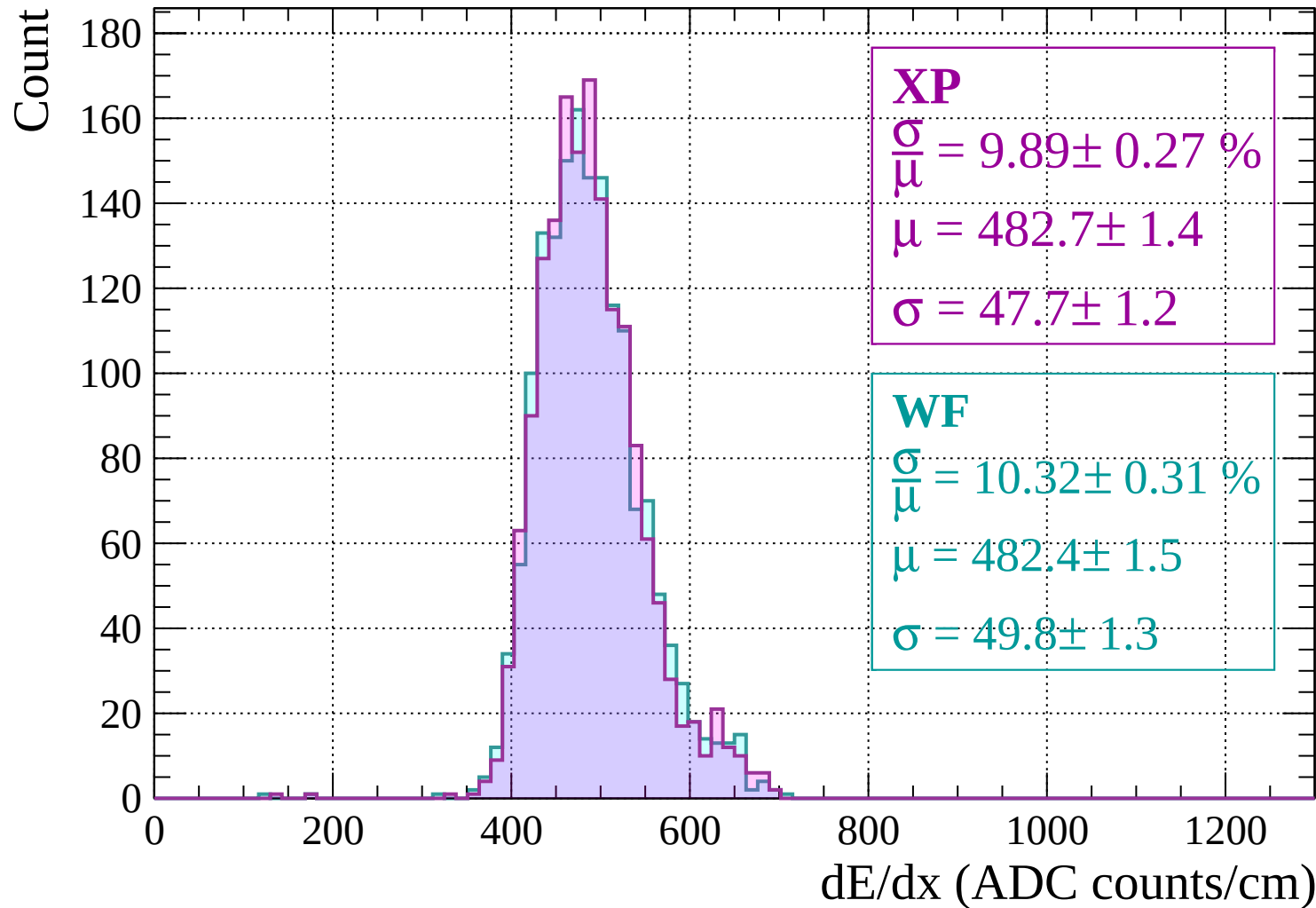


Energy loss | -2000 < p < -1960

Count

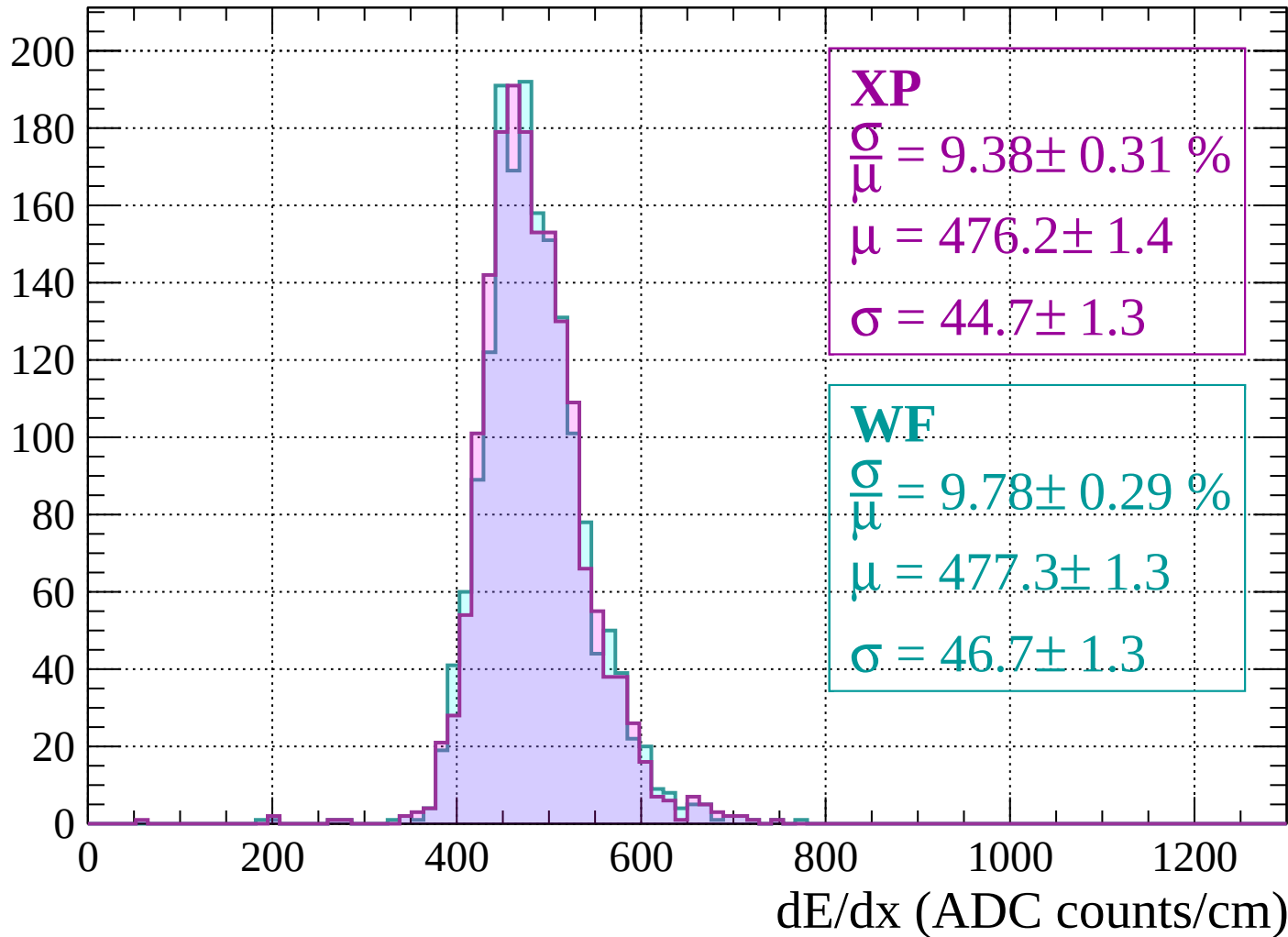


Energy loss | -1960 < p < -1920



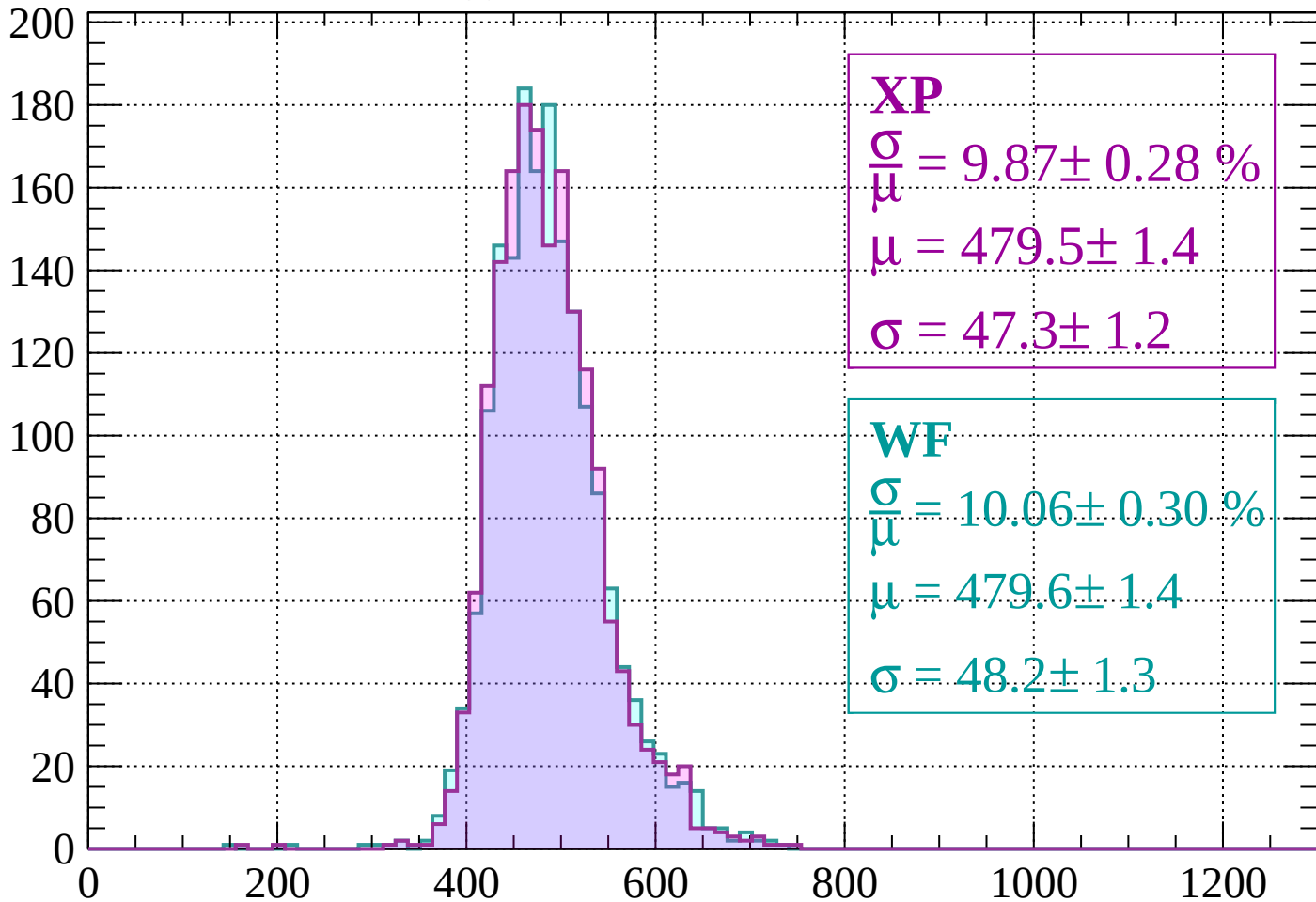
Energy loss | -1920 < p < -1880

Count



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.87 \pm 0.28 \%$$

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 47.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.06 \pm 0.30 \%$$

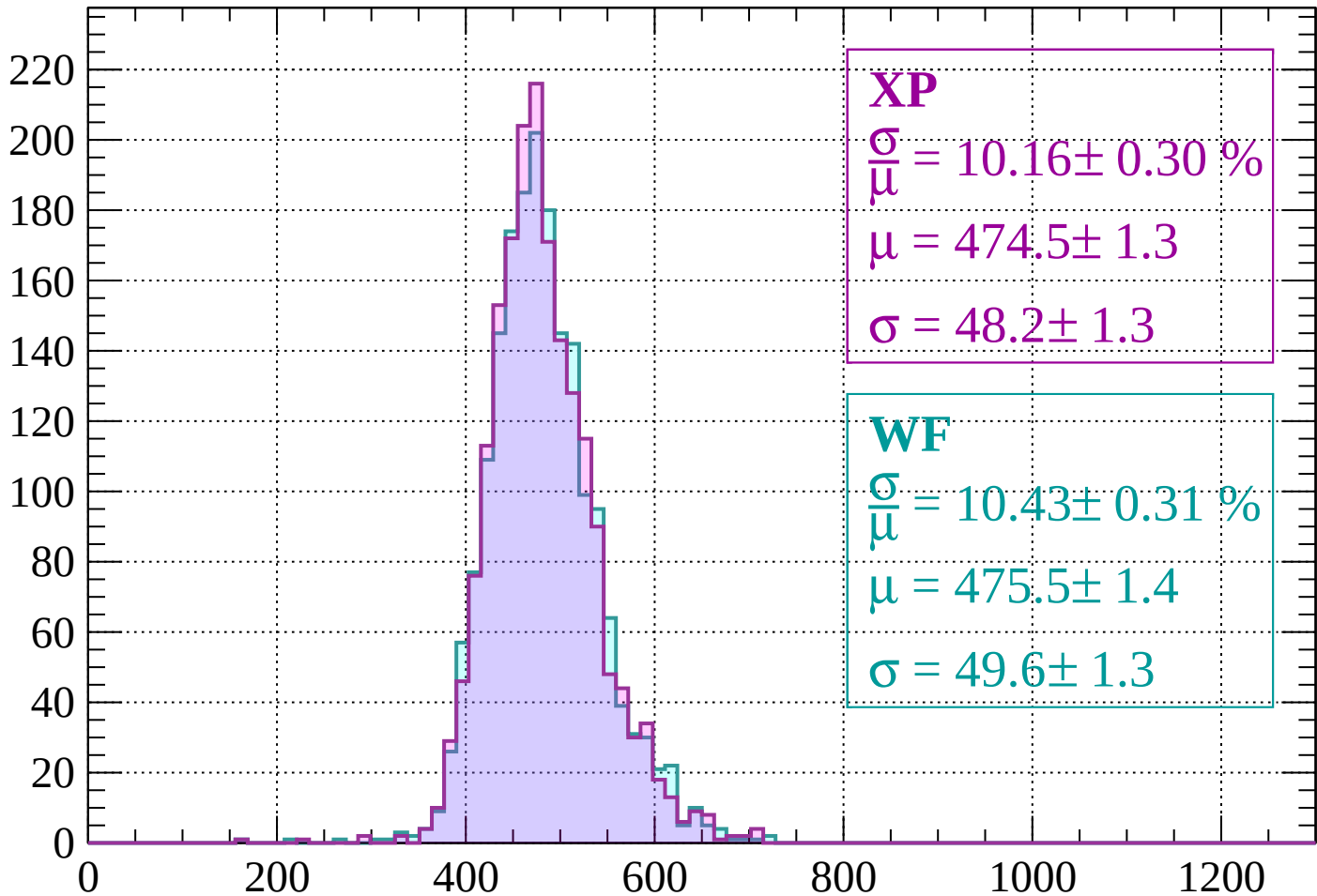
$$\mu = 479.6 \pm 1.4$$

$$\sigma = 48.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1840 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 10.16 \pm 0.30 \%$$

$$\mu = 474.5 \pm 1.3$$

$$\sigma = 48.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.43 \pm 0.31 \%$$

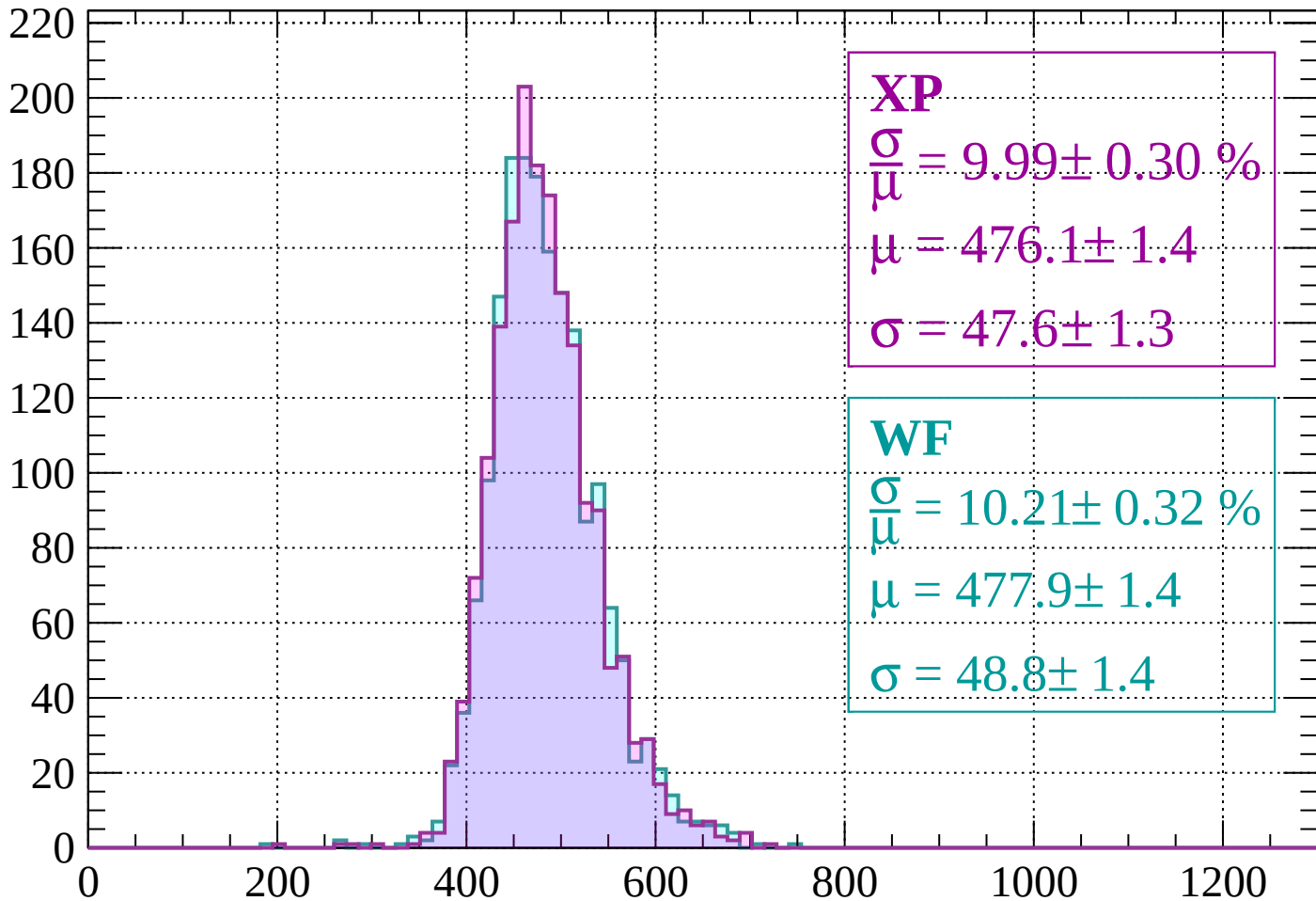
$$\mu = 475.5 \pm 1.4$$

$$\sigma = 49.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.99 \pm 0.30 \%$$

$$\mu = 476.1 \pm 1.4$$

$$\sigma = 47.6 \pm 1.3$$

WF

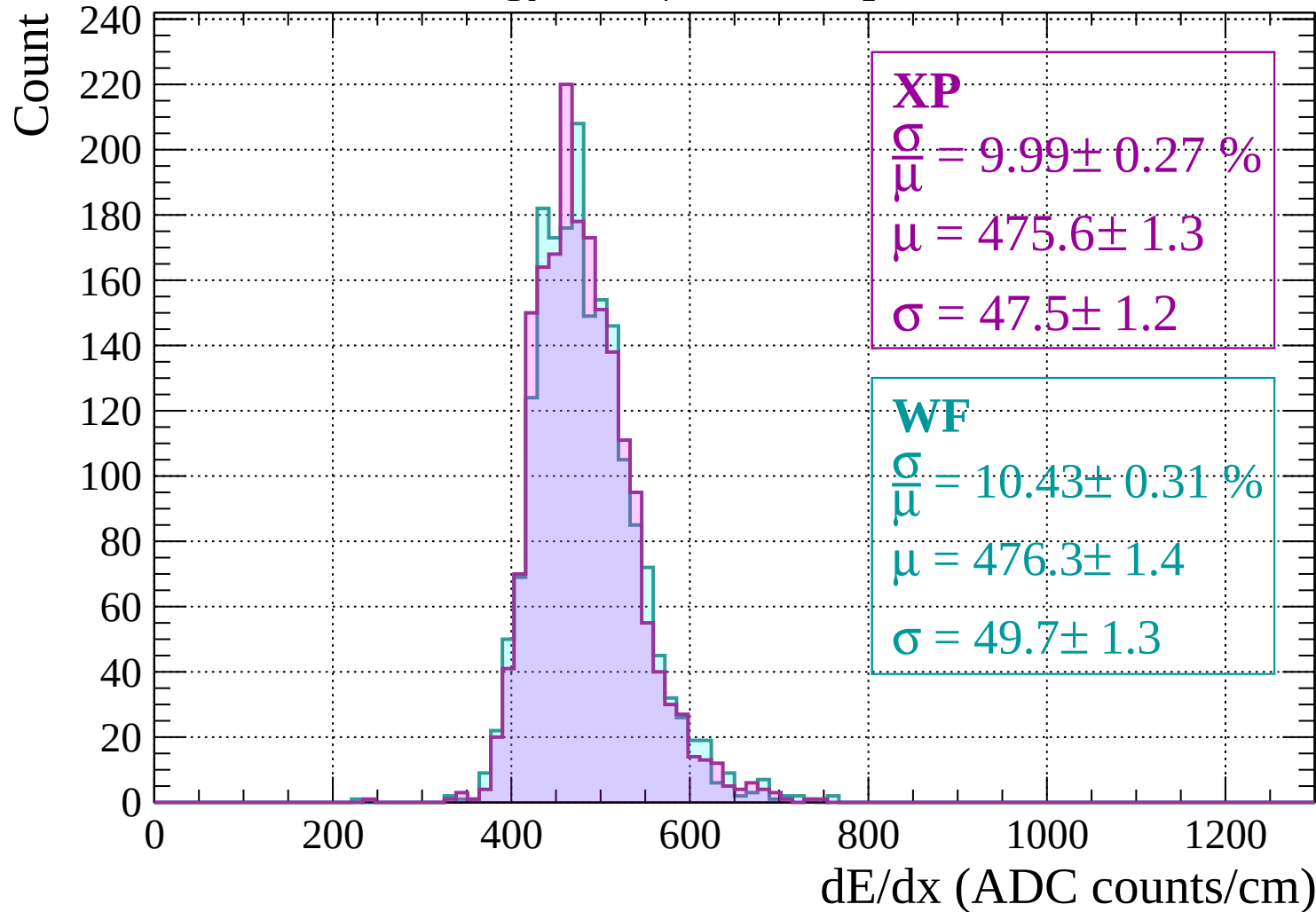
$$\frac{\sigma}{\mu} = 10.21 \pm 0.32 \%$$

$$\mu = 477.9 \pm 1.4$$

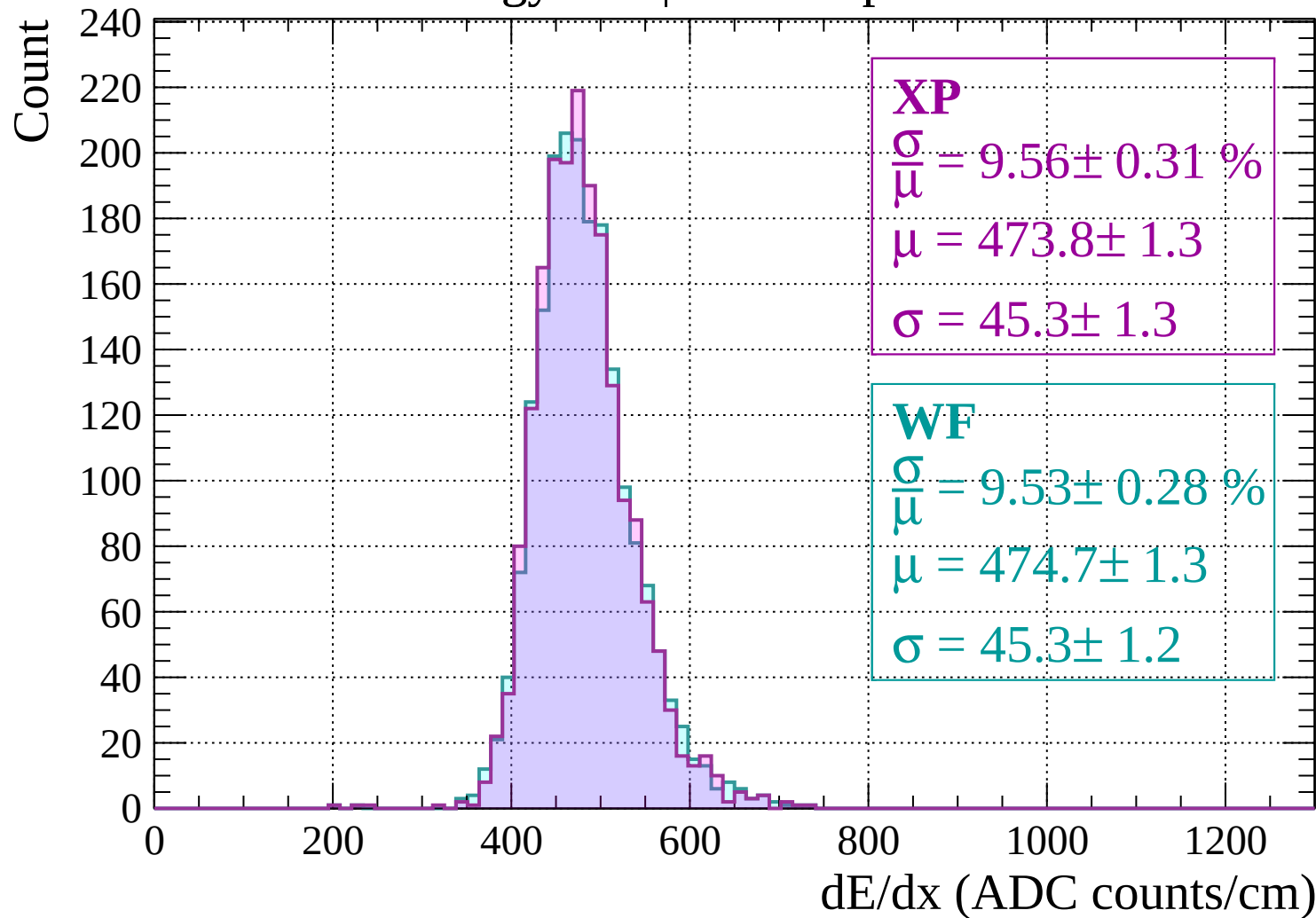
$$\sigma = 48.8 \pm 1.4$$

dE/dx (ADC counts/cm)

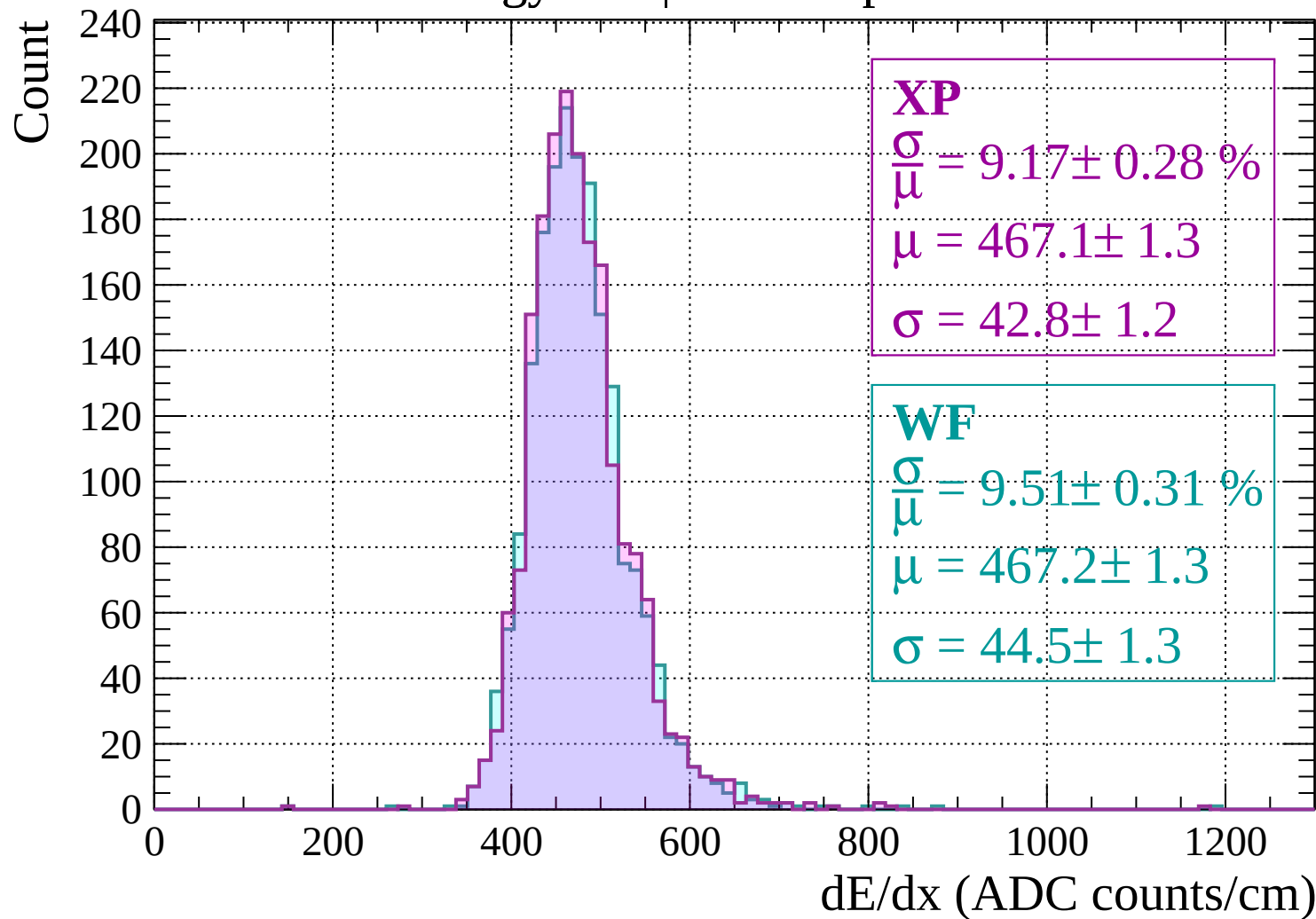
Energy loss | -1760 < p < -1720



Energy loss | -1720 < p < -1680

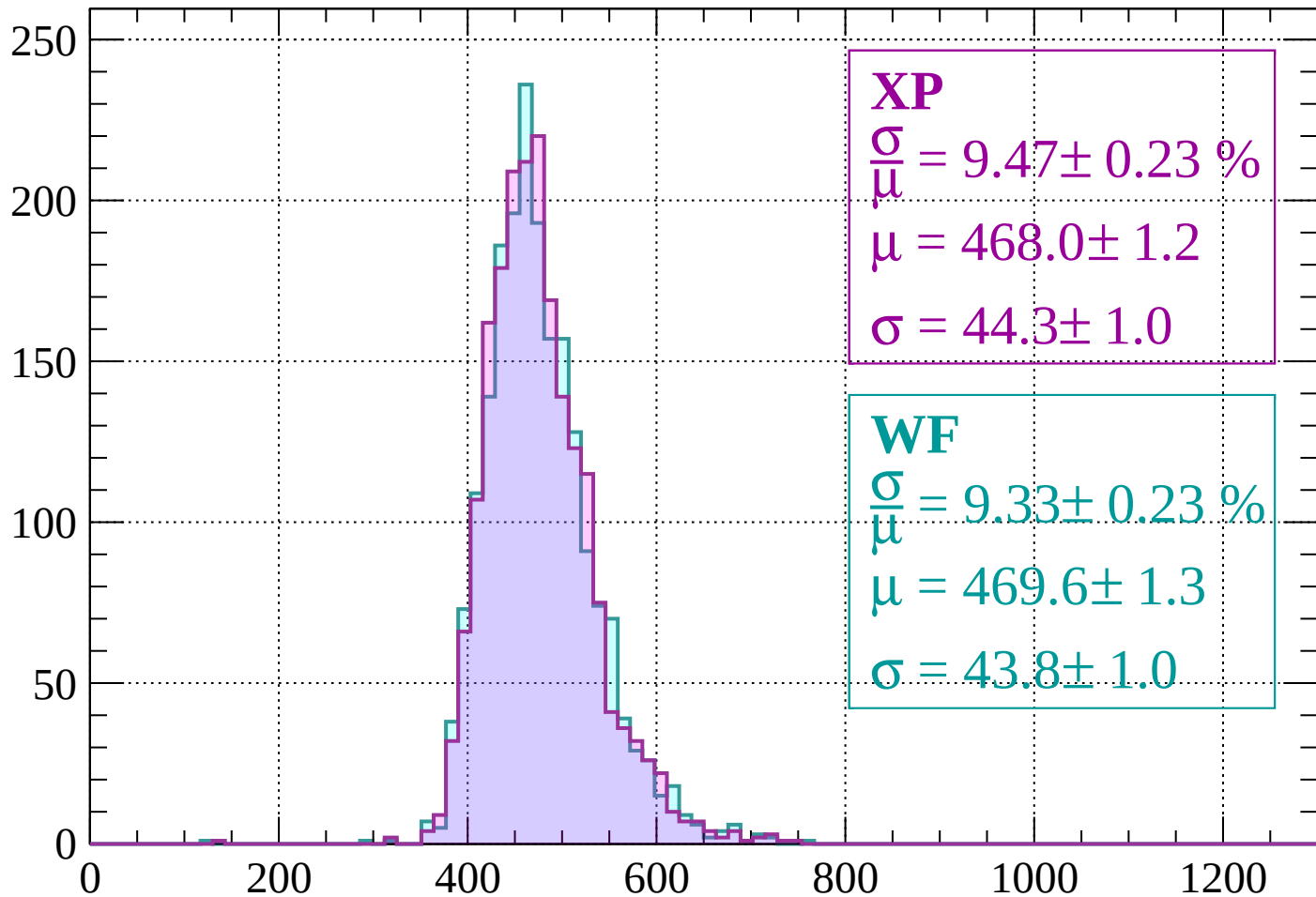


Energy loss | -1680 < p < -1640



Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.47 \pm 0.23 \%$$

$$\mu = 468.0 \pm 1.2$$

$$\sigma = 44.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.23 \%$$

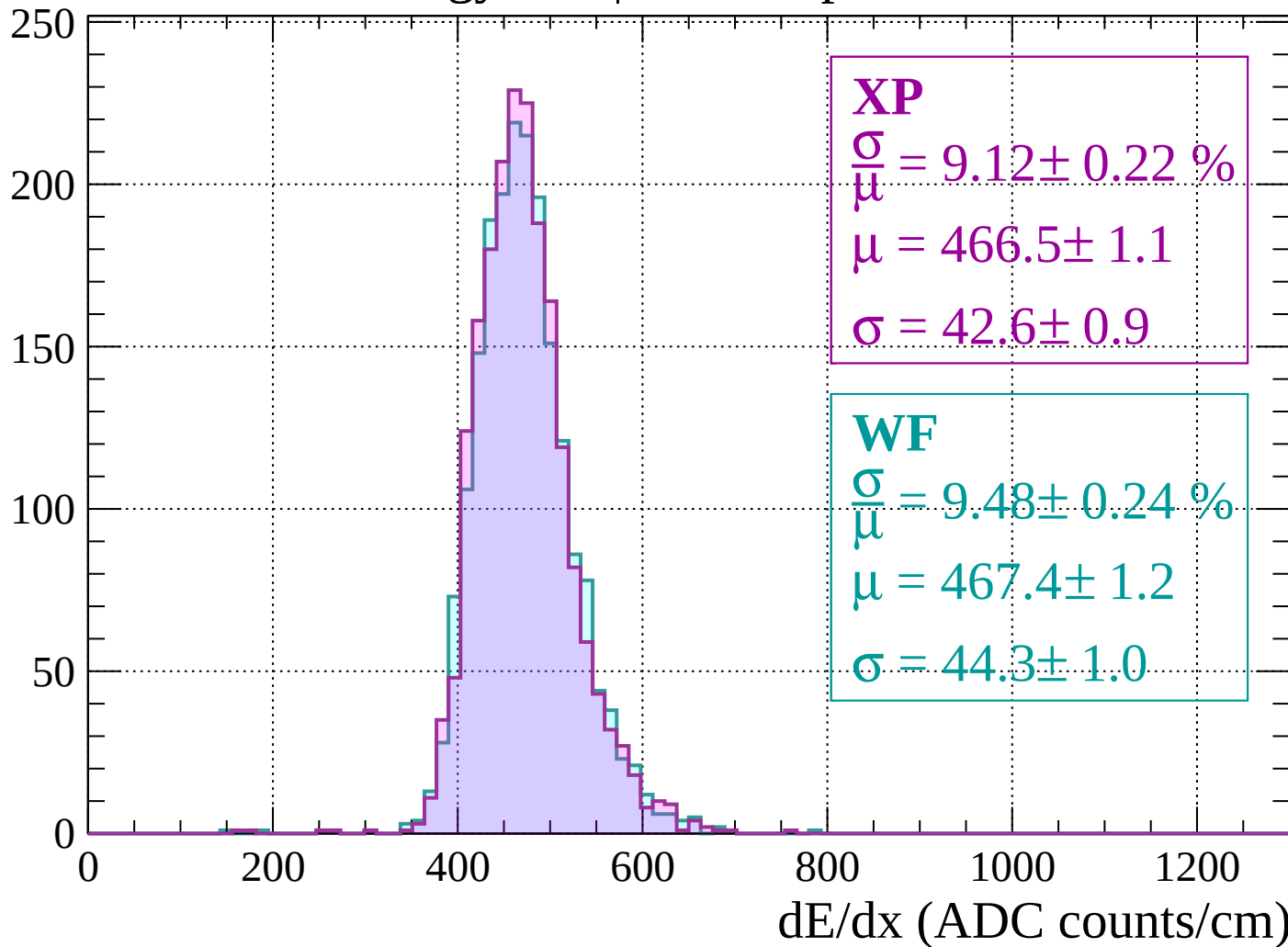
$$\mu = 469.6 \pm 1.3$$

$$\sigma = 43.8 \pm 1.0$$

dE/dx (ADC counts/cm)

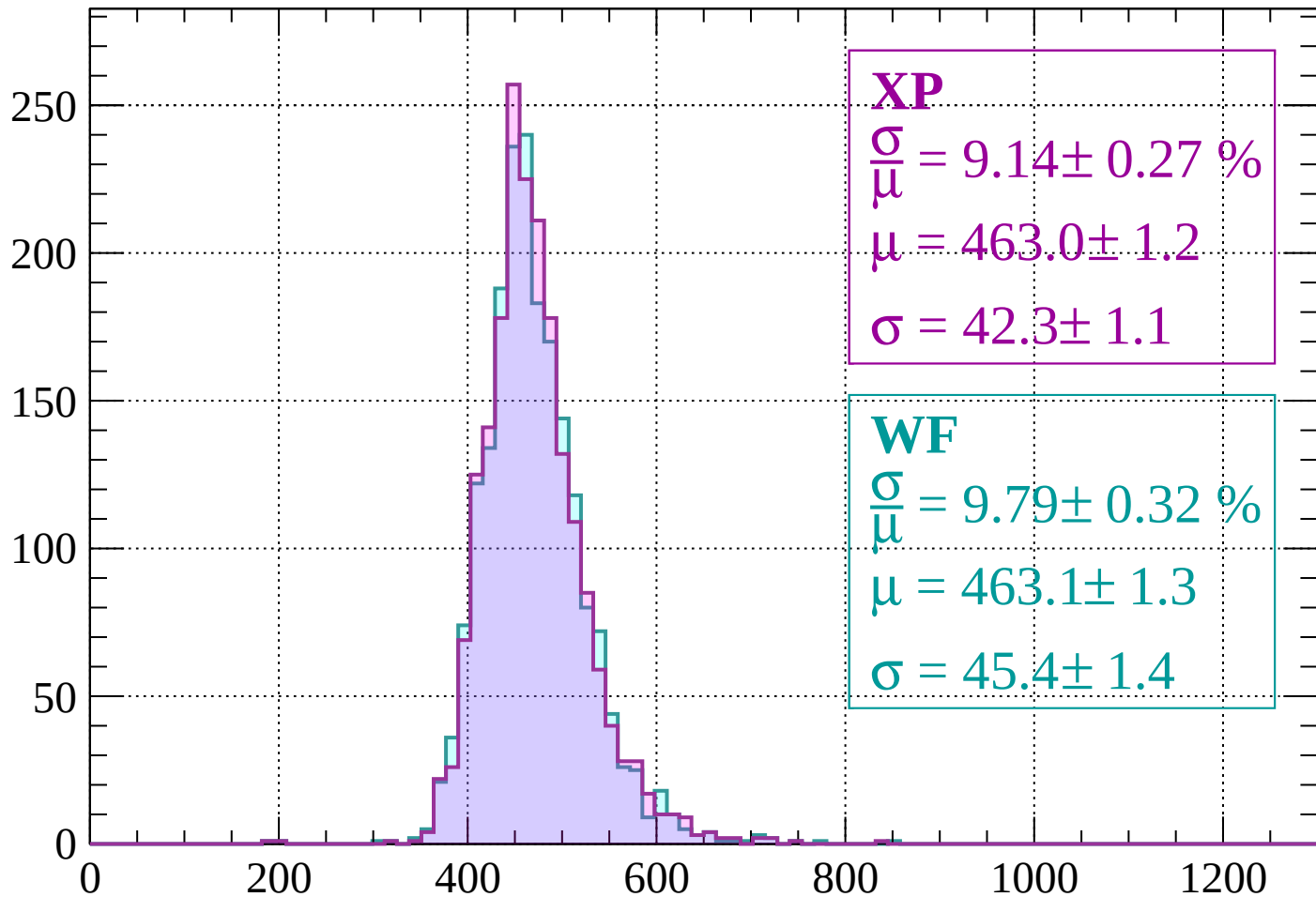
Energy loss | -1600 < p < -1560

Count



Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 9.14 \pm 0.27 \%$$

$$\mu = 463.0 \pm 1.2$$

$$\sigma = 42.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.32 \%$$

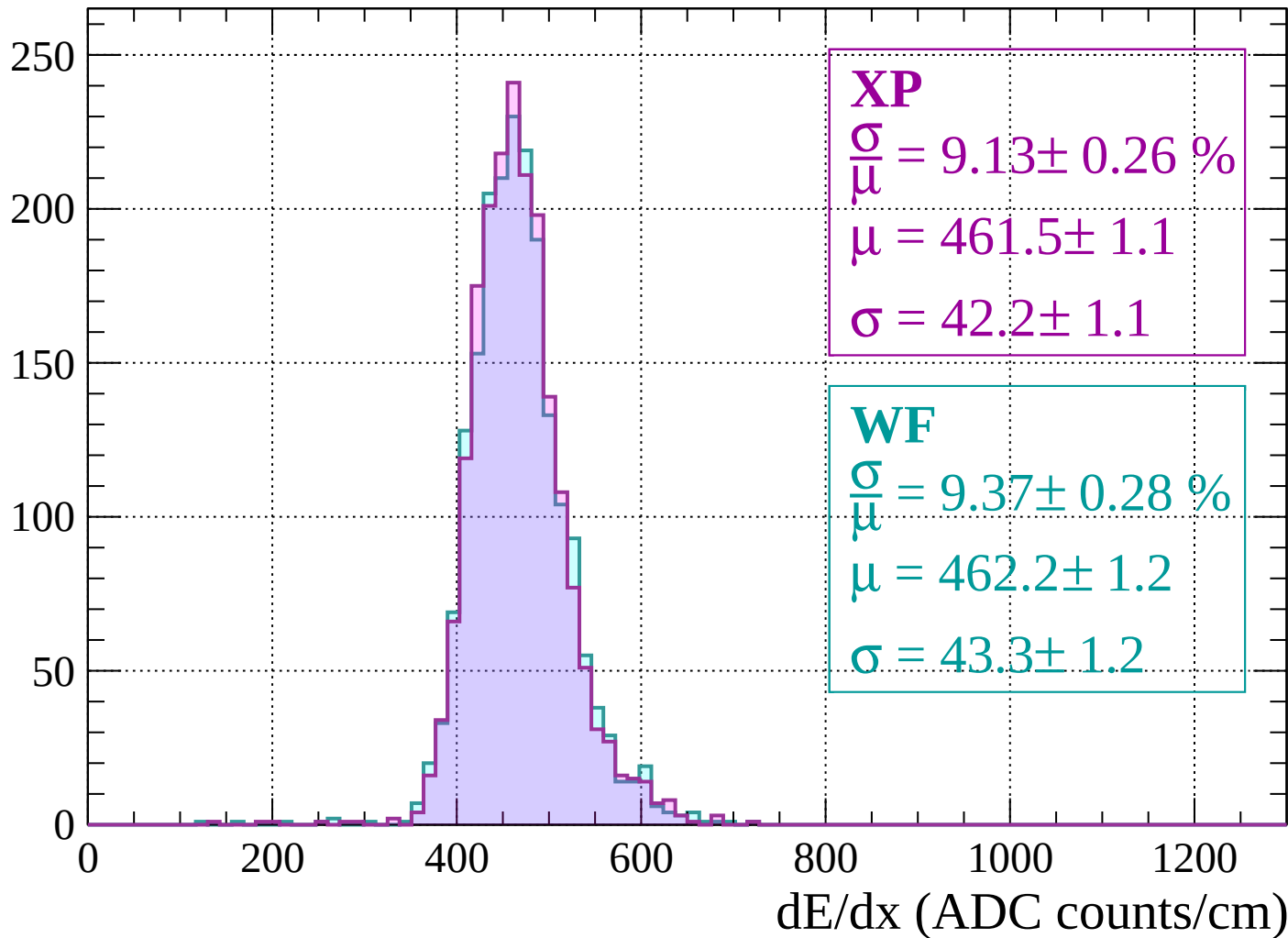
$$\mu = 463.1 \pm 1.3$$

$$\sigma = 45.4 \pm 1.4$$

dE/dx (ADC counts/cm)

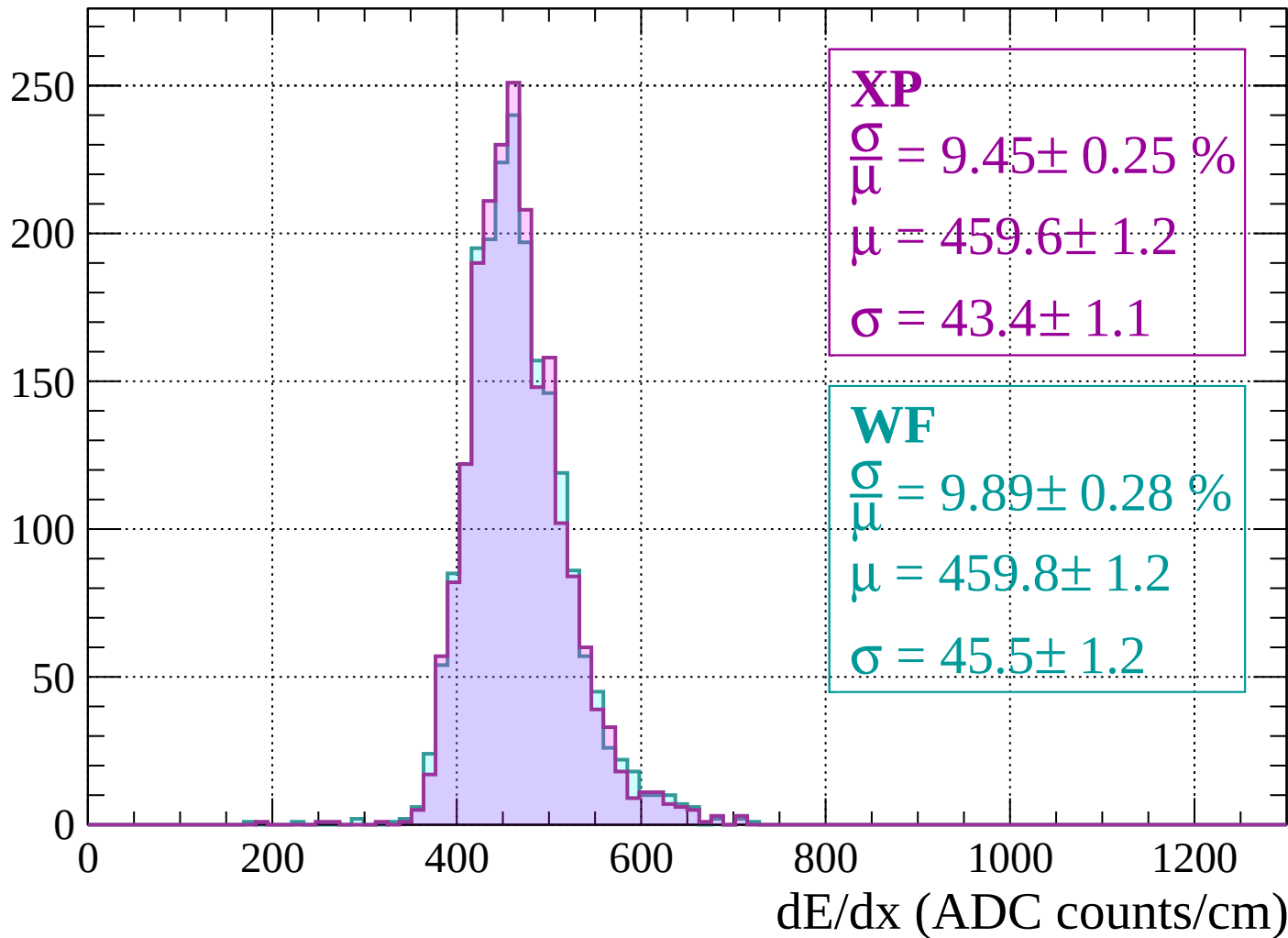
Energy loss | -1520 < p < -1480

Count



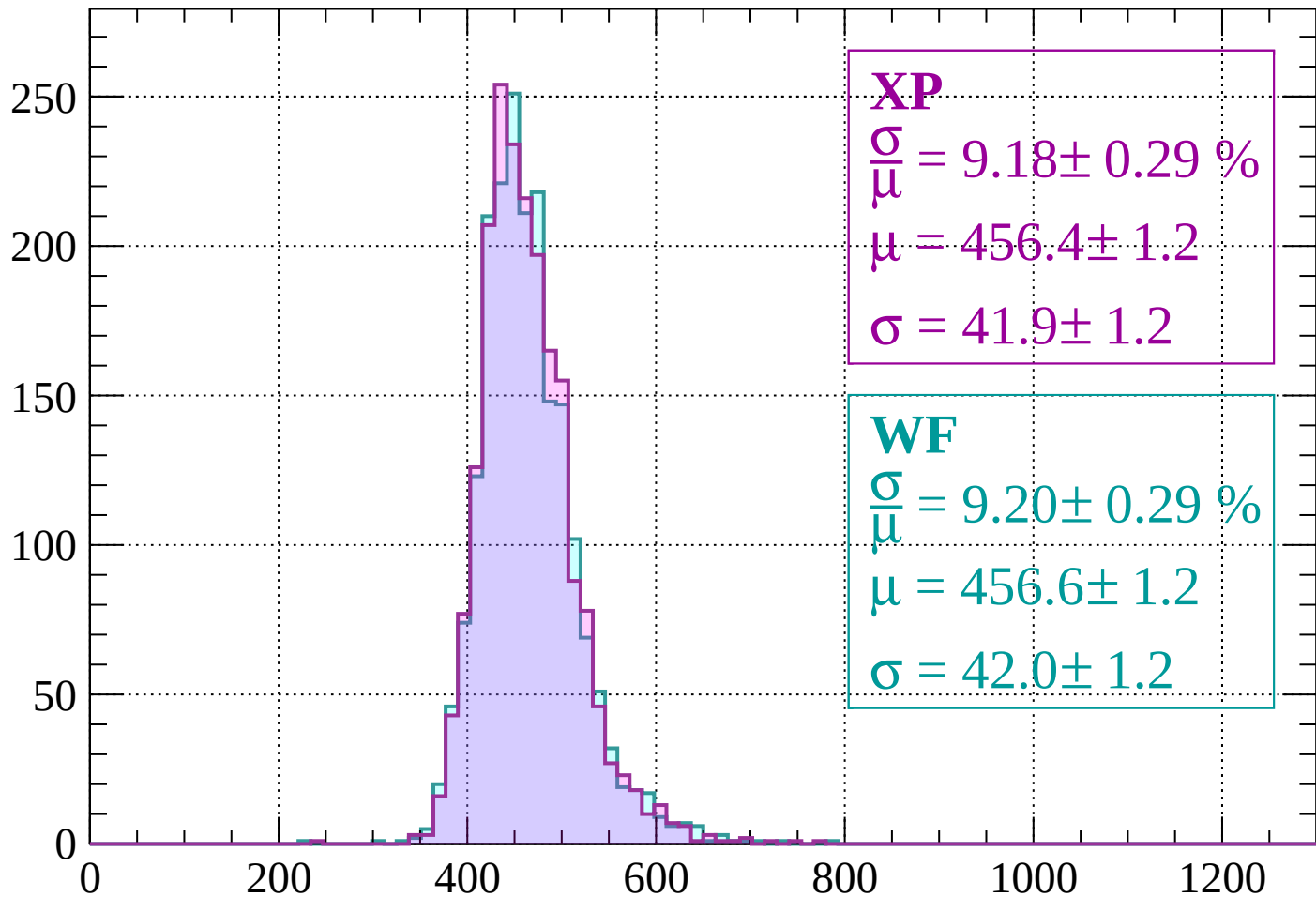
Energy loss | $-1480 < p < -1440$

Count



Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.29 \%$$

$$\mu = 456.4 \pm 1.2$$

$$\sigma = 41.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.29 \%$$

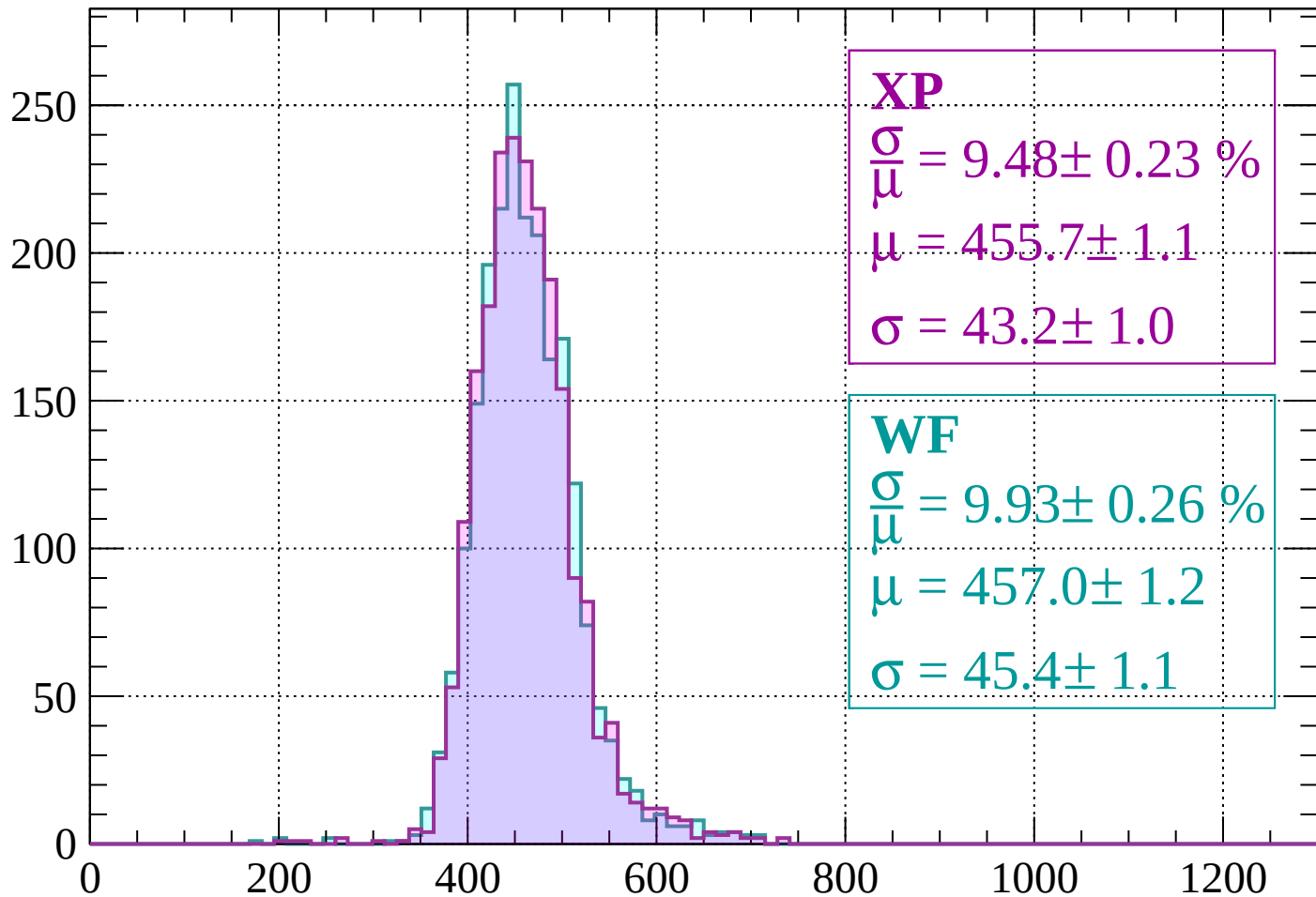
$$\mu = 456.6 \pm 1.2$$

$$\sigma = 42.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.23 \%$$

$$\mu = 455.7 \pm 1.1$$

$$\sigma = 43.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.93 \pm 0.26 \%$$

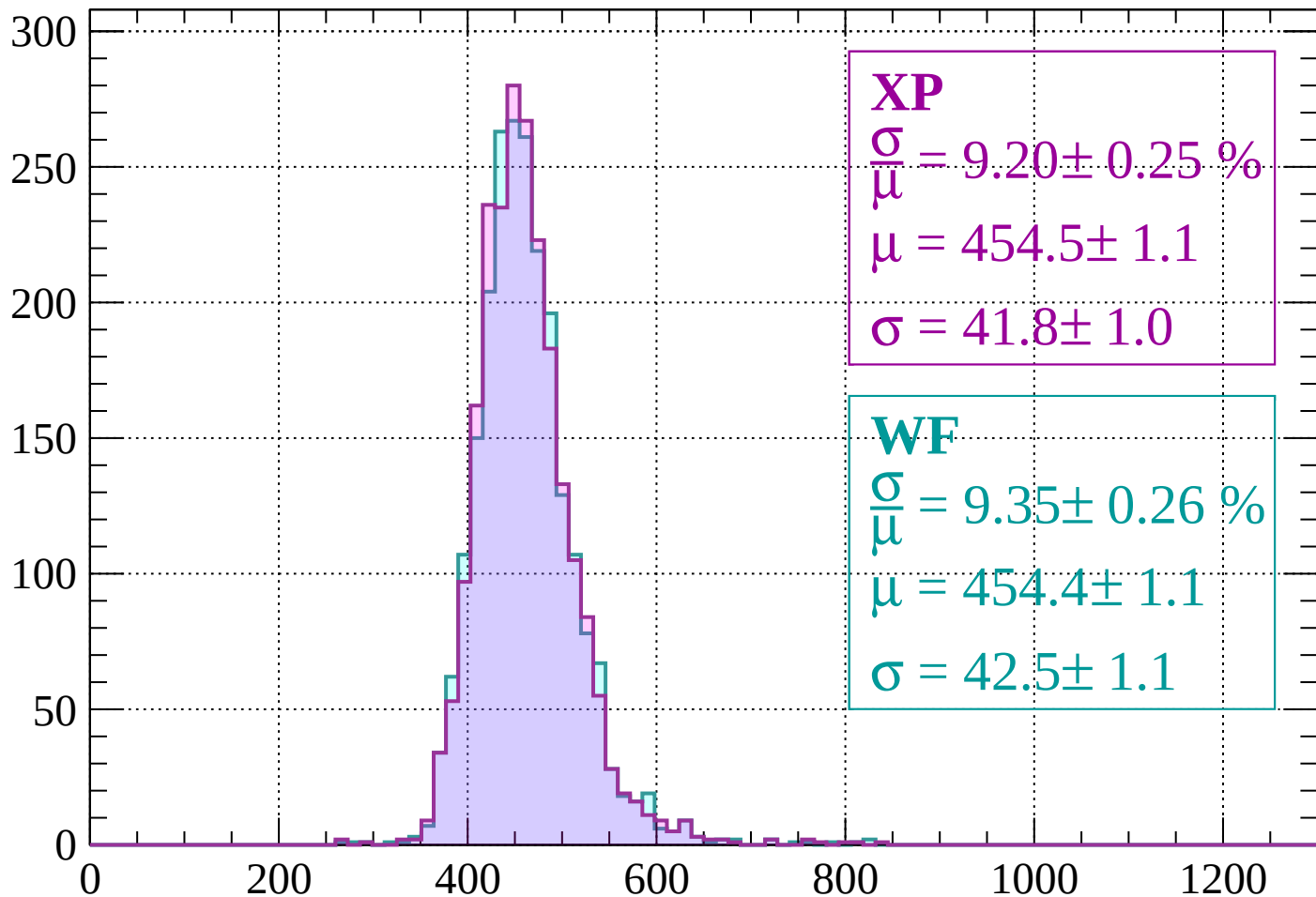
$$\mu = 457.0 \pm 1.2$$

$$\sigma = 45.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.25 \%$$

$$\mu = 454.5 \pm 1.1$$

$$\sigma = 41.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.26 \%$$

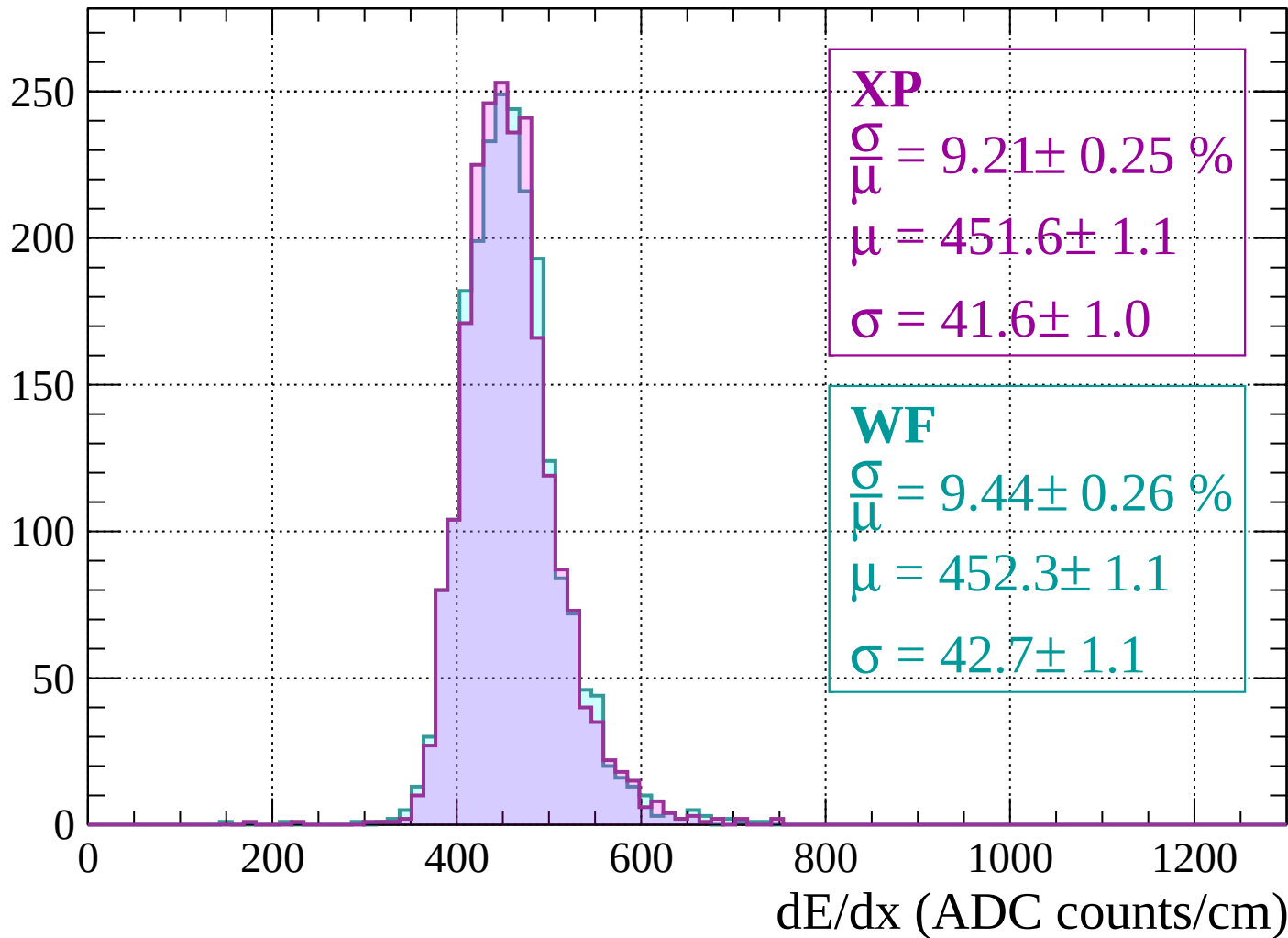
$$\mu = 454.4 \pm 1.1$$

$$\sigma = 42.5 \pm 1.1$$

dE/dx (ADC counts/cm)

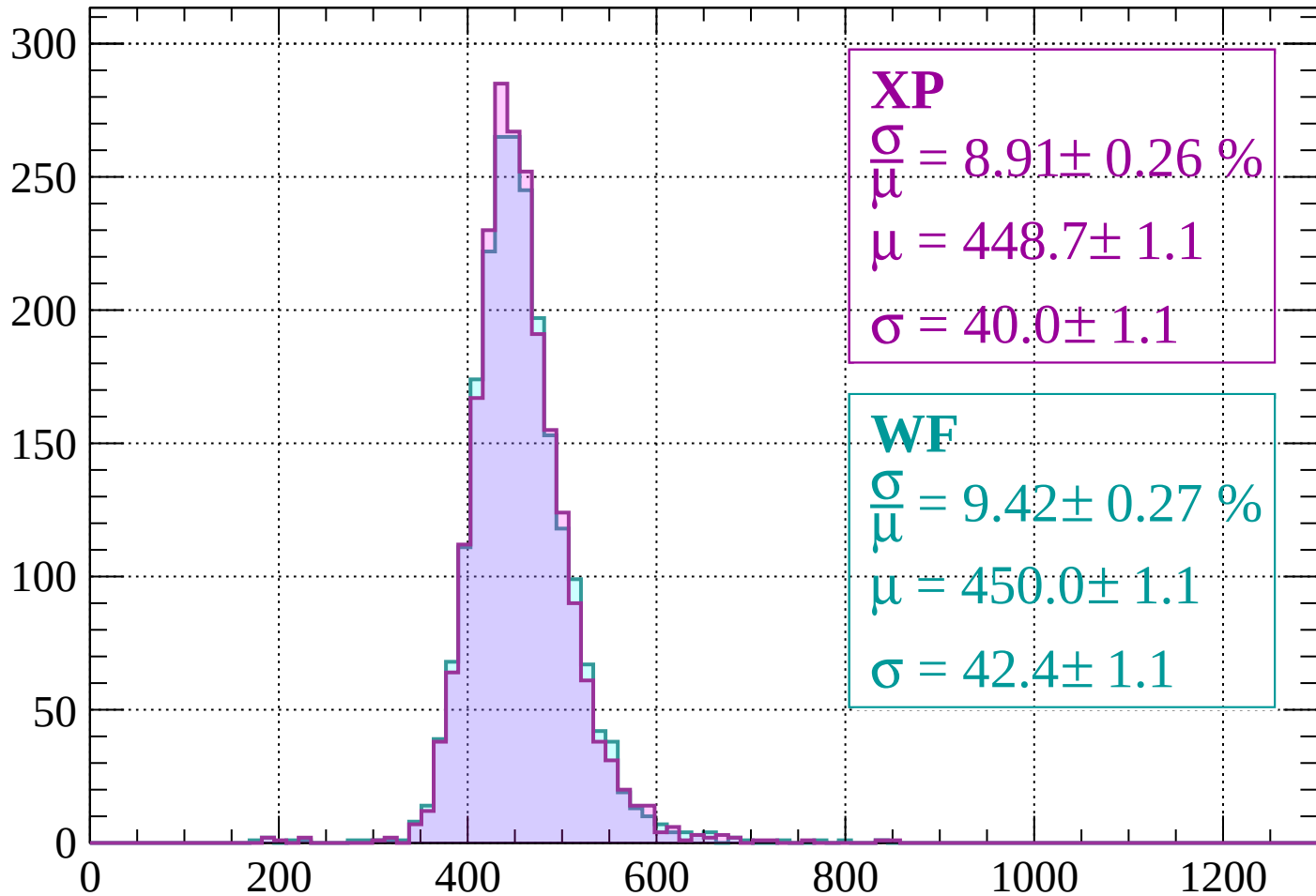
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.26 \%$$

$$\mu = 448.7 \pm 1.1$$

$$\sigma = 40.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.27 \%$$

$$\mu = 450.0 \pm 1.1$$

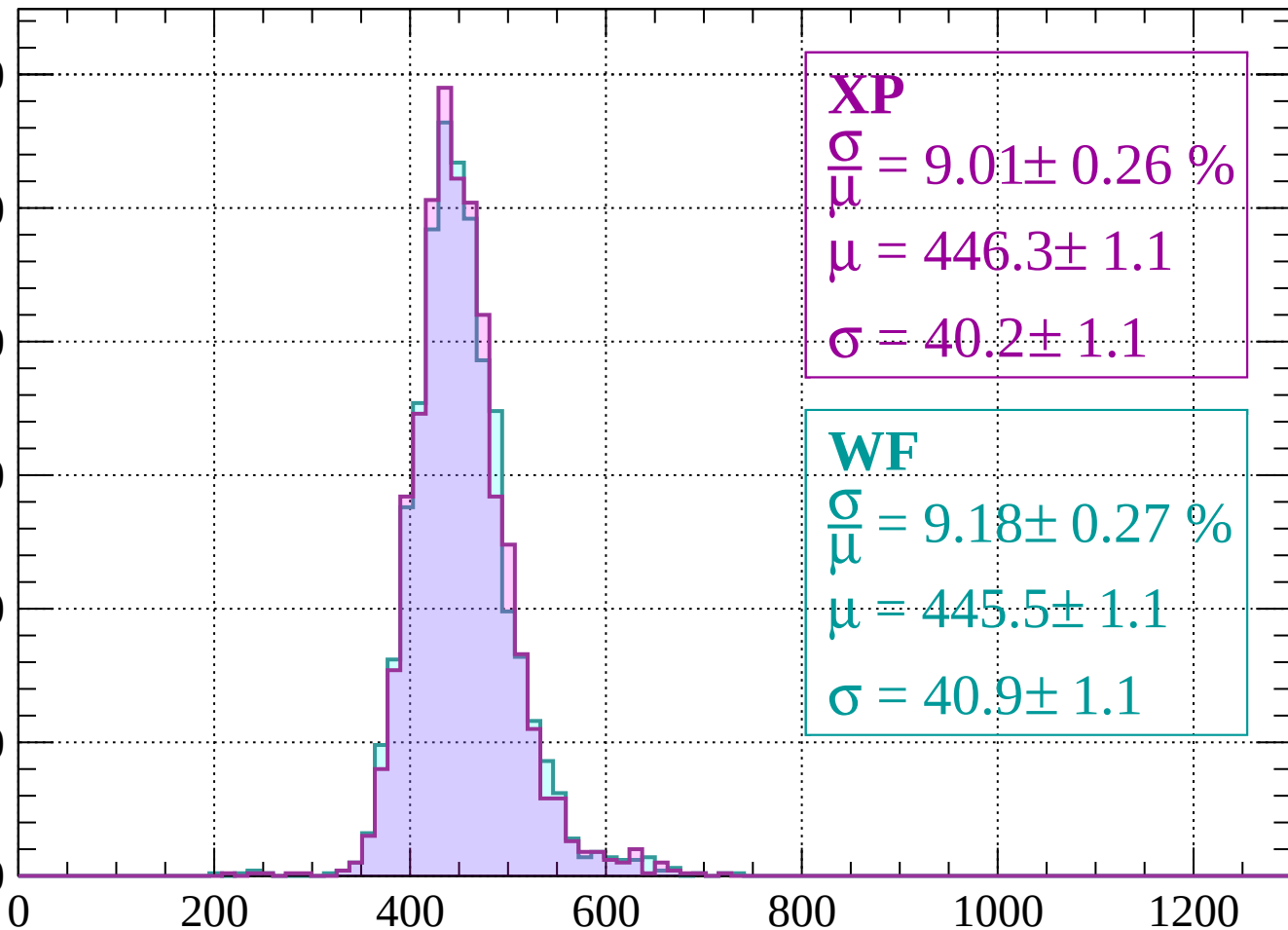
$$\sigma = 42.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.26 \%$$

$$\mu = 446.3 \pm 1.1$$

$$\sigma = 40.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.18 \pm 0.27 \%$$

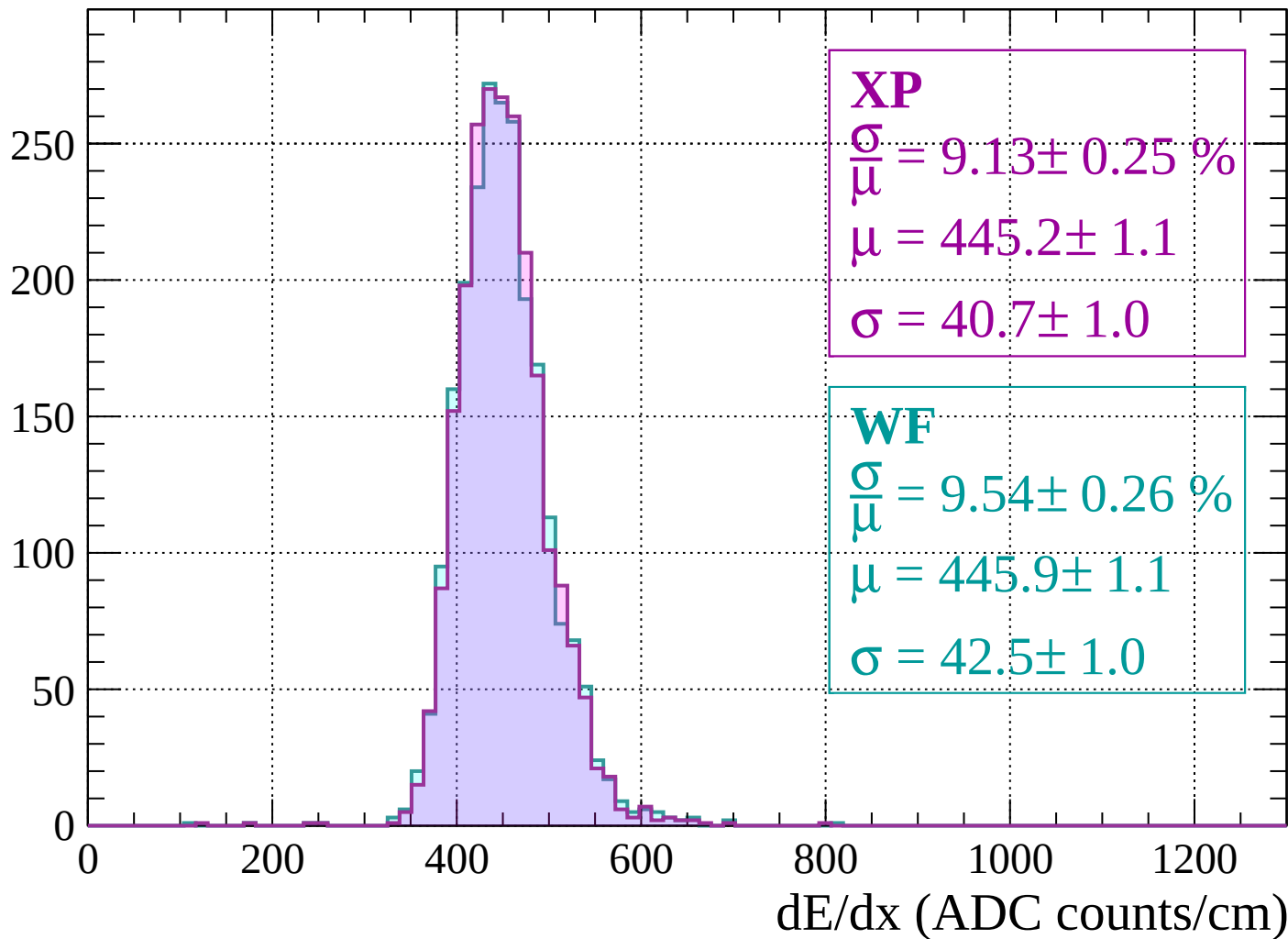
$$\mu = 445.5 \pm 1.1$$

$$\sigma = 40.9 \pm 1.1$$

dE/dx (ADC counts/cm)

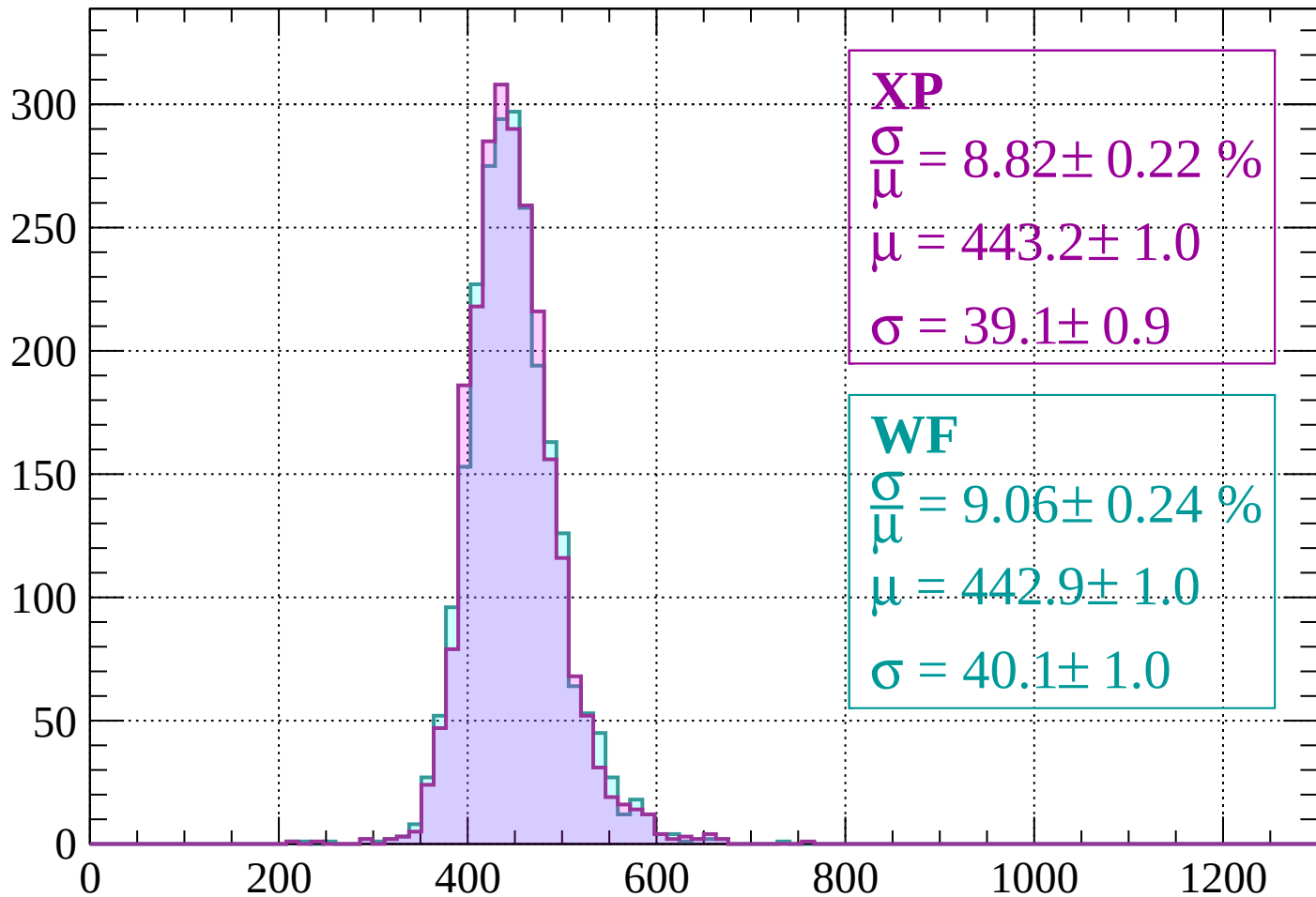
Energy loss | $-1200 < p < -1160$

Count



Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.22 \%$$

$$\mu = 443.2 \pm 1.0$$

$$\sigma = 39.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.24 \%$$

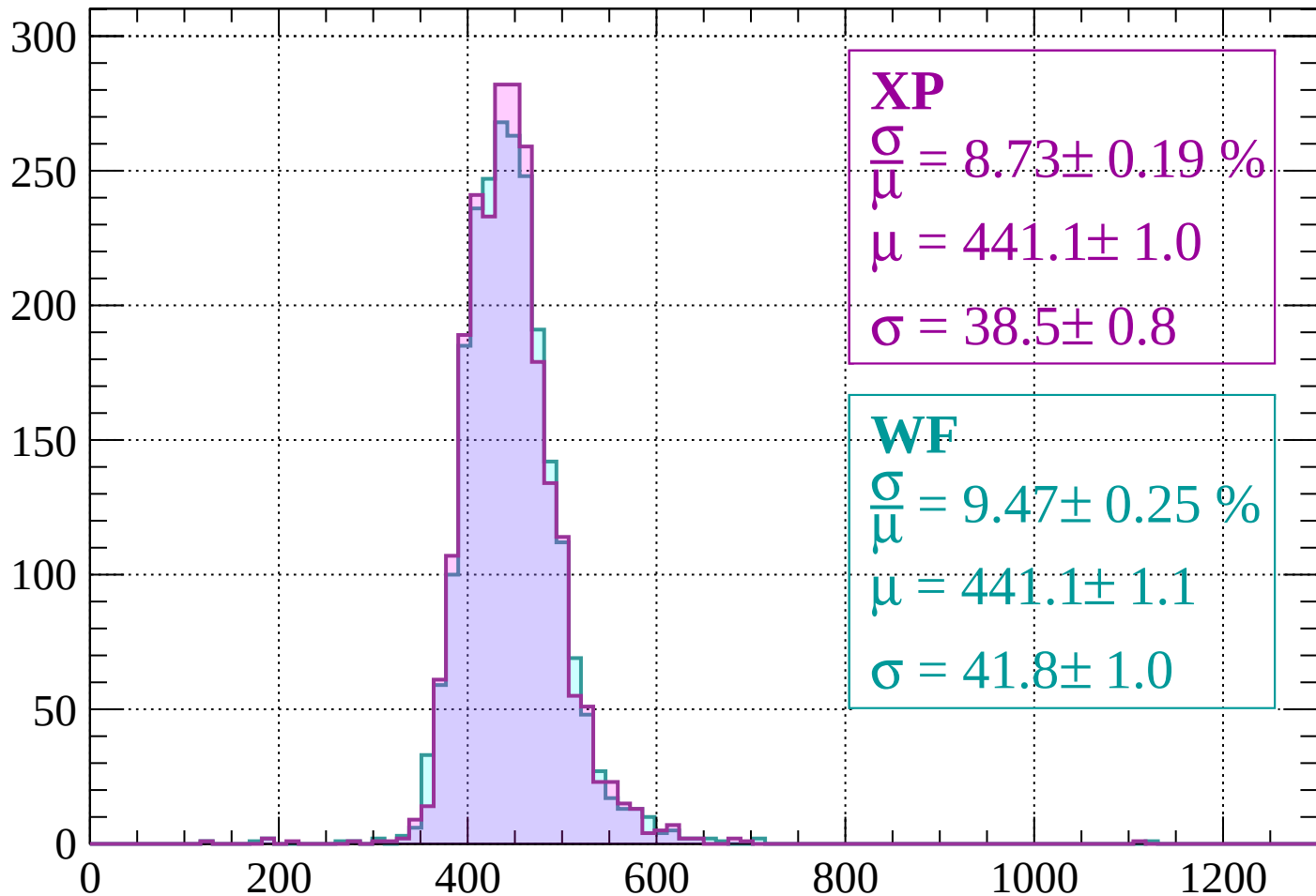
$$\mu = 442.9 \pm 1.0$$

$$\sigma = 40.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.73 \pm 0.19 \%$$

$$\mu = 441.1 \pm 1.0$$

$$\sigma = 38.5 \pm 0.8$$

WF

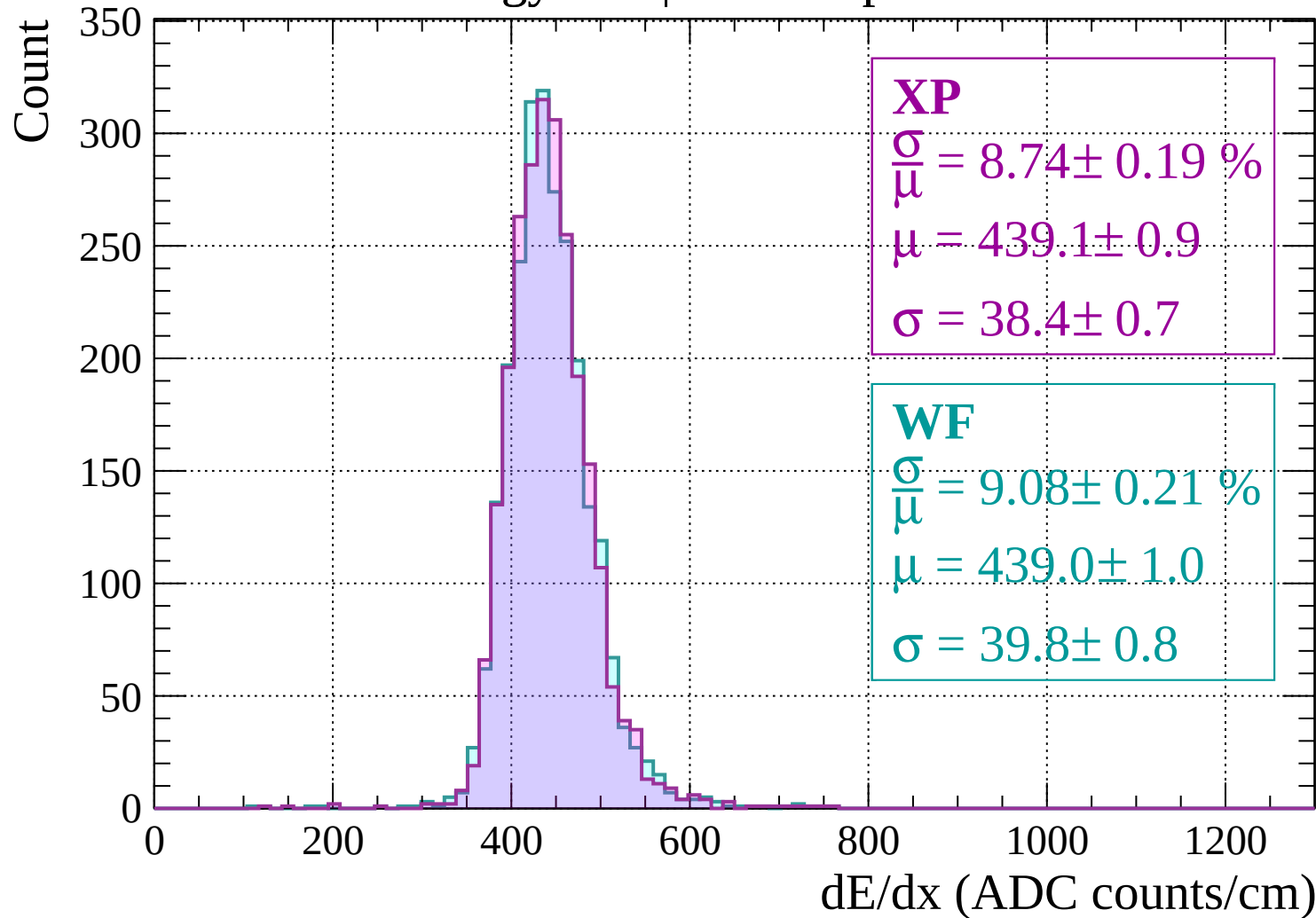
$$\frac{\sigma}{\mu} = 9.47 \pm 0.25 \%$$

$$\mu = 441.1 \pm 1.1$$

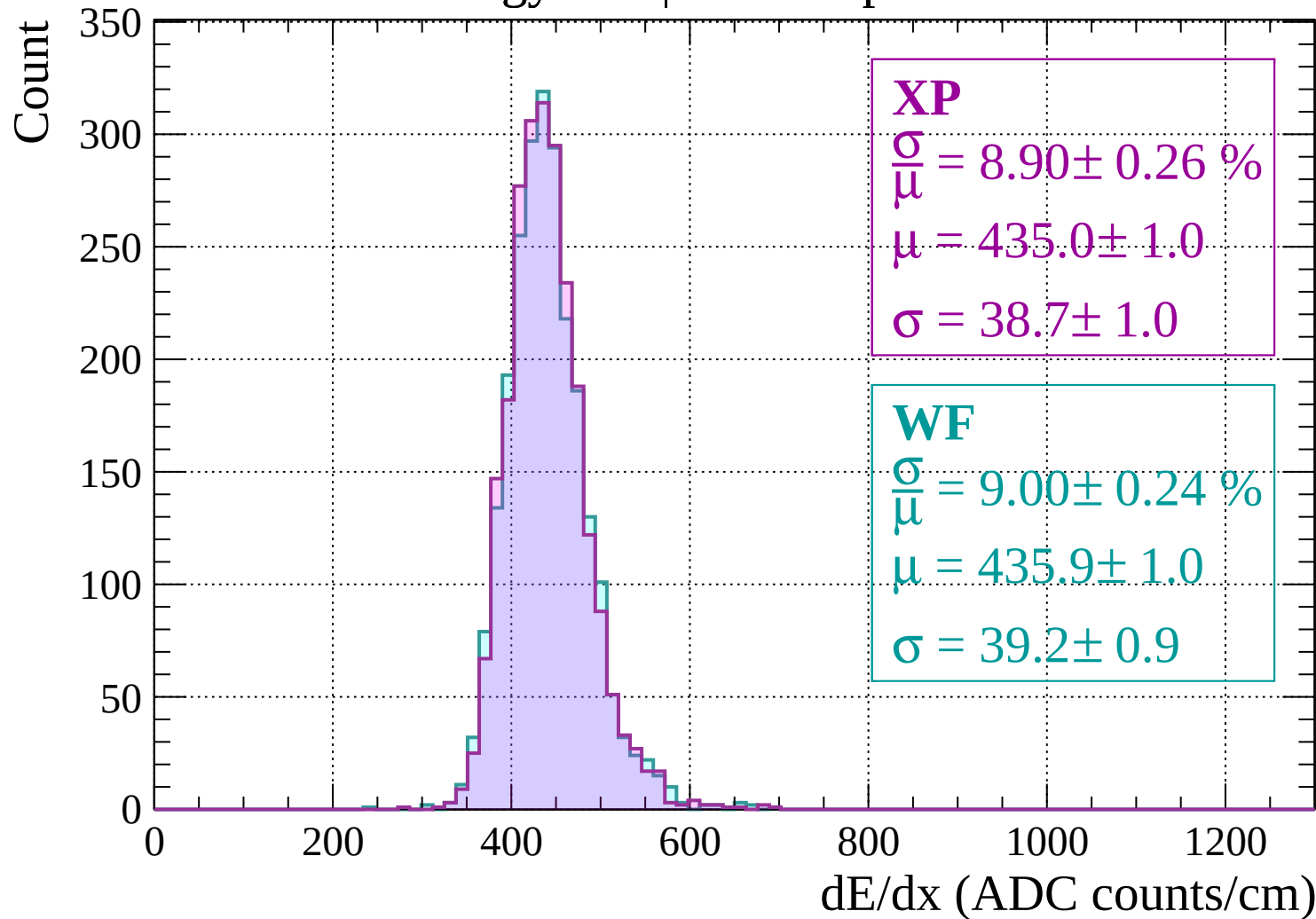
$$\sigma = 41.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

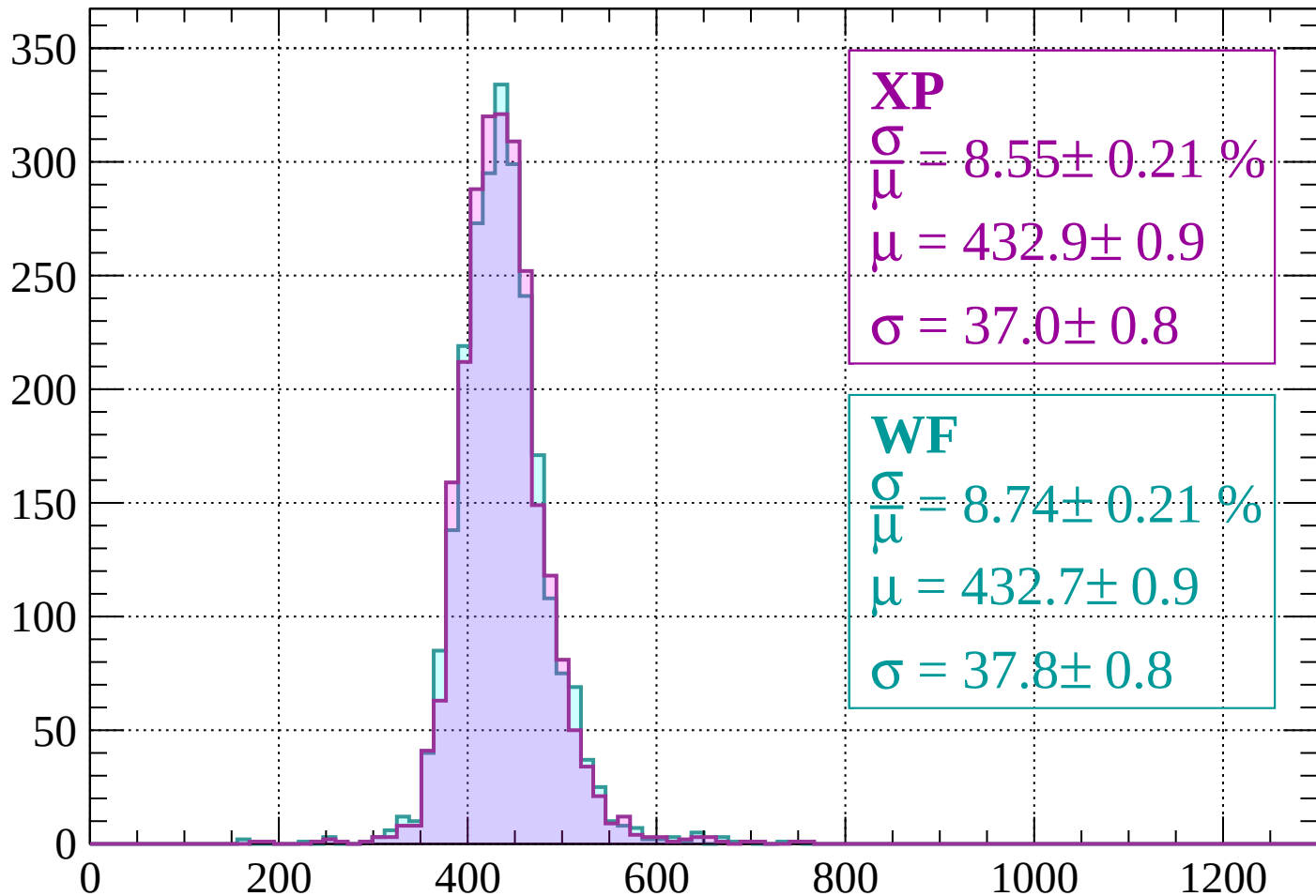


Energy loss | $-1040 < p < -1000$



Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.21 \%$$

$$\mu = 432.9 \pm 0.9$$

$$\sigma = 37.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.21 \%$$

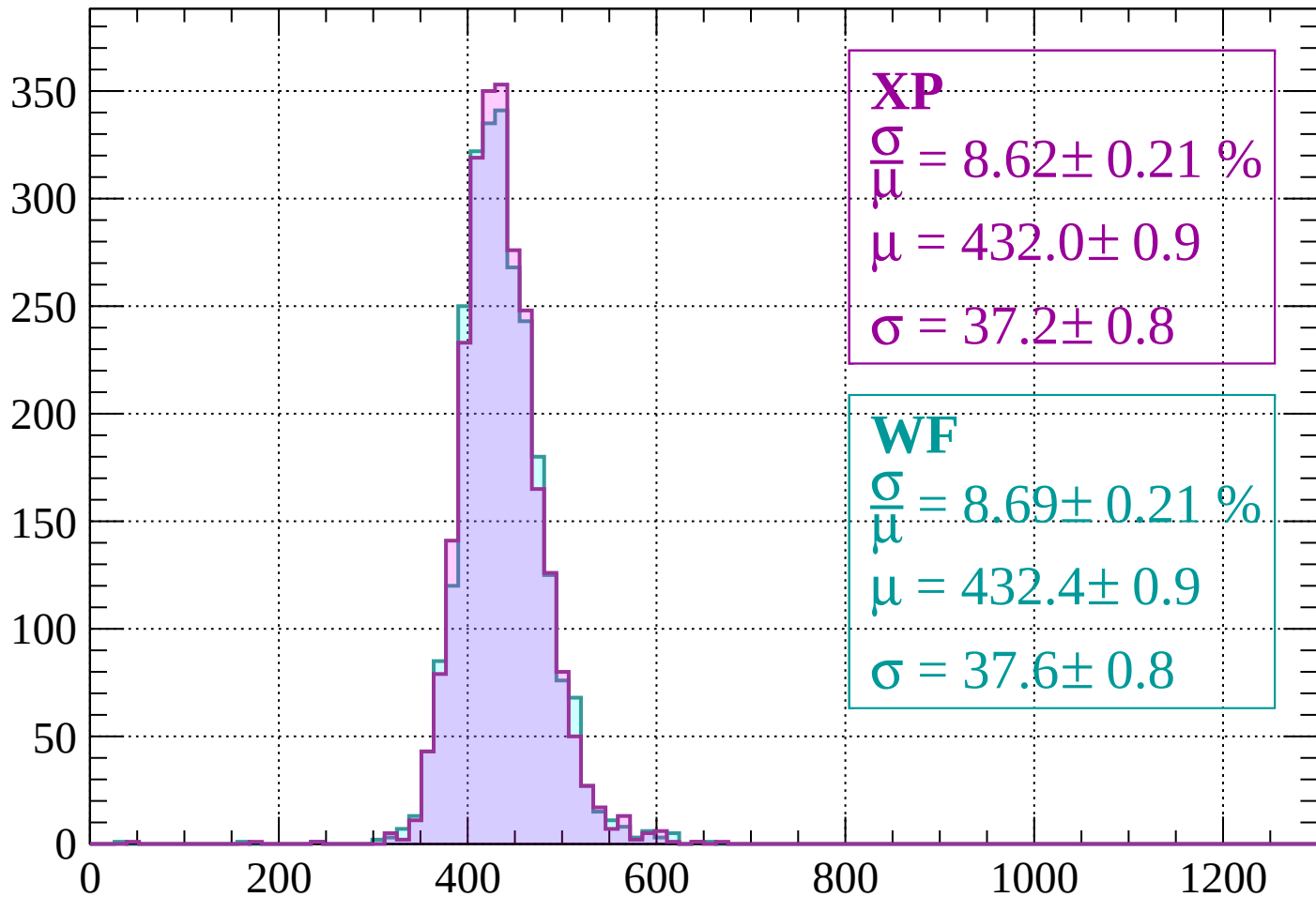
$$\mu = 432.7 \pm 0.9$$

$$\sigma = 37.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.21 \%$$

$$\mu = 432.0 \pm 0.9$$

$$\sigma = 37.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.69 \pm 0.21 \%$$

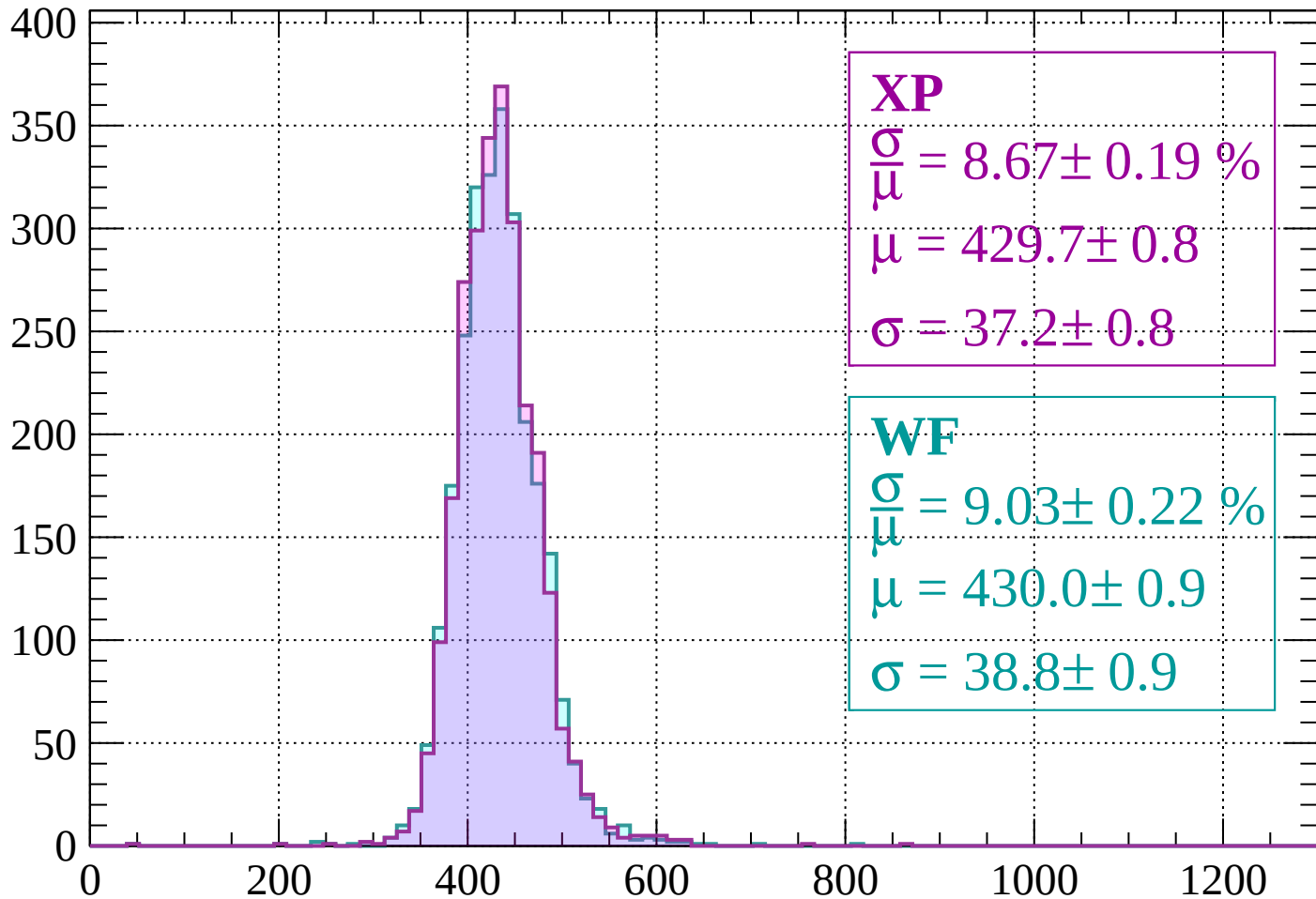
$$\mu = 432.4 \pm 0.9$$

$$\sigma = 37.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 8.67 \pm 0.19 \%$$

$$\mu = 429.7 \pm 0.8$$

$$\sigma = 37.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.22 \%$$

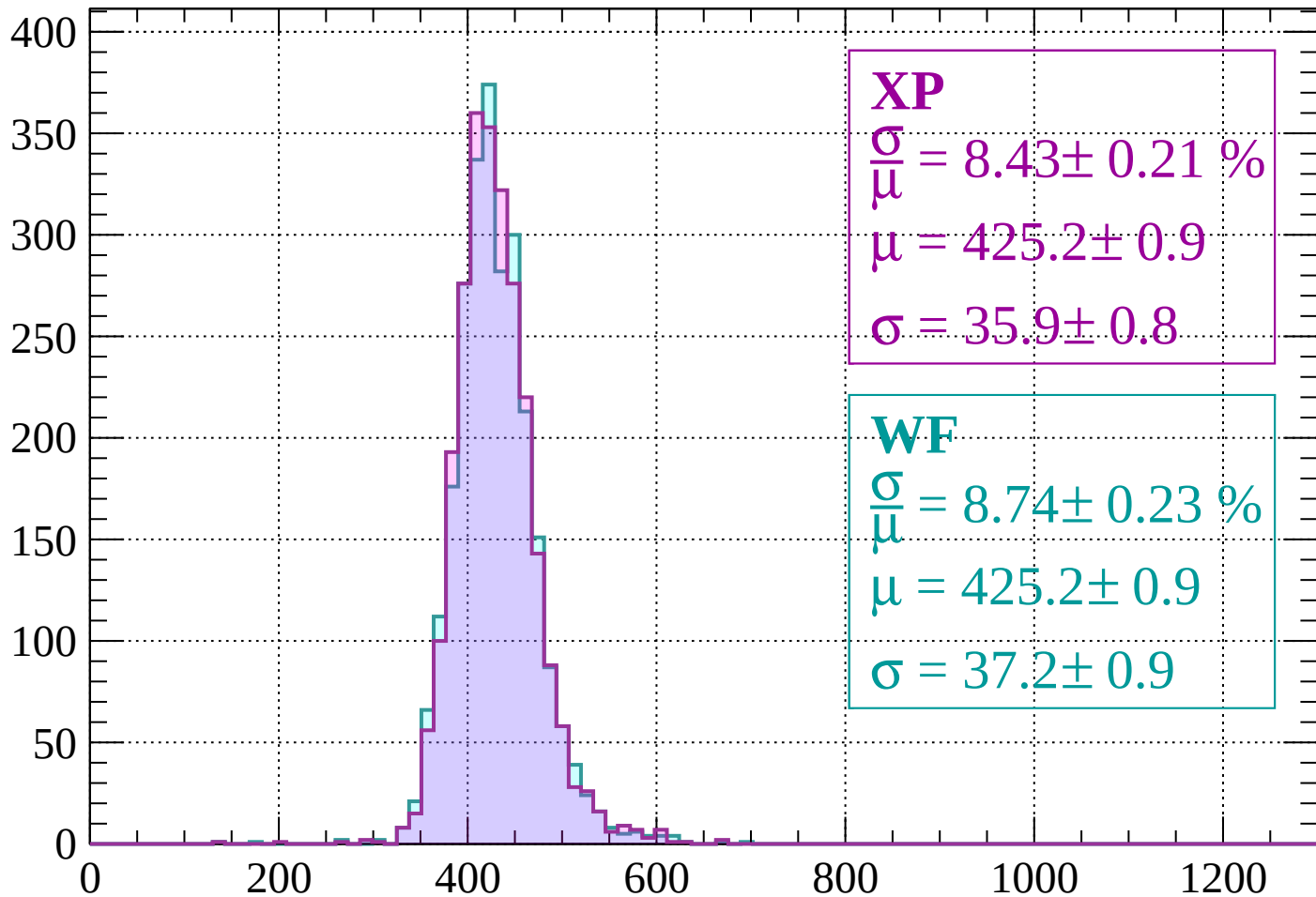
$$\mu = 430.0 \pm 0.9$$

$$\sigma = 38.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.43 \pm 0.21 \%$$

$$\mu = 425.2 \pm 0.9$$

$$\sigma = 35.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.23 \%$$

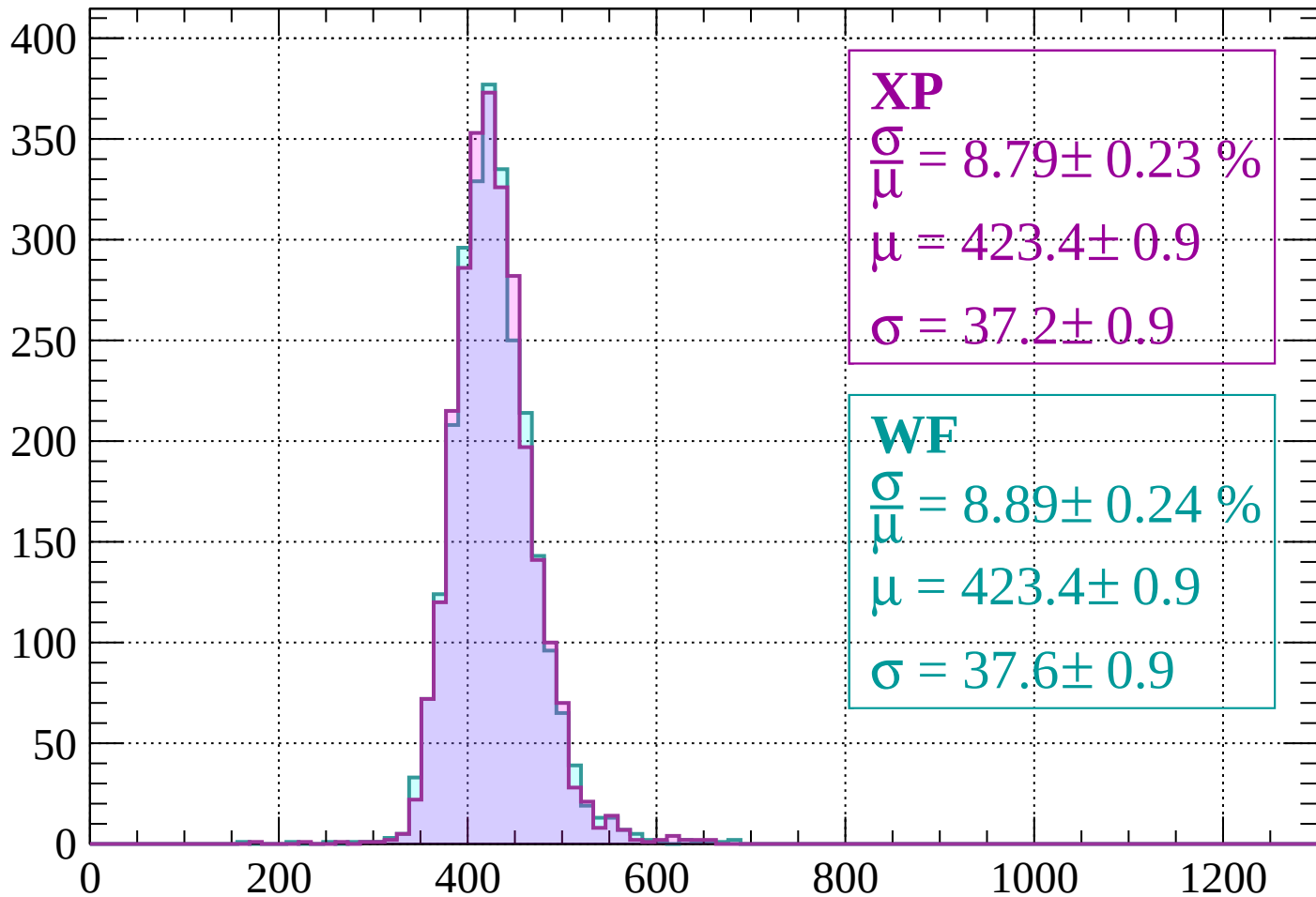
$$\mu = 425.2 \pm 0.9$$

$$\sigma = 37.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.23 \%$$

$$\mu = 423.4 \pm 0.9$$

$$\sigma = 37.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.89 \pm 0.24 \%$$

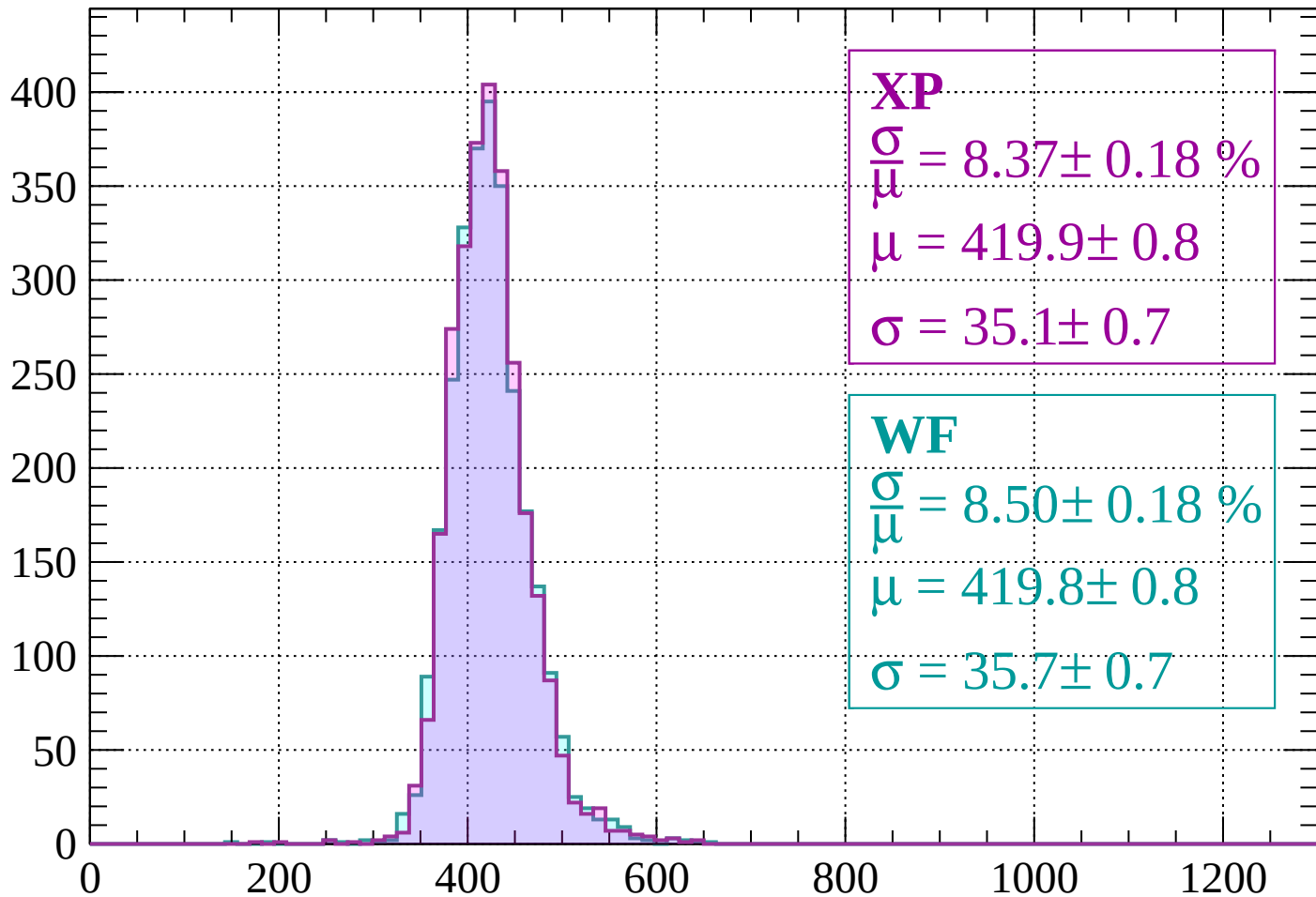
$$\mu = 423.4 \pm 0.9$$

$$\sigma = 37.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.18 \%$$

$$\mu = 419.9 \pm 0.8$$

$$\sigma = 35.1 \pm 0.7$$

WF

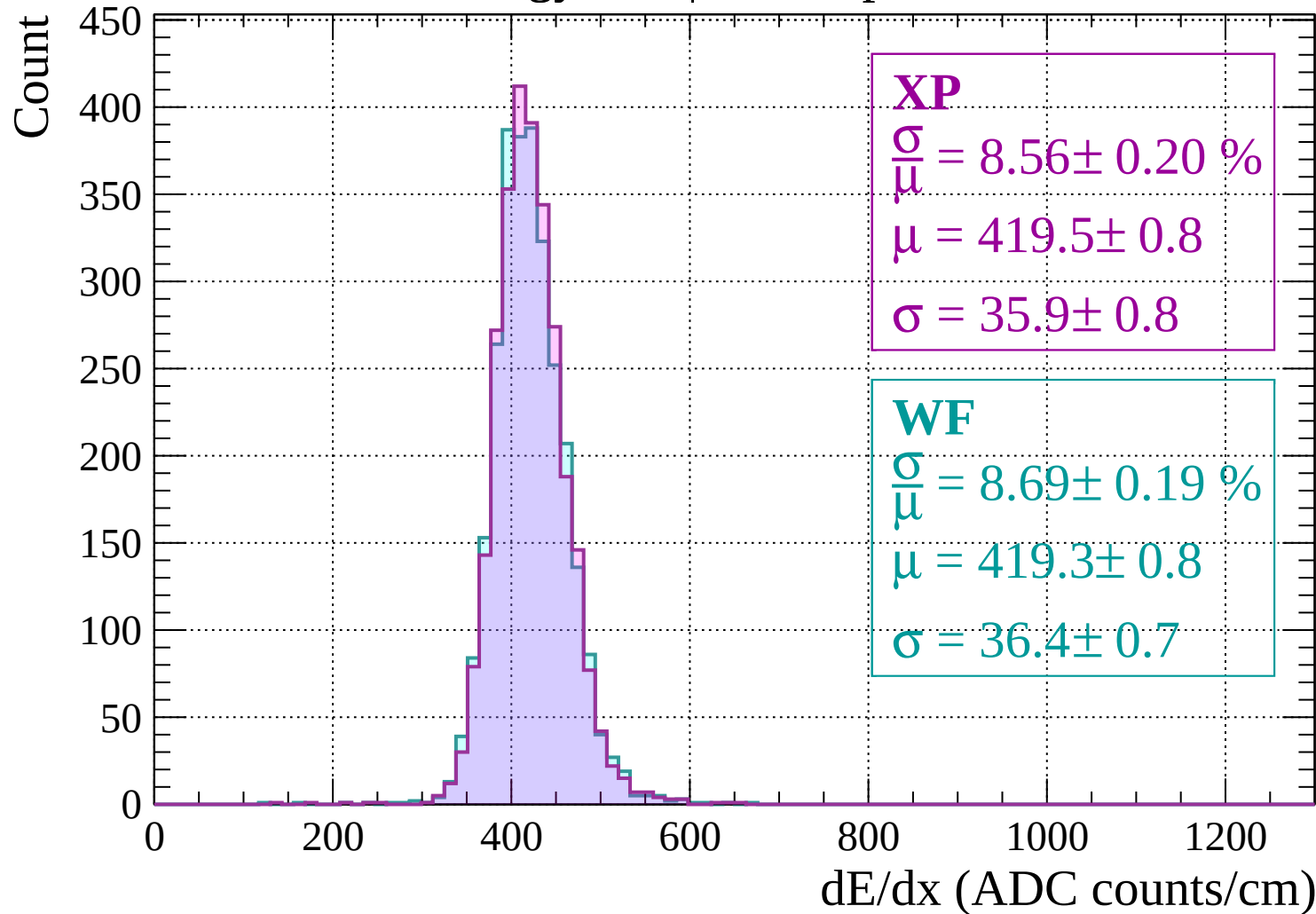
$$\frac{\sigma}{\mu} = 8.50 \pm 0.18 \%$$

$$\mu = 419.8 \pm 0.8$$

$$\sigma = 35.7 \pm 0.7$$

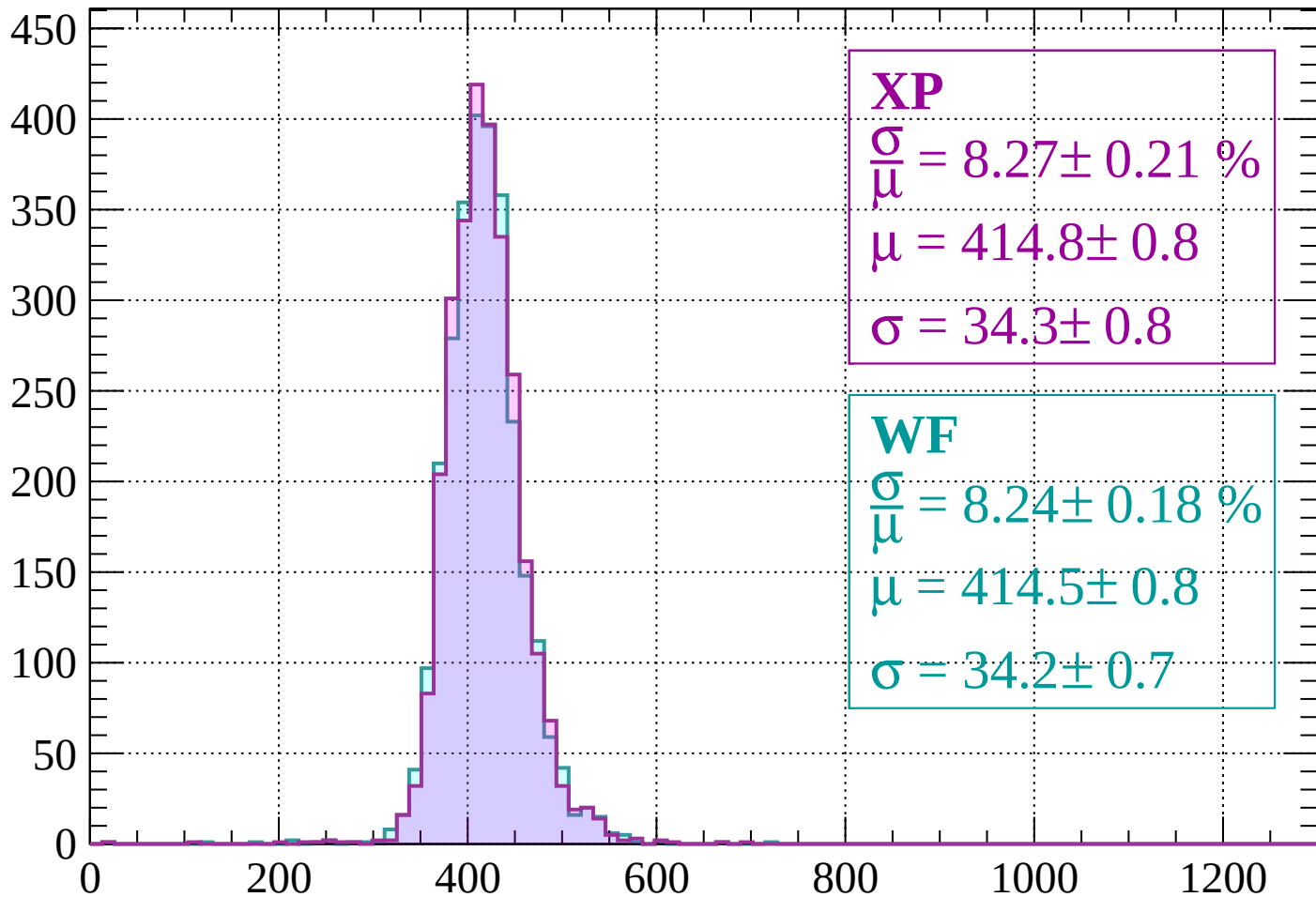
dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$



Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.27 \pm 0.21 \%$$

$$\mu = 414.8 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.18 \%$$

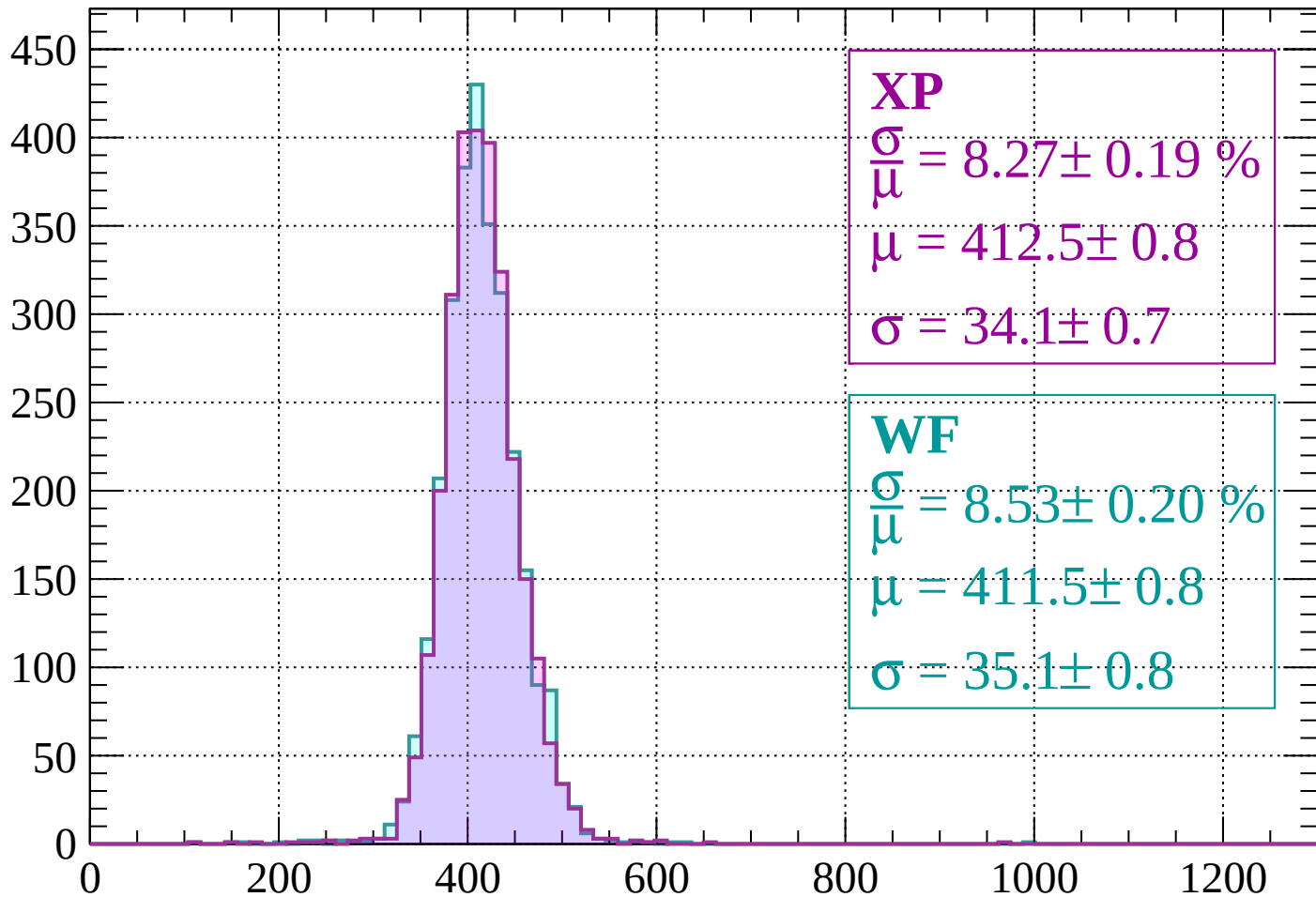
$$\mu = 414.5 \pm 0.8$$

$$\sigma = 34.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.27 \pm 0.19 \%$$

$$\mu = 412.5 \pm 0.8$$

$$\sigma = 34.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.53 \pm 0.20 \%$$

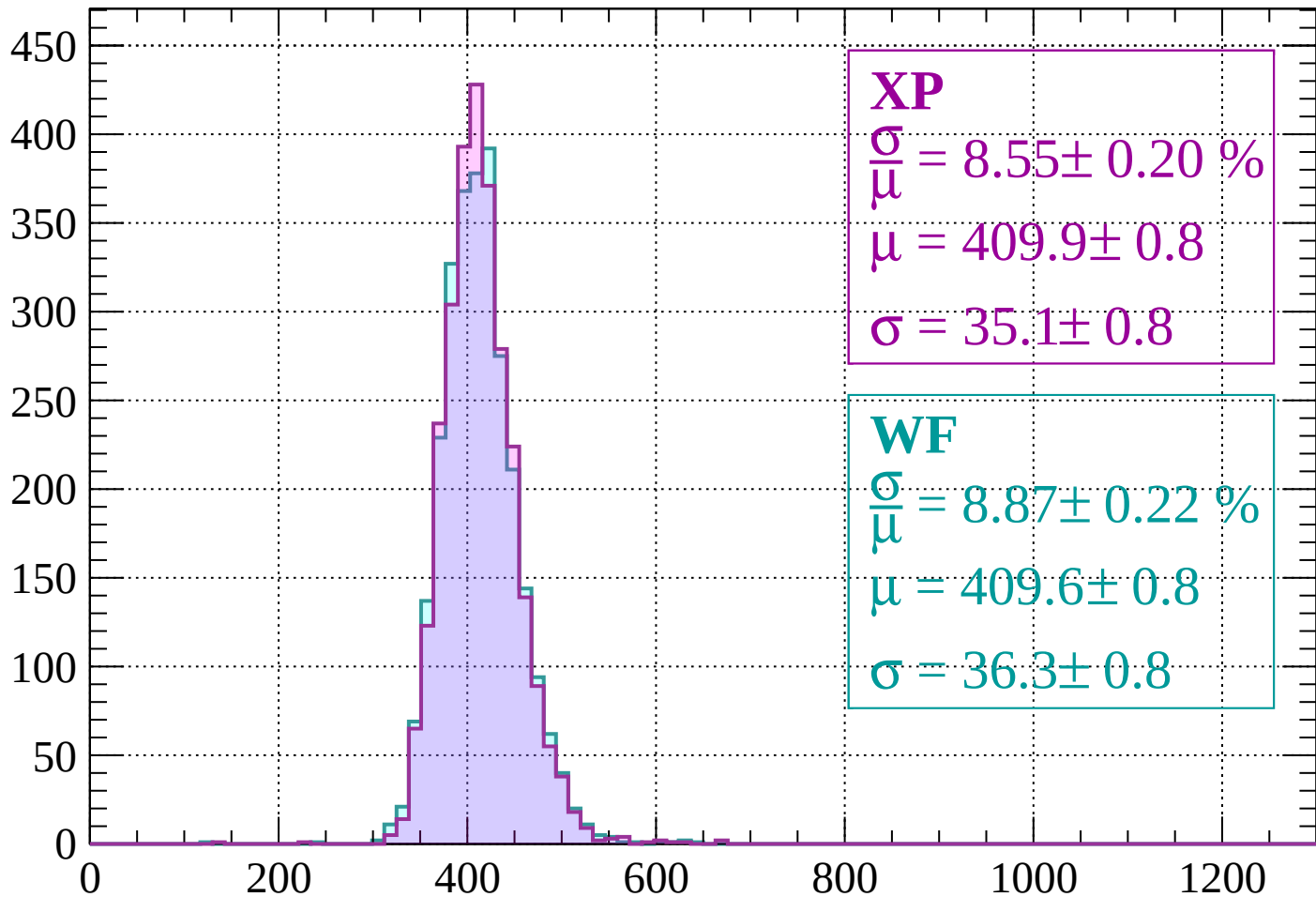
$$\mu = 411.5 \pm 0.8$$

$$\sigma = 35.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.20 \%$$

$$\mu = 409.9 \pm 0.8$$

$$\sigma = 35.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.22 \%$$

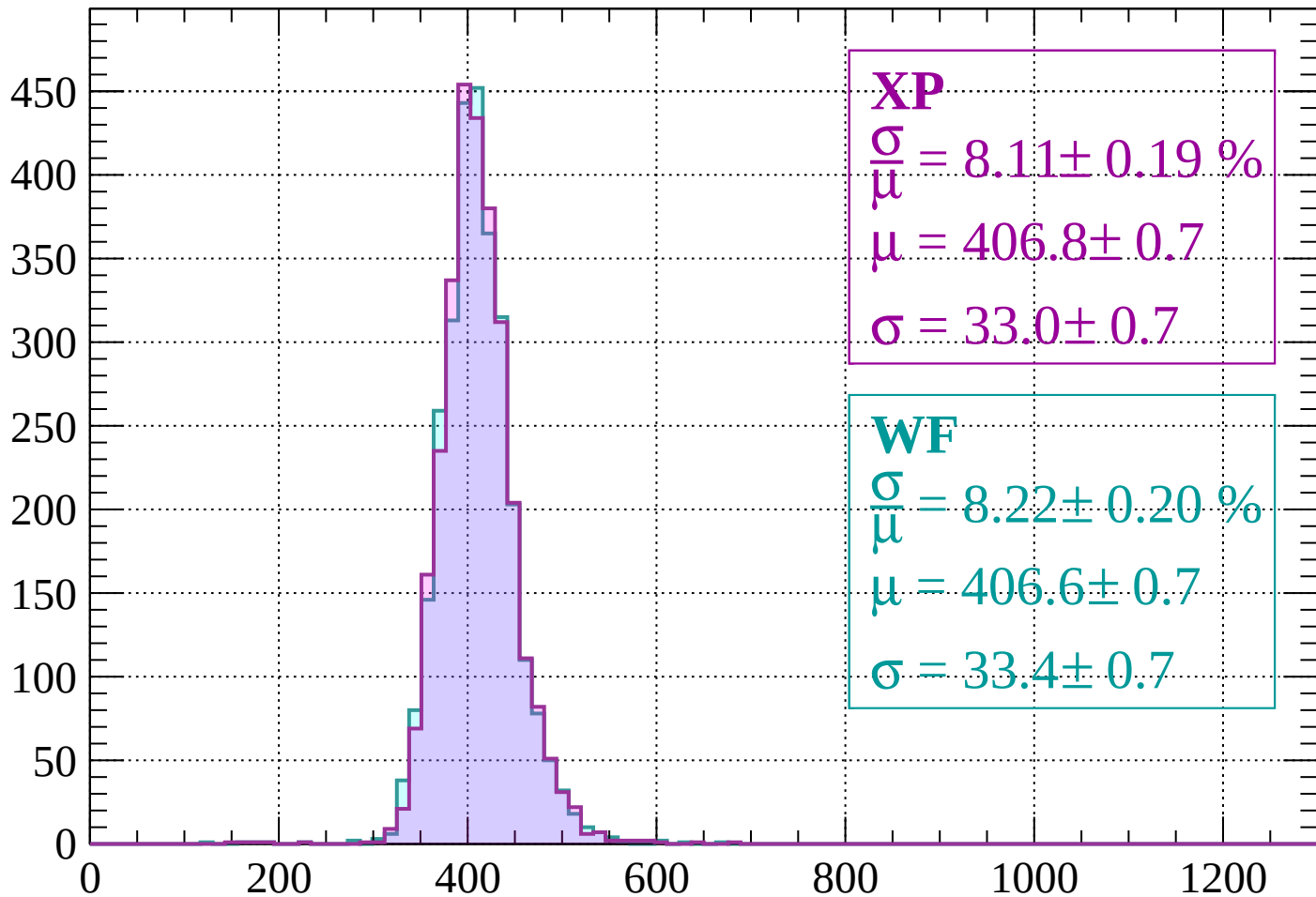
$$\mu = 409.6 \pm 0.8$$

$$\sigma = 36.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.11 \pm 0.19 \%$$

$$\mu = 406.8 \pm 0.7$$

$$\sigma = 33.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.20 \%$$

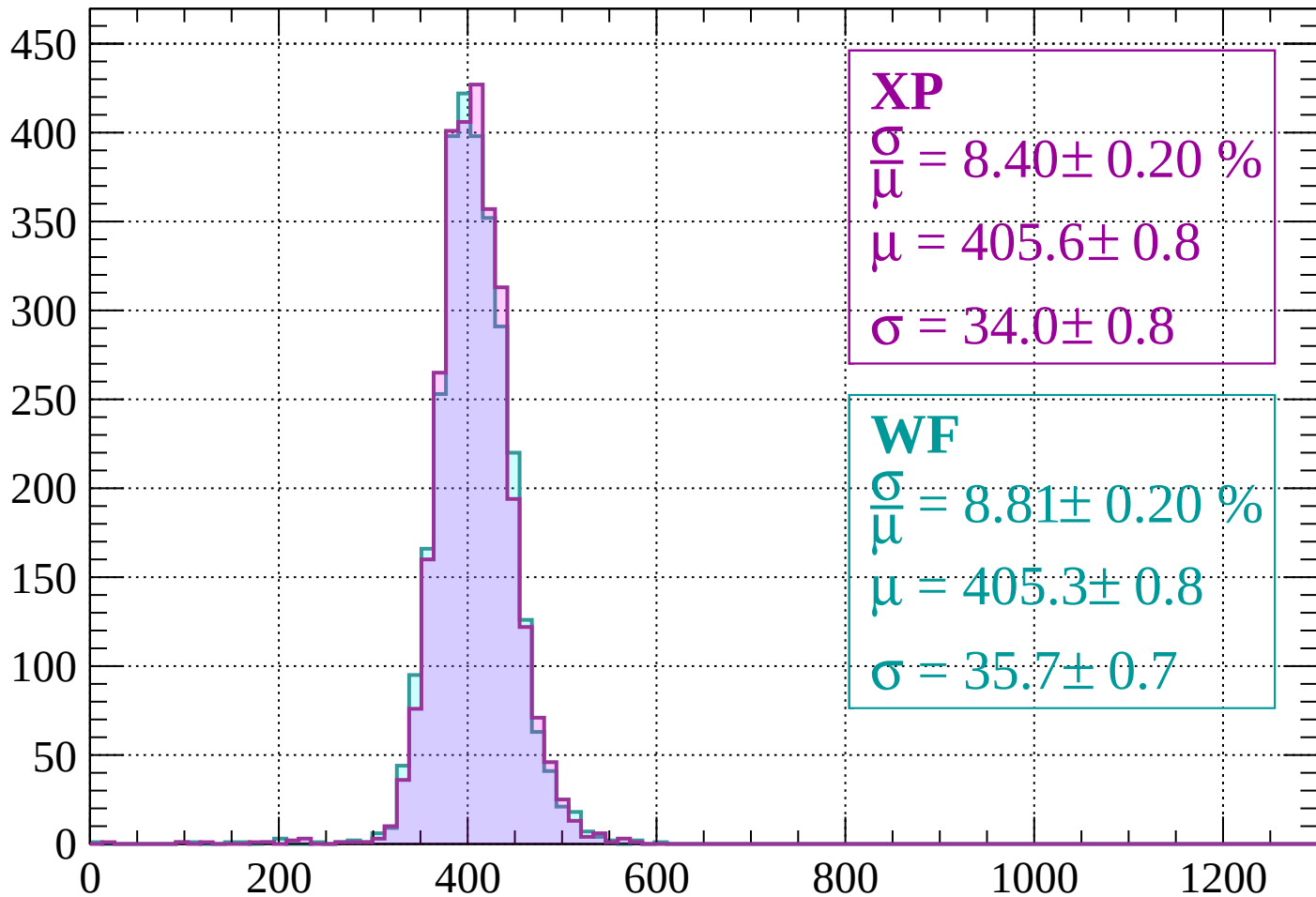
$$\mu = 406.6 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.20 \%$$

$$\mu = 405.6 \pm 0.8$$

$$\sigma = 34.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.20 \%$$

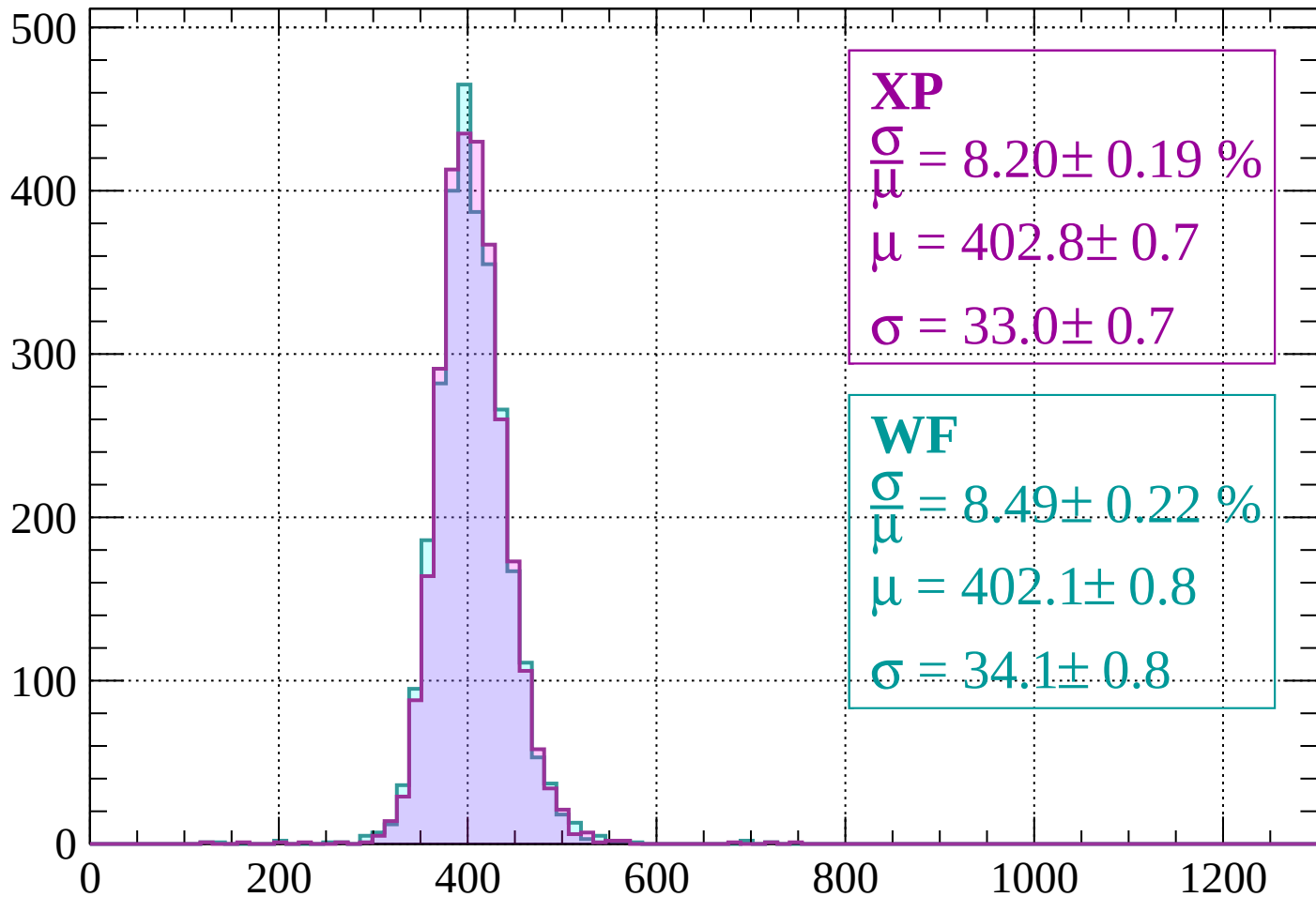
$$\mu = 405.3 \pm 0.8$$

$$\sigma = 35.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.19 \%$$

$$\mu = 402.8 \pm 0.7$$

$$\sigma = 33.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.22 \%$$

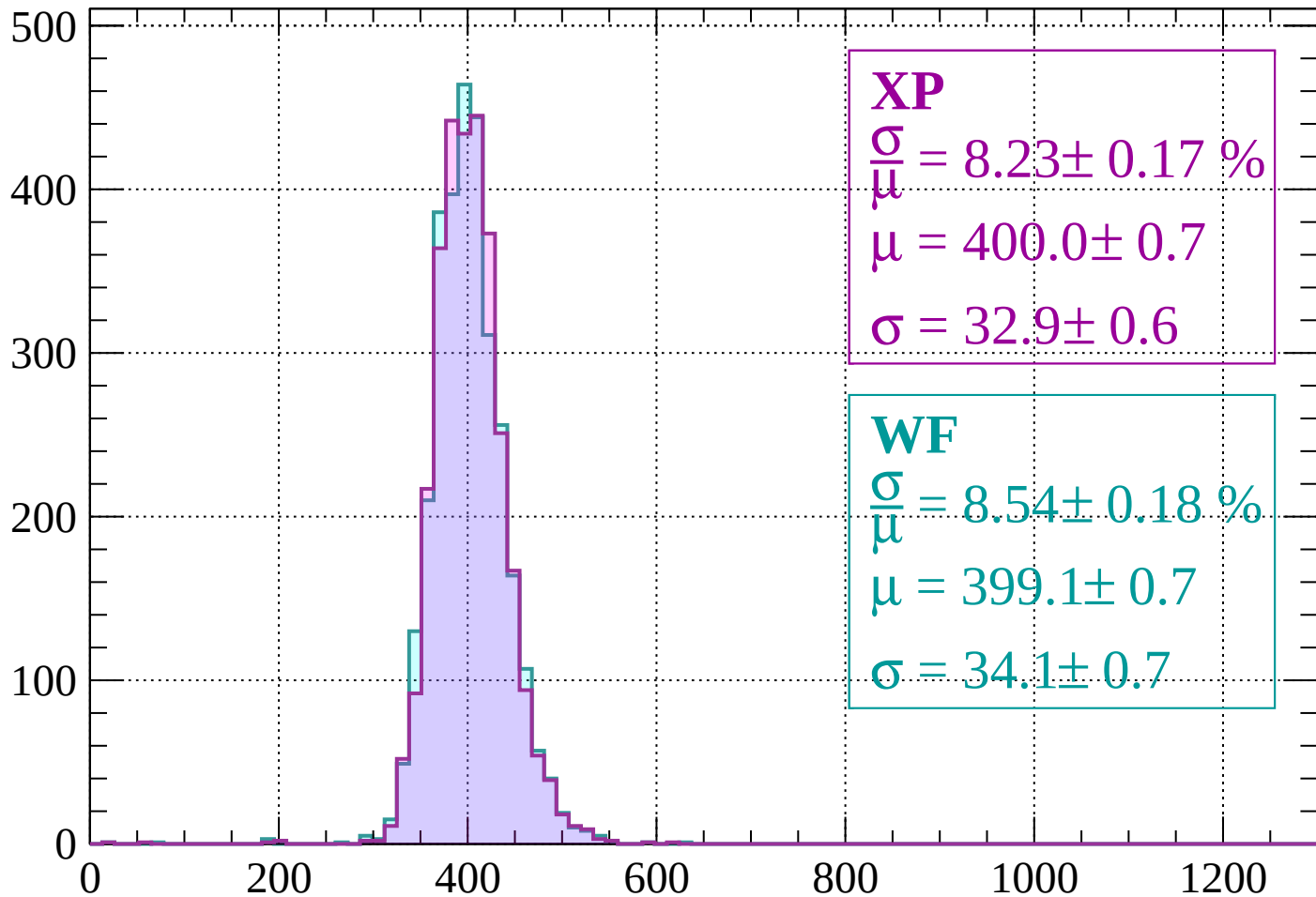
$$\mu = 402.1 \pm 0.8$$

$$\sigma = 34.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.17 \%$$

$$\mu = 400.0 \pm 0.7$$

$$\sigma = 32.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.18 \%$$

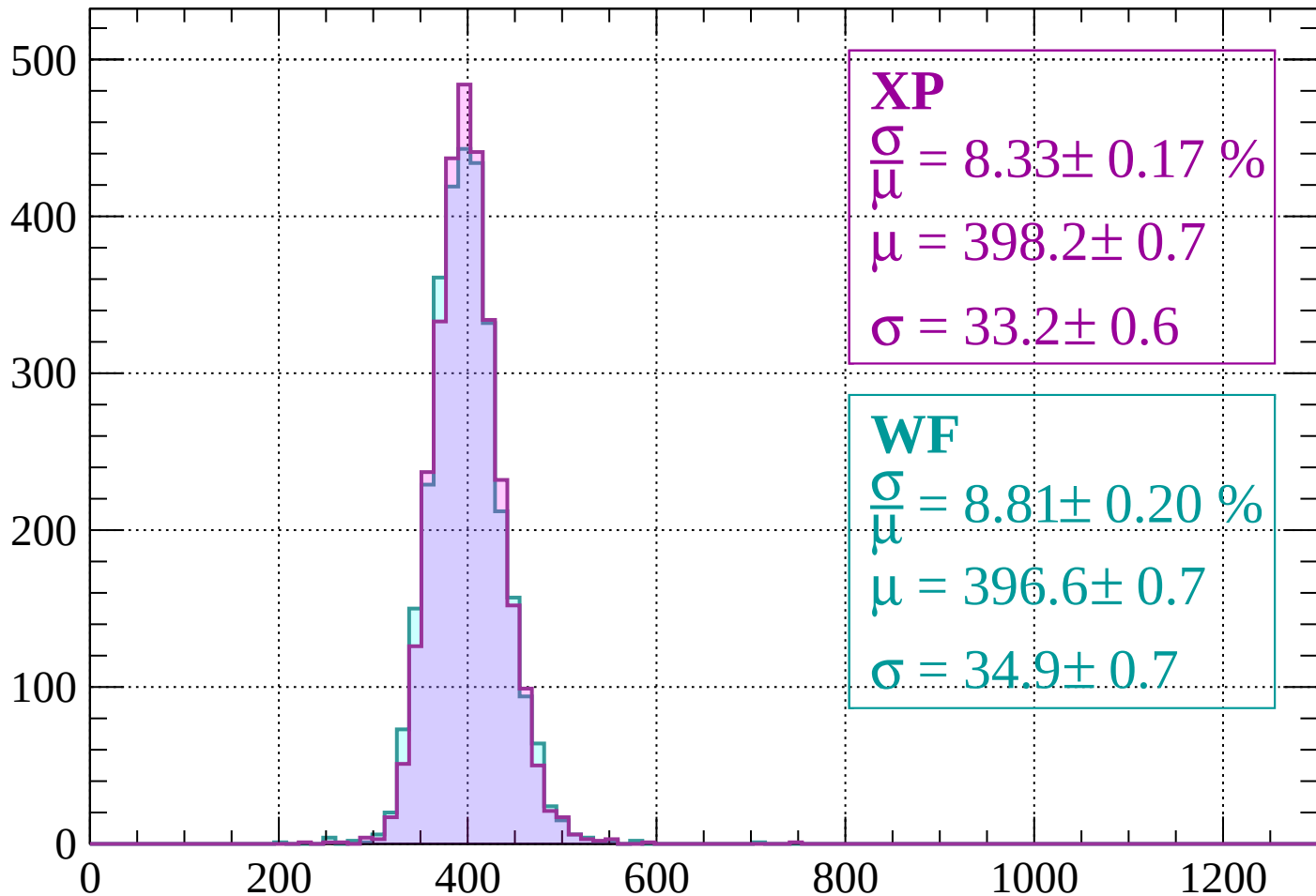
$$\mu = 399.1 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 8.33 \pm 0.17 \%$$

$$\mu = 398.2 \pm 0.7$$

$$\sigma = 33.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.20 \%$$

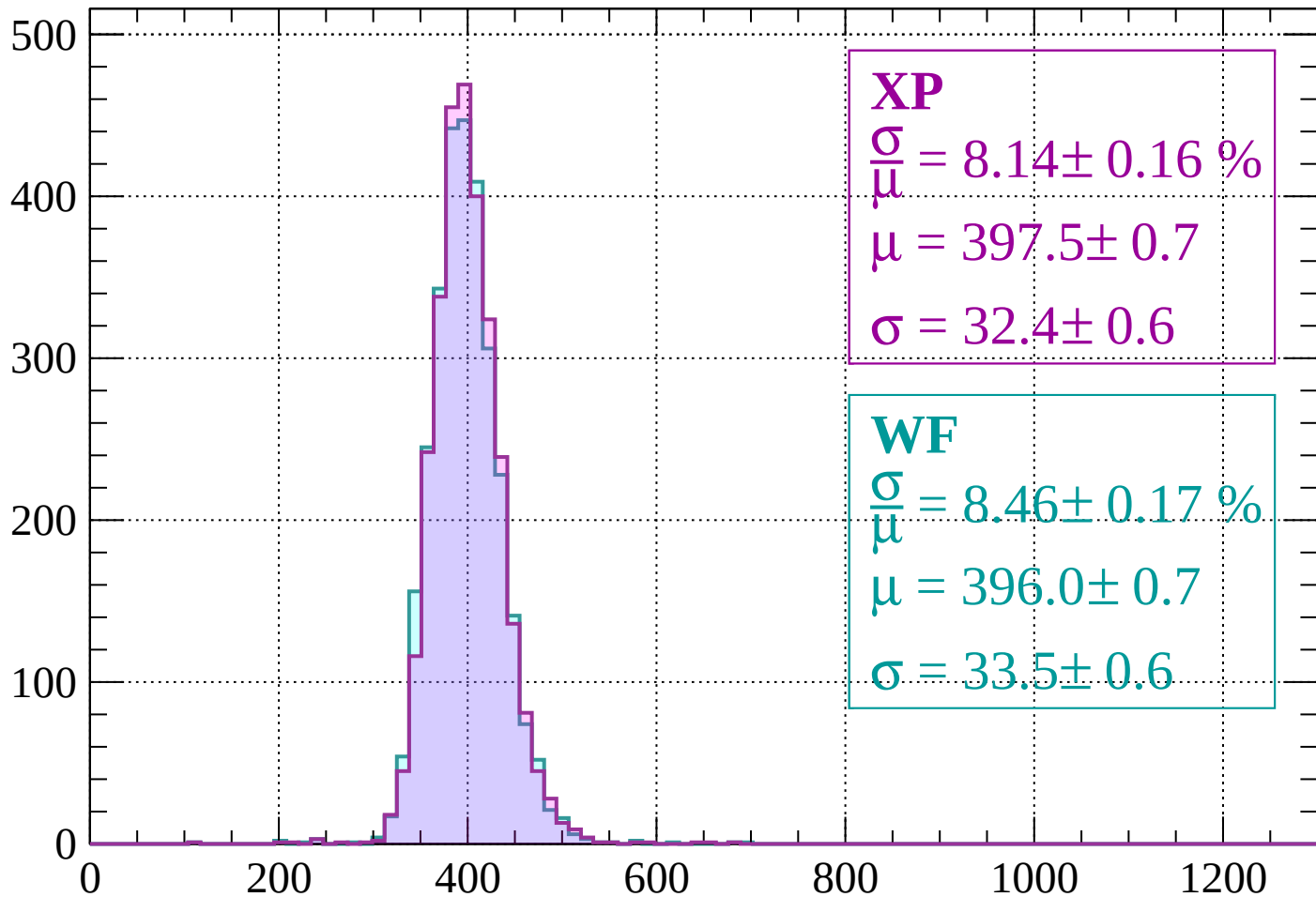
$$\mu = 396.6 \pm 0.7$$

$$\sigma = 34.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.16 \%$$

$$\mu = 397.5 \pm 0.7$$

$$\sigma = 32.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

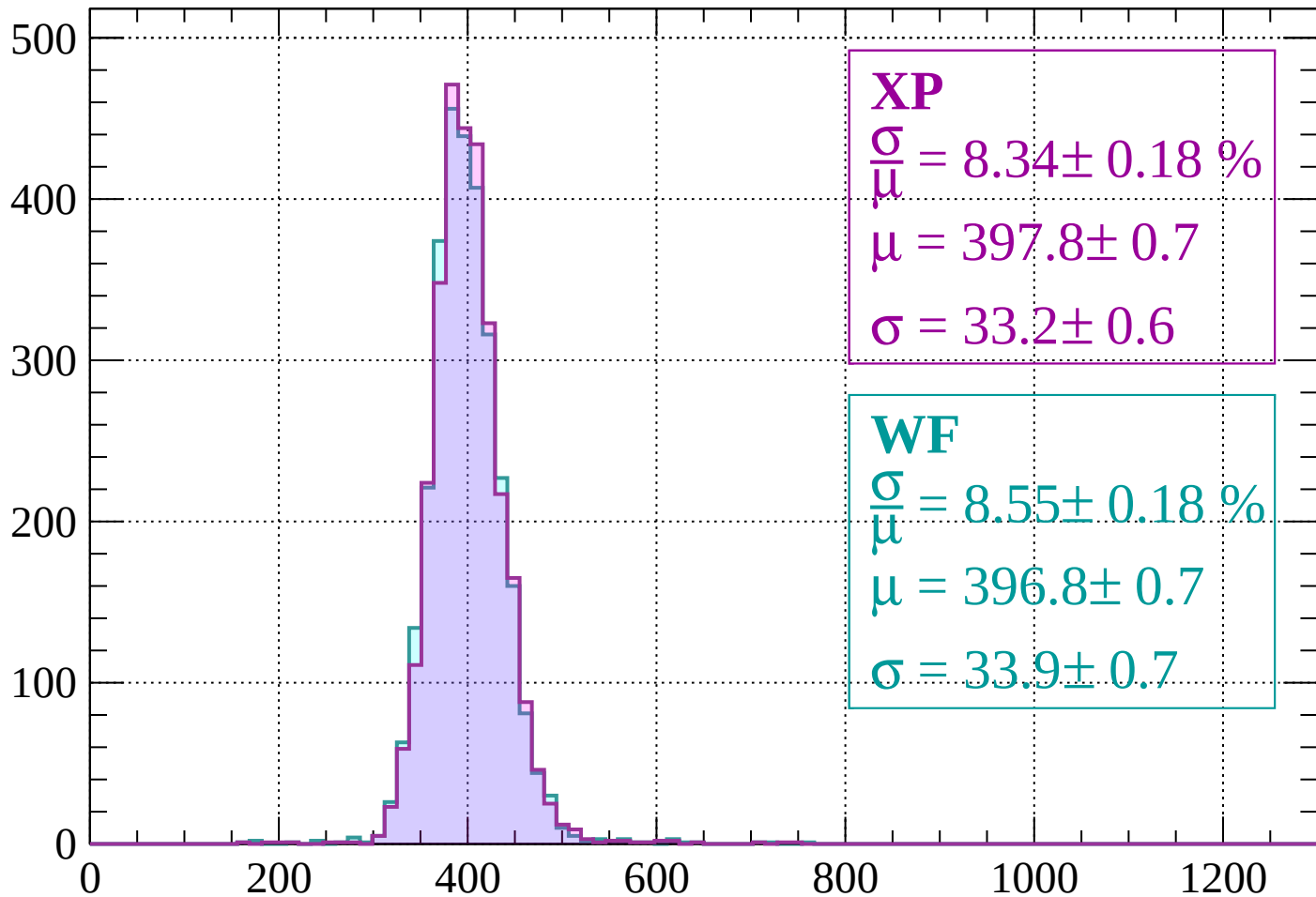
$$\mu = 396.0 \pm 0.7$$

$$\sigma = 33.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.18 \%$$

$$\mu = 397.8 \pm 0.7$$

$$\sigma = 33.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.18 \%$$

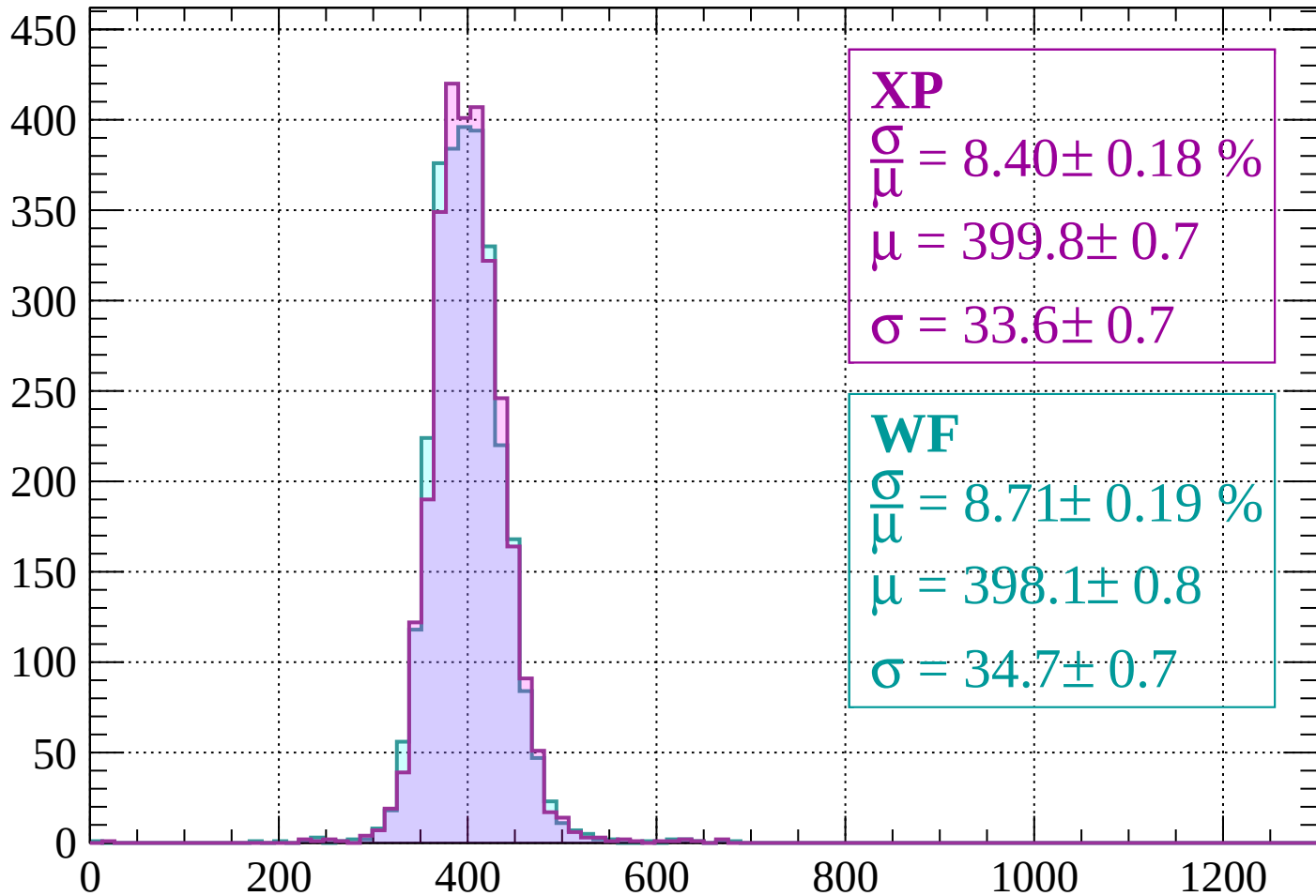
$$\mu = 396.8 \pm 0.7$$

$$\sigma = 33.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.18 \%$$

$$\mu = 399.8 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 0.19 \%$$

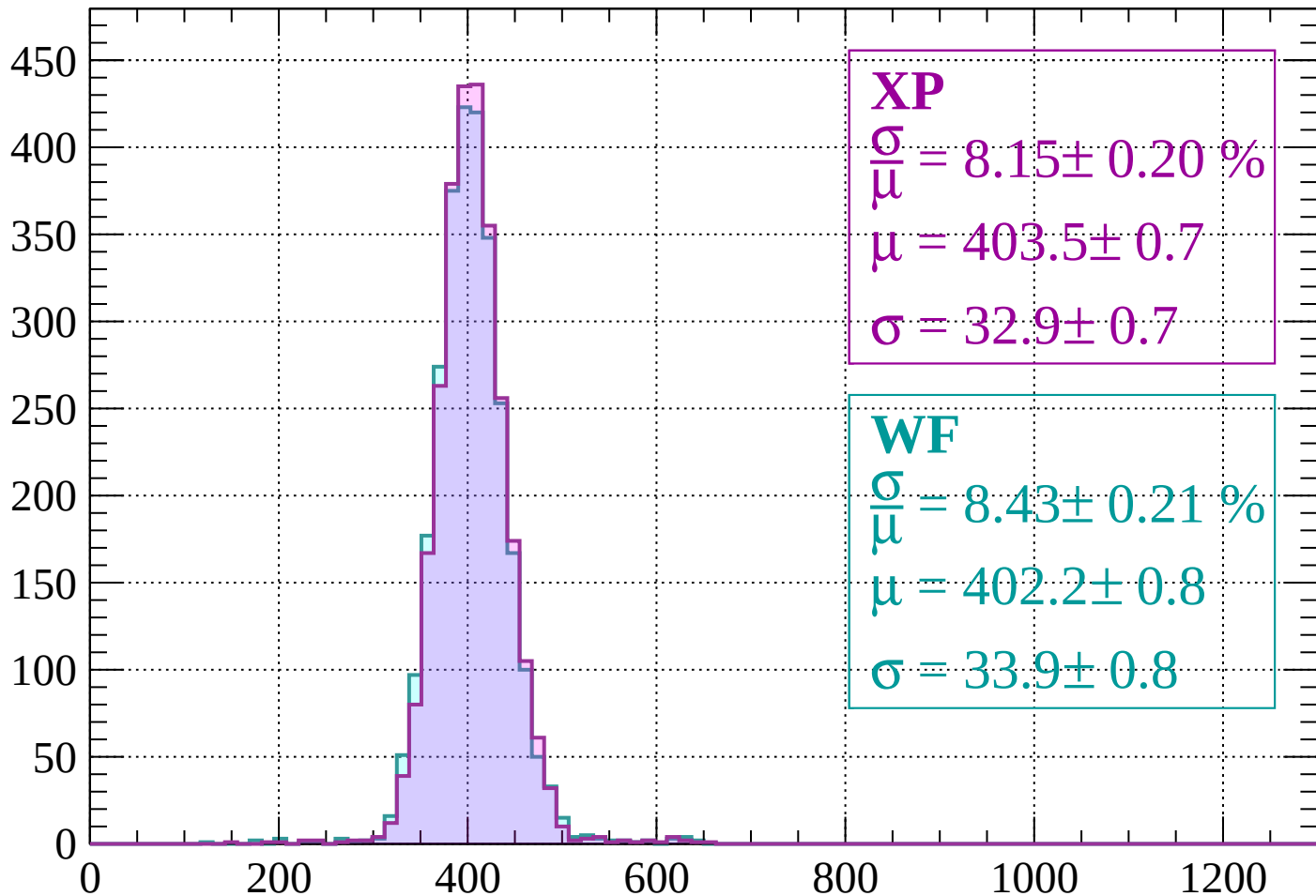
$$\mu = 398.1 \pm 0.8$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.20 \%$$

$$\mu = 403.5 \pm 0.7$$

$$\sigma = 32.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.21 \%$$

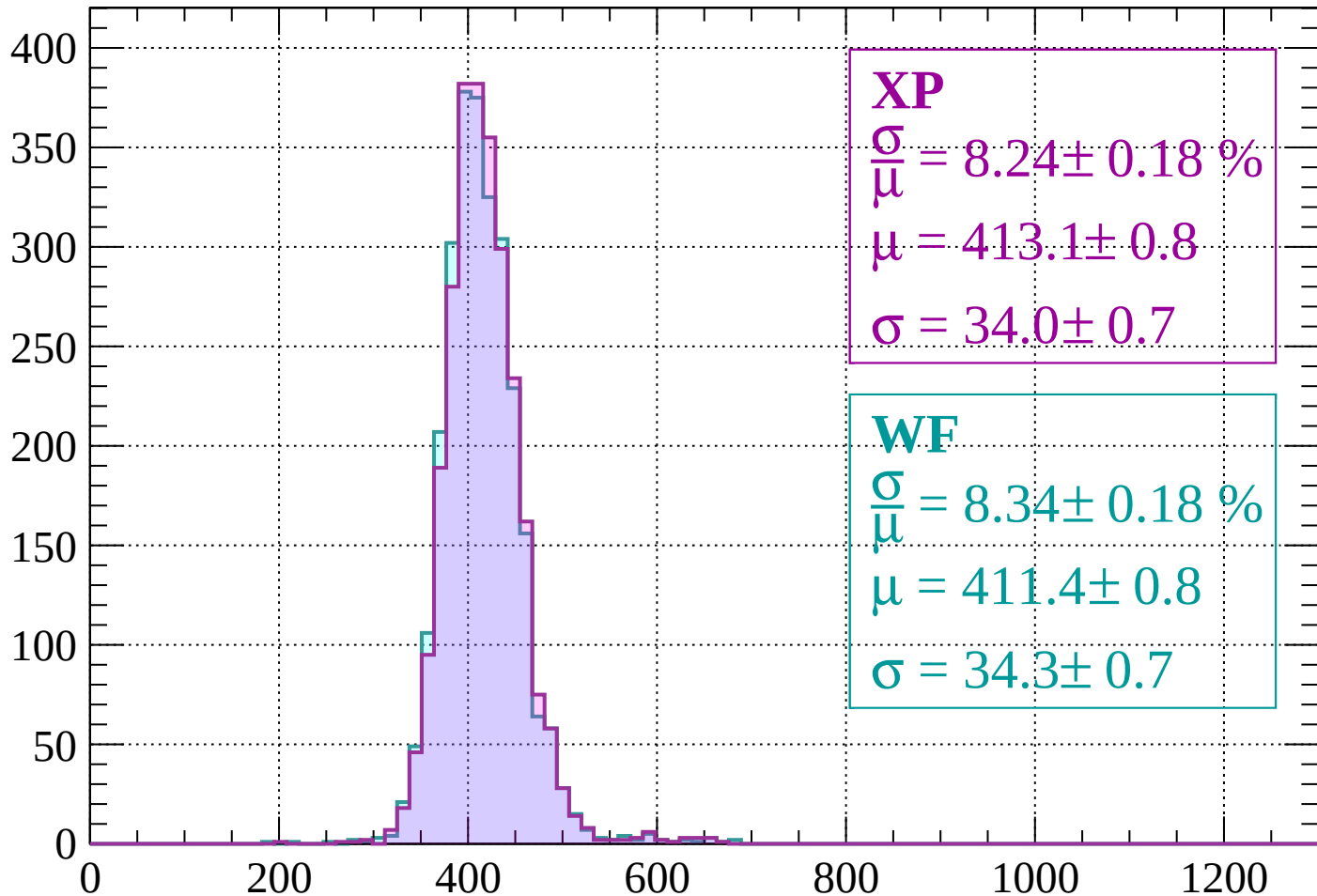
$$\mu = 402.2 \pm 0.8$$

$$\sigma = 33.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 8.24 \pm 0.18 \%$$

$$\mu = 413.1 \pm 0.8$$

$$\sigma = 34.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.34 \pm 0.18 \%$$

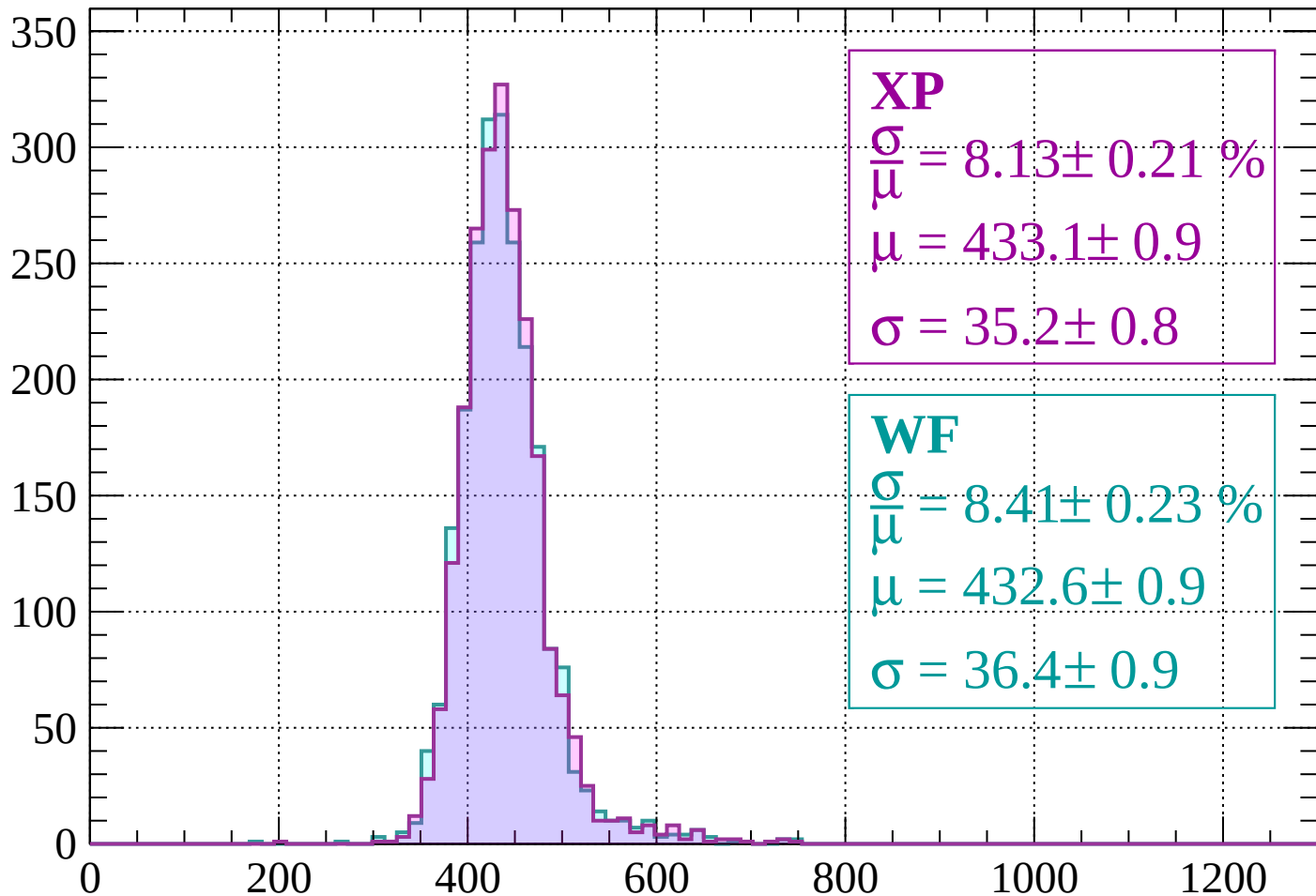
$$\mu = 411.4 \pm 0.8$$

$$\sigma = 34.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.21 \%$$

$$\mu = 433.1 \pm 0.9$$

$$\sigma = 35.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.23 \%$$

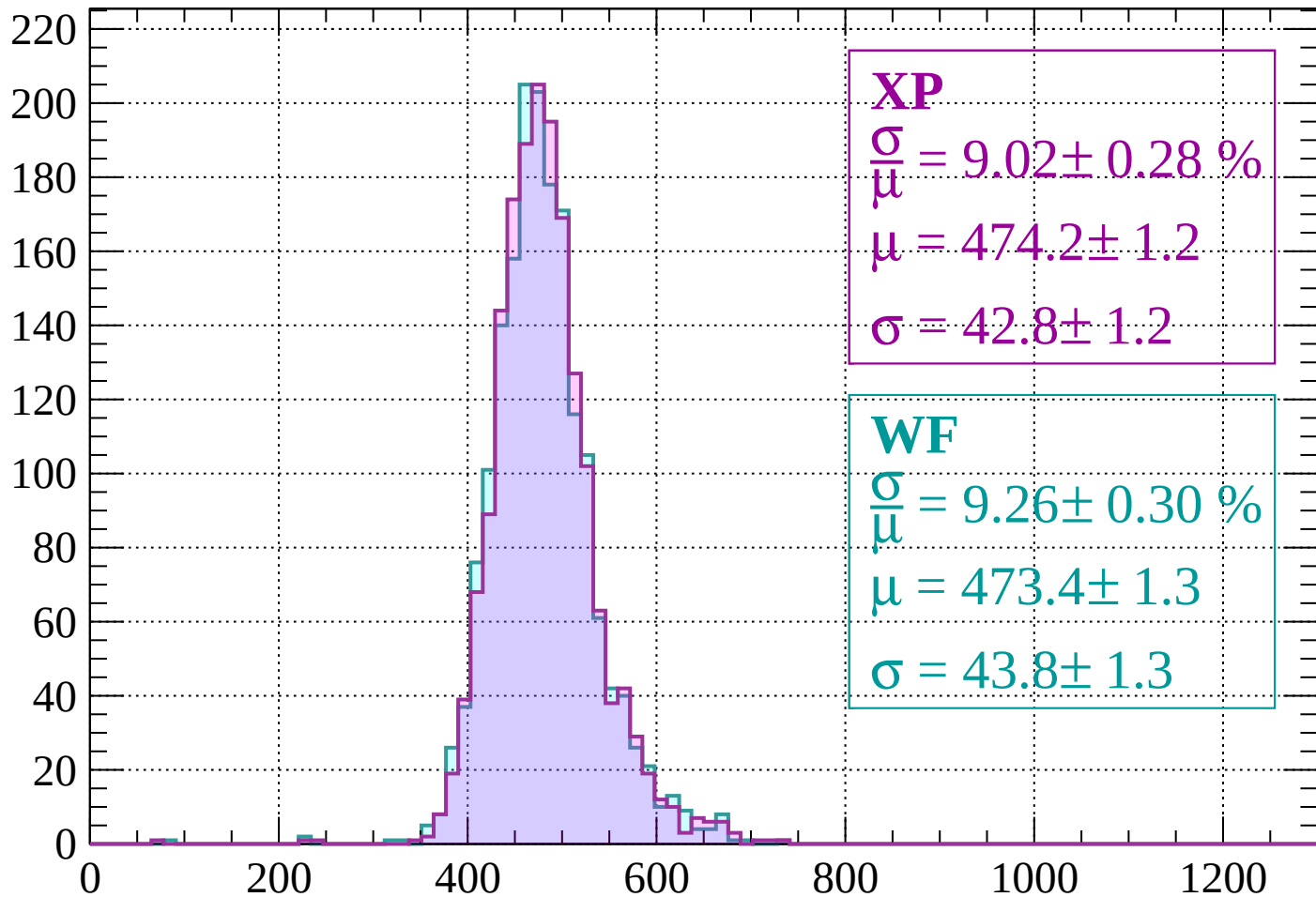
$$\mu = 432.6 \pm 0.9$$

$$\sigma = 36.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.28 \%$$

$$\mu = 474.2 \pm 1.2$$

$$\sigma = 42.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.26 \pm 0.30 \%$$

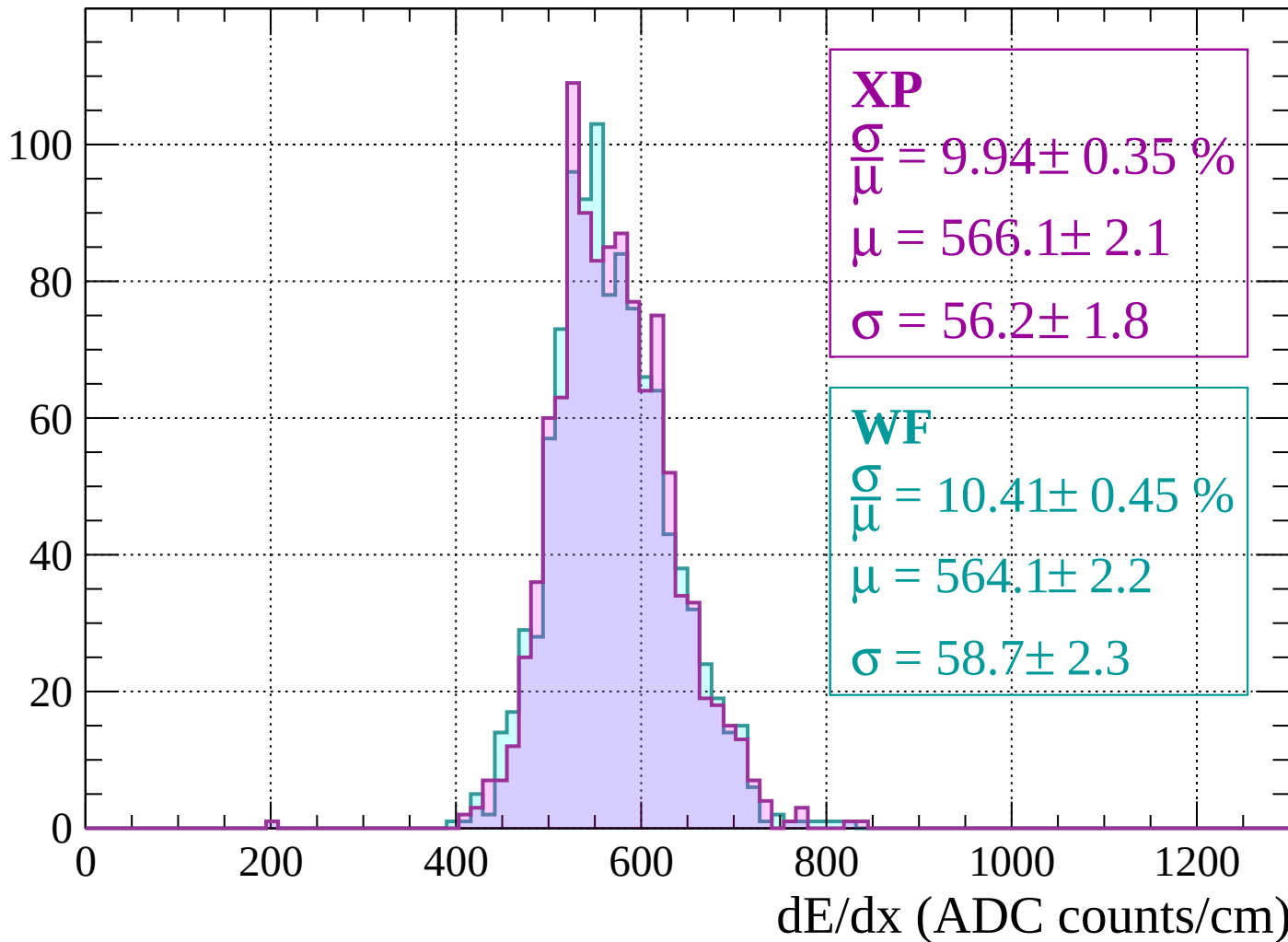
$$\mu = 473.4 \pm 1.3$$

$$\sigma = 43.8 \pm 1.3$$

dE/dx (ADC counts/cm)

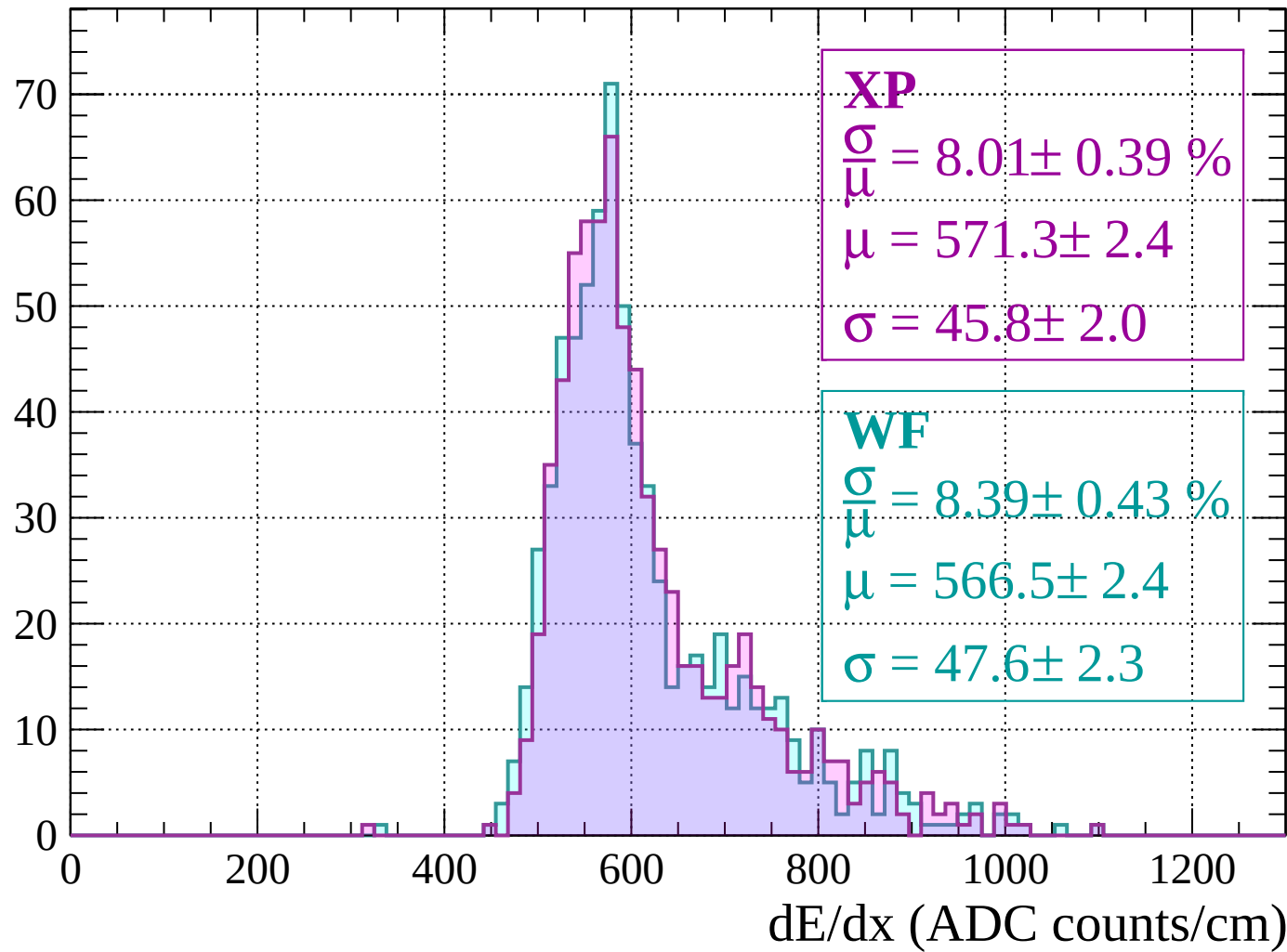
Energy loss | $-120 < p < -80$

Count



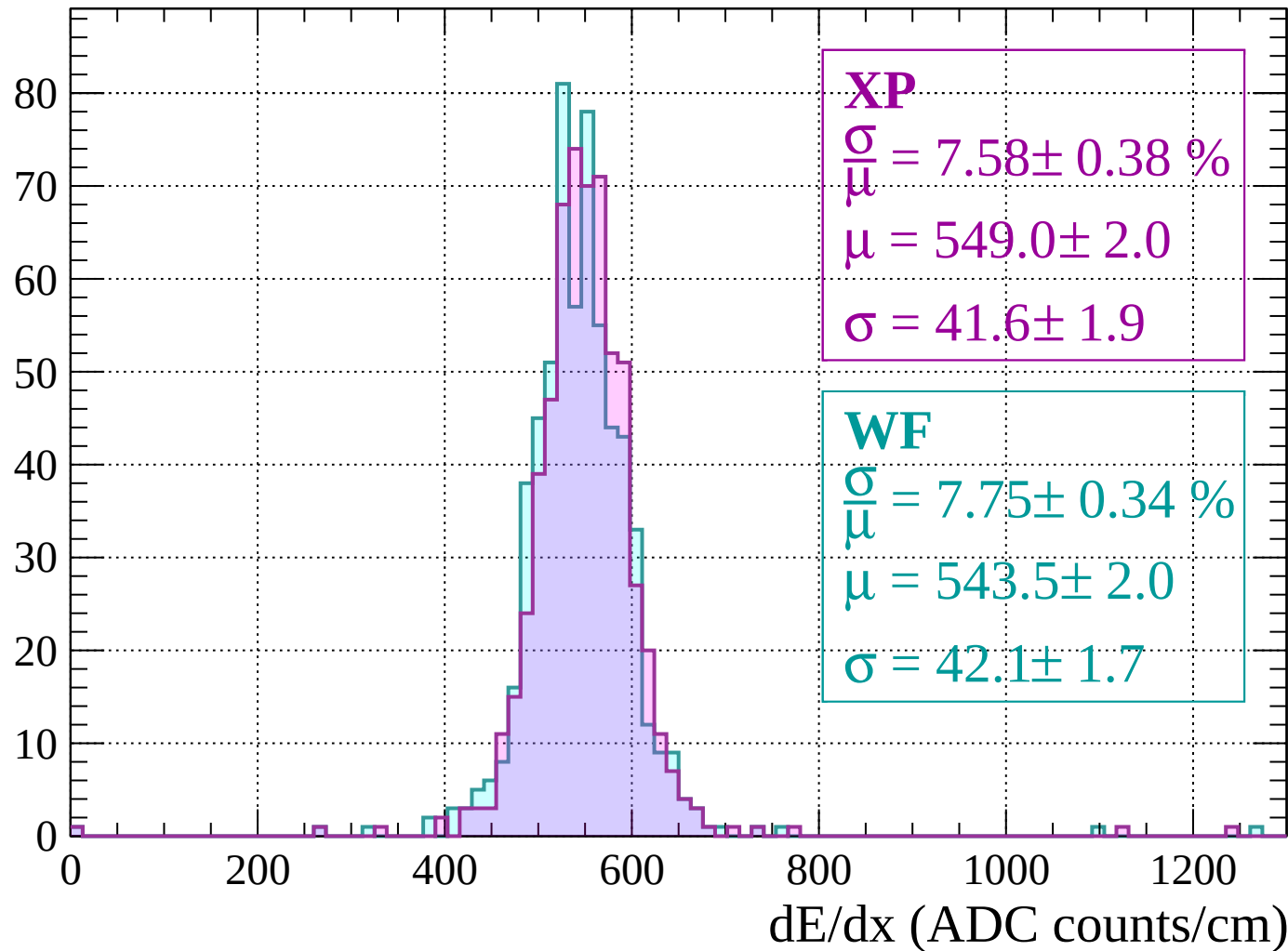
Energy loss | $-80 < p < -40$

Count



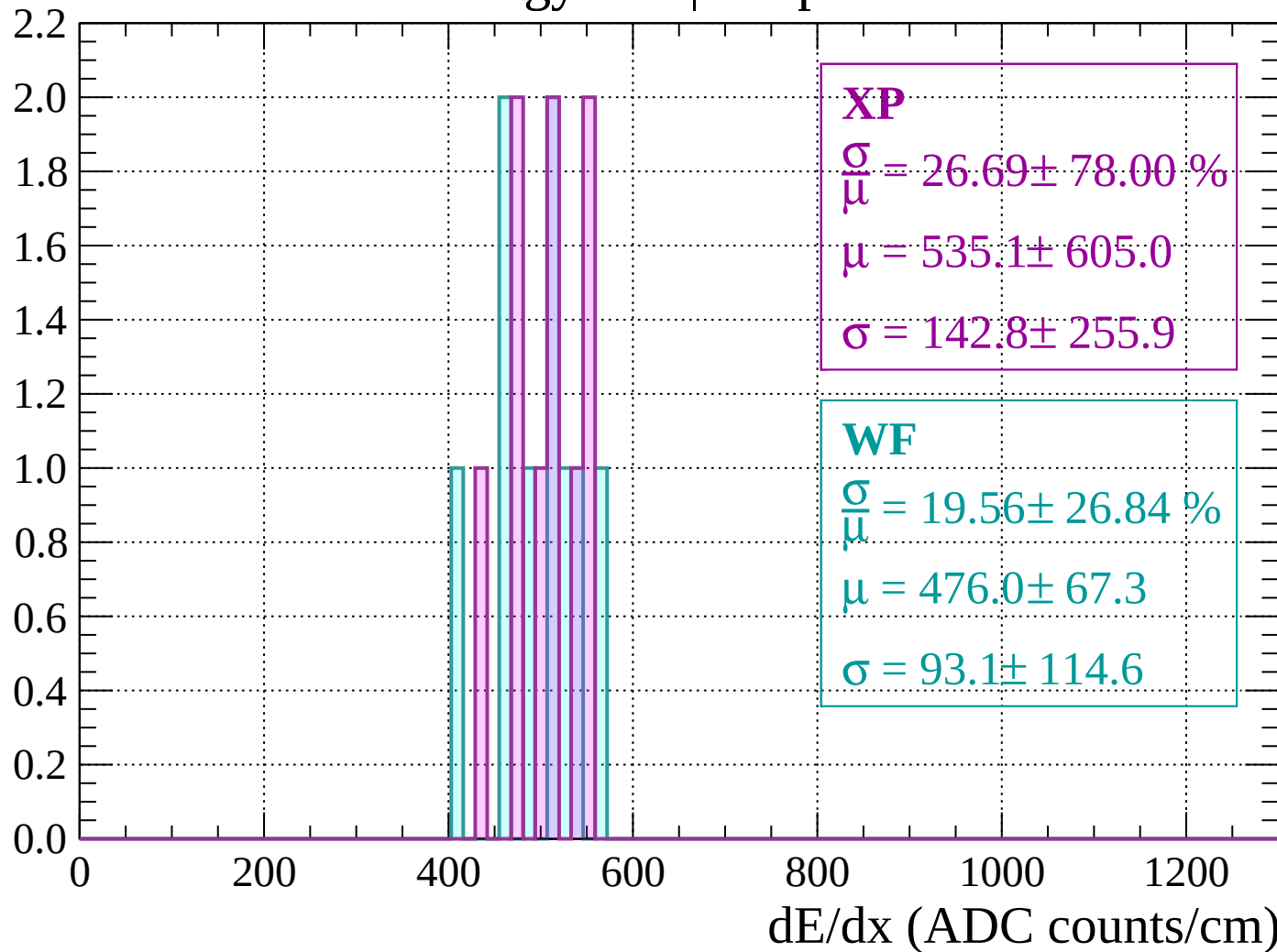
Energy loss | $-40 < p < 0$

Count



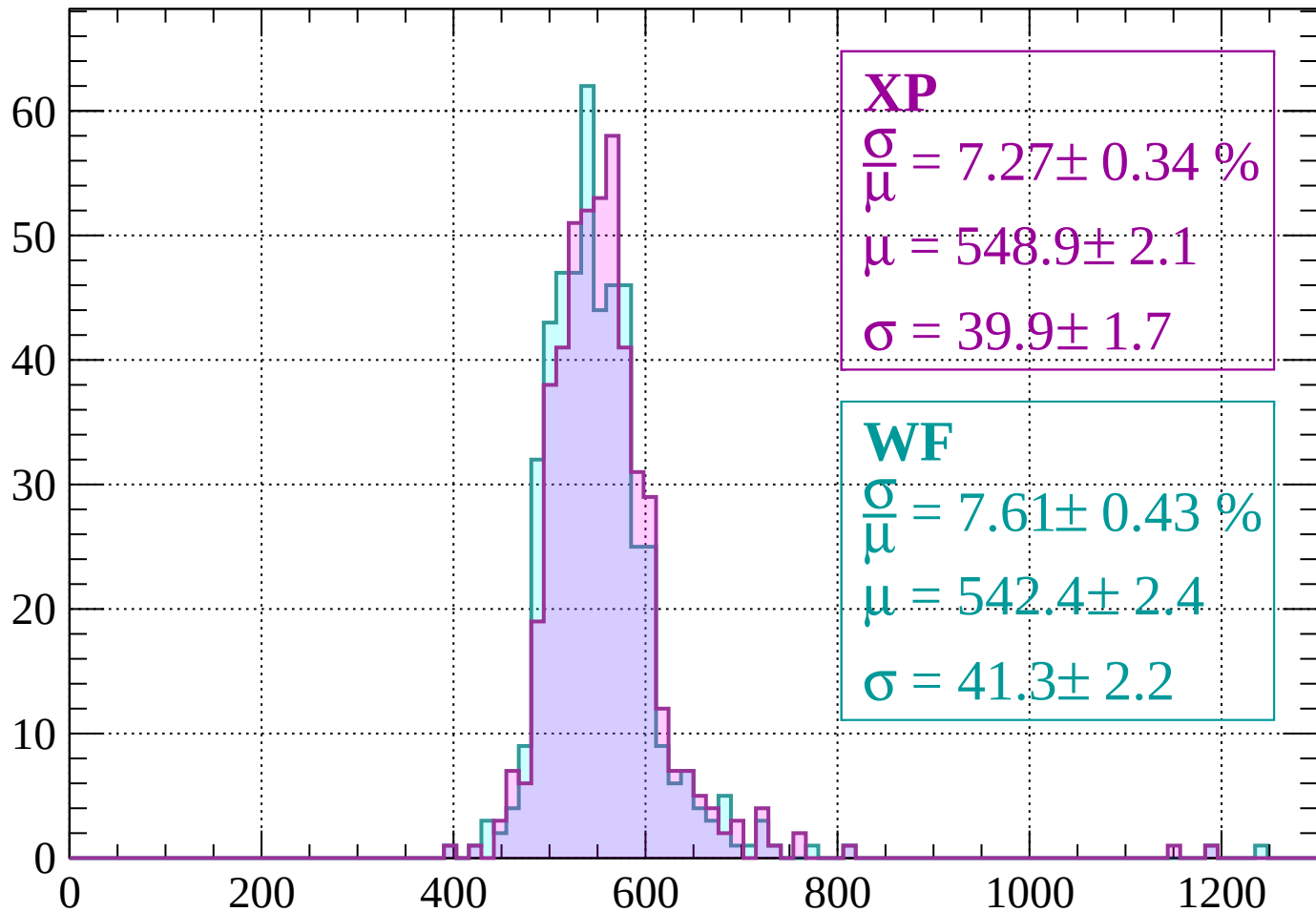
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.34 \%$$

$$\mu = 548.9 \pm 2.1$$

$$\sigma = 39.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.43 \%$$

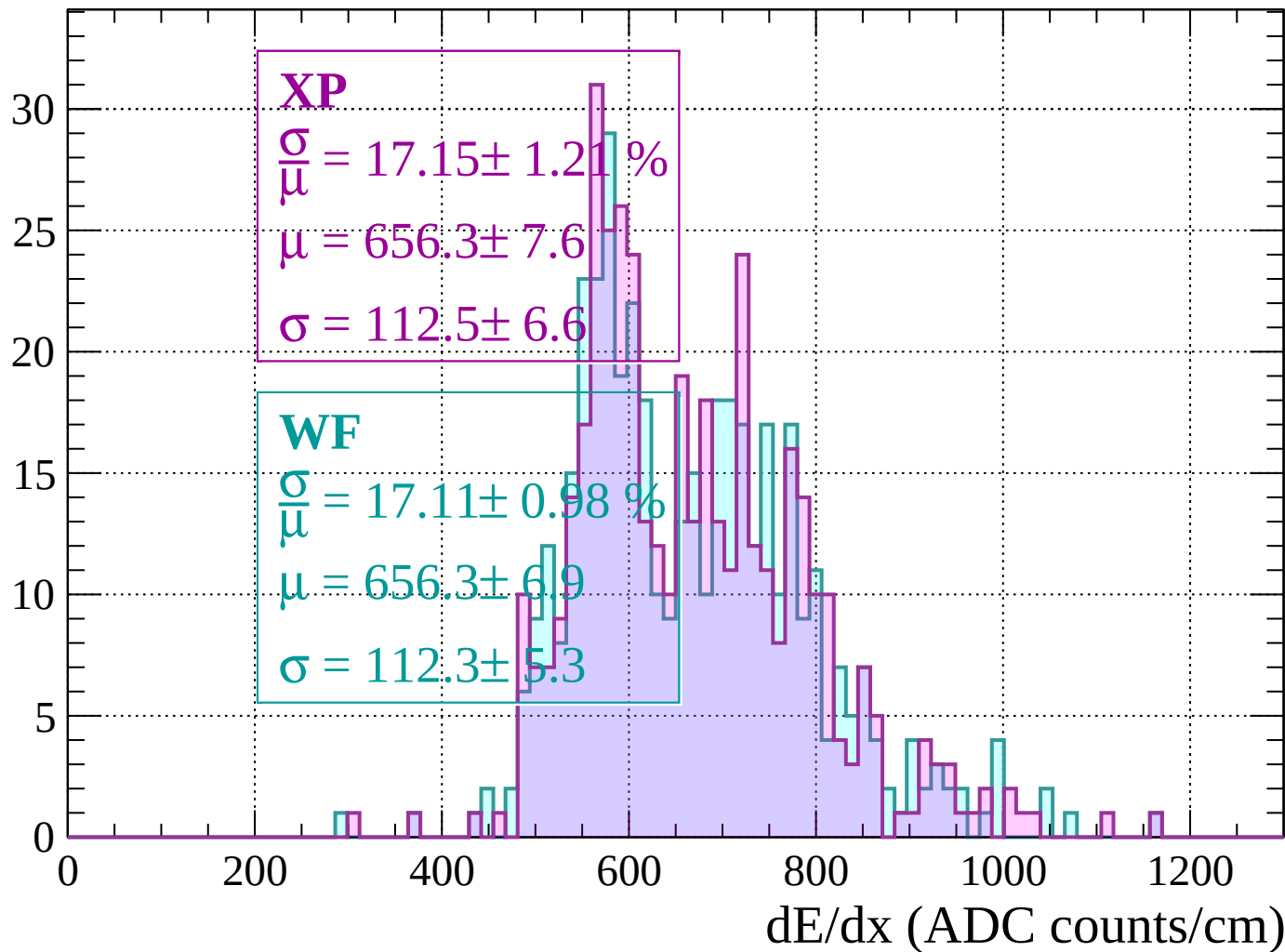
$$\mu = 542.4 \pm 2.4$$

$$\sigma = 41.3 \pm 2.2$$

dE/dx (ADC counts/cm)

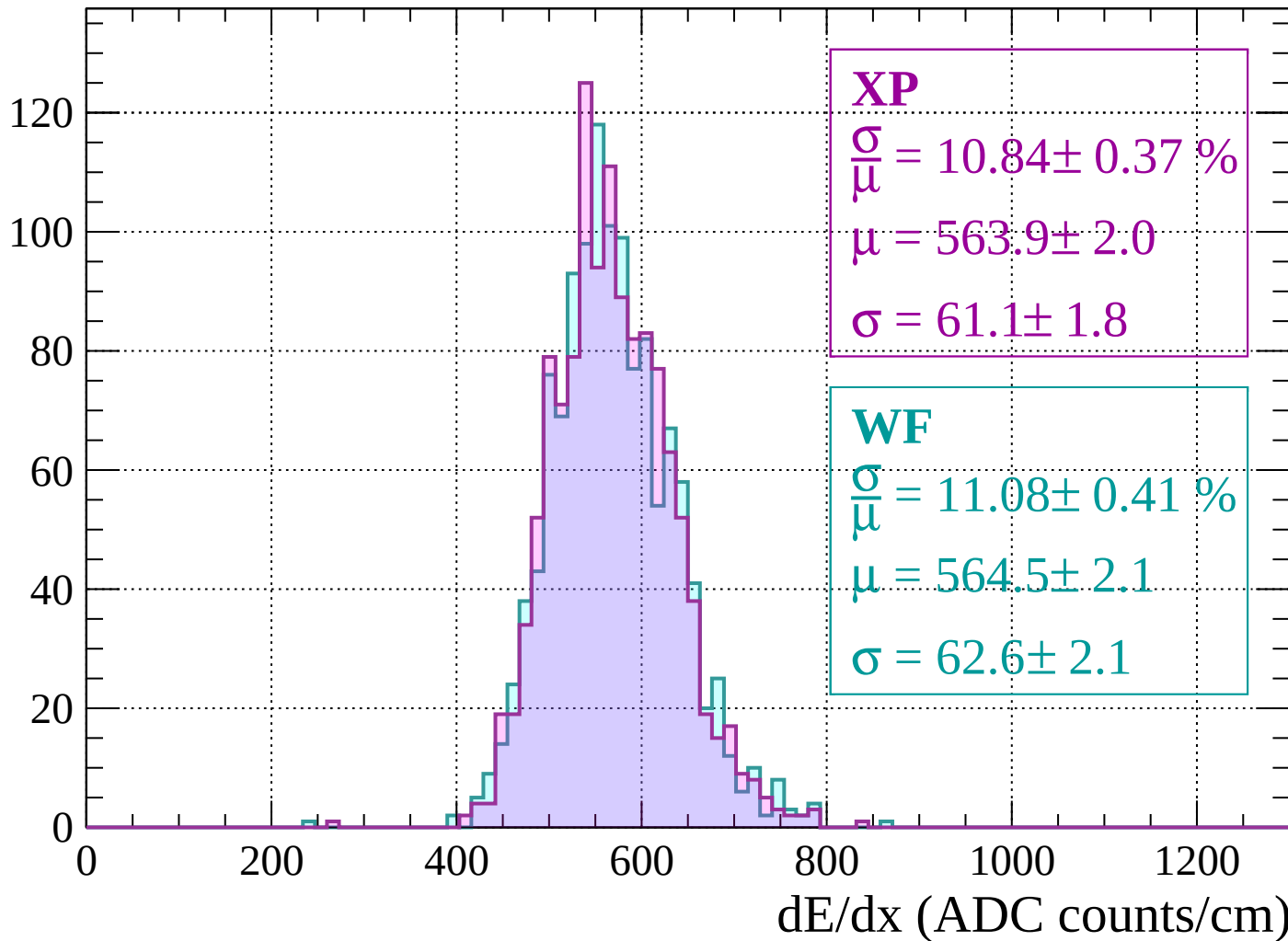
Energy loss | $80 < p < 120$

Count



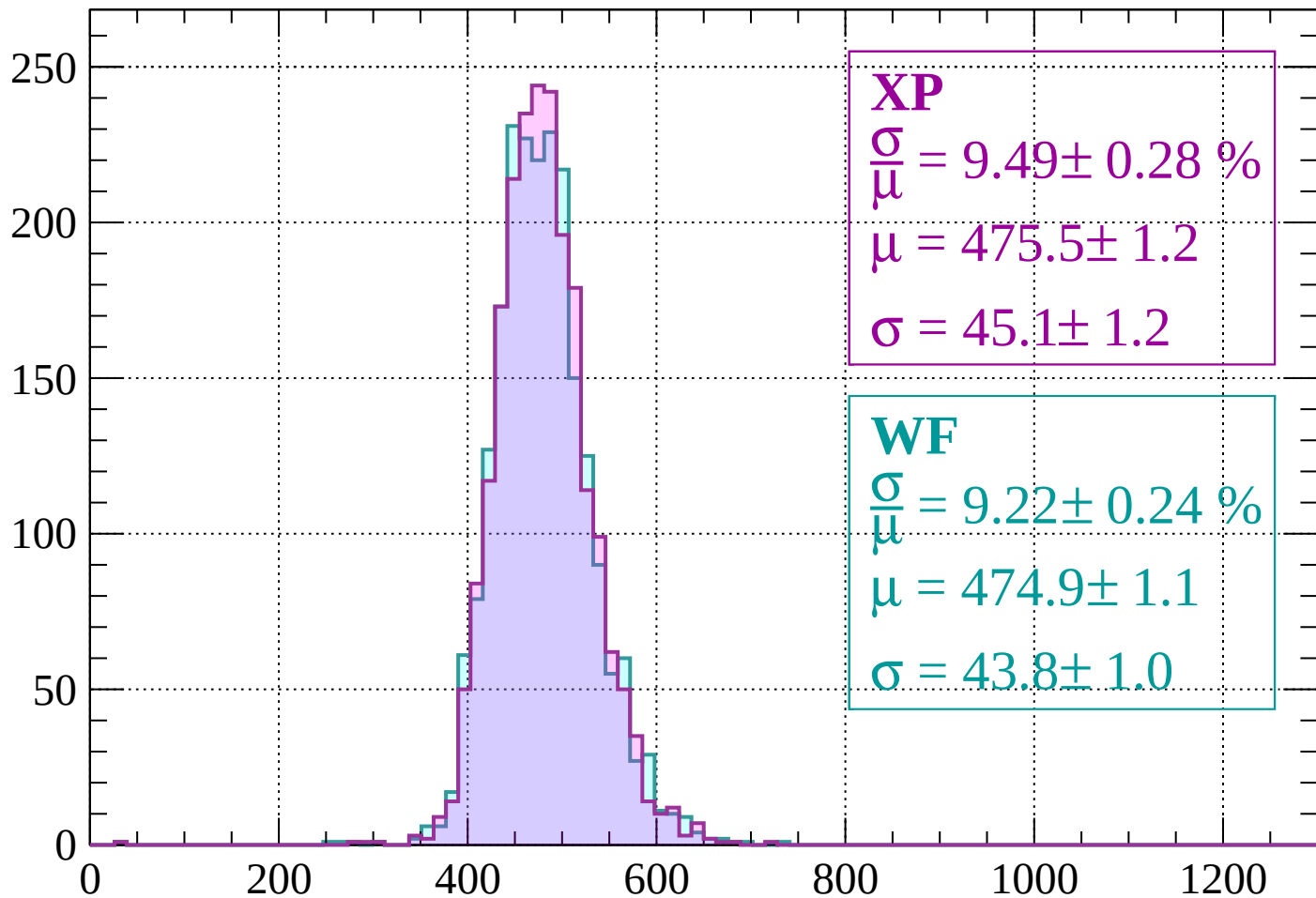
Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.28 \%$$

$$\mu = 475.5 \pm 1.2$$

$$\sigma = 45.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.24 \%$$

$$\mu = 474.9 \pm 1.1$$

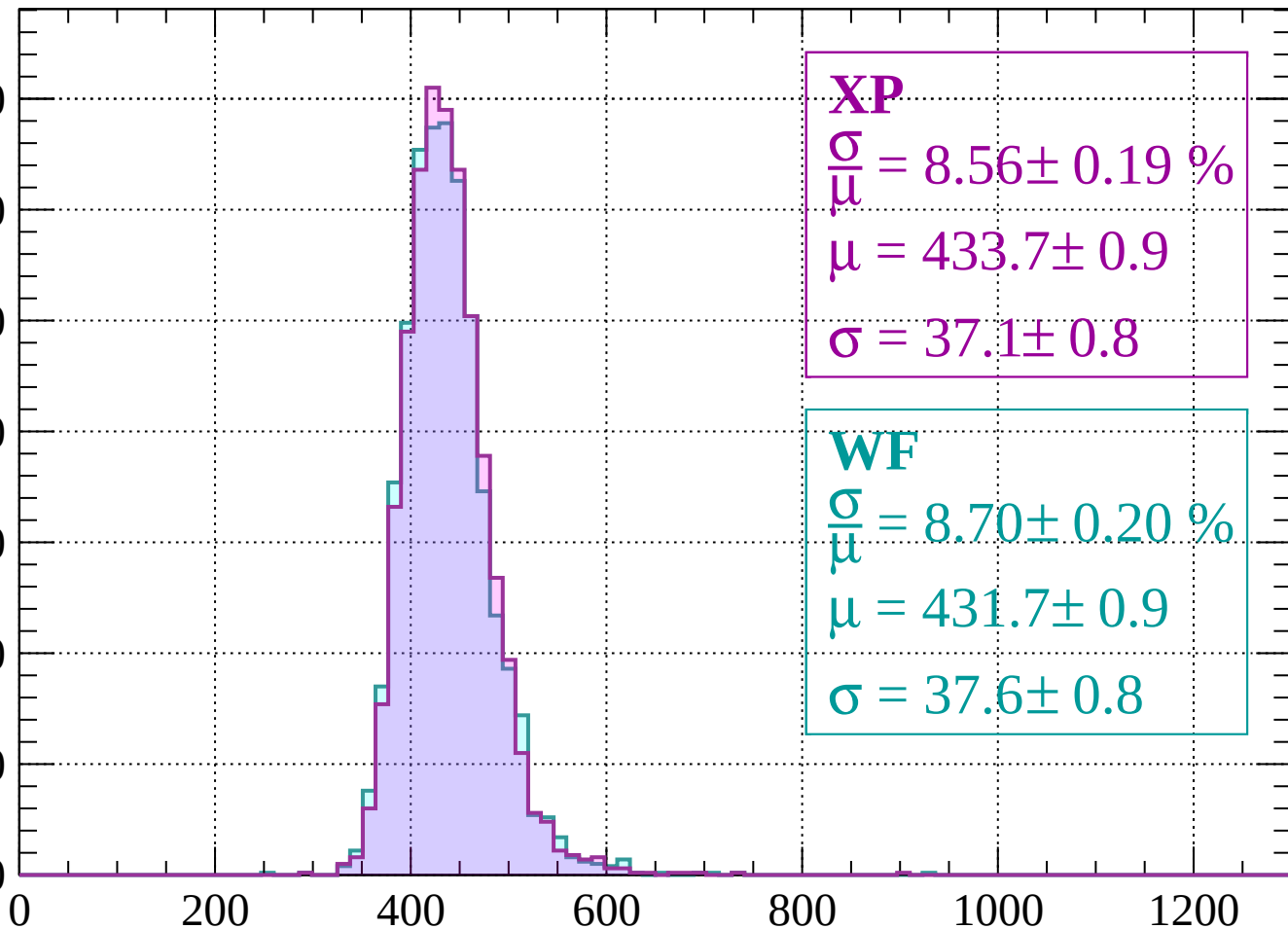
$$\sigma = 43.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.19 \%$$

$$\mu = 433.7 \pm 0.9$$

$$\sigma = 37.1 \pm 0.8$$

WF

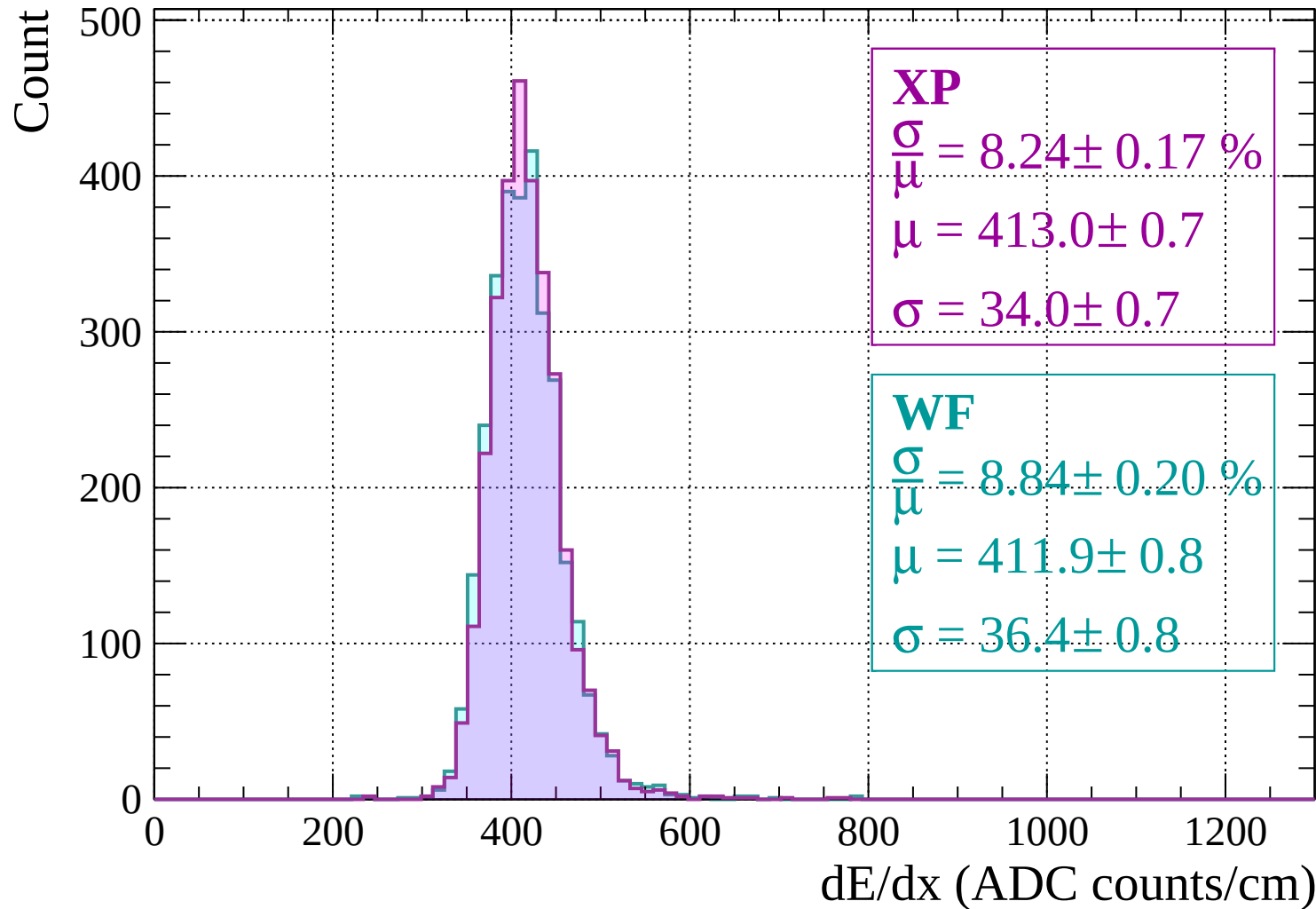
$$\frac{\sigma}{\mu} = 8.70 \pm 0.20 \%$$

$$\mu = 431.7 \pm 0.9$$

$$\sigma = 37.6 \pm 0.8$$

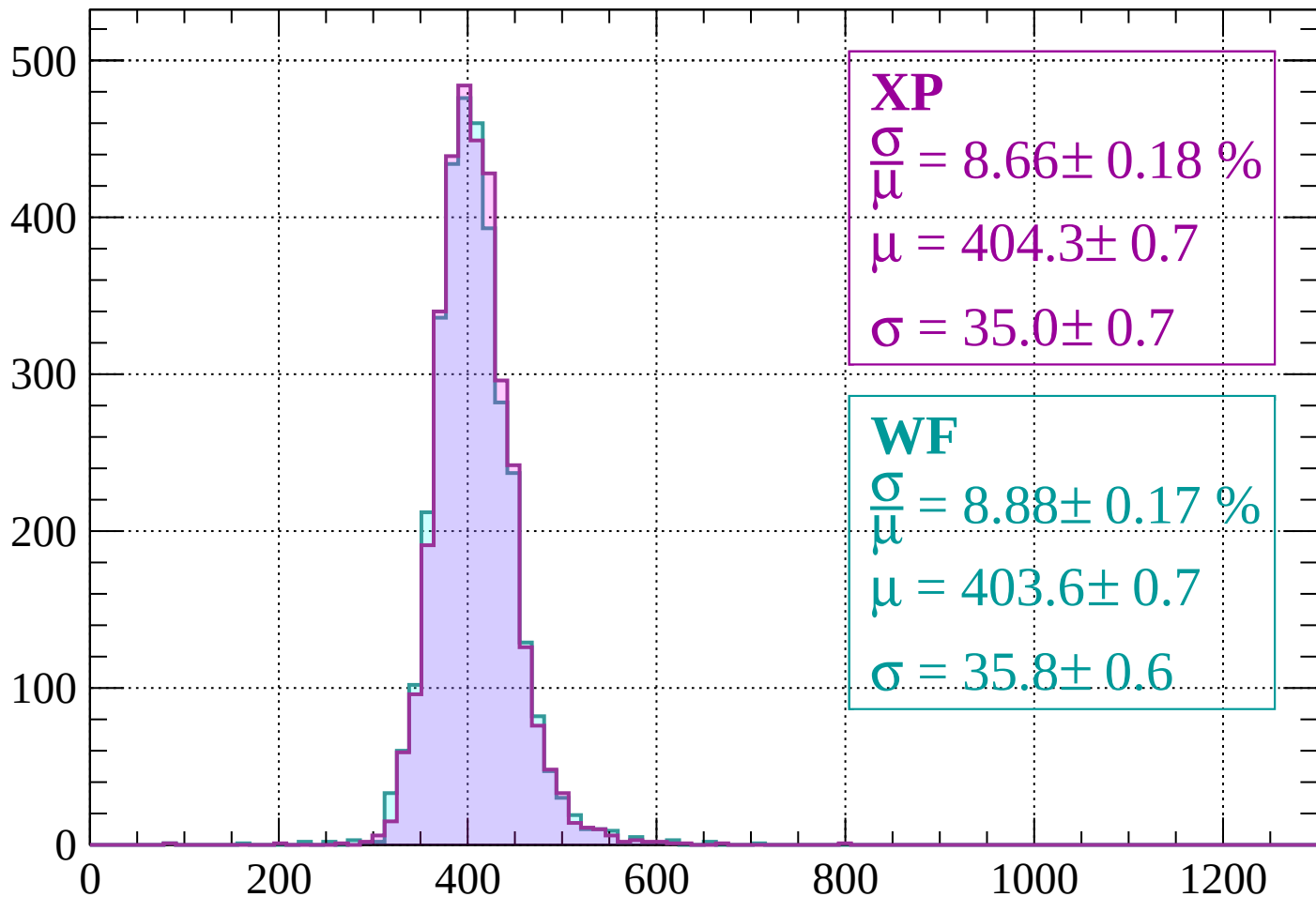
dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$



Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.66 \pm 0.18 \%$$

$$\mu = 404.3 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.88 \pm 0.17 \%$$

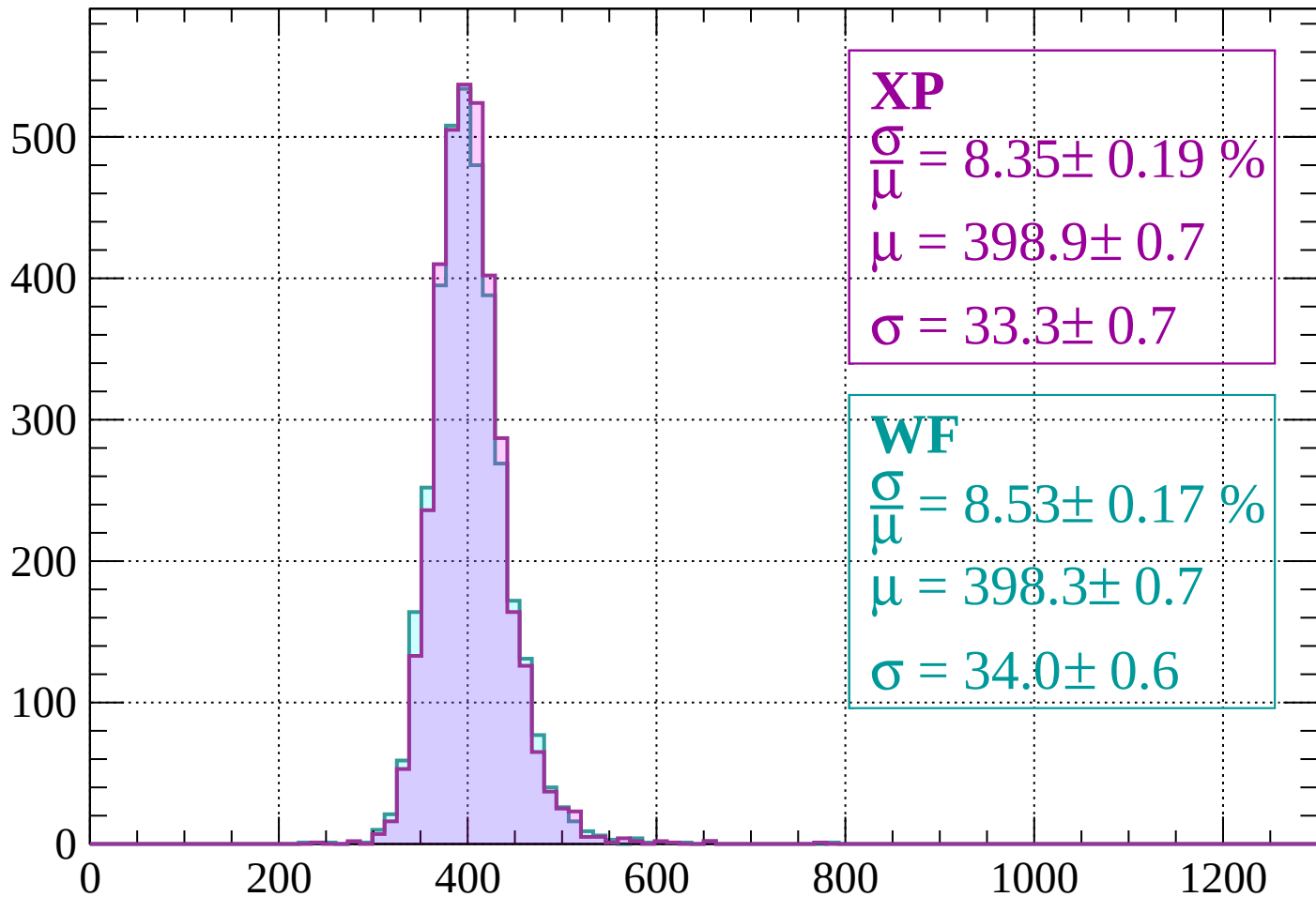
$$\mu = 403.6 \pm 0.7$$

$$\sigma = 35.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.19 \%$$

$$\mu = 398.9 \pm 0.7$$

$$\sigma = 33.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.53 \pm 0.17 \%$$

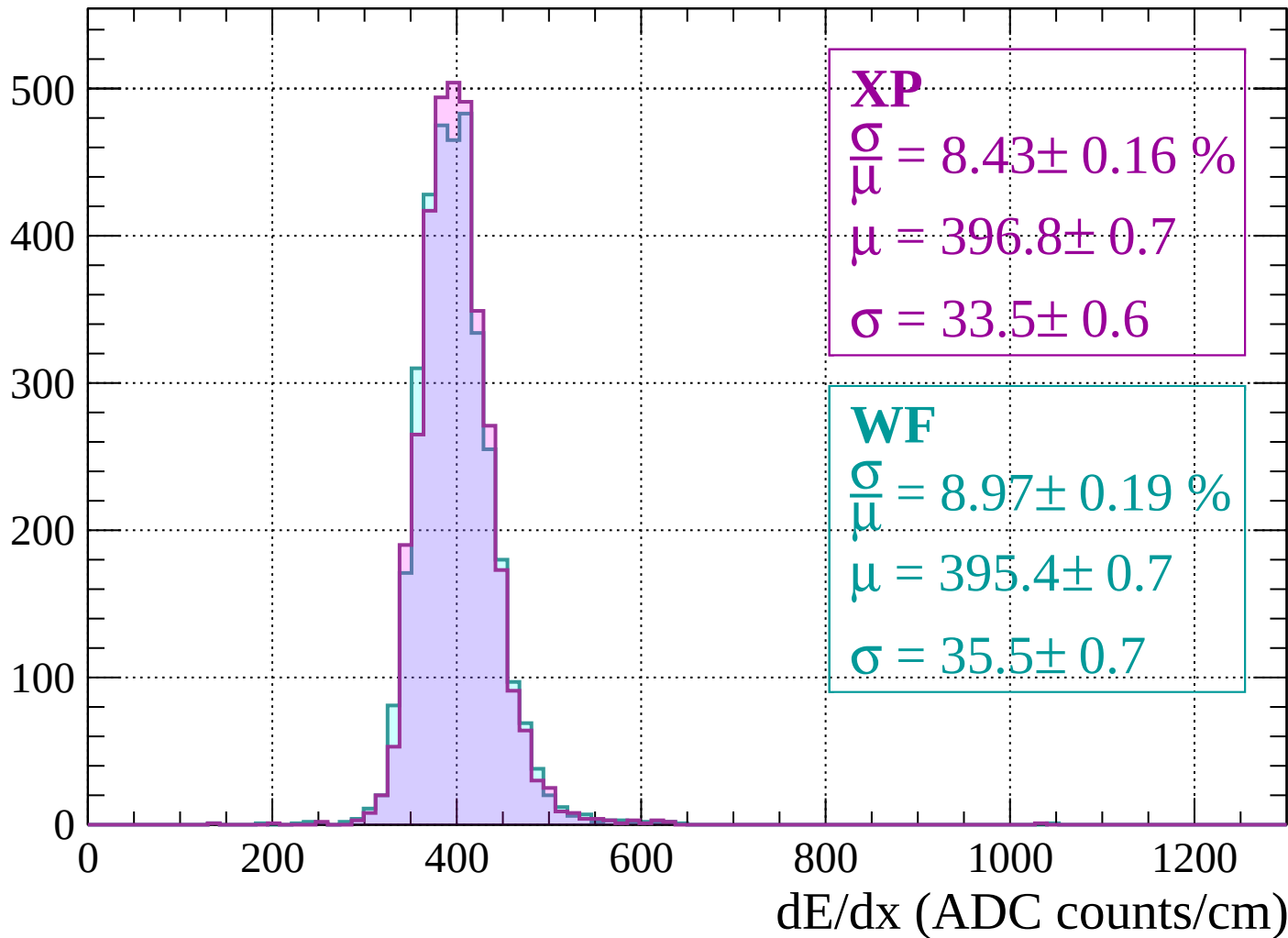
$$\mu = 398.3 \pm 0.7$$

$$\sigma = 34.0 \pm 0.6$$

dE/dx (ADC counts/cm)

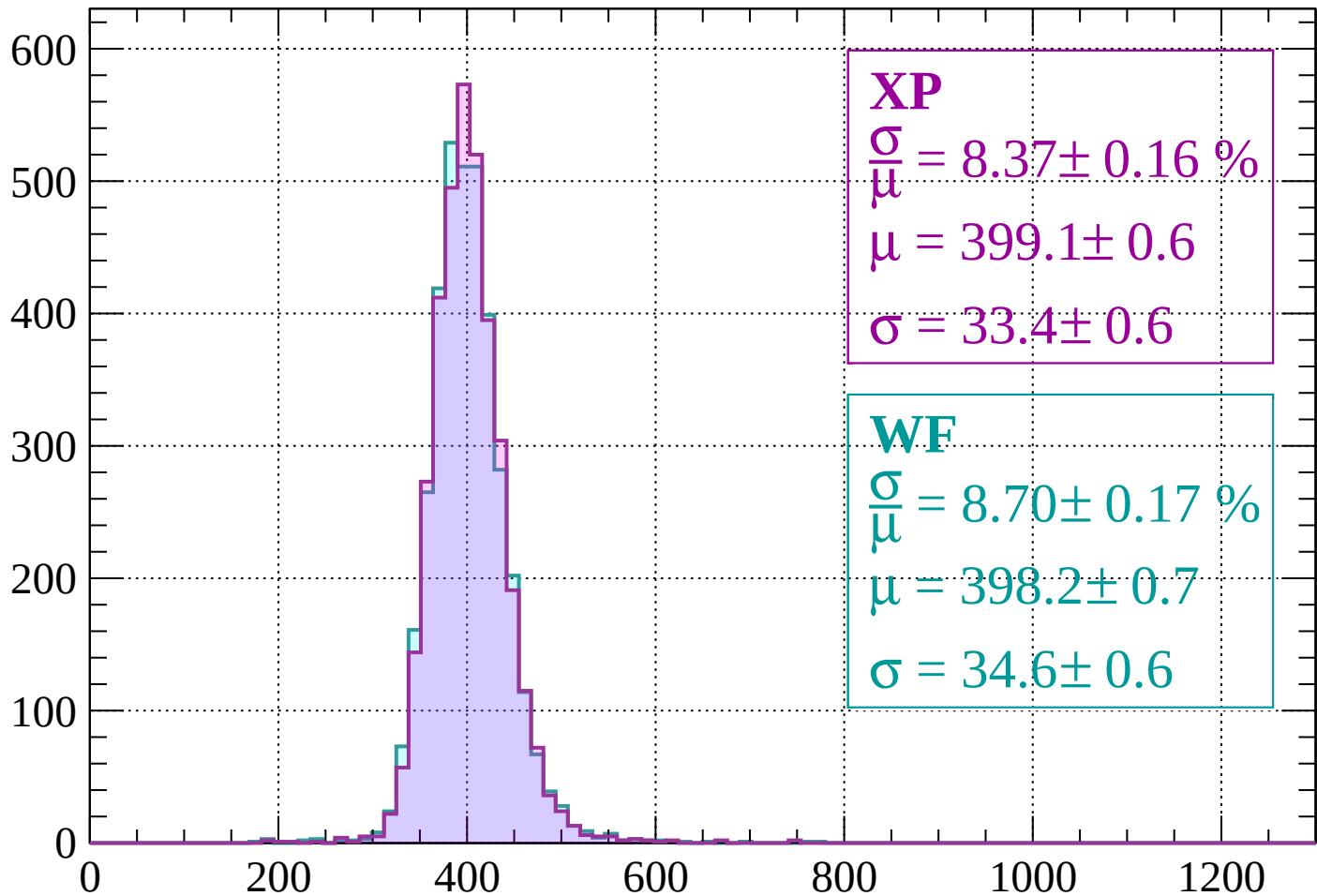
Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.16 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 33.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.17 \%$$

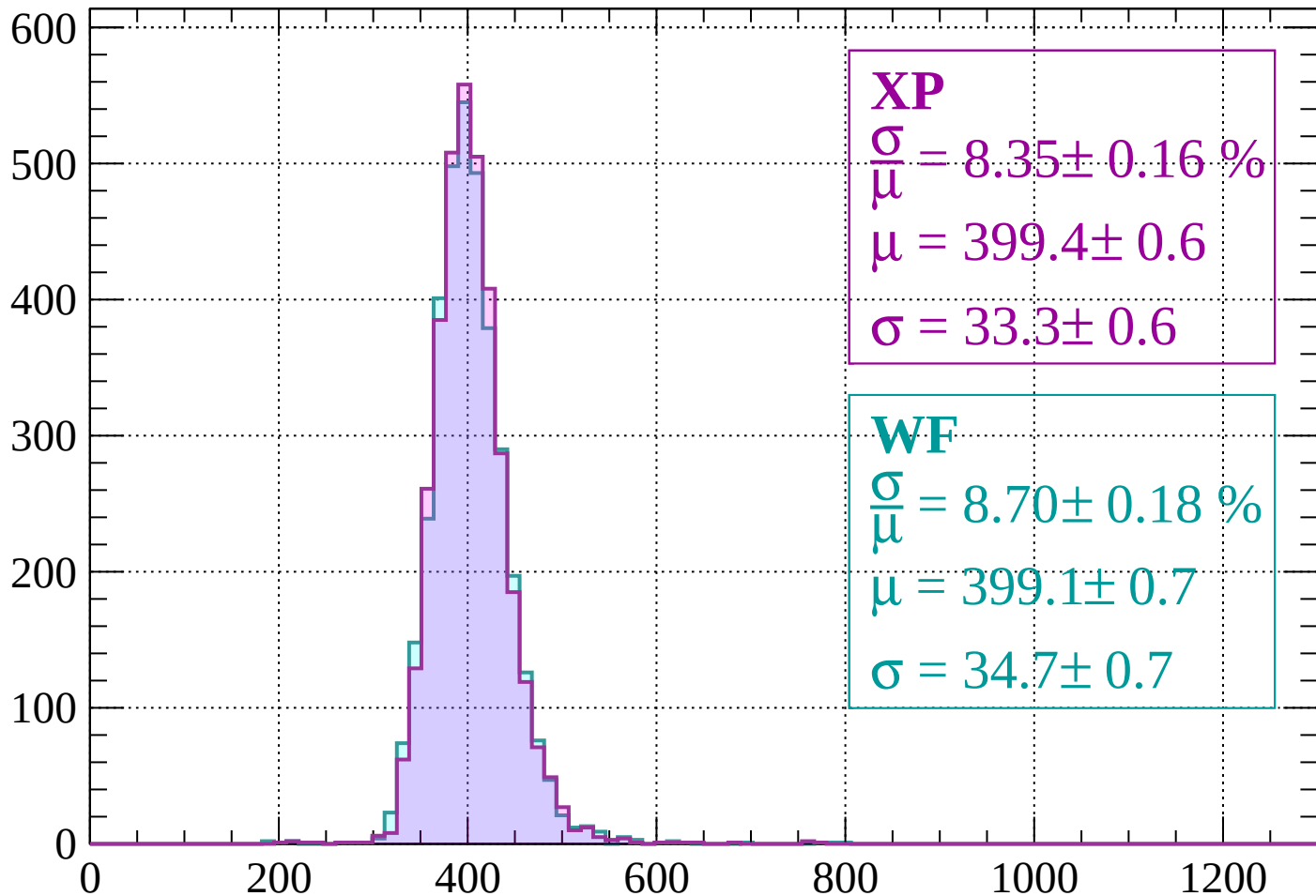
$$\mu = 398.2 \pm 0.7$$

$$\sigma = 34.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.16 \%$$

$$\mu = 399.4 \pm 0.6$$

$$\sigma = 33.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.18 \%$$

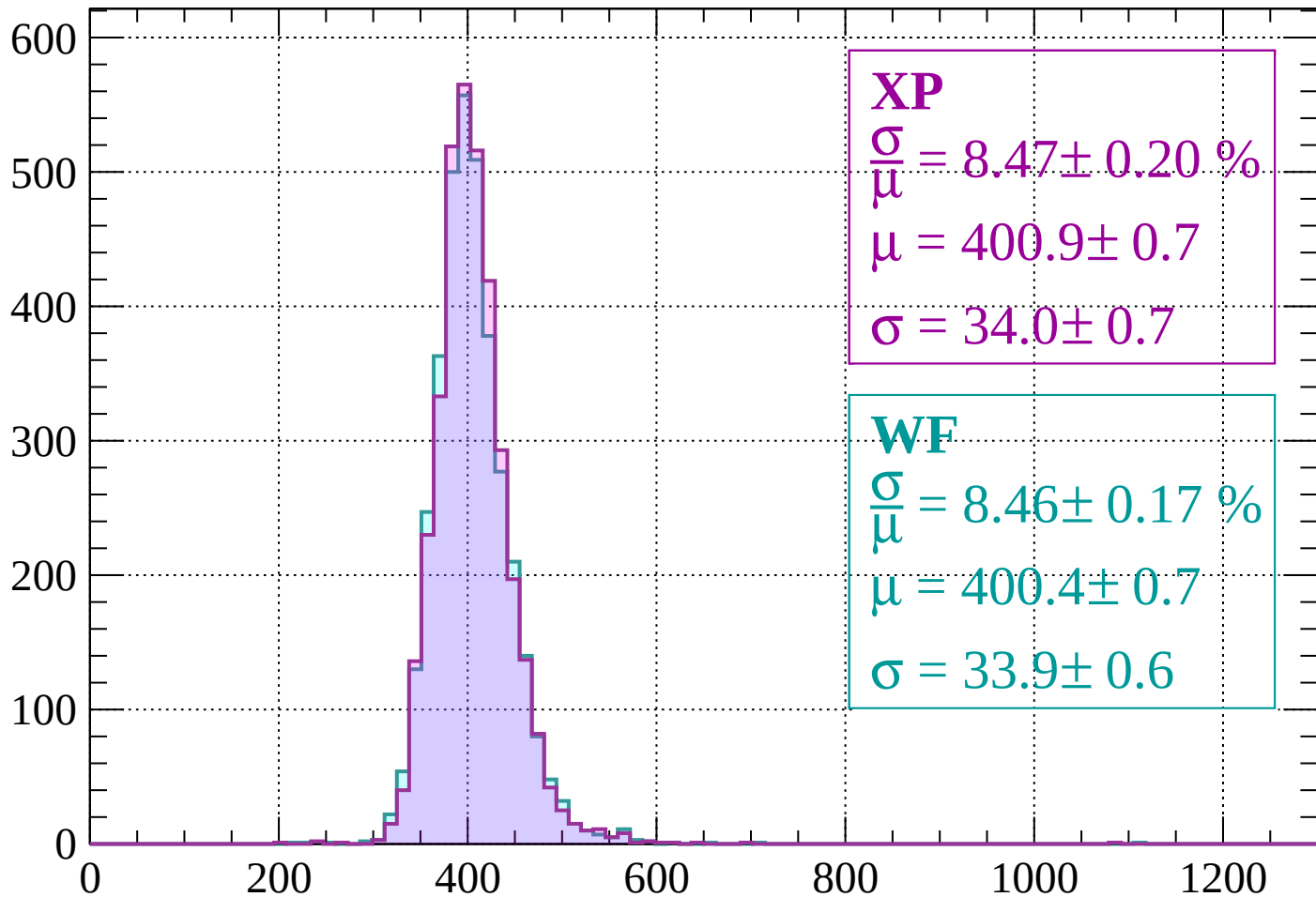
$$\mu = 399.1 \pm 0.7$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.20 \%$$

$$\mu = 400.9 \pm 0.7$$

$$\sigma = 34.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

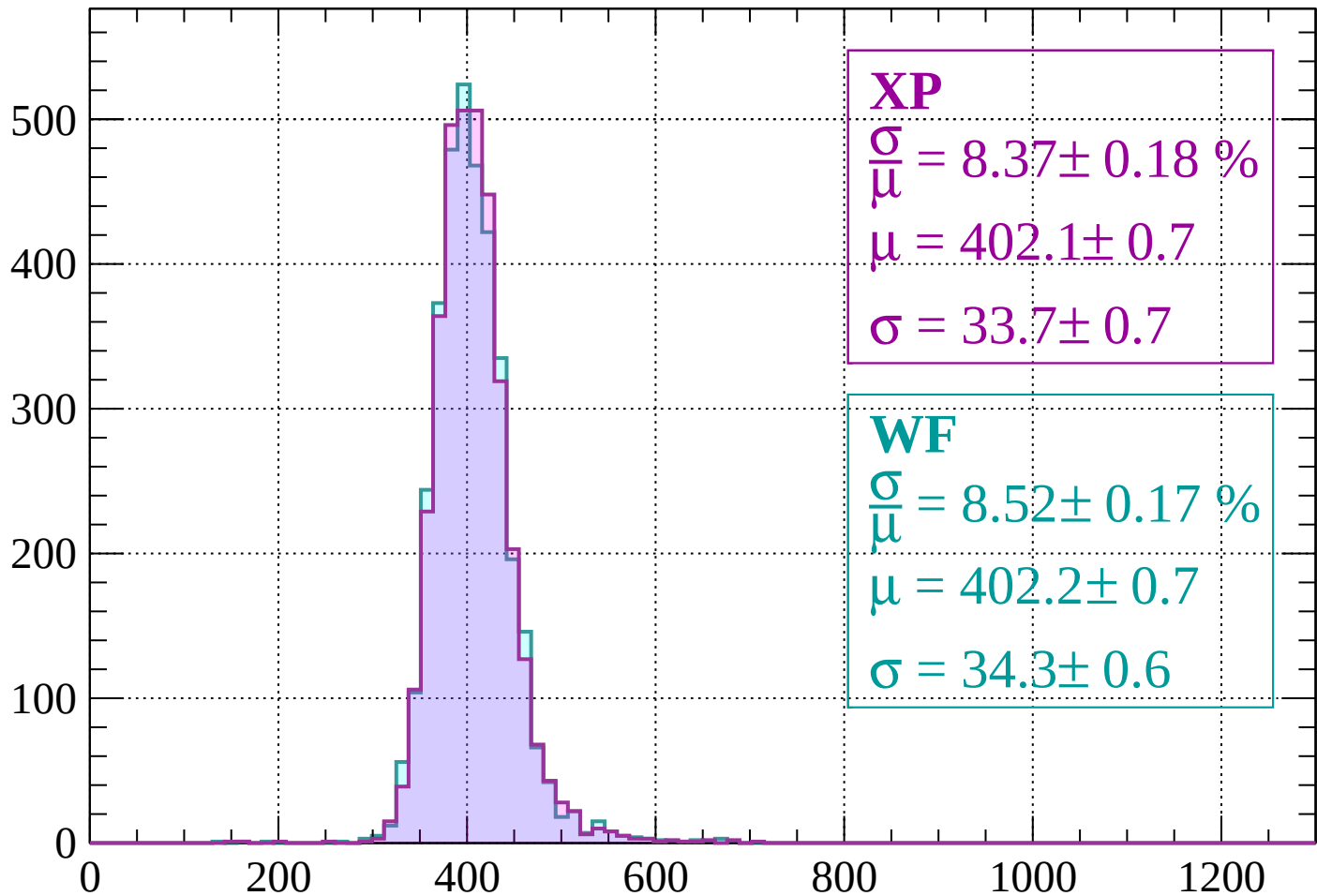
$$\mu = 400.4 \pm 0.7$$

$$\sigma = 33.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.18 \%$$

$$\mu = 402.1 \pm 0.7$$

$$\sigma = 33.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.52 \pm 0.17 \%$$

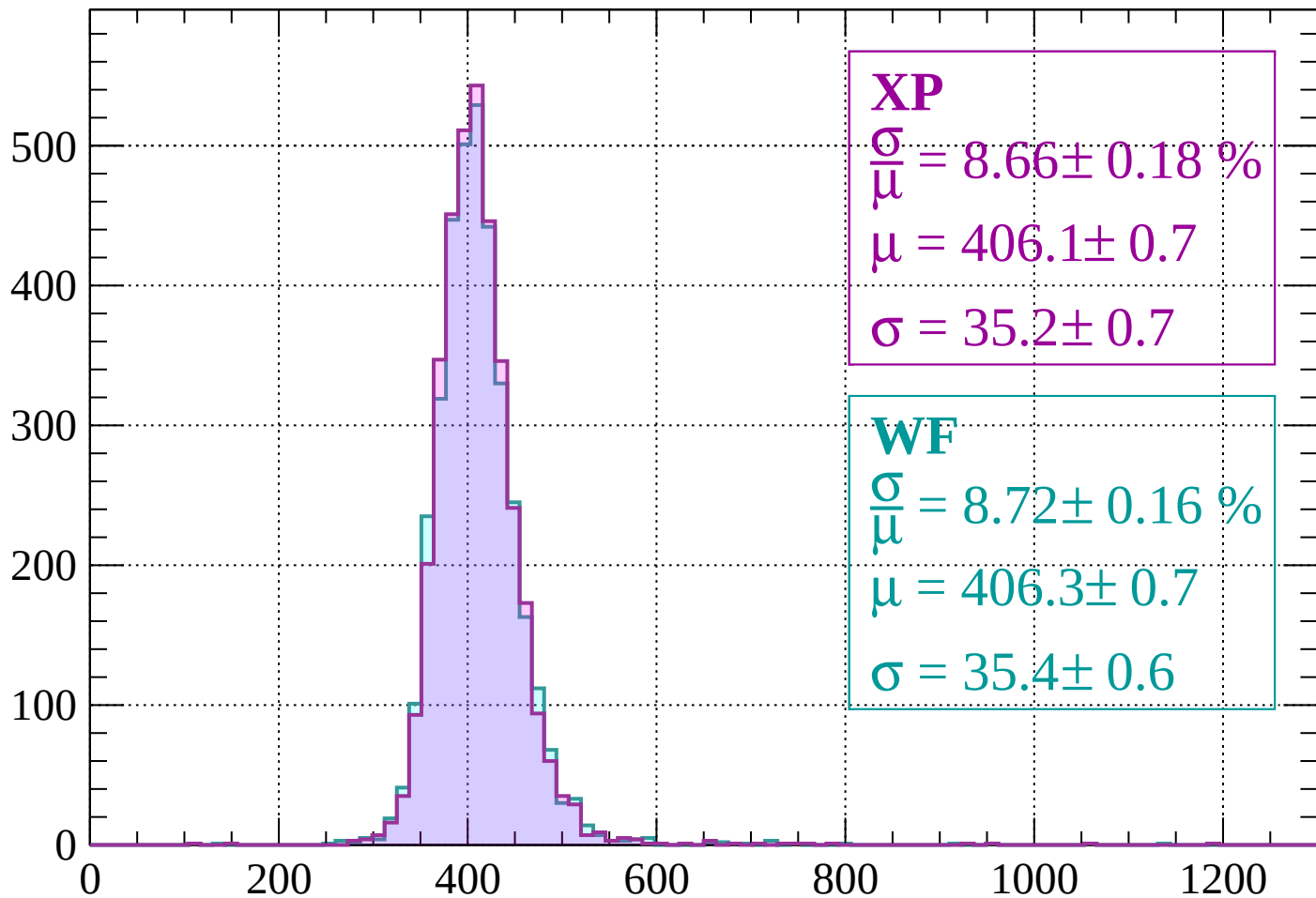
$$\mu = 402.2 \pm 0.7$$

$$\sigma = 34.3 \pm 0.6$$

dE/dx (ADC counts/cm)

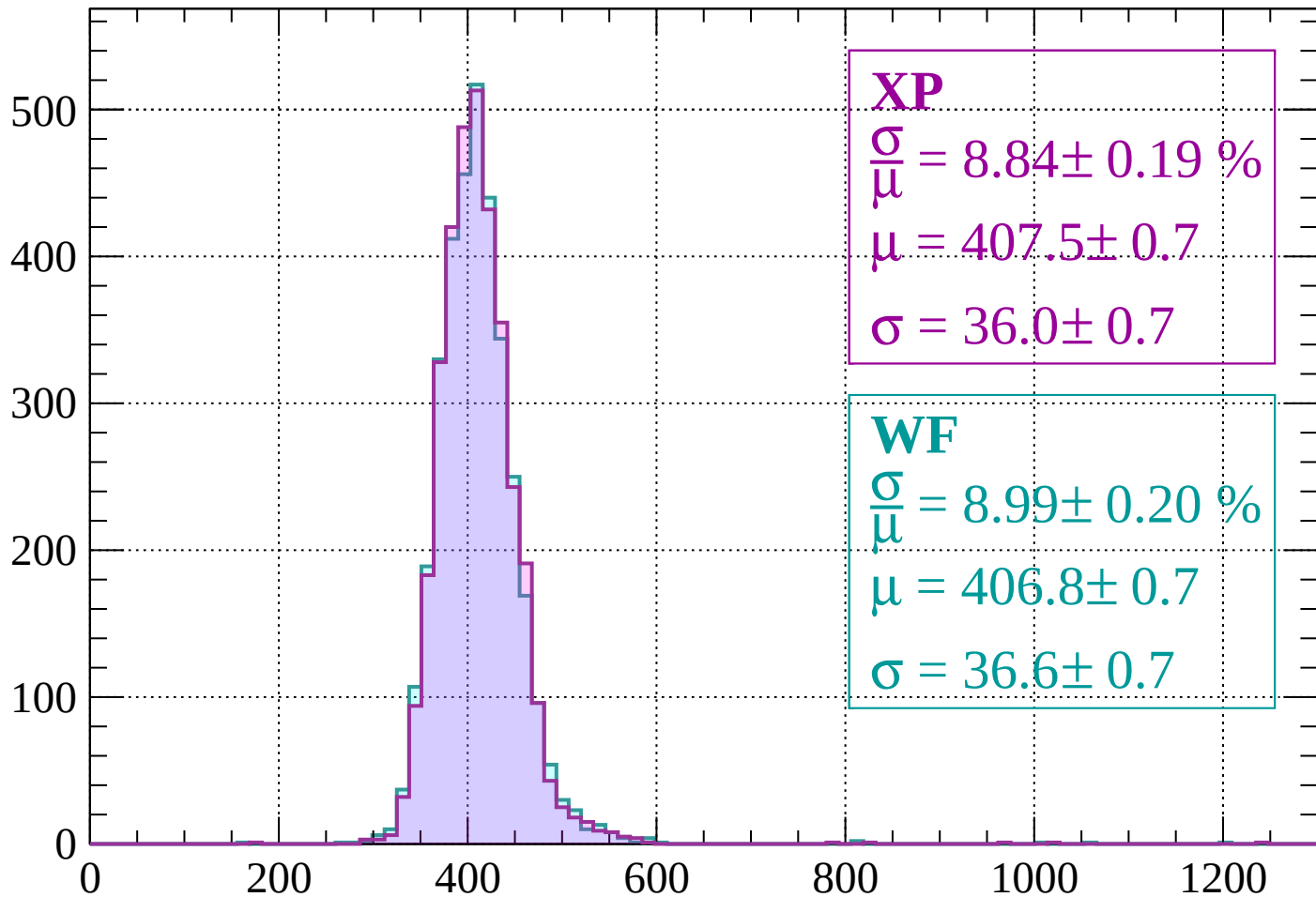
Energy loss | $560 < p < 600$

Count



Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.19 \%$$

$$\mu = 407.5 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.20 \%$$

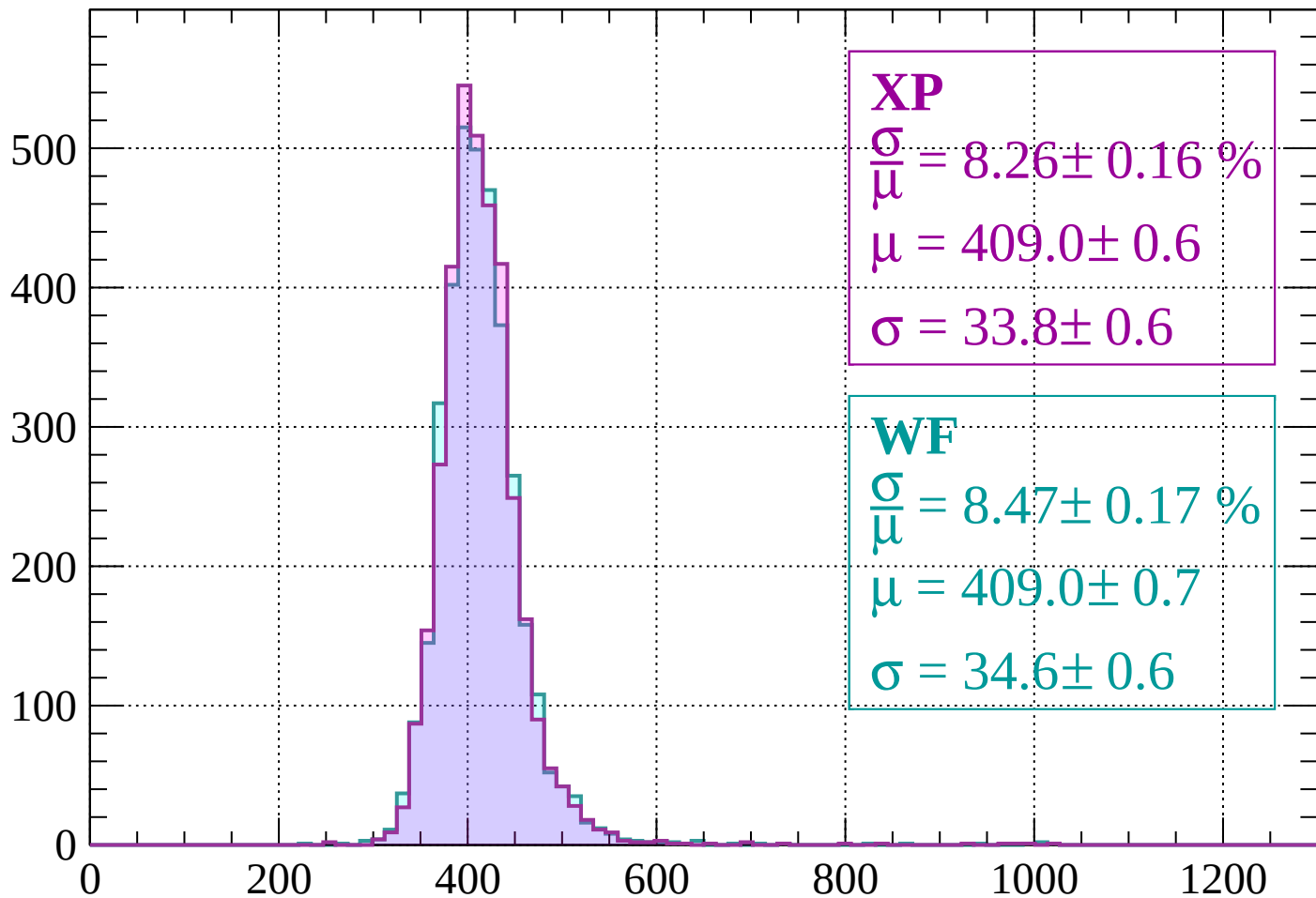
$$\mu = 406.8 \pm 0.7$$

$$\sigma = 36.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.16 \%$$

$$\mu = 409.0 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.17 \%$$

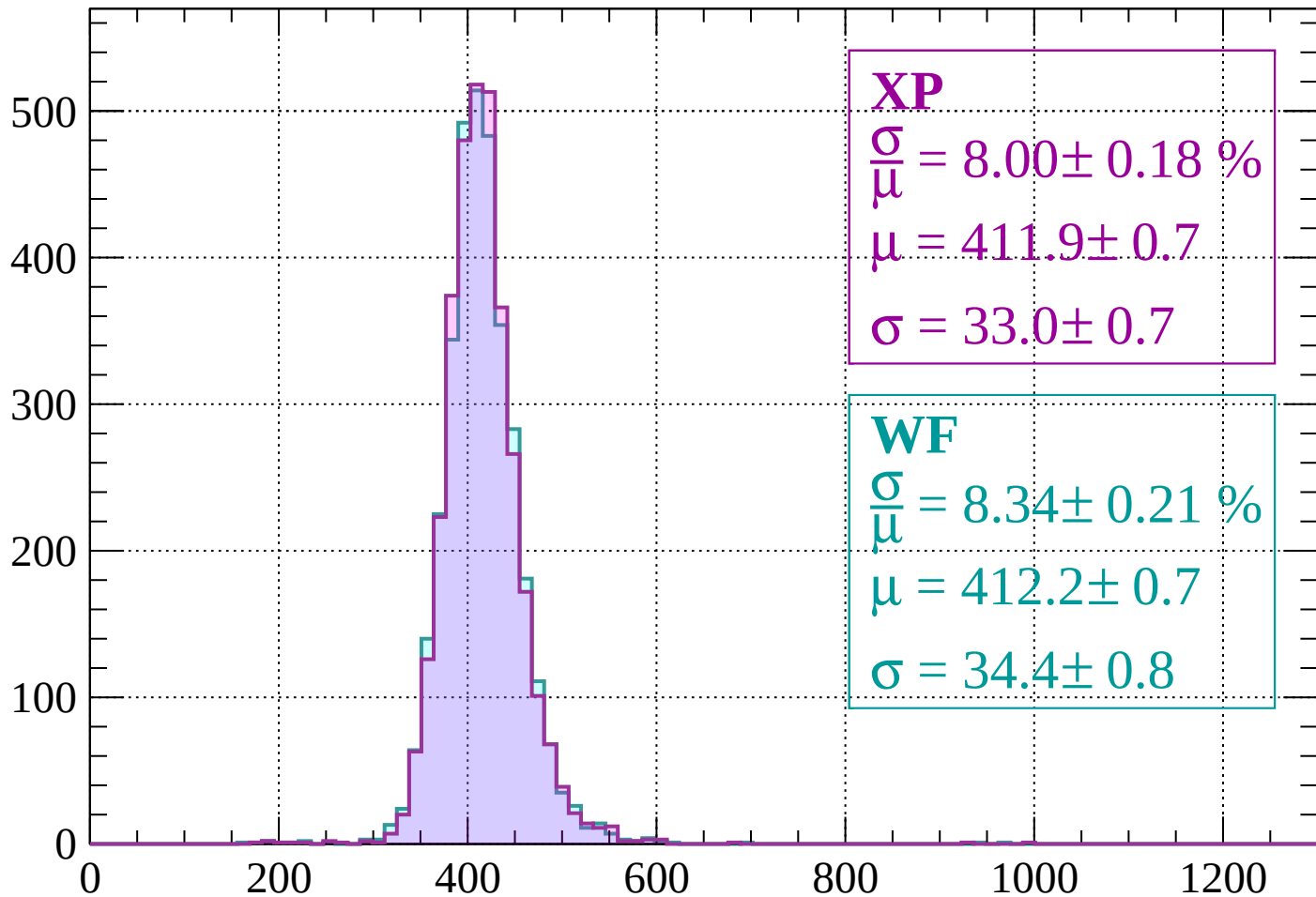
$$\mu = 409.0 \pm 0.7$$

$$\sigma = 34.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.18 \%$$

$$\mu = 411.9 \pm 0.7$$

$$\sigma = 33.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.34 \pm 0.21 \%$$

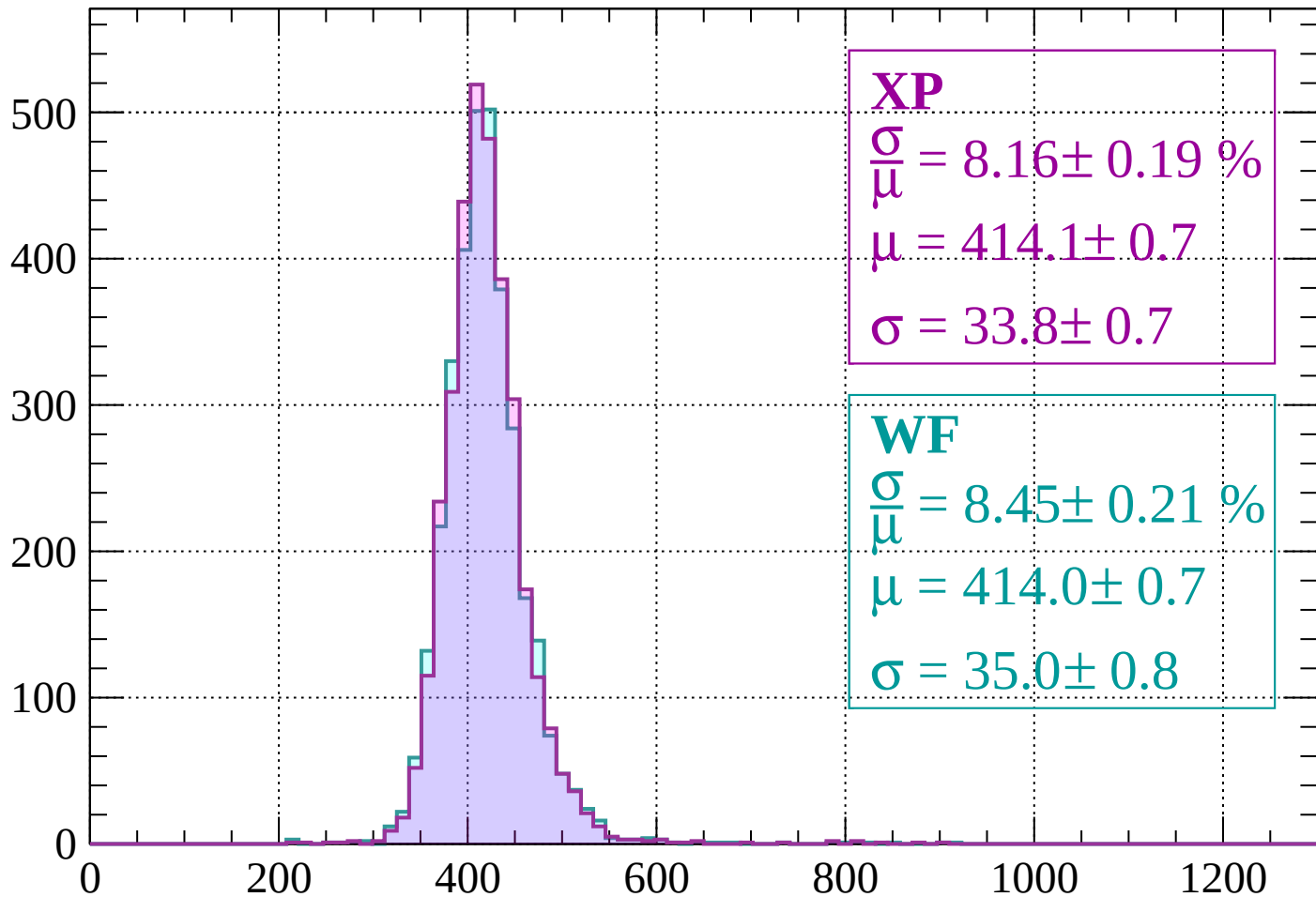
$$\mu = 412.2 \pm 0.7$$

$$\sigma = 34.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

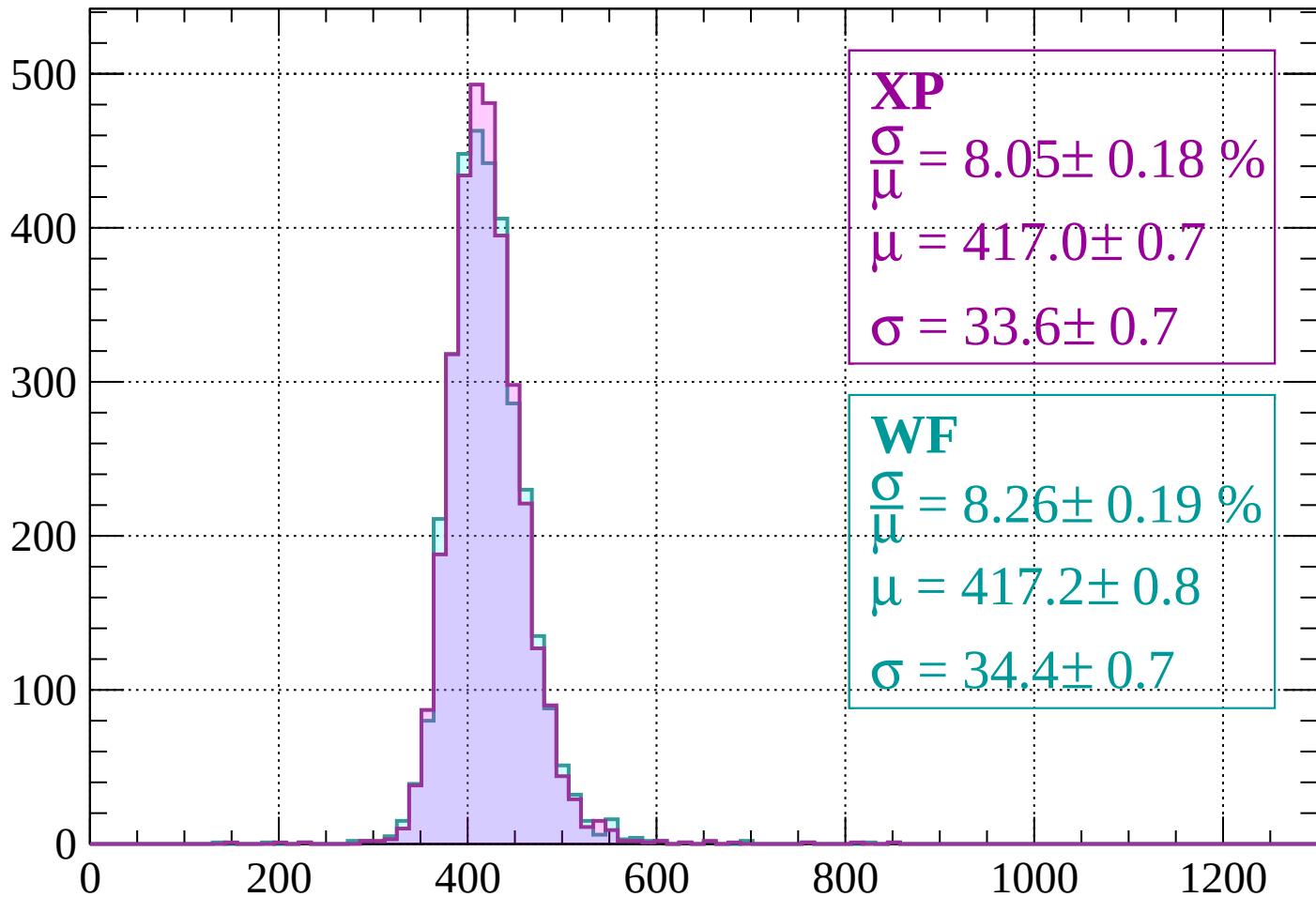
Count



dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.18 \%$$

$$\mu = 417.0 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.26 \pm 0.19 \%$$

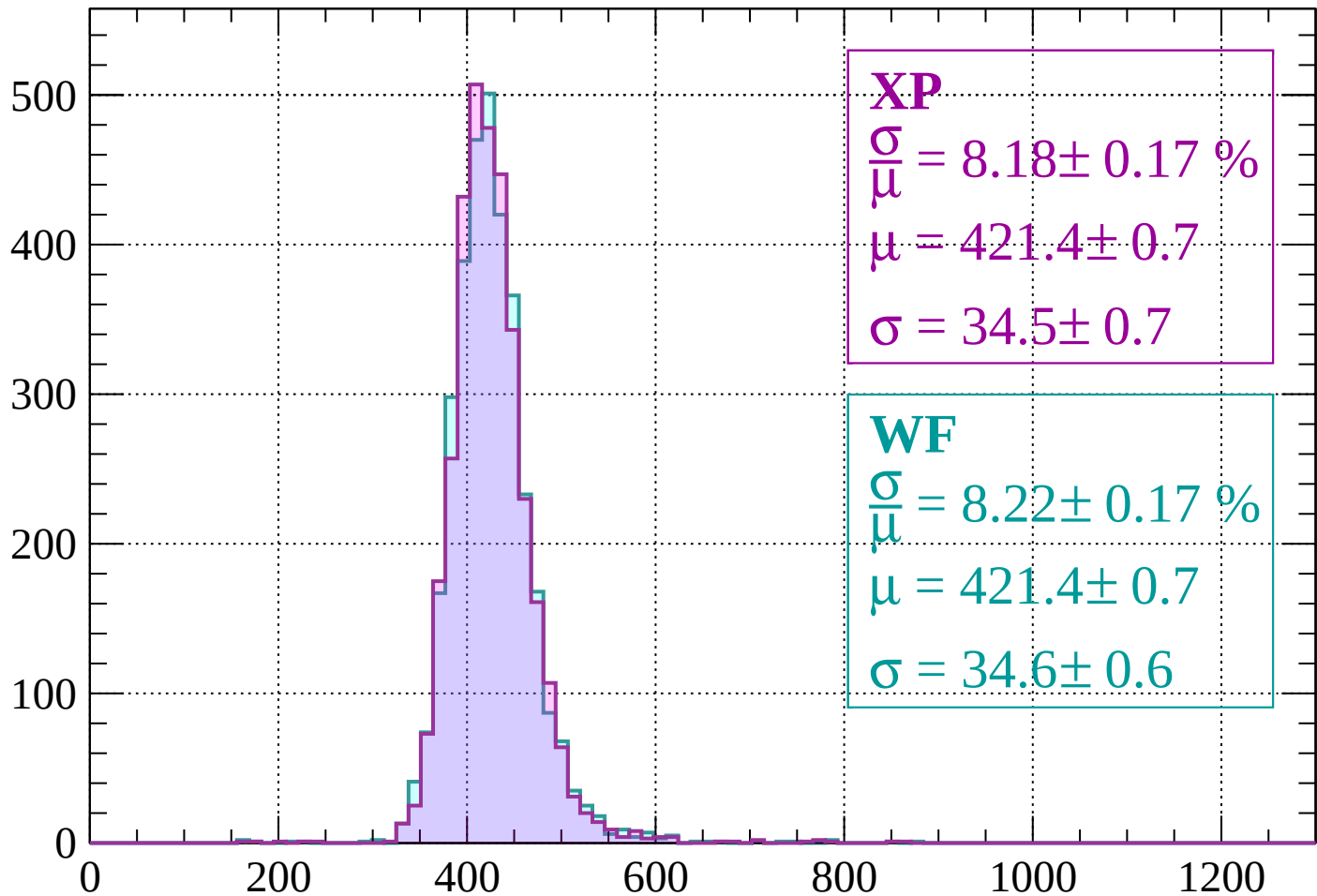
$$\mu = 417.2 \pm 0.8$$

$$\sigma = 34.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.17 \%$$

$$\mu = 421.4 \pm 0.7$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.17 \%$$

$$\mu = 421.4 \pm 0.7$$

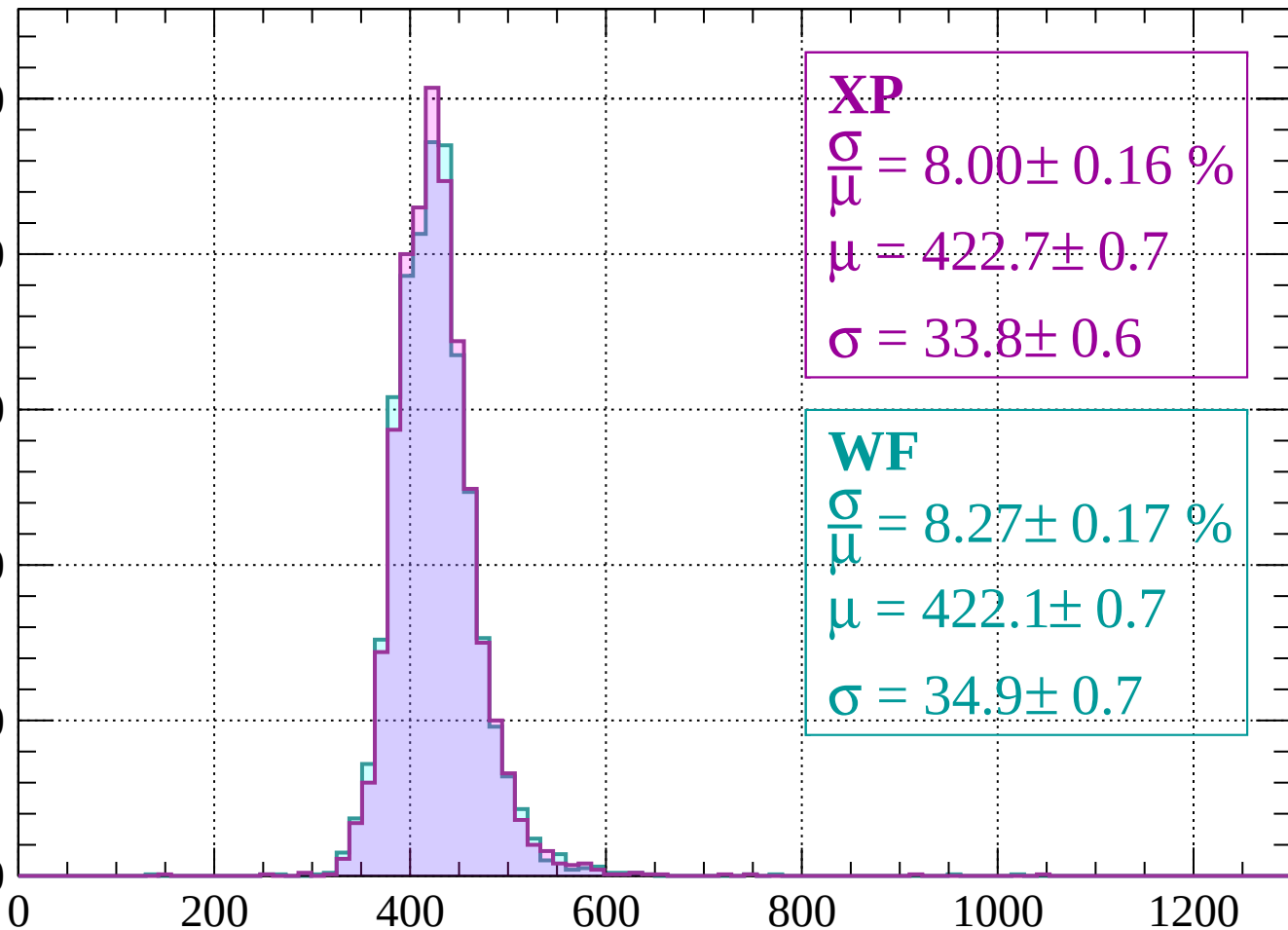
$$\sigma = 34.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.16 \%$$

$$\mu = 422.7 \pm 0.7$$

$$\sigma = 33.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.17 \%$$

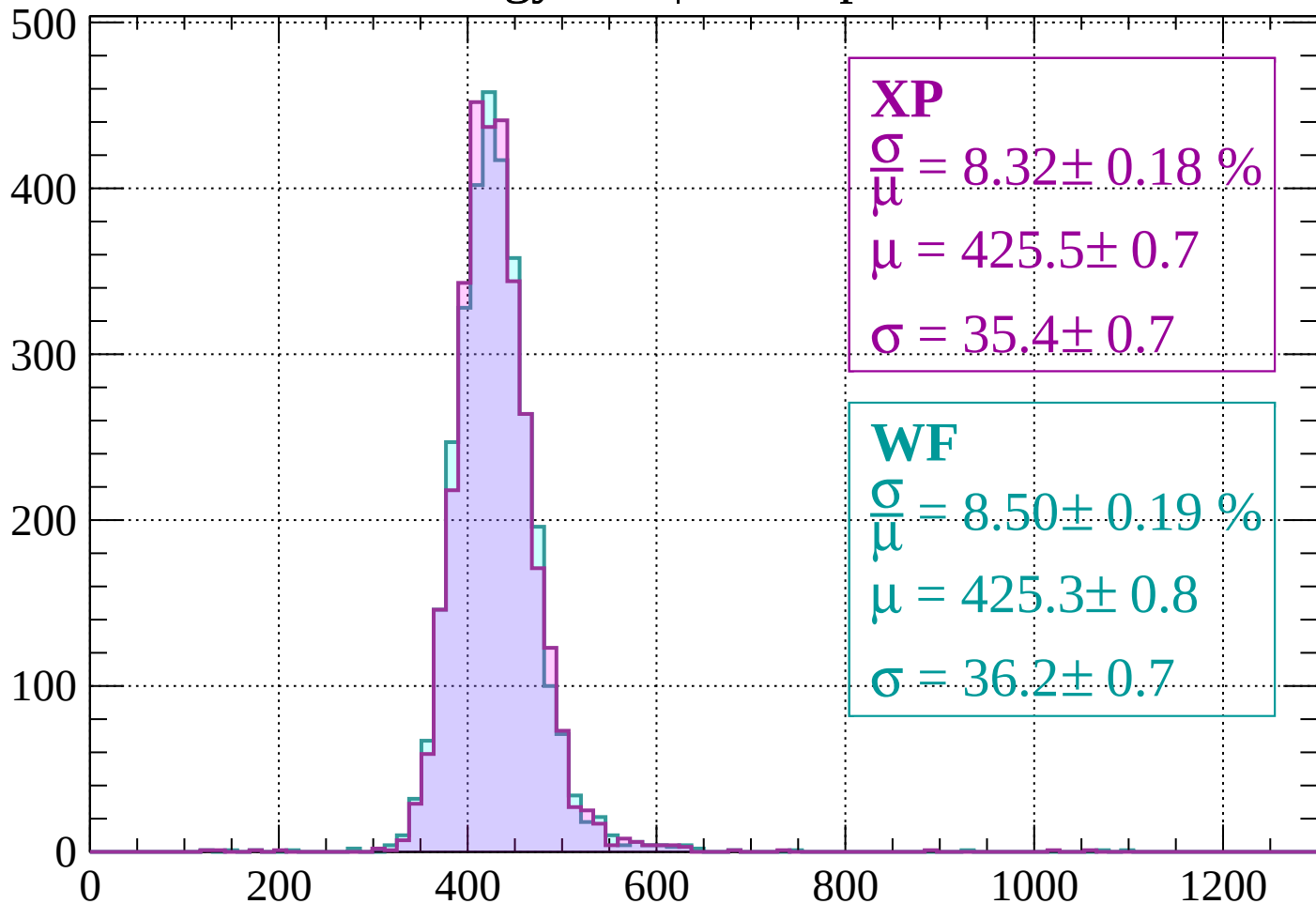
$$\mu = 422.1 \pm 0.7$$

$$\sigma = 34.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.18 \%$$

$$\mu = 425.5 \pm 0.7$$

$$\sigma = 35.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.19 \%$$

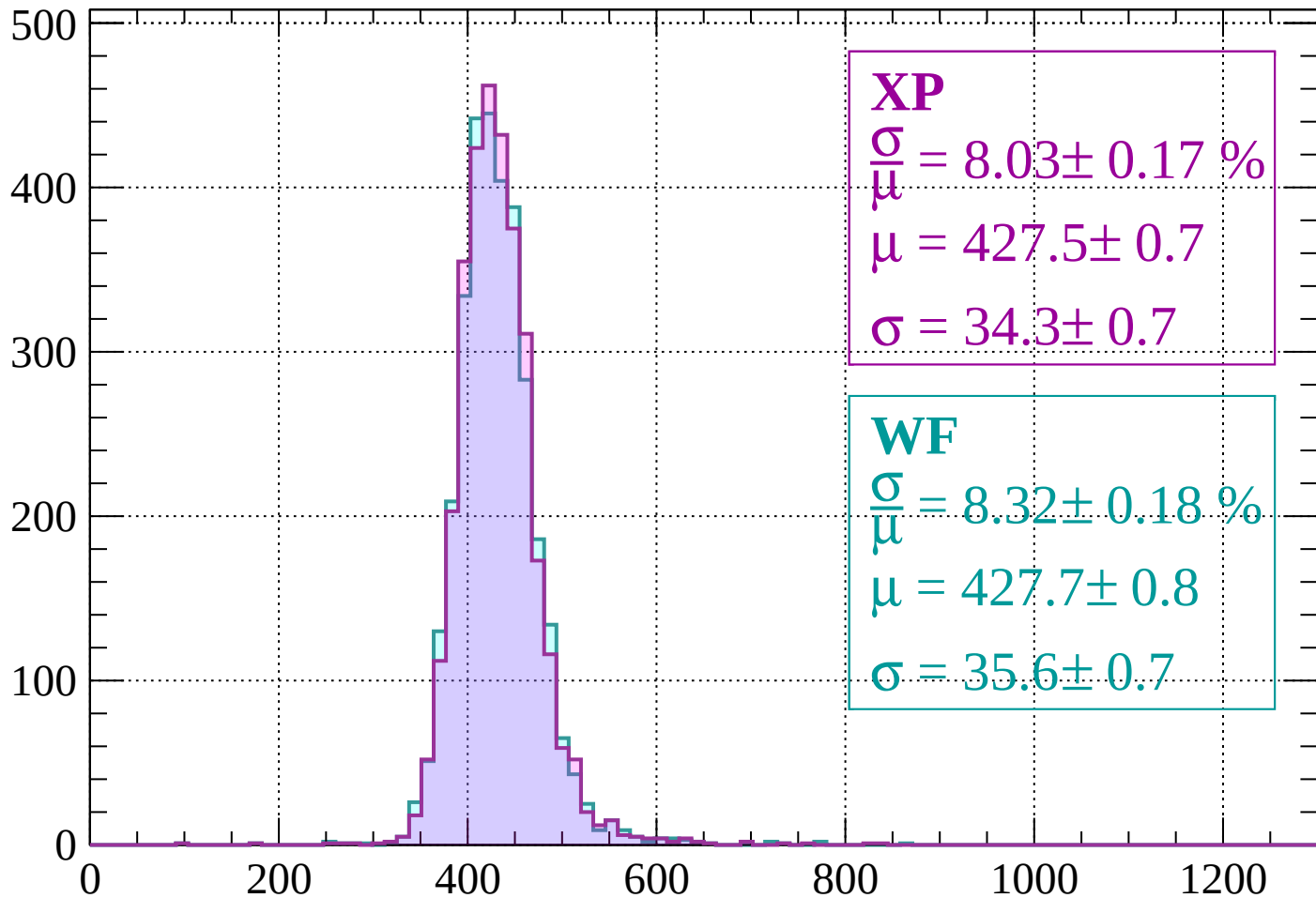
$$\mu = 425.3 \pm 0.8$$

$$\sigma = 36.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.17 \%$$

$$\mu = 427.5 \pm 0.7$$

$$\sigma = 34.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.32 \pm 0.18 \%$$

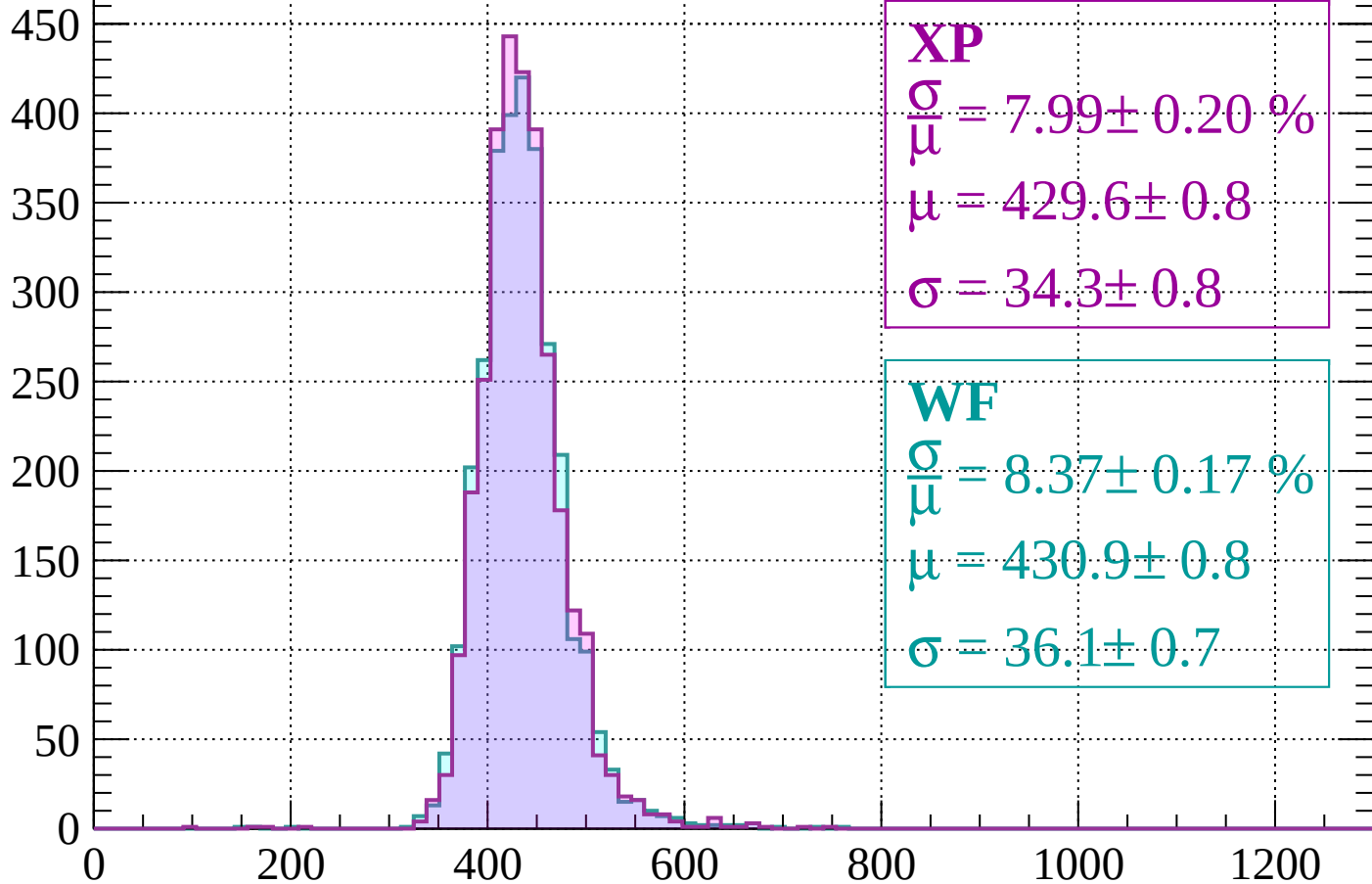
$$\mu = 427.7 \pm 0.8$$

$$\sigma = 35.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.20 \%$$

$$\mu = 429.6 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF

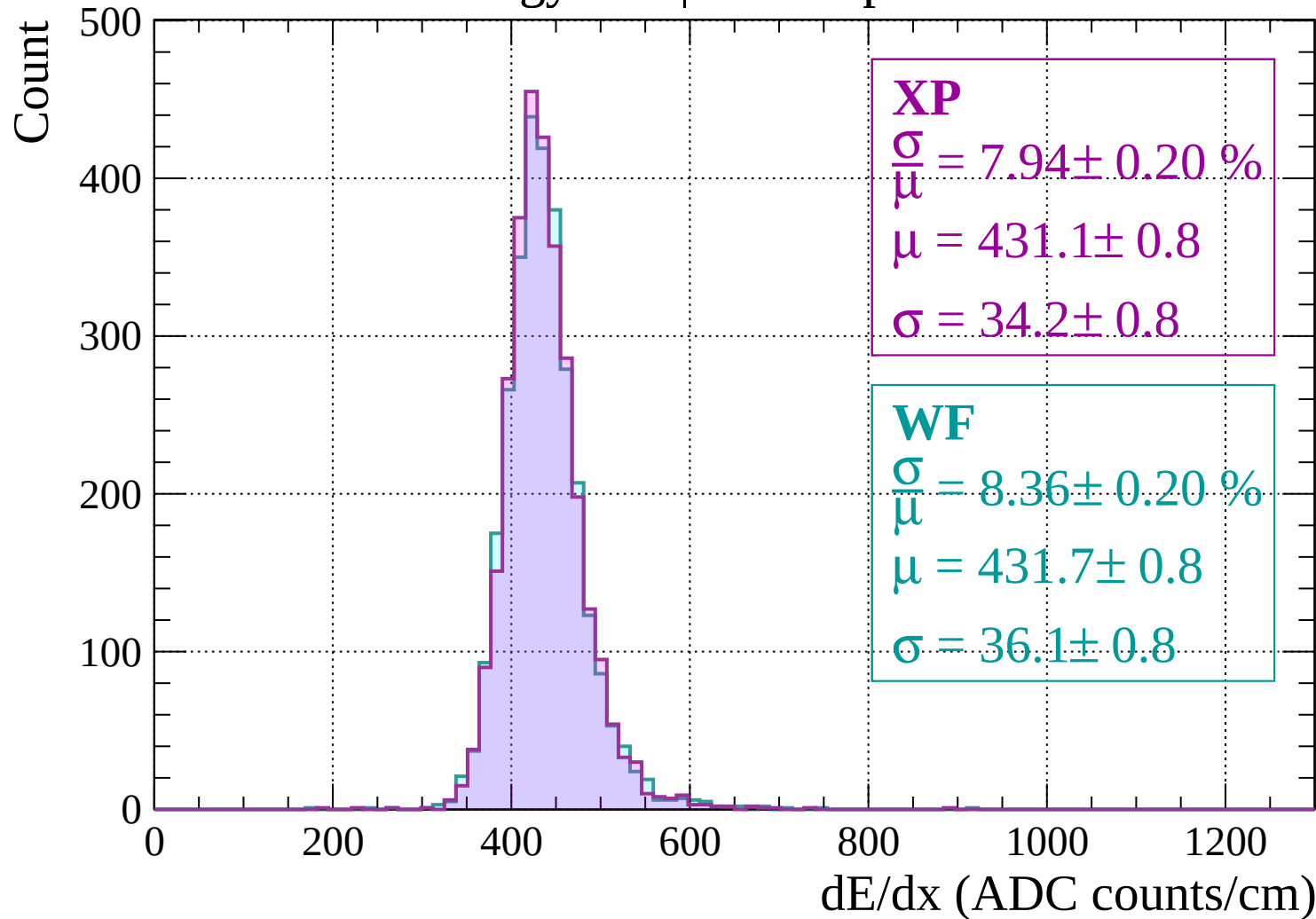
$$\frac{\sigma}{\mu} = 8.37 \pm 0.17 \%$$

$$\mu = 430.9 \pm 0.8$$

$$\sigma = 36.1 \pm 0.7$$

dE/dx (ADC counts/cm)

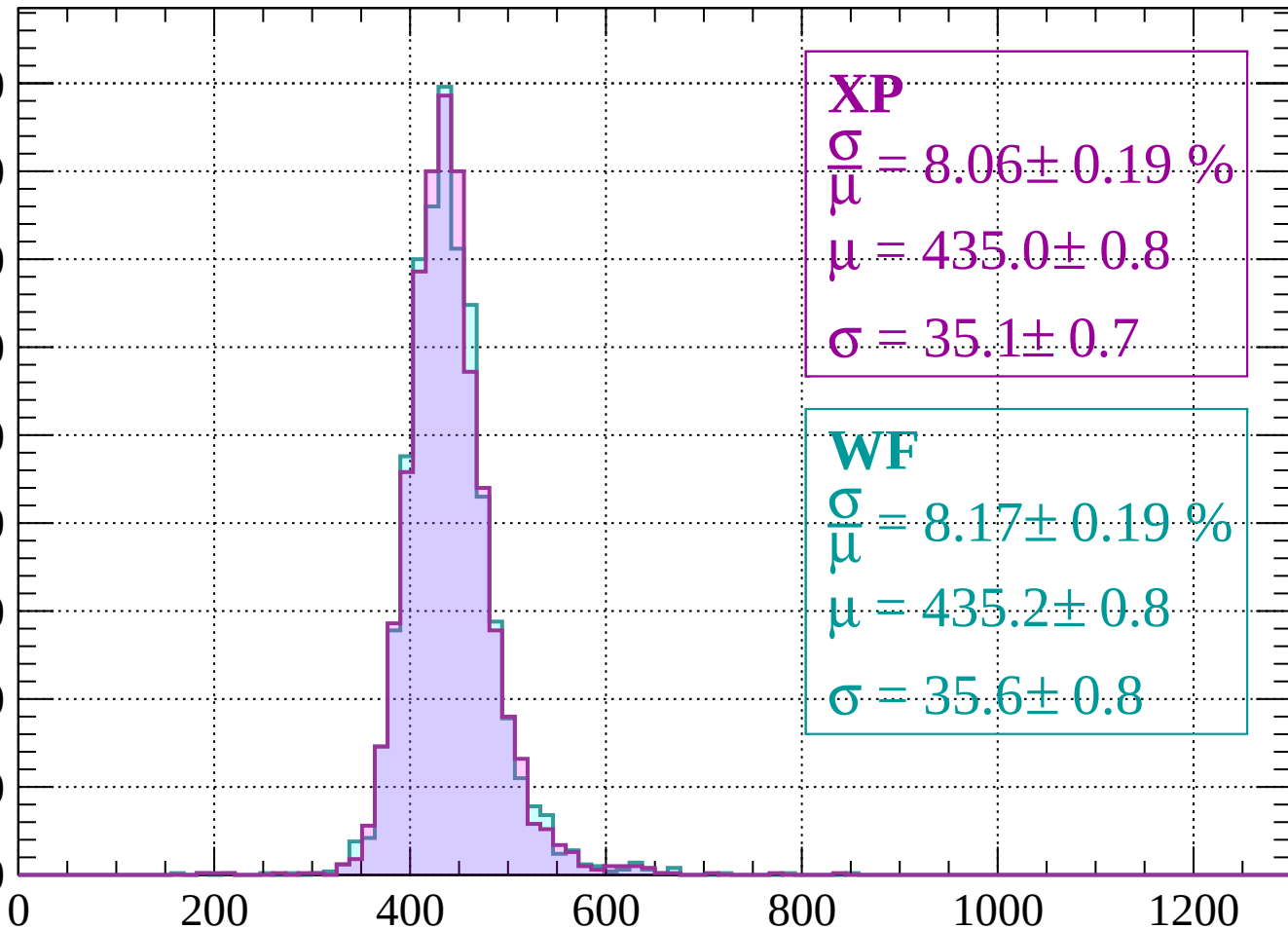
Energy loss | $1000 < p < 1040$



Energy loss | $1040 < p < 1080$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.19 \%$$

$$\mu = 435.0 \pm 0.8$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.17 \pm 0.19 \%$$

$$\mu = 435.2 \pm 0.8$$

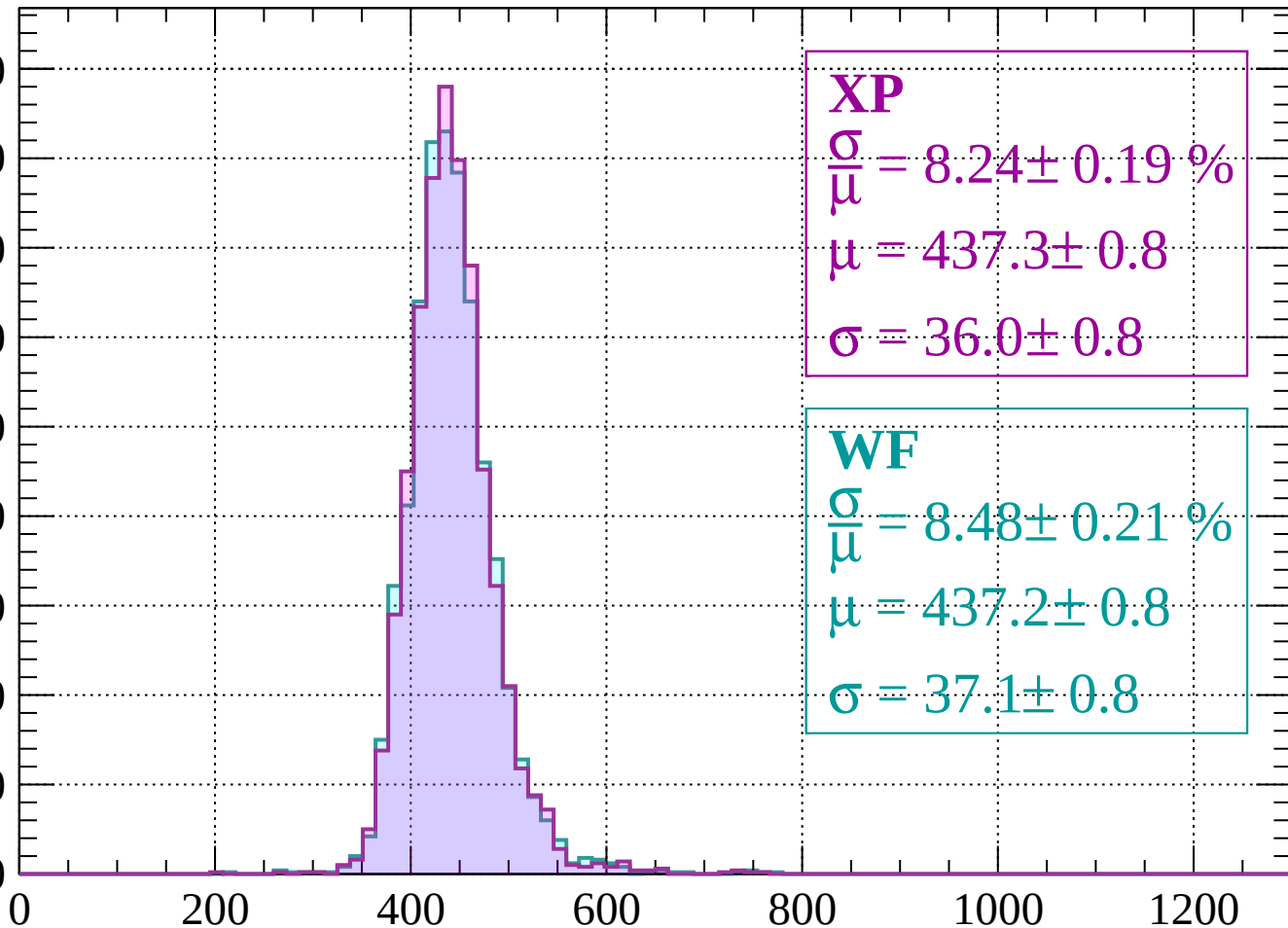
$$\sigma = 35.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\bar{\mu}} = 8.24 \pm 0.19 \%$$

$$\mu = 437.3 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.48 \pm 0.21 \%$$

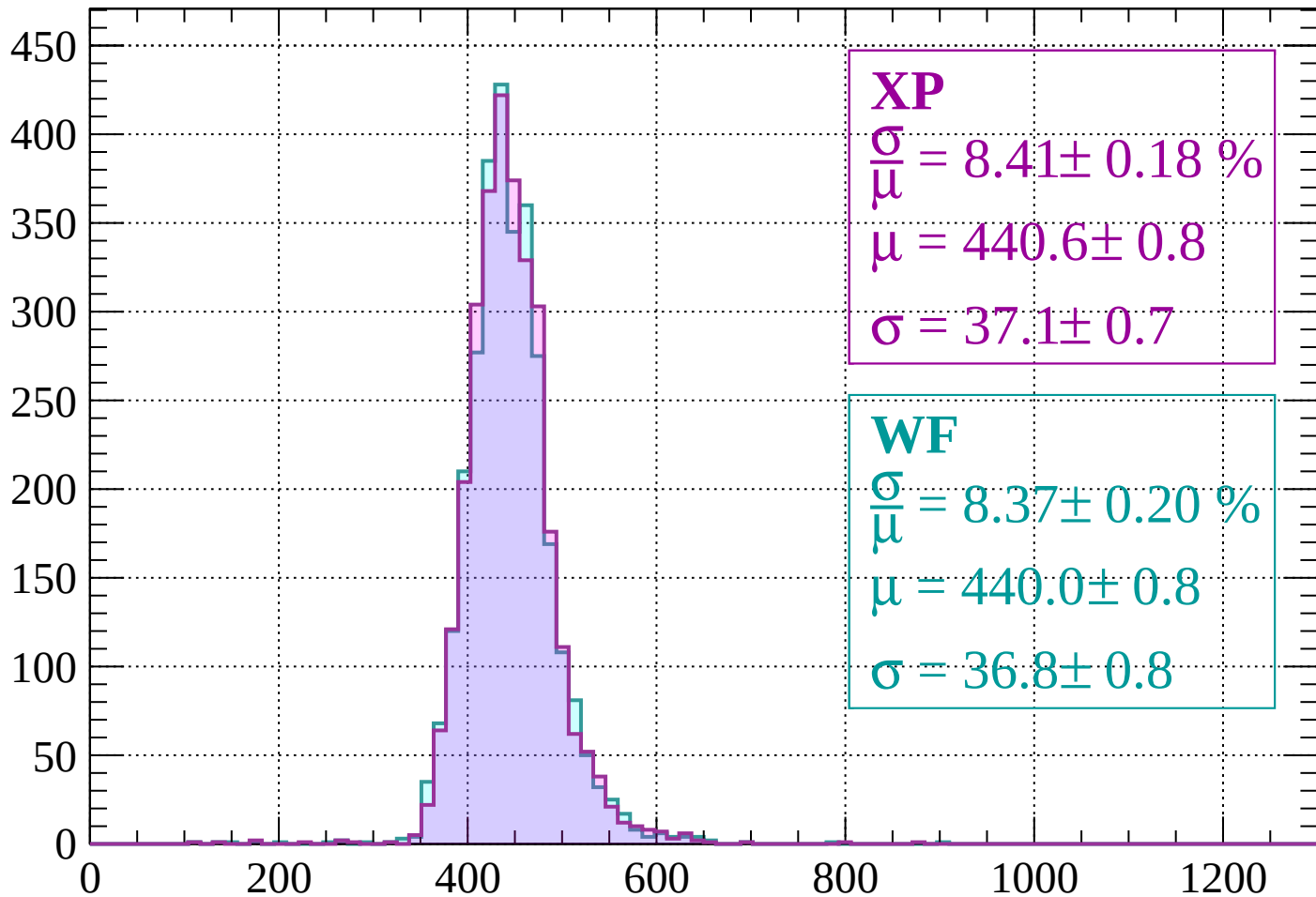
$$\mu = 437.2 \pm 0.8$$

$$\sigma = 37.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.18 \%$$

$$\mu = 440.6 \pm 0.8$$

$$\sigma = 37.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.20 \%$$

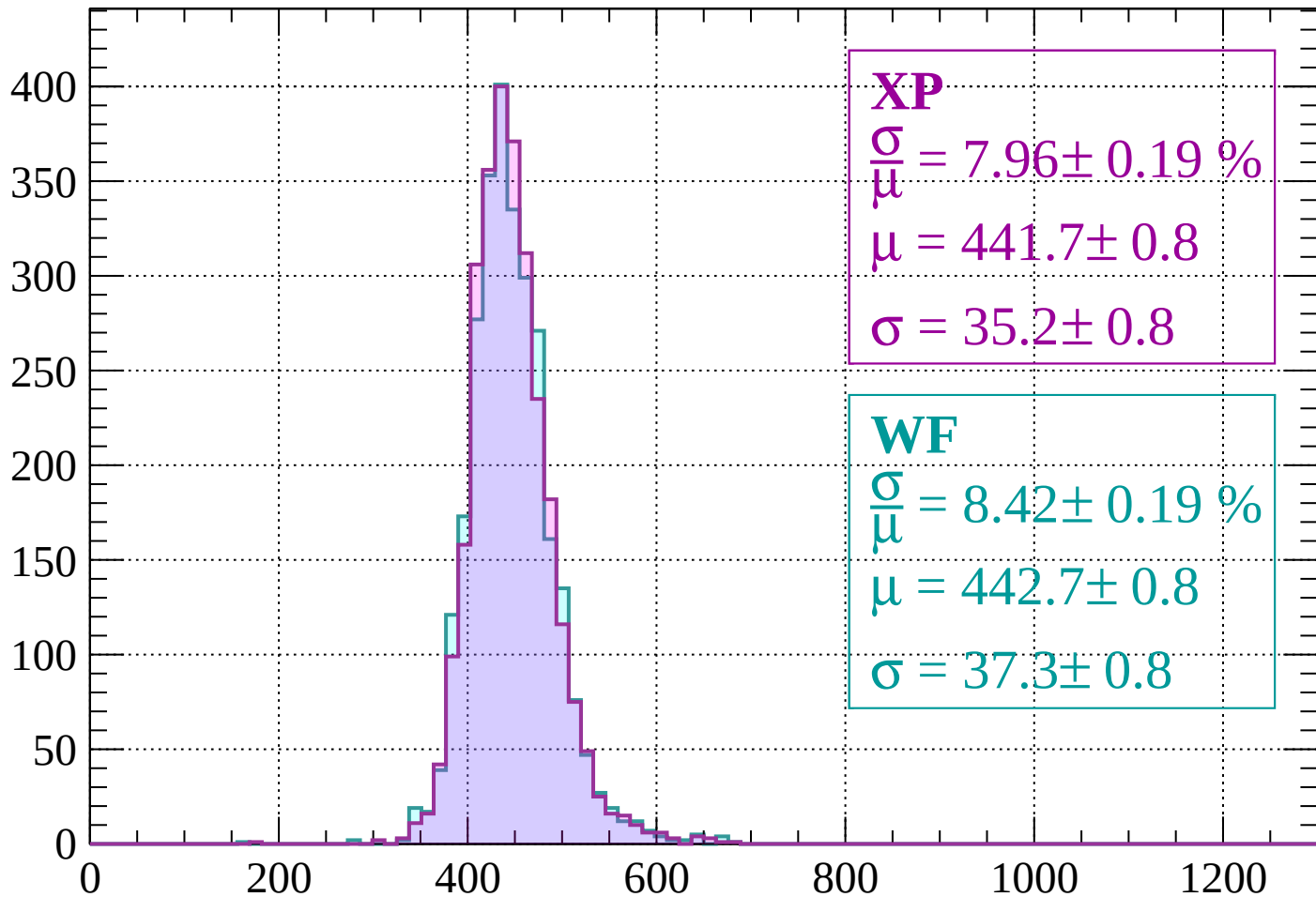
$$\mu = 440.0 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.96 \pm 0.19 \%$$

$$\mu = 441.7 \pm 0.8$$

$$\sigma = 35.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.19 \%$$

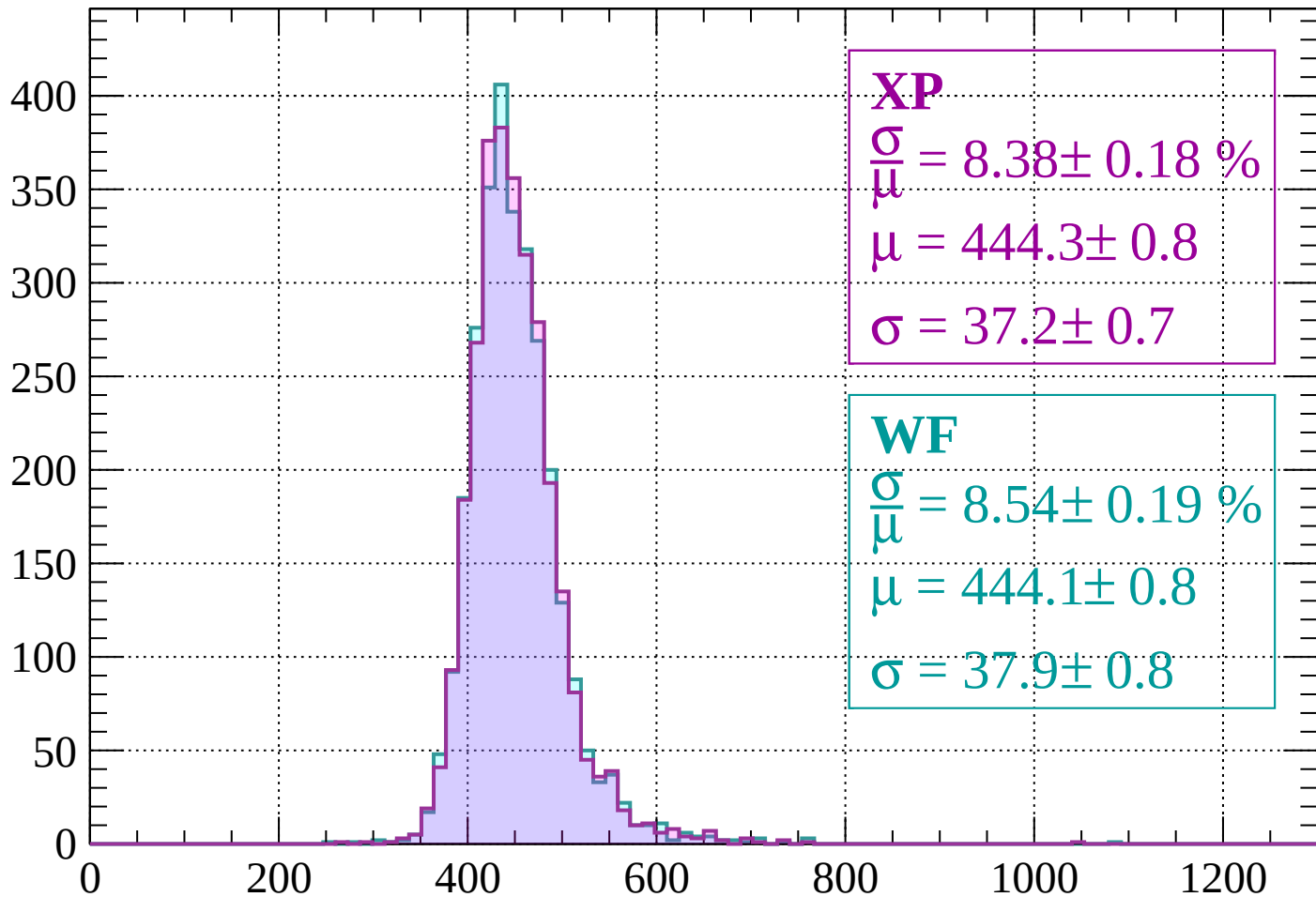
$$\mu = 442.7 \pm 0.8$$

$$\sigma = 37.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.18 \%$$

$$\mu = 444.3 \pm 0.8$$

$$\sigma = 37.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.19 \%$$

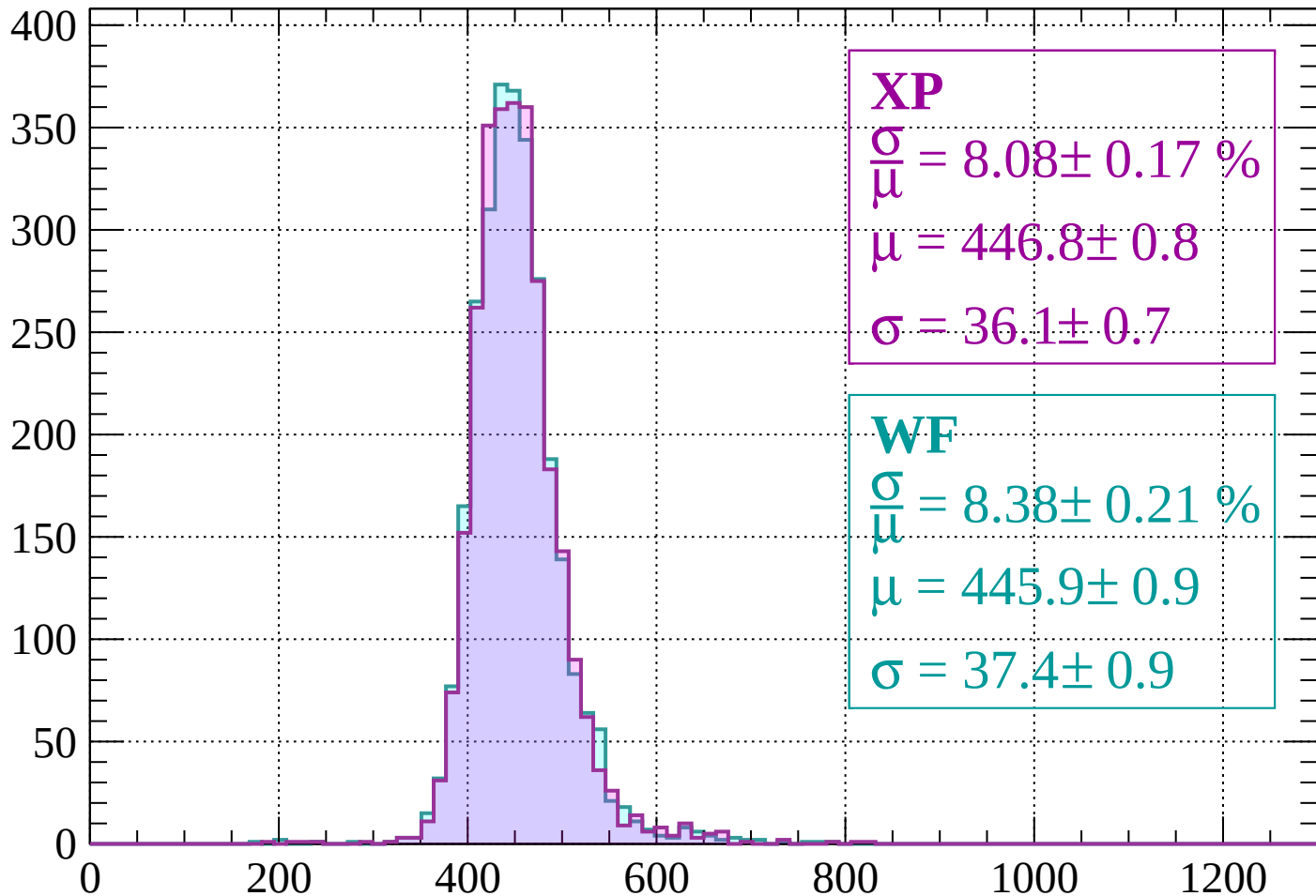
$$\mu = 444.1 \pm 0.8$$

$$\sigma = 37.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.08 \pm 0.17 \%$$

$$\mu = 446.8 \pm 0.8$$

$$\sigma = 36.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.38 \pm 0.21 \%$$

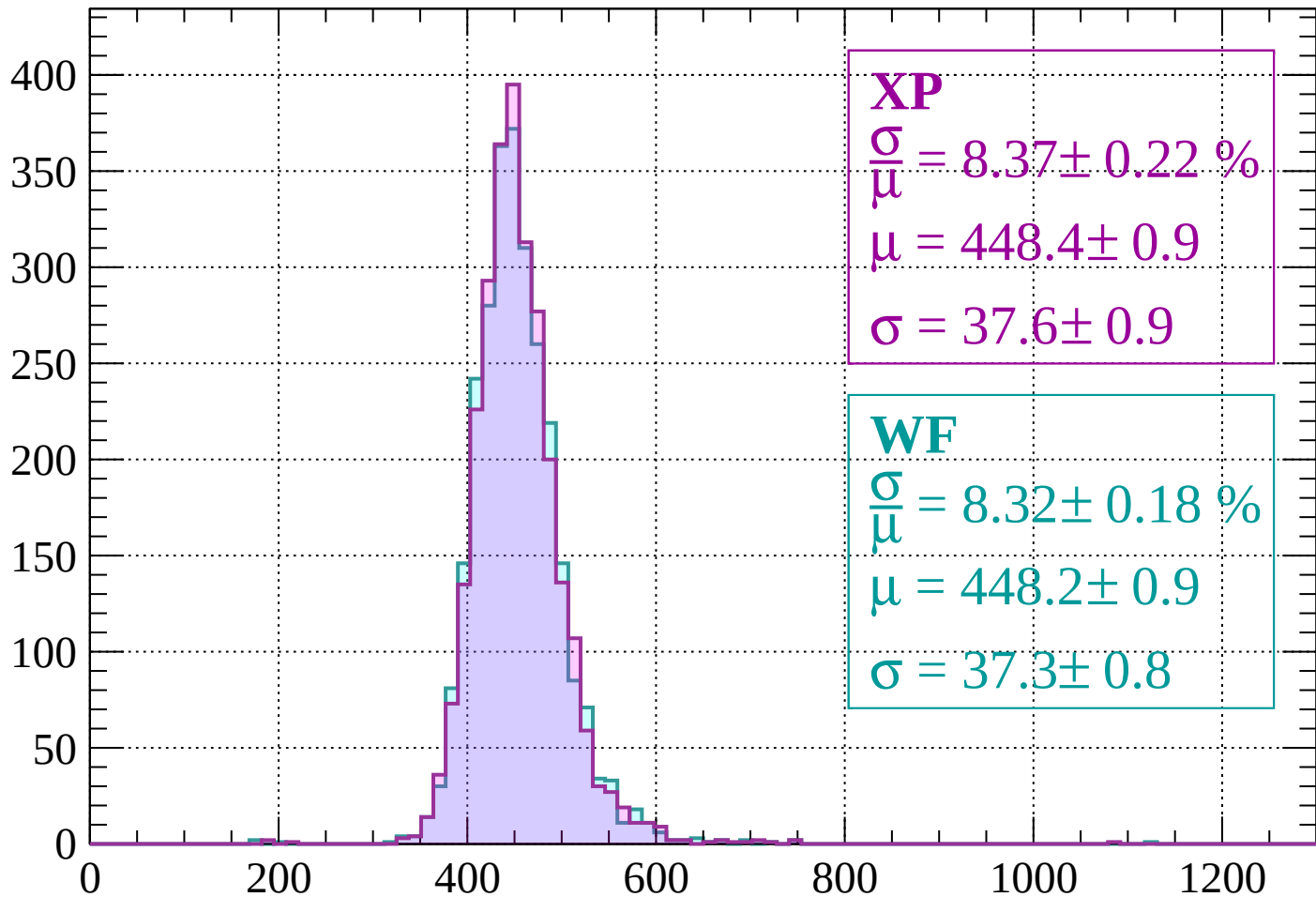
$$\mu = 445.9 \pm 0.9$$

$$\sigma = 37.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.22 \%$$

$$\mu = 448.4 \pm 0.9$$

$$\sigma = 37.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.32 \pm 0.18 \%$$

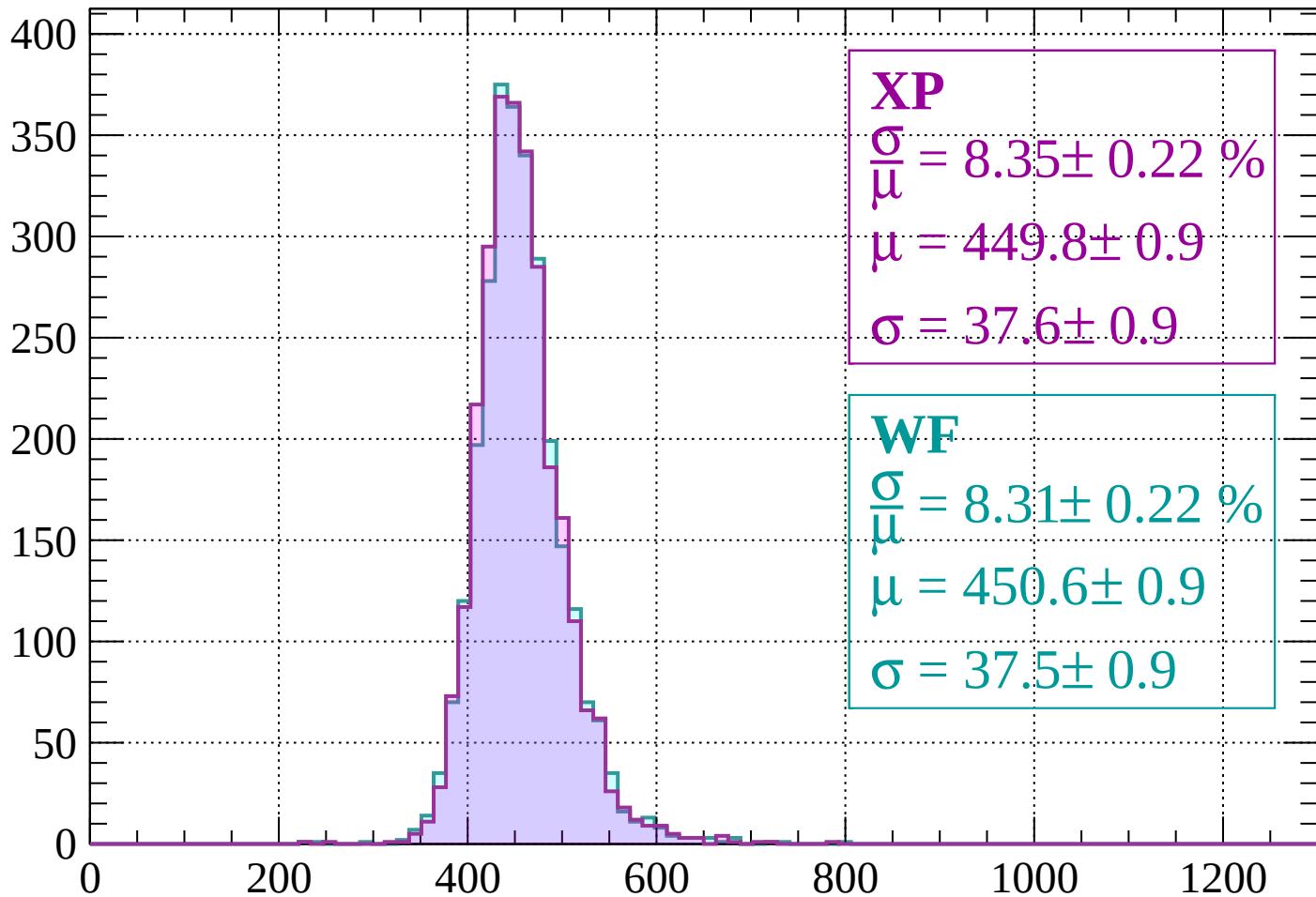
$$\mu = 448.2 \pm 0.9$$

$$\sigma = 37.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.22 \%$$

$$\mu = 449.8 \pm 0.9$$

$$\sigma = 37.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.31 \pm 0.22 \%$$

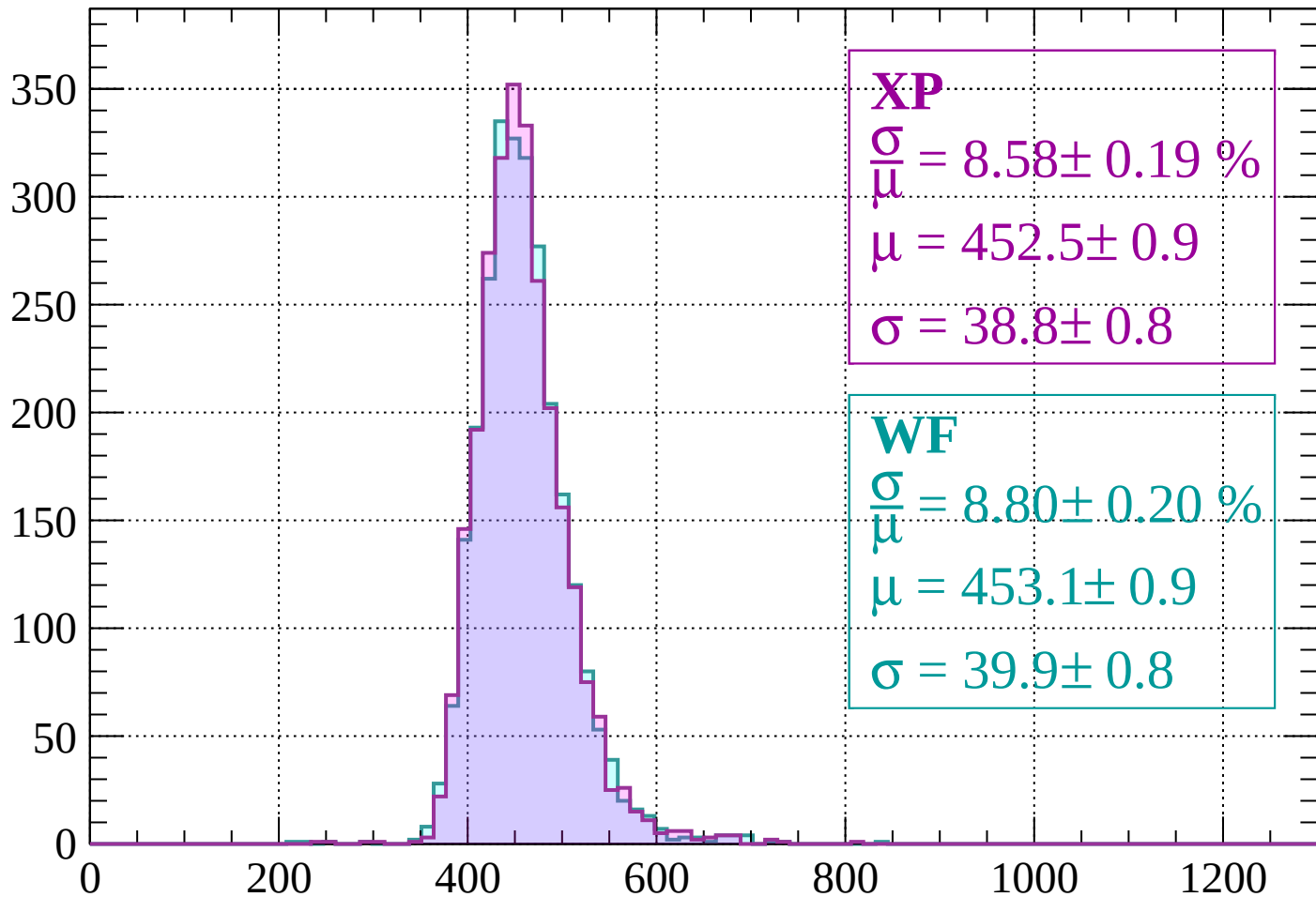
$$\mu = 450.6 \pm 0.9$$

$$\sigma = 37.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.19 \%$$

$$\mu = 452.5 \pm 0.9$$

$$\sigma = 38.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.20 \%$$

$$\mu = 453.1 \pm 0.9$$

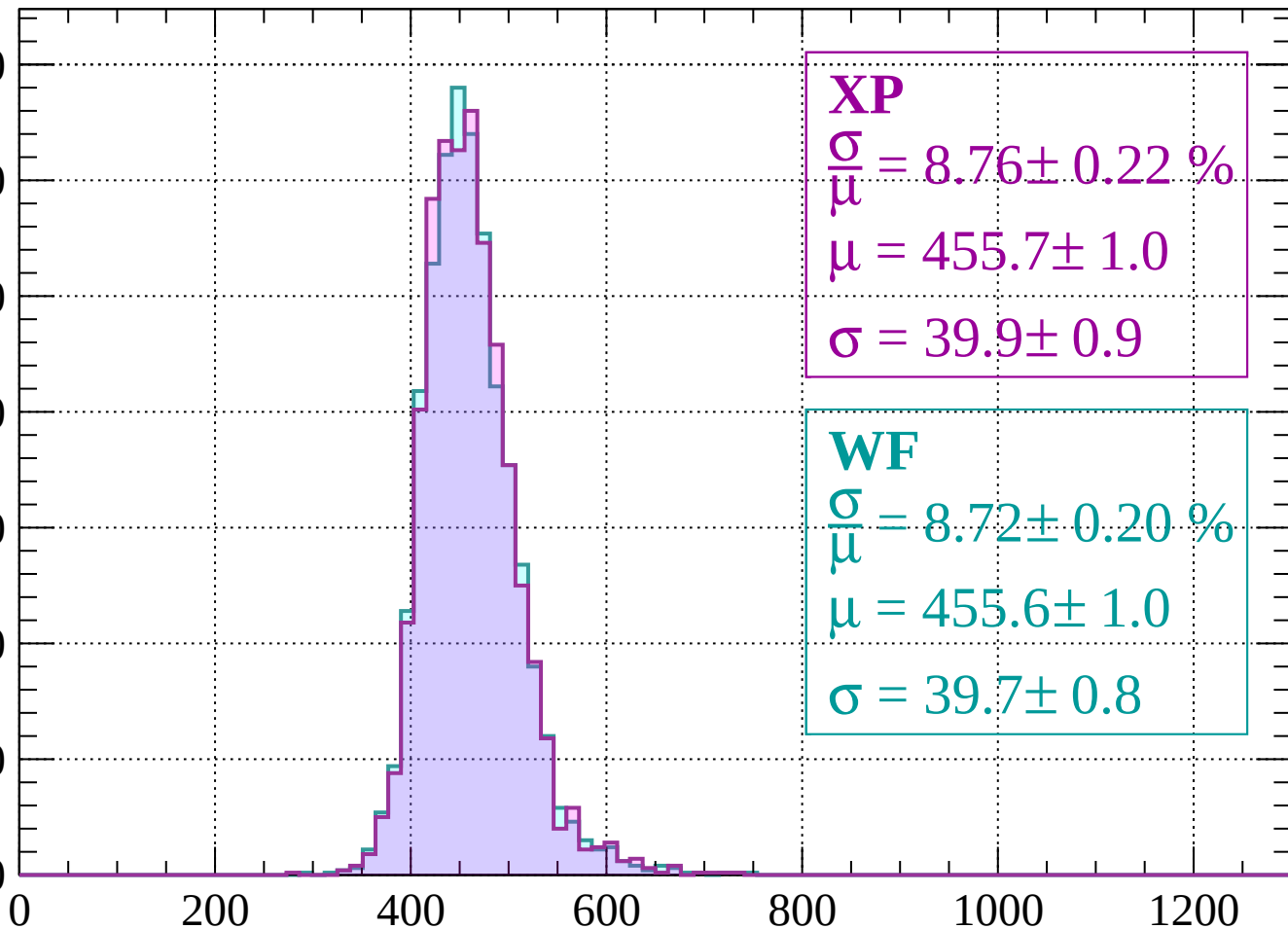
$$\sigma = 39.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.22 \%$$

$$\mu = 455.7 \pm 1.0$$

$$\sigma = 39.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.72 \pm 0.20 \%$$

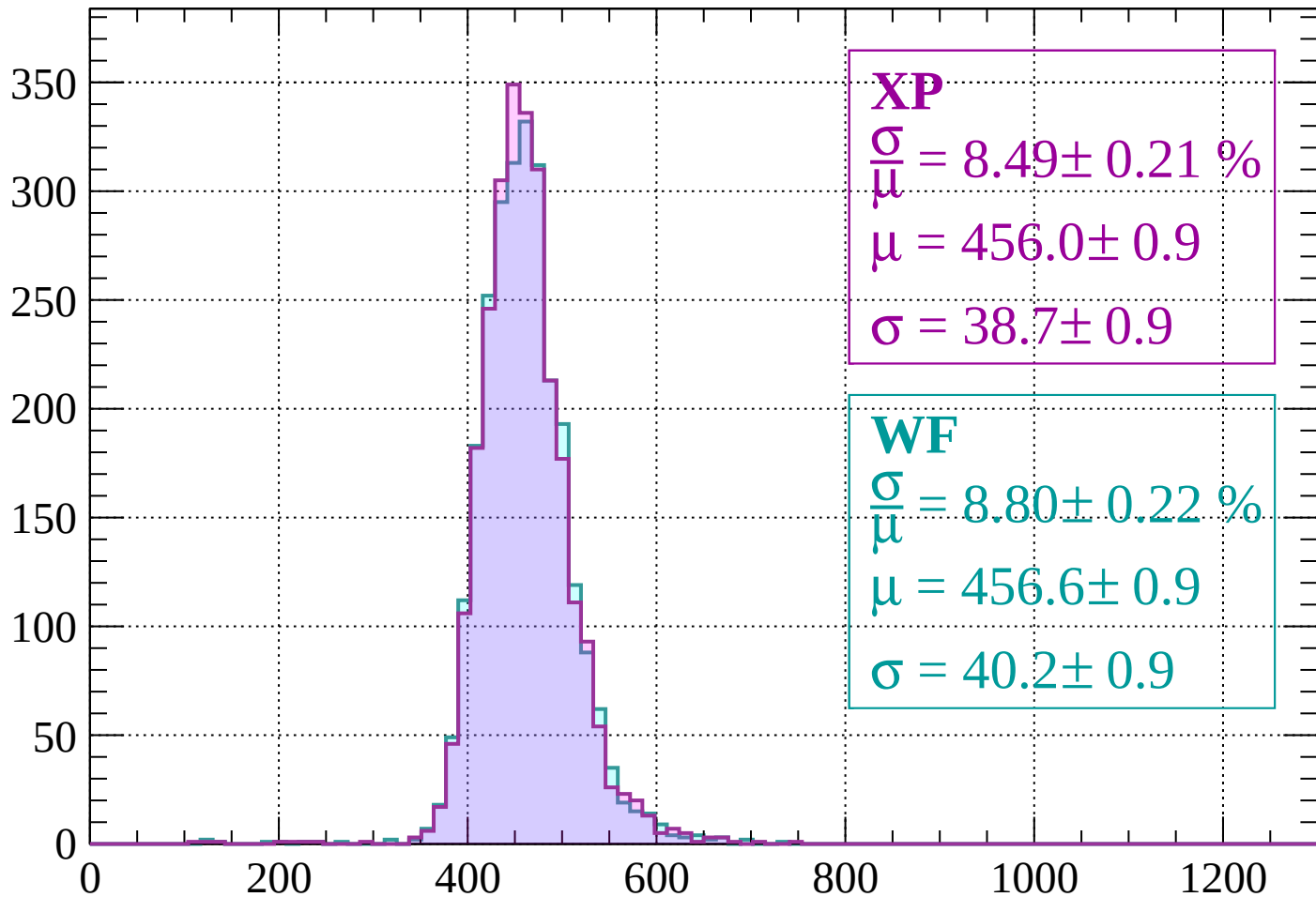
$$\mu = 455.6 \pm 1.0$$

$$\sigma = 39.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.49 \pm 0.21 \%$$

$$\mu = 456.0 \pm 0.9$$

$$\sigma = 38.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.22 \%$$

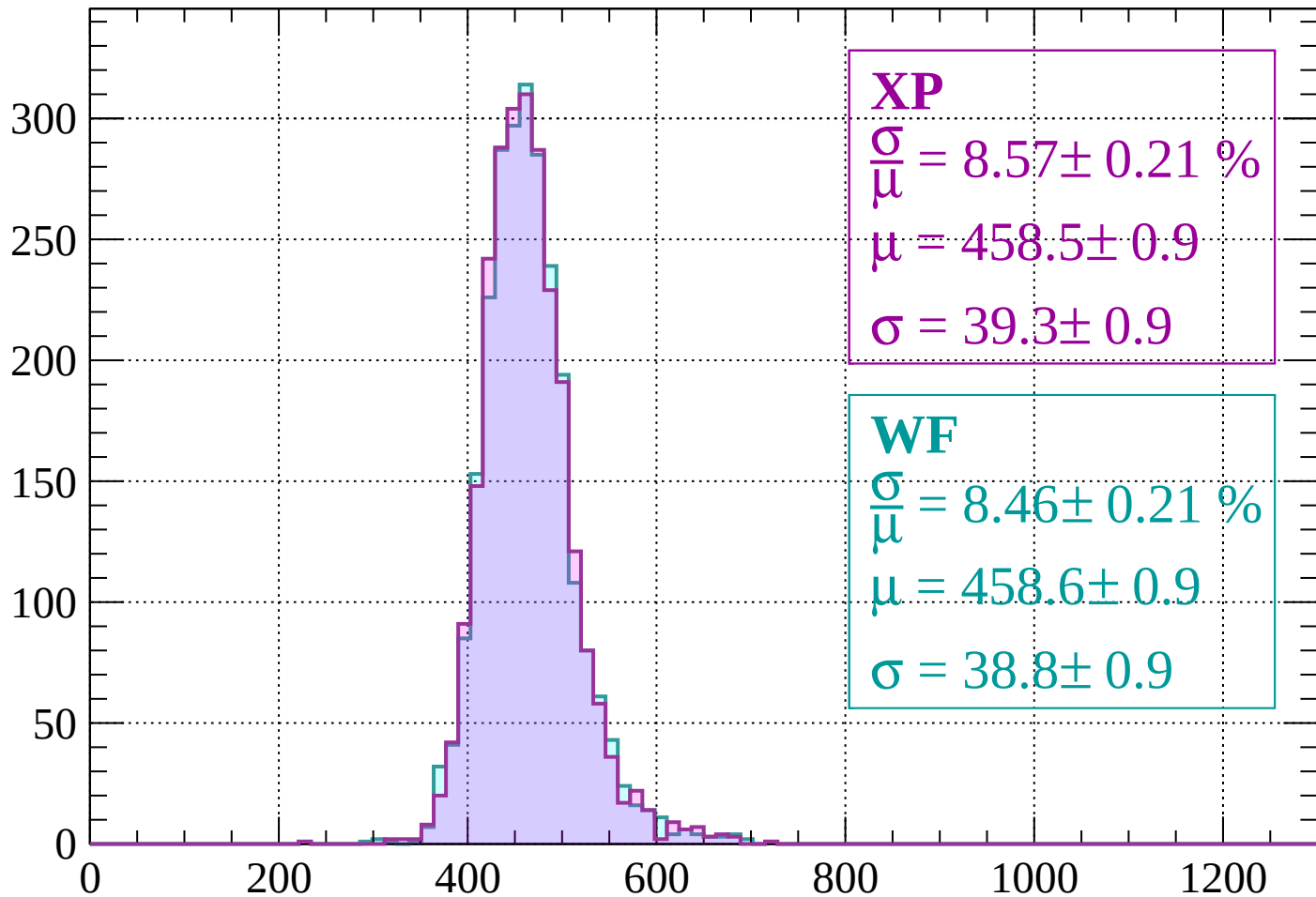
$$\mu = 456.6 \pm 0.9$$

$$\sigma = 40.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.21 \%$$

$$\mu = 458.5 \pm 0.9$$

$$\sigma = 39.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.21 \%$$

$$\mu = 458.6 \pm 0.9$$

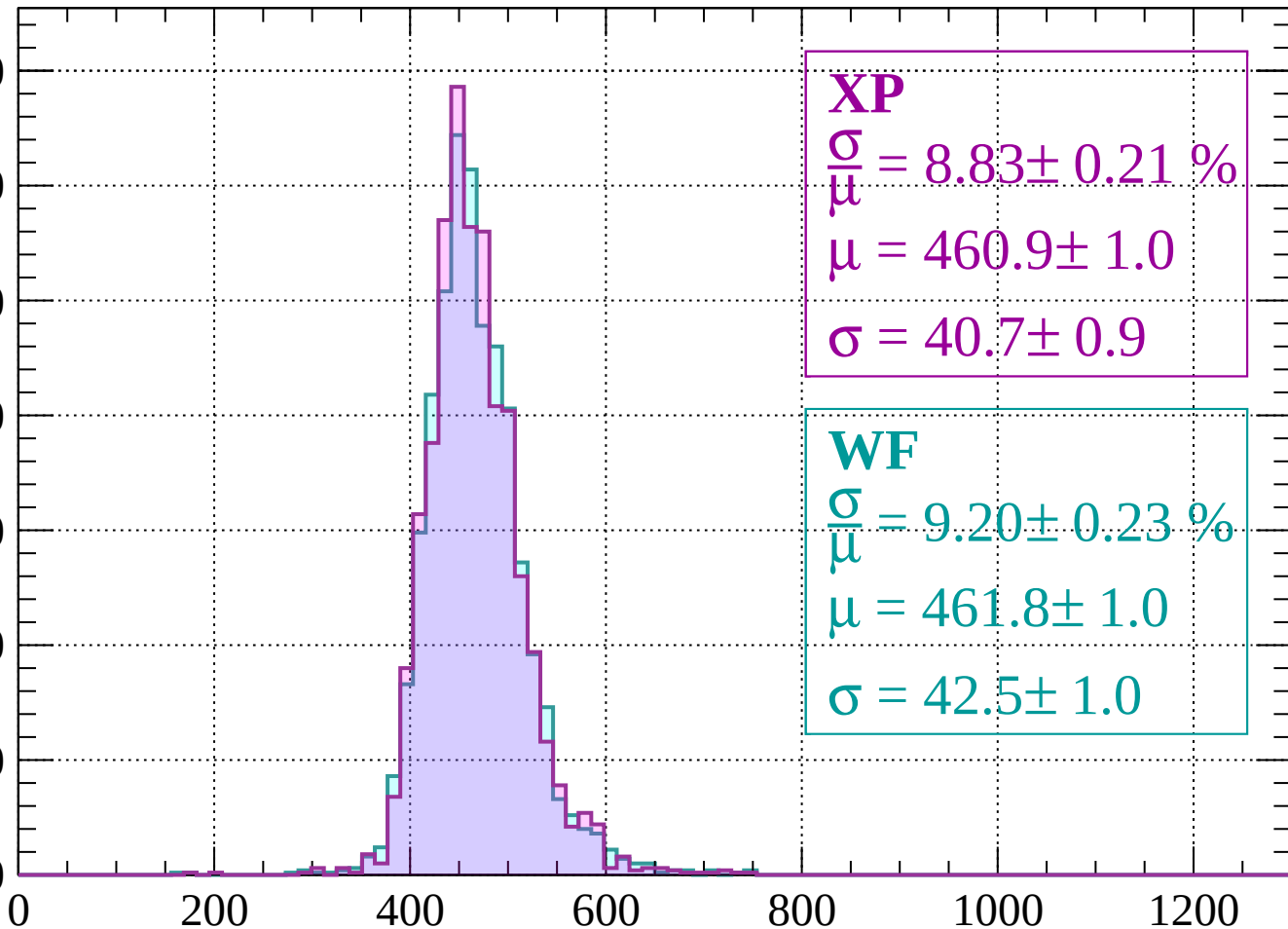
$$\sigma = 38.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.21 \%$$

$$\mu = 460.9 \pm 1.0$$

$$\sigma = 40.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.23 \%$$

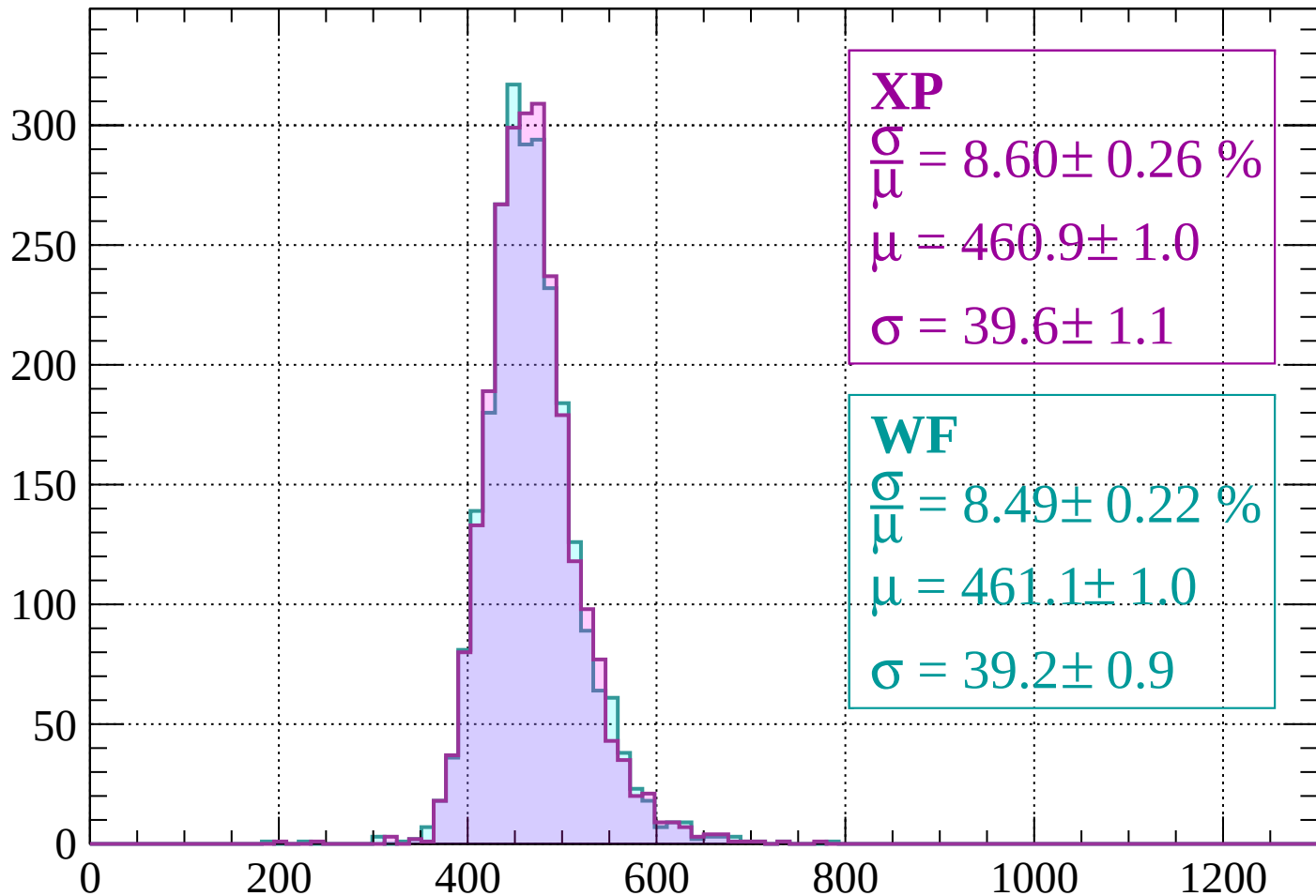
$$\mu = 461.8 \pm 1.0$$

$$\sigma = 42.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.26 \%$$

$$\mu = 460.9 \pm 1.0$$

$$\sigma = 39.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.22 \%$$

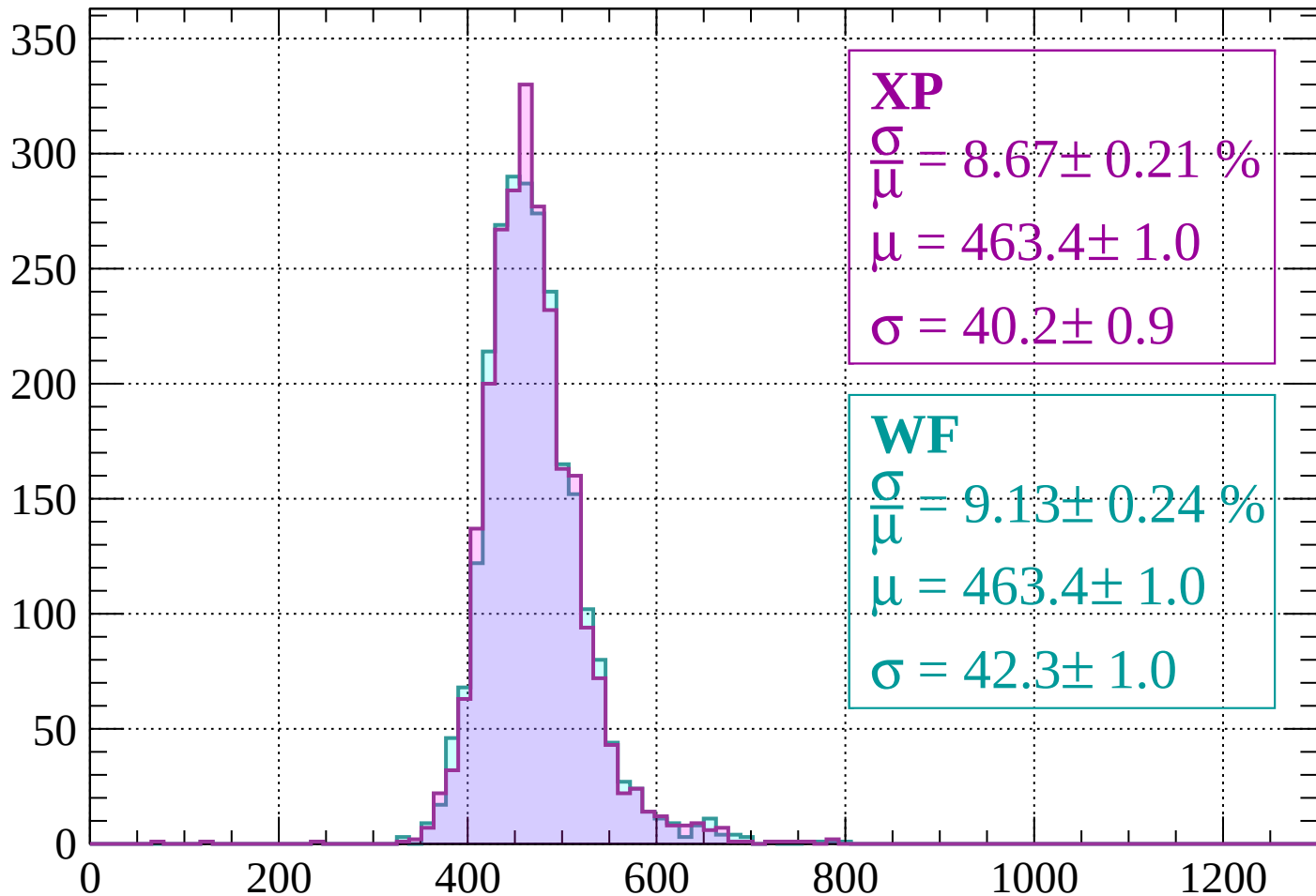
$$\mu = 461.1 \pm 1.0$$

$$\sigma = 39.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count



XP

$$\frac{\sigma}{\mu} = 8.67 \pm 0.21 \%$$

$$\mu = 463.4 \pm 1.0$$

$$\sigma = 40.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.13 \pm 0.24 \%$$

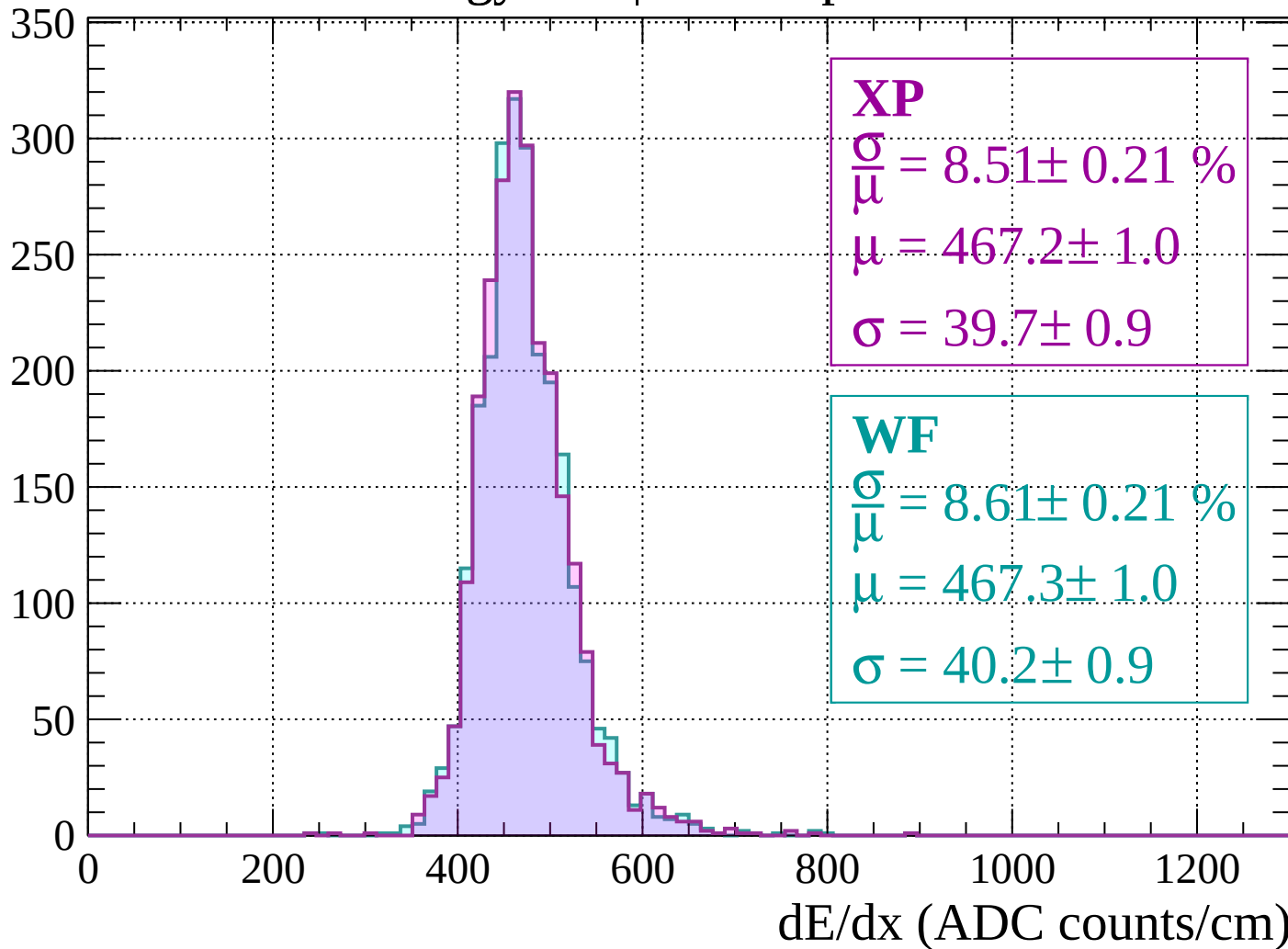
$$\mu = 463.4 \pm 1.0$$

$$\sigma = 42.3 \pm 1.0$$

dE/dx (ADC counts/cm)

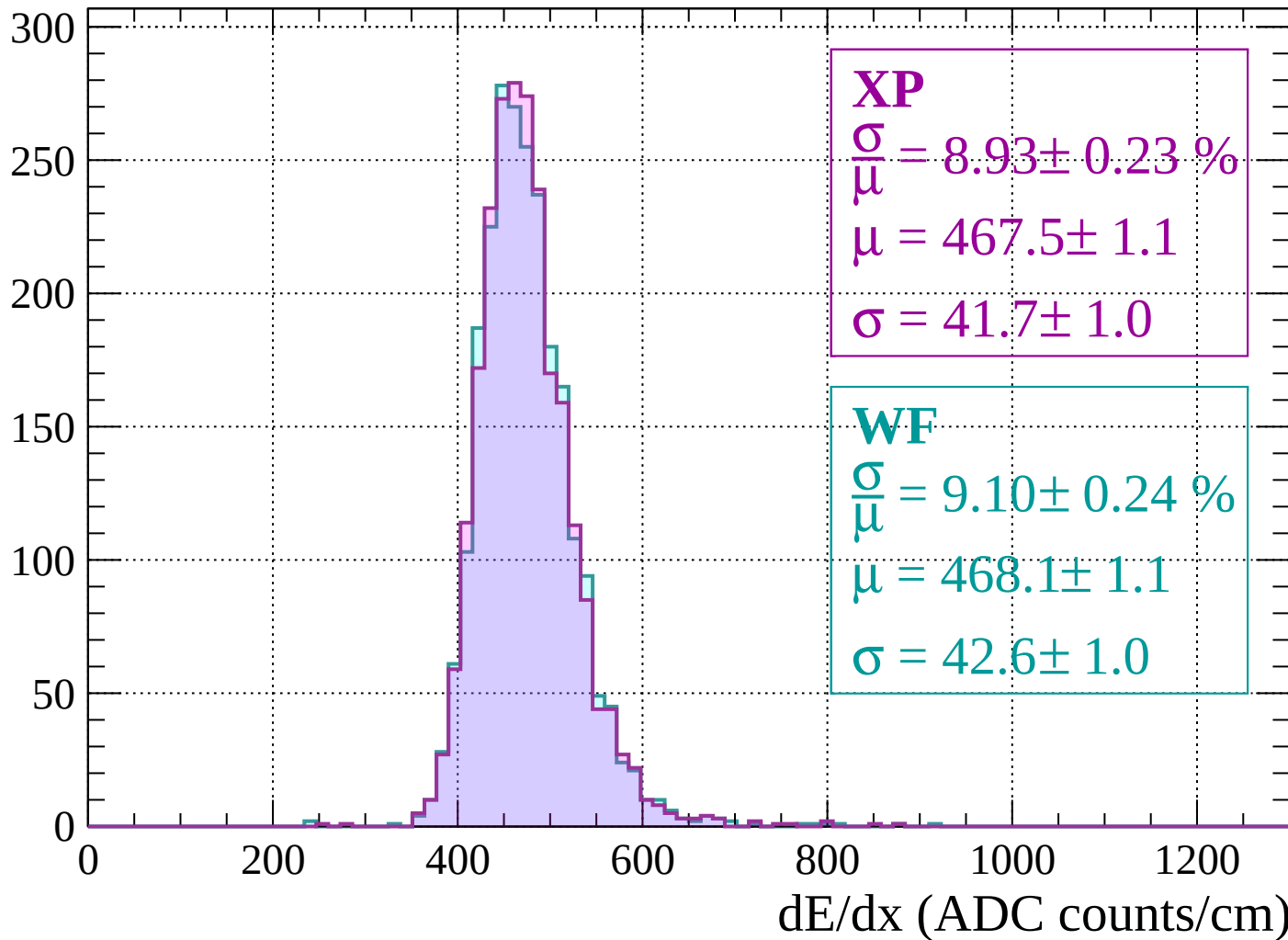
Energy loss | $1640 < p < 1680$

Count



Energy loss | $1680 < p < 1720$

Count



Energy loss | $1720 < p < 1760$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.22 \%$$

$$\mu = 469.6 \pm 1.0$$

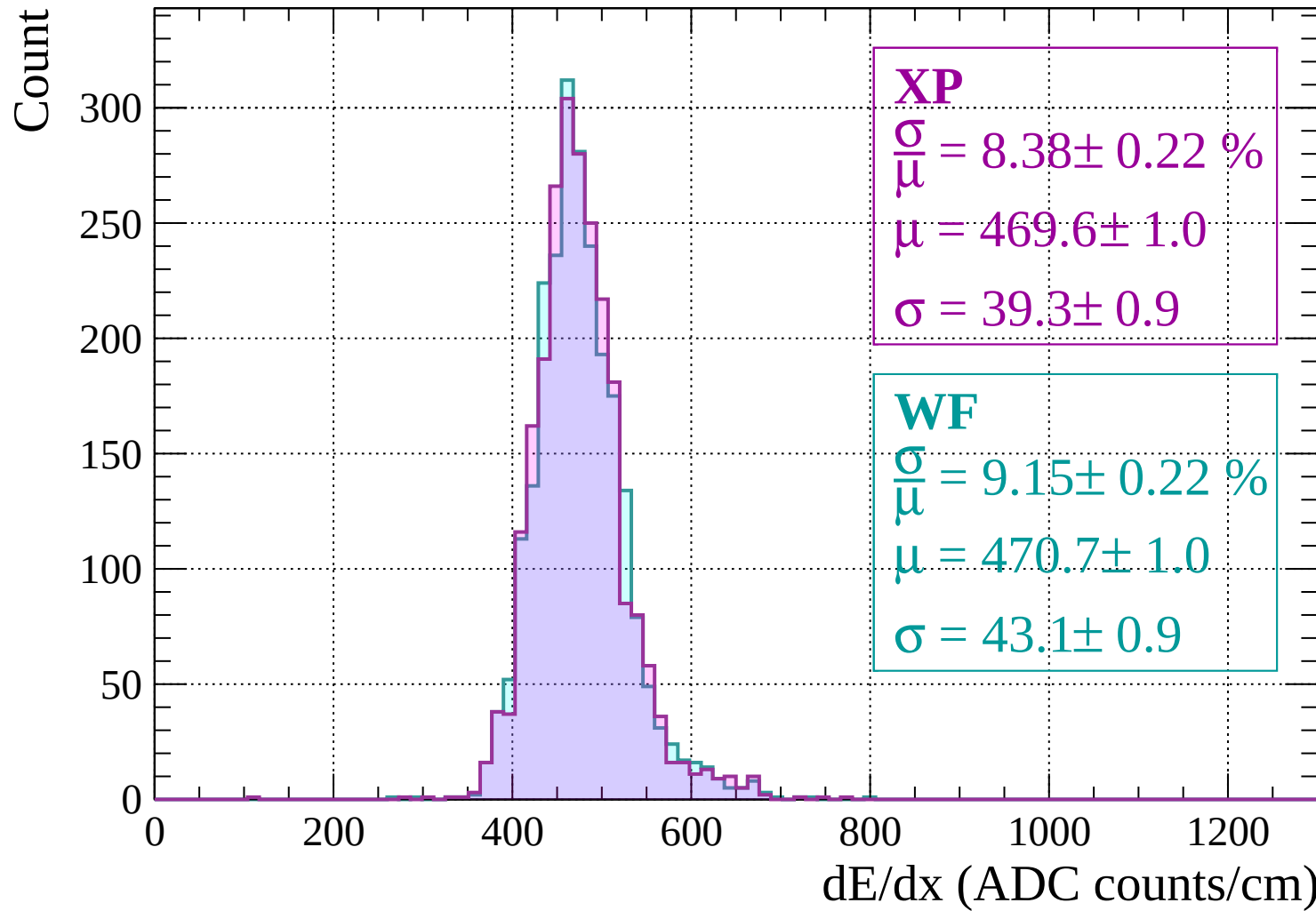
$$\sigma = 39.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.22 \%$$

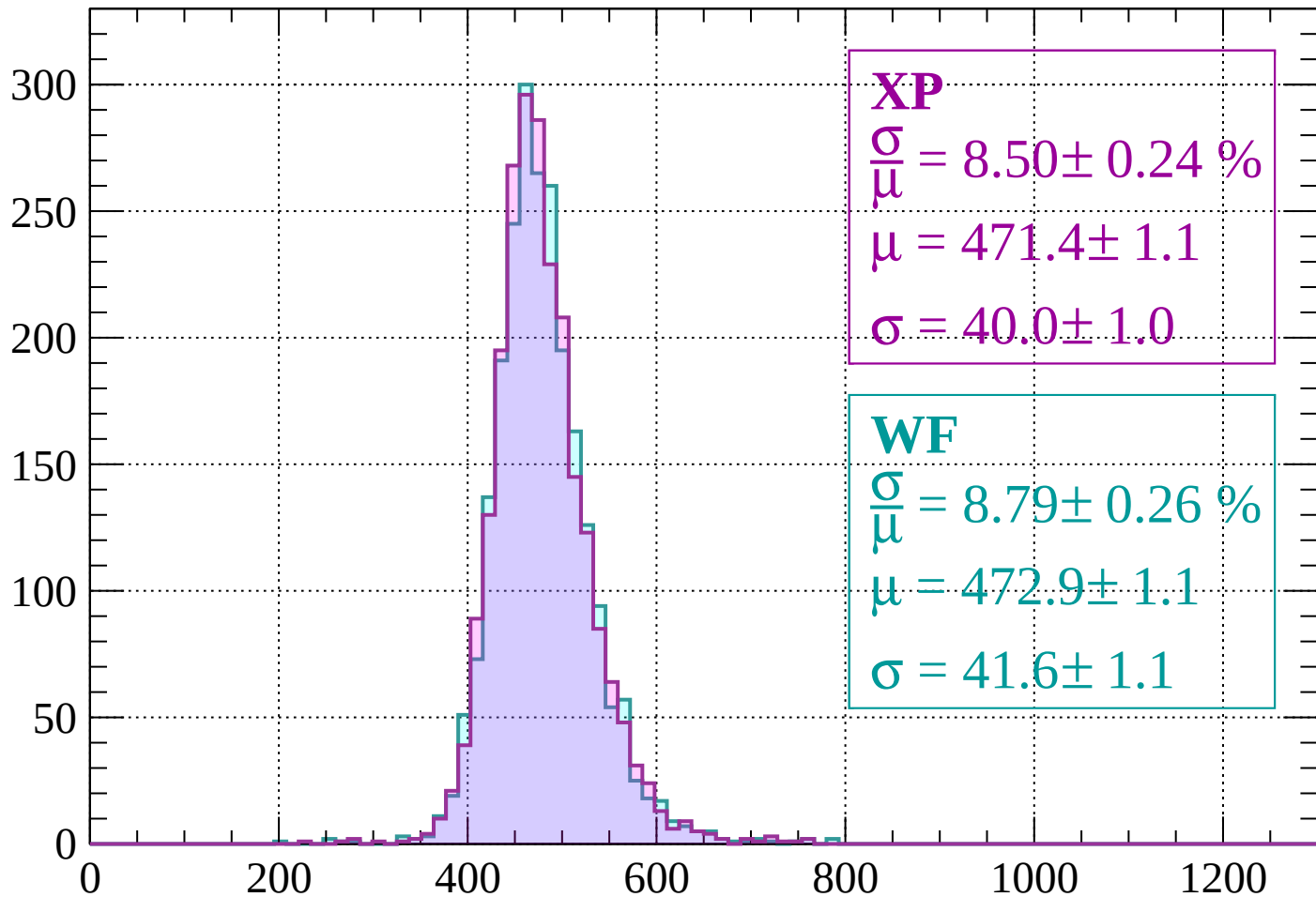
$$\mu = 470.7 \pm 1.0$$

$$\sigma = 43.1 \pm 0.9$$



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 8.50 \pm 0.24 \%$$

$$\mu = 471.4 \pm 1.1$$

$$\sigma = 40.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.79 \pm 0.26 \%$$

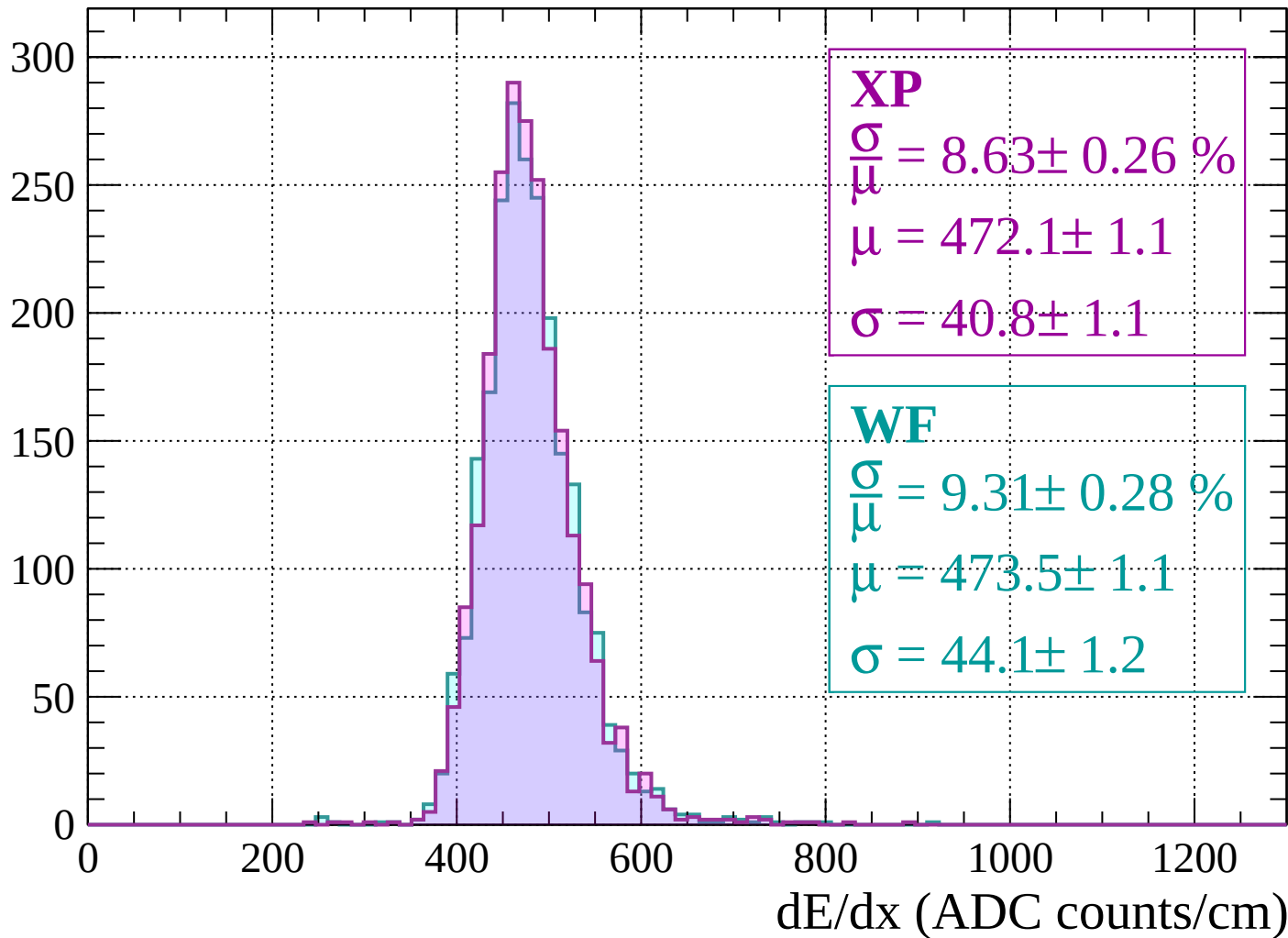
$$\mu = 472.9 \pm 1.1$$

$$\sigma = 41.6 \pm 1.1$$

dE/dx (ADC counts/cm)

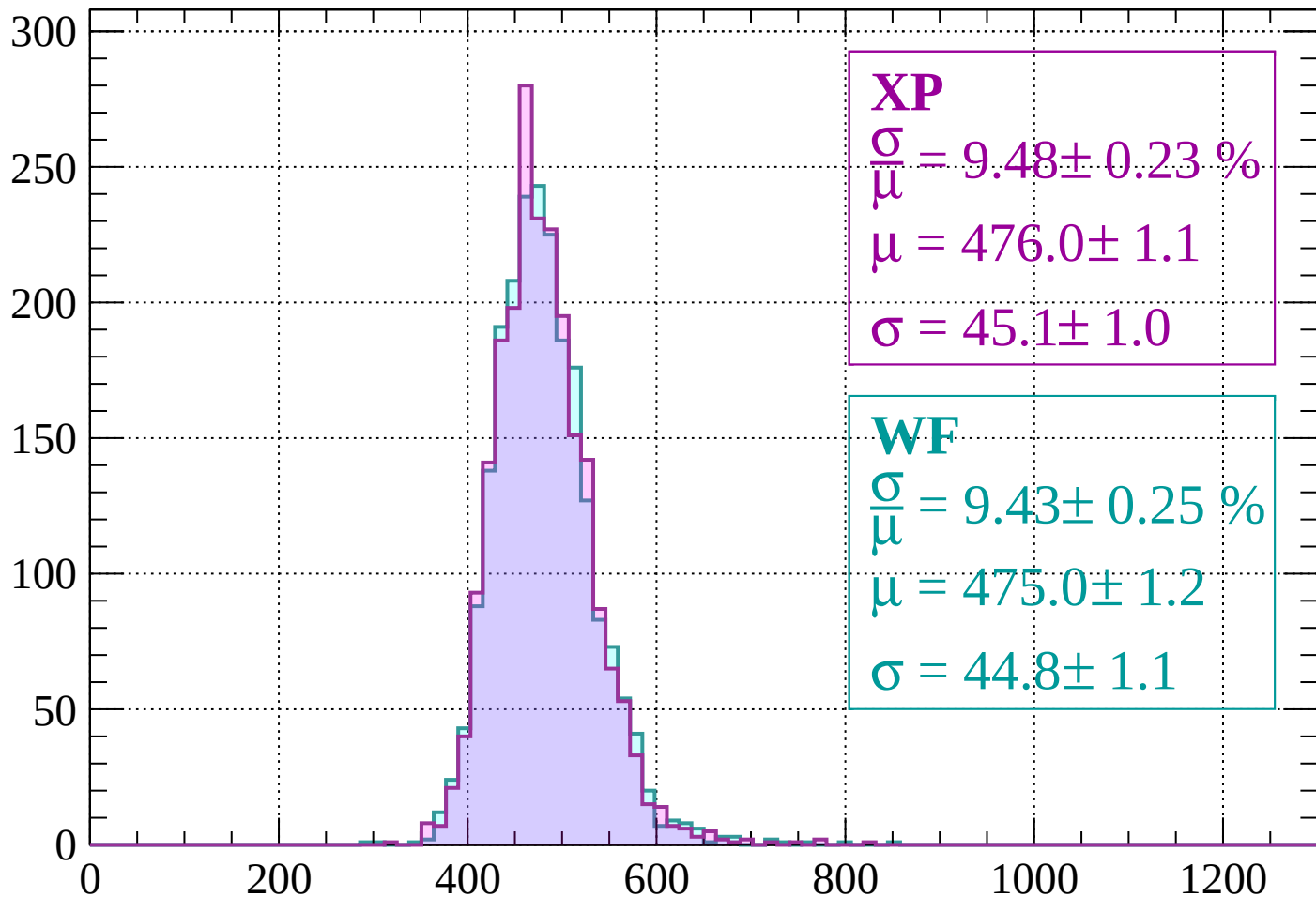
Energy loss | $1800 < p < 1840$

Count



Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.23 \%$$

$$\mu = 476.0 \pm 1.1$$

$$\sigma = 45.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.43 \pm 0.25 \%$$

$$\mu = 475.0 \pm 1.2$$

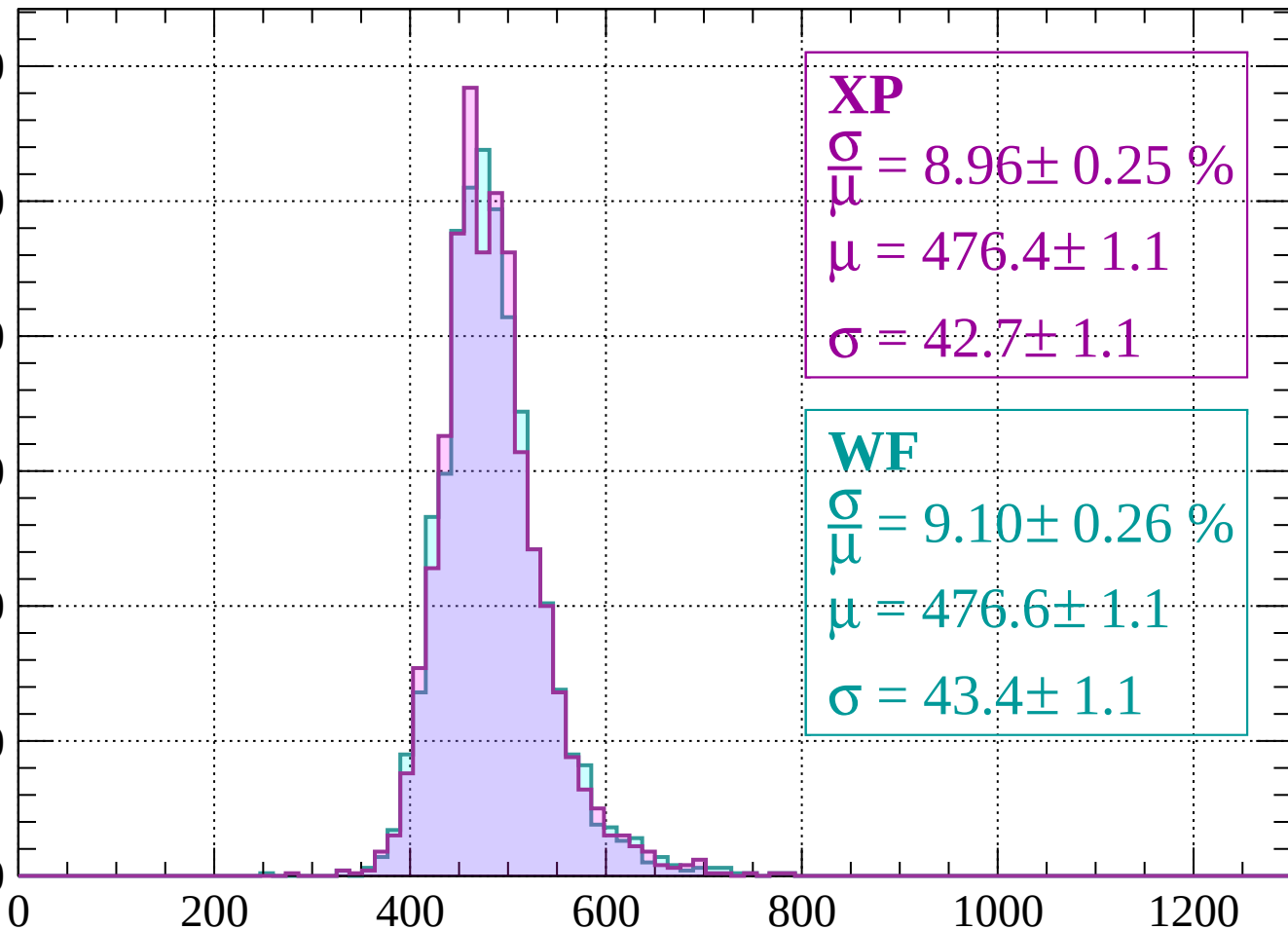
$$\sigma = 44.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.96 \pm 0.25 \%$$

$$\mu = 476.4 \pm 1.1$$

$$\sigma = 42.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.26 \%$$

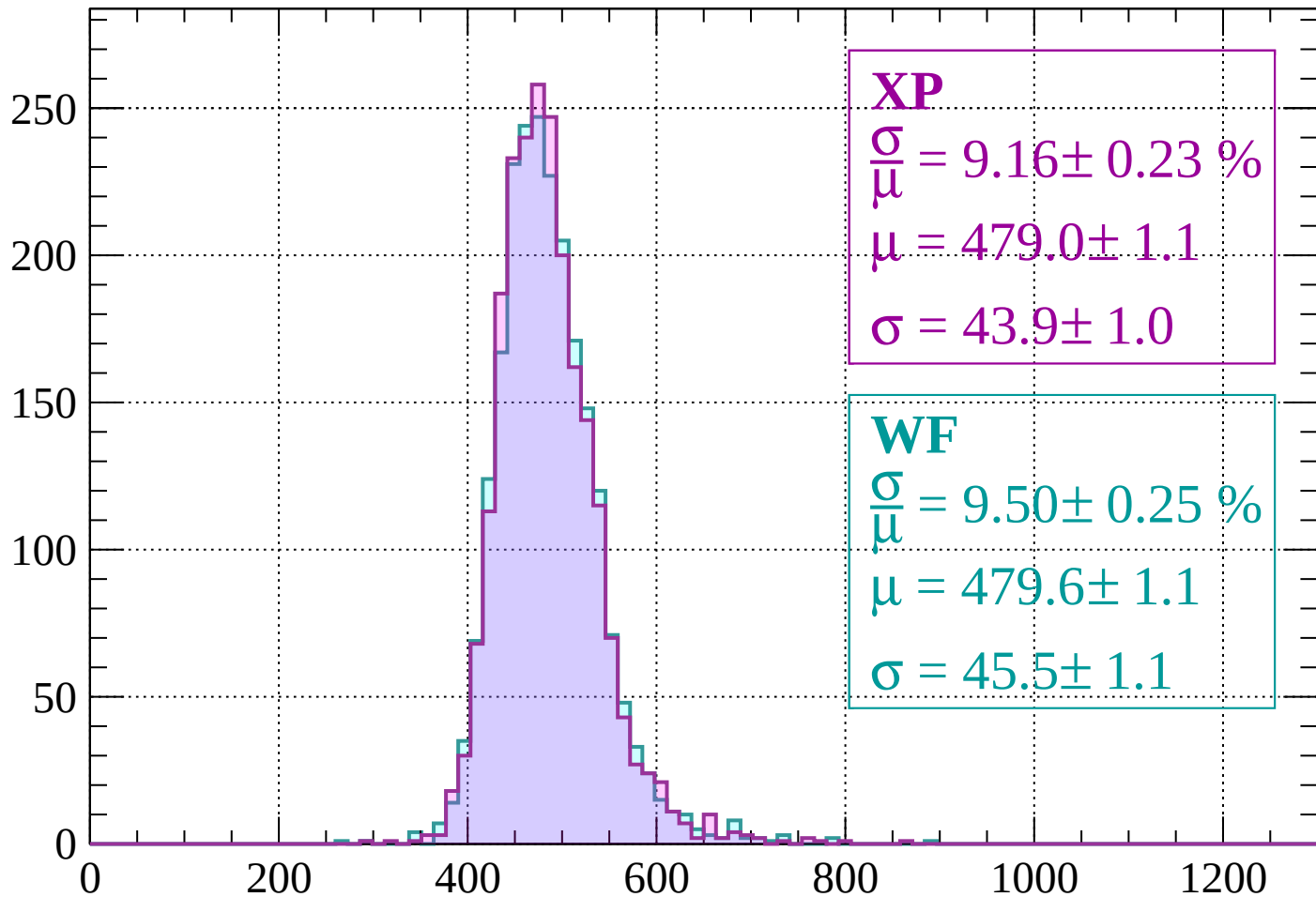
$$\mu = 476.6 \pm 1.1$$

$$\sigma = 43.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.23 \%$$

$$\mu = 479.0 \pm 1.1$$

$$\sigma = 43.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.25 \%$$

$$\mu = 479.6 \pm 1.1$$

$$\sigma = 45.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1960 < p < 2000$

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.63 \pm 0.23 \%$$

$$\mu = 480.3 \pm 1.1$$

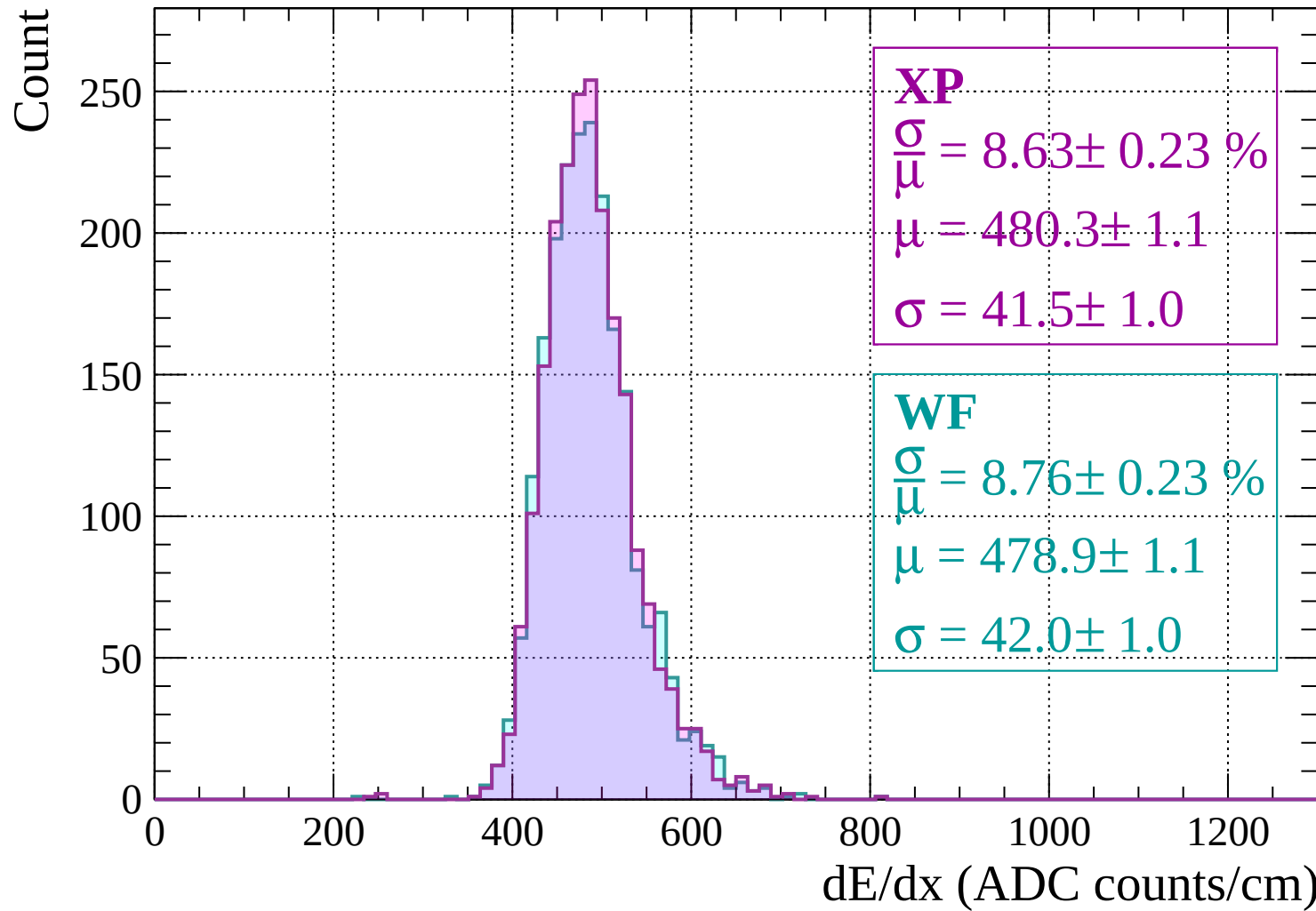
$$\sigma = 41.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.23 \%$$

$$\mu = 478.9 \pm 1.1$$

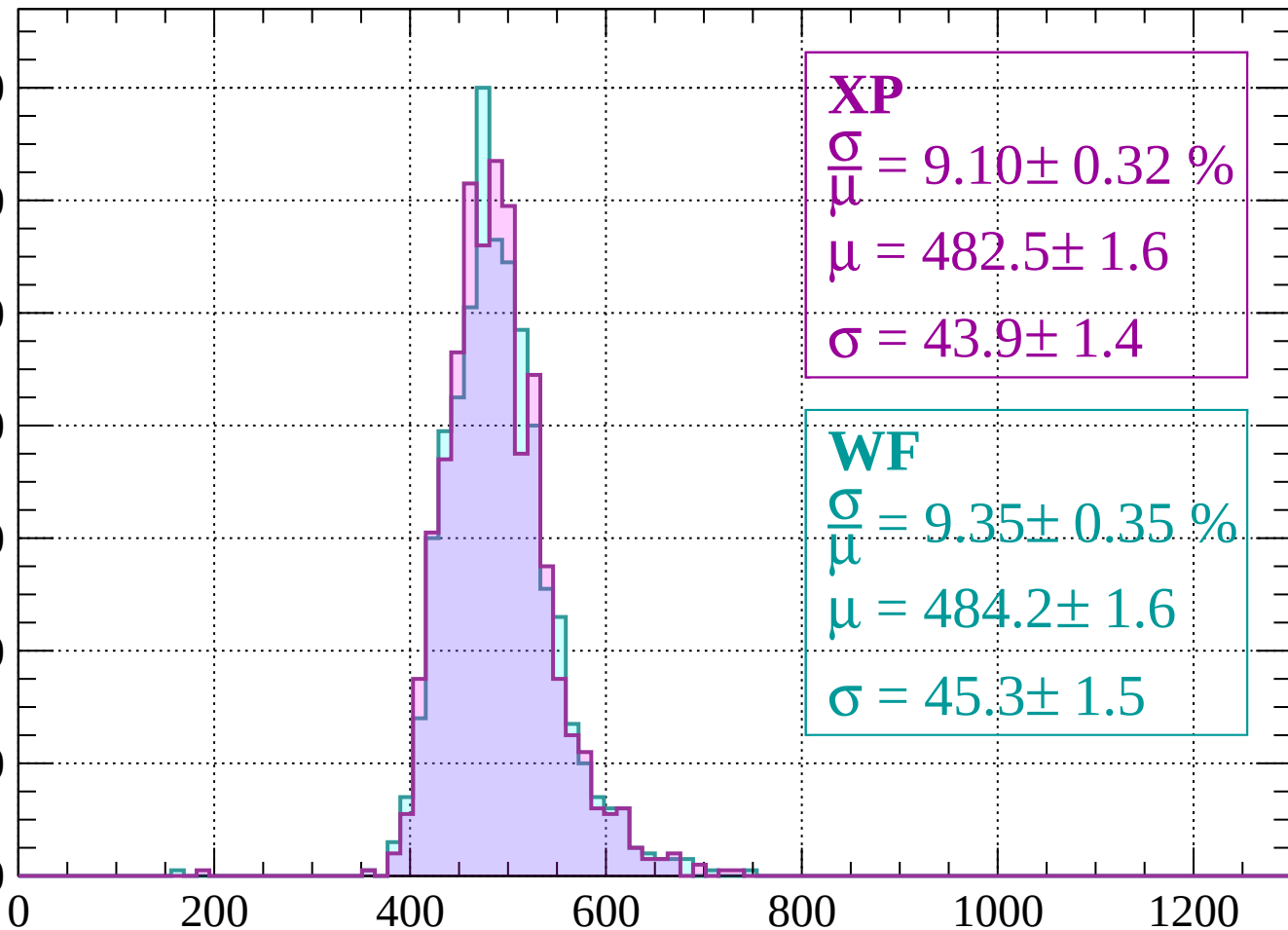
$$\sigma = 42.0 \pm 1.0$$



Energy loss | $2000 < p < 2040$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.32 \%$$

$$\mu = 482.5 \pm 1.6$$

$$\sigma = 43.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.35 \%$$

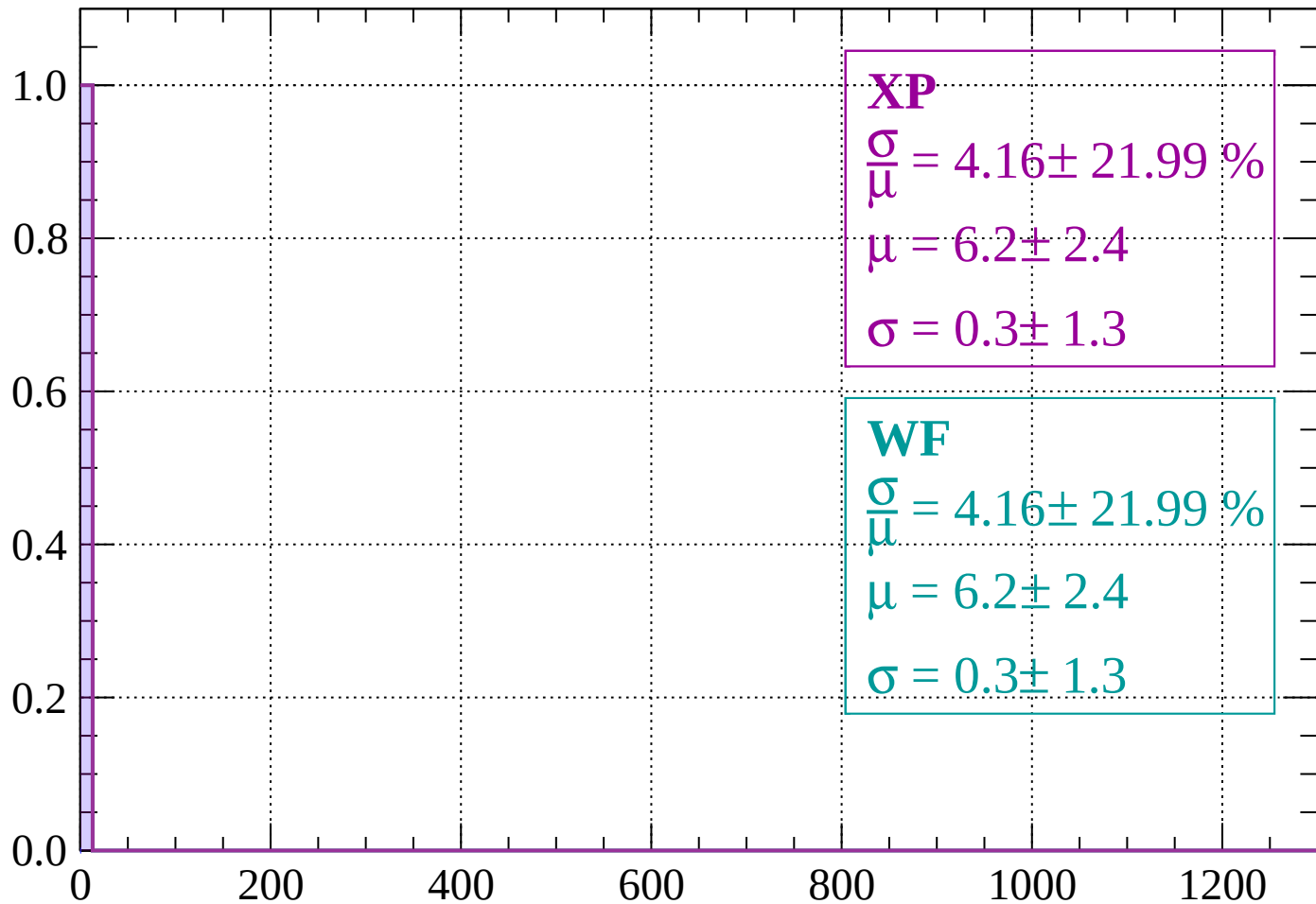
$$\mu = 484.2 \pm 1.6$$

$$\sigma = 45.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

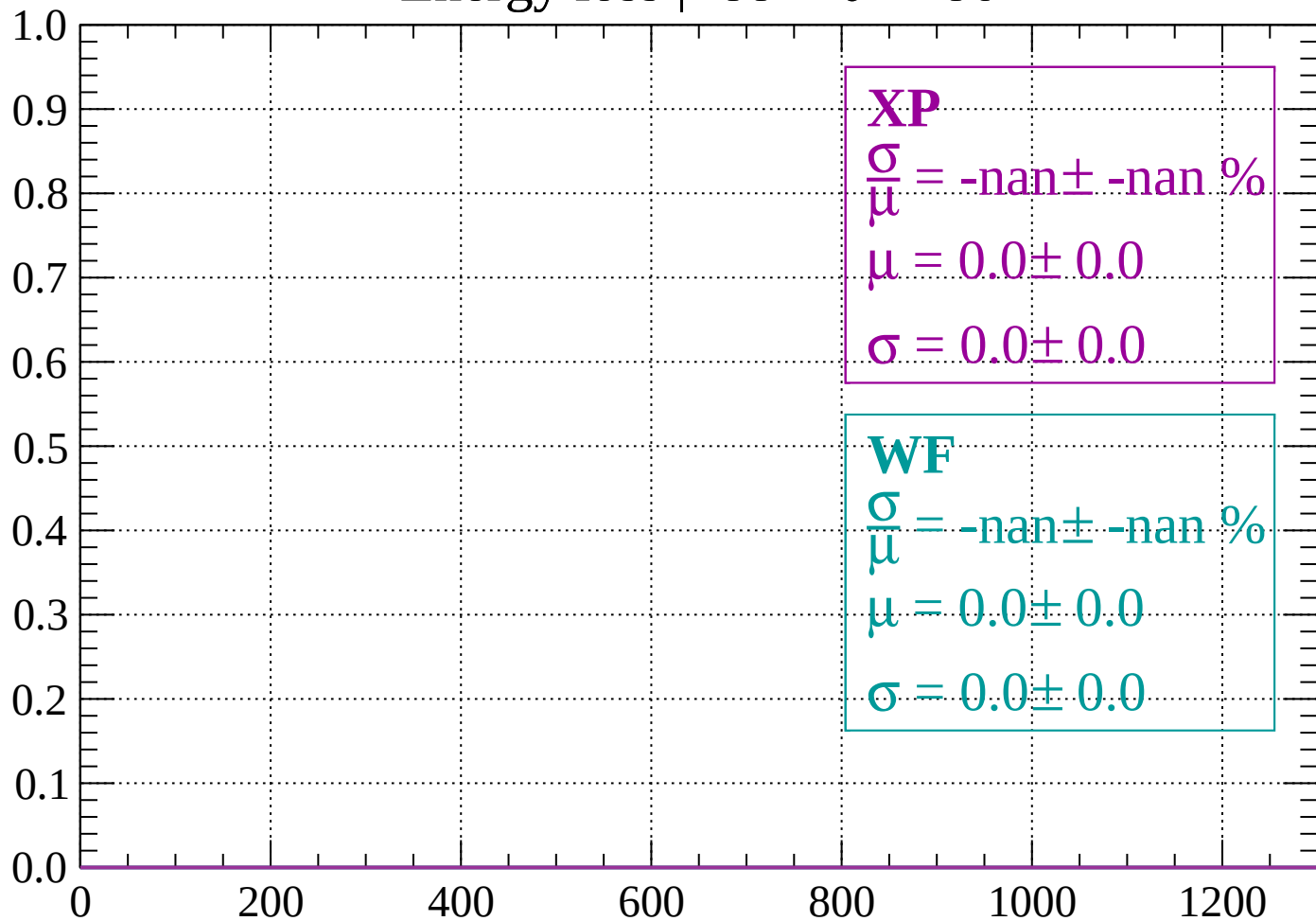
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

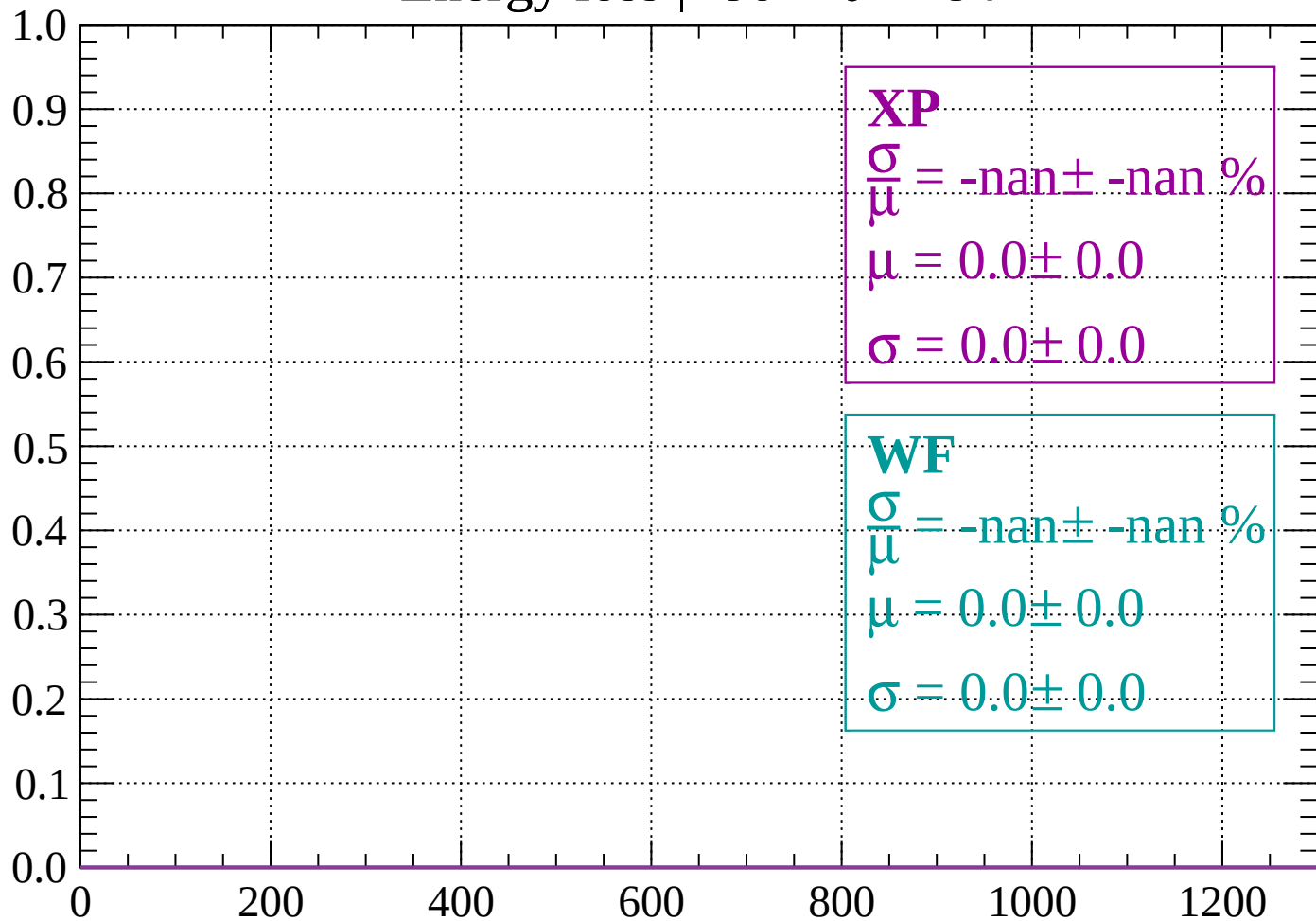
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

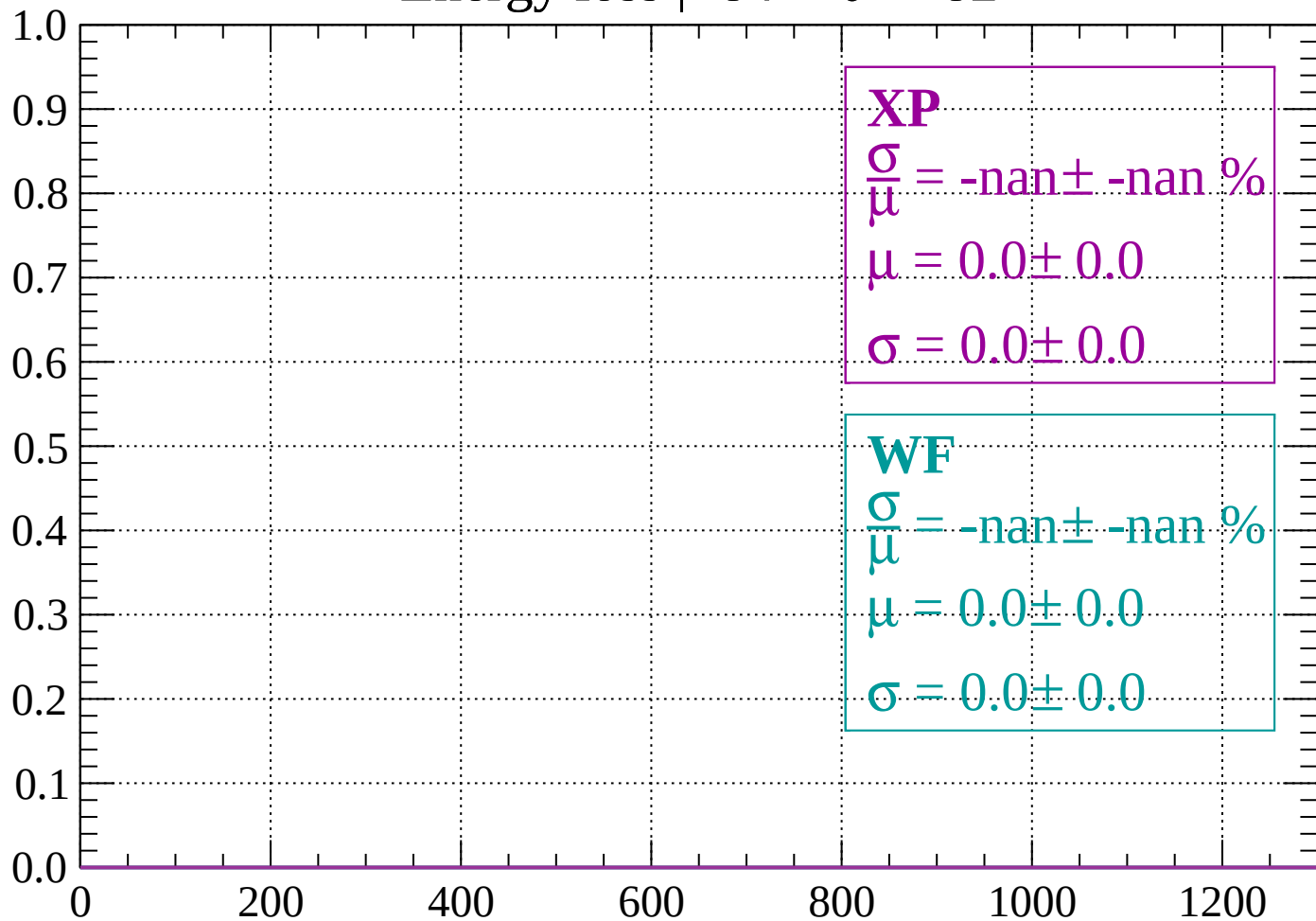
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

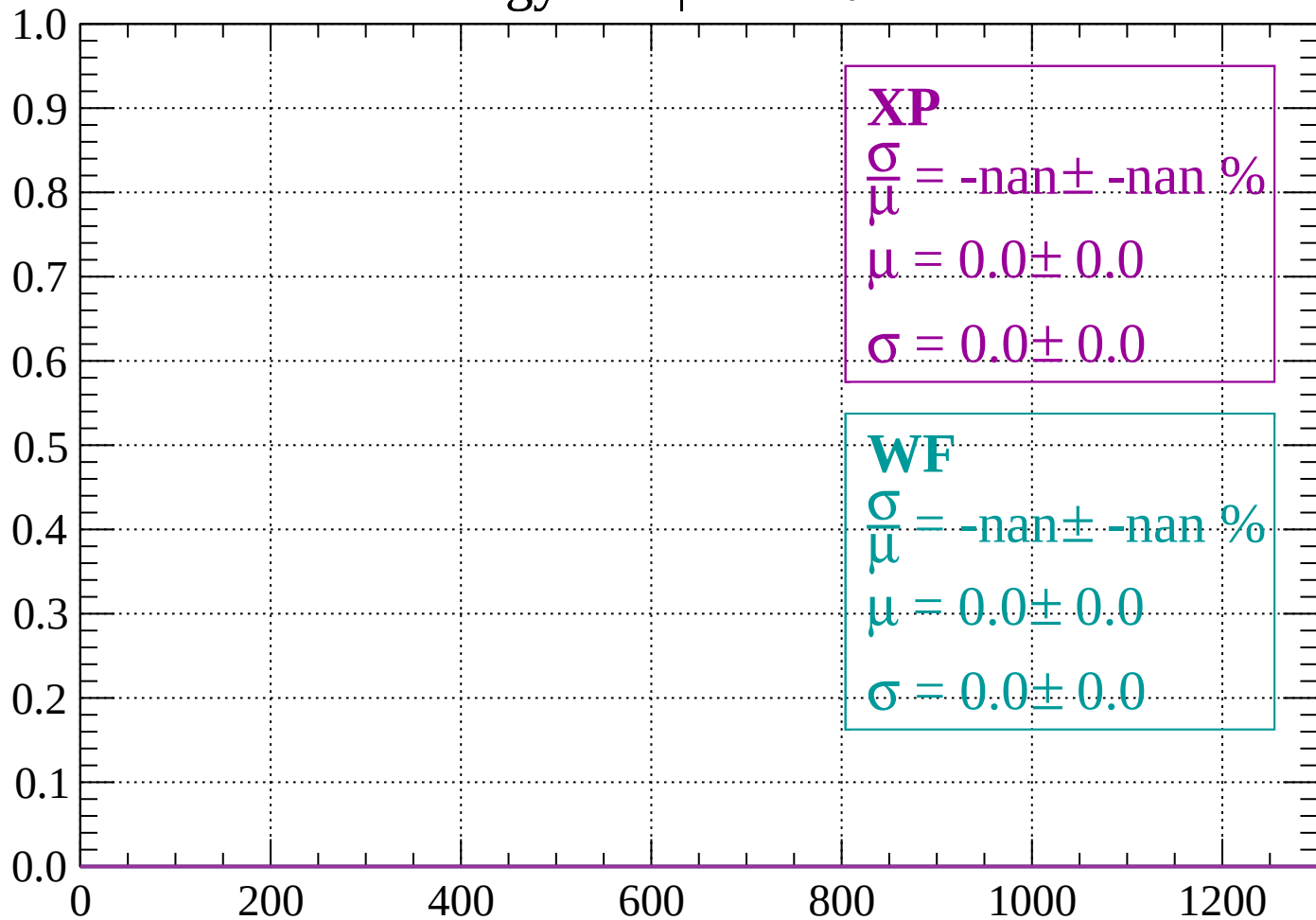
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

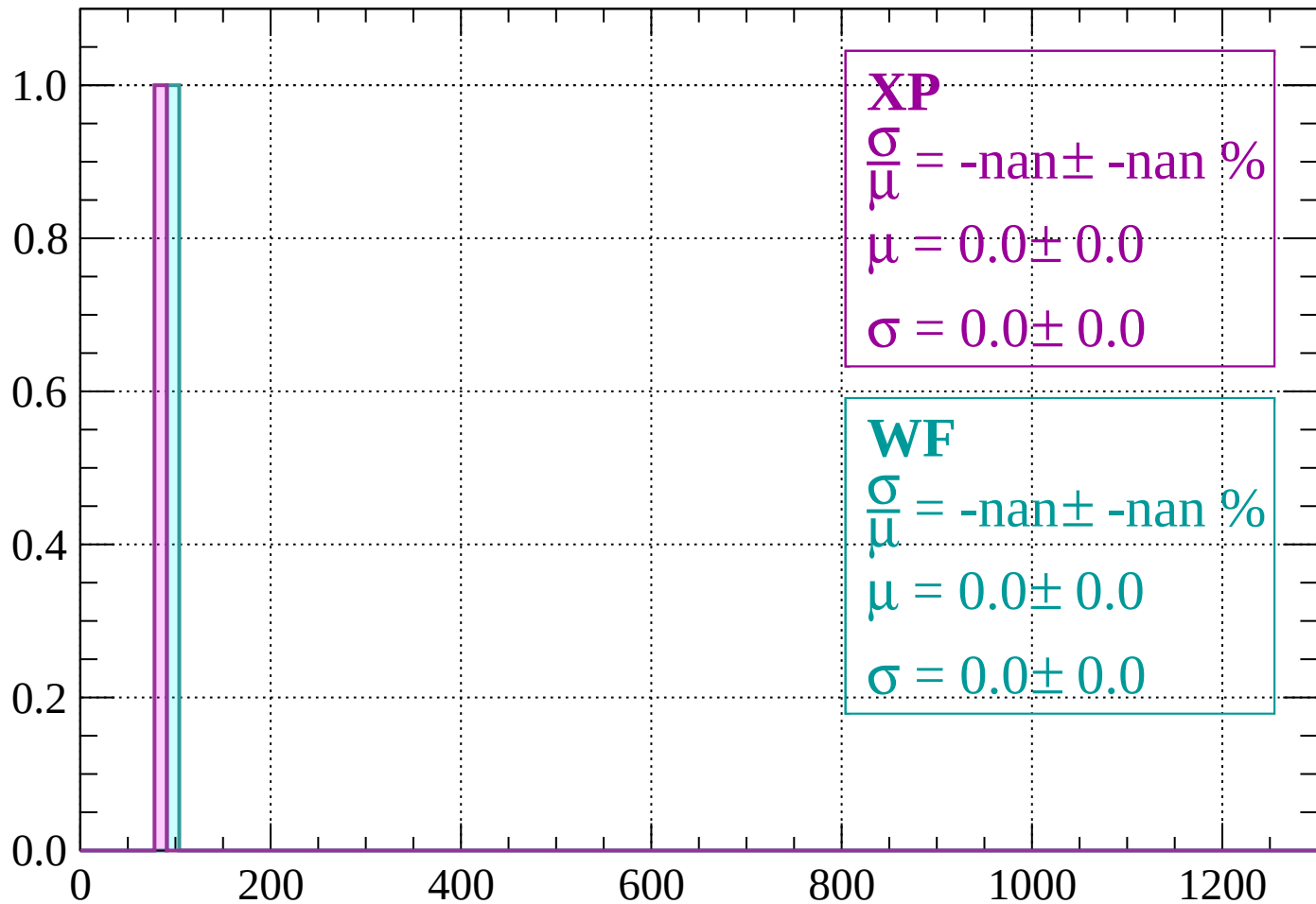
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

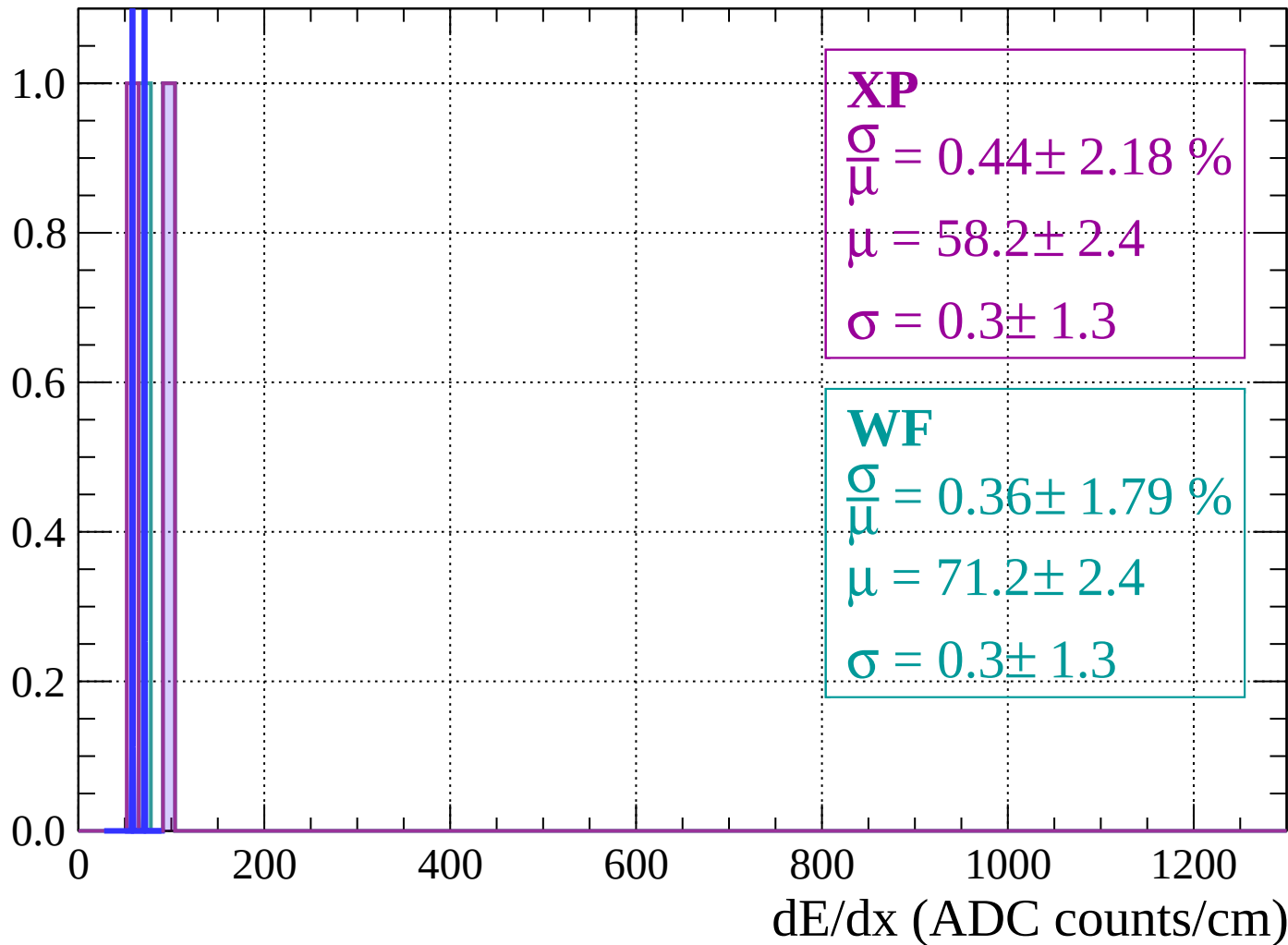
Count



dE/dx (ADC counts/cm)

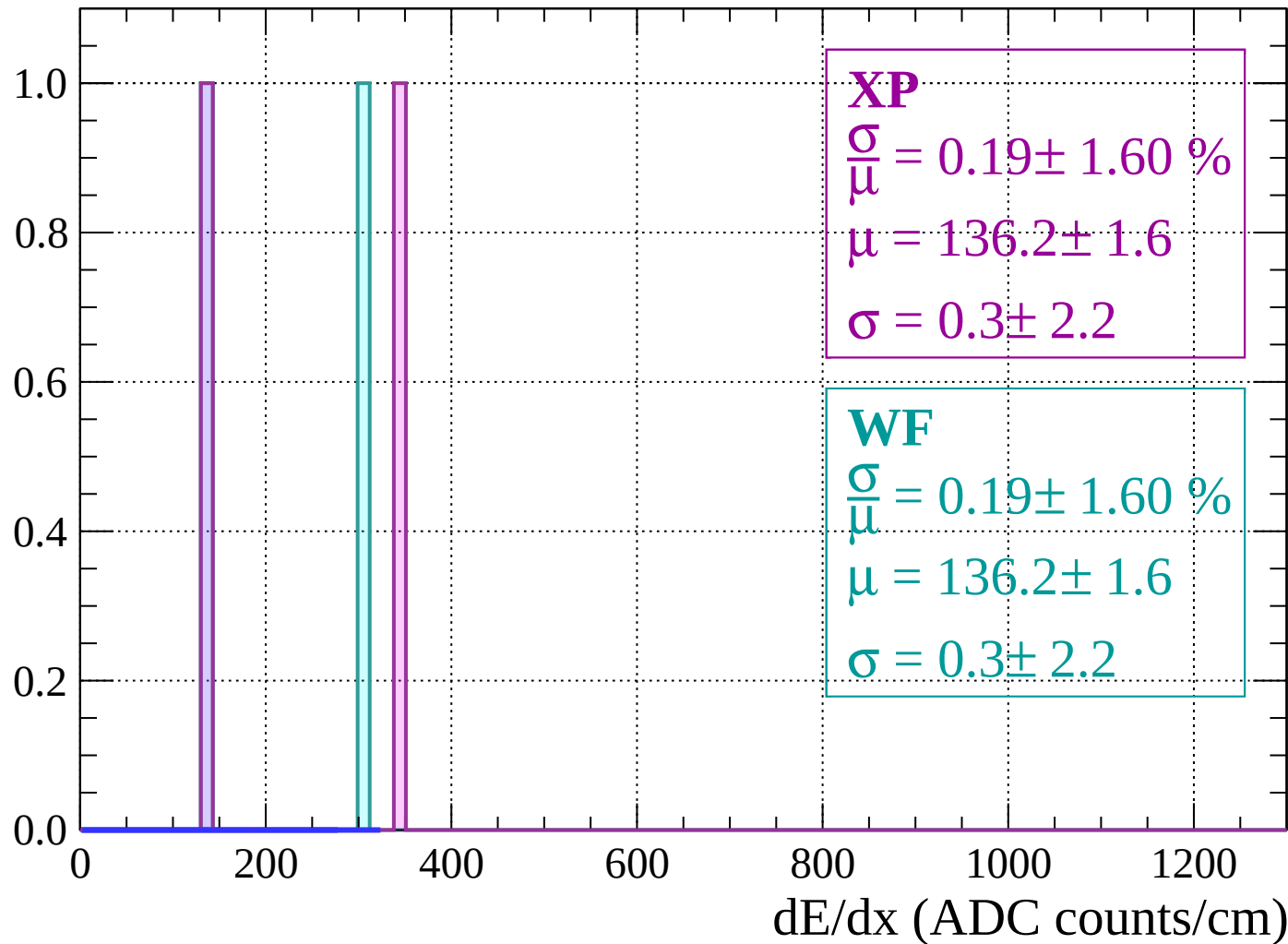
Energy loss $|-78 < \theta < -76$

Count



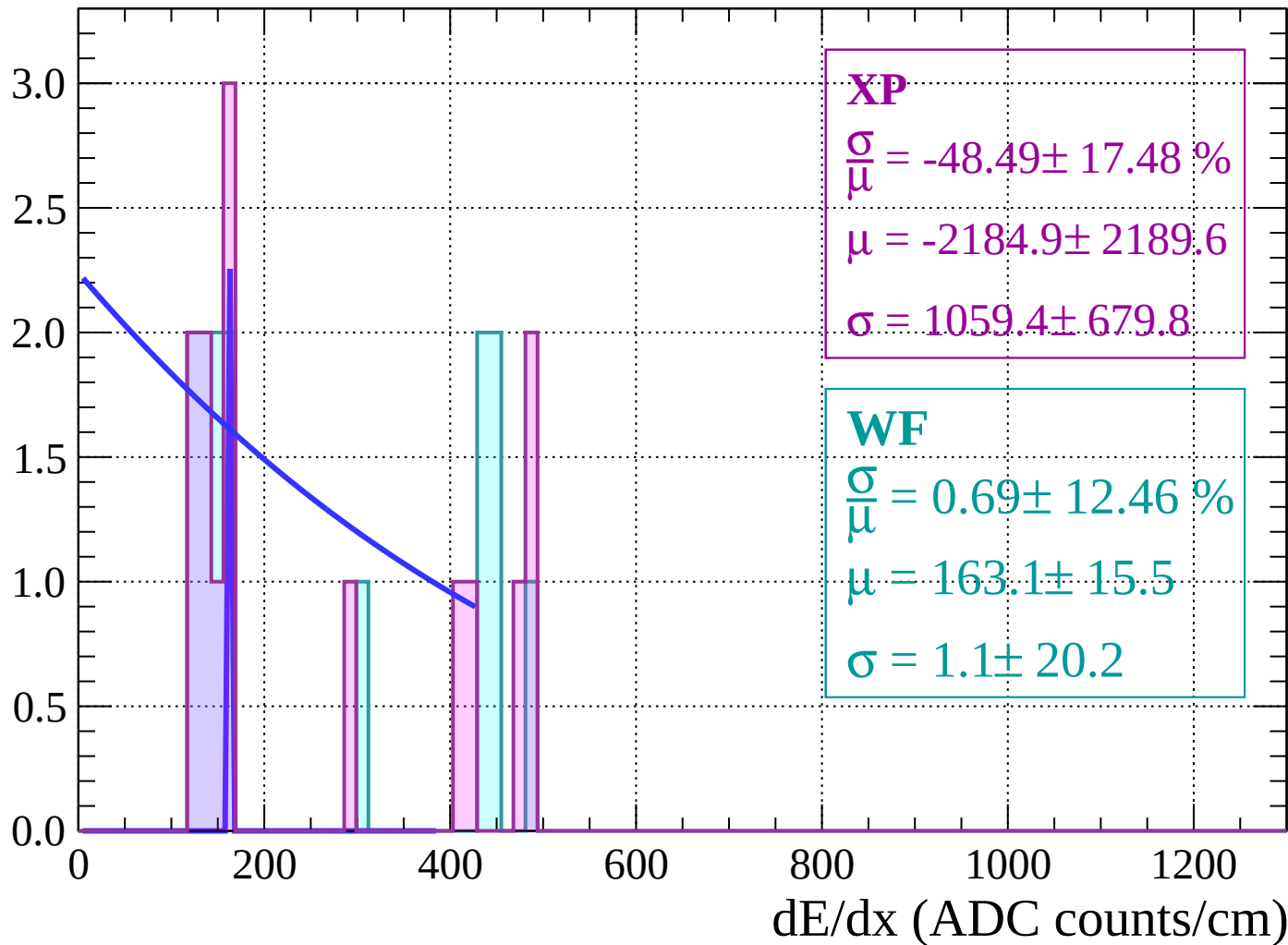
Energy loss | $-76 < \theta < -74$

Count



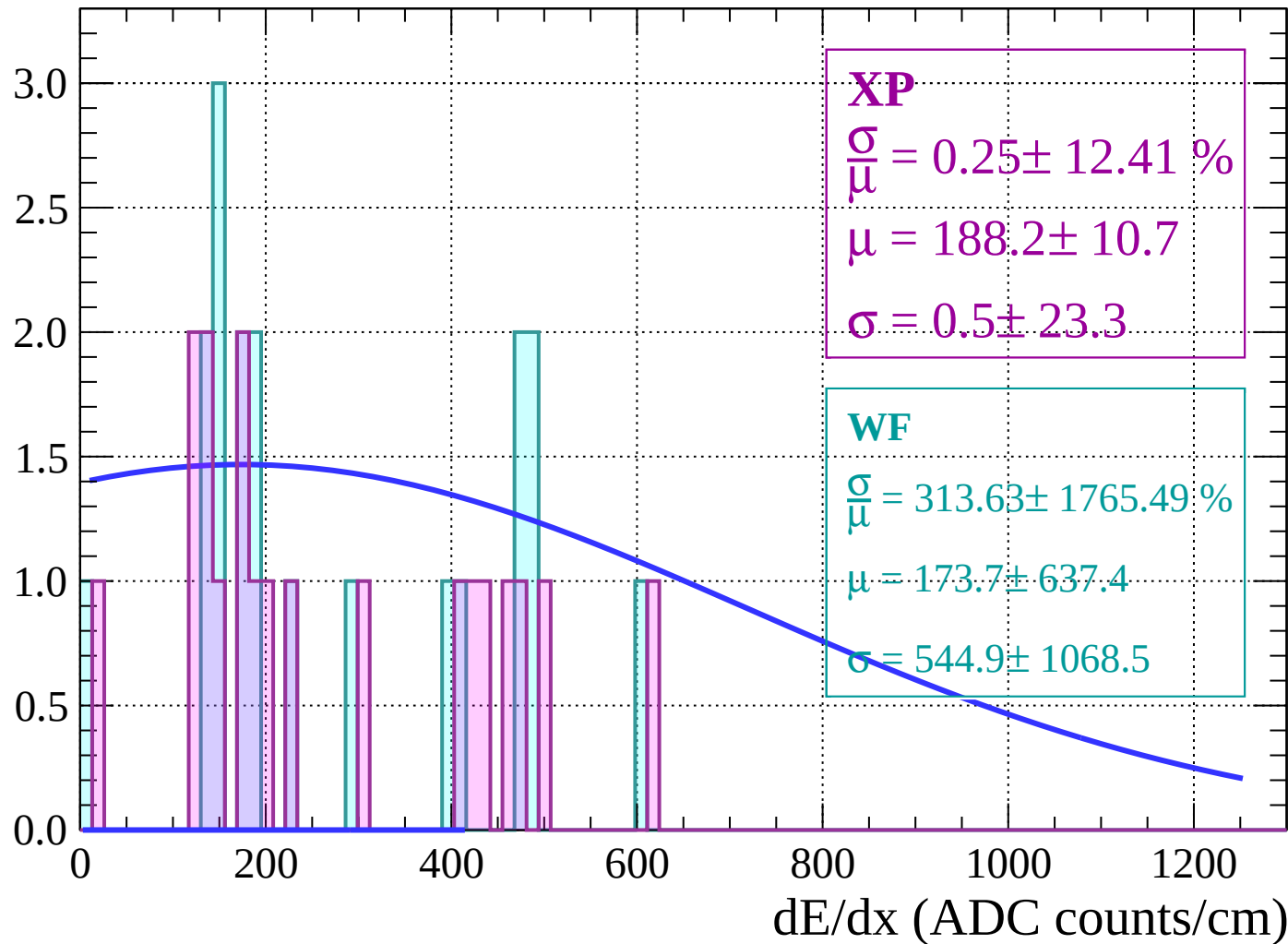
Energy loss | $-74 < \theta < -72$

Count



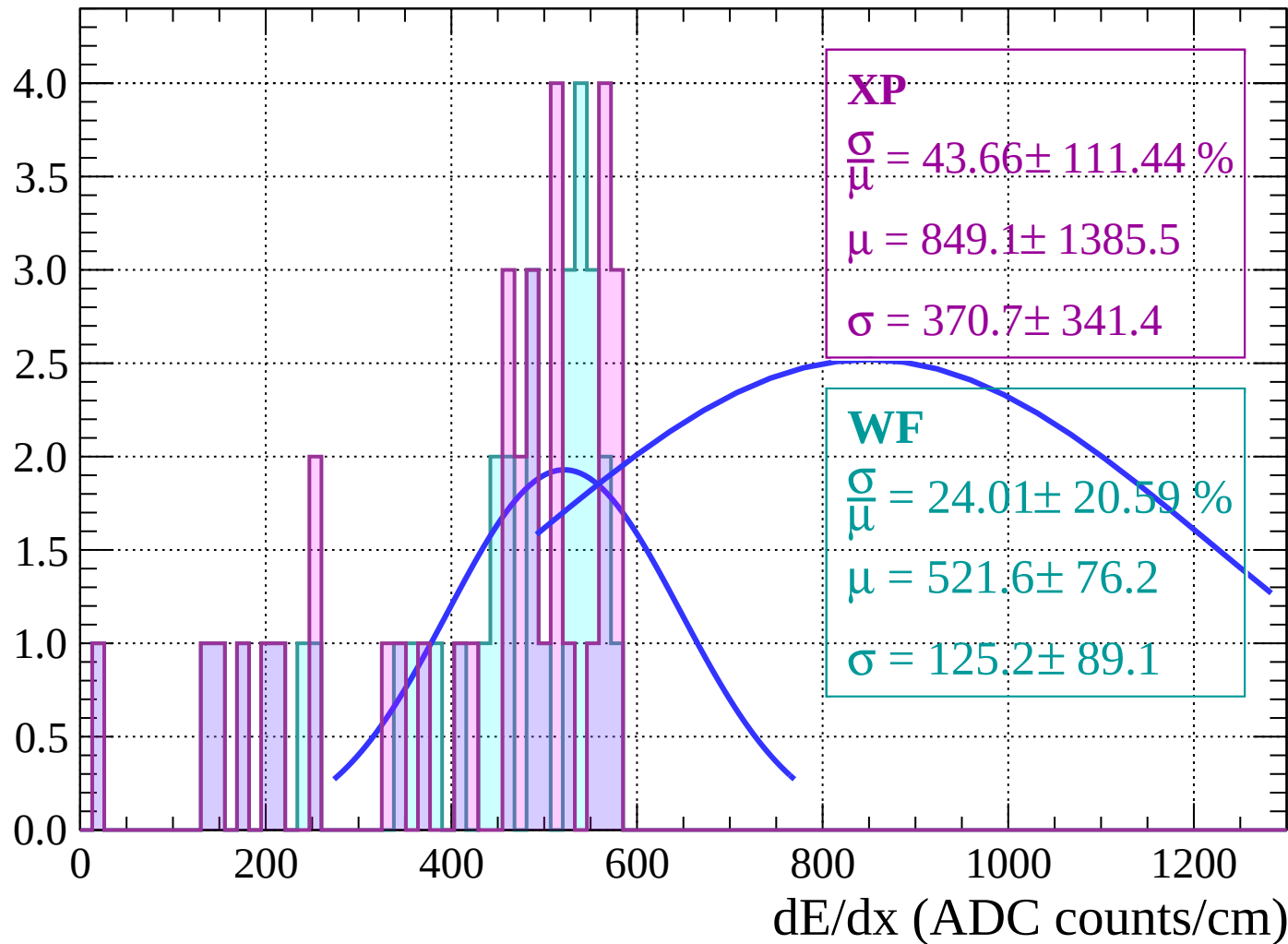
Energy loss | $-72 < \theta < -70$

Count



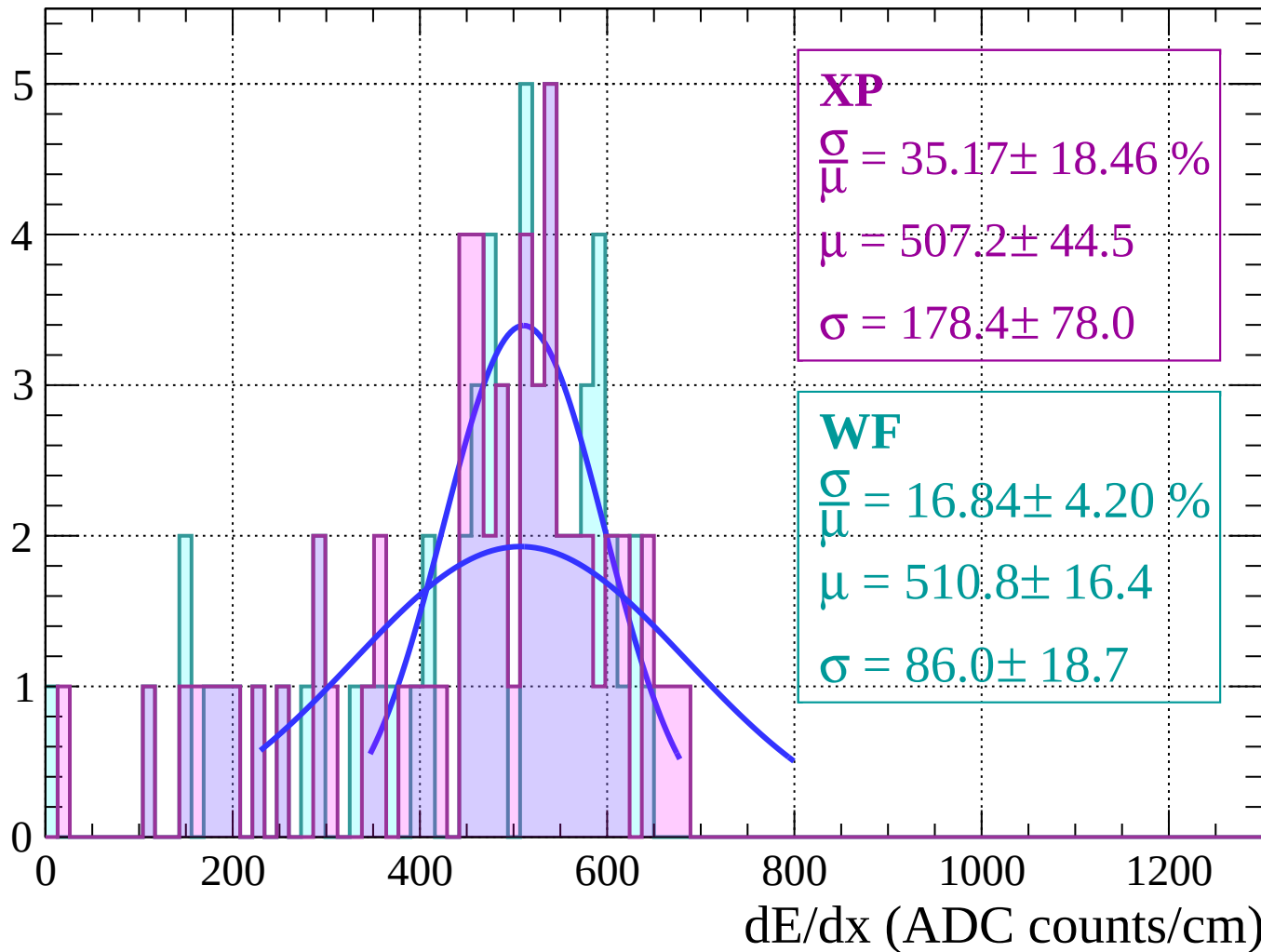
Energy loss | $-70 < \theta < -68$

Count



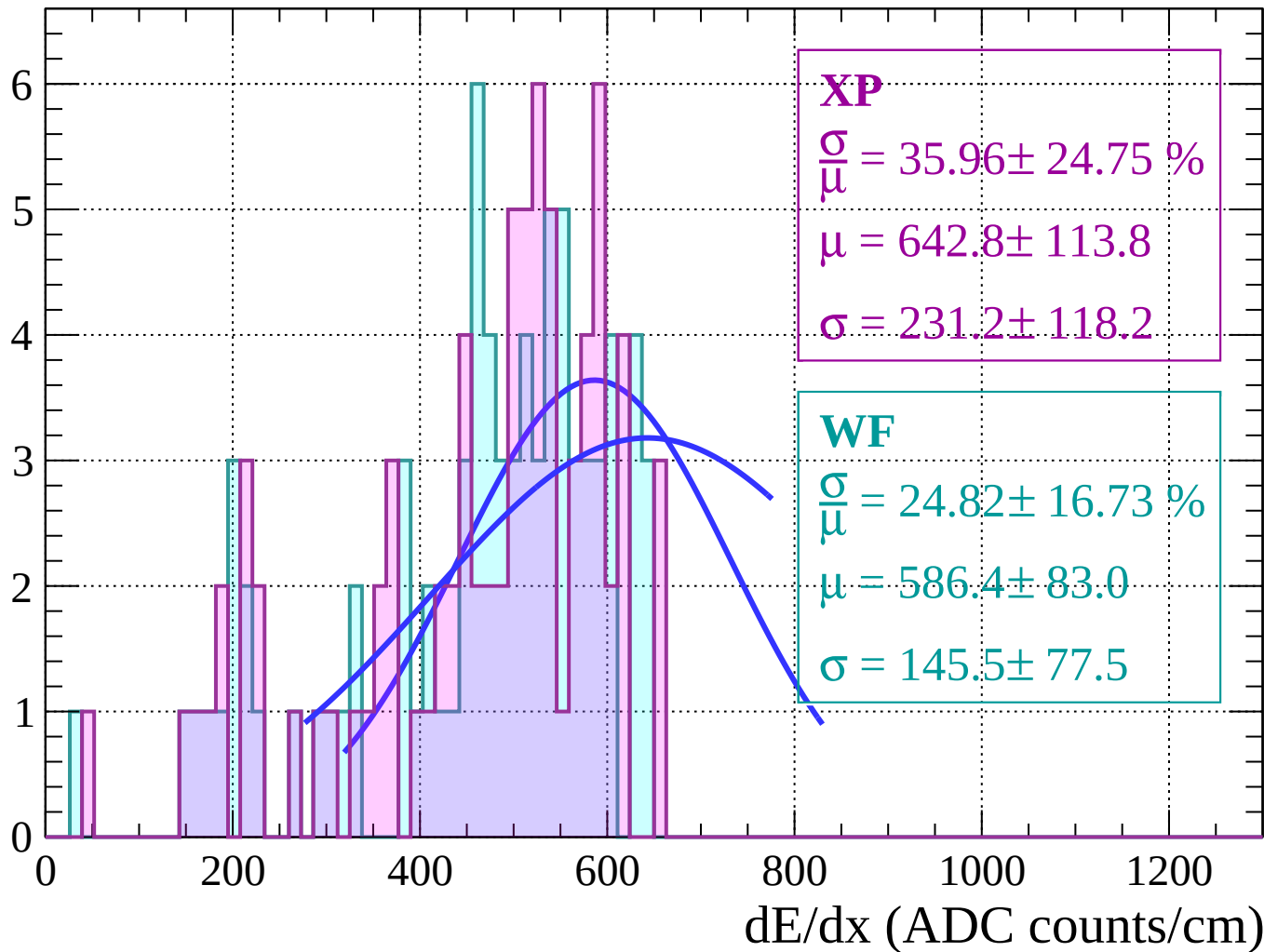
Energy loss | $-68 < \theta < -66$

Count



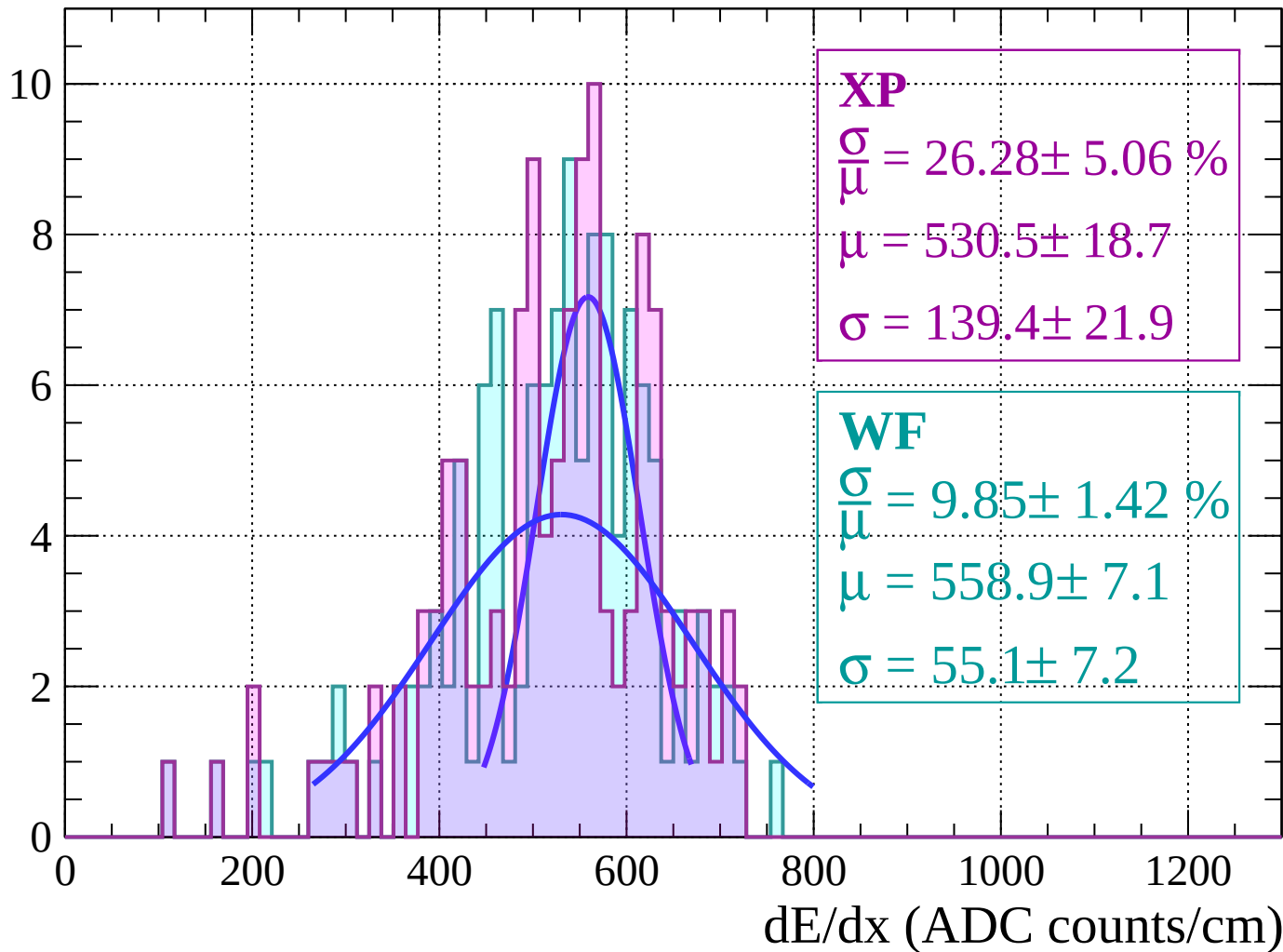
Energy loss | $-66 < \theta < -64$

Count



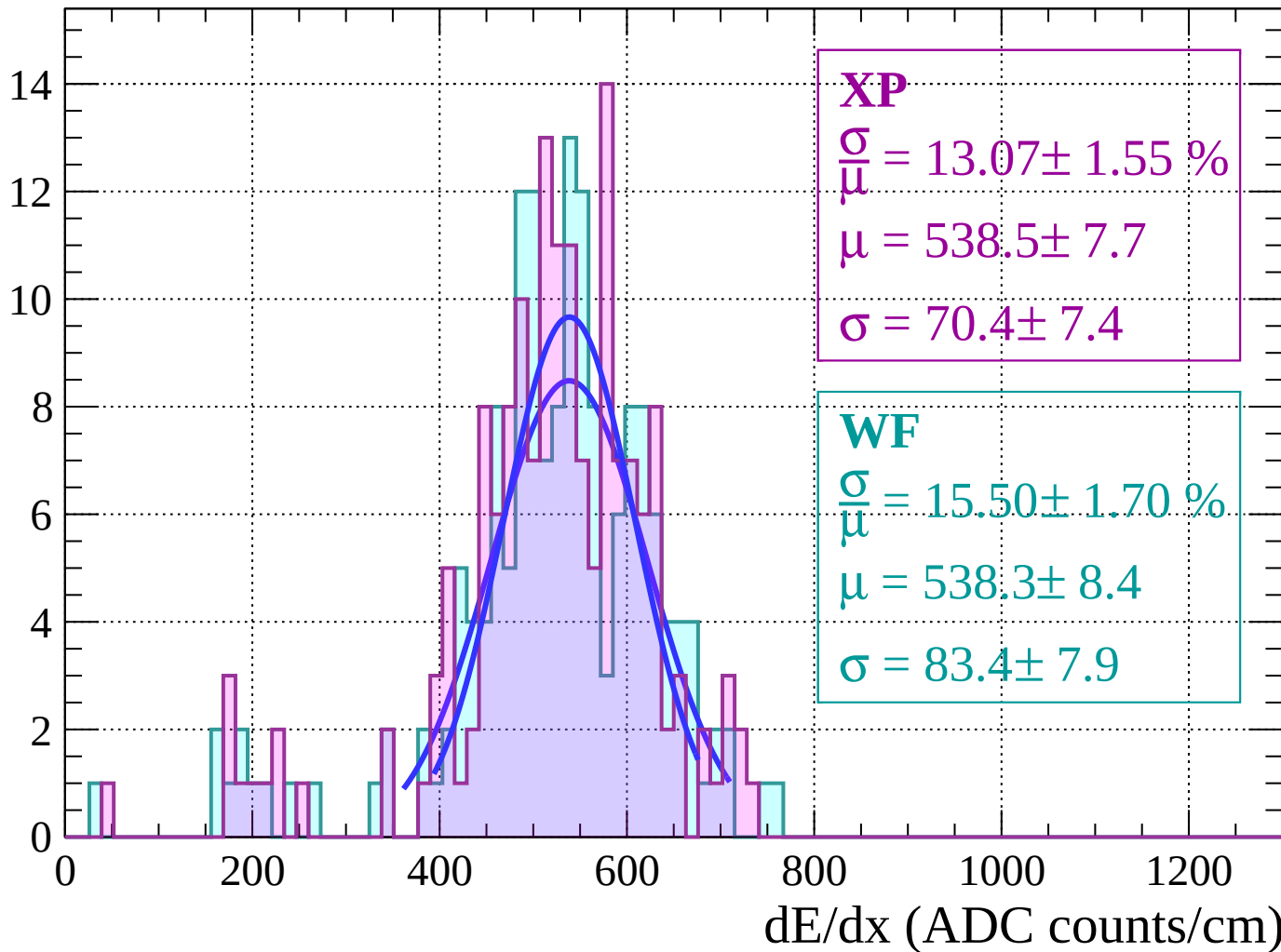
Energy loss | $-64 < \theta < -62$

Count



Energy loss | $-62 < \theta < -60$

Count



Energy loss | $-60 < \theta < -58$

Count

XP

$$\frac{\sigma}{\mu} = 15.60 \pm 1.27 \%$$

$$\mu = 547.2 \pm 6.9$$

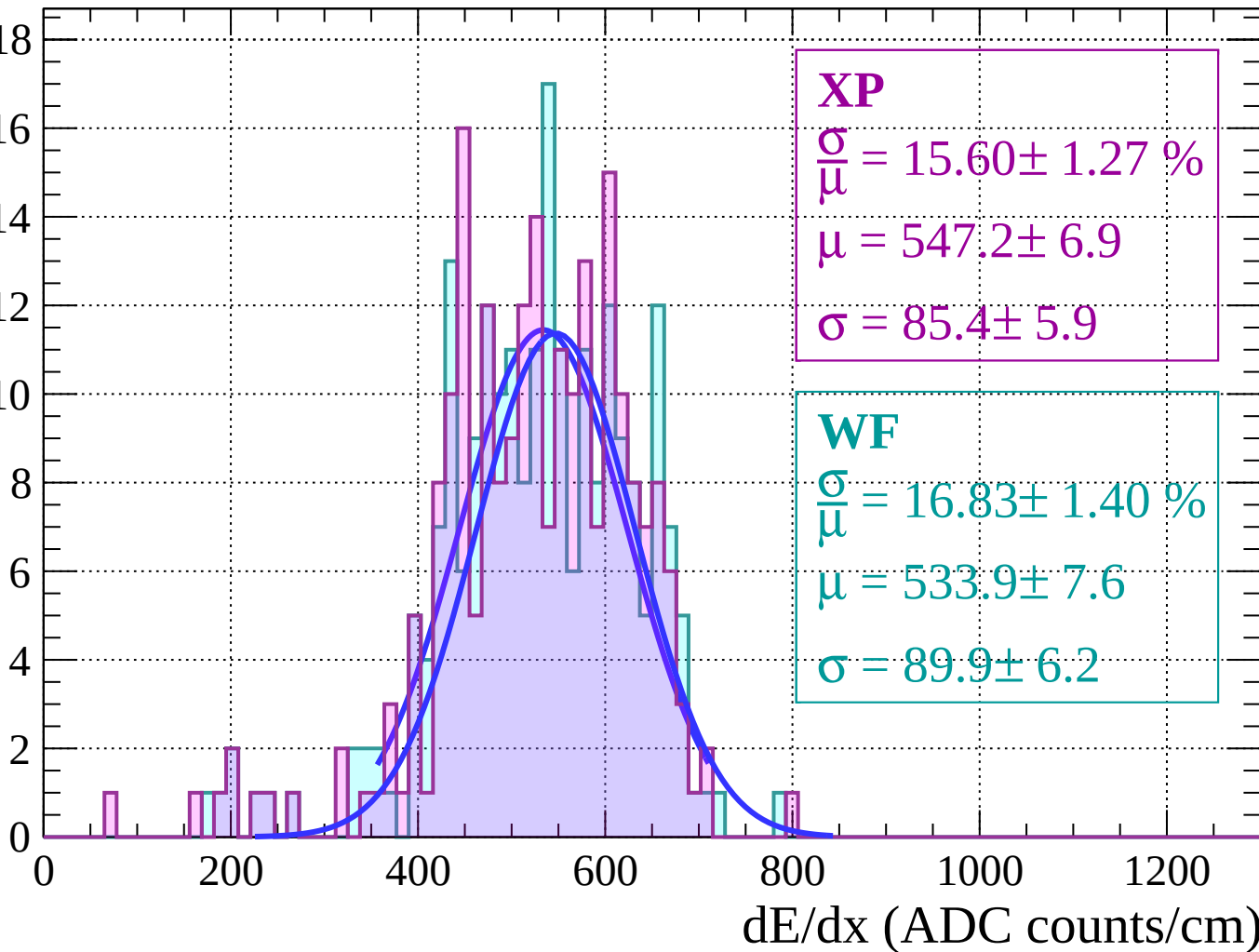
$$\sigma = 85.4 \pm 5.9$$

WF

$$\frac{\sigma}{\mu} = 16.83 \pm 1.40 \%$$

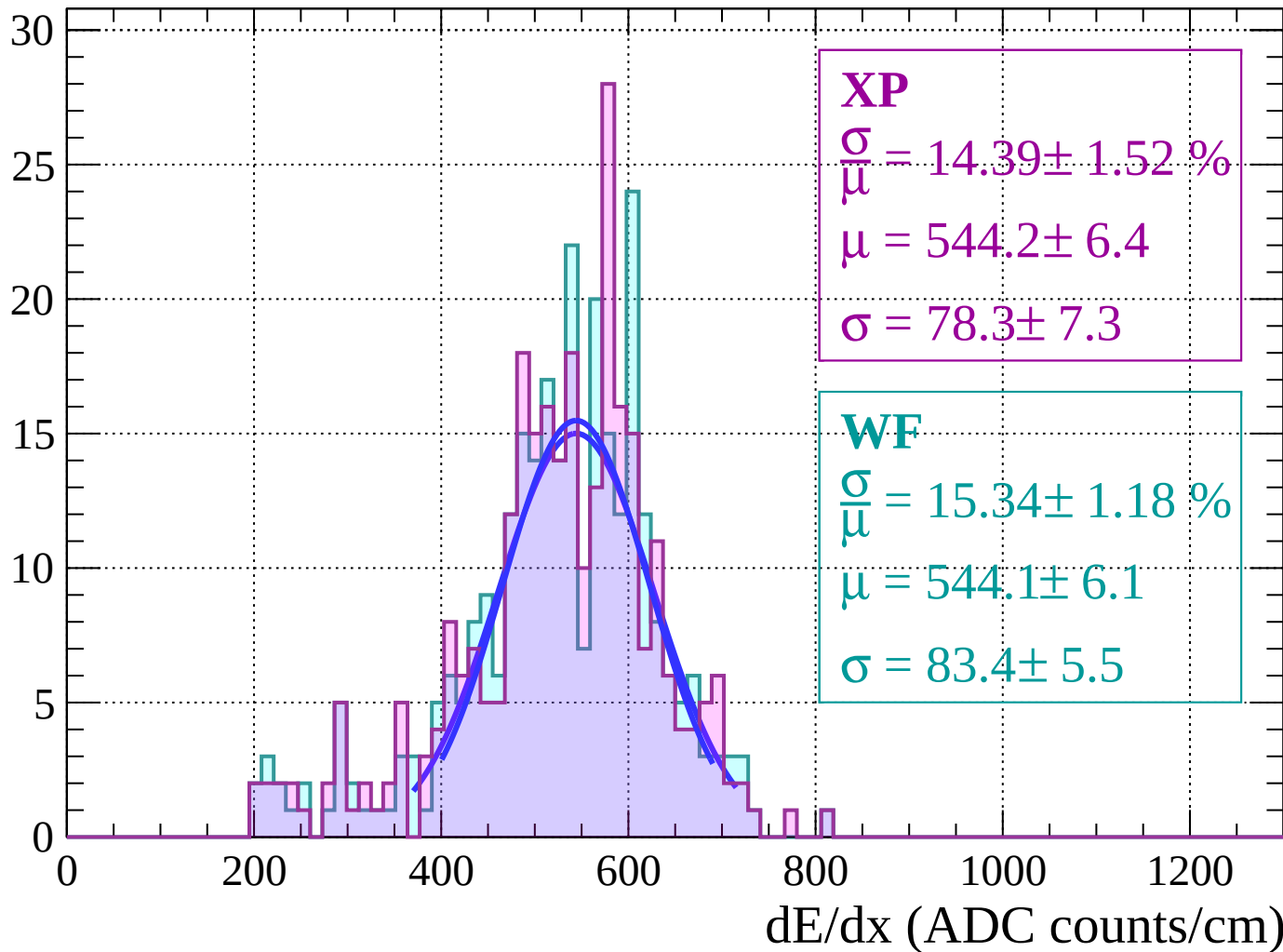
$$\mu = 533.9 \pm 7.6$$

$$\sigma = 89.9 \pm 6.2$$



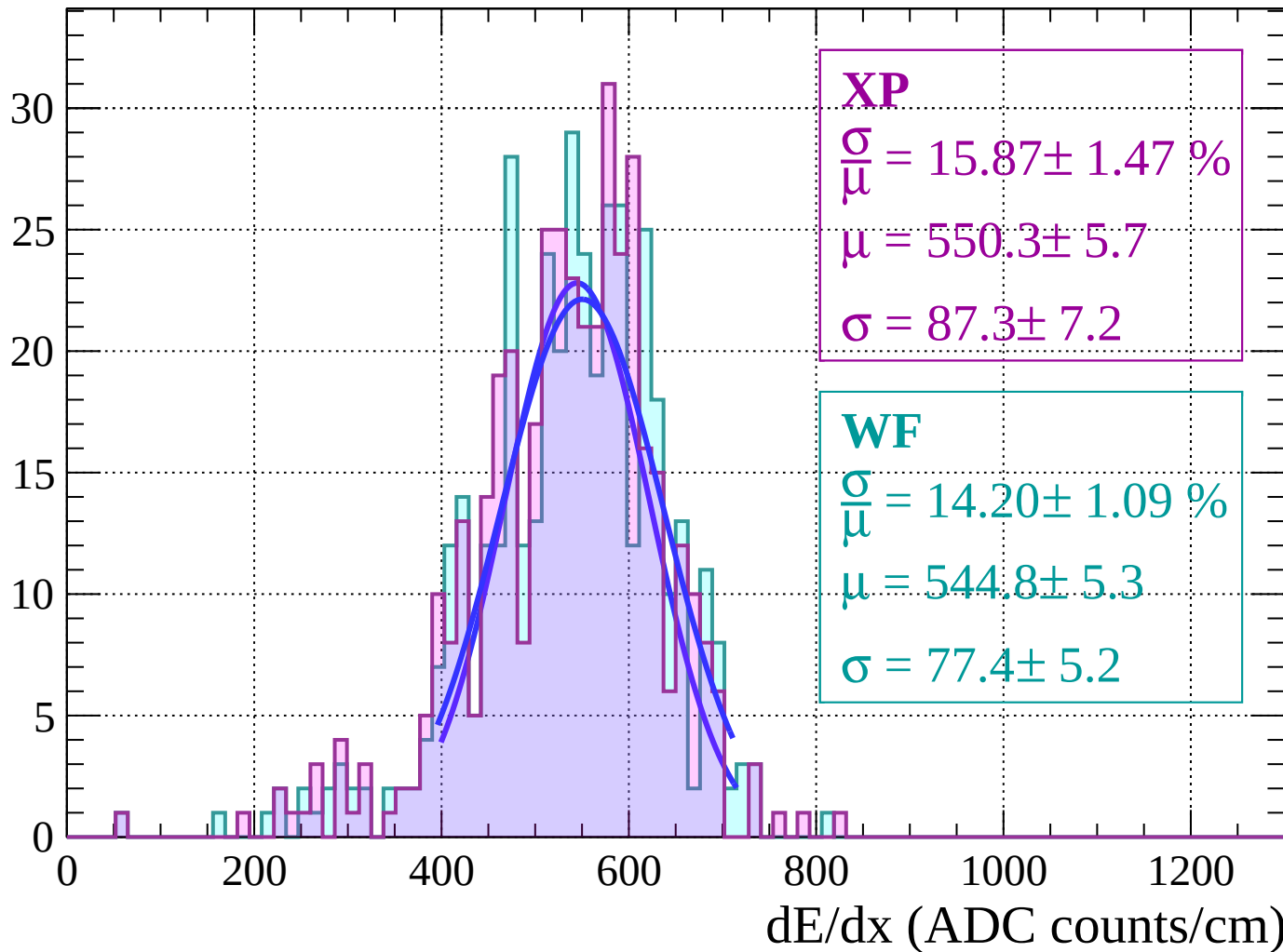
Energy loss | $-58 < \theta < -56$

Count



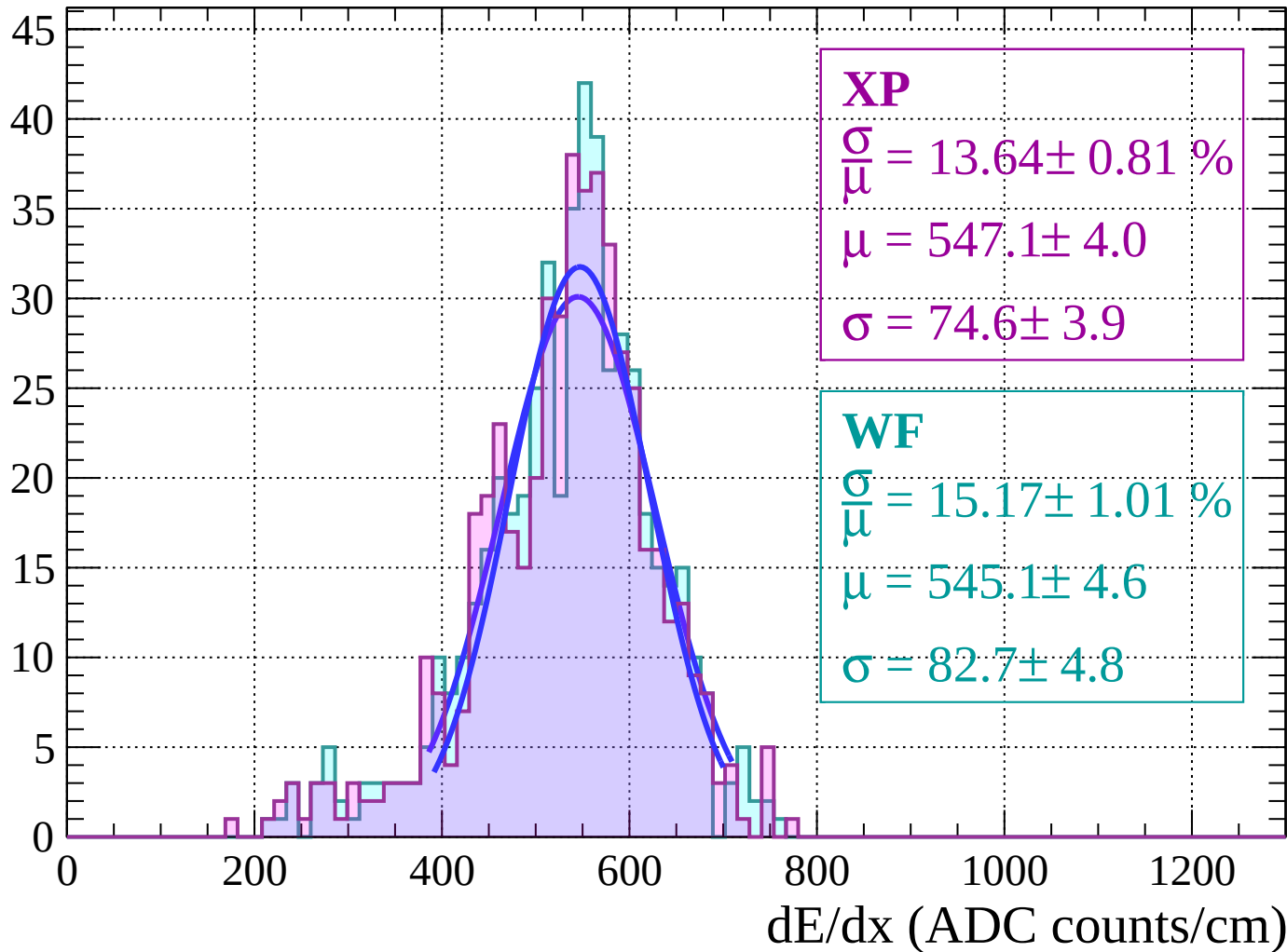
Energy loss $|-56 < \theta < -54$

Count



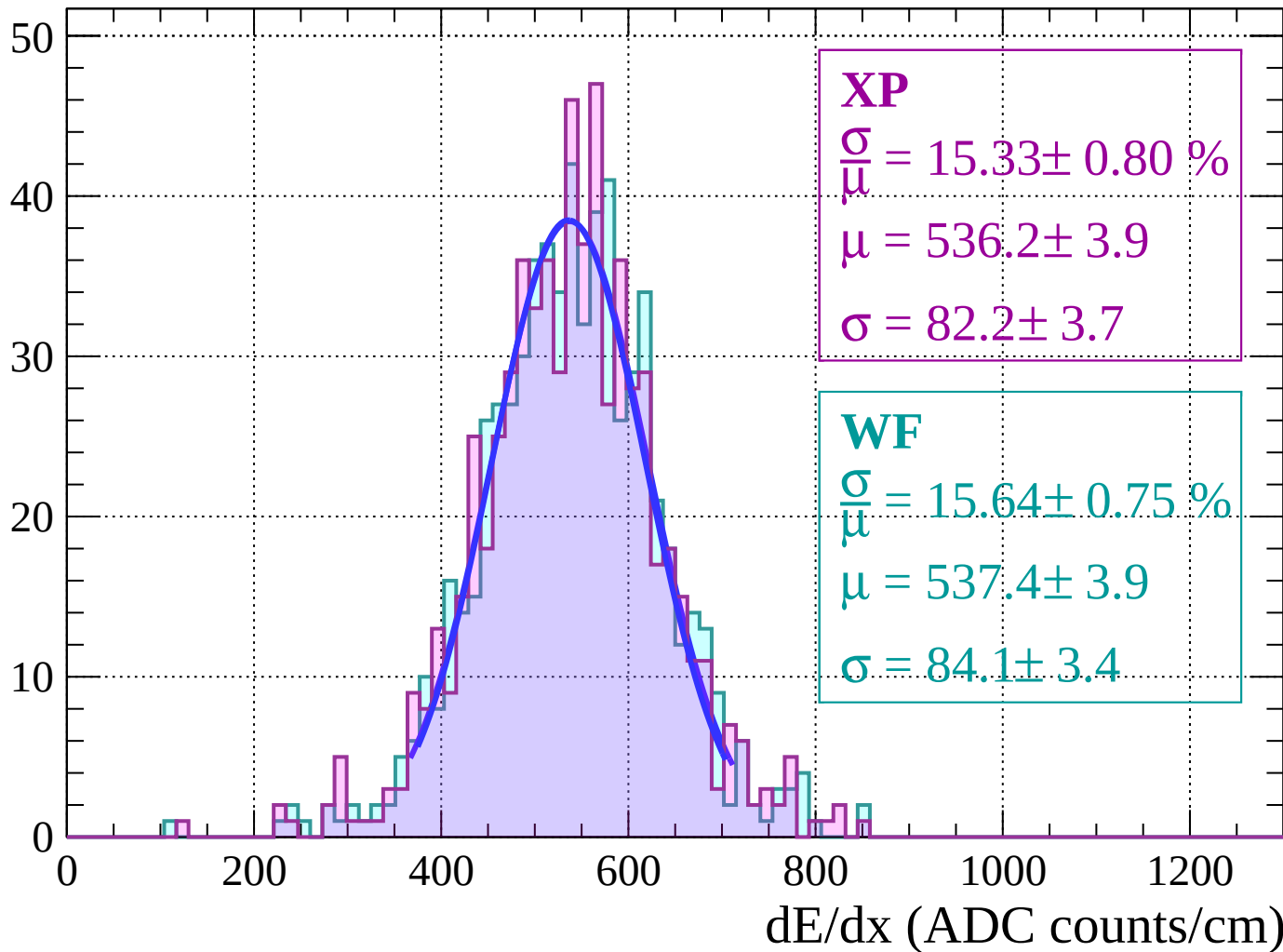
Energy loss | $-54 < \theta < -52$

Count



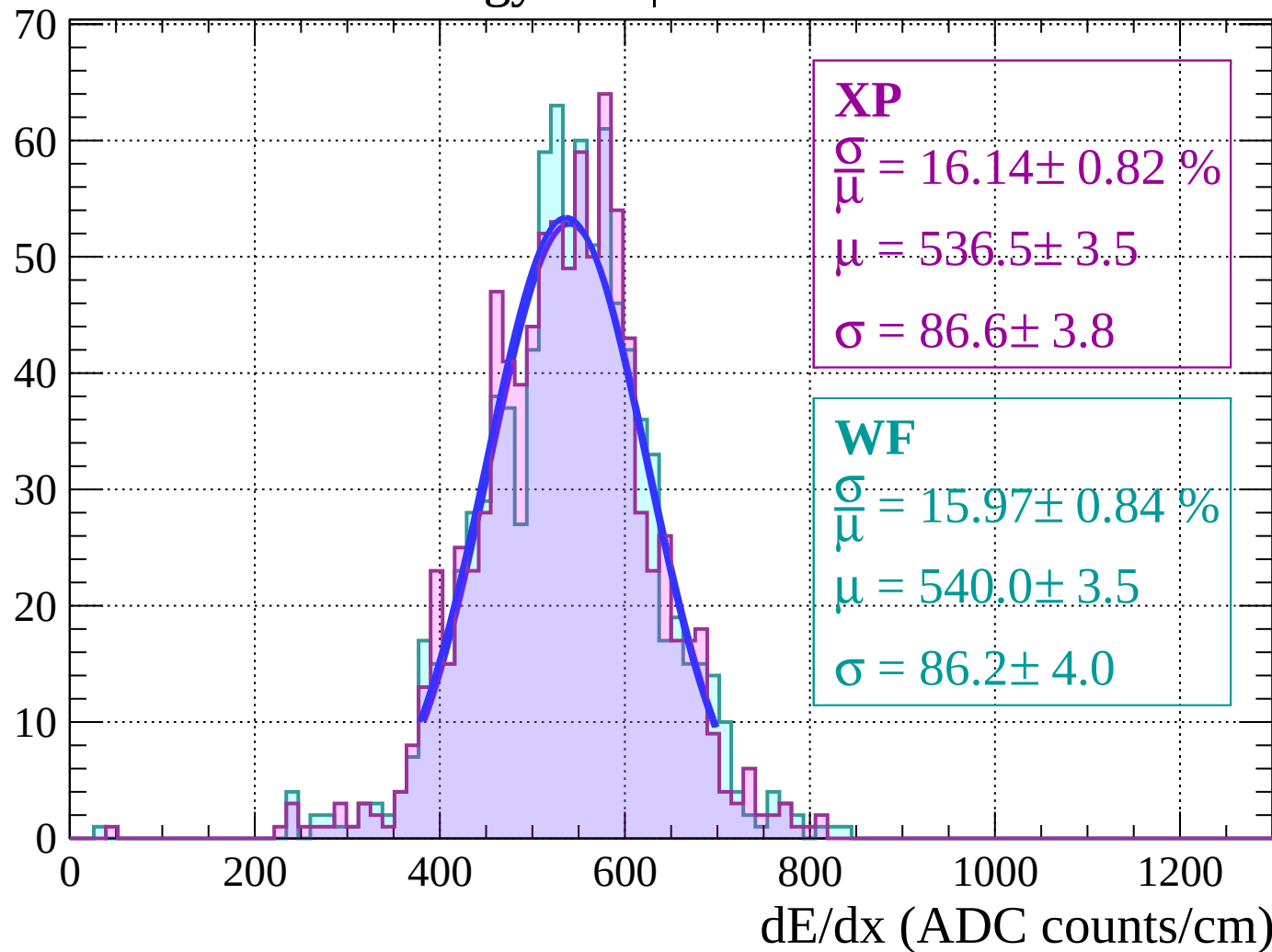
Energy loss | $-52 < \theta < -50$

Count



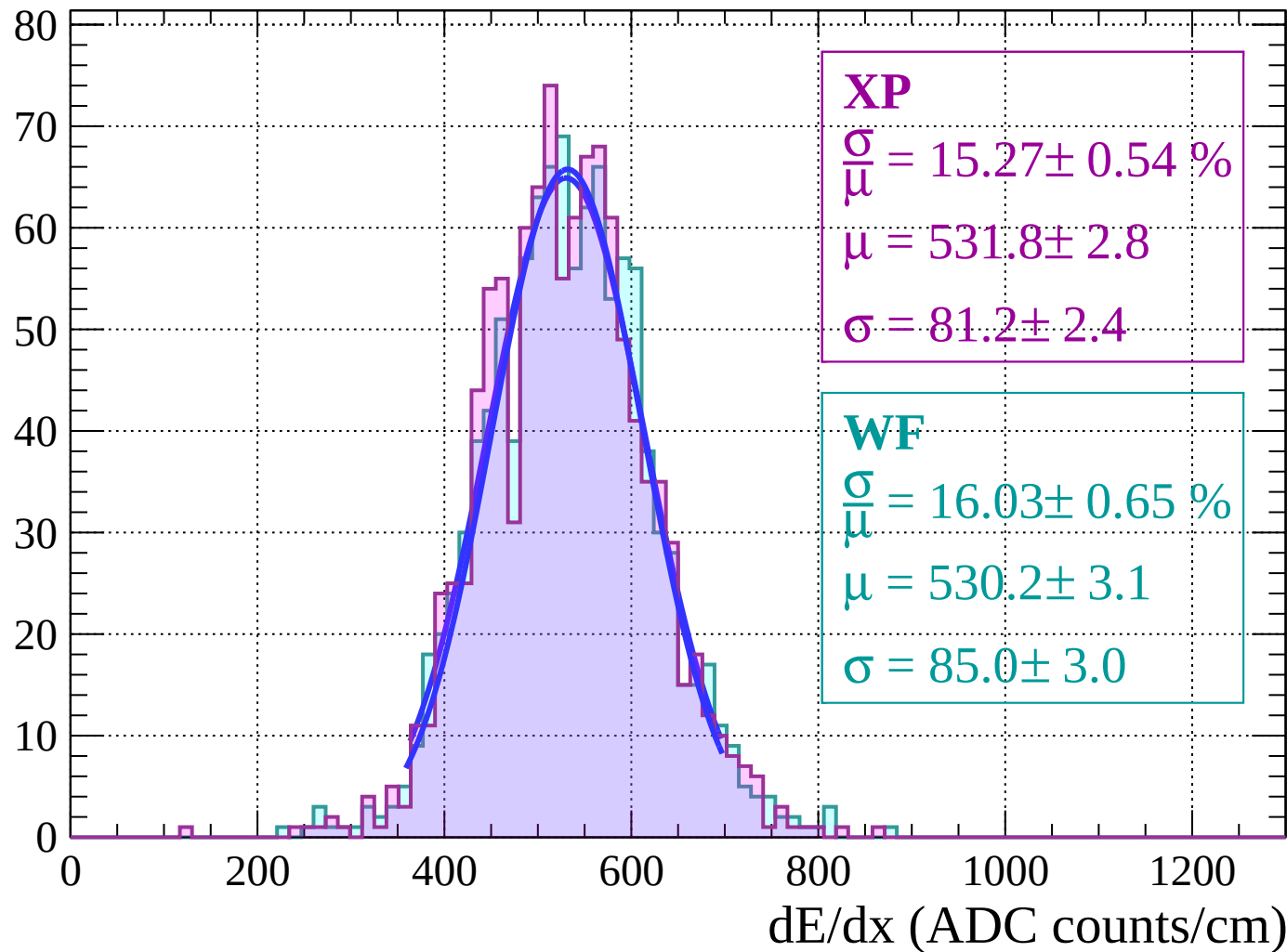
Energy loss $|-50 < \theta < -48$

Count



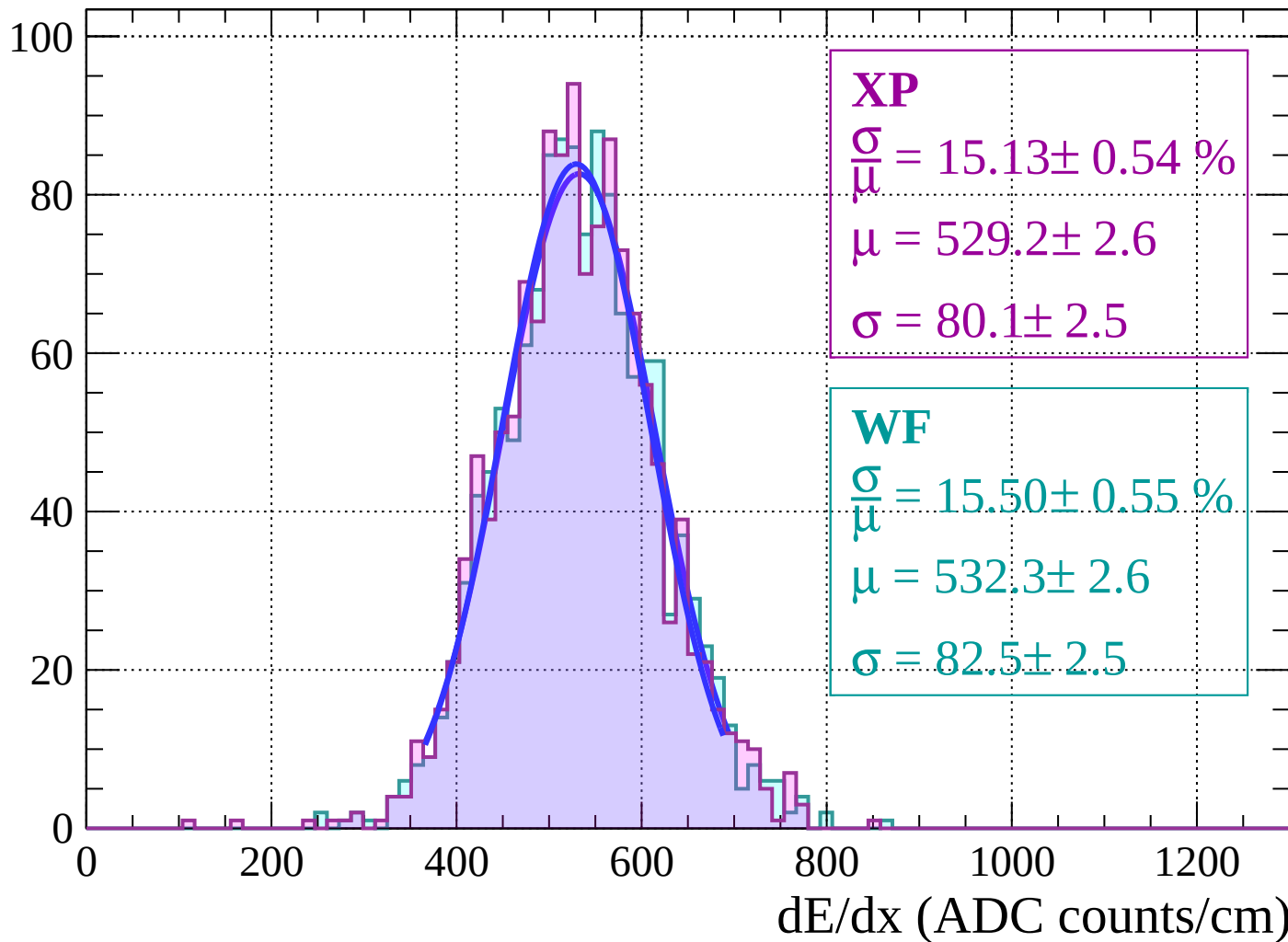
Energy loss | $-48 < \theta < -46$

Count

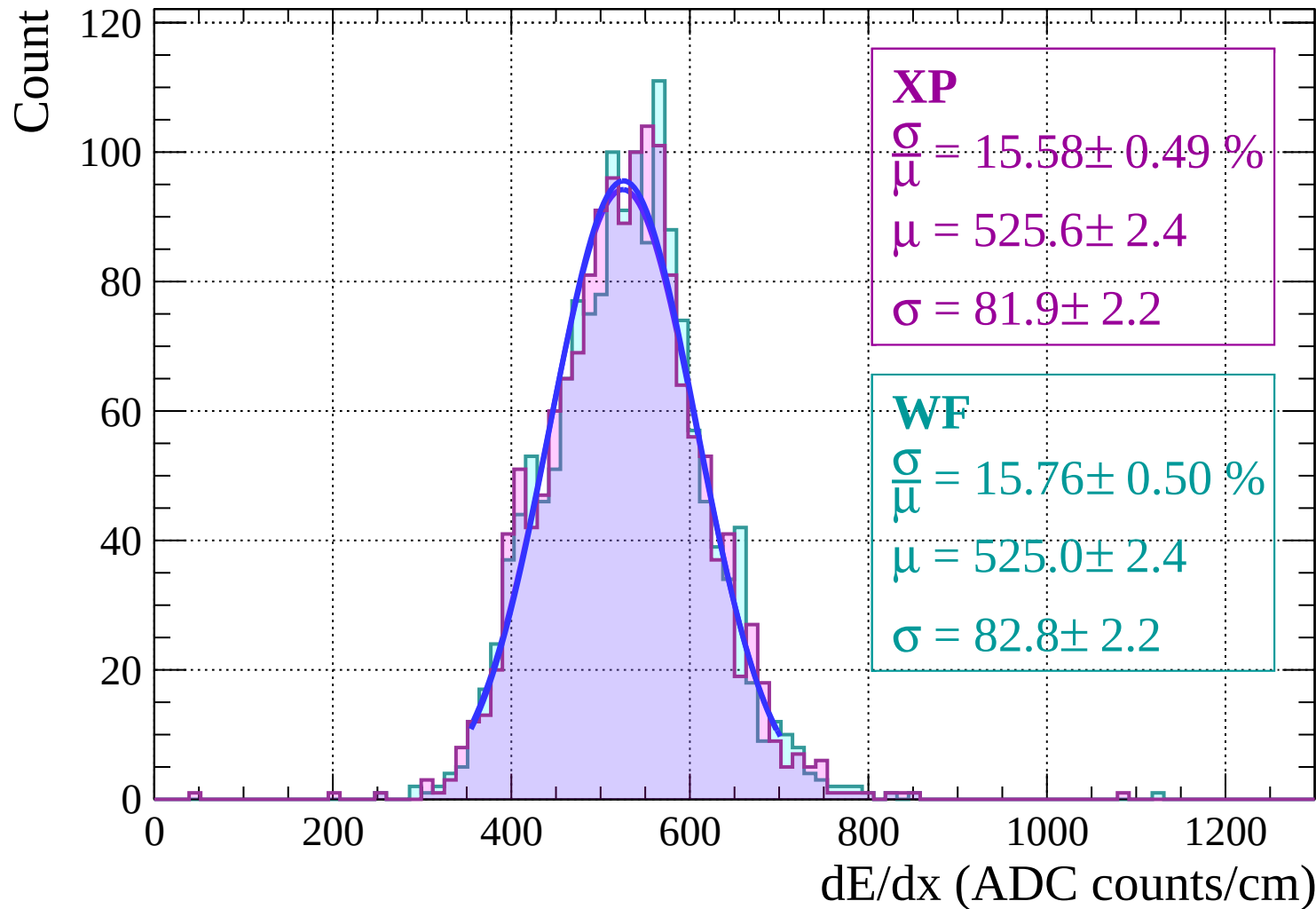


Energy loss $|-46 < \theta < -44$

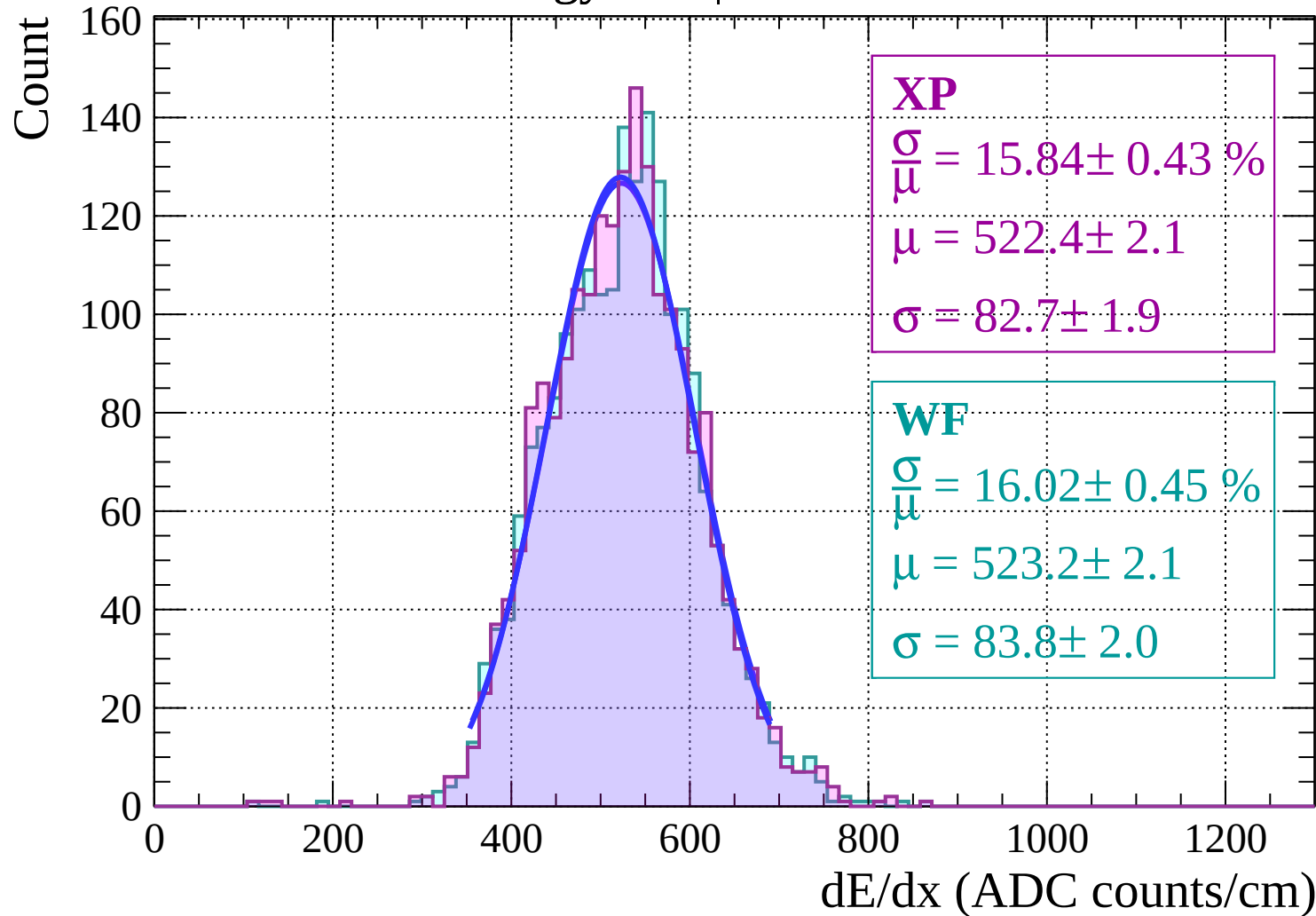
Count



Energy loss | $-44 < \theta < -42$

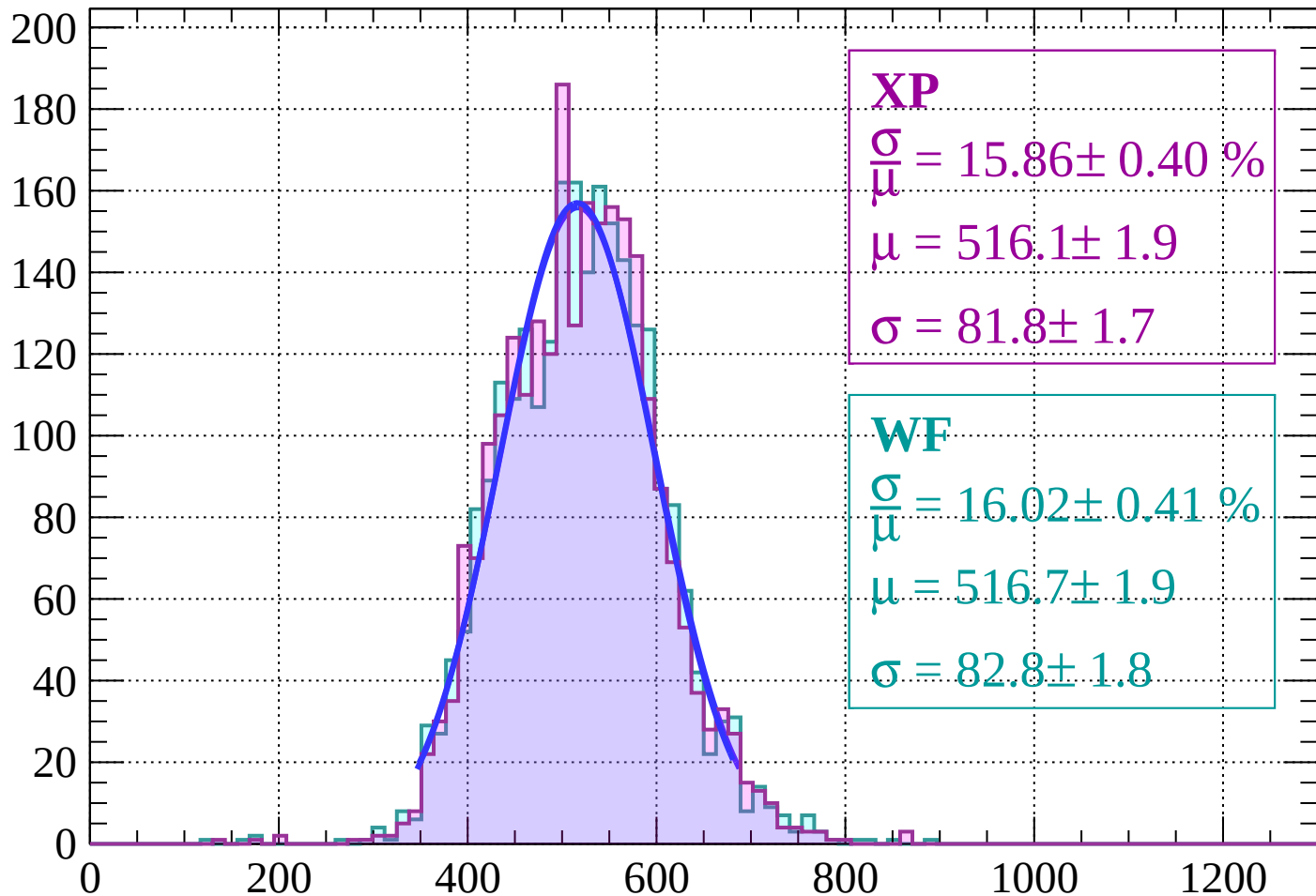


Energy loss $|-42 < \theta < -40$



Energy loss | $-40 < \theta < -38$

Count



XP

$$\frac{\sigma}{\mu} = 15.86 \pm 0.40 \%$$

$$\mu = 516.1 \pm 1.9$$

$$\sigma = 81.8 \pm 1.7$$

WF

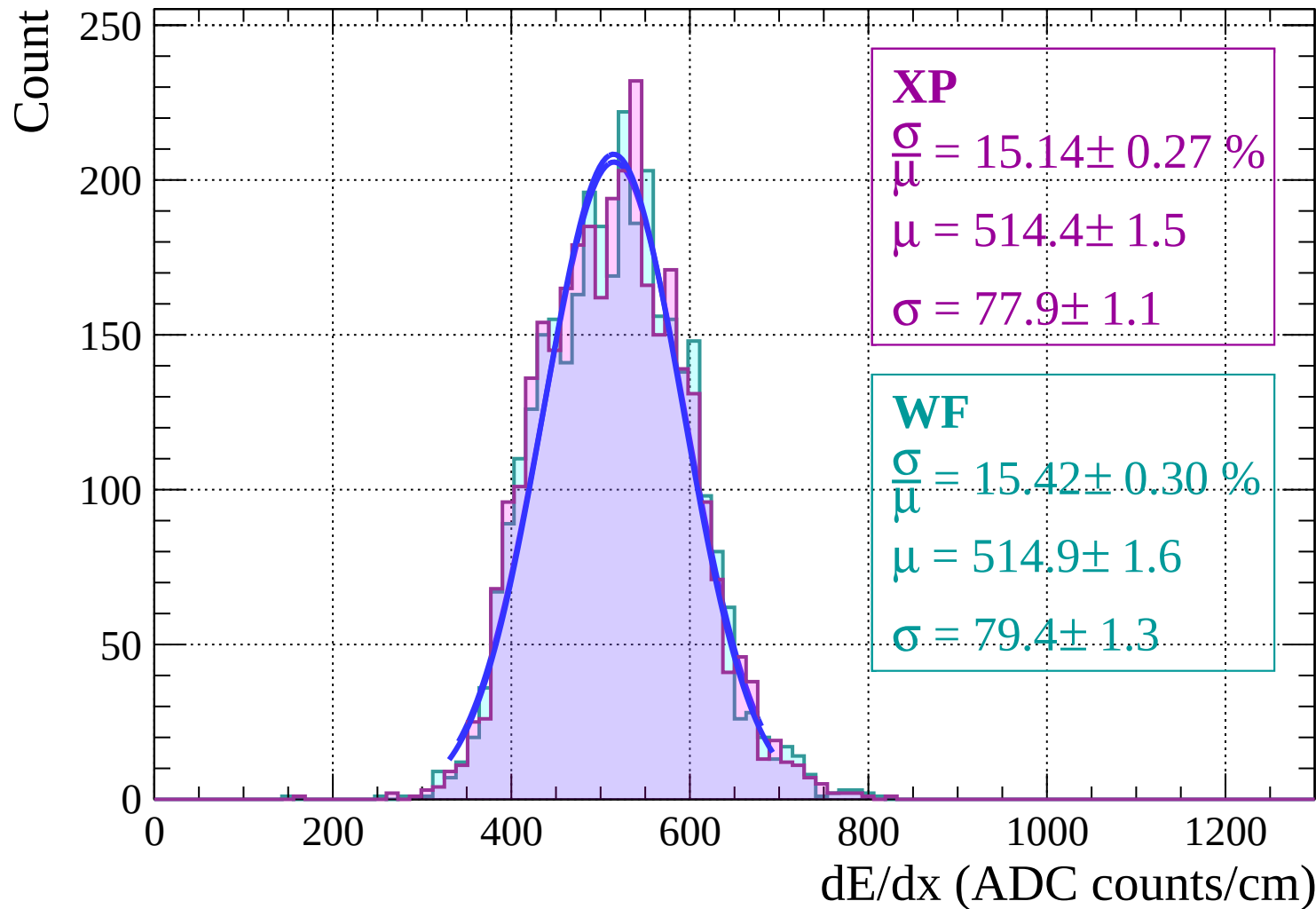
$$\frac{\sigma}{\mu} = 16.02 \pm 0.41 \%$$

$$\mu = 516.7 \pm 1.9$$

$$\sigma = 82.8 \pm 1.8$$

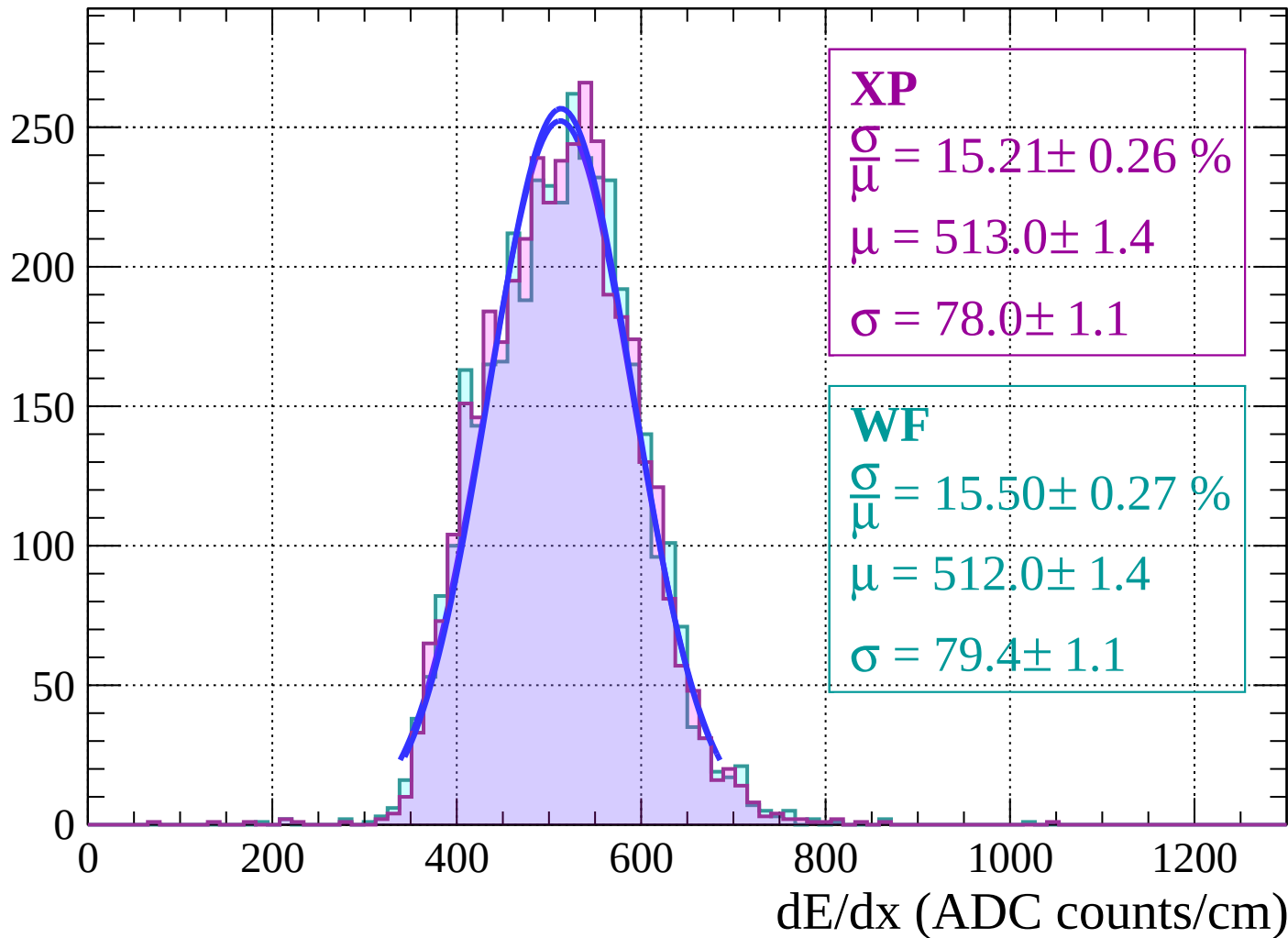
dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$



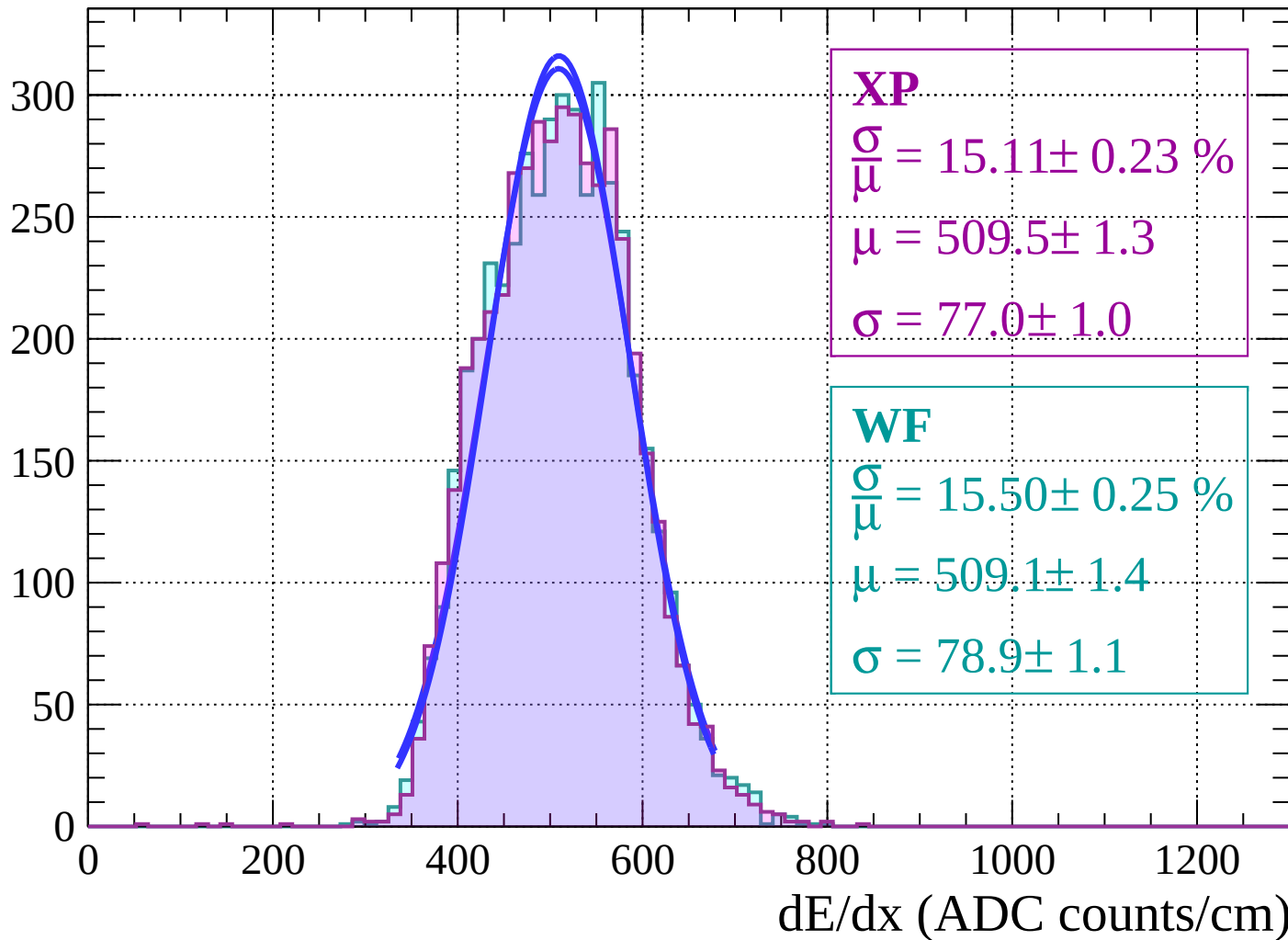
Energy loss | $-36 < \theta < -34$

Count

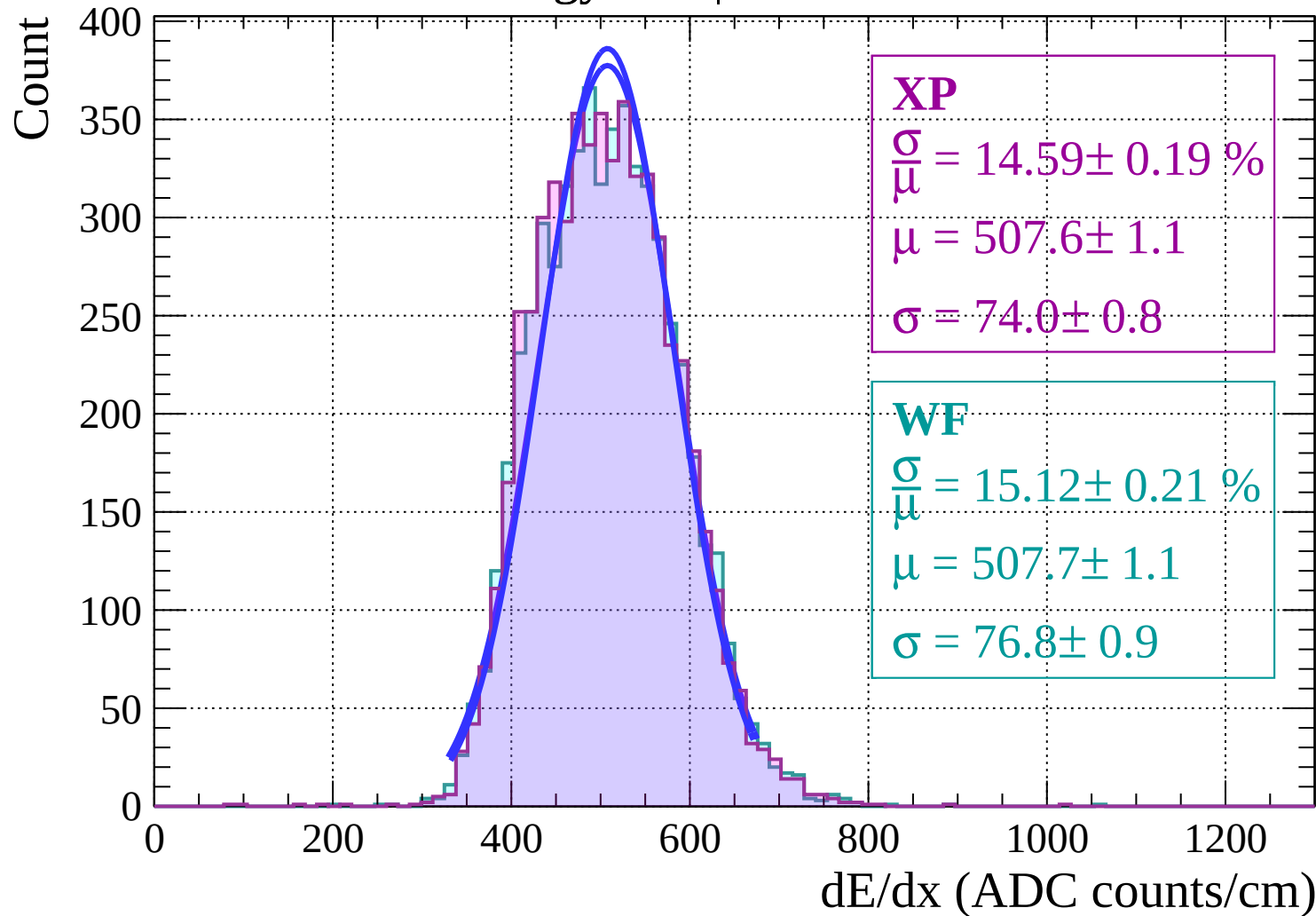


Energy loss | $-34 < \theta < -32$

Count

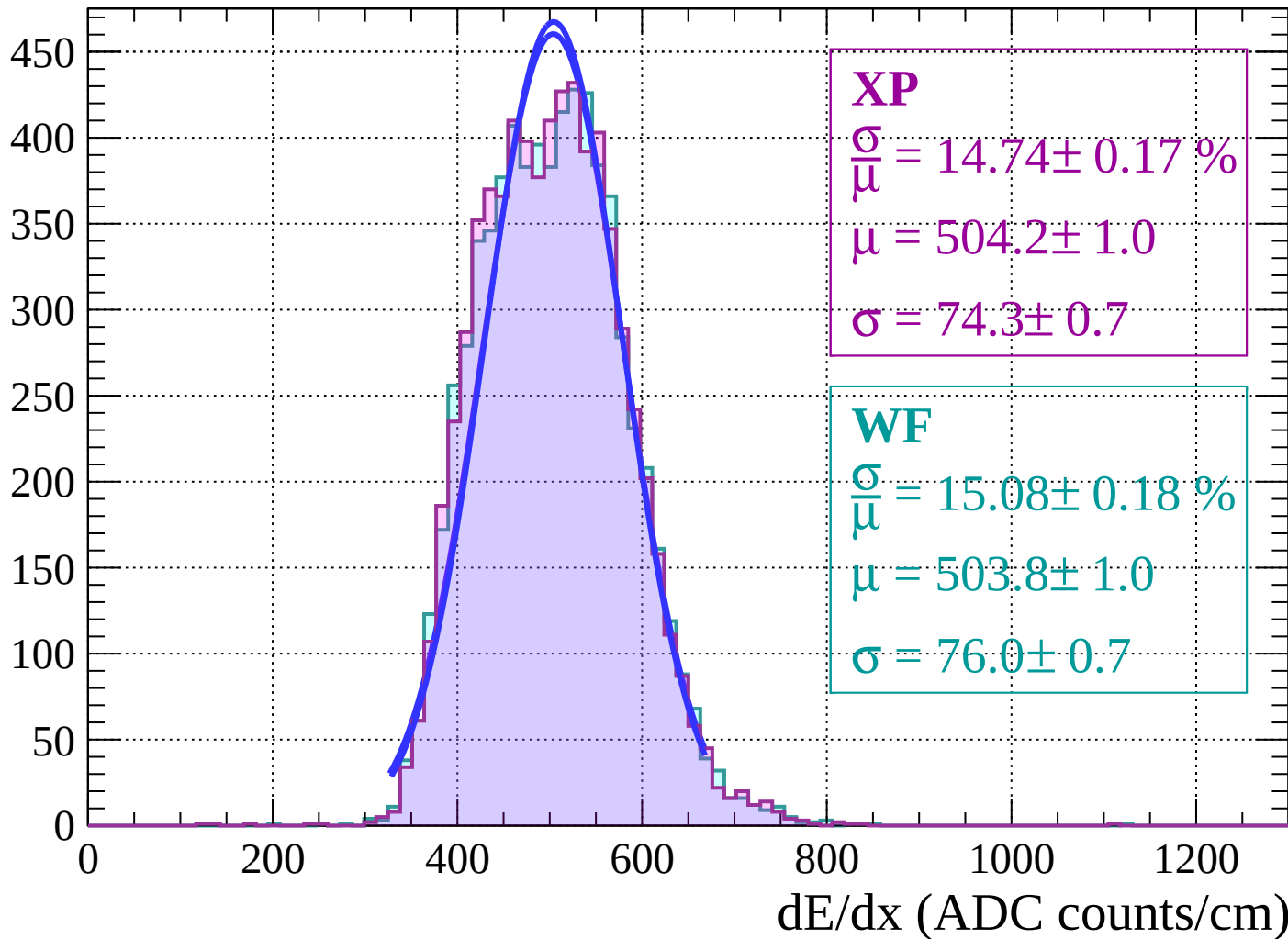


Energy loss $|-32 < \theta < -30$

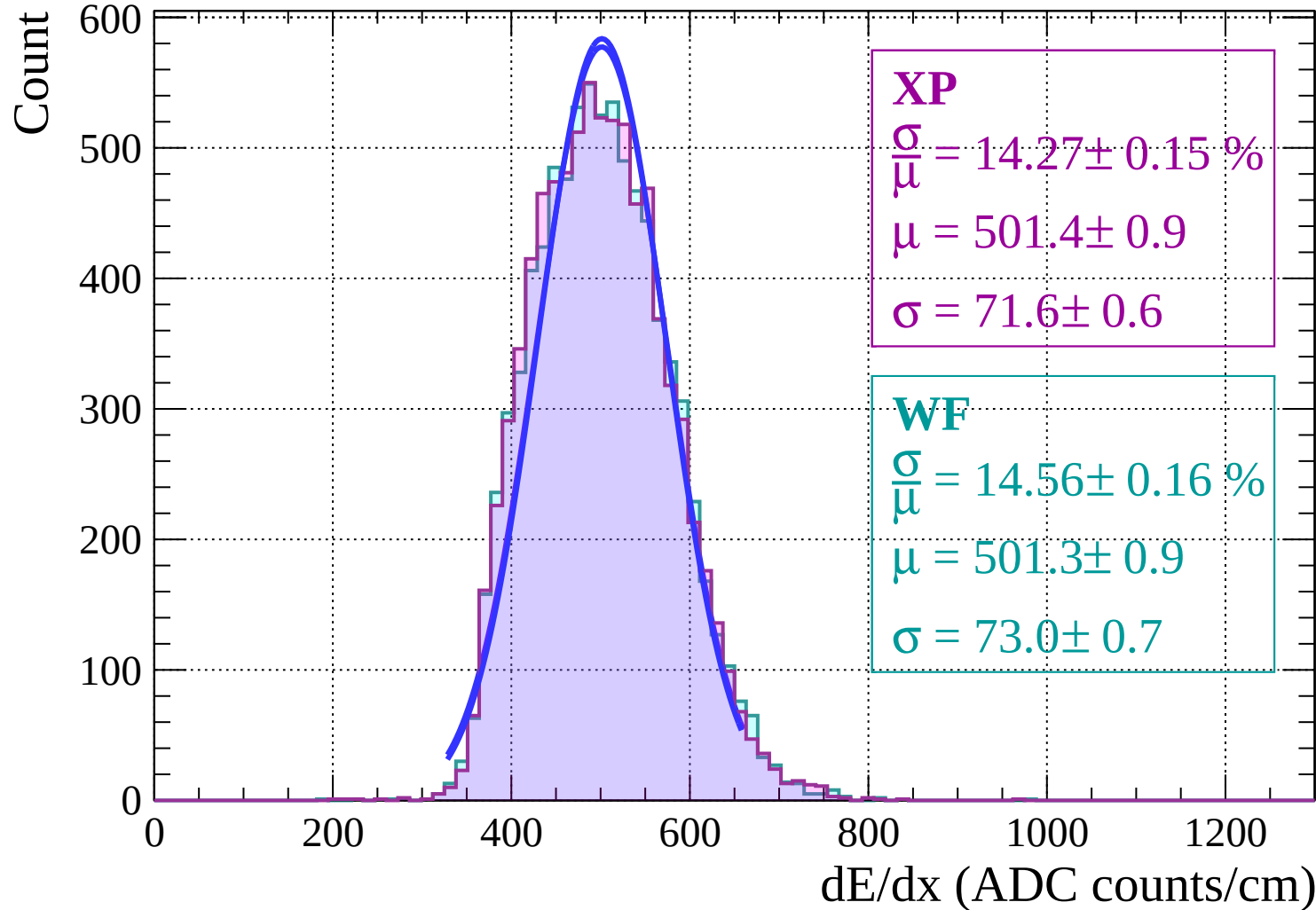


Energy loss | $-30 < \theta < -28$

Count

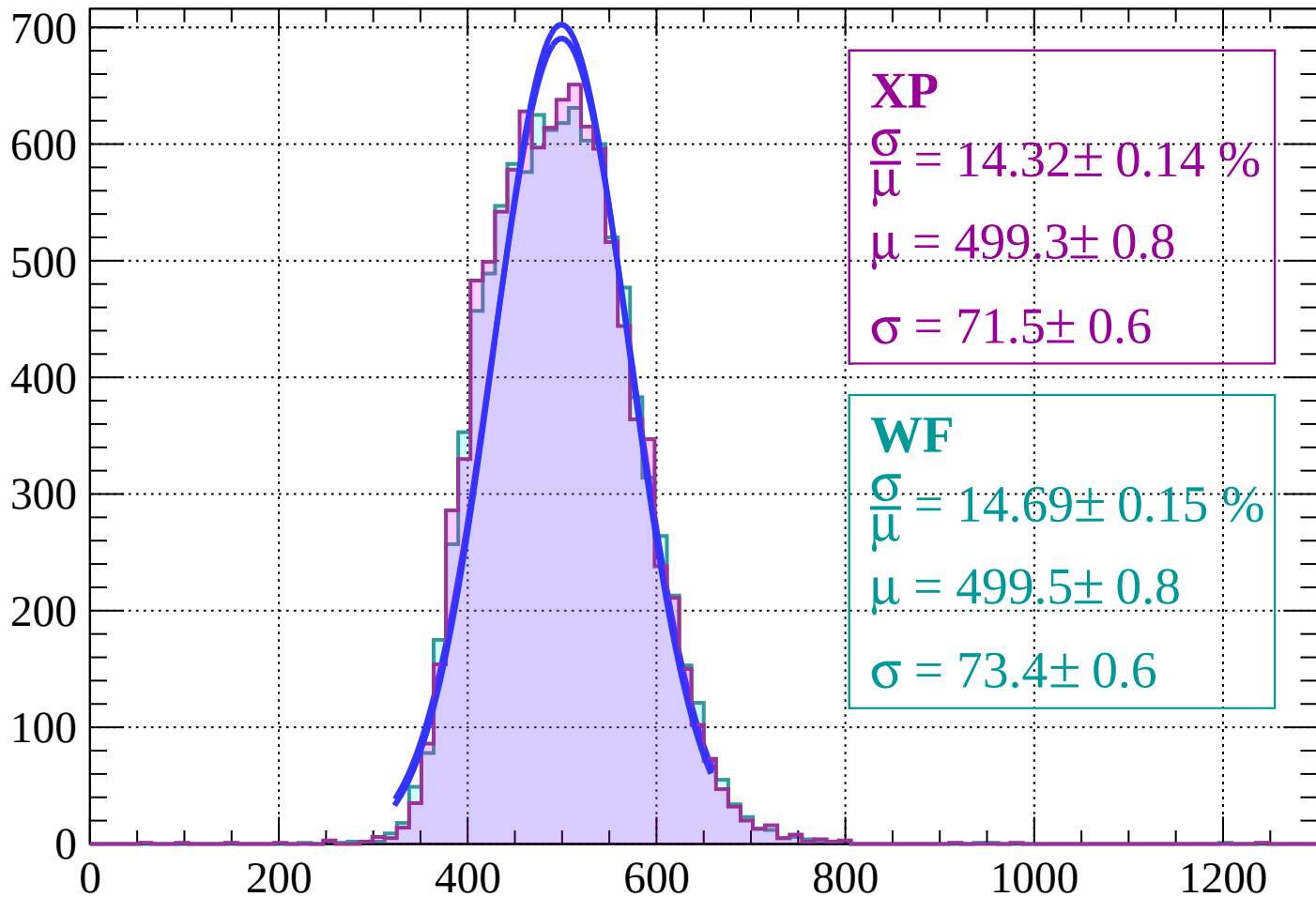


Energy loss | $-28 < \theta < -26$



Energy loss | $-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 14.32 \pm 0.14 \%$$

$$\mu = 499.3 \pm 0.8$$

$$\sigma = 71.5 \pm 0.6$$

WF

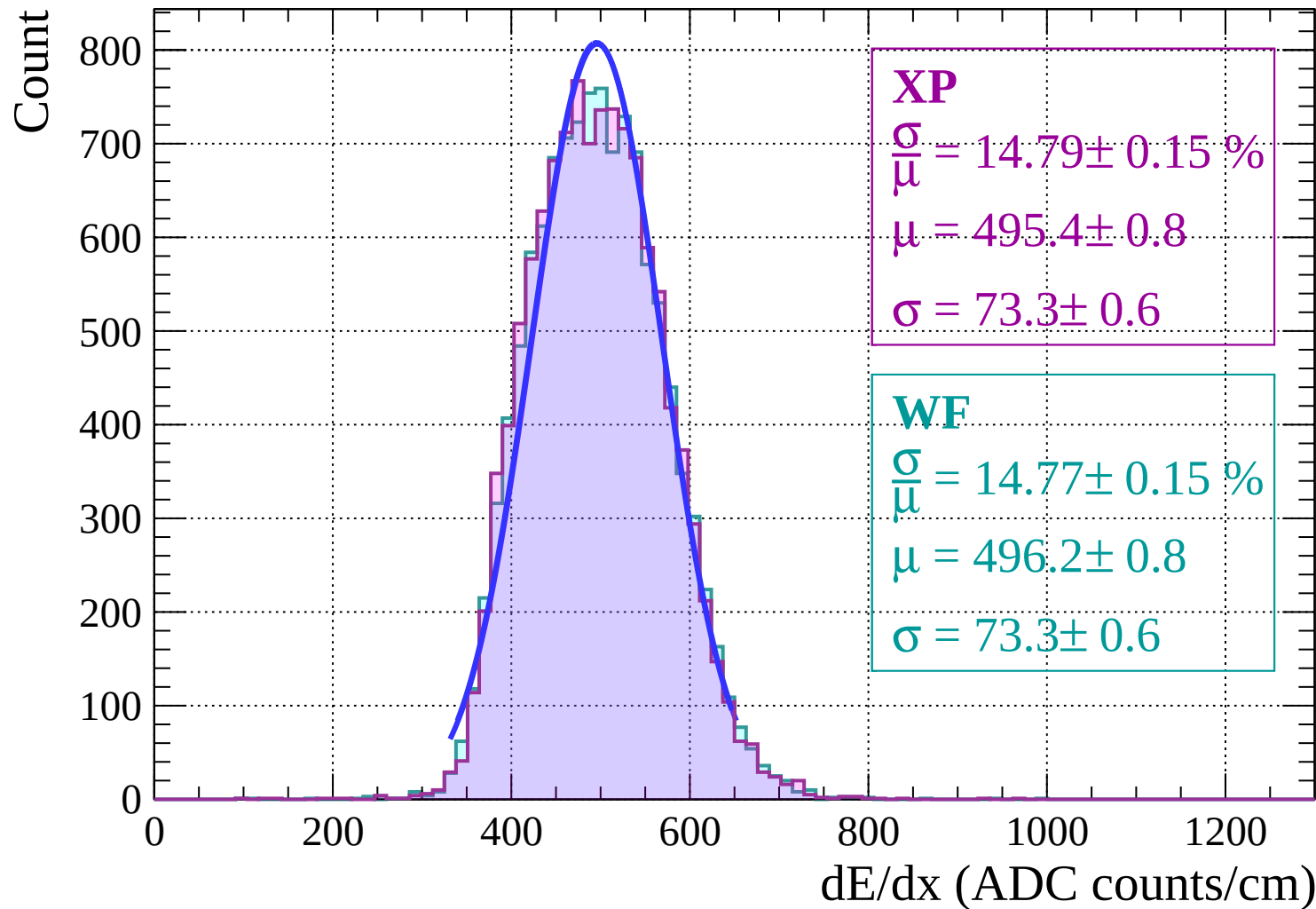
$$\frac{\sigma}{\mu} = 14.69 \pm 0.15 \%$$

$$\mu = 499.5 \pm 0.8$$

$$\sigma = 73.4 \pm 0.6$$

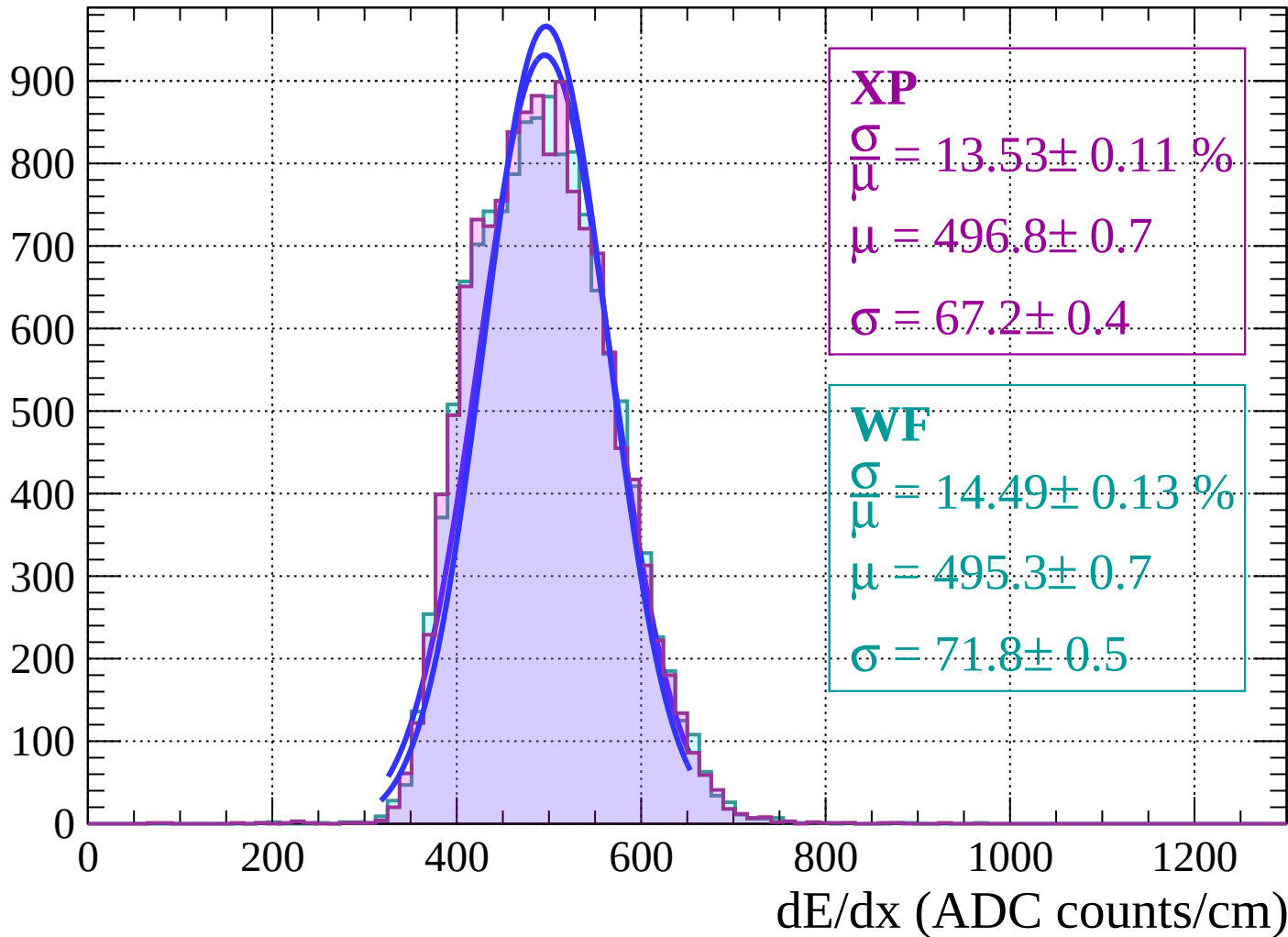
dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$



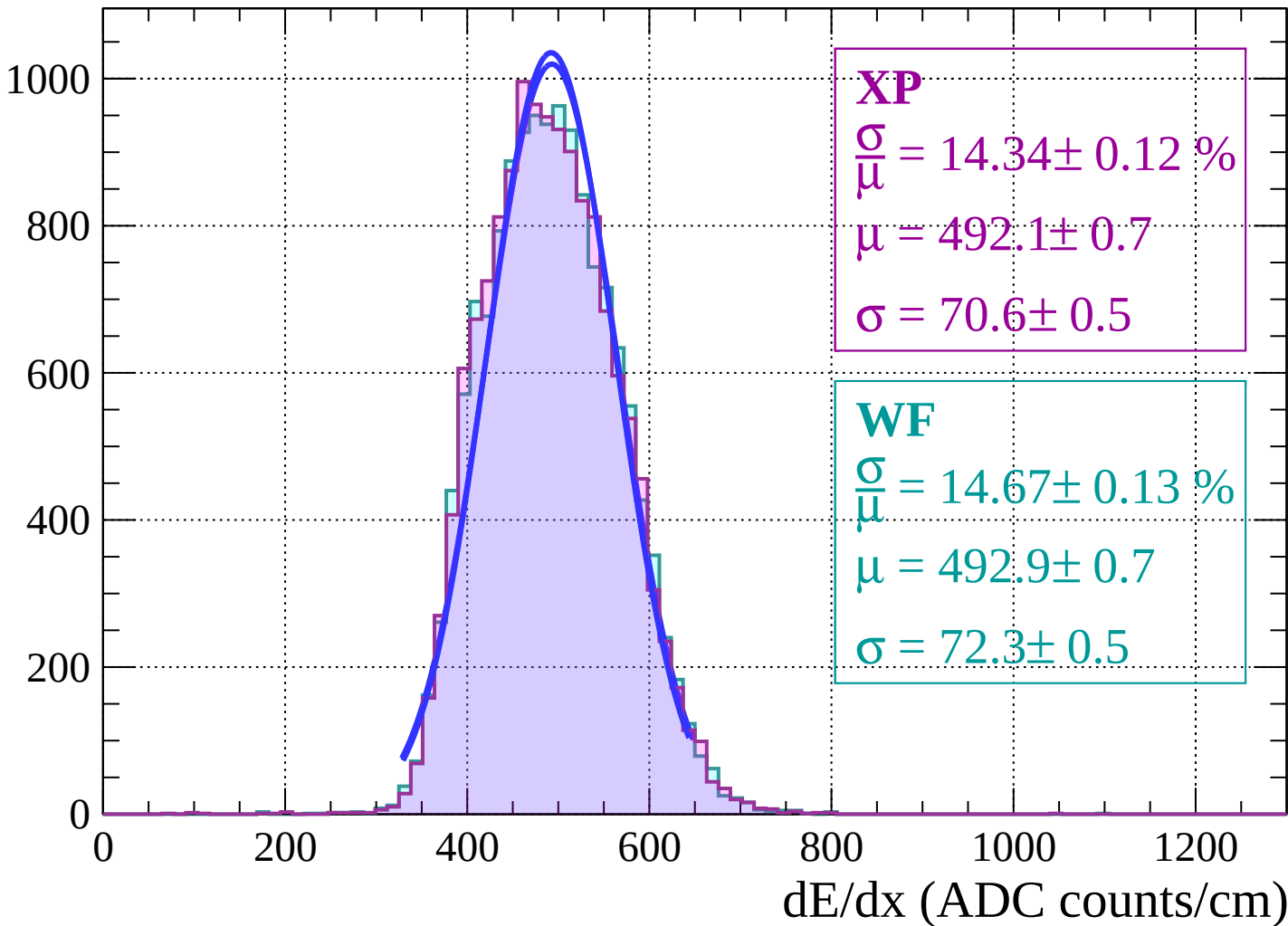
Energy loss | $-22 < \theta < -20$

Count



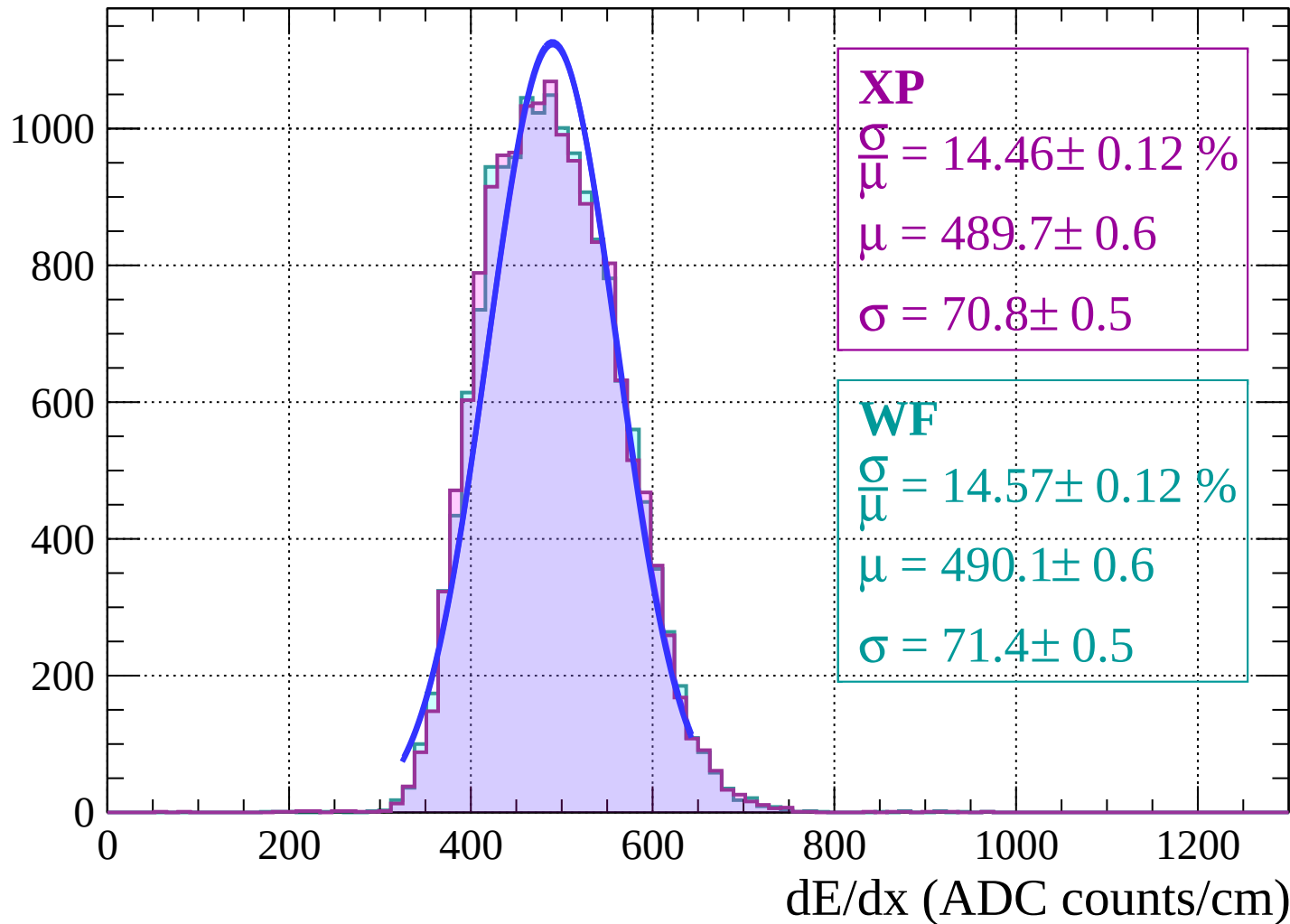
Energy loss | $-20 < \theta < -18$

Count

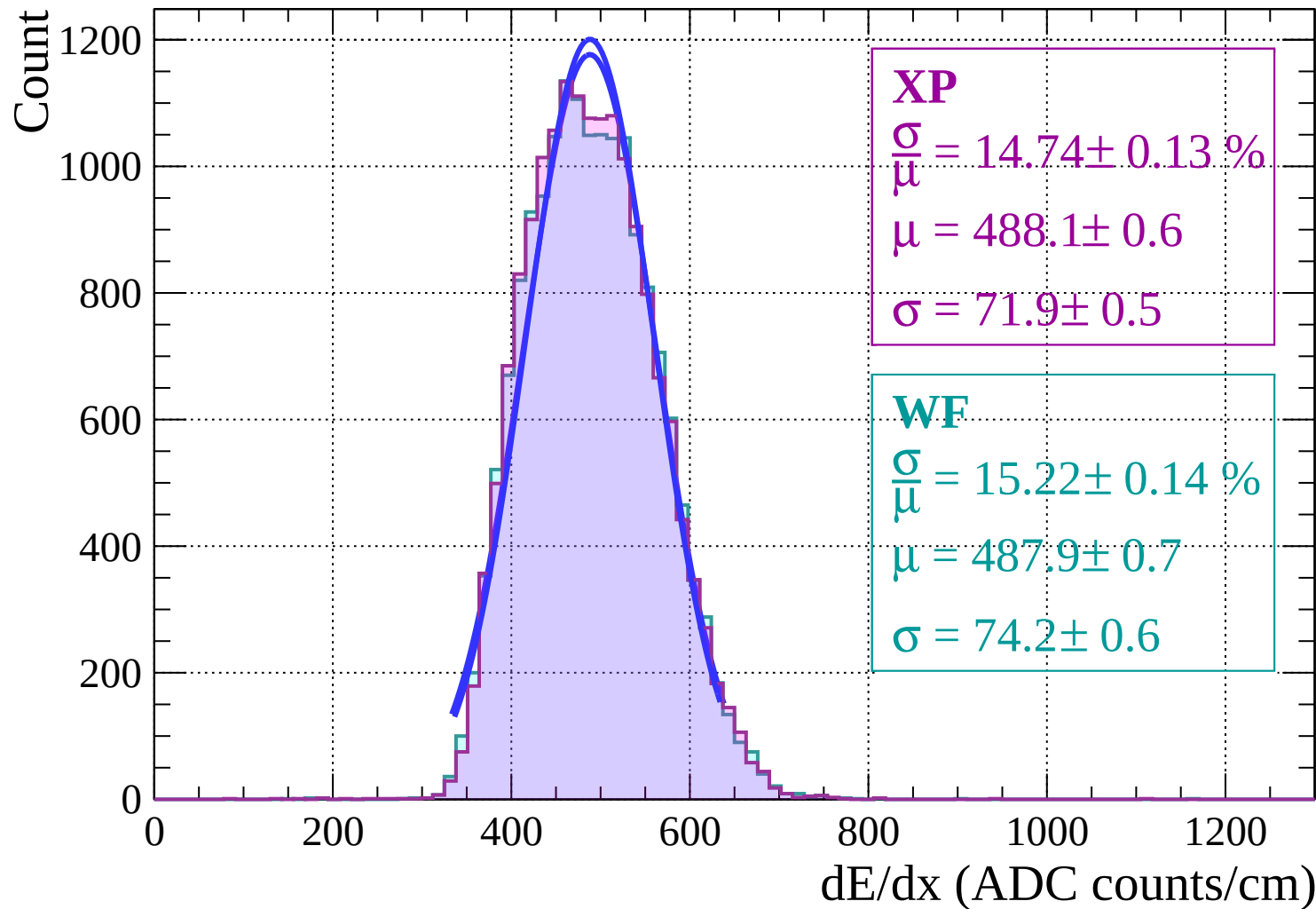


Energy loss | $-18 < \theta < -16$

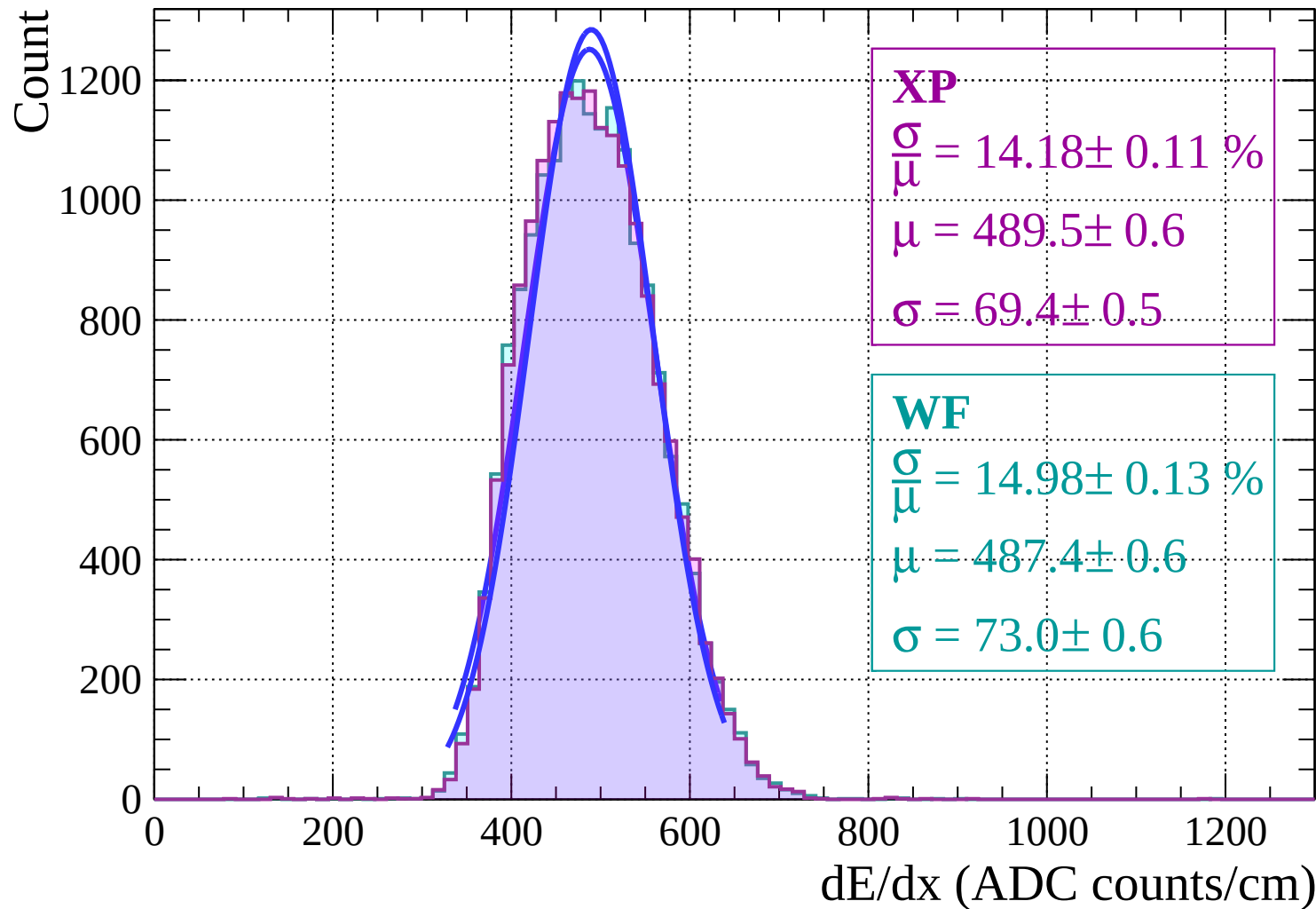
Count



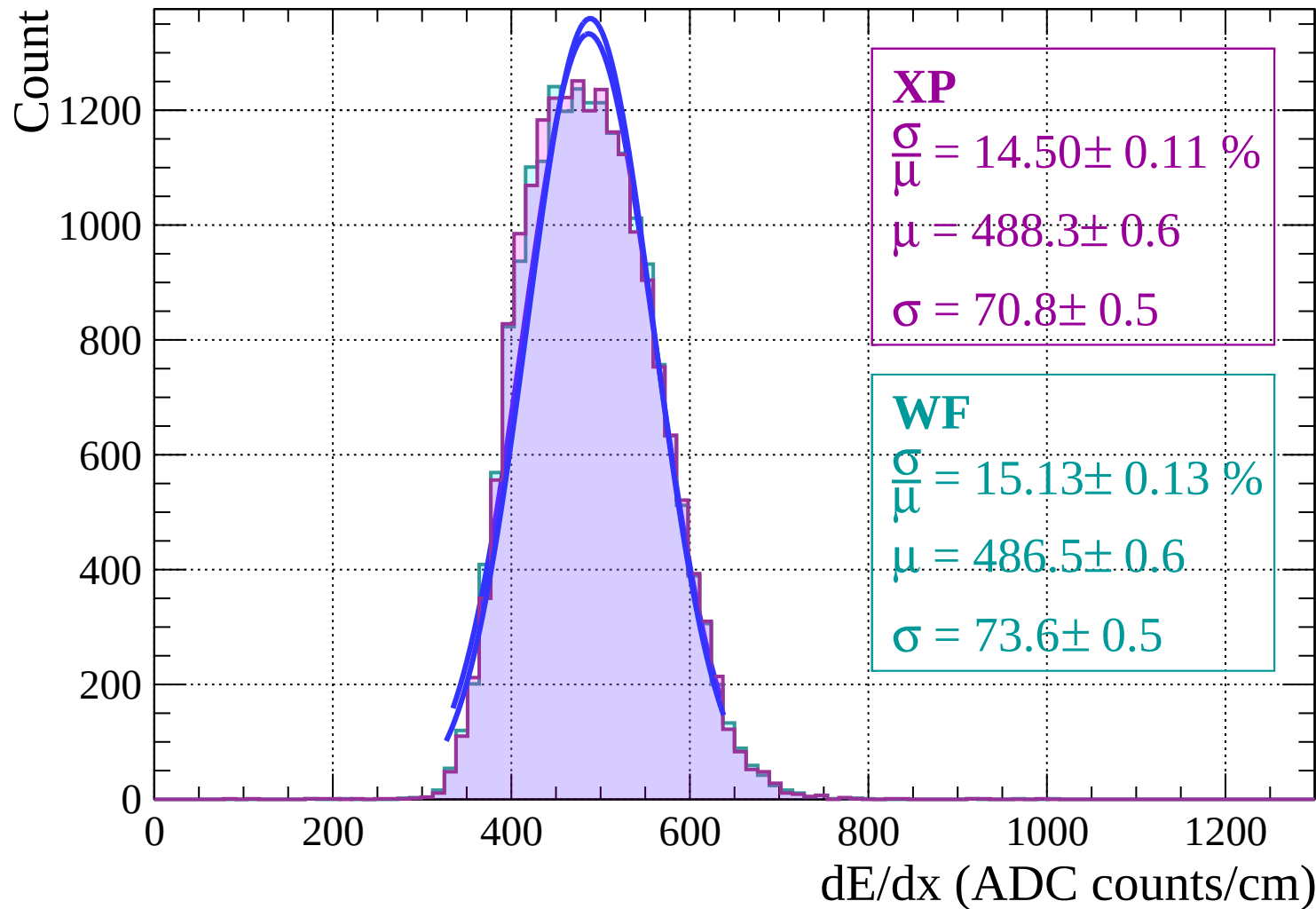
Energy loss | $-16 < \theta < -14$



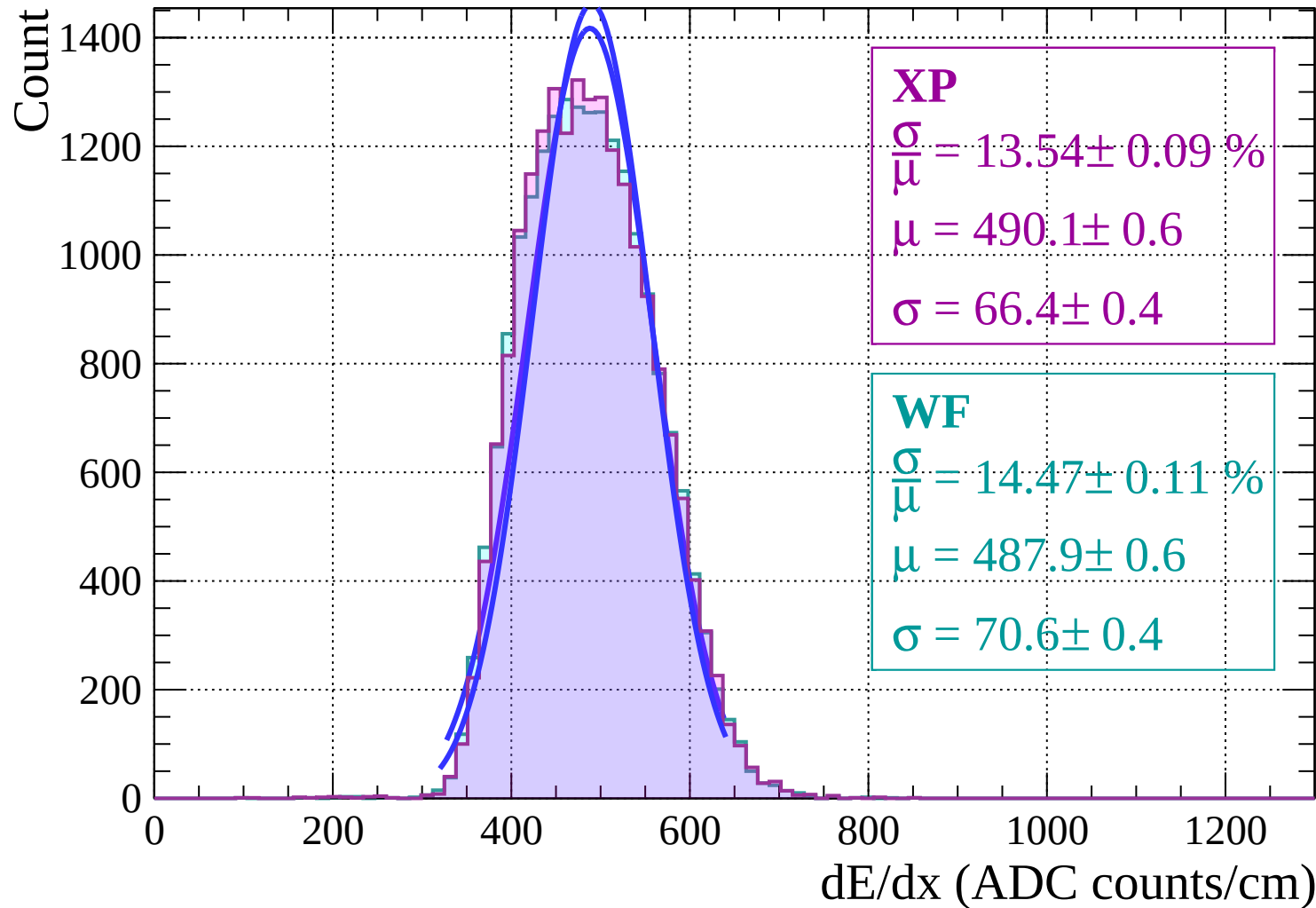
Energy loss | $-14 < \theta < -12$



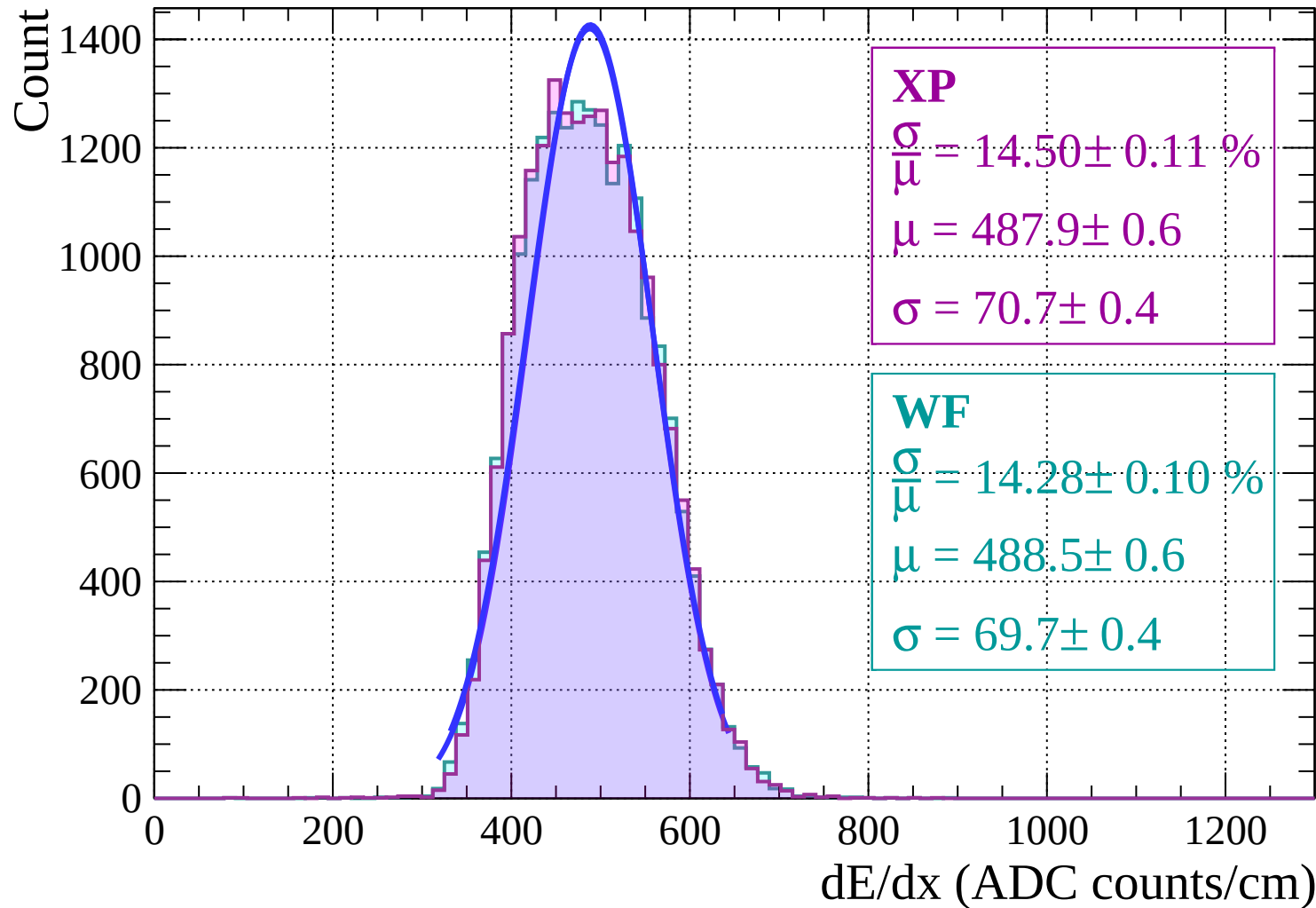
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

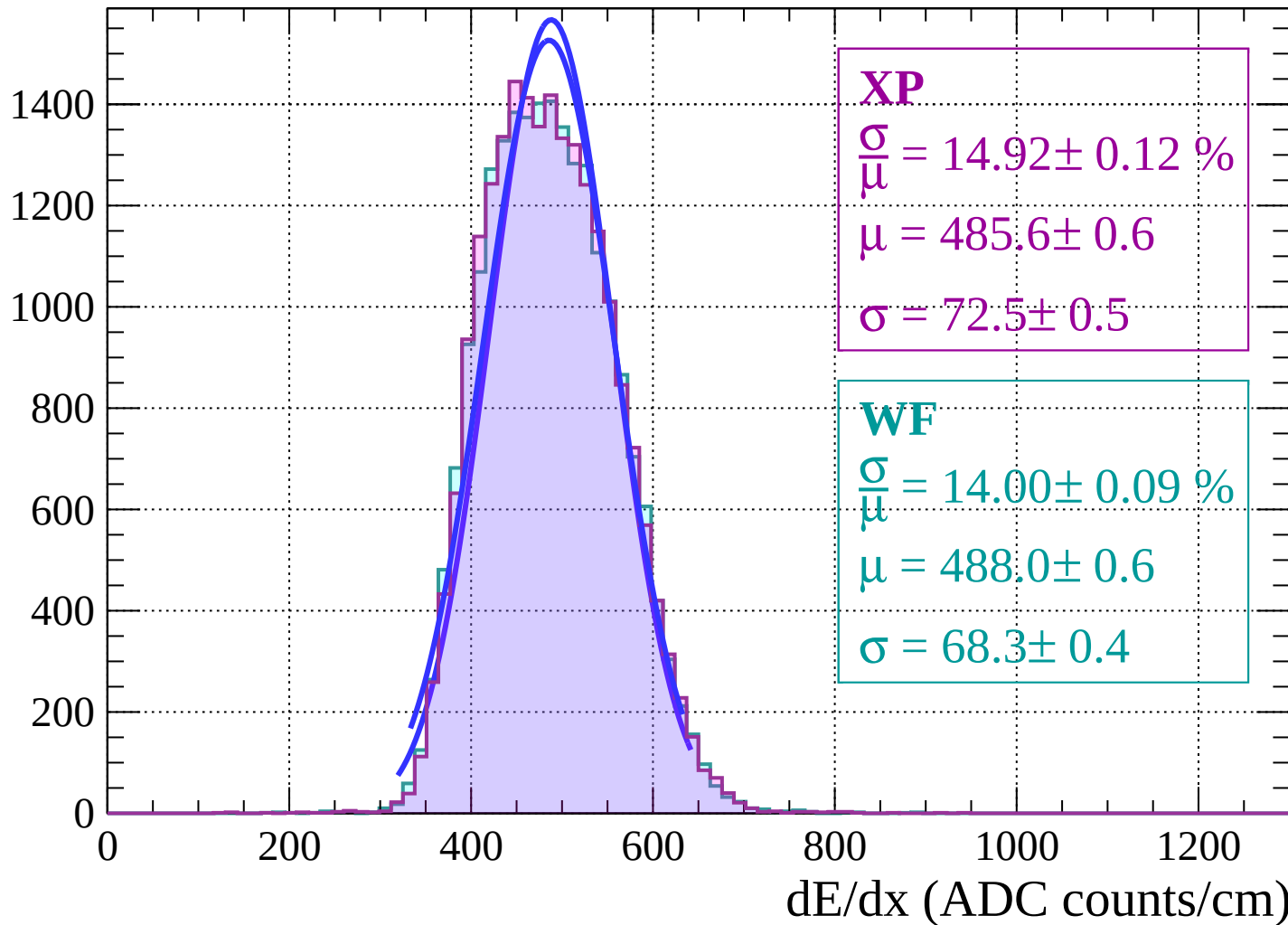


Energy loss | $-8 < \theta < -6$



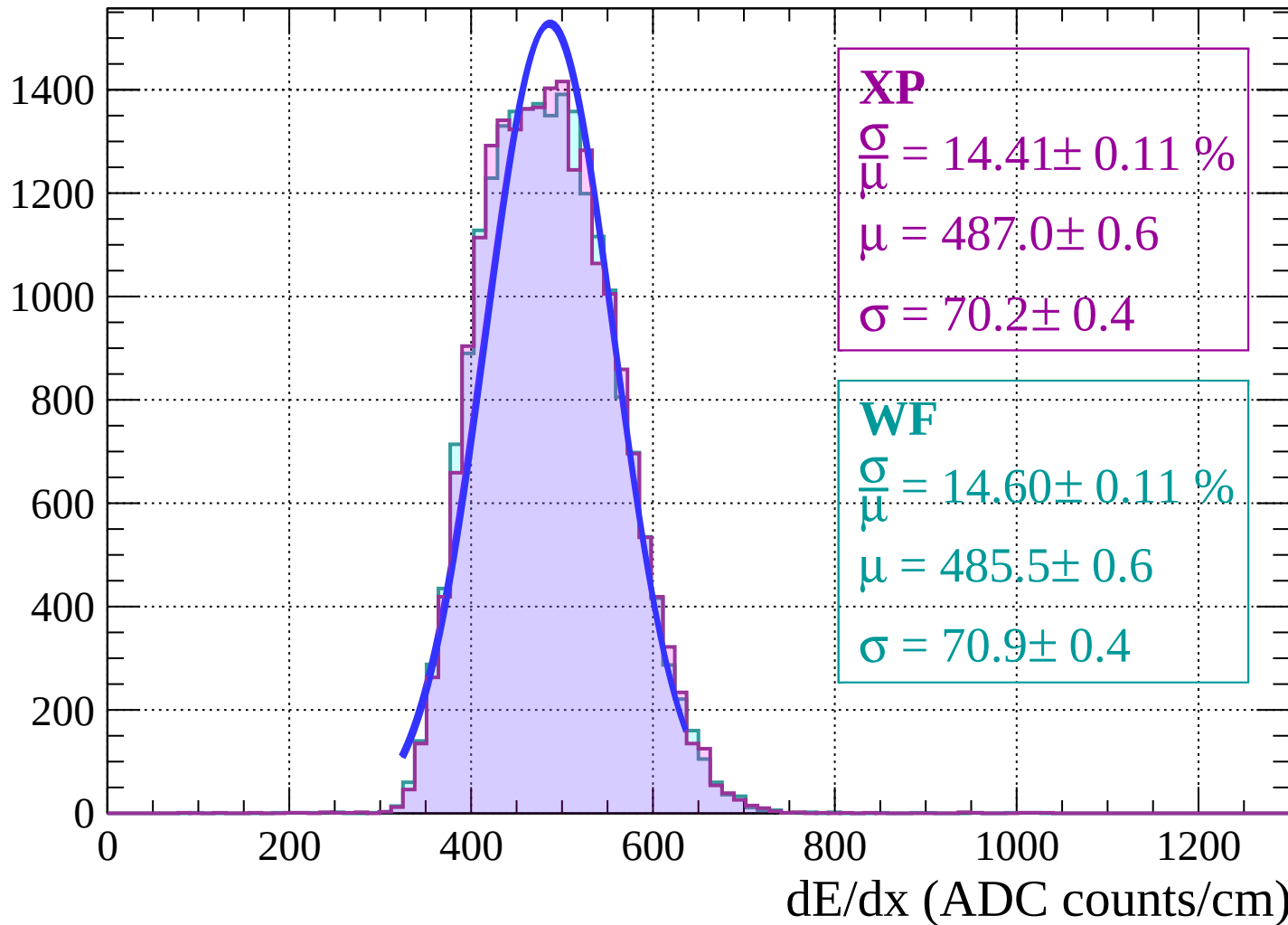
Energy loss | $-6 < \theta < -4$

Count

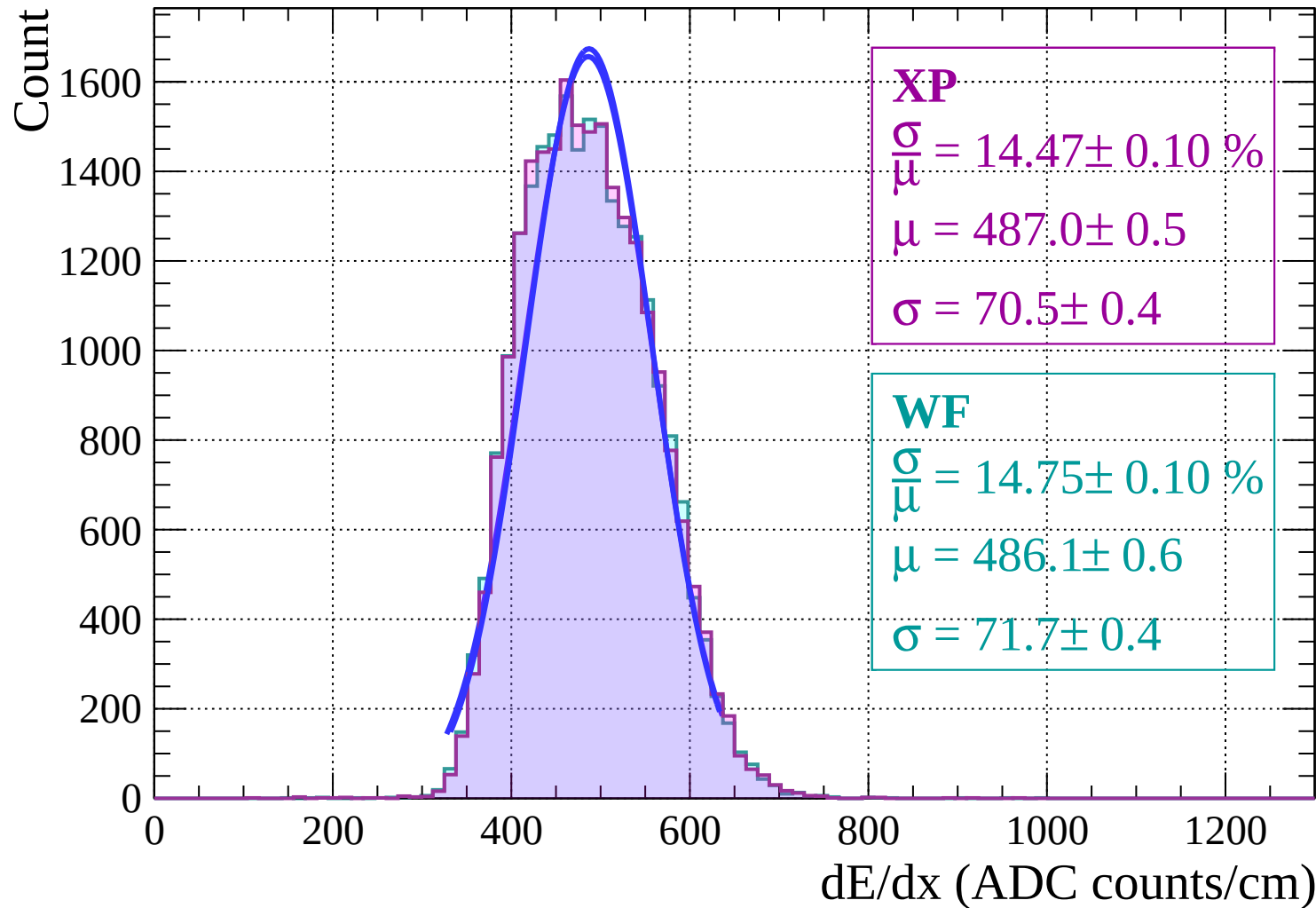


Energy loss | $-4 < \theta < -2$

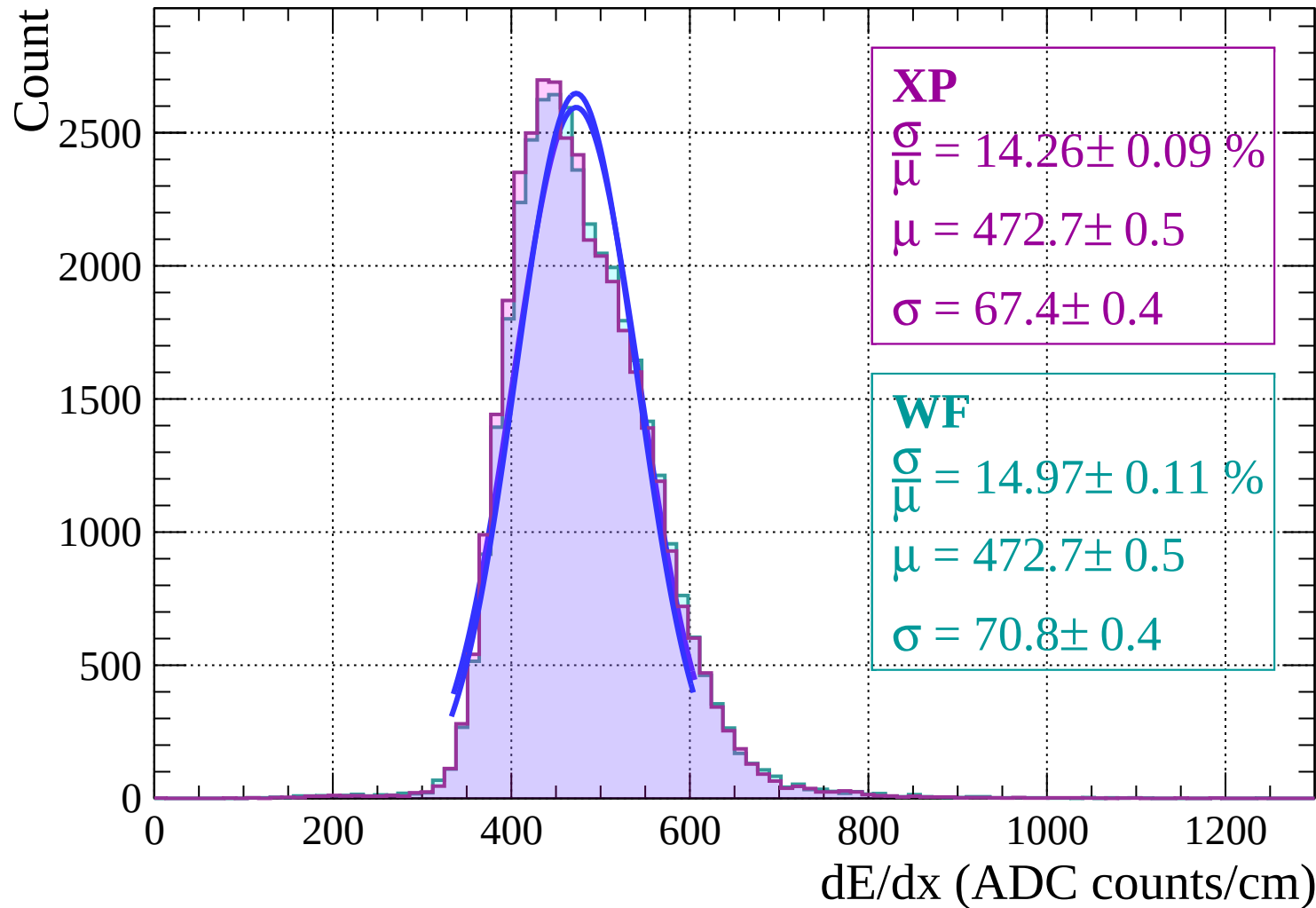
Count



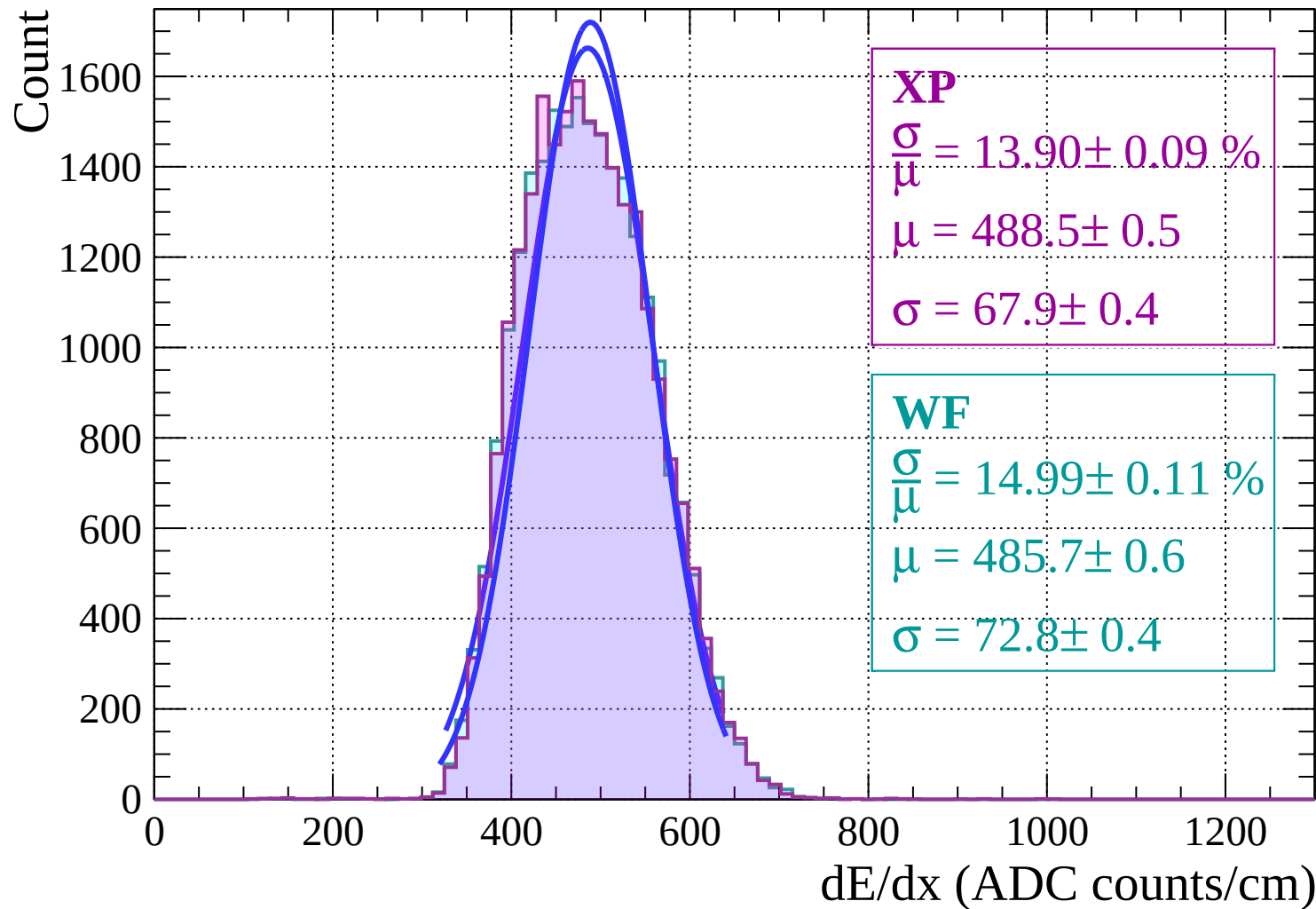
Energy loss | $-2 < \theta < 0$



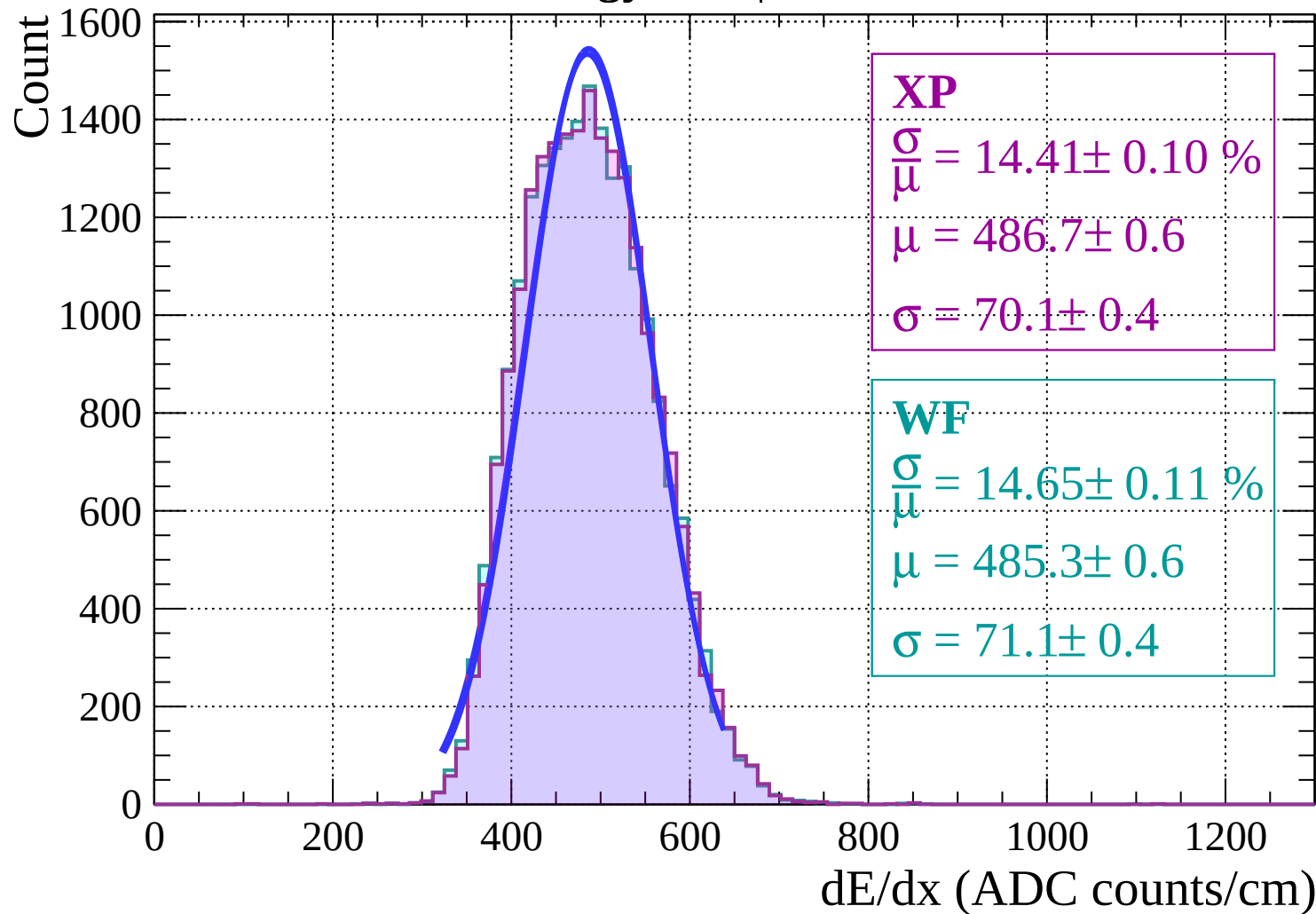
Energy loss | $0 < \theta < 2$



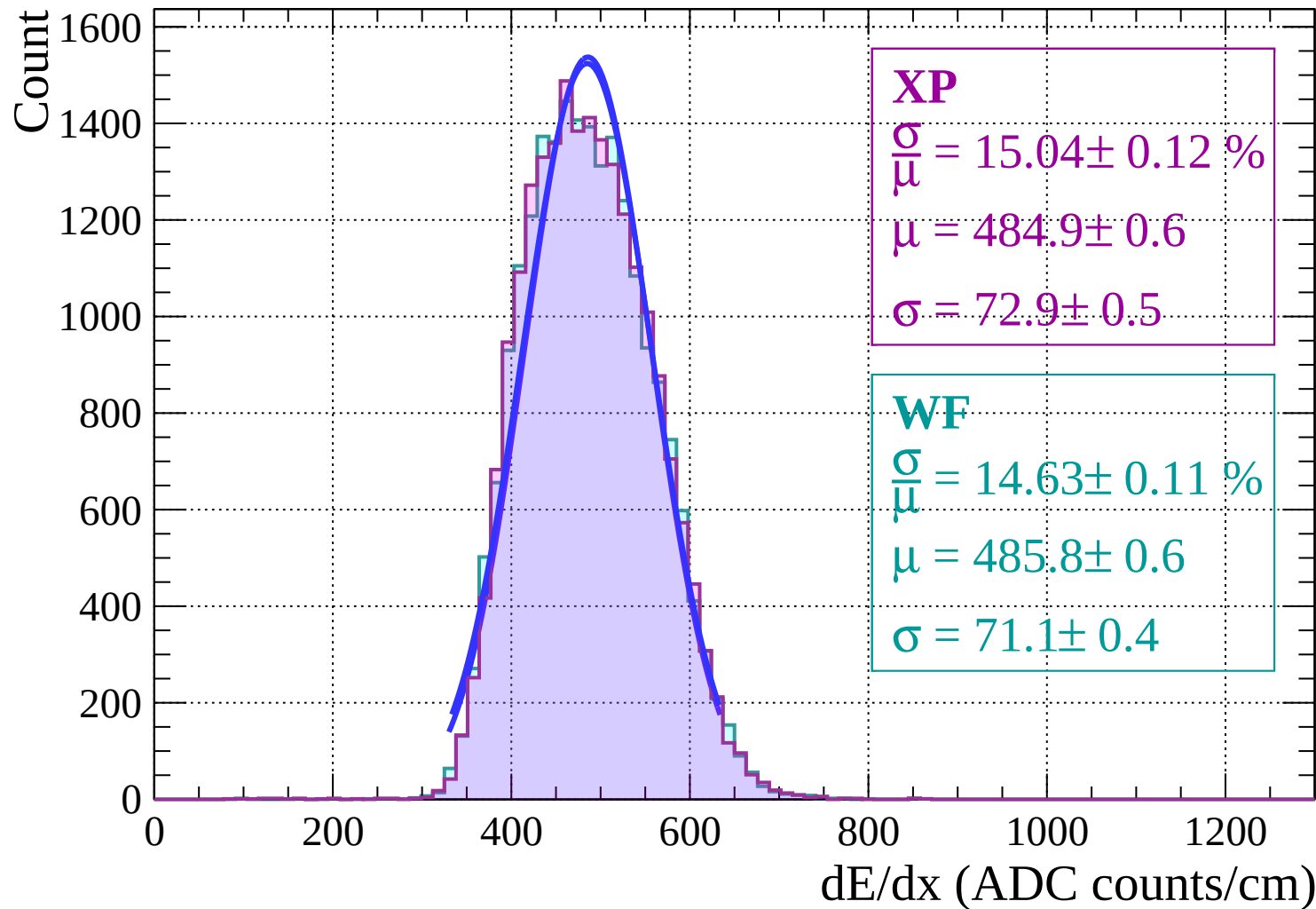
Energy loss | $2 < \theta < 4$



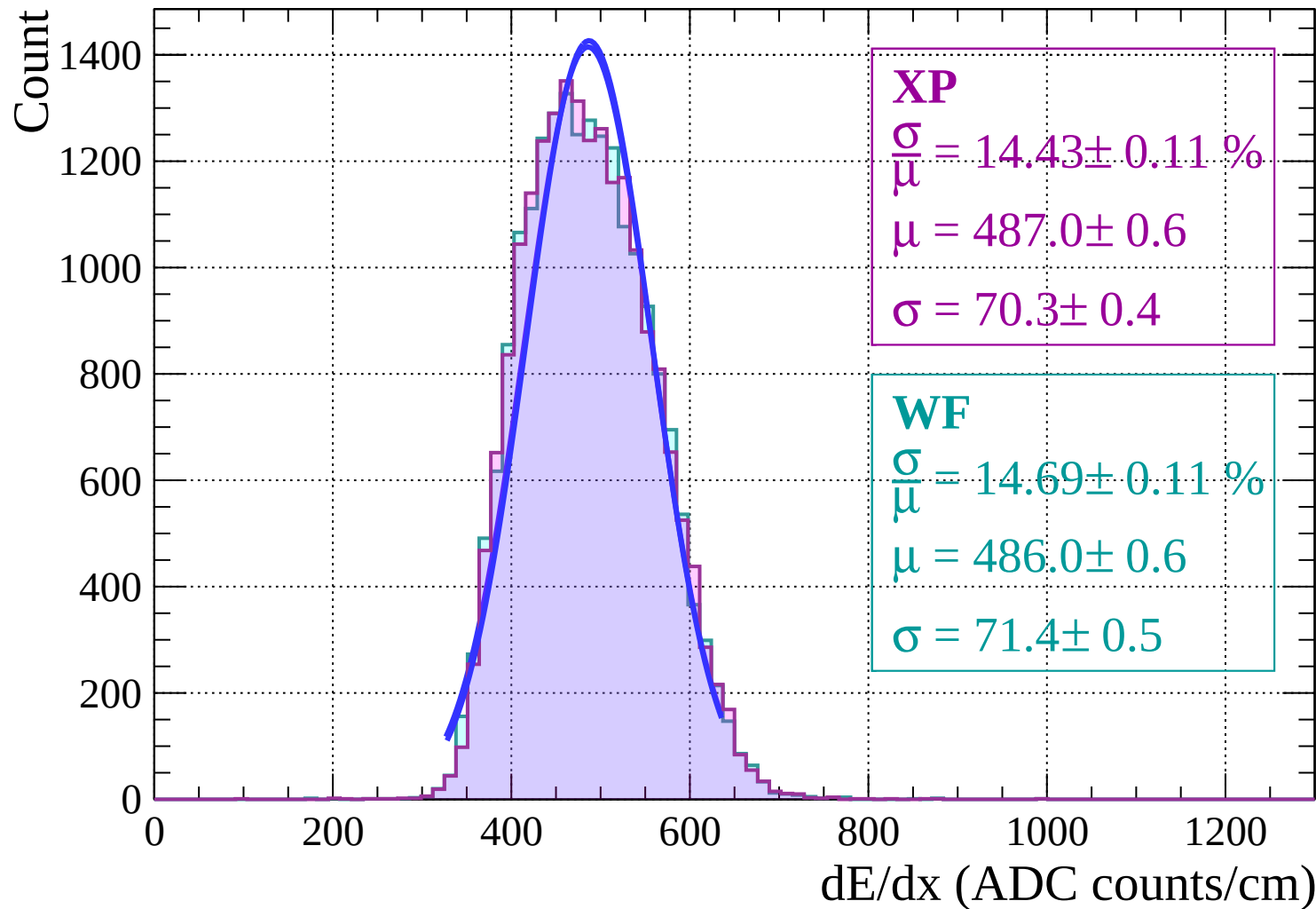
Energy loss | $4 < \theta < 6$



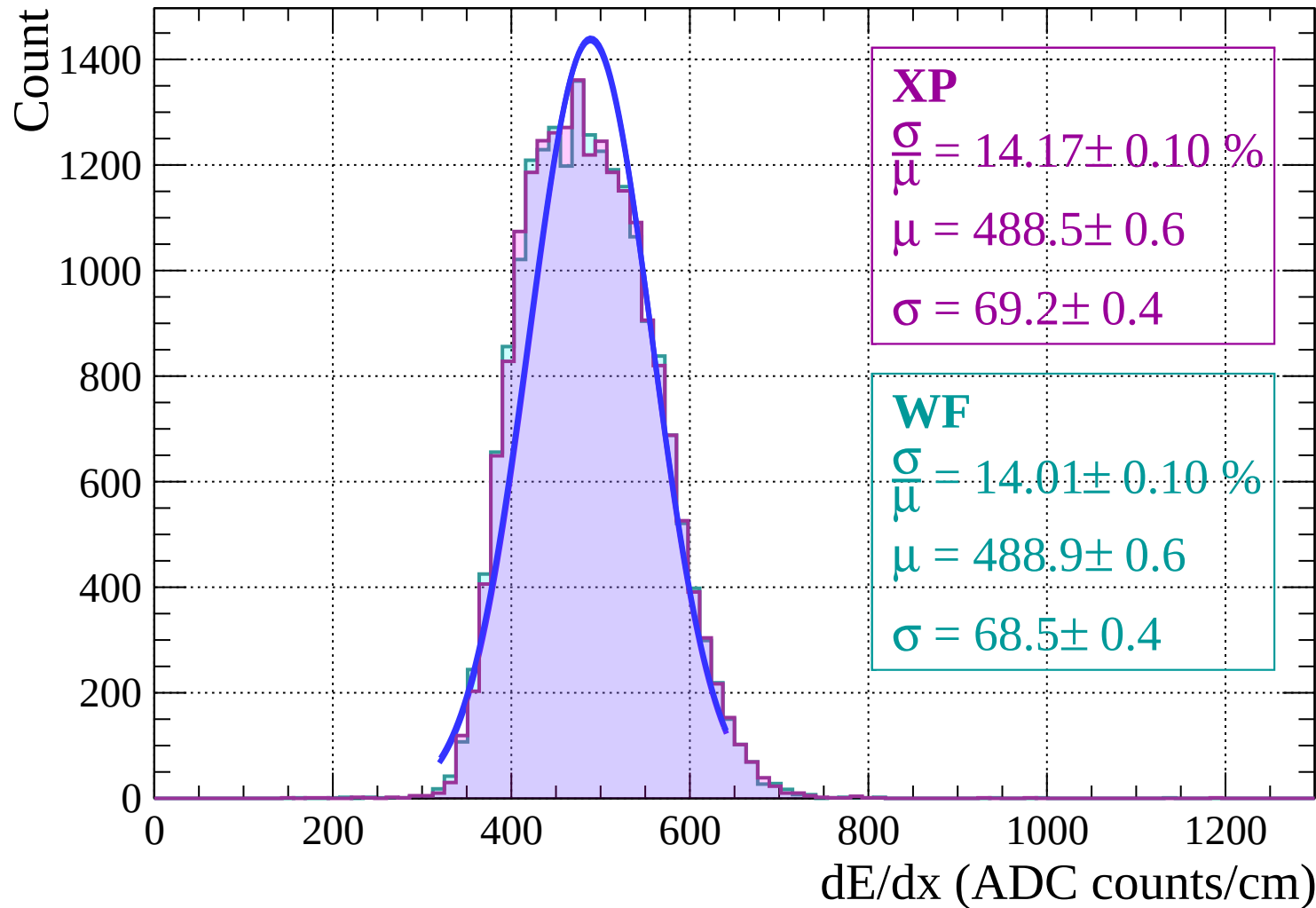
Energy loss | $6 < \theta < 8$



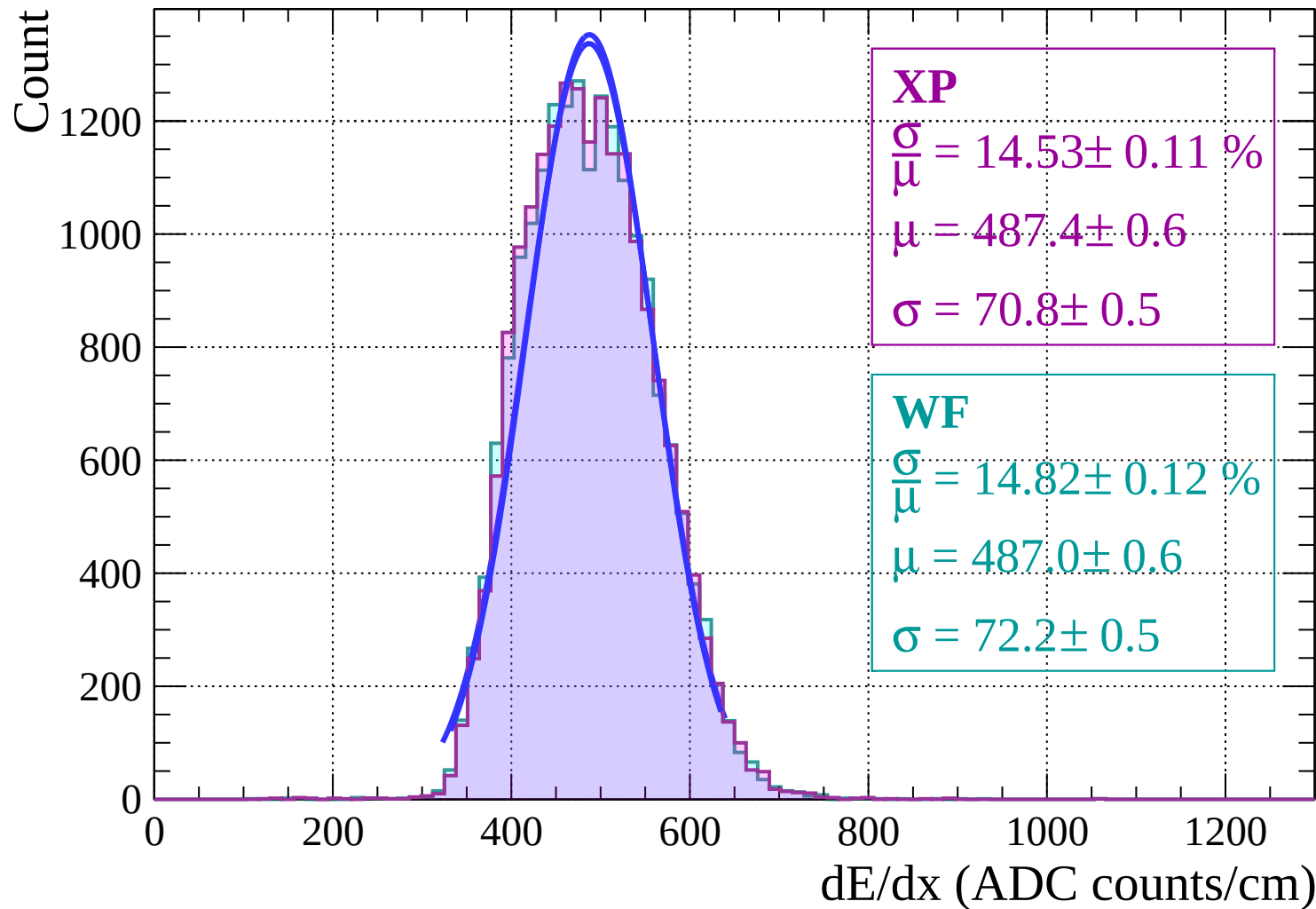
Energy loss | $8 < \theta < 10$



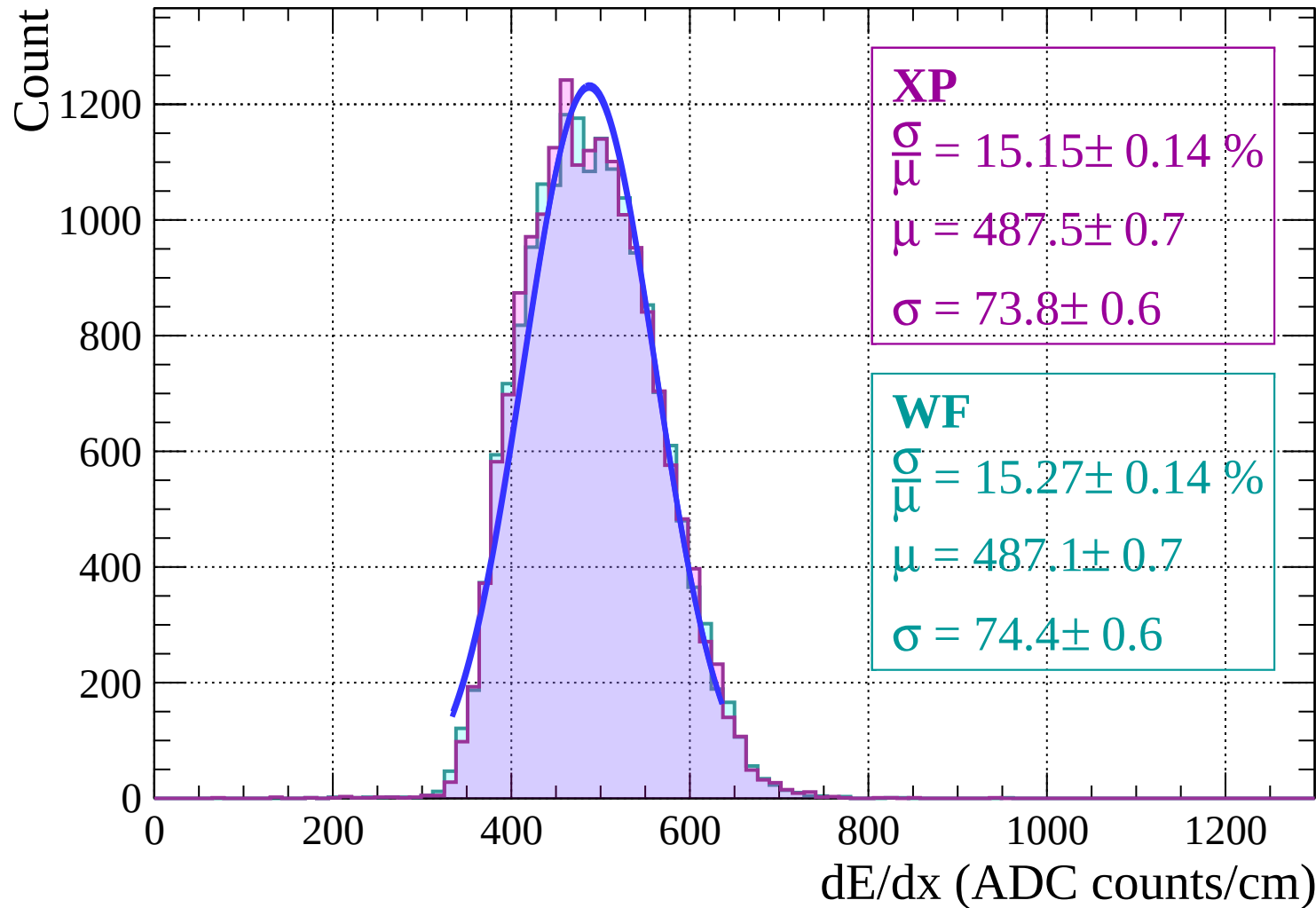
Energy loss | $10 < \theta < 12$



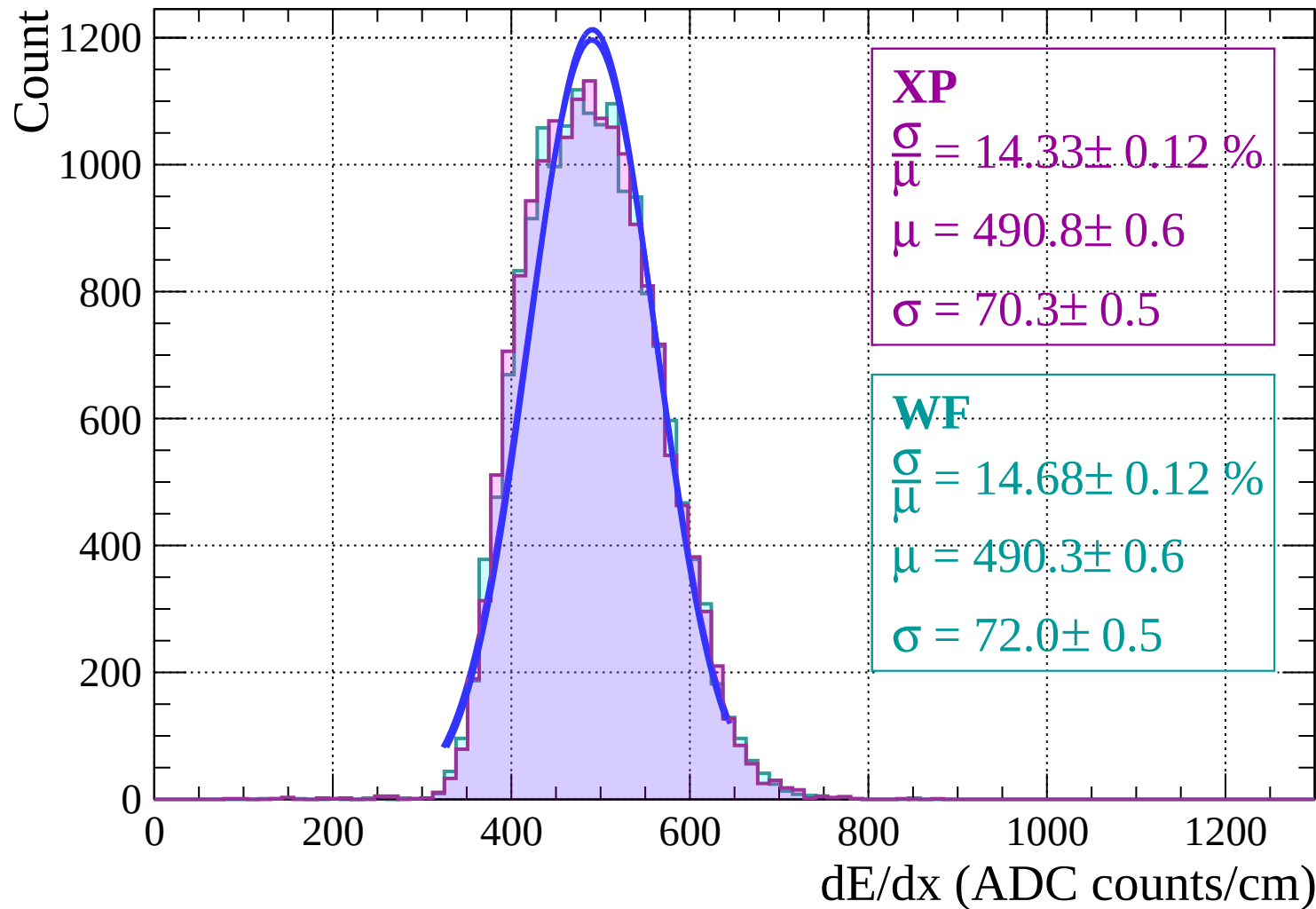
Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$



Energy loss | $18 < \theta < 20$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 14.96 \pm 0.14 \%$$

$$\mu = 490.8 \pm 0.7$$

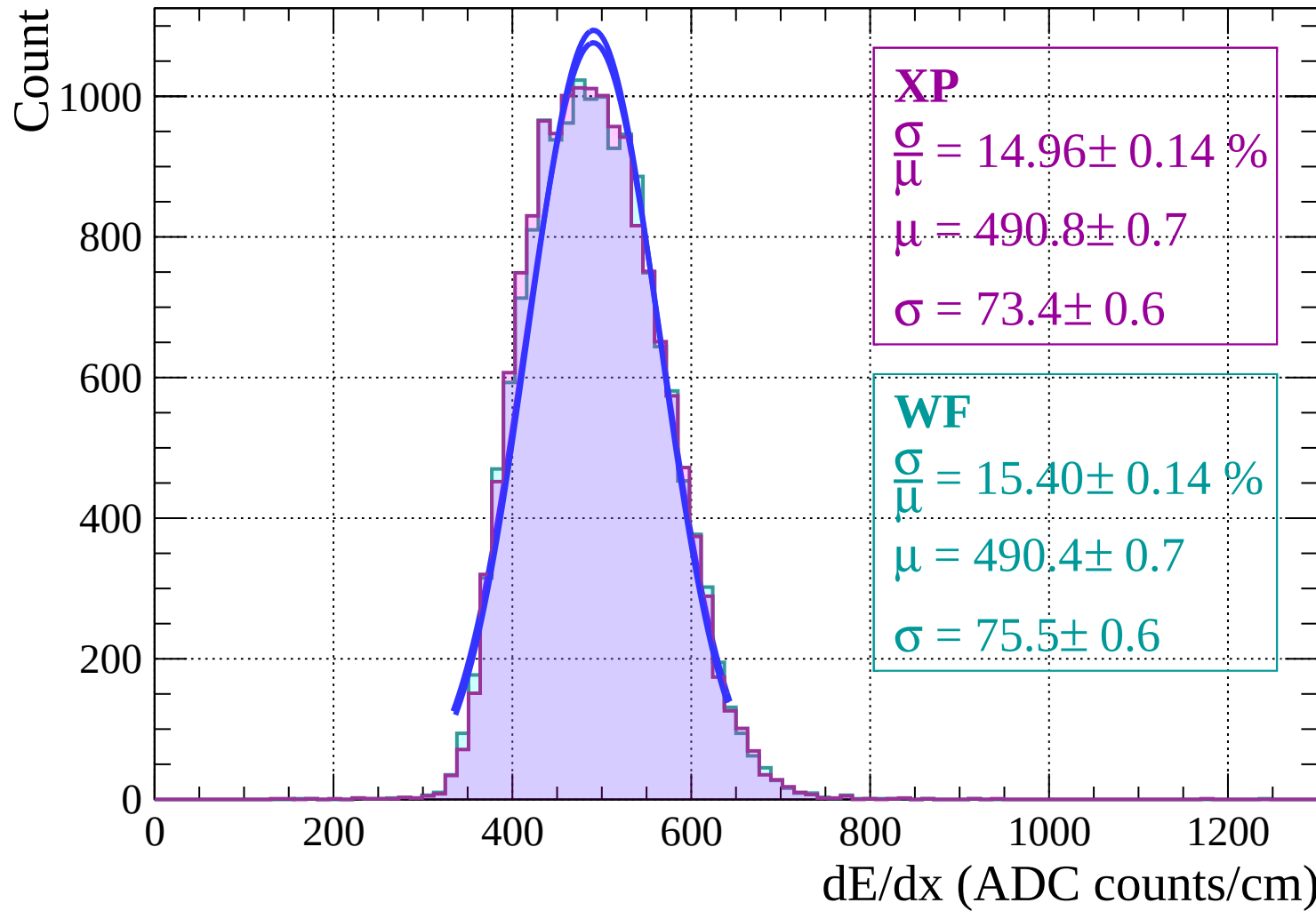
$$\sigma = 73.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.40 \pm 0.14 \%$$

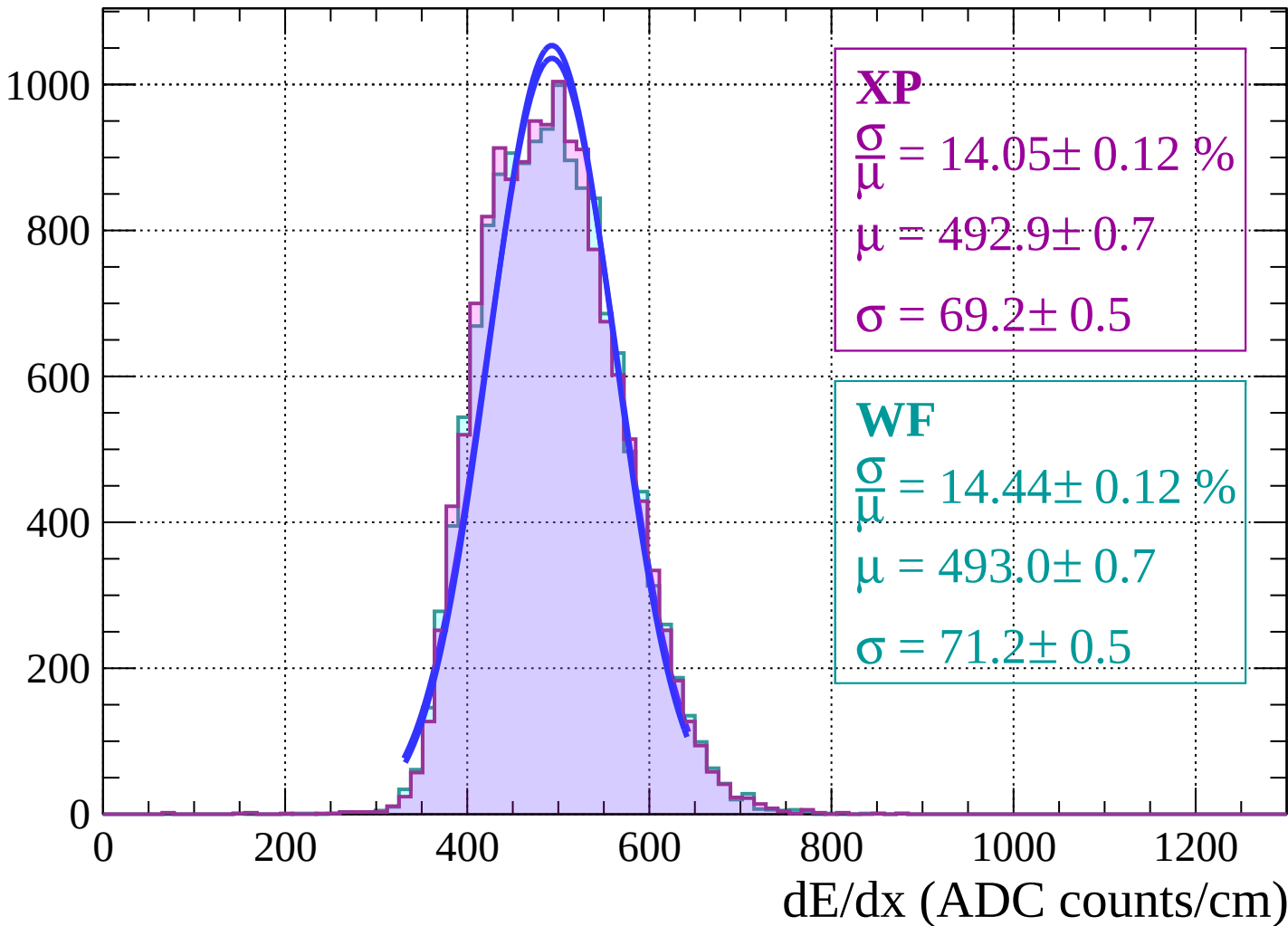
$$\mu = 490.4 \pm 0.7$$

$$\sigma = 75.5 \pm 0.6$$



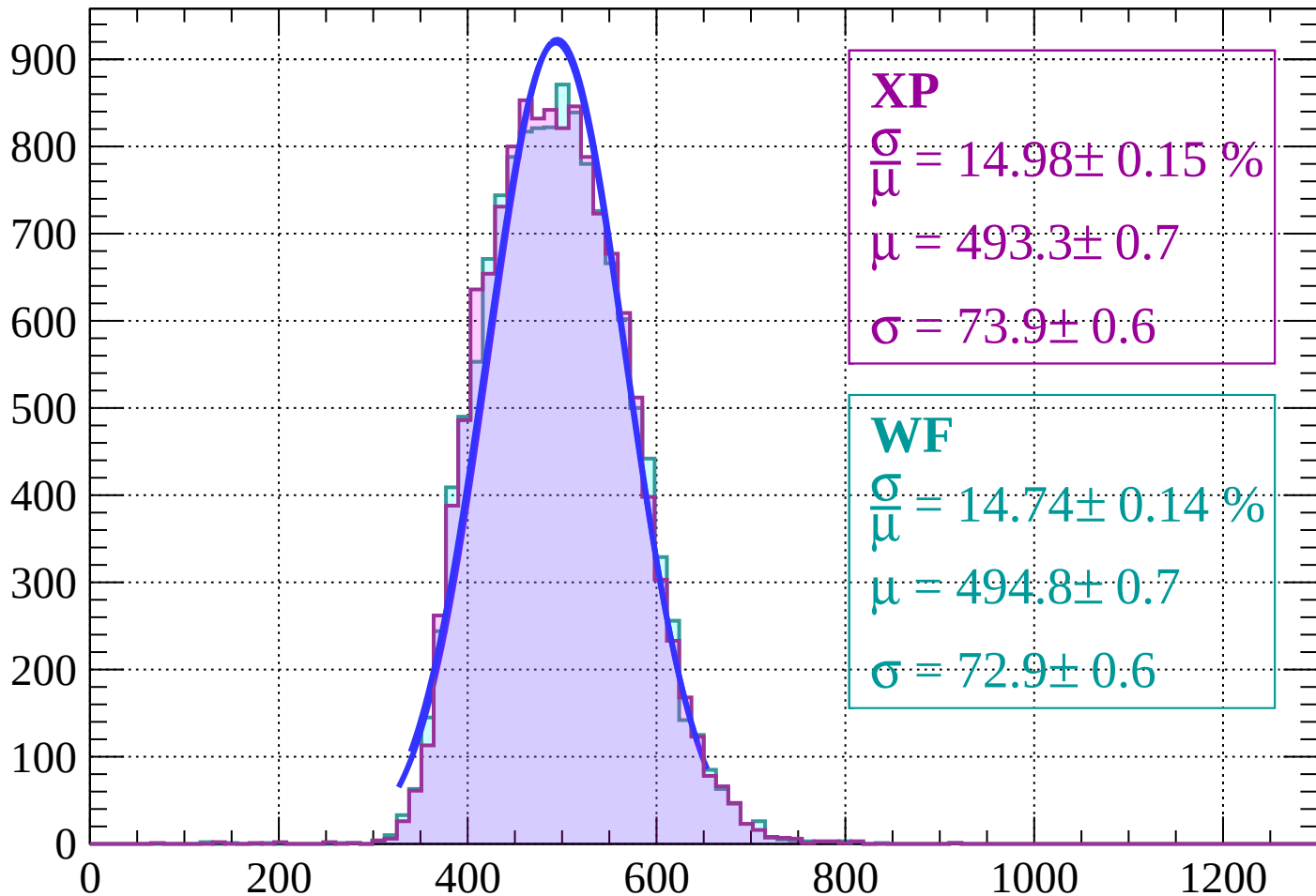
Energy loss | $20 < \theta < 22$

Count



Energy loss | $22 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 14.98 \pm 0.15 \%$$

$$\mu = 493.3 \pm 0.7$$

$$\sigma = 73.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 14.74 \pm 0.14 \%$$

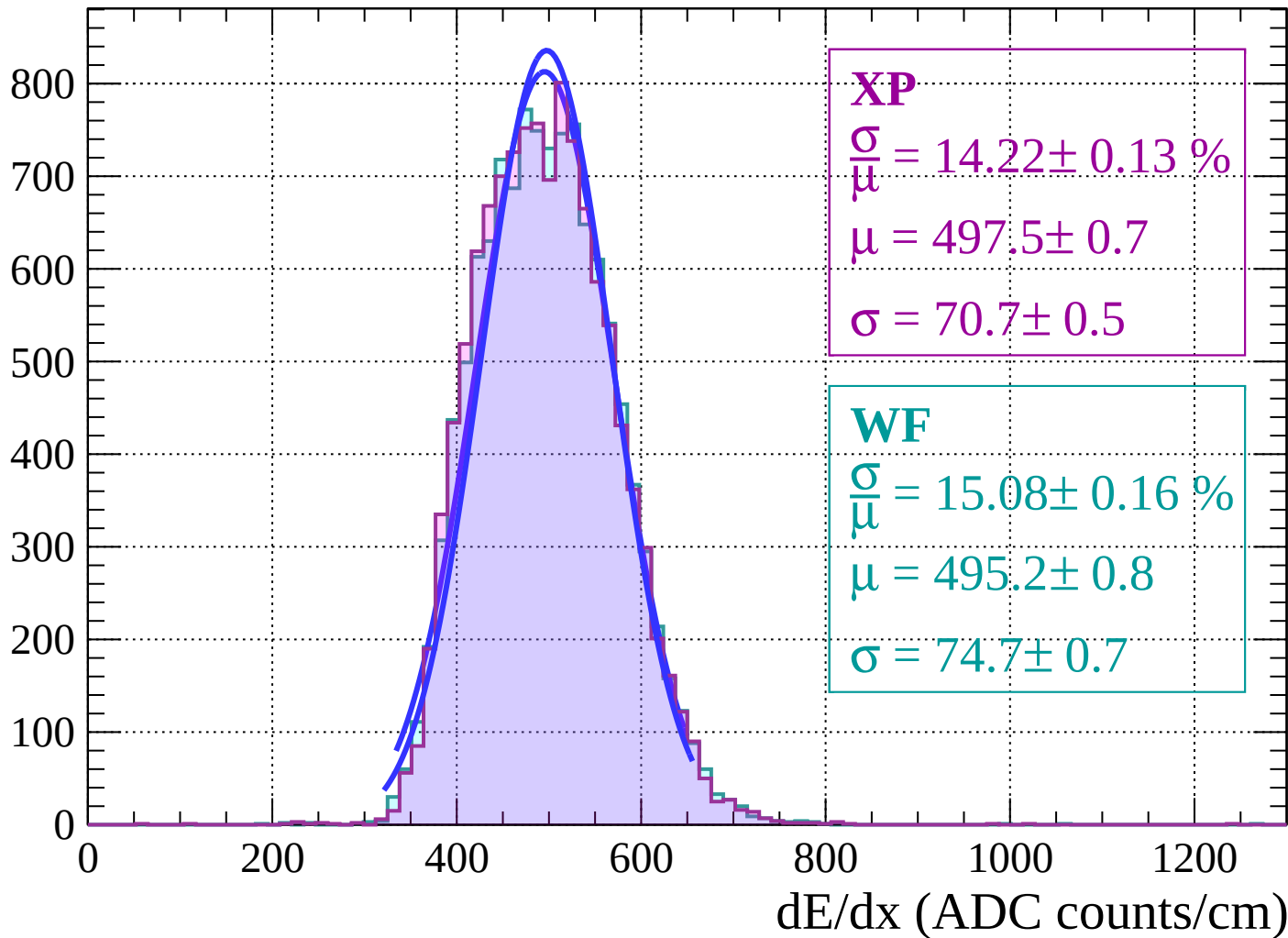
$$\mu = 494.8 \pm 0.7$$

$$\sigma = 72.9 \pm 0.6$$

dE/dx (ADC counts/cm)

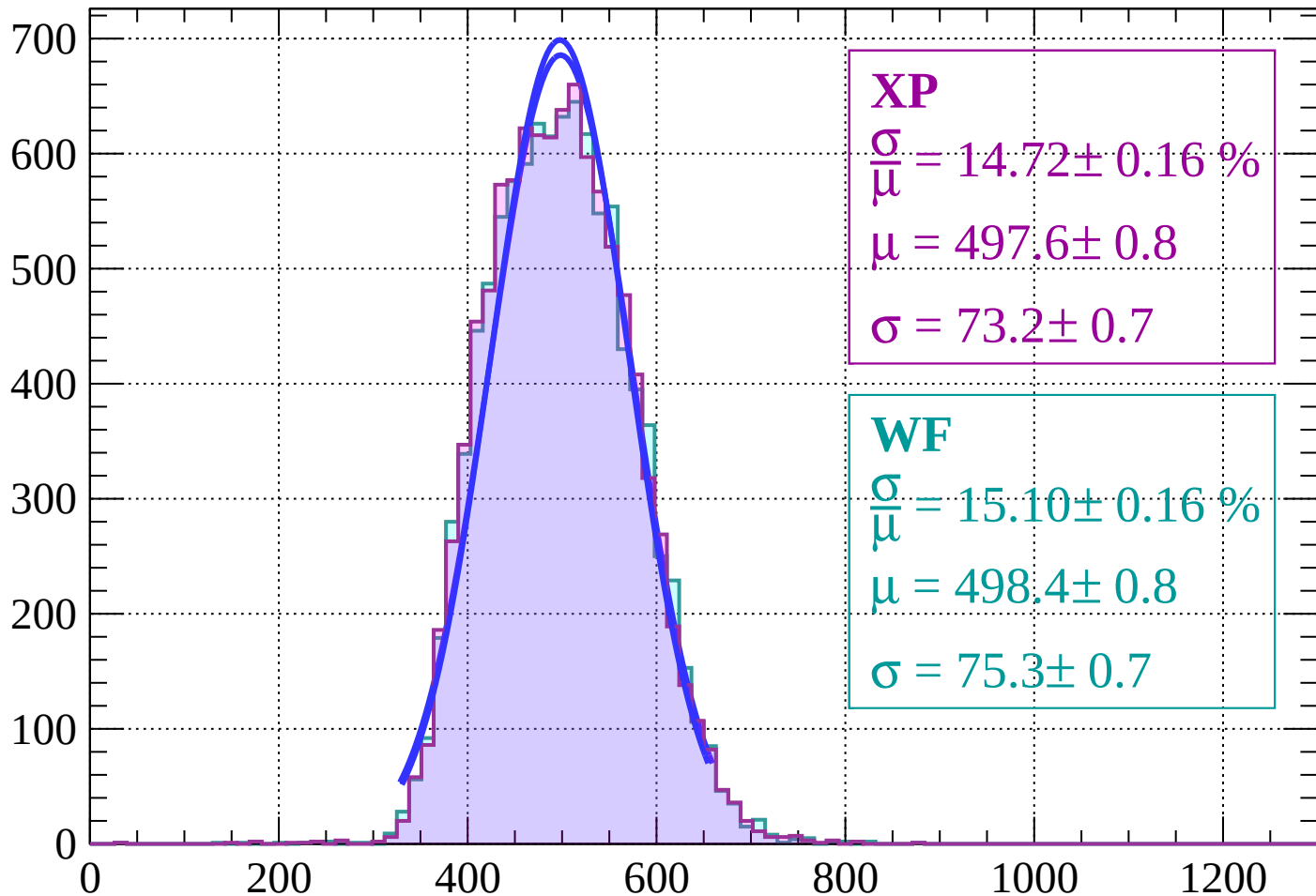
Energy loss | $24 < \theta < 26$

Count



Energy loss | $26 < \theta < 28$

Count



XP

$$\frac{\sigma}{\mu} = 14.72 \pm 0.16 \%$$

$$\mu = 497.6 \pm 0.8$$

$$\sigma = 73.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 15.10 \pm 0.16 \%$$

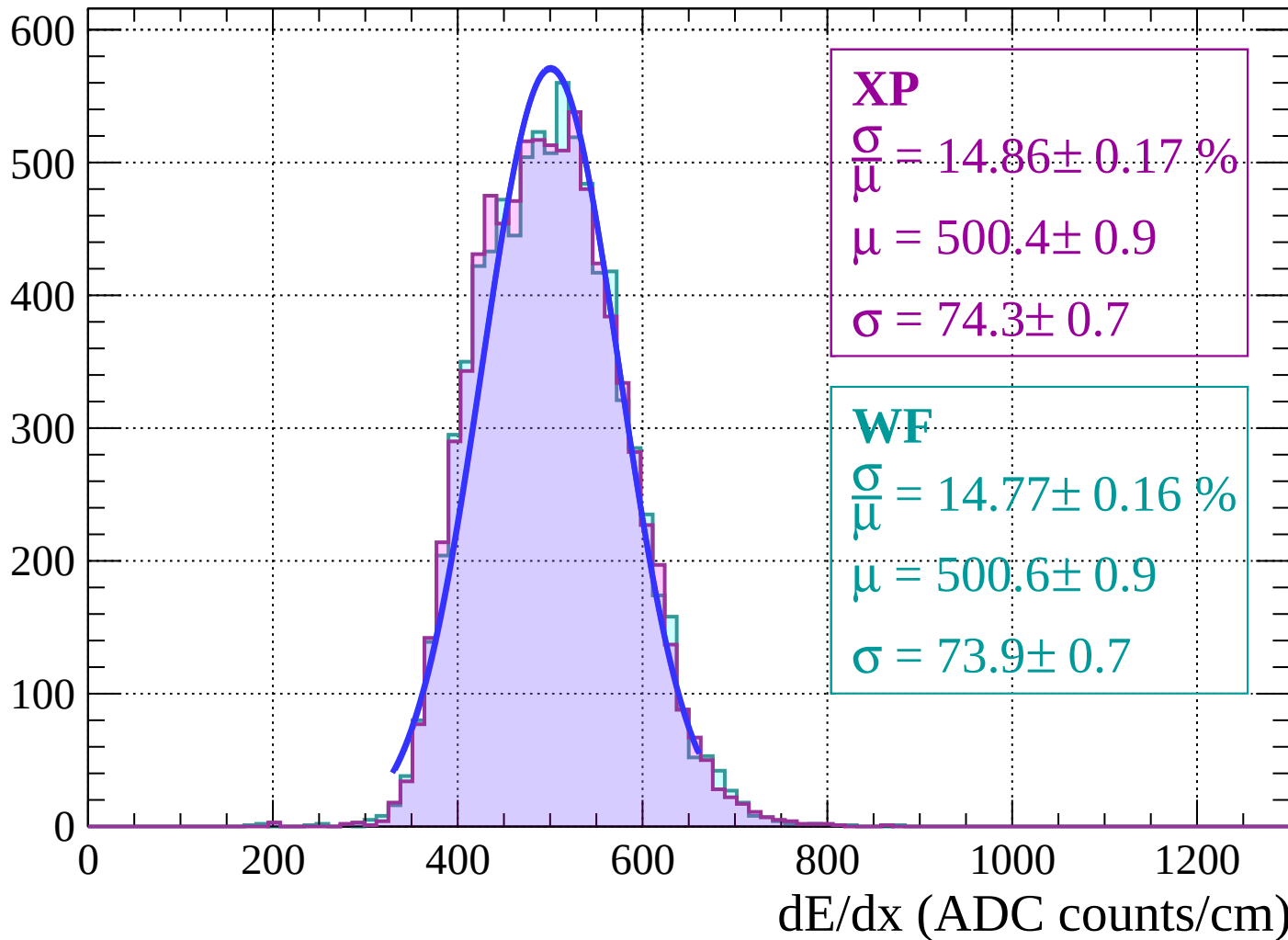
$$\mu = 498.4 \pm 0.8$$

$$\sigma = 75.3 \pm 0.7$$

dE/dx (ADC counts/cm)

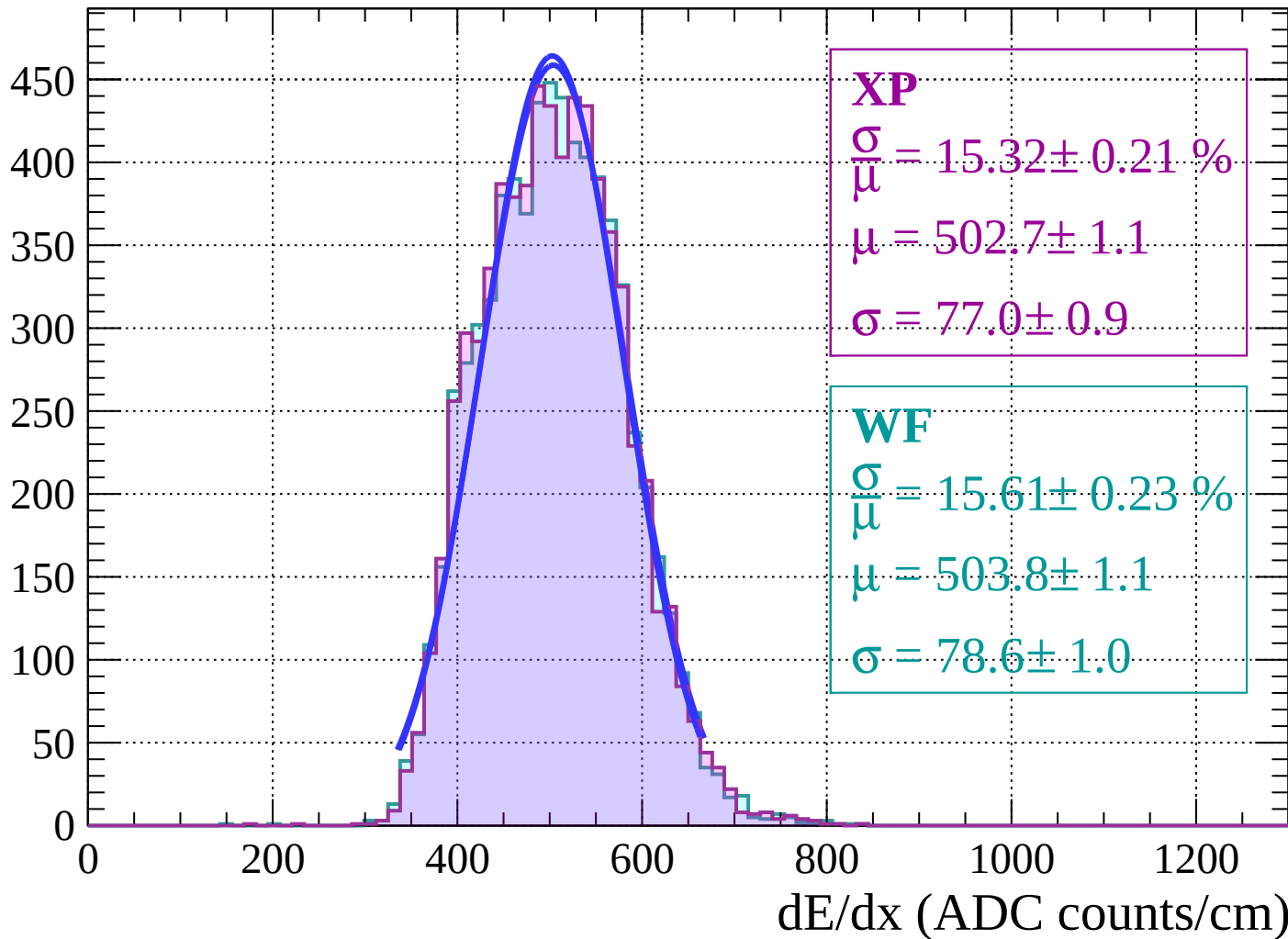
Energy loss | $28 < \theta < 30$

Count



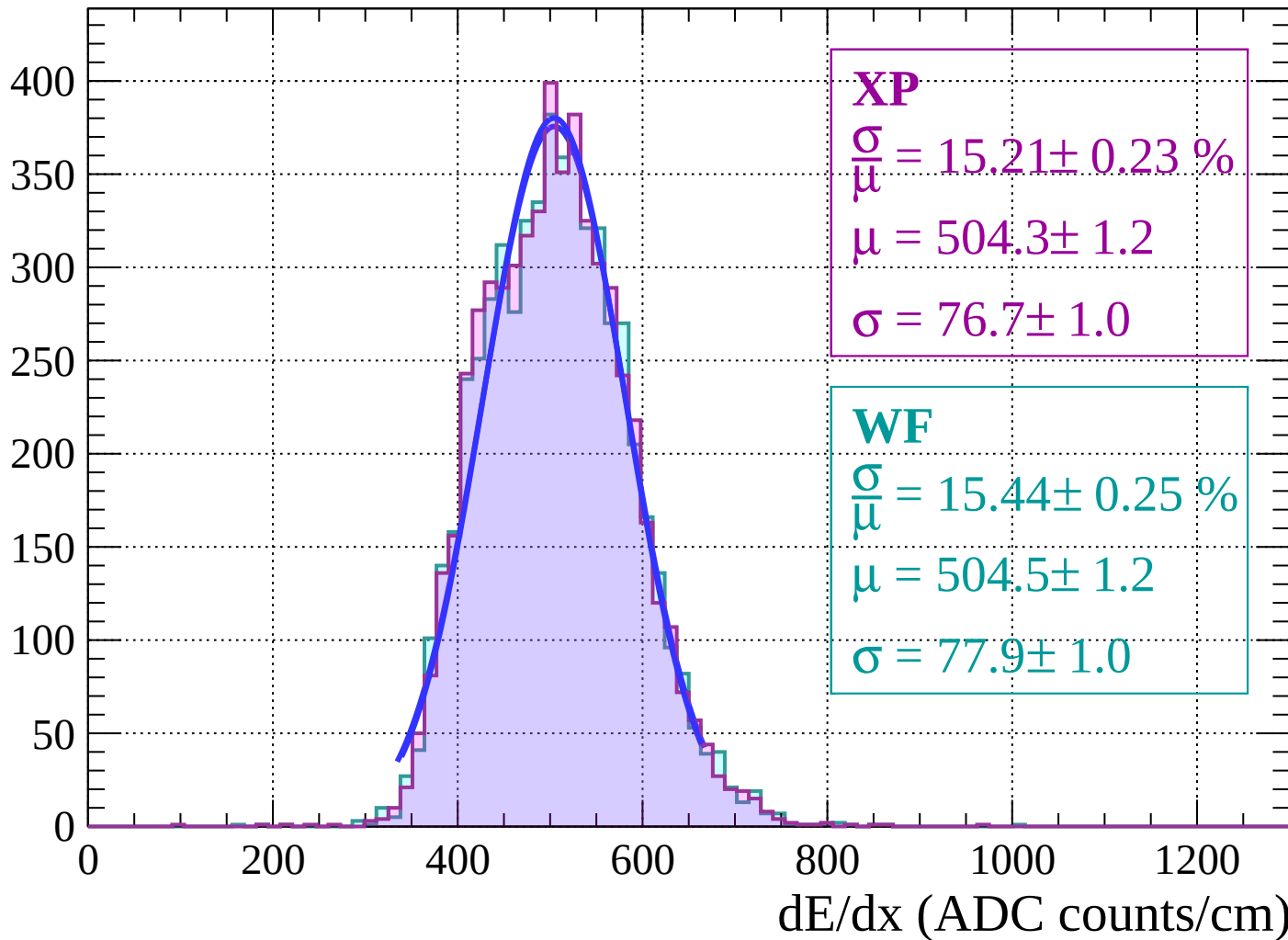
Energy loss | $30 < \theta < 32$

Count



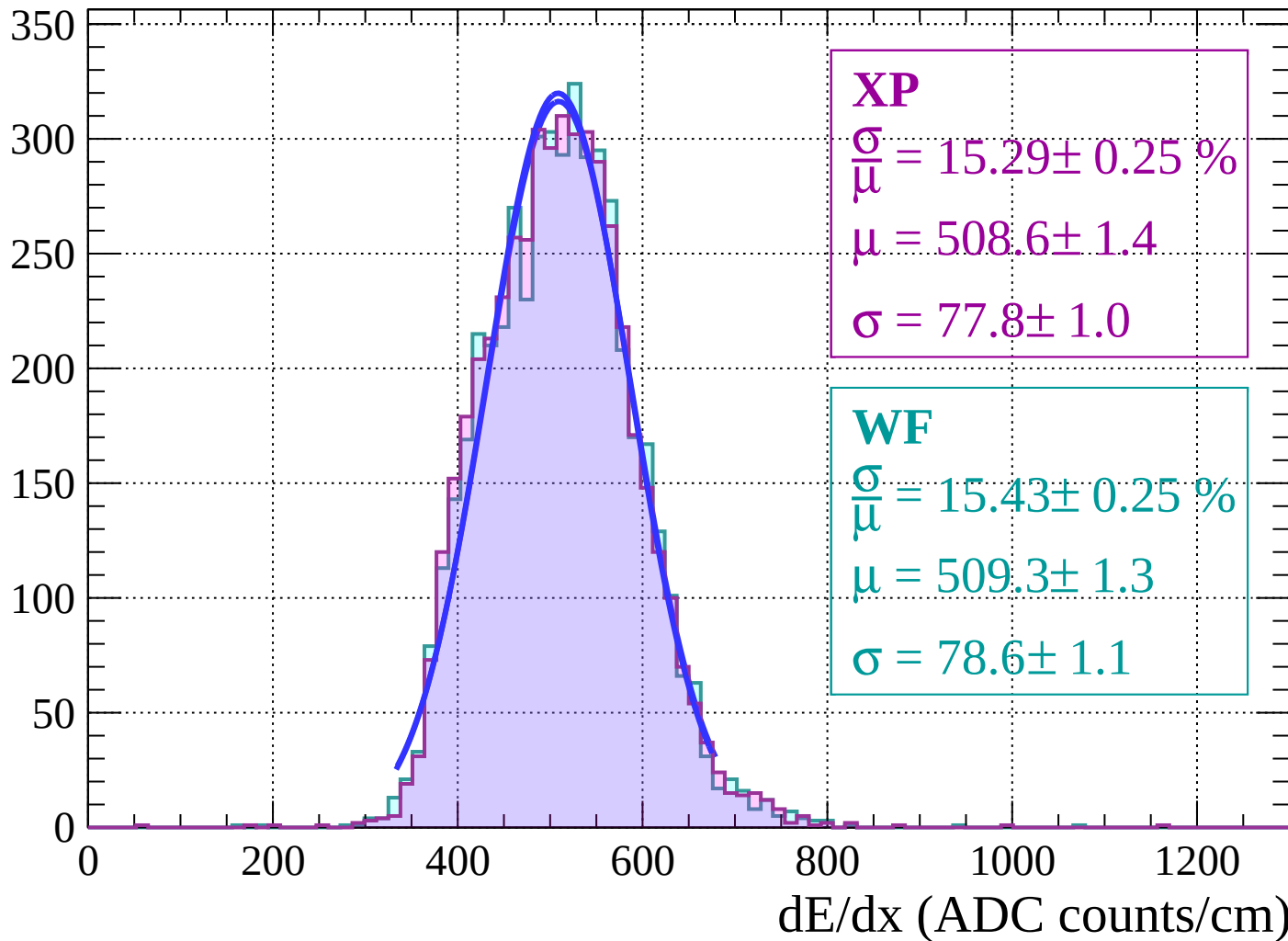
Energy loss | $32 < \theta < 34$

Count



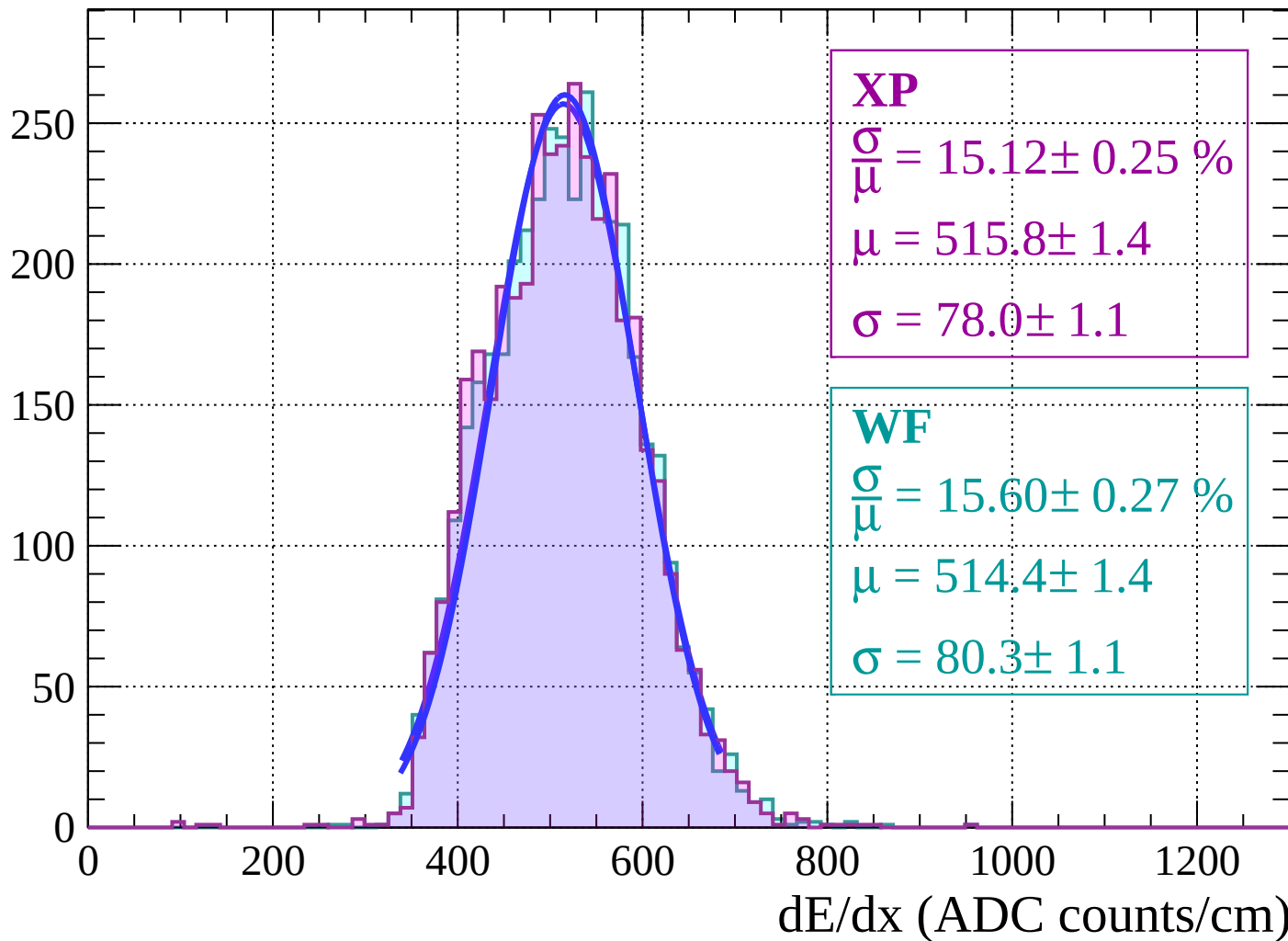
Energy loss | $34 < \theta < 36$

Count



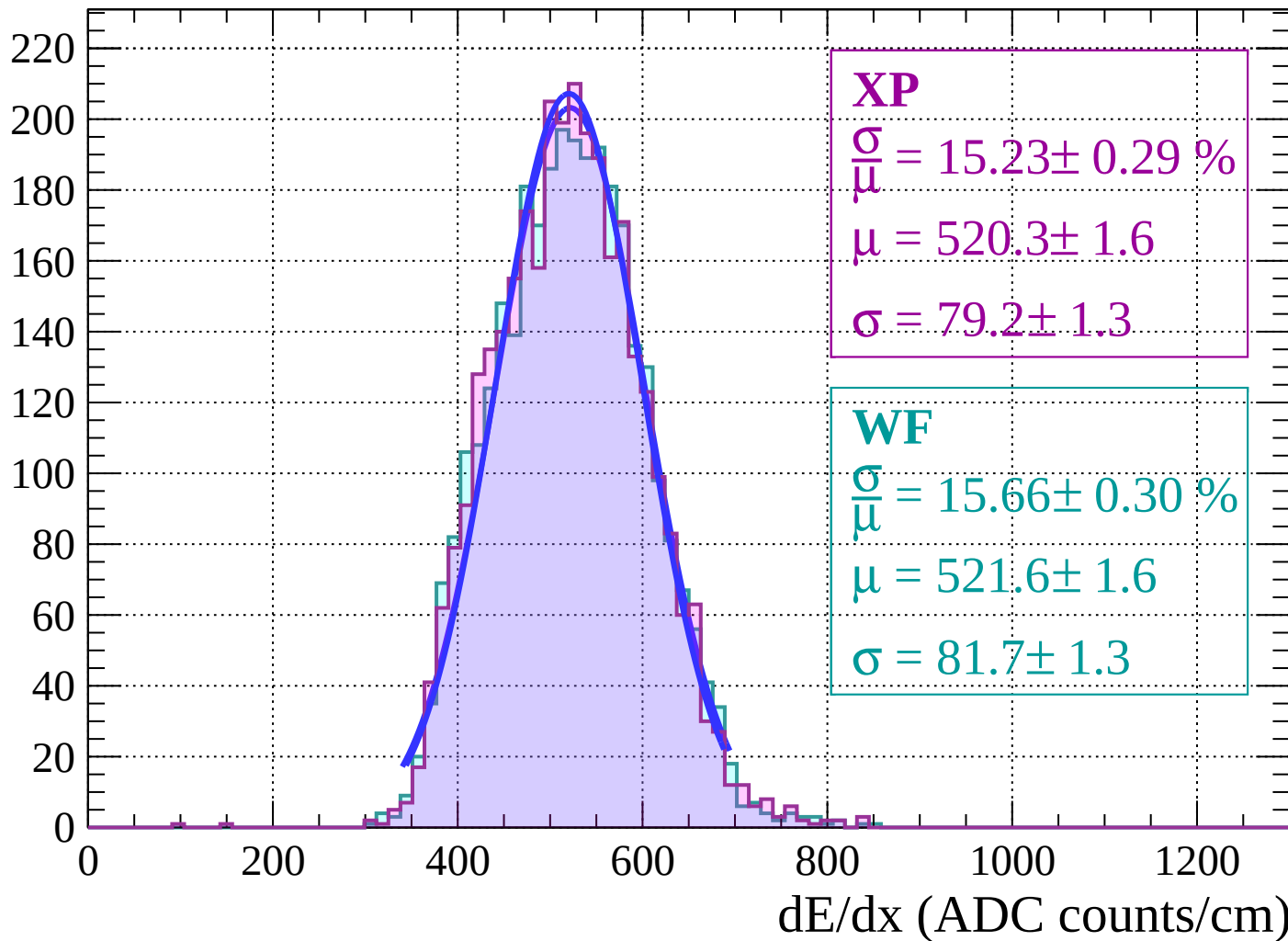
Energy loss | $36 < \theta < 38$

Count



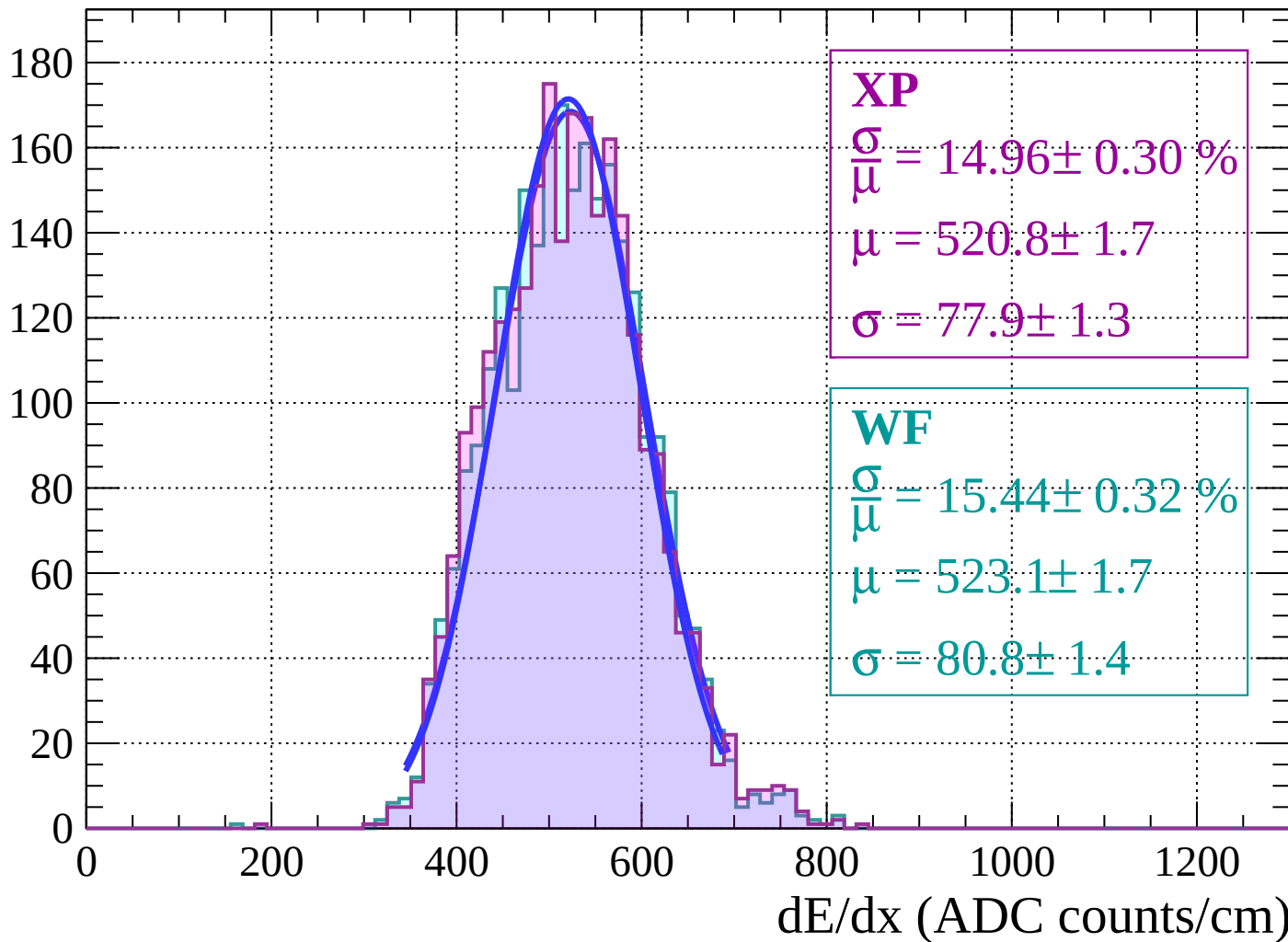
Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

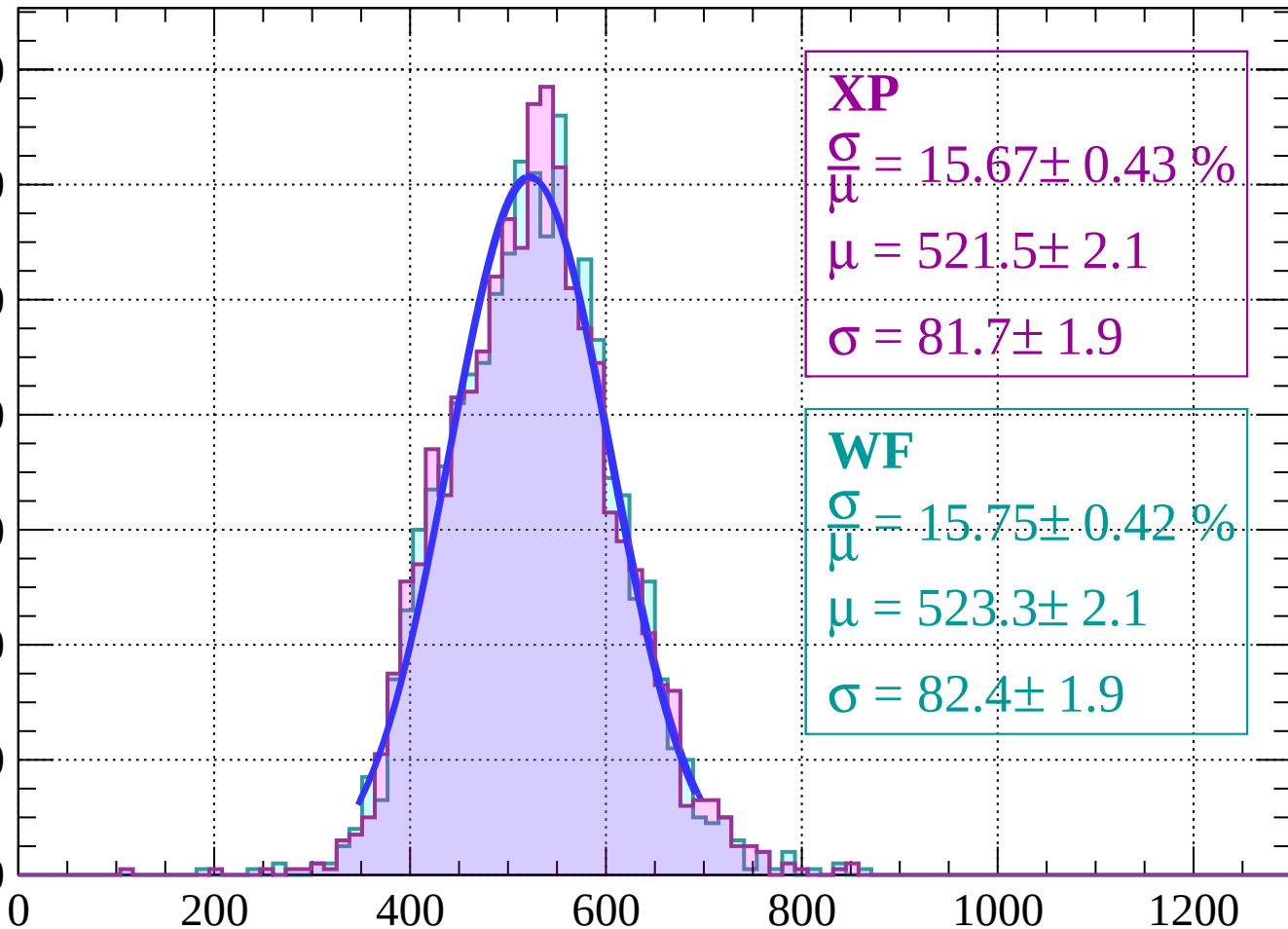
Count



Energy loss | $42 < \theta < 44$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 15.67 \pm 0.43 \%$$

$$\mu = 521.5 \pm 2.1$$

$$\sigma = 81.7 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 15.75 \pm 0.42 \%$$

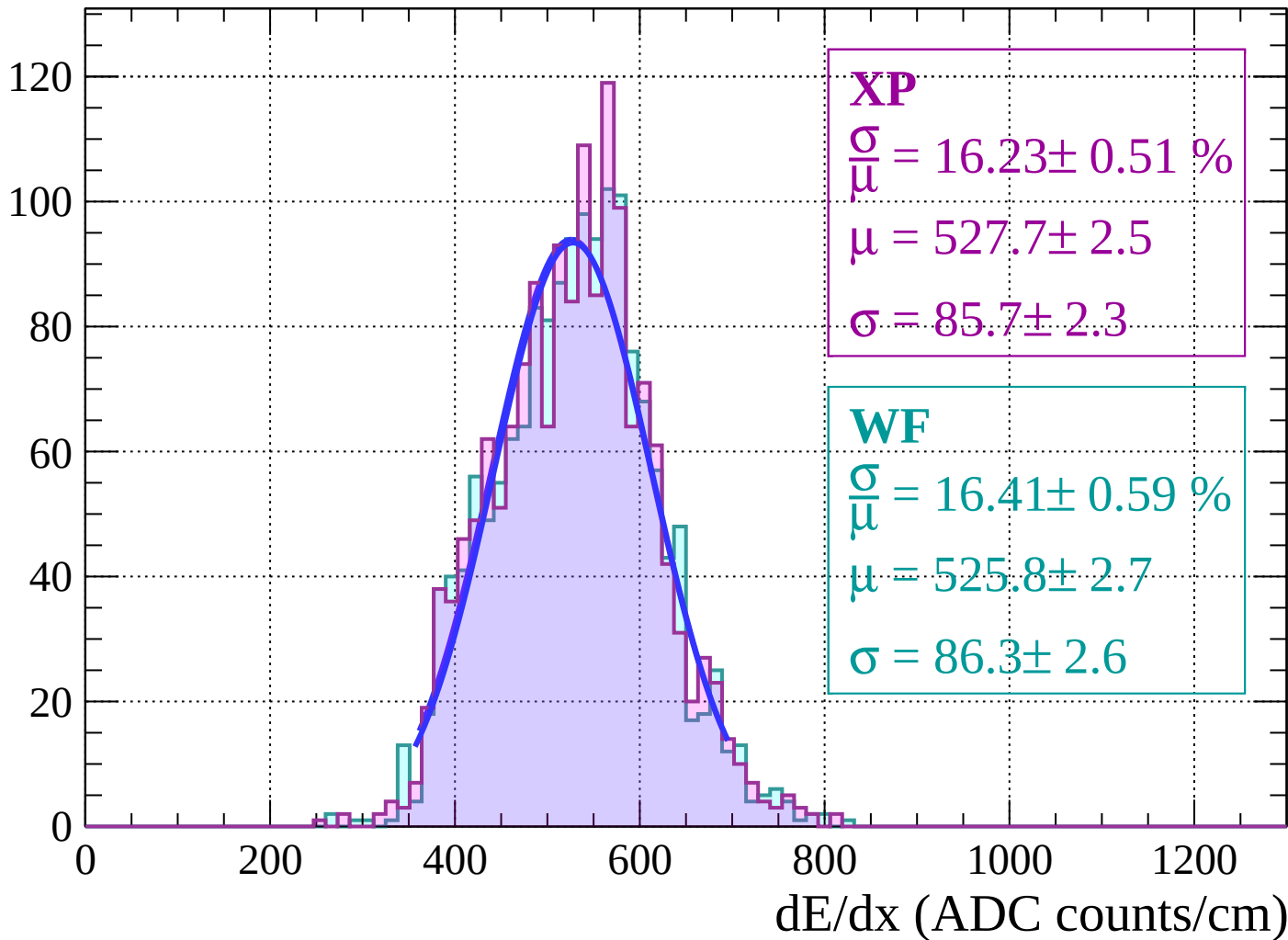
$$\mu = 523.3 \pm 2.1$$

$$\sigma = 82.4 \pm 1.9$$

dE/dx (ADC counts/cm)

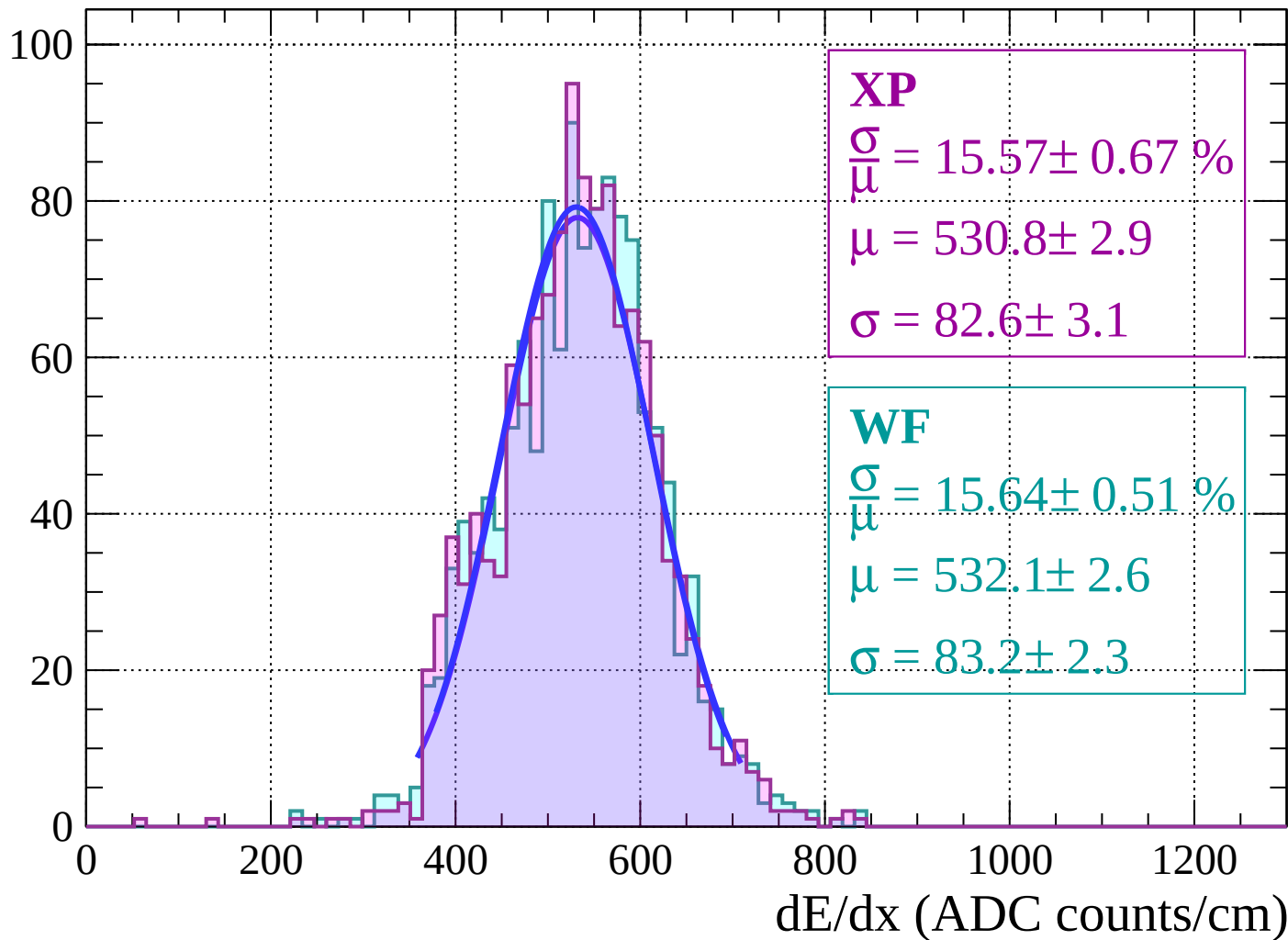
Energy loss | $44 < \theta < 46$

Count



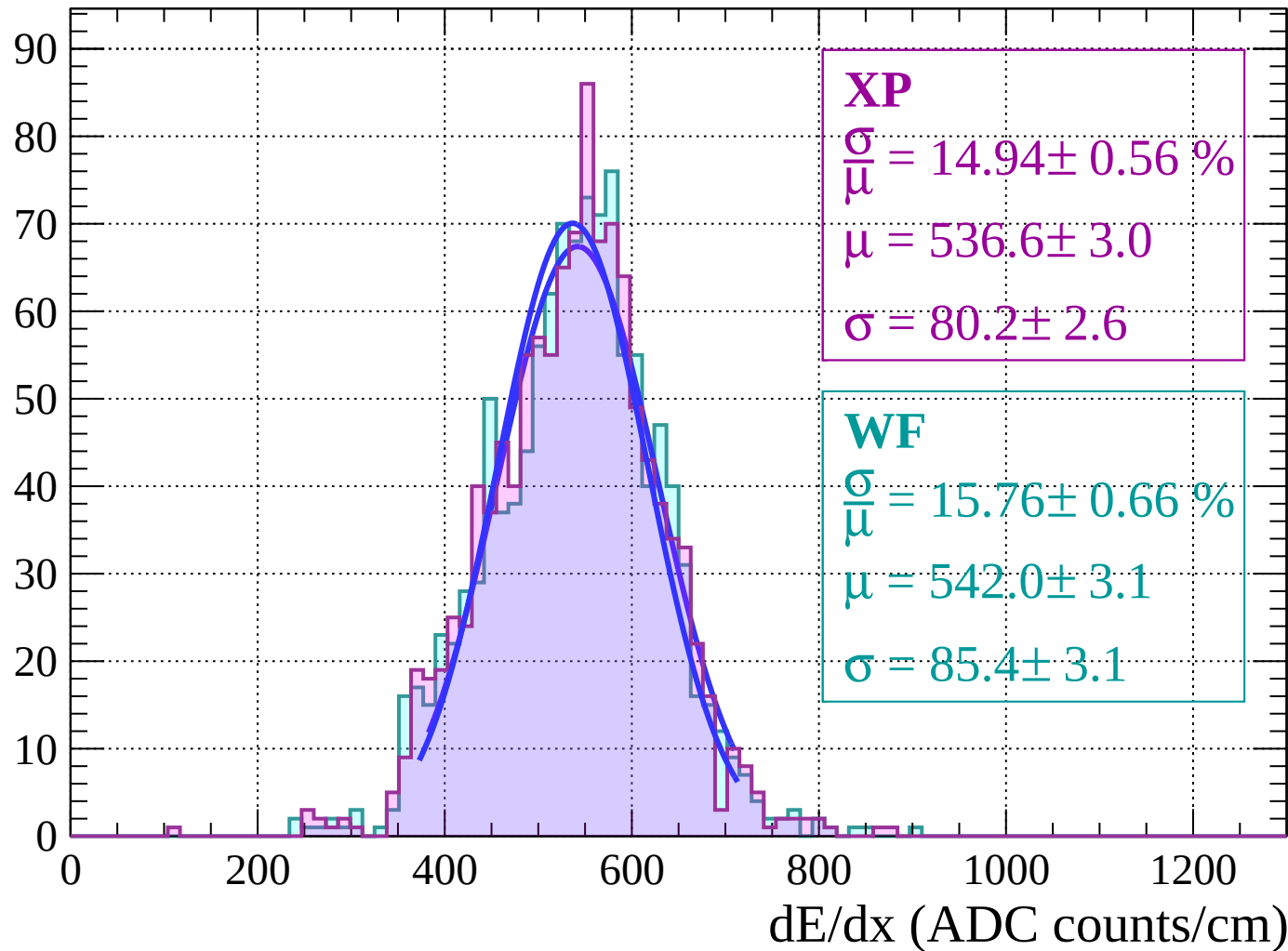
Energy loss | $46 < \theta < 48$

Count



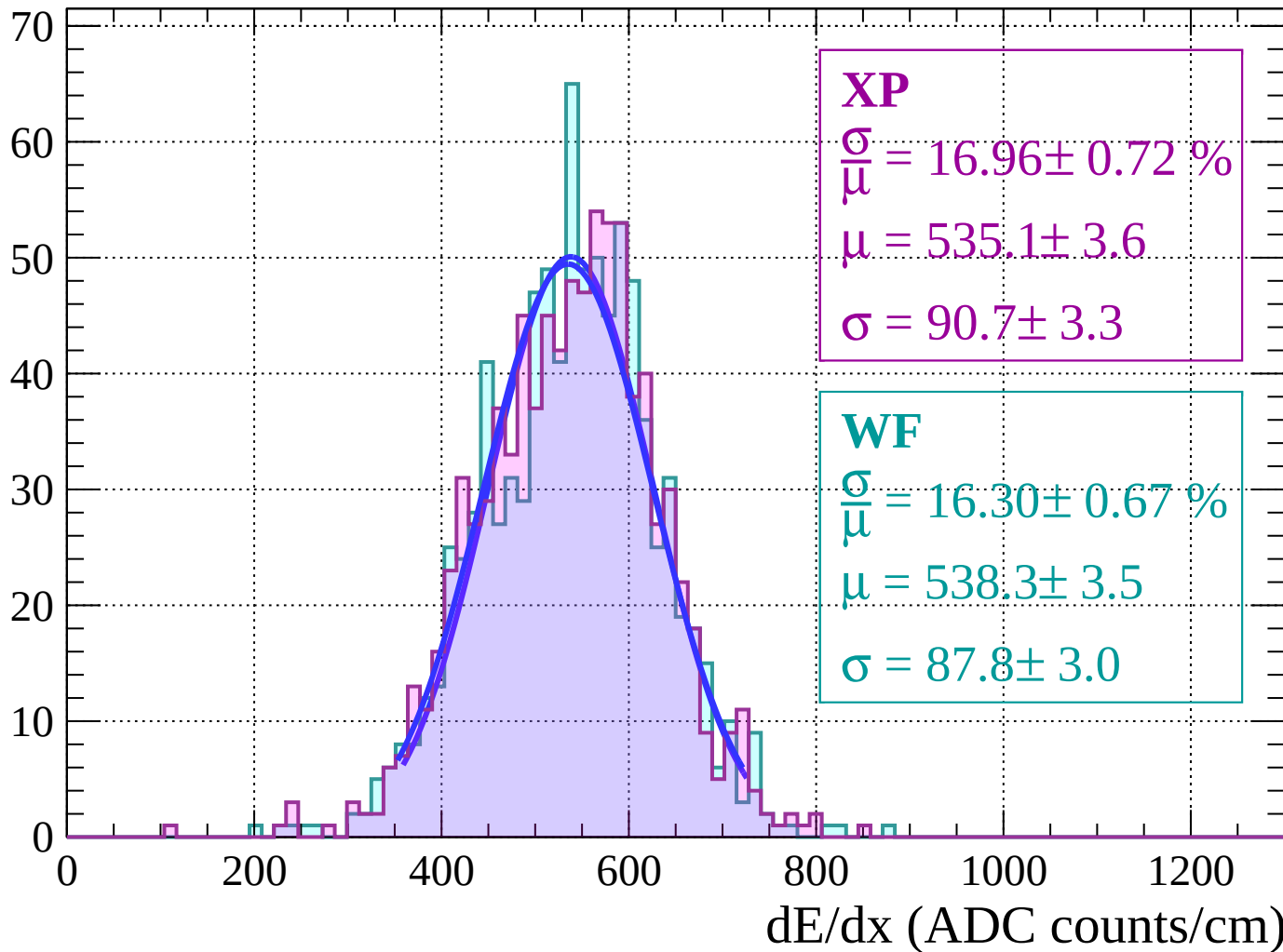
Energy loss | $48 < \theta < 50$

Count



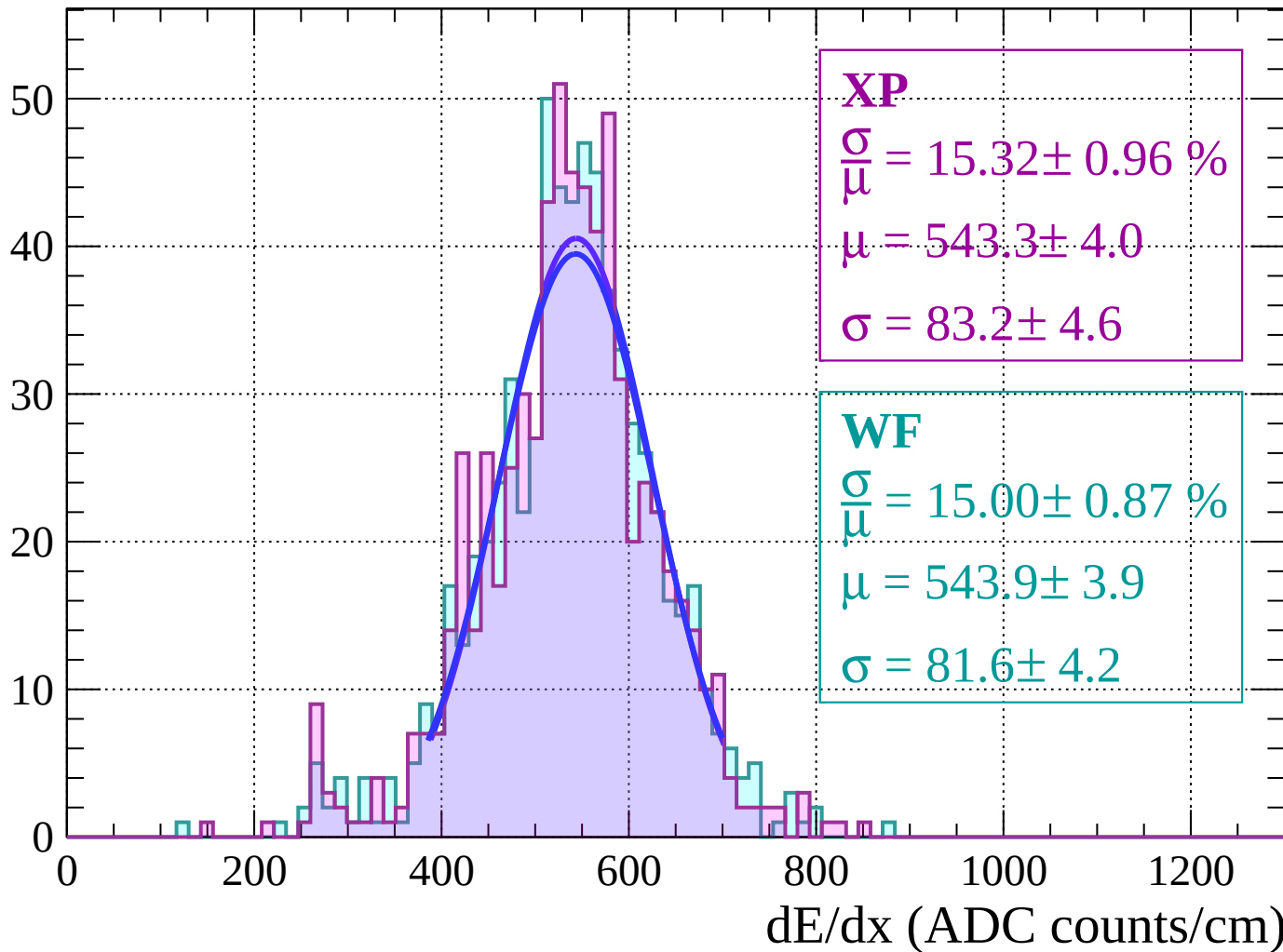
Energy loss | $50 < \theta < 52$

Count



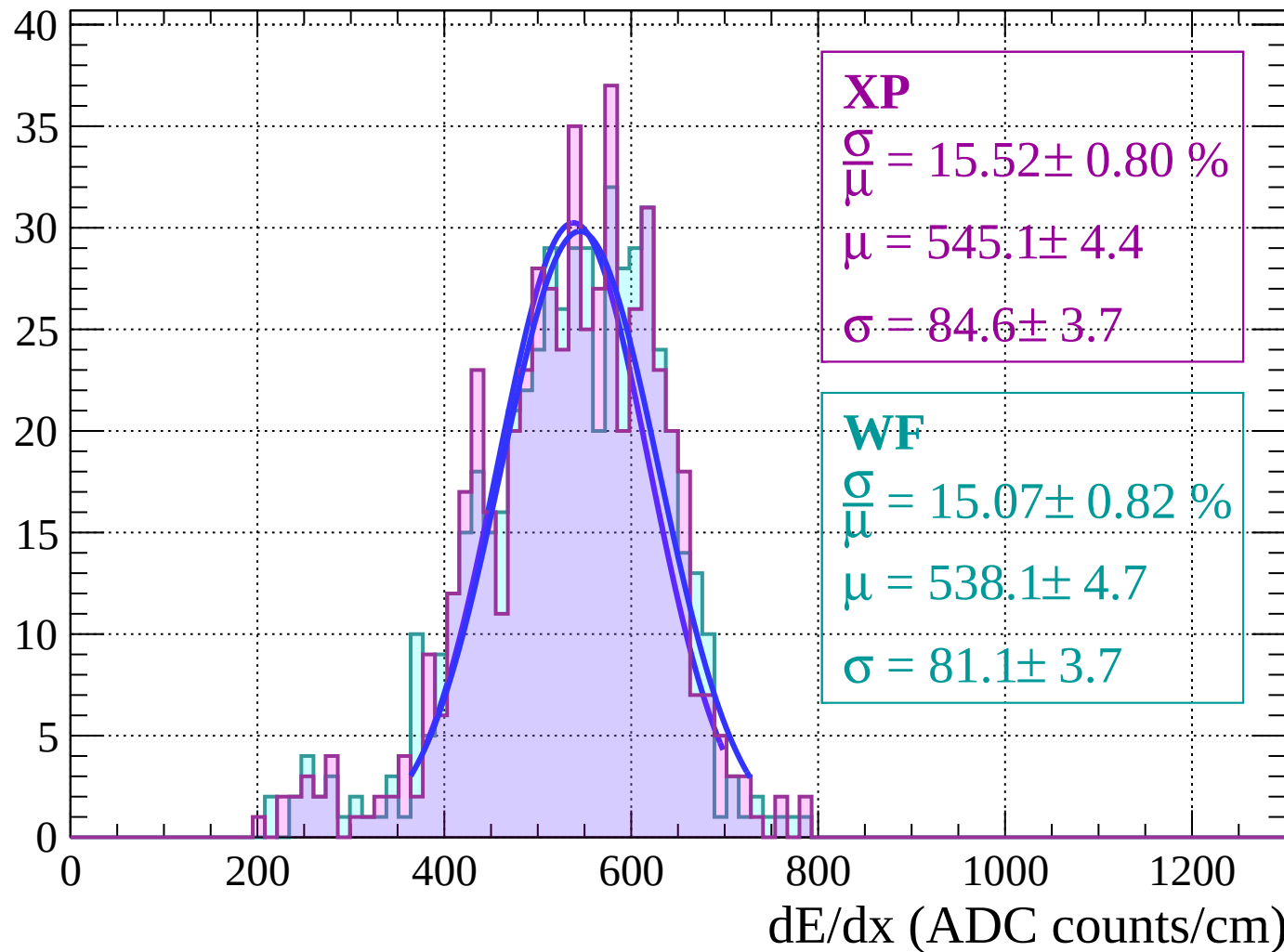
Energy loss | $52 < \theta < 54$

Count



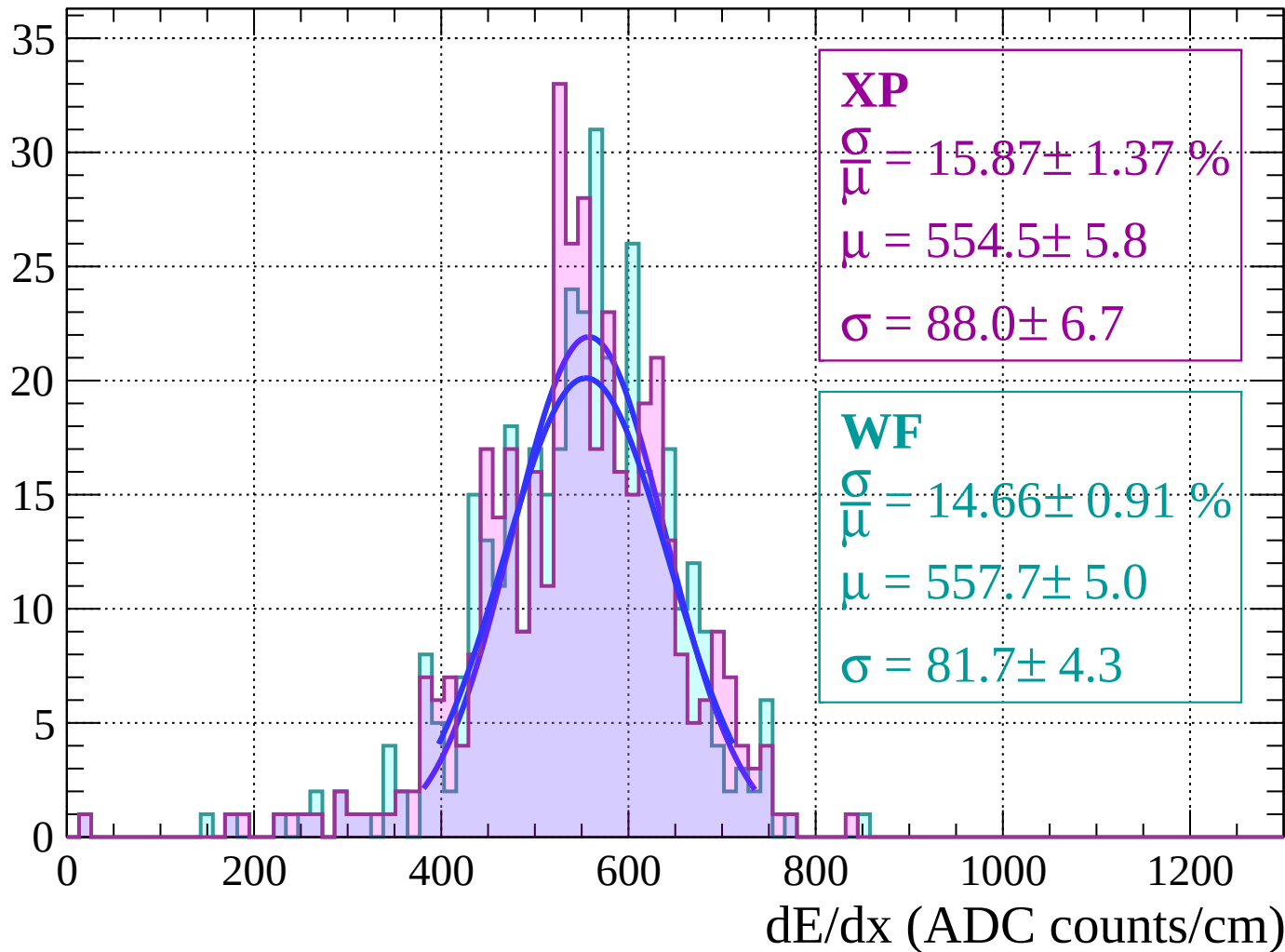
Energy loss | $54 < \theta < 56$

Count



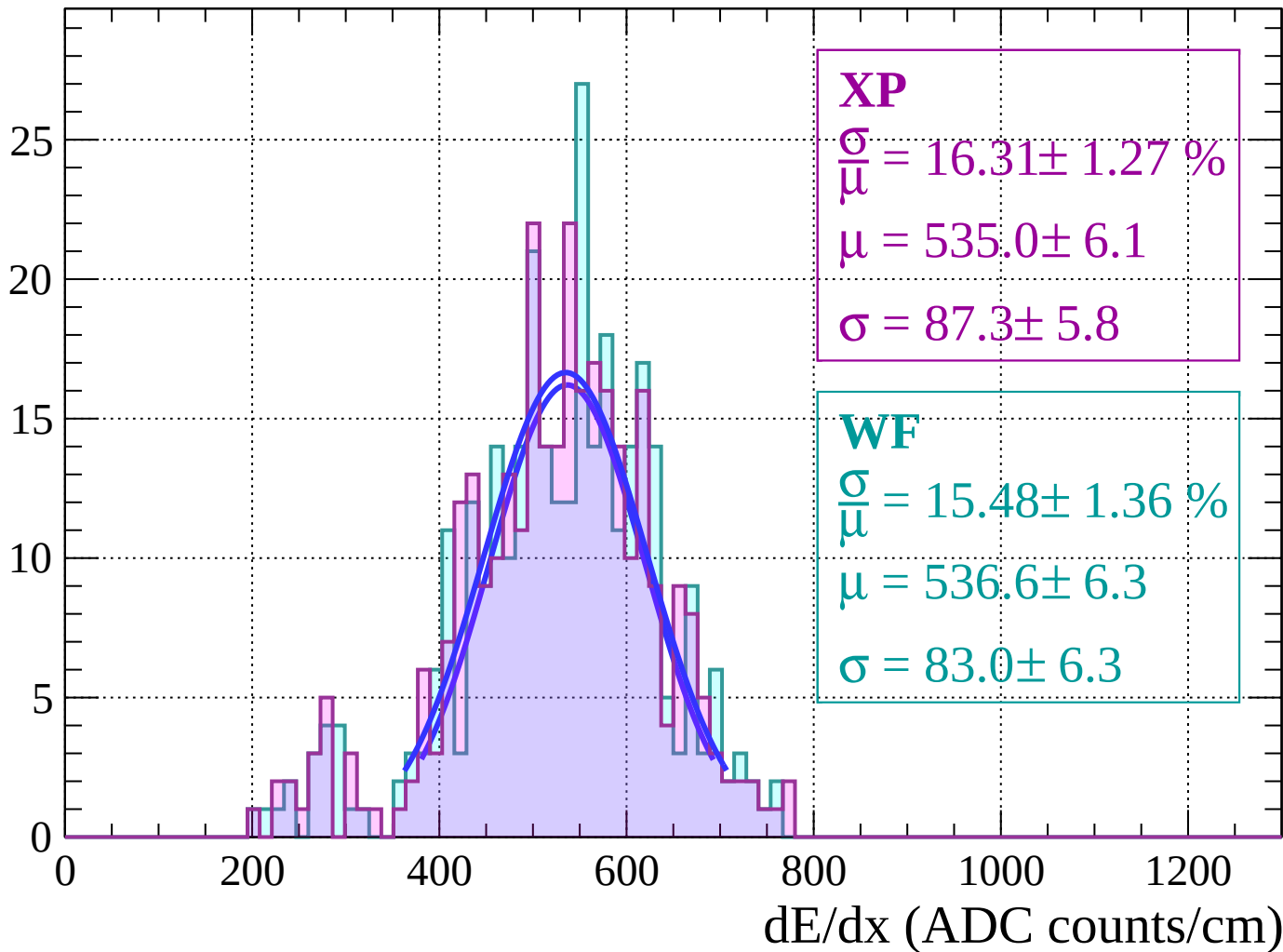
Energy loss | $56 < \theta < 58$

Count



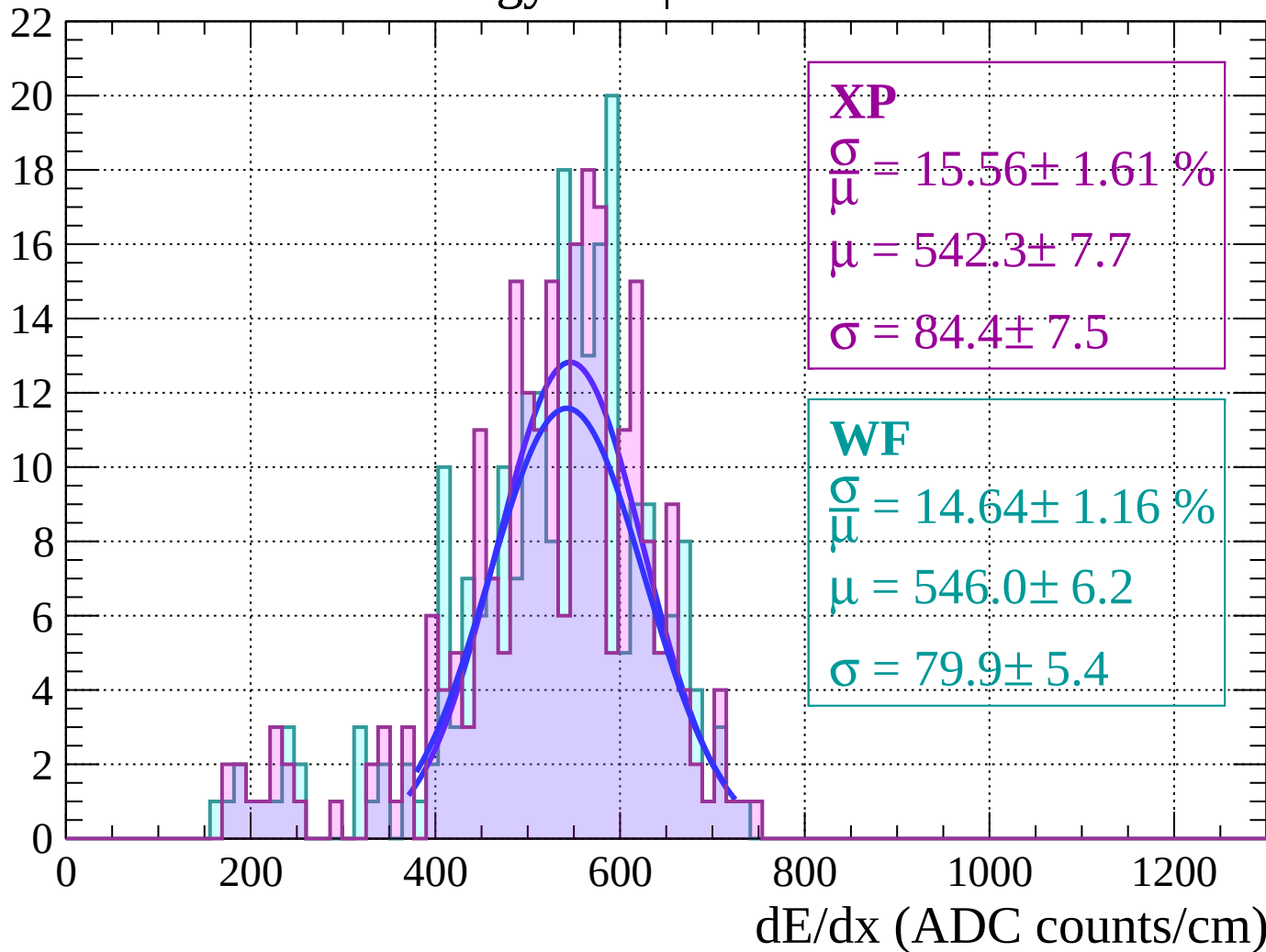
Energy loss | $58 < \theta < 60$

Count



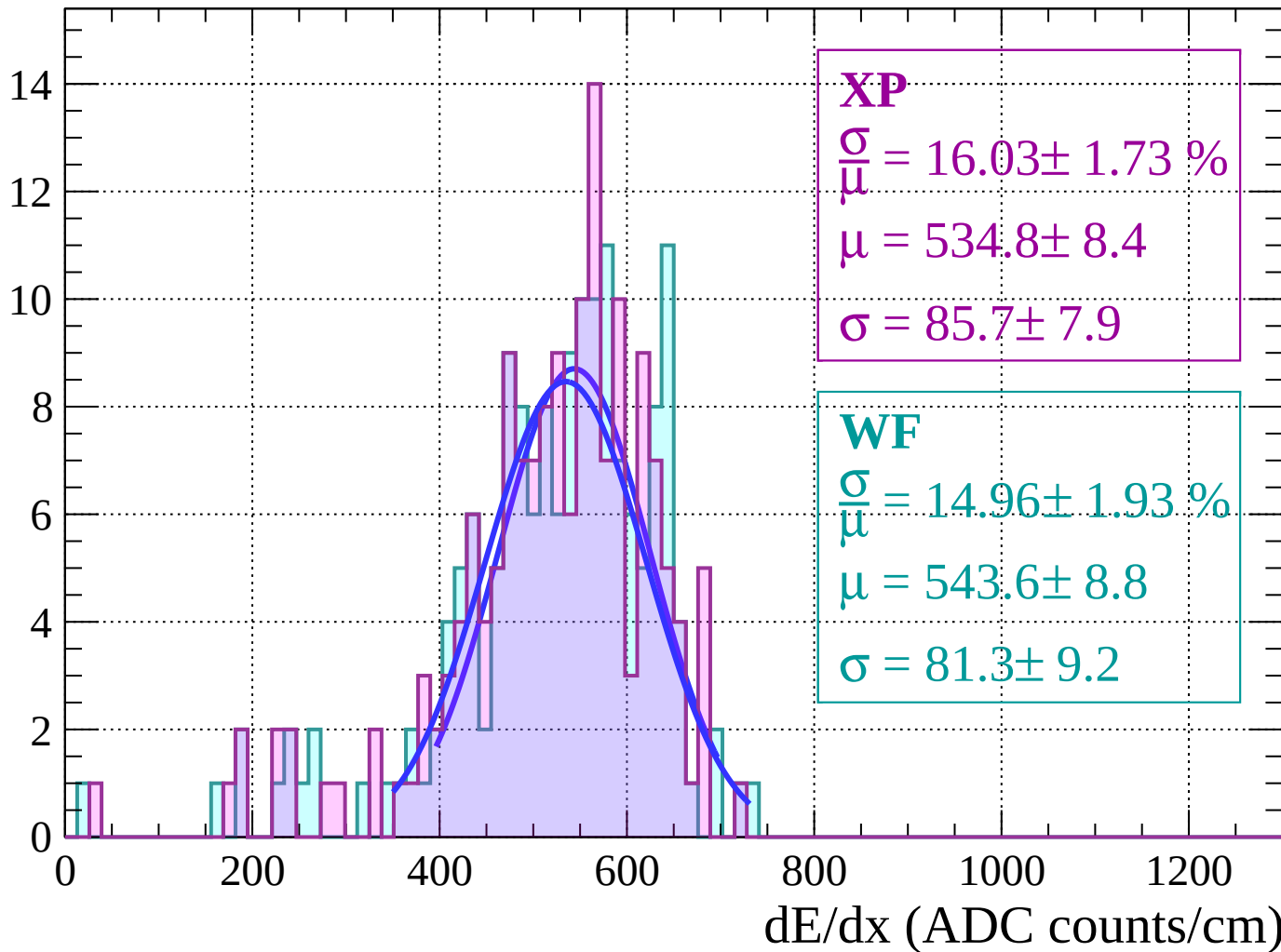
Energy loss | $60 < \theta < 62$

Count



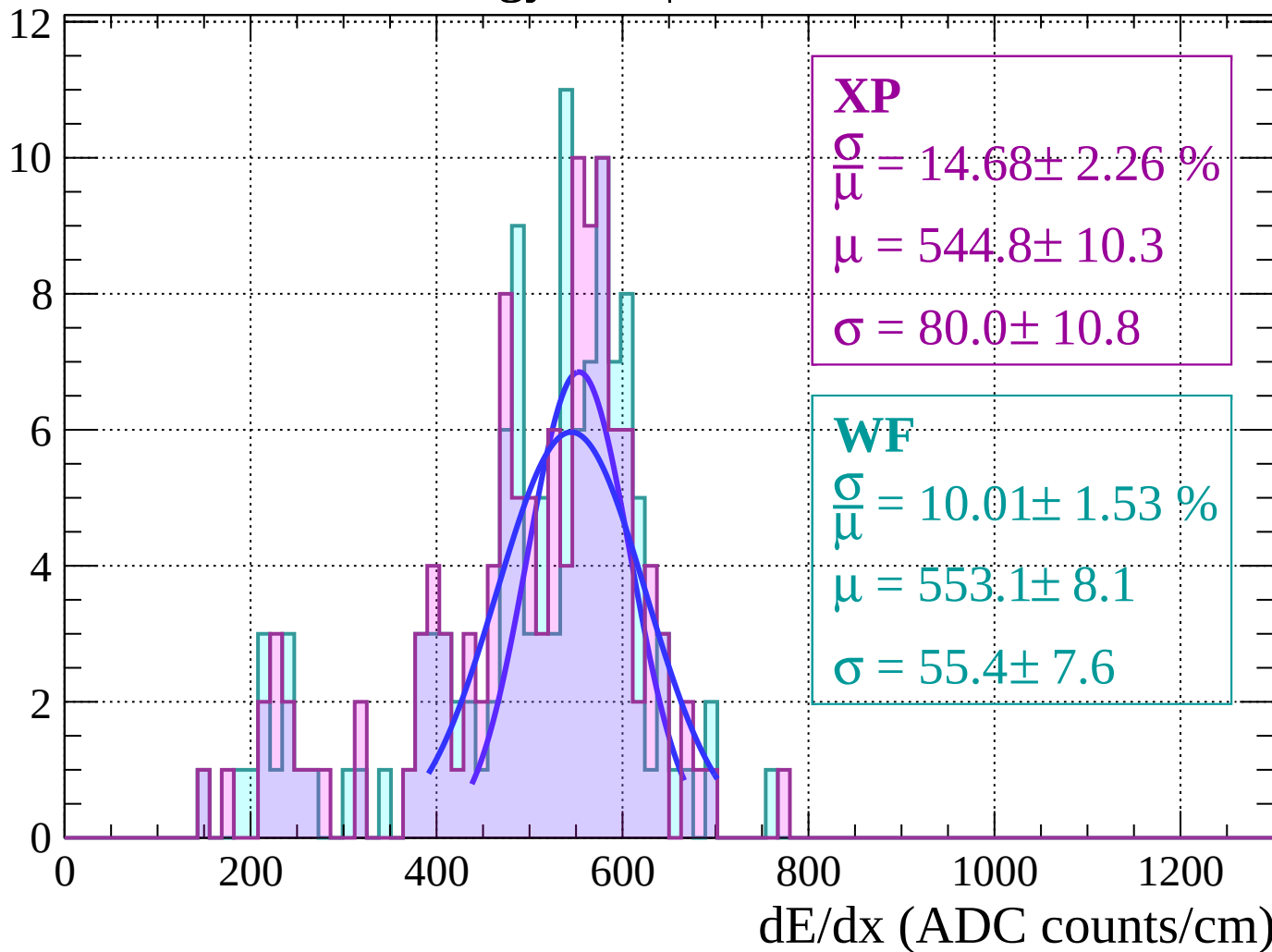
Energy loss | $62 < \theta < 64$

Count



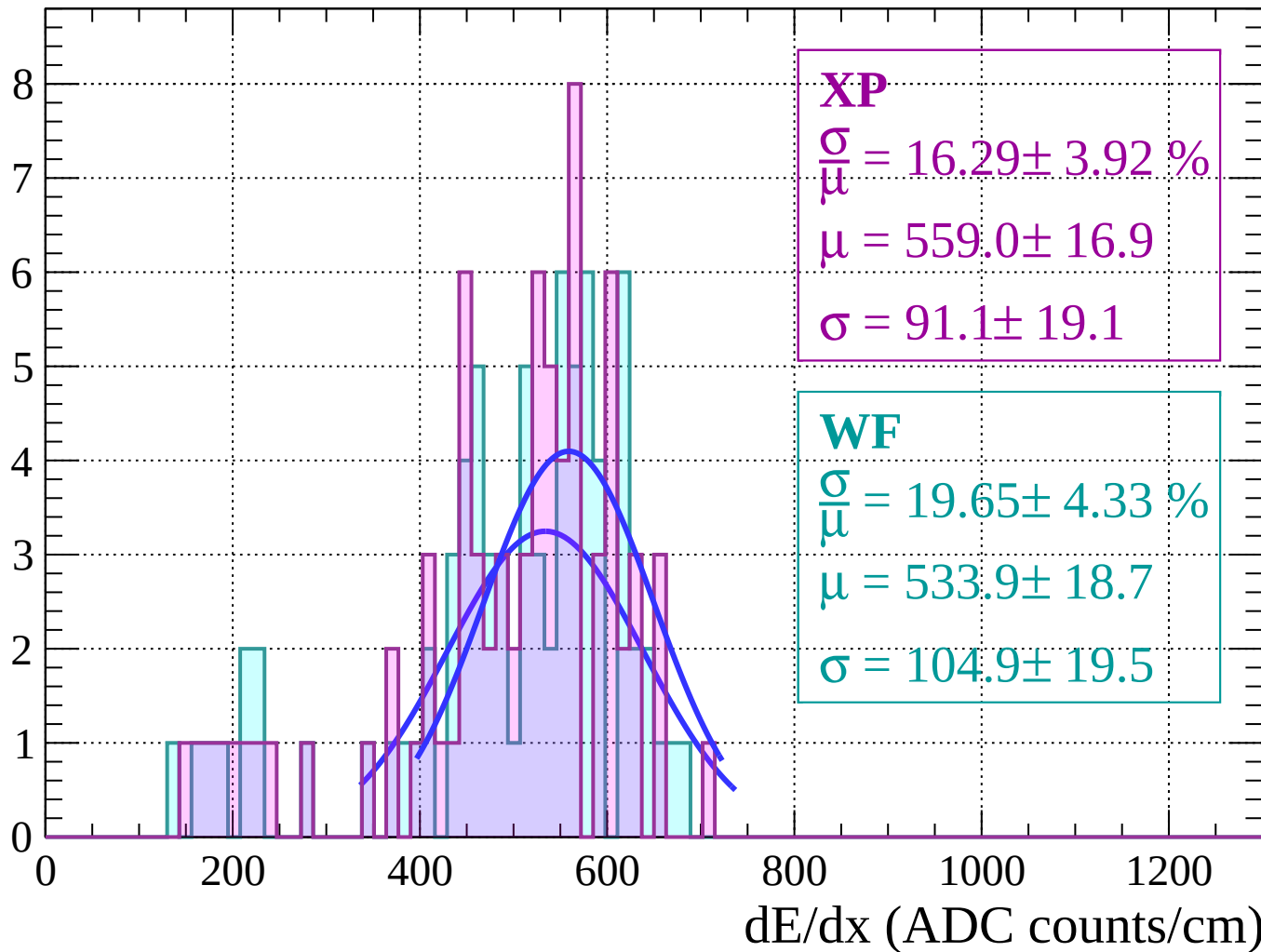
Energy loss | $64 < \theta < 66$

Count



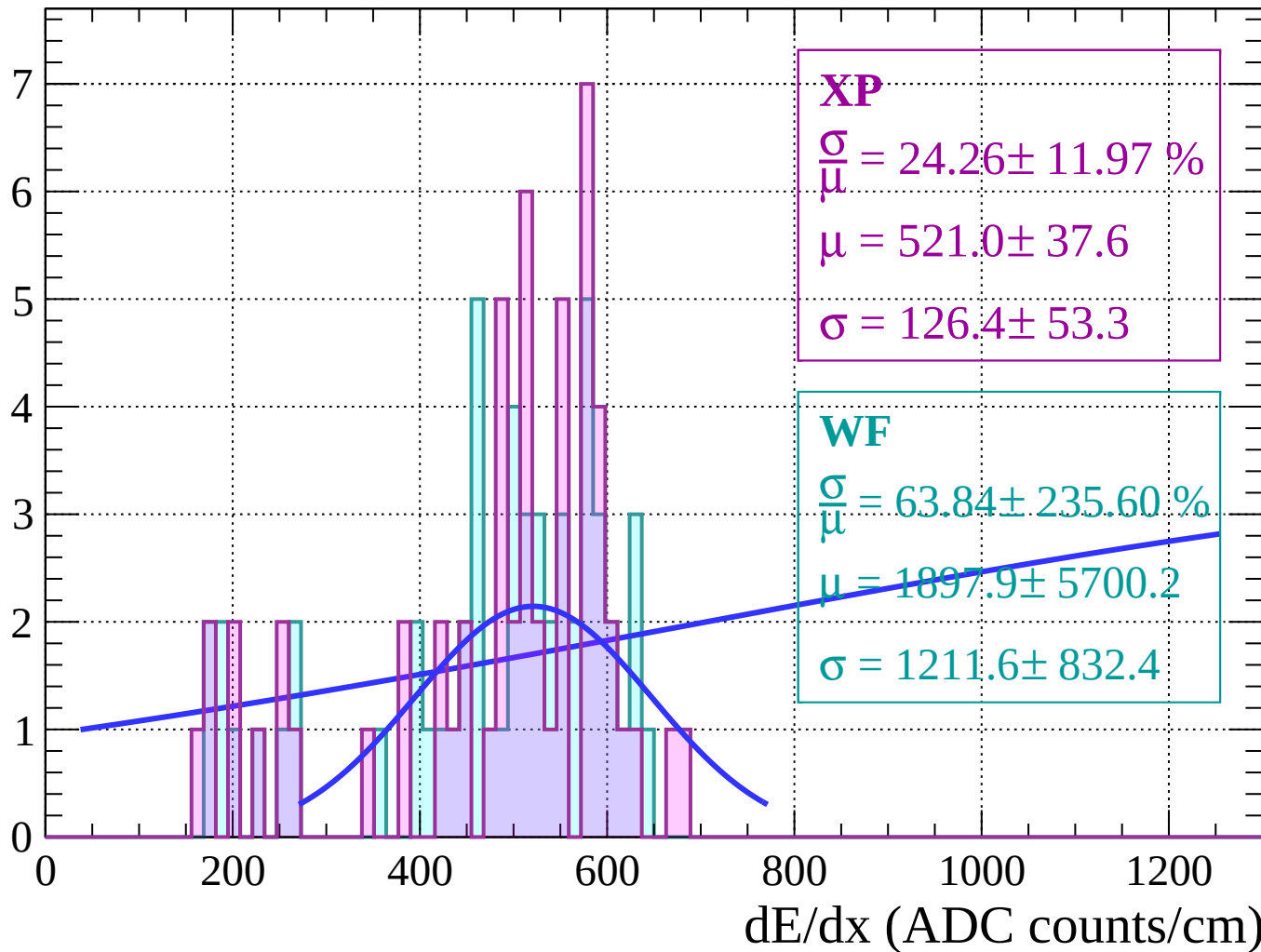
Energy loss | $66 < \theta < 68$

Count



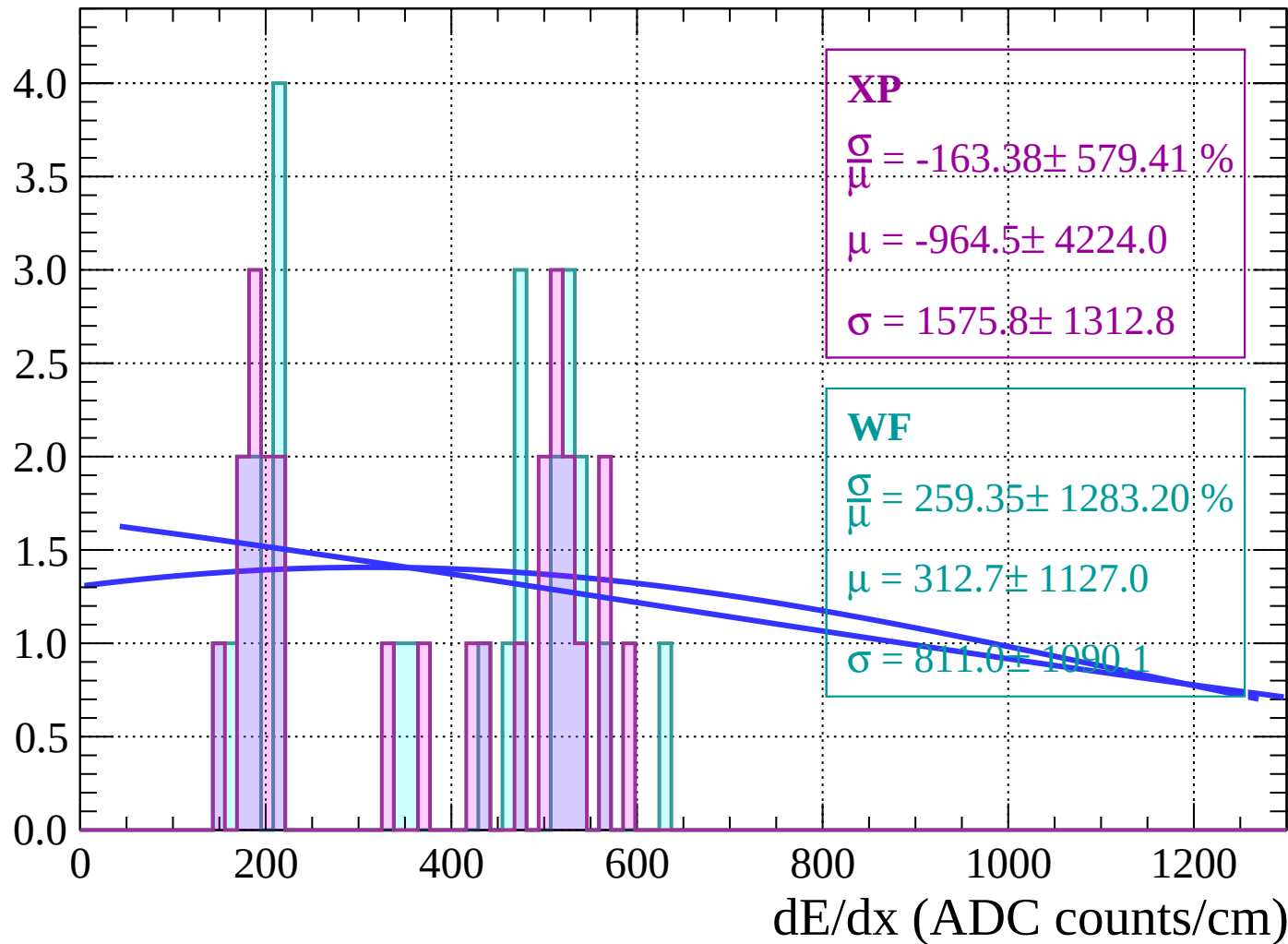
Energy loss | $68 < \theta < 70$

Count



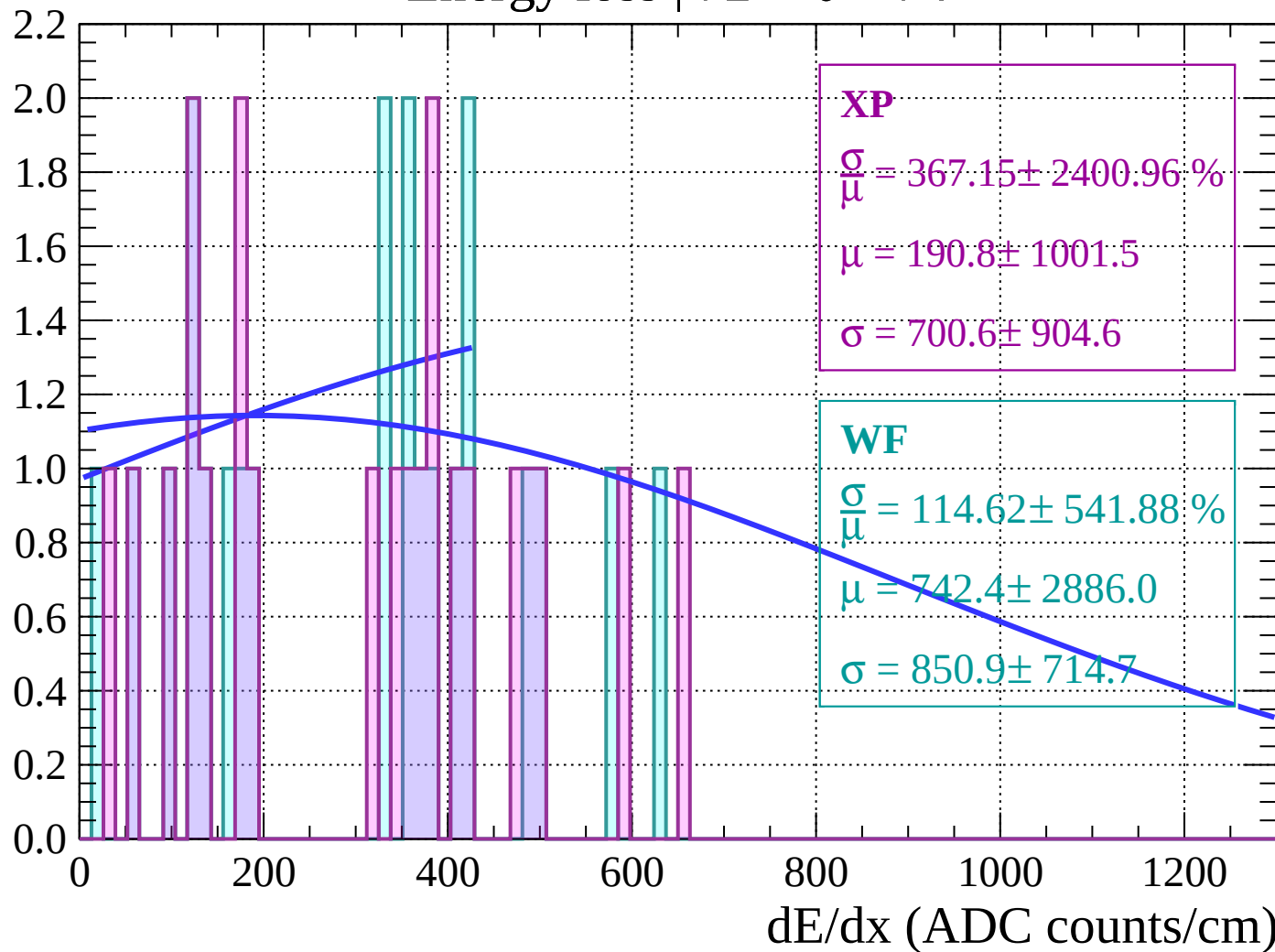
Energy loss | $70 < \theta < 72$

Count



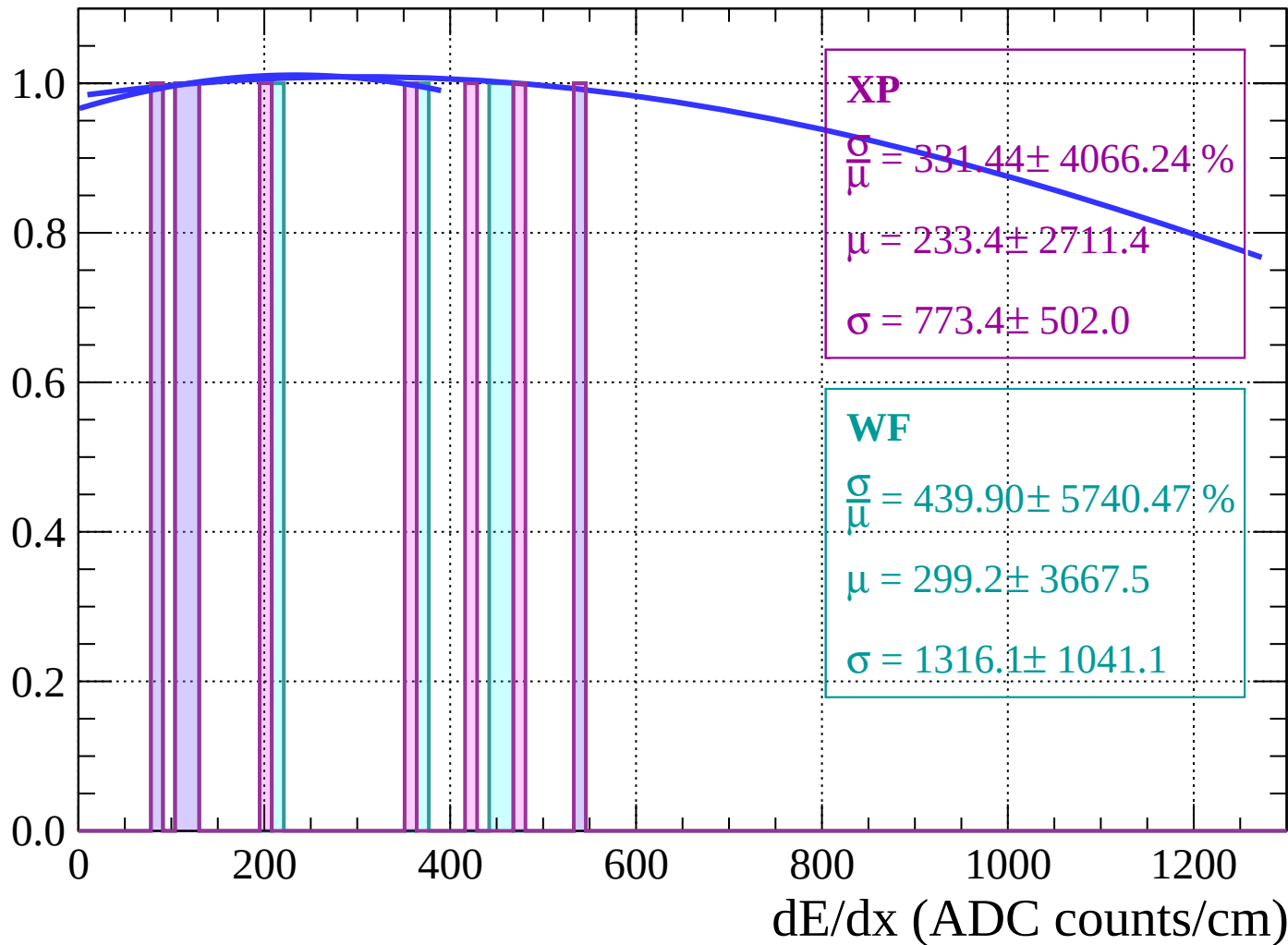
Energy loss | $72 < \theta < 74$

Count



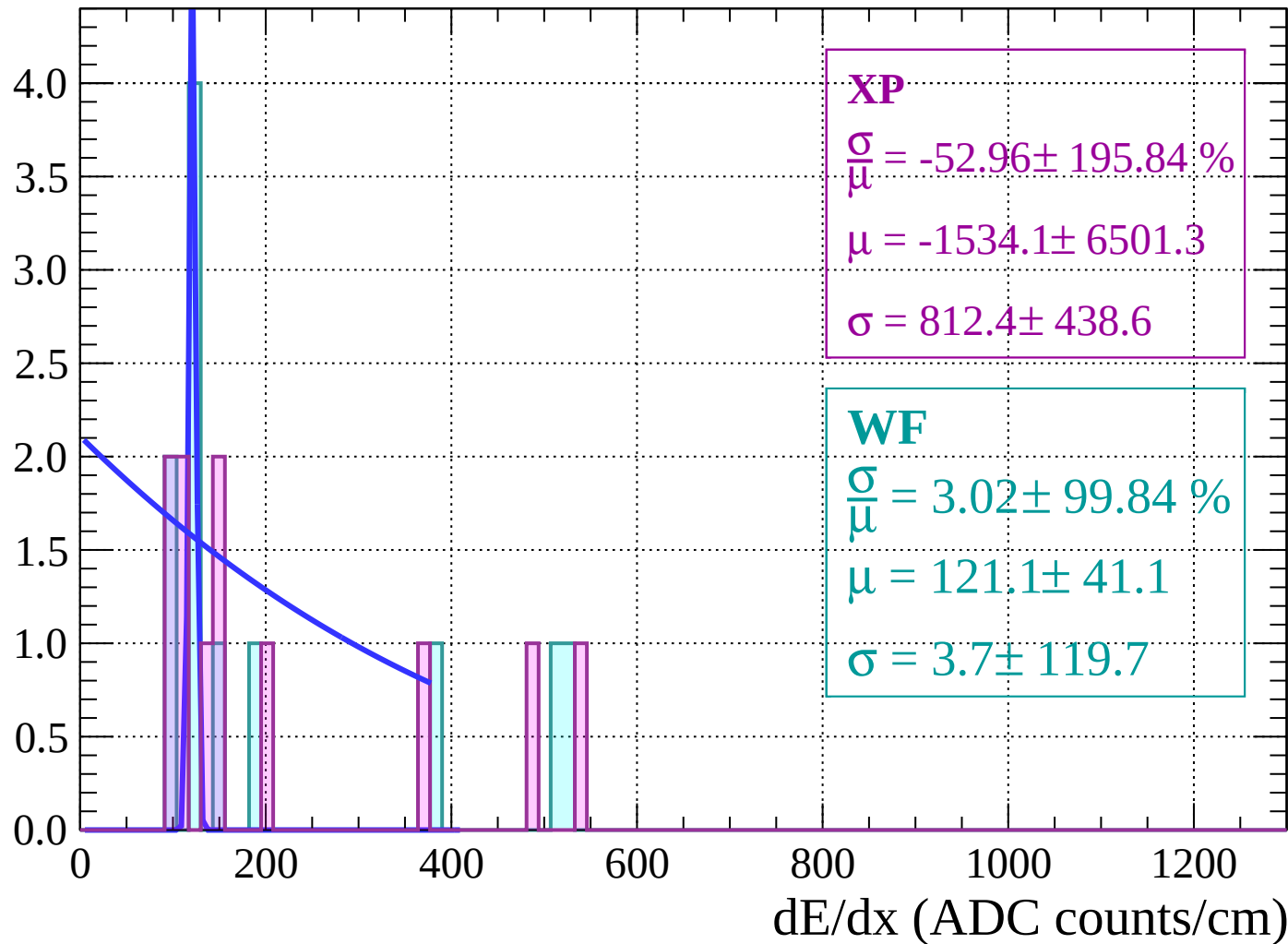
Energy loss | $74 < \theta < 76$

Count



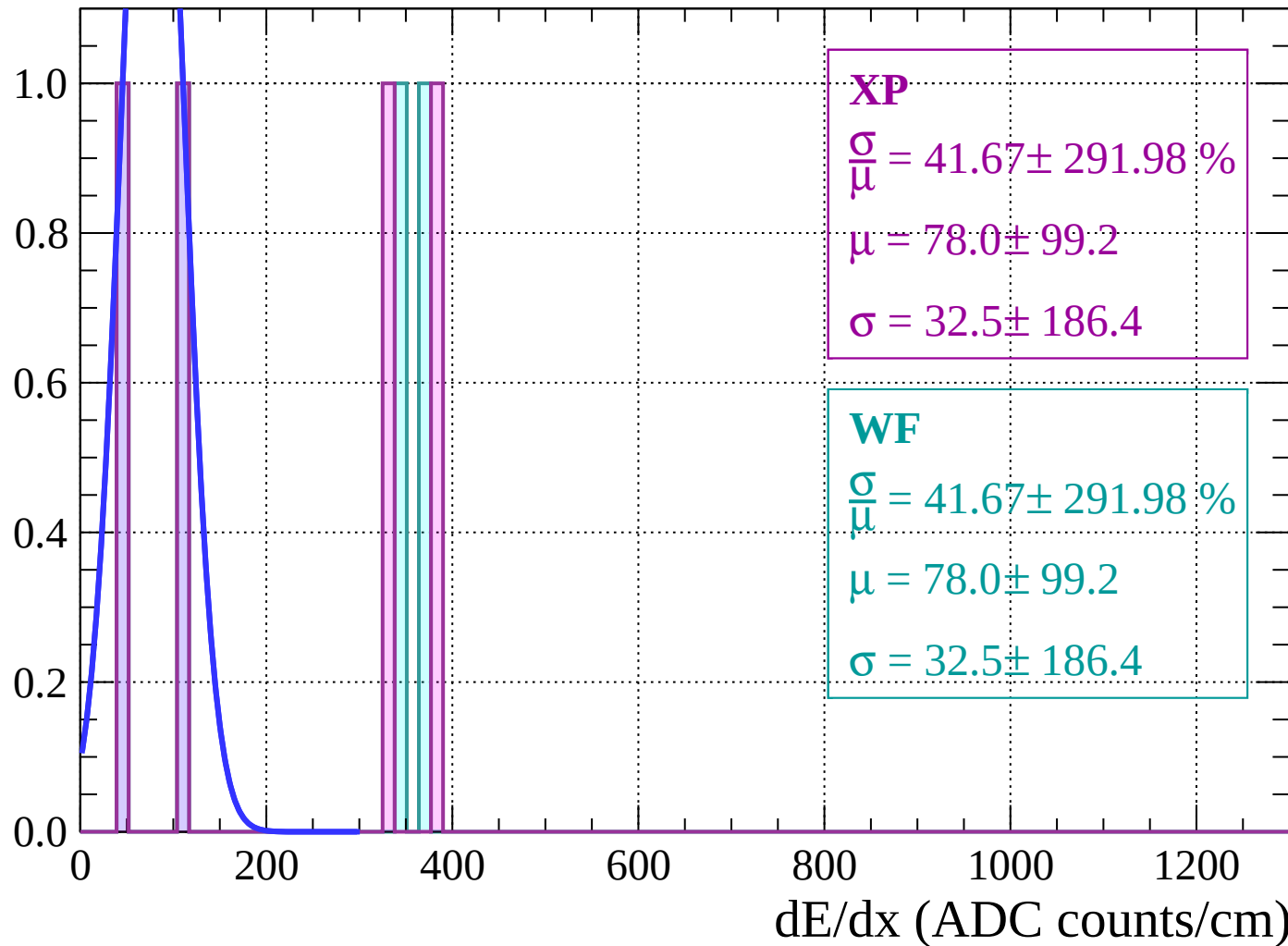
Energy loss | $76 < \theta < 78$

Count



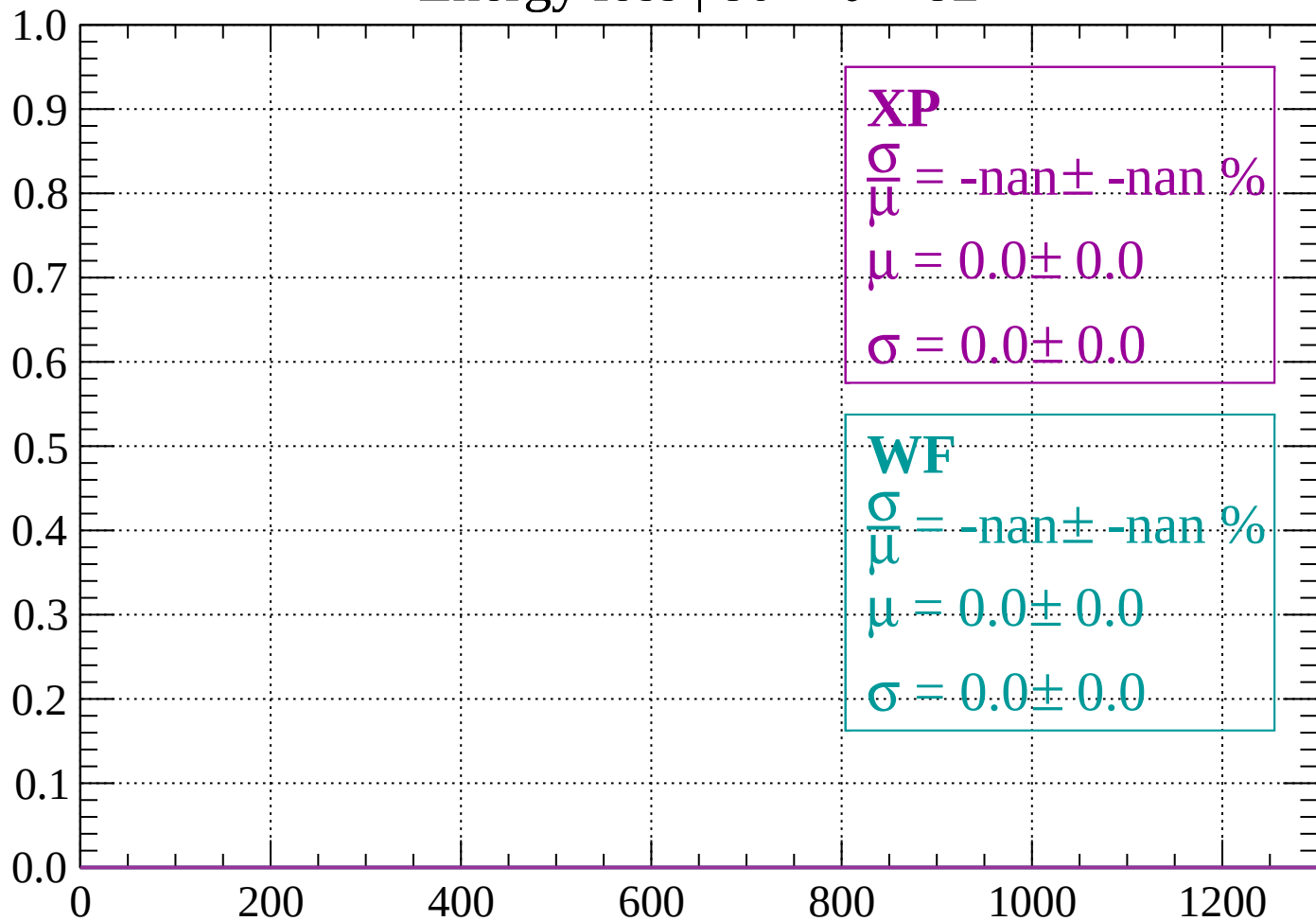
Energy loss | $78 < \theta < 80$

Count



Energy loss | $80 < \theta < 82$

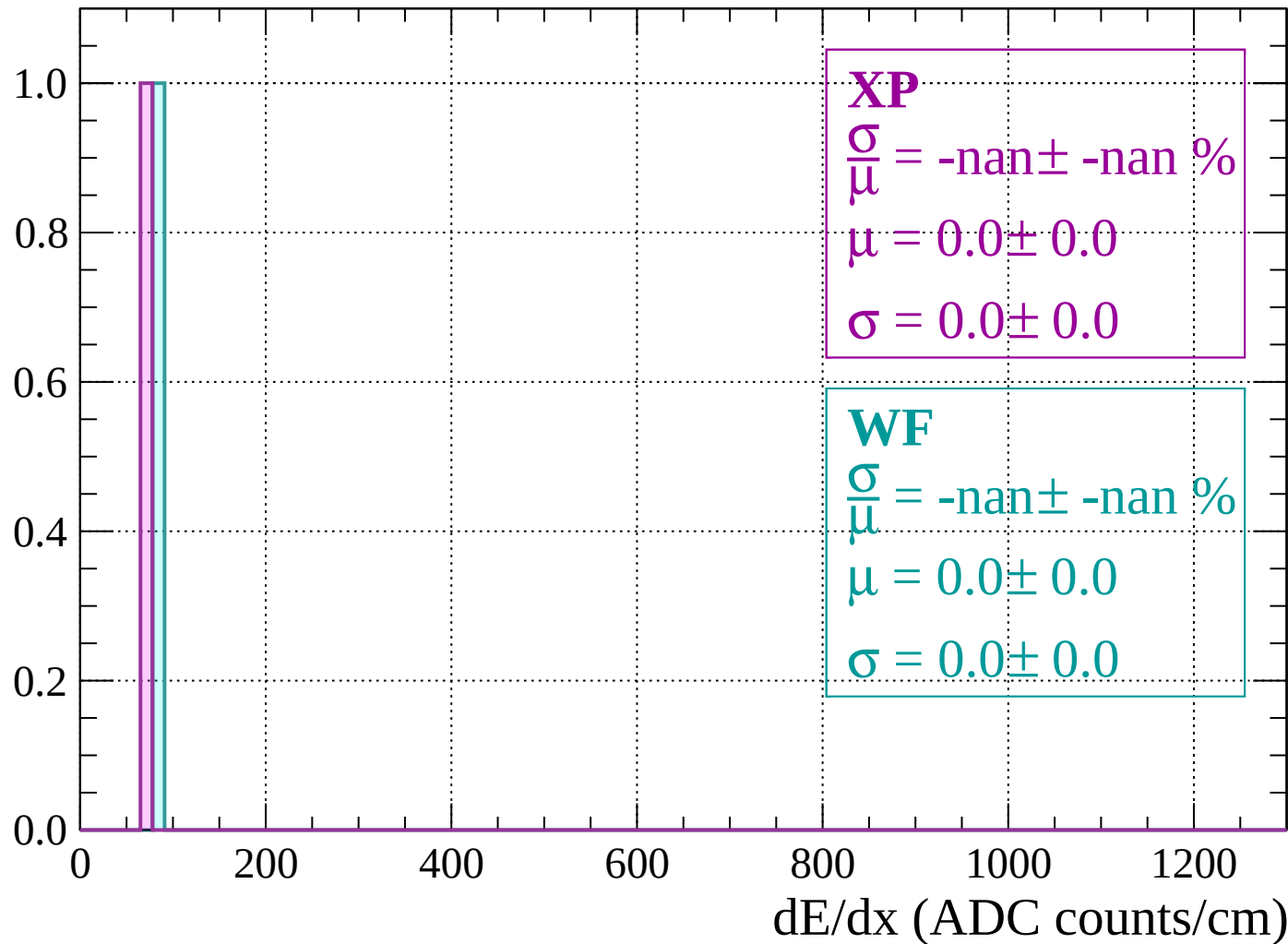
Count



dE/dx (ADC counts/cm)

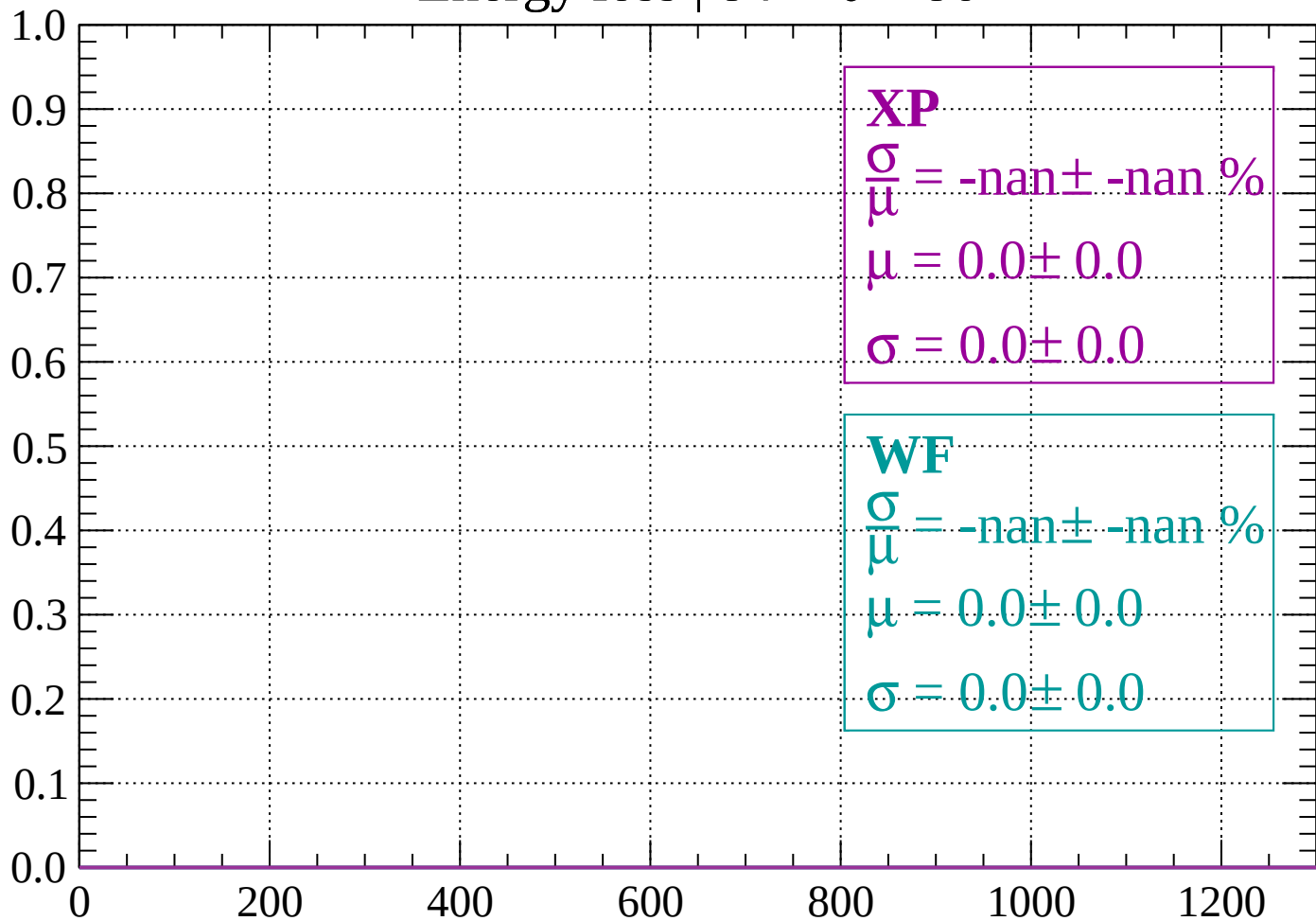
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

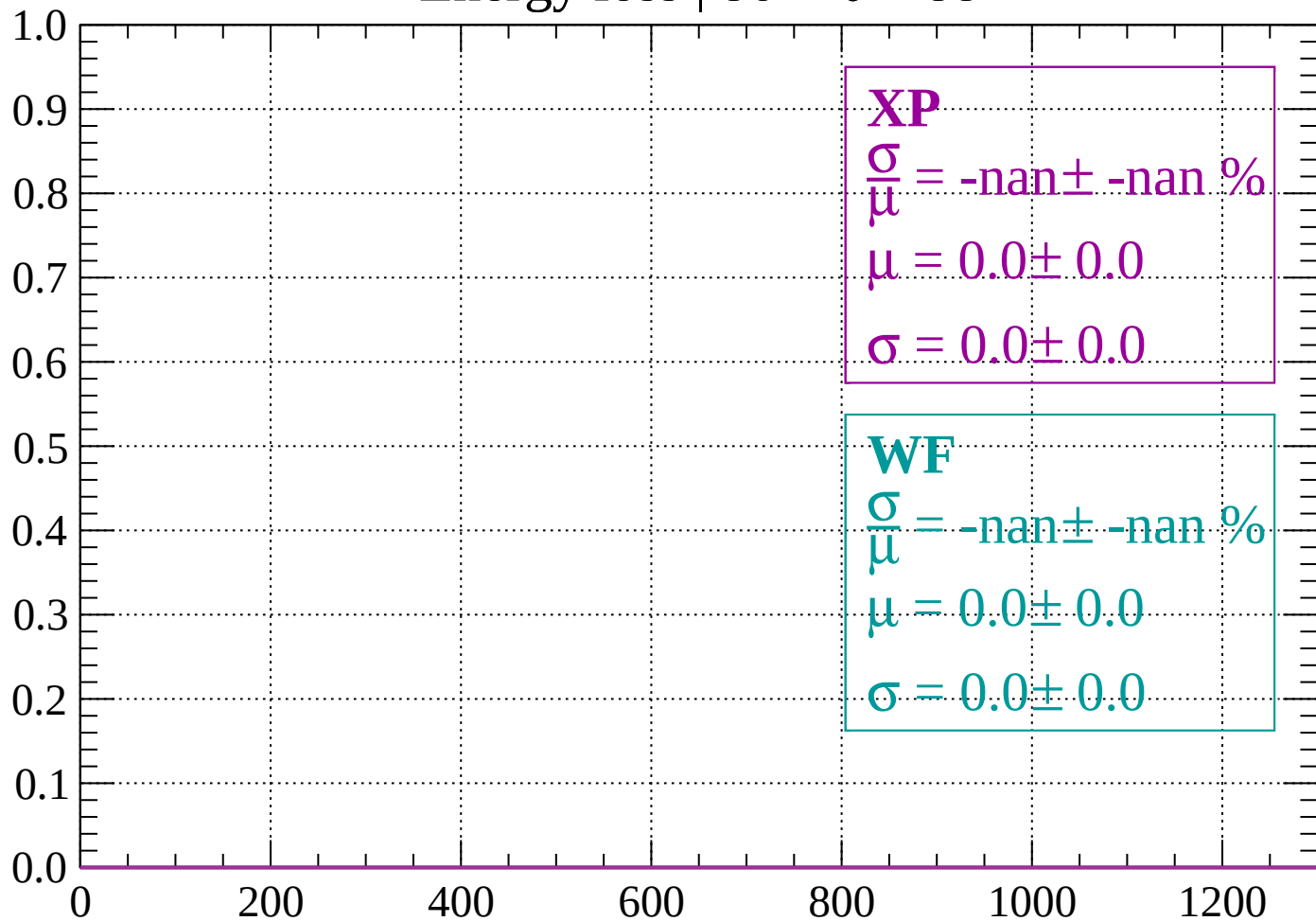
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

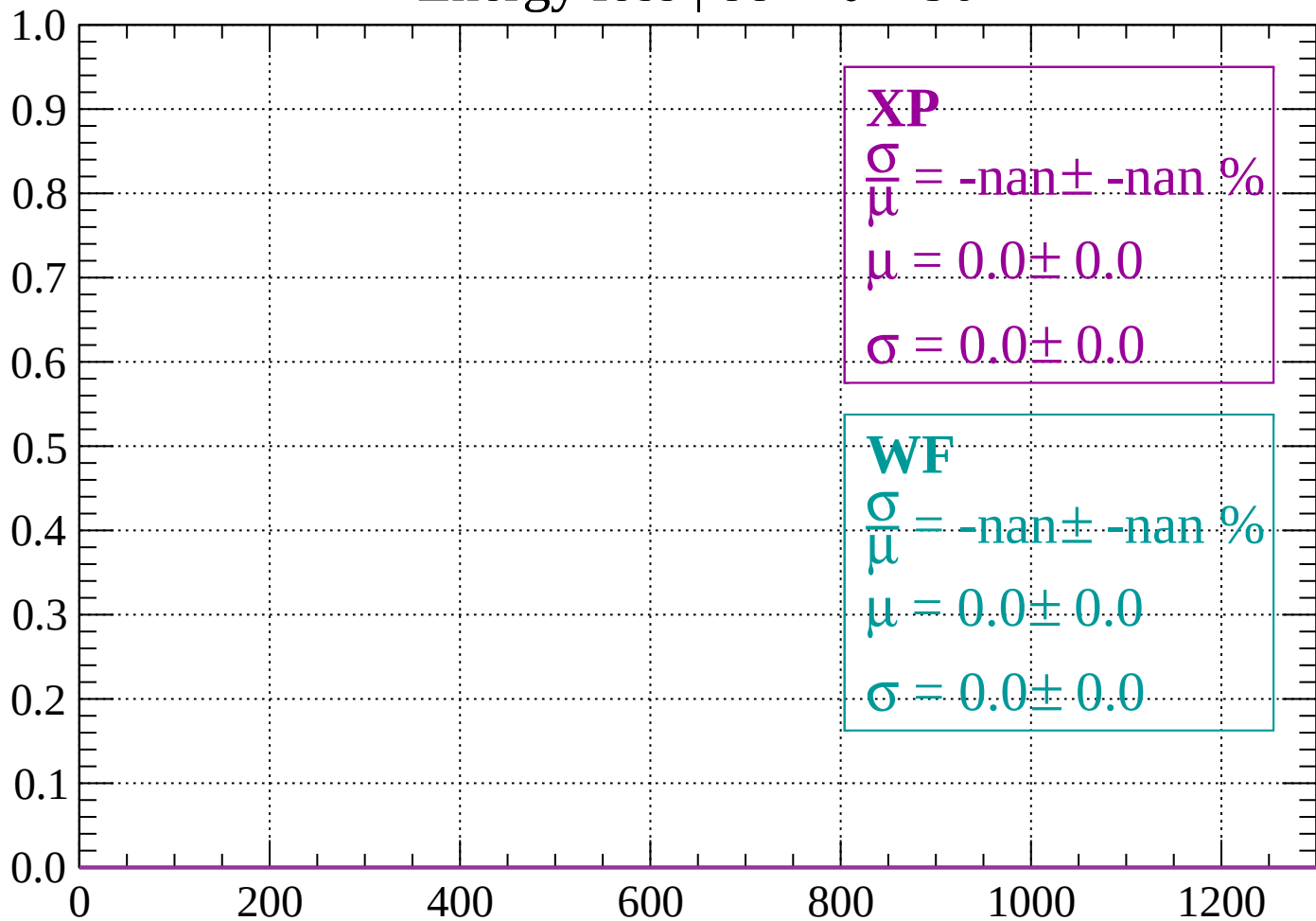
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

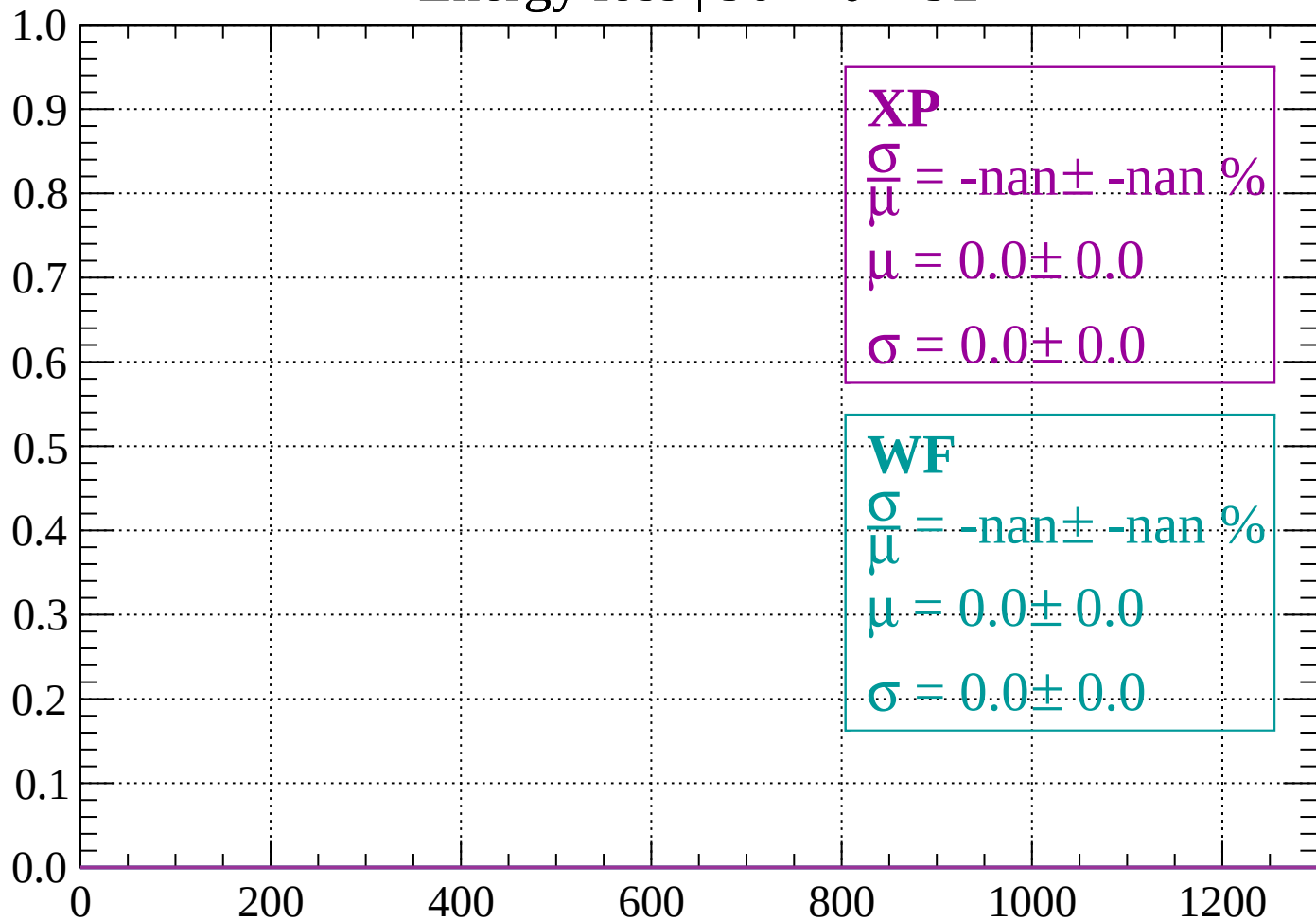
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

